Document Title:	Line by Line Action Report
Document Number:	LP1Y-200-RPT-K215-03
Security Classification:	Restricted

Line by Line Action Report

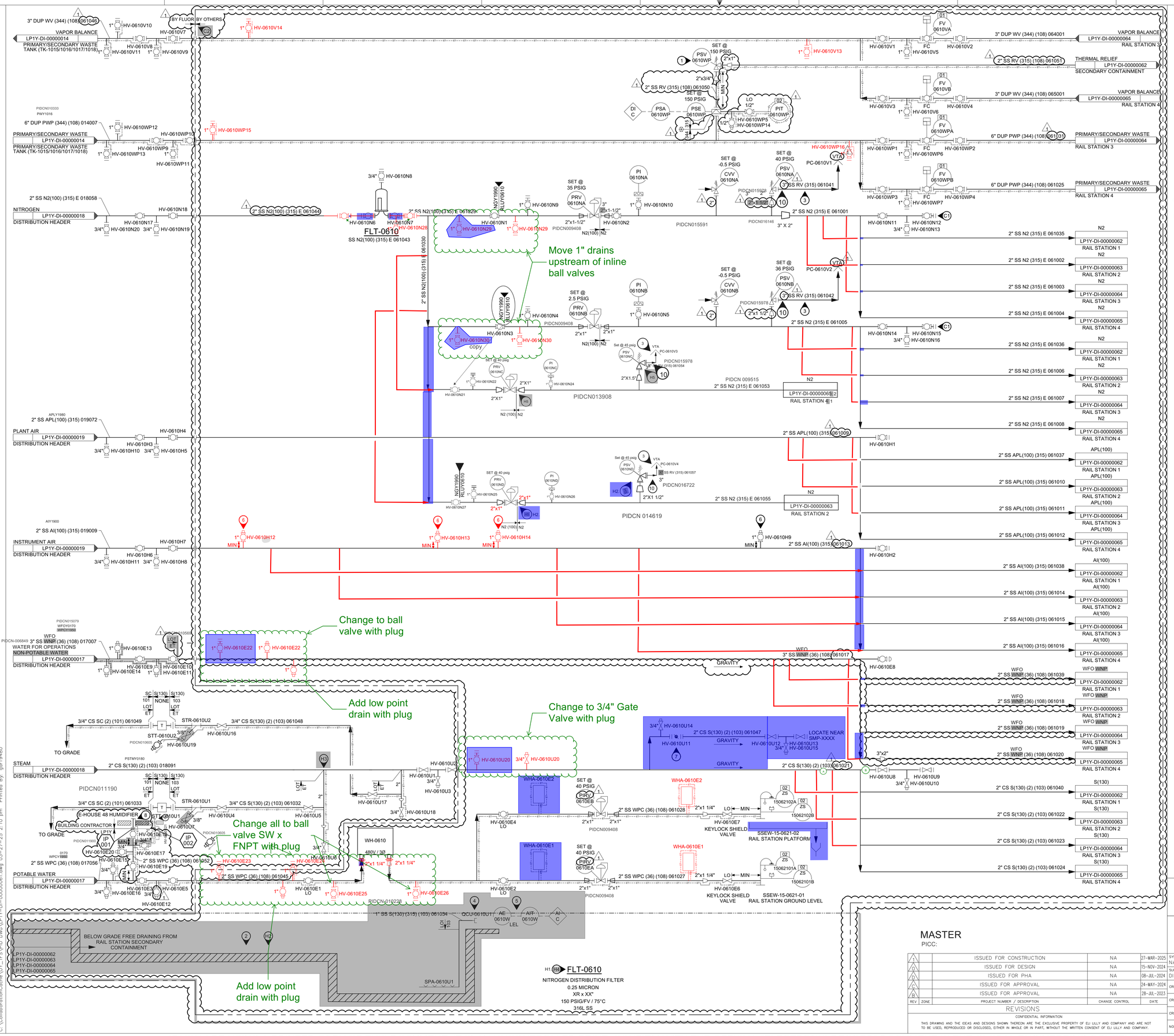
Lilly Rail Containment

Hanson Project Number:25I0007.00

A1 - DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT
Authorization to proceed does not relieve Contractor/Supplier of its responsibility or liability under the Contract and or Purchase Order.
By: Ricardo DaCosta dac54131 March 11, 2026

0	Issued for Information	MW	QR	SS	2/24/2026
Revision	**Issue Status**	**Originator**	**Reviewer**	**Approver**	**Date**

	K215 Model Review Parking Lot Items		
item	**Comments**	**Action By**	**Date entered**
1	Grounding of the rail cars	Fluor/ K215	10-Dec-25
2	Check number of bends at the pump discharge for conformance	Fluor	10-Dec-25
3	Complete pipe design as noted in LxL2 session and comments	K215	10-Dec-25
4	Include emergency light and specify normal vs emergency lights.	K215	10-Dec-25



NOTES:
1. PSV TO BE ELECTRICALLY TRACED.
2. COMMON SECONDARY CONTAINMENT TO BE CLEANED OUT BY VACUUM TRUCK.
3. VENT OUTLET TO BE MINIMUM 6.5 FEET ABOVE STANDING SURFACE. DELETED. PIDCN-007401 by Marissa
4. STEAM CONNECTION FOR SUMP WINTERIZATION.
5. ANALYZER IN PLACE TO MONITOR FOR SOLVET SPILLAGE.
6. INSTRUMENT AIR SUPPLY TO INSTRUMENTS: PIPING CONTRACTOR TO FIELD ROUTE TUBING AS REQUIRED. VALVES ARE LOCATED 30 FT AWAY FROM ONE ANOTHER.
7. BUTTERFLY VALVE, TRIPLE OFFSET, LOCATE AT ACCESSIBLE LOCATION ON HIGH POINT OF LINE.
8. ELECTRIC TRACING EXTENDED 8FT INSIDE THE BUILDING. DELETED. PIDCN-009260 by Adnan
9. ALL INSTRUMENTS TO BE PROVIDED BY FLUOR. PIDCN-012463
10. PROVIDE 1/4" WEEP HOLE AT THE LOW POINT

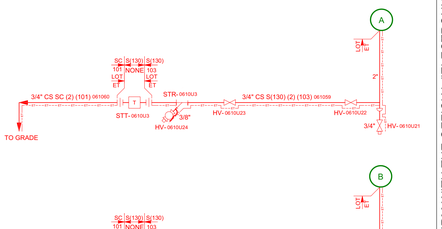
PIDCN-016352 Retain the Notes

PIDCN-017078

CONSTRUCTION NOTES:
C1. SHORT SPOOL PIECE WITH BLIND FLANGE.
C2. ALL INSTRUMENTS TO BE PROVIDED BY FLUOR. PIDCN-016168

HOLD NOTES:
H1. ALL LINE SIZES TO BE CONFIRMED PENDING DETAILED HYDRAULICS.
H2. SECONDARY CONTAINMENT DETAILS ARE TO BE CONFIRMED.
H3. NUMBER OF STEAM TRAPS AND LOCATION TO BE CONFIRMED.
H1. H4. SIZE OF NITROGEN FILTER TO BE CONFIRMED.
H5. PRV AND PSV SIZE TO BE CONFIRMED.
H2. H5. PRV AND PSV SIZE TO BE CONFIRMED.

PIDCN 014619



THE FOLLOWING PICC WERE INCLUDED IN REV 2:
PICCL-306-1605-51, PICCL-306-1605-53, PICCL-306-1605-54, PICCL-306-1605-60, PICCL-306-1605-62, PICCL-306-1605-64, PICCL-306-1605-66, PICCL-306-1605-70, PICCL-306-1605-71, PICCL-306-1605-76, PICCL-306-1605-86, PICCL-306-1605-88, PICCL-306-1605-91

PIDCN 009408 / PICCL-306-1605-51; PIDCN 010228 / PICCL-306-1605-53; PIDCN 015600 / PICCL-306-1605-54; PIDCN 011190 / PICCL-306-1605-60; PIDCN 011900 / PICCL-306-1605-62; PIDCN 009515 / PICCL-306-1605-64; PIDCN 011833, 016166, 016167, 016352 / N/A; PIDCN 012463 / PICCL-306-1605-66; PIDCN 012569 / PICCL-306-1605-70; PIDCN 009849 / PICCL-306-1605-71; PIDCN 013908 / PICCL-306-1605-76; PIDCN 015591 / PICCL-306-1605-86; PIDCN 014619 / PICCL-306-1605-88; PIDCN 015978 / PICCL-306-1605-91.

THE FOLLOWING PICC WERE INCLUDED IN REV 0:
PICCL-306-1605-29; PICCL-306-1605-30; PICCL-306-1605-31; PICCL-306-1605-33; PICCL-306-1605-34; PICCL-306-1605-35; PICCL-306-1605-36; PICCL-306-1605-37; PICCL-306-1605-38; PICCL-306-1605-39.

THE FOLLOWING PICC WERE INCLUDED IN REV 1:
PICCL-306-1605-40; PICCL-306-1605-41; PICCL-306-1605-44; PICCL-306-1605-47; PICCL-306-1605-50. PICCL-306-1605-52

REV	DATE	DES	ENG	CHK	APP	PM APP
7	27-MAR-2025	LJA	MM	JA	SSK	BS
6	15-NOV-2024	LJA	ADS	JA	SSK	BS
5	08-JUL-2024	JRD	MJP	JA	SSK	BS
4	24-MAY-2024	GNA	MJP	JA	SSK	BS
3	28-JUL-2023	CBE	MJP	JA	SSK	BS

EFDEE Engineering PC
PROJECT NO. B1LG
NODE: TFS

FLUOR

Lilly

PIPING & INSTRUMENTATION DIAGRAM
RAIL UTILITIES AND WASTE DISTRIBUTION

MASTER
PICC:

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
7		ISSUED FOR CONSTRUCTION	NA	27-MAR-2025
6		ISSUED FOR DESIGN	NA	15-NOV-2024
5		ISSUED FOR PHA	NA	08-JUL-2024
4		ISSUED FOR APPROVAL	NA	24-MAY-2024
3		ISSUED FOR APPROVAL	NA	28-JUL-2023

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SYSTEM TYPE: NA			EDIFICE/BUILDING: Y	LEVEL/FLOOR: NA
SUB DISCIPLINE: DI	LOCATION CODE: LP1	LOCATION/GRID: NA	CD CODE: YELLOW	PROJECT NAME: LP1-306-1181
CRITICAL QUALITY:		ENTERPRISE GROUP: API	DRAWING INTENTION:	DRAWING SCALE: NONE
CRITICAL BUSINESS:		EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA	MERIDIAN REV: 1
CRITICAL HSE:		VENDOR: NA	DOCUMENT ID: LP1Y-DI-00000061	
NON-CRITICAL		ENGINEERING AGENCY DOC ID: NA		

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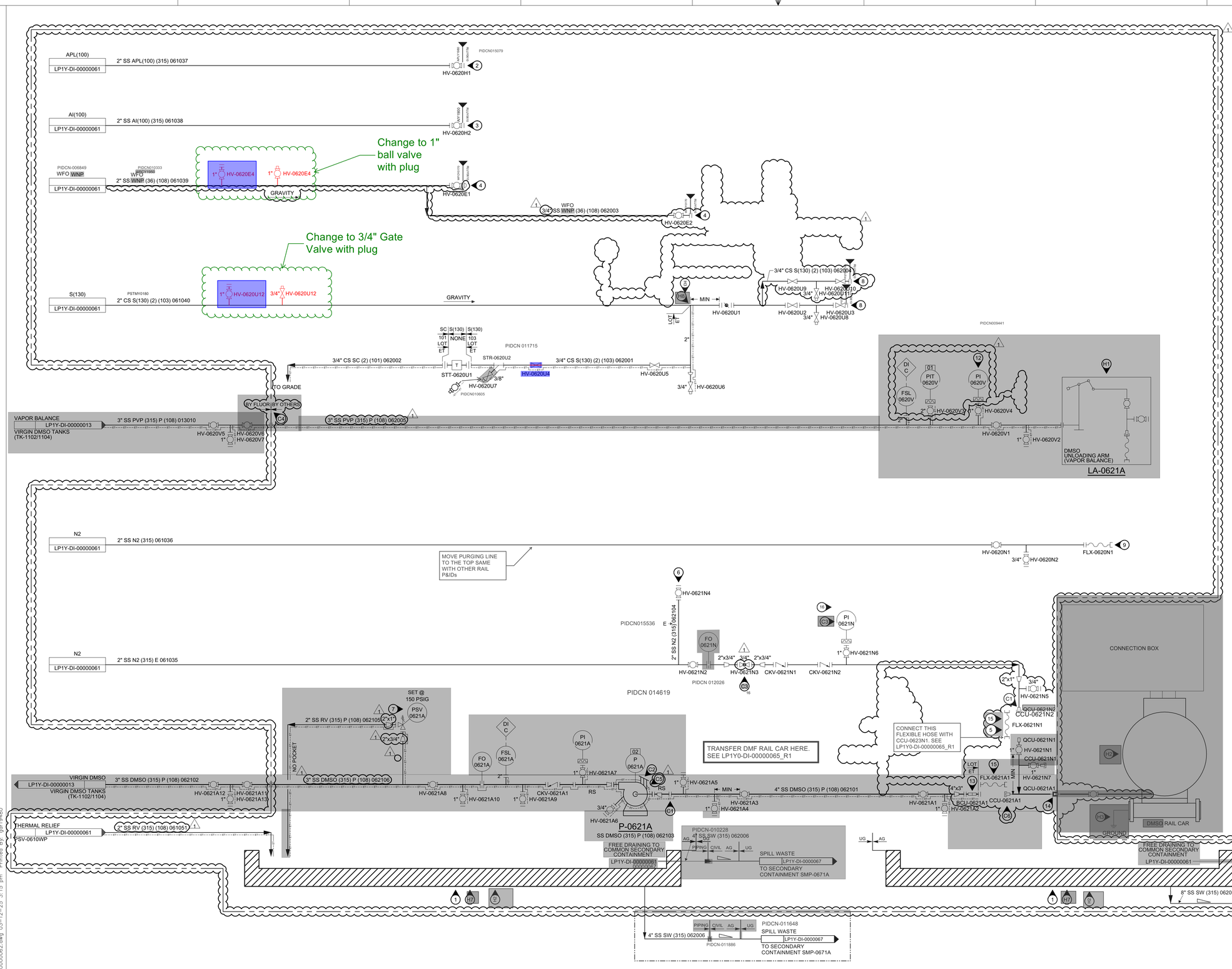

NOTES:
1. SECONDARY CONTAINMENT UNDERNEATH RAIL CAR AND PUMP TO FLOW TO COMMON SECONDARY CONTAINMENT.
2. PLANT AIR CONNECTION. LOCATE AT GRADE.
3. INSTRUMENT AIR SUPPLY TO INSTRUMENTS; PIPING CONTRACTOR TO FIELD-ROUTE TUBING AS REQUIRED.
4. NON-POTABLE WATER CONNECTION. LOCATE AT GRADE.
5. FLEXIBLE HOSE CONNECTION PROVIDED FOR TEMPORARY LINE PURGING OPERATIONS. DELETED.
6. FOR UTILITY NITROGEN. LOCATE AT GRADE.
7. PSV TO BE ELECTRICALLY TRACED DELETED.
8. STEAM CONNECTION. LOCATE AT GRADE.
9. FLEXIBLE HOSE CONNECTION FOR NITROGEN BLANKETING.
10. DELETED. PIDCN 014619
11. DELETED. FLEXIBLE HOSES IN PLACE ONLY DURING UNLOADING OPERATION.
12. PRESSURE GAUGE TO BE VISIBLE TO THE RAIL STATION OPERATOR. DELETED.
13. SAFETY BREAK-AWAY COUPLINGS TO BE INSTALLED. DELETED.
14. DRY DISCONNECT COUPLER TO BE INSTALLED. DELETED.
15. FLEXIBLE HOSES IN PLACE ONLY DURING UNLOADING OPERATION. DELETED.
16. DELETED. PRESSURE GAUGES TO BE VISIBLE AT GLOBE VALVE.
17. ALL INSTRUMENTS TO BE PROVIDED BY FLUOR. PIDCN-012463

RETAIN; DO NOT DELETE

CONSTRUCTION NOTES:
- C1. TEMPORARY STRAINER TO BE PROVIDED BY CONSTRUCTION CONTRACTOR. ARRANGE PIPING TO FACILITATE REMOVAL. STRAINER MESH SIZING TO BE CONFIRMED DURING DETAILED DESIGN.
- C2. INSULATION TO BE INSTALLED BY FIELD INSTALLATION CONTRACTOR.
- C3. PI0621N TO BE VISIBLE AT GLOBE VALVE.
- C4. ALL INSTRUMENTS TO BE PROVIDED BY FLUOR.
- C5. PUMP TO BE PROVIDED BY FLUOR.
- C6. ALL CAMLOCK CONNECTIONS TO HAVE CAP/PLUG AND CHAIN.
- C7. ALL CAMLOCK CONNECTIONS TO HAVE CAP/PLUG AND CHAIN. PIDCN-016168

HOLD NOTES:
- H1. UNLOADING ARM DETAILS TO BE CONFIRMED.
- H2. DETAILS OF INTERFACE CONNECTIONS TO RAIL CARS TO BE CONFIRMED.
- H3. GROUNDING CONFIGURATION AND INSTRUMENTATION TO BE CONFIRMED.
- H4. ALL LINE SIZES TO BE CONFIRMED PENDING DETAILED HYDRAULICS.
- H5. PUMP SIZE TO BE CONFIRMED IN DETAILED ENGINEERING.
- H6. REQUIREMENT FOR RAIL CAR HEATING USING A STEAM AND WATER MIXTURE TO BE CONFIRMED.
- H7. SECONDARY CONTAINMENT DETAILS ARE TO BE CONFIRMED.
- H8. NUMBER OF STEAM TRAPS AND LOCATION TO BE CONFIRMED.

THE FOLLOWING PICC WERE INCLUDED IN REV 2:
PICCL-306-1605-53
PICCL-306-1605-54, PICCL-306-1605-62, PICCL-306-1605-63, PICCL-306-1605-64
PICCL-306-1605-65, PICCL-306-1605-66, PICCL-306-1605-71

PIDCN 010228 / PICCL-306-1605-53;
PIDCN 010605 / PICCL-306-1605-54;
PIDCN 009441, 011648 / PICCL-306-1605-62; PIDCN 011886 / PICCL-306-1605-63;
PIDCN 012026 / PICCL-306-1605-64; PIDCN 010333, 015536, 016168 / N/A;
PIDCN 011715 / PICCL-306-1605-65; PIDCN 012463 / PICCL-306-1605-66;
PIDCN 006849 / PICCL-306-1605-71

THE FOLLOWING PICC WERE INCLUDED IN REV 0:
PICCL-306-1605-29, PICCL-306-1605-30, PICCL-306-1605-35, PICCL-306-1605-36, PICCL-306-1605-40, PICCL-306-1605-41, PICCL-306-1605-44, PICCL-306-1605-46, PICCL-306-1605-47

THE FOLLOWING PICC WERE INCLUDED IN REV 1:

3	14-MAR-2025	JRD	MM	JA	SSK	BS
2	15-NOV-2024	AV	ADS	JA	SSK	BS
1	08-JUL-2024	JDG	MJP	JA	SSK	BS
C	24-MAY-2024	GNA	MJP	JA	SSK	BS
B	28-JUL-2023	JCB	MJP	JA	SSK	BS
REV	DATE	DES	ENG	CHK	APP	PM APP

FLUOR
EFDEE Engineering PC
PROJECT NO. B1LG
NODE: TFS

Lilly

PIPING & INSTRUMENTATION DIAGRAM
RAIL STATION 1

MASTER
PICC:

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
3		ISSUED FOR CONSTRUCTION	NA	14-MAR-2025
2		ISSUED FOR DESIGN	NA	15-NOV-2024
1		ISSUED FOR PHA	NA	08-JUL-2024
C		ISSUED FOR APPROVAL	NA	24-MAY-2024
B		ISSUED FOR APPROVAL	NA	28-JUL-2023

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SYSTEM TYPE: NA		EDIFICE/BUILDING: NA	LEVEL/FLOOR: NA
SUB DISCIPLINE: DI	LOCATION CODE: LP1	LOCATION/GRID: NA	CD CODE: YELLOW	PROJECT NAME: LP1-306-1181
CRITICAL QUALITY		ENTERPRISE GROUP: API	DRAWING INTENTION	DRAWING SCALE: NONE
CRITICAL BUSINESS		EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA	MERIDIAN REV: 1
CRITICAL HSE		VENDOR: NA	DOCUMENT ID: LP1Y-DI-00000062
NON-CRITICAL		ENGINEERING AGENCY DOC ID: NA	

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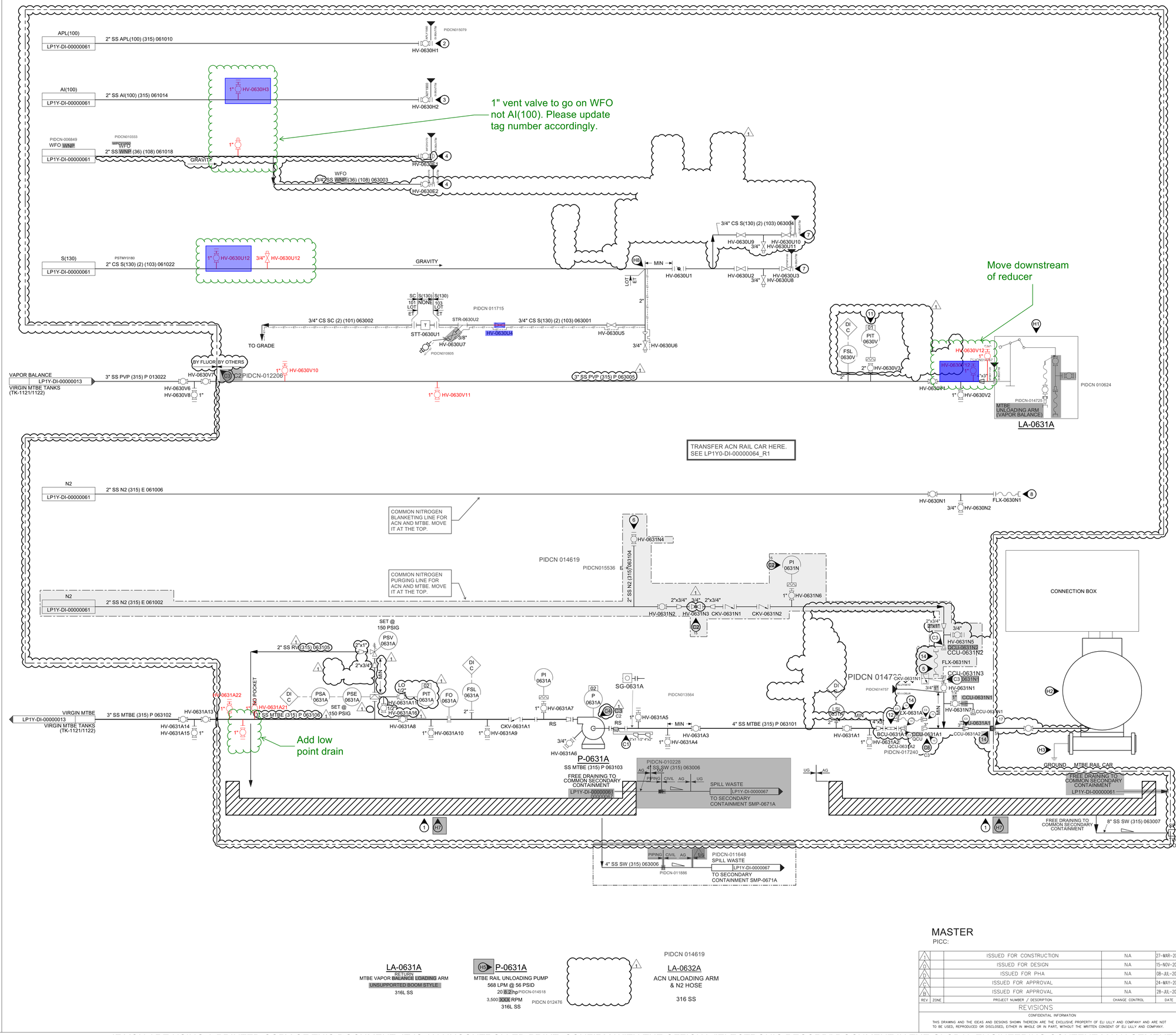

NOTES:
1. SECONDARY CONTAINMENT UNDERNEATH RAIL CAR AND PUMP TO FLOW TO COMMON SECONDARY CONTAINMENT.
2. PLANT AIR CONNECTION. LOCATE AT GRADE.
3. INSTRUMENT AIR SUPPLY TO INSTRUMENTS; PIPING CONTRACTOR TO FIELD-ROUTE TUBING AS REQUIRED. WATER FOR OPERATIONS
4. NON-POTABLE WATER CONNECTION. LOCATE AT GRADE.
5. FLEXIBLE HOSE CONNECTION PROVIDED FOR TEMPORARY LINE PURGING OPERATIONS.
6. FOR UTILITY NITROGEN. LOCATE AT GRADE.
7. STEAM CONNECTION. LOCATE AT GRADE.
8. FLEXIBLE HOSE CONNECTION FOR NITROGEN BLANKETING. COMMON FOR ACN AND MTBE.
9. DELETED
10. DELETED
11. PIT LOCAL INDICATOR TO BE VISIBLE FROM RAIL STATION OPERATOR.
12. SAFETY BREAK-AWAY COUPLINGS TO BE INSTALLED.
13. DRY DISCONNECT COUPLER TO BE INSTALLED.
14. FLEXIBLE HOSES IN PLACE ONLY DURING UNLOADING OPERATION.
15. PRESSURE GAUGES TO BE VISIBLE AT GLOBE VALVE.
16. ALL INSTRUMENTS TO BE PROVIDED BY FLUOR. PIDCN-012463
17. RAIL CAR ADAPTER TO BE THREADED CONNECTION TO RAIL CAR BOTTOM OUTLET.

PIDCN 014619

CONSTRUCTION NOTES:
C1. TEMPORARY STRAINER TO BE PROVIDED BY CONSTRUCTION CONTRACTOR. ARRANGE PIPING TO FACILITATE REMOVAL. STRAINER MESH SIZING TO BE CONFIRMED DURING DETAILED DESIGN.
C2. PI-0631N TO BE VISIBLE AT GLOBE VALVE. PIDCN-009441
C3. ALL INSTRUMENTS TO BE PROVIDED BY FLUOR. PIDCN-016168
C4. PUMP TO BE PROVIDED BY FLUOR.
C5. ALL CAMLOCK CONNECTIONS TO HAVE CAP/PLUG AND CHAIN.
C4. ALL DRY DISCONNECT COUPLERS TO HAVE DUST CAP/PLUG AND CHAIN.

HOLD NOTES:
H1. UNLOADING ARM DETAILS TO BE CONFIRMED.
H2. DETAILS OF INTERFACE CONNECTIONS TO RAIL CARS TO BE CONFIRMED.
H3. GROUNDING CONFIGURATION AND INSTRUMENTATION TO BE CONFIRMED.
H4. ALL LINE SIZES TO BE CONFIRMED PENDING DETAILED HYDRAULICS.
H5. PUMP SIZE TO BE CONFIRMED IN DETAILED ENGINEERING.
H6. REQUIREMENT FOR RAIL CAR HEATING USING A STEAM AND WATER MIXTURE TO BE CONFIRMED.
H7. SECONDARY CONTAINMENT DETAILS ARE TO BE CONFIRMED.
H8. NUMBER OF STEAM TRAPS AND LOCATION TO BE CONFIRMED.

THE FOLLOWING PICC WERE INCLUDED IN REV 2:
PICCL-306-1605-53; PICCL-306-1605-54; PICCL-306-1605-62; PICCL-306-1605-63; PICCL-306-1605-64; PICCL-306-1605-65; PICCL-306-1605-66; PICCL-306-1605-71; PICCL-306-1605-75; PICCL-306-1605-82; PICCL-306-1605-85; PICCL-306-1605-88; PICCL-306-1605-90

PIDCN 010228 / PICCL-306-1605-53; PIDCN 010605 / PICCL-306-1605-54; PIDCN 011648, 009441 / PICCL-306-1605-62; PIDCN 010333, 012206, 012476 / N/A; PIDCN 011886 / PICCL-306-1605-63; PIDCN 010624 / PICCL-306-1605-64; PIDCN 011715 / PICCL-306-1605-65; PIDCN 012463 / PICCL-306-1605-66; PIDCN 006849 / PICCL-306-1605-71; PIDCN 013564 / PICCL-306-1605-75; PIDCN 014725 / PICCL-306-1605-82; PIDCN 014518 / PICCL-306-1605-85; PIDCN 014737, 015536, 016168/ N/A; PIDCN 014619 / PICCL-306-1605-88; PIDCN 014730 / PICCL-306-1605-90

THE FOLLOWING PICC WERE INCLUDED IN REV 0:
PICCL-306-1605-29; PICCL-306-1605-30; PICCL-306-1605-31; PICCL-306-1605-35; PICCL-306-1605-36; PICCL-306-1605-37; PICCL-306-1605-38; PICCL-306-1605-39

THE FOLLOWING PICC WERE INCLUDED IN REV 1:
PICCL-306-1605-41; PICCL-306-1605-44; PICCL-306-1605-46; PICCL-306-1605-47; PICCL-306-1605-50

MASTER
PICC:

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
1		ISSUED FOR CONSTRUCTION	NA	27-MAR-2025
0		ISSUED FOR DESIGN	NA	15-NOV-2024
D		ISSUED FOR PHA	NA	08-JUL-2024
C		ISSUED FOR APPROVAL	NA	24-MAY-2024
B		ISSUED FOR APPROVAL	NA	28-JUL-2023

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REV	DATE	DES	ENG	CHK	APP	PM APP
1	27-MAR-2025	RLB	MM	JA	SSK	BS
0	15-NOV-2024	AV	ADS	JA	SSK	BS
D	08-JUL-2024	JRB	MJP	JA	SSK	BS
C	24-MAY-2024	GNA	MJP	JA	SSK	BS
B	28-JUL-2023	CBE	MJP	JA	SSK	BS

FLUOR

EFDEE Engineering PC
PROJECT NO. B1LG
NODE: TFS

Lilly

PIPING & INSTRUMENTATION DIAGRAM
RAIL STATION 2

SYSTEM TYPE: NA				EDIFICE/BUILDING: Y	LEVEL/FLOOR: NA
SUB DISCIPLINE: DI	LOCATION CODE: LP1	LOCATION/GRID: NA		CCI CODE: YELLOW	PROJECT NAME: LP1-306-1181
CRITICAL QUALITY		ENTERPRISE GROUP: API		DRAWING INTENTION	DRAWING SCALE: NONE
CRITICAL BUSINESS		EQUIPMENT CLASSIFICATION: NA		ALTERNATE DOCUMENT ID: NA	MERDIAN REV: 1
CRITICAL HSE		VENDOR: NA		DOCUMENT ID: LP1Y-DI-00000063	
NON-CRITICAL		ENGINEERING AGENCY DOC ID: NA			

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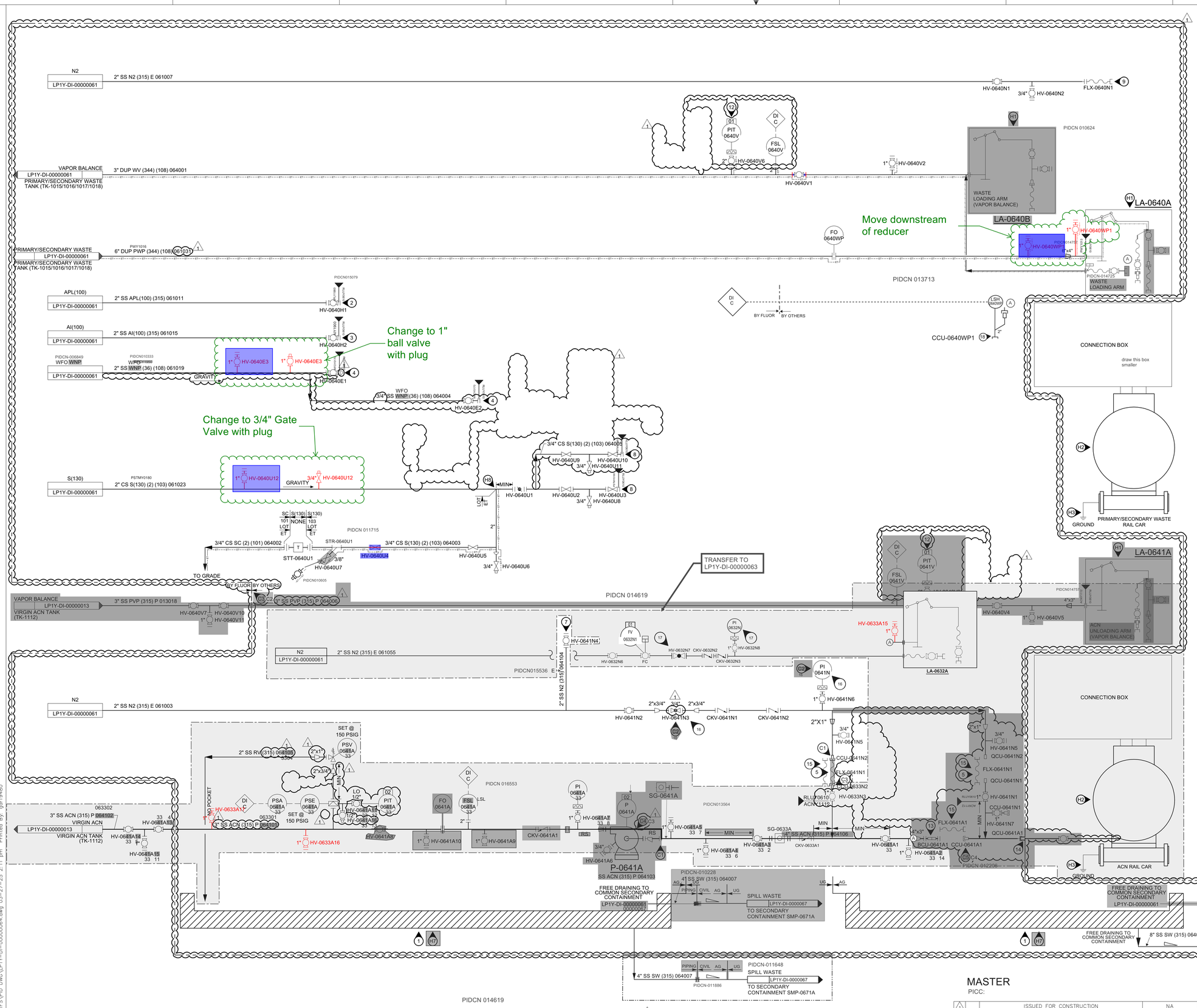

NOTES:
1. SECONDARY CONTAINMENT UNDERNEATH RAIL CAR AND PUMP TO FLOW TO COMMON SECONDARY CONTAINMENT
2. PLANT AIR CONNECTION. LOCATE AT GRADE.
3. INSTRUMENT AIR SUPPLY TO INSTRUMENTS; PIPING CONTRACTOR TO FIELD-ROUTE TUBING AS REQUIRED. , , WATER FOR OPERATIONS
4. NON-POTABLE WATER CONNECTION. LOCATE AT GRADE.
5. FLEXIBLE HOSE CONNECTION PROVIDED FOR TEMPORARY LINE PURGING OPERATIONS.
6. NITROGEN BLANKET LOADING ARM FOR PROVISIONAL INERTING IS COMMON TO THE RAIL STATION AND CAN BE USED BY SOLVENT OR WASTE RAIL CARS. DELETED
7. FOR UTILITY NITROGEN, LOCATE AT GRADE. DELETED FOR UTILITY NITROGEN. LOCATE AT GRADE.
8. STEAM CONNECTION. LOCATE AT GRADE.
9. FLEXIBLE HOSE CONNECTION FOR NITROGEN BLANKETING.
10. DELETED
11. DELETED
12. PIT LOCAL INDICATOR TO BE VISIBLE FROM RAIL STATION OPERATOR.
13. SAFETY BREAK-AWAY COUPLINGS TO BE INSTALLED. DELETED
14. DRY DISCONNECT COUPLER TO BE INSTALLED. DELETED
15. FLEXIBLE HOSES IN PLACE ONLY DURING UNLOADING OPERATION. DELETED
16. PI 0641N TO BE LOCATED TO ALLOW PRESSURE GAUGES TO BE VISIBLE AT GLOBE VALVE.

PIDCN 014619
FLEXIBLE HOSES IN PLACE ONLY DURING UNLOADING OPERATION.

CONSTRUCTION NOTES:
- C1. TEMPORARY STRAINER TO BE PROVIDED BY CONSTRUCTION CONTRACTOR; ARRANGE PIPING TO FACILITATE REMOVAL. STRAINER MESH SIZING TO BE CONFIRMED DURING DETAILED DESIGN. DELETED
- C2. PI 0641N TO BE VISIBLE AT GLOBE VALVE. PIDCN009441
- C3. ALL INSTRUMENTS TO BE PROVIDED BY FLUOR. PIDCN-016168
- C4. PUMP TO BE PROVIDED BY FLUOR.
- C5. ALL CAMLOCK CONNECTIONS TO HAVE CAPPLUG AND CHAIN
- 17. ALL INSTRUMENTS TO BE PROVIDED BY FLUOR. PIDCN-012463
- 18. Y-ADAPTER, CONNECTION ACCESSORIES, LEVEL SWITCH INCLUDING CABLE TO BE PROVIDED BY RAIL STATION VENDOR.

HOLD NOTES:
- H1. UNLOADING ARM DETAILS TO BE CONFIRMED.
- H2. DETAILS OF INTERFACE CONNECTIONS TO RAIL CARS TO BE CONFIRMED.
- H3. GROUNDING CONFIGURATION AND INSTRUMENTATION TO BE CONFIRMED.
- H4. ALL LINE SIZES TO BE CONFIRMED PENDING DETAILED HYDRAULICS.
- H5. PUMP SIZE TO BE CONFIRMED IN DETAILED ENGINEERING. DELETED
- H6. REQUIREMENT FOR RAIL CAR HEATING USING A STEAM AND WATER MIXTURE TO BE CONFIRMED.
- H7. SECONDARY CONTAINMENT DETAILS ARE TO BE CONFIRMED.
- H8. NUMBER OF STEAM TRAPS AND LOCATION TO BE CONFIRMED.

THE FOLLOWING PICC WERE INCLUDED IN REV 2:
PICCL-306-1605-53, PICCL-306-1605-54, PICCL-306-1605-62, PICCL-306-1605-63, PICCL-306-1605-64, PICCL-306-1605-65, PICCL-306-1605-66, PICCL-306-1605-74, PICCL-306-1605-75, PICCL-306-1605-77, PICCL-306-1605-82, PICCL-306-1605-88, PICCL-306-1605-96

PIDCN 010228 / PICCL-306-1605-53; PIDCN 010605 / PICCL-306-1605-54; PIDCN 011648, 009441 / PICCL-306-1605-62; PIDCN 010333, 012206, 012476, 014757, 015536, 016168 / N/A; PIDCN 011886 / PICCL-306-1605-63; PIDCN 010624 / PICCL-306-1605-64; PIDCN 011715 / PICCL-306-1605-65; PIDCN 012463 / PICCL-306-1605-66; PIDCN 006849 / PICCL-306-1605-71; PIDCN 013564 / PICCL-306-1605-75; PIDCN 013713 / PICCL-306-1605-77; PIDCN 014725 / PICCL-306-1605-82; PIDCN 014619 / PICCL-306-1605-88; PIDCN 016553 / PICCL-306-1605-96

THE FOLLOWING PICC WERE INCLUDED IN REV 0:
PICCL-306-1605-26, PICCL-306-1605-30, PICCL-306-1605-31, PICCL-306-1605-35, PICCL-306-1605-36, PICCL-306-1605-37, PICCL-306-1605-38, PICCL-306-1605-39

THE FOLLOWING PICC WERE INCLUDED IN REV 1:
PICCL-306-1605-41, PICCL-306-1605-44, PICCL-306-1605-45, PICCL-306-1605-46, PICCL-306-1605-47, PICCL-306-1605-50

MASTER
PICC:

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
1		ISSUED FOR CONSTRUCTION	NA	27-MAR-2025
0		ISSUED FOR DESIGN	NA	15-NOV-2024
D		ISSUED FOR PHA	NA	08-JUL-2024
C		ISSUED FOR APPROVAL	NA	24-MAY-2024
B		ISSUED FOR APPROVAL	NA	28-JUL-2023

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REV	DATE	DES	ENG	CHK	APP	PM APP
1	27-MAR-2025	ADM	MM	JA	SSK	BS
0	15-NOV-2024	LJA	ADS	JA	SSK	BS
D	08-JUL-2024	LJA	MJP	JA	SSK	BS
C	24-MAY-2024	PMS	MJP	JA	SSK	BS
B	28-JUL-2023	JCB	MJP	JA	SSK	BS

EFDEE Engineering PC
PROJECT NO. B1LG
NODE: TFS

FLUOR

Lilly

PIPING & INSTRUMENTATION DIAGRAM
RAIL STATION 3

SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: NA
SUB DISCIPLINE: DI	LOCATION CODE: LP1	LOCATION/GRID: NA	PROJECT NAME: LP1-306-1181
CRITICAL QUALITY		CO CODE: YELLOW	DRAWING SCALE: NONE
CRITICAL BUSINESS		ENTERPRISE GROUP: API	DRAWING INTENTION
CRITICAL HSE		EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA
NON-CRITICAL		VENDOR: NA	MERIDIAN REV: 1
		ENGINEERING AGENCY DOC ID: NA	DOCUMENT ID: LP1Y-DI-00000064

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N2
LP1Y-DI-00000061
2" SS N2 (315) E 061008
HV-0650N1
3/4" HV-0650N2
FLX-0650N1
9

VAPOR BALANCE
LP1Y-DI-00000061
PRIMARY/SECONDARY WASTE TANK (TK-1015/1016/1017/1018)
3" DUP WV (344) (108) 065001
PIT 0650V
DI C
FSL 0650V
HV-0650V6
HV-0650V1
HV-0650V3
PIDCN 010624
WASTE LOADING ARM (VAPOR BALANCE)
LA-0650B
LA-0650A
HV-0650WP1
HV-0650WP1A
6"x4"
PIDCN014727
PIDCN-014725
WASTE LOADING ARM
Move downstream of reducer
FO 0650WP
BY FLUOR
BY OTHERS
PIDCN 013713
LSH 0650WP
CCU-0650WP1
2"
CONNECTION BOX
draw it smaller
GROUND
PRIMARY/SECONDARY WASTE RAIL CAR

PRIMARY/SECONDARY WASTE
LP1Y-DI-00000061
PRIMARY/SECONDARY WASTE TANK (TK-1015/1016/1017/1018)
PIDCN010333
PWY1016
6" DUP PWP (344) (108) 061025

APL(100)
LP1Y-DI-00000061
2" SS APL(100) (315) 061012
PIDCN015079
HV-0650H1

AI(100)
LP1Y-DI-00000061
2" SS AI(100) (315) 061016
HV-0650H2
Change to 1" ball valve with plug

PIDCN-006849
WFO WNP
LP1Y-DI-00000061
2" SS WNP (36) (108) 061020
GRAVITY
HV-0650E3
HV-0650E1
WFO
3/4" SS WNP (36) (108) 065004
HV-0650E2

Change to 3/4" Gate Valve with plug

S(130)
LP1Y-DI-00000061
PSTMY0180
2" CS S(130) (2) (103) 061024
HV-0650U12
GRAVITY
3/4" CS S(130) (2) (103) 065005
HV-0650U9
HV-0650U10
3/4"
HV-0650U11
HV-0650U1
HV-0650U2
HV-0650U3
3/4"
HV-0650U8
SC S(130) 101 LOT ET
NONE
S(130) 103 LOT ET
PIDCN 011715
3/4" CS SC (2) (101) 065002
STR-0650U1
3/4" CS S(130) (2) (103) 065003
STT-0650U1
3/8"
HV-0650U4
HV-0650U7
PIDCN010605
HV-0650U5
3/4" HV-0650U6
TO GRADE

TRANSFER TO LP1Y-DI-00000062
PIDCN 014619
BY FLUOR
BY OTHERS
VAPOR BALANCE
LP1Y-DI-00000013
VIRGIN DMF TANKS (TK-1021/1022)
3" SS PVP (315) P 013004
HV-0650V7
HV-0650V10
1" HV-0650V11
3" SS PVP (315) P 065006
PIDCN 014730
PIDCN013564
PV 0622N1
HV-0651N4
H 0622N
HV-0622N2
CKV-0622N1
CKV-0622N2
N2
LP1Y-DI-00000061
2" SS N2 (315) E 061053
PIDCN-009515
PIDCN015536
2" SS N2 (315) E 061004
PI 0651N
HV-0651N6
2"x3/4"
HV-0651N2
3/4"
HV-0651N3
CKV-0651N1
CKV-0651N2
3/4"
2"X 1"
HV-0651N5
CCU-0651N2
FLX-0651N1
CCU-0651N3
HV-0651N8
RLUY0610
DMFY1022
DMFY1020
SG-0651A
23
HV-0623A15
LA-0652A
4"x3"
RS
CONNECT THIS COUPLING WITH FLX-0621N1, SEE LP1Y0-DI-00000062_R1
CONNECTION BOX
DMF UNLOADING ARM (VAPOR BALANCE)
LA-0651A
HV-0650V5
HV-0650V4
HV-0650V8
PIDCN 011648
4" SS SW (315) 065007
SPILL WASTE
LP1Y-DI-0000067
TO SECONDARY CONTAINMENT SMP-0671A
8" SS SW (315) 065008
FREE DRAINING TO COMMON SECONDARY CONTAINMENT
LP1Y-DI-00000061
GROUND
DMF RAIL CAR
SET @ 150 PSIG
062304
2" SS RV (315) 061018
PSV 0651A
PSA 0651A
PSE 0651A
PIT 0651A
FO 0651A
FSL 0651A
LSL
PI 0651A
NO POCKET
HV-0623A17
3" SS DMF (315) P 065102
VIRGIN DMF
LP1Y-DI-00000013
VIRGIN DMF TANKS (TK-1021/1022)
HV-0651A14
HV-0651A15
HV-0651A11
HV-0651A12
HV-0651A13
HV-0651A8
HV-0651A10
HV-0651A9
CKV-0651A1
HV-0651A6
HV-0651A5
HV-0651A7
HV-0651A4
HV-0651A3
HV-0651A2
HV-0651A1
SG-0651A
HV-0623A16
1" SS DMF (315) P 065106
LBCU-0651A1
HV-0623A6
CCU-0651A1
QCU-0651A1
HV-0651N7
1" CCU-0651N1
HV-0651N1
QCU-0651N1
FLX-0651A1
P-0651A
SS DMF (315) P 065103
FREE DRAINING TO COMMON SECONDARY CONTAINMENT
LP1Y-DI-00000061
00000067
PIDCN-010228
4" SS SW (315) 065007
SPILL WASTE
LP1Y-DI-0000067
TO SECONDARY CONTAINMENT SMP-0671A
PIDCN-011886
PIDCN-011648
4" SS SW (315) 065007
SPILL WASTE
LP1Y-DI-0000067
TO SECONDARY CONTAINMENT SMP-0671A
PIDCN-012206
PIDCN 016518

NOTES:
1. SECONDARY CONTAINMENT UNDERNEATH RAIL CAR AND PUMP TO FLOW TO COMMON SECONDARY CONTAINMENT.
2. PLANT AIR CONNECTION. LOCATE AT GRADE.
3. INSTRUMENT AIR SUPPLY TO INSTRUMENTS; PIPING CONTRACTOR TO FIELD-ROUTE TUBING AS REQUIRED. WATER FOR OPERATIONS PIDCN 014619
4. NON-POTABLE WATER CONNECTION. LOCATE AT GRADE.
5. FLEXIBLE HOSE CONNECTION PROVIDED FOR TEMPORARY LINE PURGING OPERATIONS.
6. NITROGEN LOADING ARM IS COMMON TO THE RAIL STATION AND CAN BE USED BY SOLVENT OR WASTE RAIL CARS. DELETED
7. FOR UTILITY NITROGEN, LOCATE AT GRADE. DELETED FOR UTILITY NITROGEN. LOCATE AT GRADE.
8. STEAM CONNECTION. LOCATE AT GRADE.
9. FLEXIBLE HOSE CONNECTION FOR NITROGEN BLANKETING.
10. HOT WATER FOR CONNECTION TO RAIL CAR TO MAINTAIN SOLVENT TEMPERATURE.
11. DELETED
12. PIT LOCAL INDICATOR TO BE VISIBLE FROM RAIL STATION OPERATOR.
13. SAFETY BREAK-AWAY COUPLINGS TO BE INSTALLED. DELETED
14. DRY DISCONNECT COUPLER TO BE INSTALLED. DELETED
15. FLEXIBLE HOSES IN PLACE ONLY DURING UNLOADING OPERATION. DELETED
16. VENT TO BE LOCATED AT GRADE. DELETED
17. PRESSURE GAUGE TO BE VISIBLE AT GLOBE VALVE.
18. ALL INSTRUMENTS TO BE PROVIDED BY FLUOR. PIDCN-012463
19. Y-ADAPTER, CONNECTION ACCESSORIES, LEVEL SWITCH INCLUDING CABLE TO BE PROVIDED BY RAIL STATION VENDOR. FLEXIBLE HOSES IN PLACE ONLY DURING UNLOADING OPERATION.

CONSTRUCTION NOTES:
C1. TEMPORARY STRAINER TO BE PROVIDED BY CONSTRUCTION CONTRACTOR. ARRANGE PIPING TO FACILITATE REMOVAL. STRAINER MESH SIZING TO BE CONFIRMED DURING DETAILED DESIGN. DELETED
C2. PI-0651N TO BE VISIBLE AT GLOBE VALVE. PIDCN-003941
C3. ALL INSTRUMENTS TO BE PROVIDED BY FLUOR. PIDCN-016168
C4. PUMP TO BE PROVIDED BY FLUOR. DELETED
C5. ALL CAMLOCK CONNECTIONS TO HAVE CAP/PLUG AND CHAIN.

HOLD NOTES:
H1. UNLOADING ARM DETAILS TO BE CONFIRMED.
H2. DETAILS OF INTERFACE CONNECTIONS TO RAIL CARS TO BE CONFIRMED.
H3. GROUNDING CONFIGURATION AND INSTRUMENTATION TO BE CONFIRMED.
H4. ALL LINE SIZES TO BE CONFIRMED PENDING DETAILED HYDRAULICS.
H5. PUMP SIZE TO BE CONFIRMED IN DETAILED ENGINEERING. DELETED
H6. REQUIREMENT FOR RAIL CAR HEATING USING A STEAM AND WATER MIXTURE TO BE CONFIRMED.
H7. SECONDARY CONTAINMENT DETAILS ARE TO BE CONFIRMED.
H8. NUMBER OF STEAM TRAPS AND LOCATION TO BE CONFIRMED.

THE FOLLOWING PICC WERE INCLUDED IN REV 2:
PICCL-306-1605-53, PICCL-306-1605-54, PICCL-306-1605-62, PICCL-306-1605-63, PICCL-306-1605-64, PICCL-306-1605-65, PICCL-306-1605-66, PICCL-306-1605-71, PICCL-306-1605-75, PICCL-306-1605-77, PICCL-306-1605-82, PICCL-306-1605-88, PICCL-306-1605-90

PIDCN 010228 / PICCL-306-1605-53; PIDCN 010605 / PICCL-306-1605-54; PIDCN 011648, 009441 / PICCL-306-1605-62; PIDCN 010333, 012206, 014757, 015536, 016168, 016518 / N/A; PIDCN 011886 / PICCL-306-1605-63; PIDCN 009515, 010624 / PICCL-306-1605-64; PIDCN 011715 / PICCL-306-1605-65; PIDCN 012463 / PICCL-306-1605-66; PIDCN 006849 / PICCL-306-1605-71; PIDCN 013564 / PICCL-306-1605-75; PIDCN 013713 / PICCL-306-1605-77; PIDCN 014725 / PICCL-306-1605-82; PIDCN 014619 / PICCL-306-1605-88; PIDCN 014730 / PICCL-306-1605-90

THE FOLLOWING PICC WERE INCLUDED IN REV 0:
PICCL-306-1605-29; PICCL-306-1605-30; PICCL-306-1605-35; PICCL-306-1605-36; PICCL-306-1605-39
THE FOLLOWING PICC WERE INCLUDED IN REV 1:
PICCL-306-1605-41; PICCL-306-1605-44; PICCL-306-1605-46; PICCL-306-1605-47; PICCL-306-1605-50

LA-0650A
PRIMARY/SECONDARY WASTE LOADING ARM
4" PRIMARY/SECONDARY VAPOR RETURN ARM
UNSUPPORTED BOOM STYLE
DUPLEX 316L SS

LA-0650B
PRIMARY/SECONDARY WASTE VAPOR BALANCE LOADING ARM
UNSUPPORTED BOOM STYLE
316L SS

LA-0651A
DMF VAPOR BALANCE LOADING ARM
UNSUPPORTED BOOM STYLE
316L SS

H8 P-0651A
DMF RAIL UNLOADING PUMP
568 LPM @ 71 PSID
10.3 hp
XXX RPM
316L SS

LA-0652A
DMF UNLOADING ARM & N2 HOSE
316 SS
PIDCN 014619

MASTER
PICC:

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
1		ISSUED FOR CONSTRUCTION	NA	27-MAR-2025
0		ISSUED FOR DESIGN	NA	15-NOV-2024
D		ISSUED FOR PHA	NA	08-JUL-2024
C		ISSUED FOR APPROVAL	NA	24-MAY-2024
B		ISSUED FOR APPROVAL	NA	28-JUL-2023

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REV	DATE	DES	END	CHK	APP	PM APP
1	27-MAR-2025	MTT	MM	JA	SSK	BS
0	15-NOV-2024	JDG	ADS	JA	SSK	BS
D	08-JUL-2024	LJA	MJP	JA	SSK	BS
C	24-MAY-2024	PMS	MJP	JA	SSK	BS
B	28-JUL-2023	JCB	MJP	JA	SSK	BS

EFDEE Engineering PC
PROJECT NO. B1LG
NODE: TFS

FLUOR

Lilly

PIPING & INSTRUMENTATION DIAGRAM
RAIL STATION 4

SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: NA
SUB DISCIPLINE: DI	LOCATION CODE: LP1	LOCATION/GRID: NA	PROJECT NAME: LP1-306-1181
CRITICAL QUALITY		ENTERPRISE GROUP: API	DRAWING SCALE: NONE
CRITICAL BUSINESS		EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA / MERIDIAN REV: 1
CRITICAL HSE		VENDOR: NA	DOCUMENT ID: LP1Y-DI-00000065
NON-CRITICAL		ENGINEERING AGENCY DOC ID: NA	

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NOTES:
1. PSV TO BE ELECTRICALLY TRACED.
2. COMMON SECONDARY CONTAINMENT TO BE CLEANED OUT BY VACUUM TRUCK.
3. VENT OUTLET TO BE MINIMUM 6.5 FEET ABOVE STANDING SURFACE. DELETED. PIDCN-007401 by Marissa
4. STEAM CONNECTION FOR SUMP WINTERIZATION. PIDCN-016352 Retain the Notes
5. ANALYZER IN PLACE TO MONITOR FOR SOLVET SPILLAGE.
6. INSTRUMENT AIR SUPPLY TO INSTRUMENTS: PIPING CONTRACTOR TO FIELD ROUTE TUBING AS REQUIRED. VALVES ARE LOCATED 30 FT AWAY FROM ONE ANOTHER.
7. BUTTERFLY VALVE, TRIPLE OFFSET, LOCATE AT ACCESSIBLE LOCATION ON HIGH POINT OF LINE.
8. ELECTRIC TRACING EXTENDED 8FT INSIDE THE BUILDING. DELETED. PIDCN-009260 by Adnan
9. ALL INSTRUMENTS TO BE PROVIDED BY FLUOR. PIDCN-012463
10. PROVIDE 1/4" WEEP HOLE AT THE LOW POINT
PIDCN-017078

CONSTRUCTION NOTES:
C1. SHORT SPOOL PIECE WITH BLIND FLANGE.
C2. ALL INSTRUMENTS TO BE PROVIDED BY FLUOR. PIDCN-016168

HOLD NOTES:
H1. ALL LINE SIZES TO BE CONFIRMED PENDING DETAILED HYDRAULICS.
H2. SECONDARY CONTAINMENT DETAILS ARE TO BE CONFIRMED.
H3. NUMBER OF STEAM TRAPS AND LOCATION TO BE CONFIRMED.
H1.H4. SIZE OF NITROGEN FILTER TO BE CONFIRMED.
H5. PRV AND PSV SIZE TO BE CONFIRMED.
H2.H5. PRV AND PSV SIZE TO BE CONFIRMED.
PIDCN 014619

THE FOLLOWING PICC WERE INCLUDED IN REV 2:
PICCL-306-1605-51, PICCL-306-1605-53, PICCL-306-1605-54, PICCL-306-1605-55, PICCL-306-1605-62, PICCL-306-1605-64, PICCL-306-1605-66, PICCL-306-1605-70, PICCL-306-1605-71, PICCL-306-1605-76, PICCL-306-1605-86, PICCL-306-1605-88, PICCL-306-1605-91, PICCL-306-1605-97

PIDCN 009408 / PICCL-306-1605-51; PIDCN 010228 / PICCL-306-1605-53; PIDCN 015605 / PICCL-306-1605-54; PIDCN 011190 / PICCL-306-1605-60; PIDCN 011900 / PICCL-306-1605-62; PIDCN 005515 / PICCL-306-1605-64; PIDCN 010333 / PICCL-306-1605-66; PIDCN 016352, 015078, 015079/ N/A; PIDCN 012463 / PICCL-306-1605-70; PIDCN 012582 / PICCL-306-1605-71; PIDCN 005849 / PICCL-306-1605-76; PIDCN 013908 / PICCL-306-1605-86; PIDCN 015591 / PICCL-306-1605-86; PIDCN 014619 / PICCL-306-1605-88; PIDCN 015978 / PICCL-306-1605-91; PIDCN 016722 / PICCL-306-1605-97

THE FOLLOWING PICC WERE INCLUDED IN REV 0:
PICCL-306-1605-29; PICCL-306-1605-30; PICCL-306-1605-31; PICCL-306-1605-33; PICCL-306-1605-34; PICCL-306-1605-35; PICCL-306-1605-36; PICCL-306-1605-37; PICCL-306-1605-38; PICCL-306-1605-39.

THE FOLLOWING PICC WERE INCLUDED IN REV 1:
PICCL-306-1605-40; PICCL-306-1605-41; PICCL-306-1605-44; PICCL-306-1605-47; PICCL-306-1605-50. PICCL-306-1605-52

REV	DATE	DES	ENG	CHK	APP	PM APP
1	27-MAR-2025	LJA	MM	JA	SSK	BS
0	15-NOV-2024	LJA	ADS	JA	SSK	BS
D	08-JUL-2024	JRD	MJP	JA	SSK	BS
C	24-MAY-2024	GNA	MJP	JA	SSK	BS
B	28-JUL-2023	CBE	MJP	JA	SSK	BS

EFDEE Engineering PC
PROJECT NO. B1LG
NODE: TFS

FLUOR

Lilly

PIPING & INSTRUMENTATION DIAGRAM
RAIL UTILITIES AND WASTE DISTRIBUTION

MASTER
PICC:

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
1		ISSUED FOR CONSTRUCTION	NA	27-MAR-2025
0		ISSUED FOR DESIGN	NA	15-NOV-2024
D		ISSUED FOR PHA	NA	08-JUL-2024
C		ISSUED FOR APPROVAL	NA	24-MAY-2024
B		ISSUED FOR APPROVAL	NA	28-JUL-2023

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SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: NA
SUB DISCIPLINE: DI	LOCATION CODE: LP1	LOCATION/GRID: NA	CD CODE: YELLOW
CRITICAL QUALITY:		ENTERPRISE GROUP: API	PROJECT NAME: LP1-306-1181
CRITICAL BUSINESS:		EQUIPMENT CLASSIFICATION: NA	DRAWING INTENTION:
CRITICAL HSE:		VENDOR: NA	DRAWING SCALE: NONE
NON-CRITICAL:		ENGINEERING AGENCY DOC ID: NA	ALTERNATE DOCUMENT ID: NA
			MERIDIAN REV: 1
			DOCUMENT ID: LP1Y-DI-00000061

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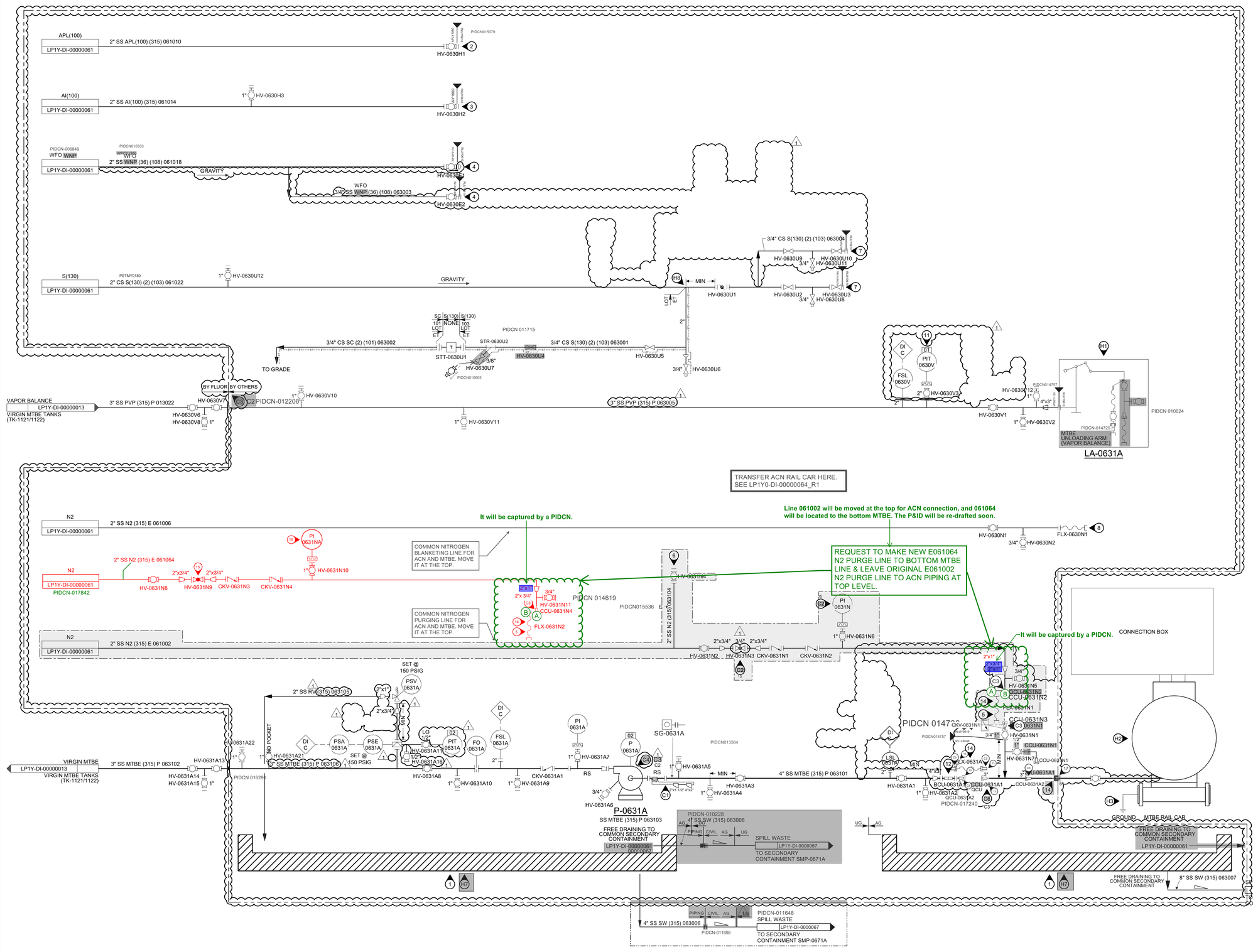

NOTES:
1. SECONDARY CONTAINMENT UNDERNEATH RAIL CAR AND PUMP TO FLOW TO COMMON SECONDARY CONTAINMENT.
2. PLANT AIR CONNECTION. LOCATE AT GRADE.
3. INSTRUMENT AIR SUPPLY TO INSTRUMENTS; PIPING CONTRACTOR TO FIELD-ROUTE TUBING AS REQUIRED. WATER FOR OPERATIONS
4. NON-POTABLE WATER CONNECTION. LOCATE AT GRADE.
5. FLEXIBLE HOSE CONNECTION PROVIDED FOR TEMPORARY LINE PURGING OPERATIONS.
6. FOR UTILITY NITROGEN. LOCATE AT GRADE.
7. STEAM CONNECTION. LOCATE AT GRADE.
8. FLEXIBLE HOSE CONNECTION FOR NITROGEN BLANKETING. COMMON FOR ACN AND MTBE.
9. DELETED.
10. DELETED.
11. PIT LOCAL INDICATOR TO BE VISIBLE FROM RAIL STATION OPERATOR.
12. SAFETY BREAK-AWAY COUPLINGS TO BE INSTALLED.
13. DRY DISCONNECT COUPLER TO BE INSTALLED.
14. FLEXIBLE HOSES IN PLACE ONLY DURING UNLOADING OPERATION. PIDCN 014619
15. PRESSURE GAUGES TO BE VISIBLE AT GLOBE VALVE.
16. ALL INSTRUMENTS TO BE PROVIDED BY FLUOR. PIDCN-012463
17. RAIL CAR ADAPTER TO BE THREADED CONNECTION TO RAIL CAR BOTTOM OUTLET.

CONSTRUCTION NOTES:
C1. TEMPORARY STRAINER TO BE PROVIDED BY CONSTRUCTION CONTRACTOR. ARRANGE PIPING TO FACILITATE REMOVAL. STRAINER MESH SIZING TO BE CONFIRMED DURING DETAILED DESIGN.
C2. PI-0631N TO BE VISIBLE AT GLOBE VALVE. PIDCN-009441
C3. ALL INSTRUMENTS TO BE PROVIDED BY FLUOR. PIDCN-016168
C4. PUMP TO BE PROVIDED BY FLUOR.
C5. ALL CAMLOCK CONNECTIONS TO HAVE CAP/PLUG AND CHAIN.
C4. ALL DRY DISCONNECT COUPLERS TO HAVE DUST CAP/PLUG AND CHAIN.

HOLD NOTES:
H1. UNLOADING ARM DETAILS TO BE CONFIRMED.
H2. DETAILS OF INTERFACE CONNECTIONS TO RAIL CARS TO BE CONFIRMED.
H3. GROUNDING CONFIGURATION AND INSTRUMENTATION TO BE CONFIRMED.
H4. ALL LINE SIZES TO BE CONFIRMED PENDING DETAILED HYDRAULICS.
H5. PUMP SIZE TO BE CONFIRMED IN DETAILED ENGINEERING.
H6. REQUIREMENT FOR RAIL CAR HEATING USING A STEAM AND WATER MIXTURE TO BE CONFIRMED.
H7. SECONDARY CONTAINMENT DETAILS ARE TO BE CONFIRMED.
H8. NUMBER OF STEAM TRAPS AND LOCATION TO BE CONFIRMED.

THE FOLLOWING PICC WERE INCLUDED IN REV 2:
PICCL-306-1605-53, PICCL-306-1605-54, PICCL-306-1605-62, PICCL-306-1605-63, PICCL-306-1605-64, PICCL-306-1605-65, PICCL-306-1605-66, PICCL-306-1605-71, PICCL-306-1605-75, PICCL-306-1605-82, PICCL-306-1605-85, PICCL-306-1605-88, PICCL-306-1605-90

PIDCN 010228 / PICCL-306-1605-53; PIDCN 010605 / PICCL-306-1605-54; PIDCN 011648, 009441 / PICCL-306-1605-62; PIDCN 010333, 012206, 012478, 015078, 017240, 014757, 015536, 016168 / N/A; PIDCN 011886 / PICCL-306-1605-63; PIDCN 010624 / PICCL-306-1605-64; PIDCN 011715 / PICCL-306-1605-65; PIDCN 012463 / PICCL-306-1605-66; PIDCN 006849 / PICCL-306-1605-71; PIDCN 013564 / PICCL-306-1605-75; PIDCN 014725 / PICCL-306-1605-82; PIDCN 014518 / PICCL-306-1605-85; PIDCN 014619 / PICCL-306-1605-88; PIDCN 014730 / PICCL-306-1605-90

THE FOLLOWING PICC WERE INCLUDED IN REV 0:
PICCL-306-1605-29, PICCL-306-1605-30, PICCL-306-1605-31, PICCL-306-1605-35, PICCL-306-1605-36, PICCL-306-1605-39, PICCL-306-1605-40

THE FOLLOWING PICC WERE INCLUDED IN REV 1:
PICCL-306-1605-41, PICCL-306-1605-44, PICCL-306-1605-46, PICCL-306-1605-47, PICCL-306-1605-50

LA-0631A
RETURN
MTBE VAPOR BALANCE LOADING ARM
UNSUPPORTED BOOM STYLE
316L SS

H5 P-0631A
MTBE RAIL UNLOADING PUMP
568 LPM @ 56 PSID
20 8.2 hp PIDCN-014518
3,500 XXX RPM PIDCN 012476
316L SS

PIDCN 014619
LA-0632A
ACN UNLOADING ARM & N2 HOSE
316 SS

MASTER
PICC:

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
1		ISSUED FOR CONSTRUCTION	NA	27-MAR-2025
0		ISSUED FOR DESIGN	NA	15-NOV-2024
D		ISSUED FOR PHA	NA	08-JUL-2024
C		ISSUED FOR APPROVAL	NA	24-MAY-2024
B		ISSUED FOR APPROVAL	NA	28-JUL-2023

REVISIONS

CONFIDENTIAL INFORMATION

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REV	DATE	DES	END	CHK	APP	PM APP
1	27-MAR-2025	RLB	MM	JA	SSK	BS
0	15-NOV-2024	AV	ADS	JA	SSK	BS
D	08-JUL-2024	JRB	MJP	JA	SSK	BS
C	24-MAY-2024	GNA	MJP	JA	SSK	BS
B	28-JUL-2023	CBE	MJP	JA	SSK	BS

FLUOR
EFDEE Engineering PC
PROJECT NO. B1LG
NODE: TFS

Lilly

PIPING & INSTRUMENTATION DIAGRAM
RAIL STATION 2

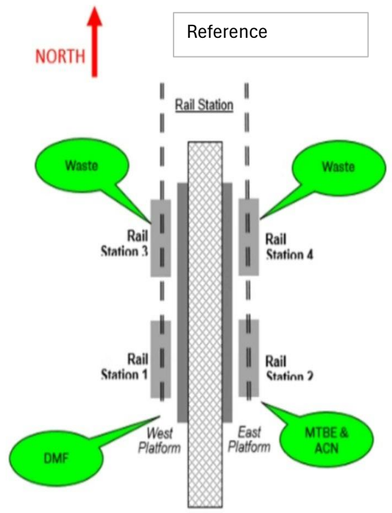
SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: NA
SUB DISCIPLINE: DI	LOCATION CODE: LP1	LOCATION/GRID: NA	PROJECT NAME: LP1-306-1181
CRITICAL QUALITY:	ENTERPRISE GROUP: API	DRAWING INTENTION: CD CODE: YELLOW	DRAWING SCALE: NONE
CRITICAL BUSINESS:	EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA	MERIDIAN REV: 1
CRITICAL HSE:	VENDOR: NA	DOCUMENT ID: LP1Y-DI-00000063	
NON-CRITICAL:	ENGINEERING AGENCY DOC ID: NA		

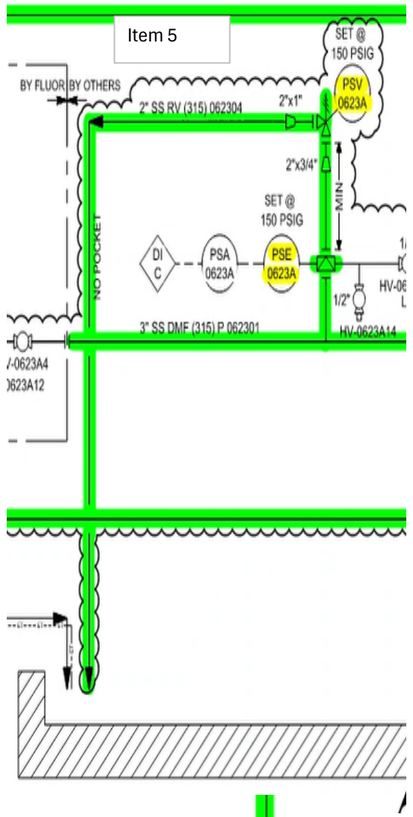
IF YOU ARE USING A PRINTED COPY OF THIS DOCUMENT, IT IS AN UNCONTROLLED PRINT. CONFIRM WITH THE OFFICIAL "ELECTRONIC RECORD" DOCUMENT IN THE GLOBAL ENGINEERING CONTENT MANAGEMENT SYSTEM.

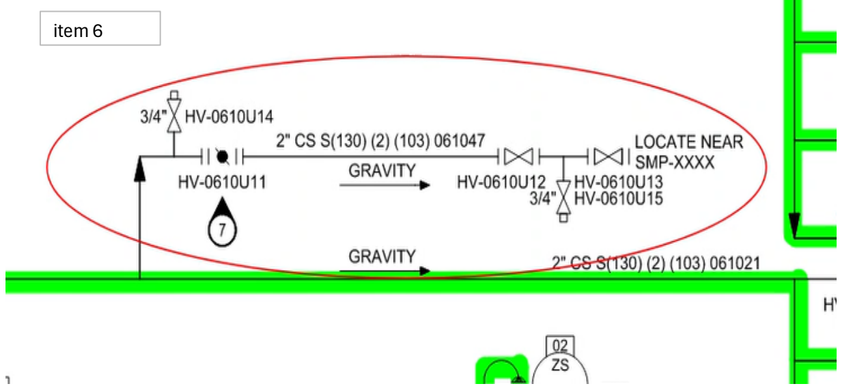
C:\CollaborationCache\OY_TFS\PID DWG\LP1Y-DI-00000063.dwg 03-27-25 2:11 pm Printed By: gpr19480

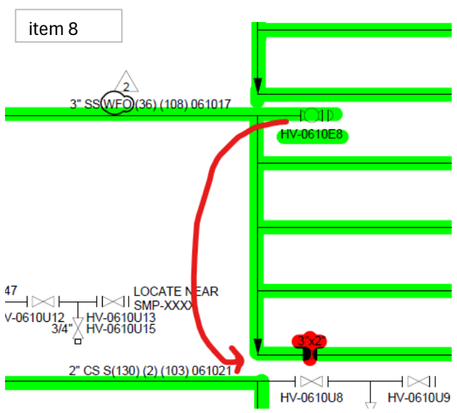
IF YOU ARE USING A PRINTED COPY OF THIS DOCUMENT, IT IS AN UNCONTROLLED PRINT. CONFIRM WITH THE OFFICIAL "ELECTRONIC RECORD" DOCUMENT IN THE GLOBAL ENGINEERING CONTENT MANAGEMENT SYSTEM.

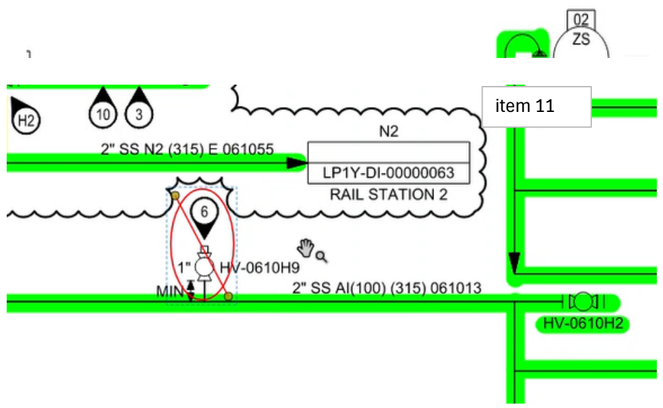
K215 LxL 2 Parking Lot Items				Screen shot for reference
item	Comments	Action By	Date entered	
1	Global comment: Complete pipe support design for scope	Hanson	8-Dec-25	
2	Global comment: Complete modeling of steam lines (drip legs, steam traps & branch connections) for design scope based on Fluor pipe spec guideline.	Hanson	8-Dec-25	
3	Global comment: Model missing instruments as shown in the marked up PIDs	Hanson	8-Dec-25	
4	Model flange on 3" DUP WV 064001 line at station 3. Duplicate flange design on all other stations as shown in P&IDs.	Hanson	8-Dec-25	
5	Global comment: Complete drain line to pump containment area once option is confirmed by Lilly.	Hanson	8-Dec-25	
6	Confirm the deletion of steam line at 2"CS 130 061047 line	Fluor	8-Dec-25	
7	Inititate PIDCN for safety shower (reverse position of water hammer)	Fluor	8-Dec-25	
8	Relocate HV-0610E8 to the end of main line in model and update P&ID to match.	Hanson/ Fluor	8-Dec-25	
9	Sequence of branches off the main WFO header to match model. Hanson to make up P&ID with the new sequence.	Hanson/ Fluor	8-Dec-25	
10	Modeling on system drains on service lines to be completed	Hanson	8-Dec-25	
11	Keep 1" root valve, every 30' in the main instrument air header.	Hanson	8-Dec-25	
12	Need low point drain on N2 lines	Hanson	8-Dec-25	
13	Adjust spacing for PI0631N to accomomdate camlock and HV design on 2" SS N2 61022 line.	Hanson	8-Dec-25	
14	Adjust PSV spacing/ clash based with globe valve handwheel on N2 press off line, all stations.	Hanson	8-Dec-25	
15	Vapor balance for MTBE, PVP line, rotate high point drain 180 degrees.	Hanson	8-Dec-25	
16	Make LSL minimum distance as shown on PID. Put pipe support on opposite to achieve requirement.	Hanson	8-Dec-25	
17	Global comment: Hanson to update model to show all PSV inlet line number the same as process line number.	Hanson	8-Dec-25	
18	MTBE bottom unloading connection to be updatedand modeled as per email by Fluor Process	Fluor/ Hanson	8-Dec-25	



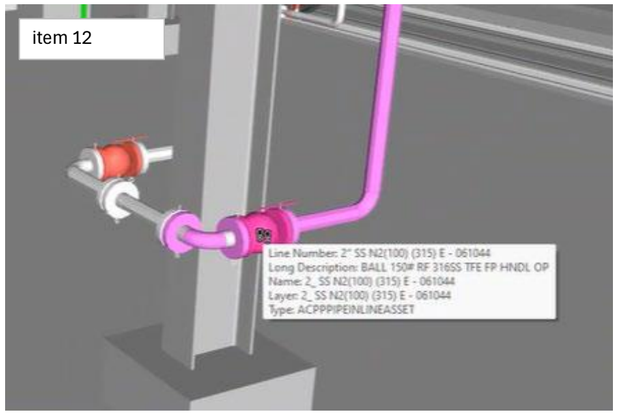




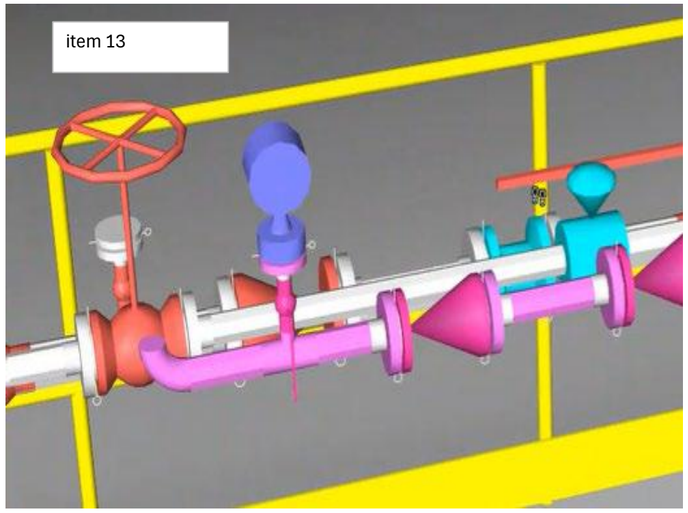


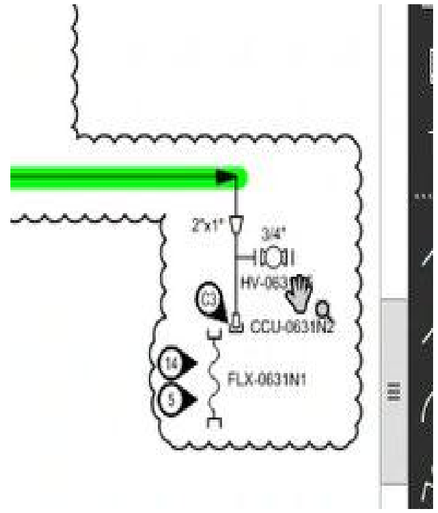


item 12

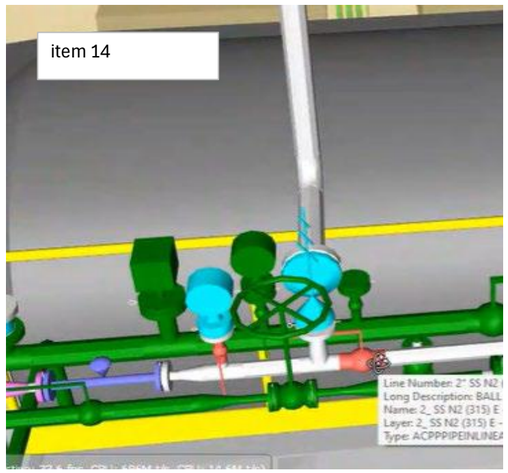


item 13

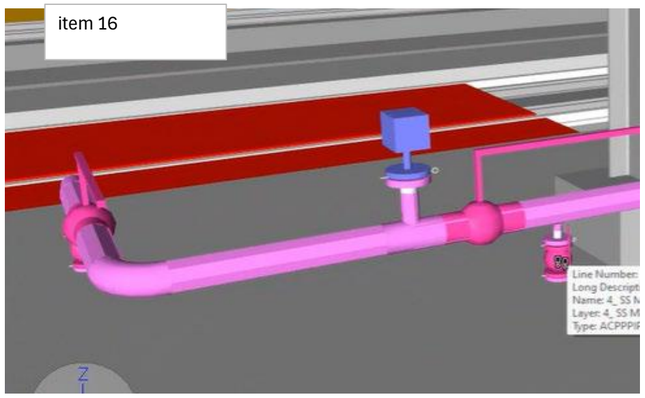




item 14



item 16



Issue detail

RPT_20260106_Lilly-LxL MR3-Hanson-Actions_Update 20260223

Created on	Feb 23, 2026, 4:55 PM EST
Created by	**Devin Robertson** (Hanson Professional Services)
Total items	4
Sorted by	ID (Ascending)

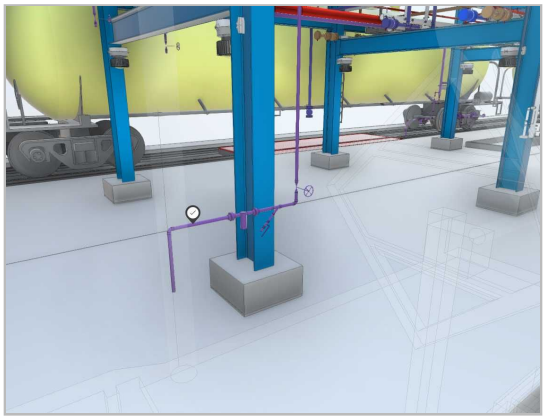
Contents

#109: LxL #3 - MRA 1 - Shorten Steam Leg to Grade - Typ. 3
#110: LxL #3 - MRA 2: Remove Ball Valve 4
#111: LxL #3 - MRA 3: New Steam Leg Line Number 5
#112: LxL #3 - MRA 4: Keep BV on Steam Legs - Additional 6

Issue detail

#109: LxL #3 - MRA 1 - Shorten Steam Leg to Grade - Typ.

Status	Closed
Type	CF Design > Client Feedback



Standard fields

Description	Shorten drip leg, or wrap around the column similar to the truck station
Assigned to	Andrew Worthy (Hanson Professional Services)
Created by	Devin Robertson (Hanson Professional Services)
Created on	Jan 6, 2026
Location	—
Location details	—
Due date	—
Start date	—
Placement	LP10Y-TFS-M0-000-000-00-HNS-P3D.dwg
Root cause	—

Comments

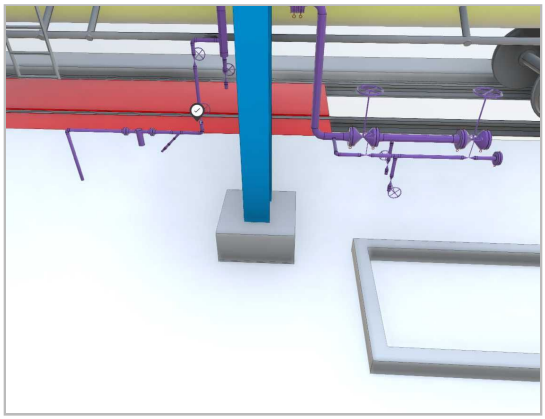
Andrew Worthy
Feb 22, 2026, 6:40 PM EST

Completed. @Devin Robertson , please review.

Issue detail

#110: LxL #3 - MRA 2: Remove Ball Valve

Status	Closed
Type	CF Design > Client Feedback



Standard fields

Description	Remove Ball Valve from model and P&ID. Ball valve above serves the purpose to close off leg.
Assigned to	Andrew Worthy (Hanson Professional Services)
Created by	Devin Robertson (Hanson Professional Services)
Created on	Jan 6, 2026
Location	—
Location details	—
Due date	—
Start date	—
Placement	LP10Y-TFS-M0-000-000-00-HNS-P3D.dwg
Root cause	—

Comments

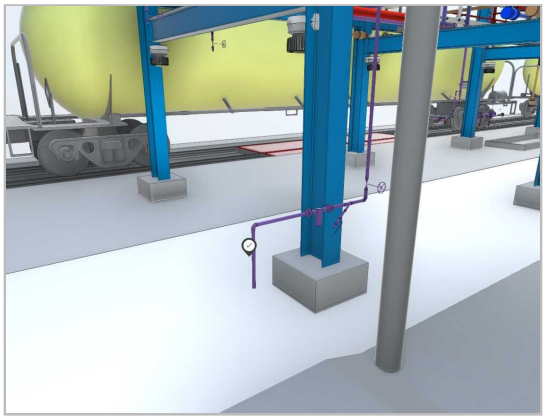
Andrew Worthy
Feb 22, 2026, 6:40 PM EST

Completed. @Devin Robertson , please review.

Issue detail

#111: LxL #3 - MRA 3: New Steam Leg Line Number

Status	Closed
Type	CF Design > Client Feedback



Standard fields

Description	Steam leg line number should change to condensate piping. New line number required from Fluor.
Assigned to	Andrew Worthy (Hanson Professional Services)
Created by	Devin Robertson (Hanson Professional Services)
Created on	Jan 6, 2026
Location	—
Location details	—
Due date	—
Start date	—
Placement	LP10Y-TFS-M0-000-000-00-HNS-P3D.dwg
Root cause	—

Comments

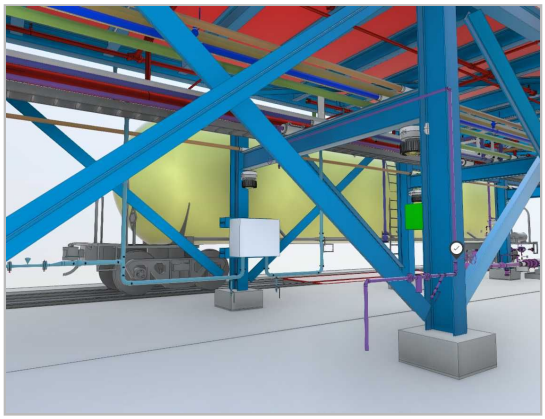
Andrew Worthy
Feb 22, 2026, 6:41 PM EST

Completed. @Devin Robertson , please review.

Issue detail

#112: LxL #3 - MRA 4: Keep BV on Steam Legs - Additional

Status	**Closed**
Type	CF Design > Client Feedback



Standard fields

Description	Keep second ball valve for additional steam drip legs here and at the end of the rack as length of piping dictates the necessity to have.
Assigned to	Andrew Worthy (Hanson Professional Services)
Created by	Devin Robertson (Hanson Professional Services)
Created on	Jan 6, 2026
Location	—
Location details	—
Due date	—
Start date	—
Placement	LP10Y-TFS-S0-000-000-00-HNS.rvt
Root cause	—

Comments

Andrew Worthy
Feb 22, 2026, 6:41 PM EST

Completed. @Devin Robertson , please review.

Issue detail

MR#2 & LxL#2 Parking Lot Items Report

Created on	Feb 23, 2026, 5:06 PM EST
Created by	**Andrew Worthy** (Hanson Professional Services)
Total items	15
Sorted by	ID (Ascending)
Filtered by	Status (Closed) Created by (Andrew Worthy (Hanson Professional Services))

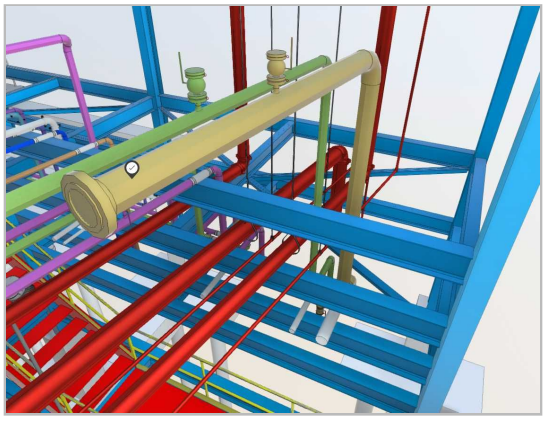
Contents

#102: MR#2 - Parking Lot Item 3 - Completion of Pipe Design 3
#104: LxL#2 - Parking Lot Item 1 - Complete Pipe Support Modeling 4
#105: LxL#2 - Parking Lot Item 2 - Complete Steam Lines 5
#106: LxL#2 - Parking Lot Item 3 - Model missing instruments 6
#107: LxL#2 - Parking Lot Item 4 - Model Flange on WV 7
#108: LxL#2 - Parking Lot Item 5 - Complete Modeling of Drain Lines to Containment (RV) 8
#118: LxL#2 - Parking Lot Item 9 - WFO branches in model to match P&IDs 9
#119: LxL#2 - Parking Lot Item 11 - 1" Valve Every 30' on AI Header 10
#120: LxL#2 - Parking Lot Item 12 - Low Point Drain on N2 Header 11
#121: LxL#2 - Parking Lot Item 13 - Proper Spacing for PI, Drain HV, and Camlock 12
#122: LxL#2 - Parking Lot Item 14 - Clash Between PSV and Globe Valve Handwheel (typ.) 13
#123: LxL#2 - Parking Lot Item 15 - Rotate High Point Vent on PVP 14
#124: LxL#2 - Parking Lot Item 16 - LSL Min Distance 15
#126: LxL#2 - Parking Lot Item 17 - PSV Inlet Line 16
#127: LxL#2 - Parking Lot Item 18 - Complete Modeling of MTBE Unloading Connection 17

Issue detail

#102: MR#2 - Parking Lot Item 3 - Completion of Pipe Design

Status	Closed
Type	DWC Design > Work to Complete



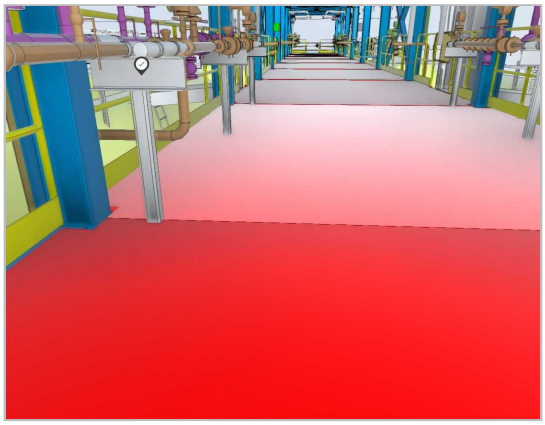
Standard fields

Description	Close out this item once all piping design is complete.
Assigned to	Andrew Worthy (Hanson Professional Services)
Created by	Andrew Worthy (Hanson Professional Services)
Created on	Dec 18, 2025
Location	—
Location details	—
Due date	—
Start date	—
Placement	LP10Y-TFS-M0-000-000-00-HNS-P3D.dwg
Root cause	—

Issue detail

#104: LxL#2 - Parking Lot Item 1 - Complete Pipe Support Modeling

Status	Closed
Type	DWC Design > Work to Complete



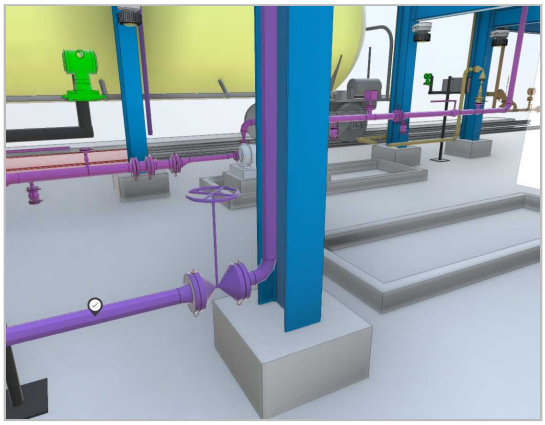
Standard fields

Description	Close out once all pipe supports have been modeled.
Assigned to	Andrew Worthy (Hanson Professional Services)
Created by	Andrew Worthy (Hanson Professional Services)
Created on	Dec 18, 2025
Location	—
Location details	—
Due date	—
Start date	—
Placement	LP10Y-TFS-M0-000-000-00-HNS-P3D.dwg
Root cause	—

Issue detail

#105: LxL#2 - Parking Lot Item 2 - Complete Steam Lines

Status	Closed
Type	DWC Design > Work to Complete



Standard fields

Description	—
Assigned to	Andrew Worthy (Hanson Professional Services)
Created by	Andrew Worthy (Hanson Professional Services)
Created on	Dec 18, 2025
Location	—
Location details	—
Due date	—
Start date	—
Placement	LP10Y-TFS-M0-000-000-00-HNS-P3D.dwg
Root cause	—

Comments

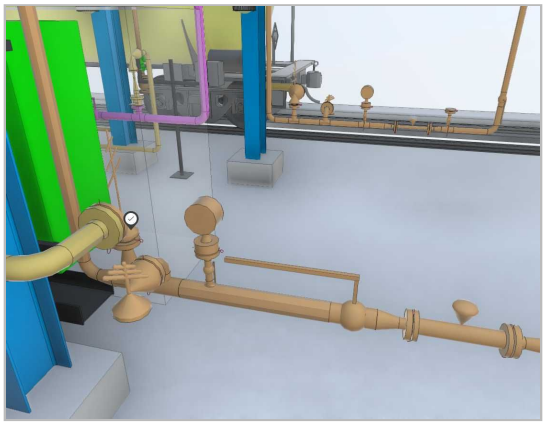
Devin Robertson
Jan 5, 2026, 4:38 PM EST

Drew completed Wednesday 12/31/25

Issue detail

#106: LxL#2 - Parking Lot Item 3 - Model missing instruments

Status	Closed
Type	DWC Design > Work to Complete



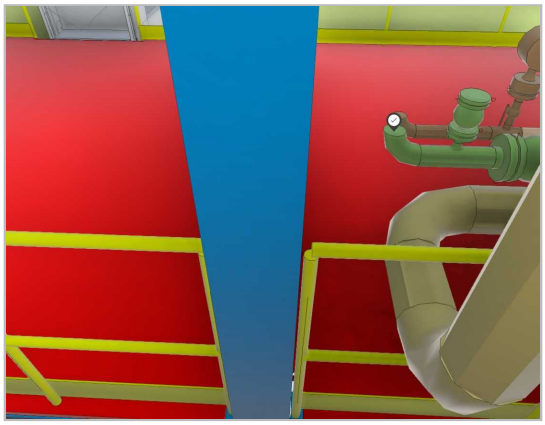
Standard fields

Description	Model missing instruments as shown in the marked up P&IDs
Assigned to	Andrew Worthy (Hanson Professional Services)
Created by	Andrew Worthy (Hanson Professional Services)
Created on	Dec 18, 2025
Location	—
Location details	—
Due date	—
Start date	—
Placement	LP10Y-TFS-M0-000-000-00-HNS-P3D.dwg
Root cause	—

Issue detail

#107: LxL#2 - Parking Lot Item 4 - Model Flange on WV

Status	Closed
Type	DWC Design > Work to Complete



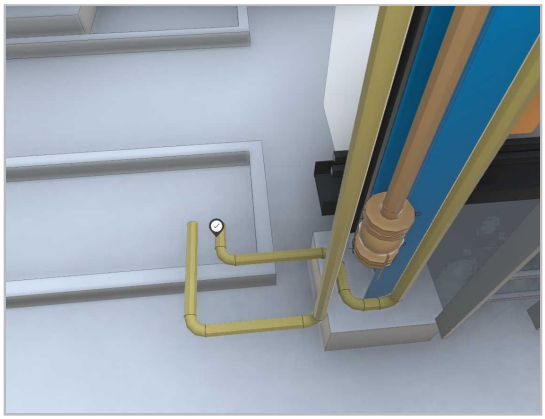
Standard fields

Description	Model Flange on WV line. Make sure all flanges to hoses are modeled.
Assigned to	Andrew Worthy (Hanson Professional Services)
Created by	Andrew Worthy (Hanson Professional Services)
Created on	Dec 18, 2025
Location	—
Location details	—
Due date	—
Start date	—
Placement	LP10Y-TFS-M0-000-000-00-HNS-P3D.dwg
Root cause	—

Issue detail

#108: LxL#2 - Parking Lot Item 5 - Complete Modeling of Drain Lines to Containment (RV)

Status	Closed
Type	DWC Design > Work to Complete



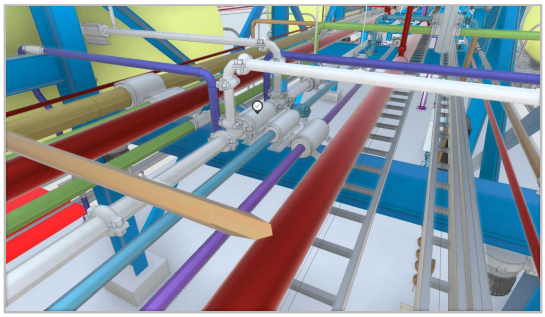
Standard fields

Description	Complete the modeling of all RV lines to containment once containment decision has been finalized.
Assigned to	Andrew Worthy (Hanson Professional Services)
Created by	Andrew Worthy (Hanson Professional Services)
Created on	Dec 18, 2025
Location	—
Location details	—
Due date	—
Start date	—
Placement	LP10Y-TFS-M0-000-000-00-HNS-P3D.dwg
Root cause	—

Issue detail

#118: LxL#2 - Parking Lot Item 9 - WFO branches in model to match P&IDs

Status	Closed
Type	DWC Design > Work to Complete



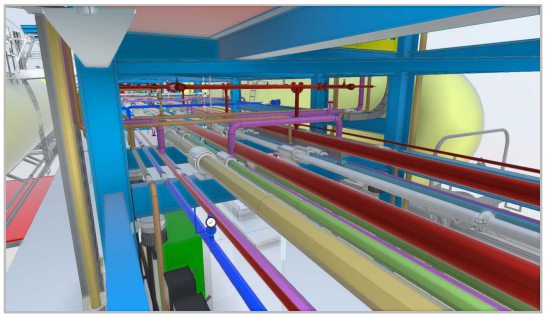
Standard fields

Description	—
Assigned to	Andrew Worthy (Hanson Professional Services)
Created by	Andrew Worthy (Hanson Professional Services)
Created on	Feb 22, 2026
Location	—
Location details	—
Due date	—
Start date	—
Placement	LP10Y-TFS-M0-000-000-00-HNS-P3D.dwg
Root cause	—

Issue detail

#119: LxL#2 - Parking Lot Item 11 - 1" Valve Every 30' on AI Header

Status	Closed
Type	DWC Design > Work to Complete



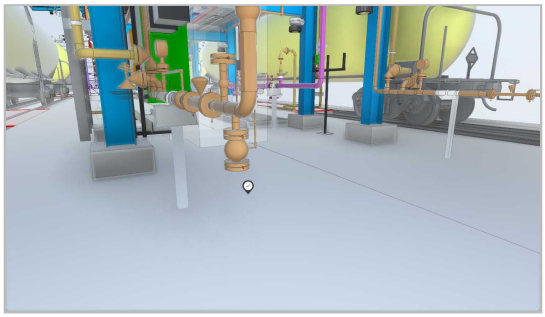
Standard fields

Description	—
Assigned to	Andrew Worthy (Hanson Professional Services)
Created by	Andrew Worthy (Hanson Professional Services)
Created on	Feb 22, 2026
Location	—
Location details	—
Due date	—
Start date	—
Placement	LP10Y-TFS-M0-000-000-00-HNS-P3D.dwg
Root cause	—

Issue detail

#120: LxL#2 - Parking Lot Item 12 - Low Point Drain on N2 Header

Status	Closed
Type	DWC Design > Work to Complete



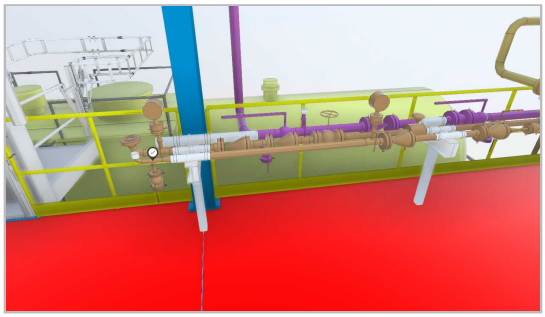
Standard fields

Description	—
Assigned to	Andrew Worthy (Hanson Professional Services)
Created by	Andrew Worthy (Hanson Professional Services)
Created on	Feb 22, 2026
Location	—
Location details	—
Due date	—
Start date	—
Placement	LP10Y-TFS-E0-000-000-00-HNS.rvt
Root cause	—

Issue detail

#121: LxL#2 - Parking Lot Item 13 - Proper Spacing for PI, Drain HV, and Camlock

Status	Closed
Type	DWC Design > Work to Complete



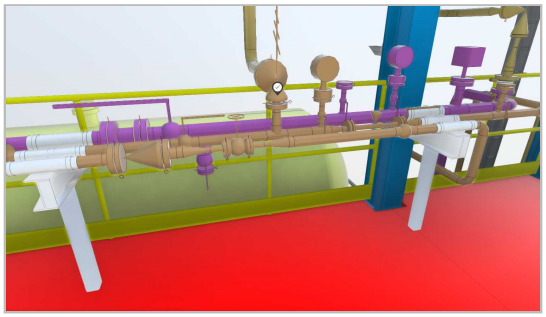
Standard fields

Description	—
Assigned to	Andrew Worthy (Hanson Professional Services)
Created by	Andrew Worthy (Hanson Professional Services)
Created on	Feb 22, 2026
Location	—
Location details	—
Due date	—
Start date	—
Placement	LP10Y-TFS-M0-000-000-00-HNS-P3D.dwg
Root cause	—

Issue detail

#122: LxL#2 - Parking Lot Item 14 - Clash Between PSV and Globe Valve Handwheel (typ.)

Status	Closed
Type	DWC Design > Work to Complete



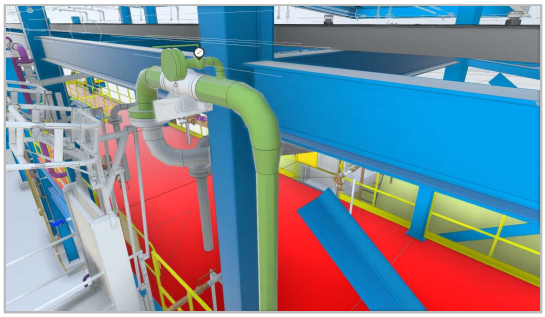
Standard fields

Description	—
Assigned to	Andrew Worthy (Hanson Professional Services)
Created by	Andrew Worthy (Hanson Professional Services)
Created on	Feb 22, 2026
Location	—
Location details	—
Due date	—
Start date	—
Placement	LP10Y-TFS-M0-000-000-00-HNS-P3D.dwg
Root cause	—

Issue detail

#123: LxL#2 - Parking Lot Item 15 - Rotate High Point Vent on PVP

Status	Closed
Type	DWC Design > Work to Complete



Standard fields

Description	—
Assigned to	—
Created by	Andrew Worthy (Hanson Professional Services)
Created on	Feb 22, 2026
Location	—
Location details	—
Due date	—
Start date	—
Placement	LP10Y-TFS-M0-000-000-00-HNS-P3D.dwg
Root cause	—

Issue detail

#124: LxL#2 - Parking Lot Item 16 - LSL Min Distance

Status	Closed
Type	DWC Design > Work to Complete

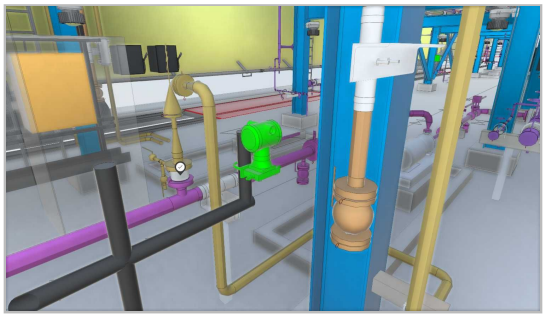
Standard fields

Description	—
Assigned to	Andrew Worthy (Hanson Professional Services)
Created by	Andrew Worthy (Hanson Professional Services)
Created on	Feb 22, 2026
Location	—
Location details	—
Due date	—
Start date	—
Placement	LP10Y-TFS-M0-000-000-00-HNS-P3D.dwg
Root cause	—

Issue detail

#126: LxL#2 - Parking Lot Item 17 - PSV Inlet Line

Status	Closed
Type	DWC Design > Work to Complete



Standard fields

Description	—
Assigned to	—
Created by	Andrew Worthy (Hanson Professional Services)
Created on	Feb 23, 2026
Location	—
Location details	—
Due date	—
Start date	—
Placement	LP10Y-TFS-M0-000-000-00-HNS-P3D.dwg
Root cause	—

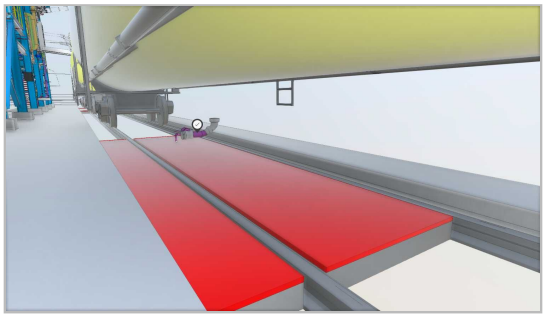
Comments

Andrew Worthy Feb 23, 2026, 5:02 PM EST	Model matches P&IDs and PSV Inlet line up to Rupture Disk has service line number.

Issue detail

#127: LxL#2 - Parking Lot Item 18 - Complete Modeling of MTBE Unloading Connection

Status	Closed
Type	DWC Design > Work to Complete



Standard fields

Description	—
Assigned to	Andrew Worthy (Hanson Professional Services)
Created by	Andrew Worthy (Hanson Professional Services)
Created on	Feb 23, 2026
Location	—
Location details	—
Due date	—
Start date	—
Placement	LP10Y-TFS-M0-000-000-00-HNS-P3D.dwg
Root cause	—

Document Title:	Structural Calculations – Unloading Rack
Document Number:	LP1Y-215-CAL-K215-01
Security Classification:	Restricted

A1 - DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT
Authorization to proceed does not relieve Contractor/Supplier of its responsibility or liability under the Contract and or Purchase Order.
By: Ricardo DaCosta dac54131 March 11, 2026

Structural Calculations – Unloading Rack

Lilly Rail Containment

Hanson Project Number:25I0007.00

Note: This calculation package includes structural calculations for the above ground unloading rack steel framing, concrete piers supporting the steel columns, and the pile caps and tie beams below grade. It does not include any calculations relating to the slab-on-grade or sidewalks as that portion of the design was on hold at the time of IFC submittal of this calculation package.

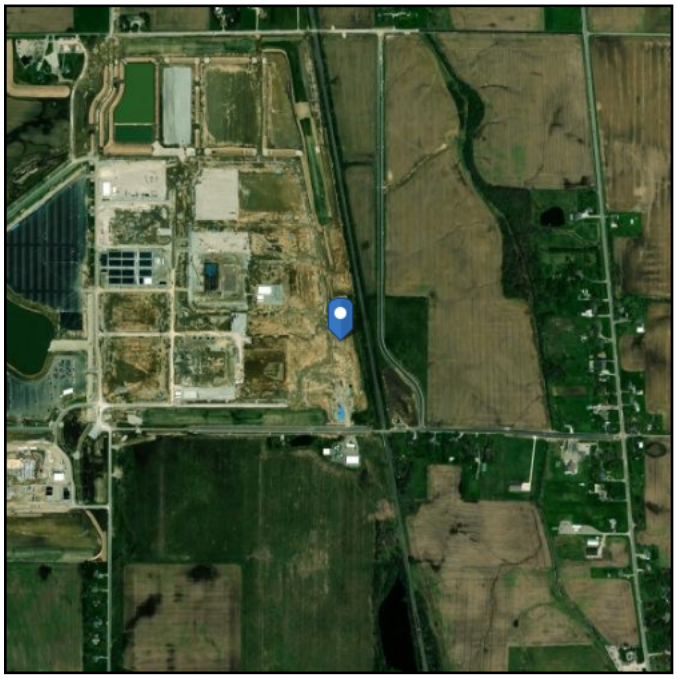
RYNE M. FIORITO REGISTERED PE12100177 STATE OF INDIANA PROFESSIONAL ENGINEER

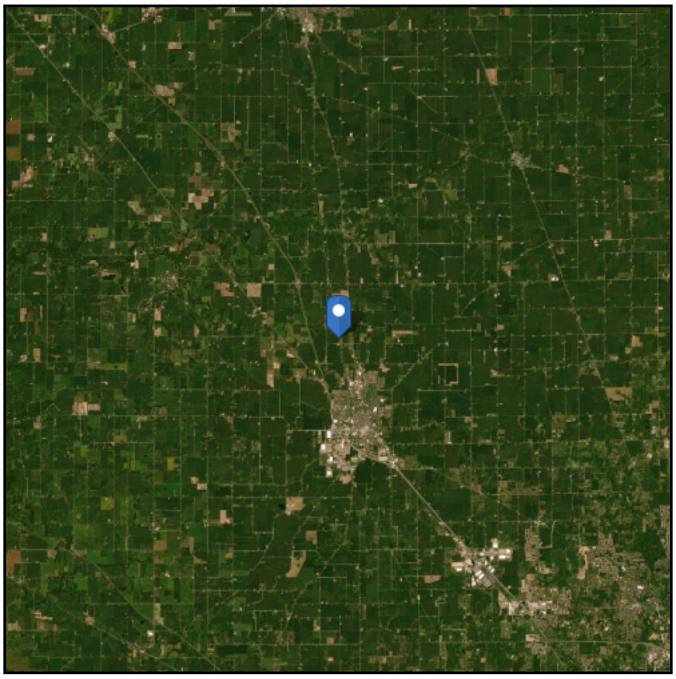
Revision	Issue Status	Originator	Reviewer	Approver	Date
0	Issued for Construction	SJS	FCB	RMF	02/20/26
A	Issued for Review	SJS	FCB	RMF	10/14/25

ASCE Hazards Report

Address:
No Address at This Location

Standard:	ASCE/SEI 7-10	**Latitude:**	40.096314
Risk Category:	IV	**Longitude:**	-86.488684
Soil Class:	C - Very Dense Soil and Soft Rock	**Elevation:**	931.6142823379028 ft (NAVD 88)





Wind

Results:

Wind Speed	120 Vmph
10-year MRI	76 Vmph
25-year MRI	84 Vmph
50-year MRI	90 Vmph
100-year MRI	96 Vmph

Data Source: ASCE/SEI 7-10, Fig. 26.5-1B and Figs. CC-1–CC-4, and Section 26.5.2, incorporating errata of March 12, 2014

Date Accessed: Fri Aug 15 2025

Value provided is 3-second gust wind speeds at 33 ft above ground for Exposure C Category, based on linear interpolation between contours. Wind speeds are interpolated in accordance with the 7-10 Standard. Wind speeds correspond to approximately a 3% probability of exceedance in 50 years (annual exceedance probability = 0.000588, MRI = 1,700 years).

Site is not in a hurricane-prone region as defined in ASCE/SEI 7-10 Section 26.2.

Seismic

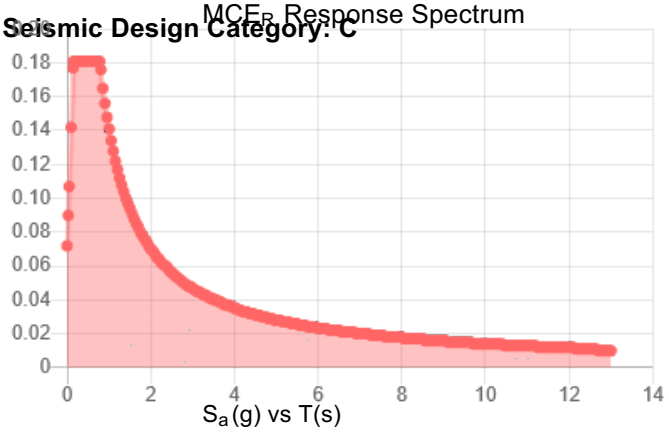
Site Soil Class: C - Very Dense Soil and Soft Rock

Results:

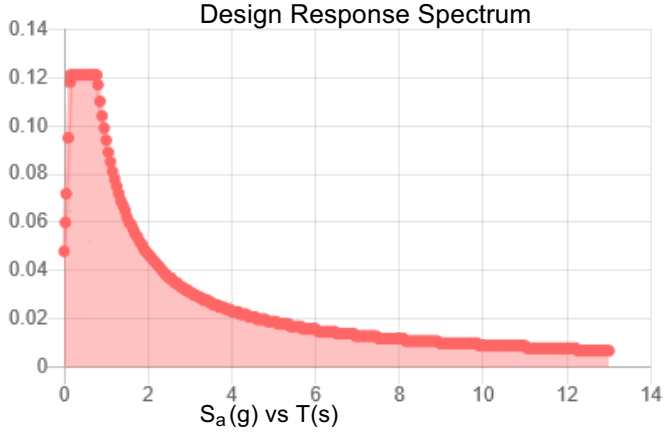
S_S :	0.151	S_{D1} :	0.094
S_1 :	0.083	T_L :	12
F_a :	1.2	PGA :	0.069
F_v :	1.7	PGA_M :	0.083
S_{MS} :	0.181	F_{PGA} :	1.2
S_{M1} :	0.141	I_e :	1.5
S_{DS} :	0.121		

Seismic Design Category: C

MCE_R Response Spectrum



Design Response Spectrum



Data Accessed: Fri Aug 15 2025

Date Source:

USGS Seismic Design Maps based on ASCE/SEI 7-10, incorporating Supplement 1 and errata of March 31, 2013, and ASCE/SEI 7-10 Table 1.5-2. Additional data for site-specific ground motion procedures in accordance with ASCE/SEI 7-10 Ch. 21 are available from USGS.

Ice

Results:

Ice Thickness:	1.00 in.
Concurrent Temperature:	5 F
Gust Speed	40 mph
Data Source:	Standard ASCE/SEI 7-10, Figs. 10-2 through 10-8
Date Accessed:	Fri Aug 15 2025

Ice thicknesses on structures in exposed locations at elevations higher than the surrounding terrain and in valleys and gorges may exceed the mapped values.

Values provided are equivalent radial ice thicknesses due to freezing rain with concurrent 3-second gust speeds, for a 50-year mean recurrence interval, and temperatures concurrent with ice thicknesses due to freezing rain. Thicknesses for ice accretions caused by other sources shall be obtained from local meteorological studies. Ice thicknesses in exposed locations at elevations higher than the surrounding terrain and in valleys and gorges may exceed the mapped values.

Snow

Results:

Ground Snow Load, p_g:	20 lb/ft^2
Mapped Elevation:	931.6 ft
Data Source:	ASCE/SEI 7-10, Fig. 7-1.
Date Accessed:	Fri Aug 15 2025

Values provided are ground snow loads. In areas designated "case study required," extreme local variations in ground snow loads preclude mapping at this scale. Site-specific case studies are required to establish ground snow loads at elevations not covered.

Snow load values are mapped to a 0.5 mile resolution. This resolution can create a mismatch between the mapped elevation and the site-specific elevation in topographically complex areas. Engineers should consult the local authority having jurisdiction in locations where the reported ‘elevation’ and ‘mapped elevation’ differ significantly from each other.

Rain

Results:

15-minute Precipitation Intensity: 7.86 in./h

60-minute Precipitation Intensity: 4.03 in./h

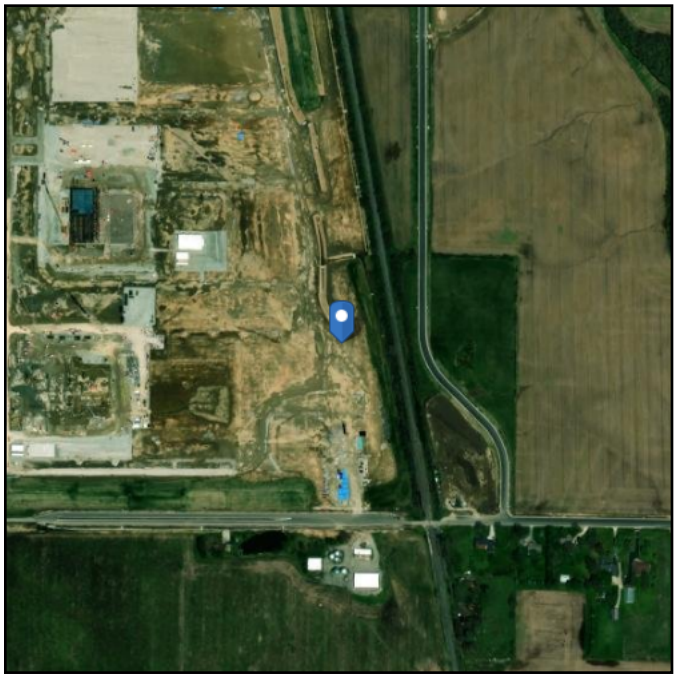
Data Source: NOAA National Weather Service, Precipitation Frequency Data Server, Atlas 14 (https://www.nws.noaa.gov/oh/hdsc/)

Date Accessed: Fri Aug 15 2025

Flood

Results:

Flood Zone Categorization:	X (unshaded)
Base Flood Elevation:	Refer to map for local elevations and interpolate according to the Authority Having Jurisdiction.
Data Source:	FEMA National Flood Hazard Layer - Effective Flood Hazard Layer for US, where modernized (https://msc.fema.gov/portal/search)
Date Accessed:	Fri Aug 15 2025
FIRM Panel:	If available, download FIRM panel here
Insurance Study Note:	Download FEMA Flood Insurance Study for this area here



ASCE
AMERICAN SOCIETY OF CIVIL ENGINEERS

Tornado

Results:

	RC = IV (MRI = 3,000 years)	MRI = 10,000 years	MRI = 100,000 years	MRI = 1,000,000 years	MRI = 10,000,000 years
Effective Plan Area (ft^2)	Tornado Speed (mph)	Tornado Speed (mph)	Tornado Speed (mph)	Tornado Speed (mph)	Tornado Speed (mph)
A_e = 1	V_T = 74	V_T = 104	V_T = 152	V_T = 188	V_T = 229
A_e = 2,000	V_T = 77	V_T = 105	V_T = 154	V_T = 193	V_T = 233
A_e = 10,000	V_T = 82	V_T = 112	V_T = 154	V_T = 194	V_T = 234
A_e = 40,000	V_T = 85	V_T = 114	V_T = 161	V_T = 197	V_T = 236
A_e = 100,000	V_T = 92	V_T = 121	V_T = 164	V_T = 203	V_T = 239
A_e = 250,000	V_T = 96	V_T = 124	V_T = 167	V_T = 205	V_T = 245
A_e = 1,000,000	V_T = 111	V_T = 134	V_T = 175	V_T = 215	V_T = 253
A_e = 4,000,000	V_T = 123	V_T = 144	V_T = 185	V_T = 224	V_T = 262

To select the appropriate tornado hazard map, the effective plan area, Ae, of the building, other structure, or facility, shall be determined in accordance with Section 32.5.4 and shall be rounded up to the next available mapped Ae. Alternatively, linear interpolation of tornado speed between maps using the logarithm of the effective plan area size is permitted, per Section 32.5.1.

Data Source: ASCE/SEI Standard 7-22, Figs. 32.5-1, 32.5-2, and G.2-1 through -4

Date Accessed: Fri Aug 15 2025

The ASCE Hazard Tool is provided for your convenience, for informational purposes only, and is provided “as is” and without warranties of any kind. The location data included herein has been obtained from information developed, produced, and maintained by third party providers; or has been extrapolated from maps incorporated in the ASCE standard. While ASCE has made every effort to use data obtained from reliable sources or methodologies, ASCE does not make any representations or warranties as to the accuracy, completeness, reliability, currency, or quality of any data provided herein. Any third-party links provided by this Tool should not be construed as an endorsement, affiliation, relationship, or sponsorship of such third-party content by or from ASCE.

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Unloading Rack Structural Design Criteria

Project No: 25i0007

Location: Lebanon, IN

Codes: IBC 2012
Indiana Building Code Amendments (09/09/2020)
ASCE 7- 10
AISC Steel Manual 14th Edition
ACI 318- 11

Risk Category: IV

Loads:

Dead Loads	PLATFORM GRATING	10	psf
	MISC MEP LOADS	15	psf
	GUARDRAIL W/ TOE PLATE	15	lbs/ft
	LADDER W/O CAGE	11	lbs/ft
	PIPING/CABLE TRAYS	40	psf
	METAL ROOF	3	psf
	GANGWAY	1000	lbs
	STAIRS	15	psf
Live Loads	PLATFORM LIVE LOAD	150	psf
	PLATFORM LIVE LOAD (CONC.)	2000	lbs
	GANGWAY LIVE LOAD	500	lbs
	STAIR LIVE LOAD	100	psf
	STAIR LIVE LOAD (CONC.)	200	lbs
	HANDRAIL LIVE LOAD	50	plf
	HANDRAIL LIVE LOAD (CONC.)	300	lbs
	ROOF LIVE LOAD	40	psf
	MIN. ROOF LIVE LOAD	20	psf
Snow Loads	GROUND SNOW LOAD	20	psf
	EXPOSURE FACTOR, CE	0.9	fully exposed
	THERMAL FACTOR, C(T)	1.2	Unheated
	IMPORTANCE FACTOR, IS	1.2	
Wind Loads	BASIC WIND SPEED	120	mph
	WIND EXP. CAT.	C	

Seismic Load		(Analysis: Eq. Lat. Force)
	Importance Factor	1.5
	S_s	0.151
	S_1	0.083
	Soil Site Class	C
	Structural System	Steel Systems Not Specifically Detailed for Seismic Resistance
	Resp. Mod. Coeff. R	3
	Seismic Des. Cat.	C
	Cs	0.0604

Ice Load	ICE THICKNESS	1	inches
	CONCURRENT TEMP.	55 °F	
	GUST SPEED	40	mph
	IMPORTANCE FACTOR	1.25	

Tekla Tedds
Hanson Professional Services, Inc
1300 W Commerce Dr.
Suite 200
Peoria, IL 61615

Project				Job Ref.	
Fluor Eli Lilly LP1 Unloading Rack					
Section				Sheet no./rev.	
				1	
Calc. by	Date	Chk'd by	Date	App'd by	Date
SJS	9/11/2025	RMF	02/20/2026		

SNOW LOADING

In accordance with ASCE7-10

Tedds calculation version 1.0.12

Building details

Roof type	Hip and gable
Width of roof (left on elevation)	b_1 = **22.50** ft
Width of roof (right on elevation)	b_2 = **22.50** ft
Slope of roof (left on elevation)	α_1 = **5.00** deg
Slope of roof (right on elevation)	α_2 = **5.00** deg

Ground snow load

Ground snow load (Figure 7-1)	p_g = **20.00** lb/ft^2
Density of snow	γ = min(0.13 × p_g / 1ft + 14lb/ft^3, 30lb/ft^3) = **16.60** lb/ft^3
Surface roughness category (Sect. 26.7)	C
Exposure condition (Table 7-2)	Fully exposed
Exposure factor (Table 7-2)	C_e = **0.90**
Thermal condition (Table 7-3)	Unheated structures
Thermal factor (Table 7-3)	C_t = **1.20**
Importance category (Table 1.5-1)	IV
Importance factor (Table 1.5-2)	I_s = **1.20**
Min snow load for low slope roofs (Sect 7.3.4)	p_{f_min} = I_s × p_g = **24.00** lb/ft^2
Flat roof snow load (Sect 7.3)	p_f = 0.7 × C_e × C_t × I_s × p_g = **18.14** lb/ft^2

Cold roof slope factor (C_t > 1.0)

Roof surface type	Slippery
Ventilation	Ventilated
Thermal resistance (R-value)	R = **26.00** °F h ft^2 / Btu
Roof slope factor - left Fig 7-2c (dashed line)	C_{s_l} = **1.00**
Roof slope factor - right Fig 7-2c (dashed line)	C_{s_r} = **1.00**

Hip and gable roof loads

Balanced sloped snow load - left (Cl.7.4)	p_{s_l} = max(C_{s_l} × p_f, p_{f_min}) = **24.00** lb/ft^2
Balanced sloped snow load - right (Cl.7.4)	p_{s_r} = max(C_{s_r} × p_f, p_{f_min}) = **24.00** lb/ft^2
Slope of left roof	S_l = 1 / tan(α_1) = **11.43**
Slope of right roof	S_r = 1 / tan(α_2) = **11.43**
Unbalanced load - left roof windward	p_{s_lw} = 0.3 × p_f × C_{s_l} = **5.44** lb/ft^2
Unbalanced load - right roof leeward	p_{s_rl} = p_f × C_{s_r} = **18.14** lb/ft^2
Length eaves to ridge for drift height	$l_{u_ww_l}$ = b_1 = **22.50** ft
Drift height	h_{dr_r} = 0.43 × (max($l_{u_ww_l}$, 20 ft) × 1ft^2)$^{1/3}$ × (p_g / 1 lb/ft^2 + 10)$^{1/4}$ - 1.5 × 1ft = **1.34** ft
Rectangular surcharge to part leeward side	$p_{s_rl_sur}$ = h_{dr_r} × γ / √(S_r) = **6.58** lb/ft^2
Length of rectangular surcharge	$l_{u_rl_sur}$ = min(8 / 3 × h_{dr_r} × √(S_r), b_2) = **12.09** ft
Unbalanced load - left roof leeward	p_{s_ll} = p_f × C_{s_l} = **18.14** lb/ft^2
Unbalanced load - right roof windward	p_{s_rw} = 0.3 × p_f × C_{s_r} = **5.44** lb/ft^2
Length eaves to ridge for drift height	$l_{u_ww_r}$ = b_2 = **22.50** ft

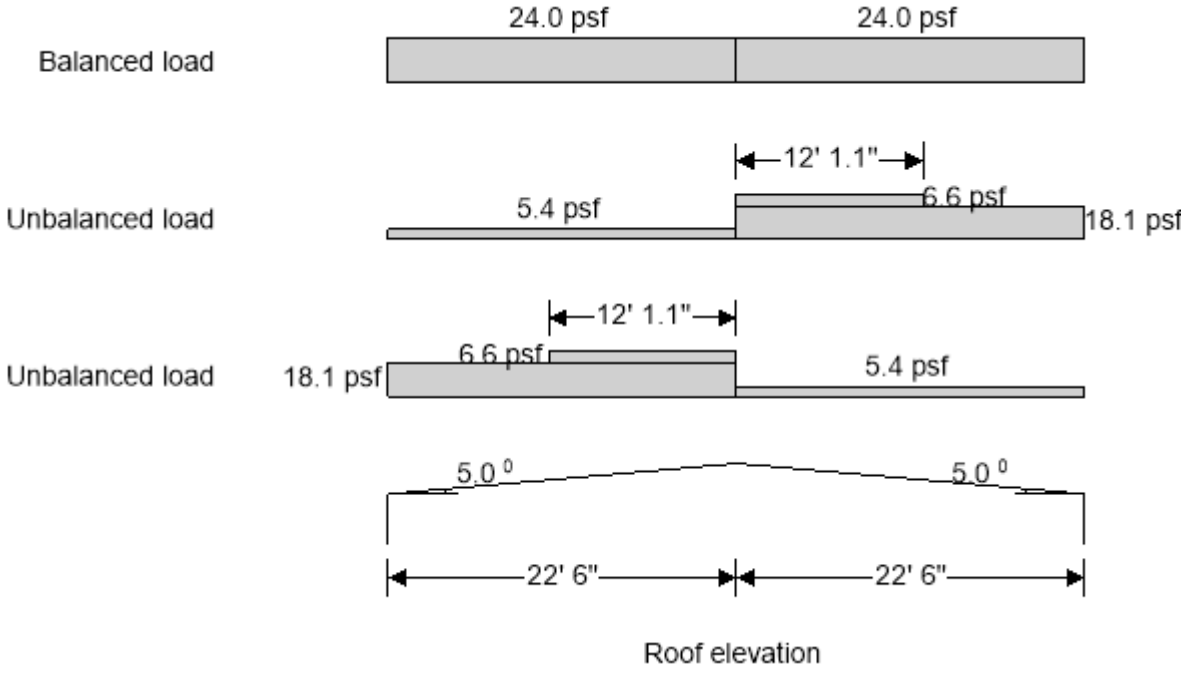
Tekla Tedds
Hanson Professional Services, Inc
1300 W Commerce Dr.
Suite 200
Peoria, IL 61615

Project				Job Ref.	
Fluor Eli Lilly LP1 Unloading Rack					
Section				Sheet no./rev.	
				2	
Calc. by	Date	Chk'd by	Date	App'd by	Date
SJS	9/11/2025	RMF	02/20/2026		

Drift height $h_{dr_l} = 0.43 \times (\max(l_{u_ww_r}, 20\ ft) \times 1ft^2)^{1/3} \times (p_g / 1\ lb/ft^2 + 10)^{1/4} - 1.5 \times 1ft =$ **1.34** ft

Rectangular surcharge to part leeward side $p_{s_ll_sur} = h_{dr_l} \times \gamma / \sqrt{(S_l)} =$ **6.58** lb/ft^2

Length of rectangular surcharge $l_{u_ll_sur} = \min(8 / 3 \times h_{dr_l} \times \sqrt{(S_l)}, b_1) =$ **12.09** ft



Roof elevation

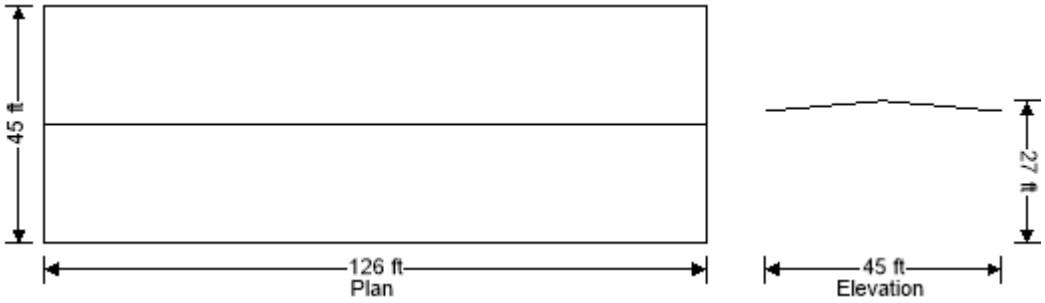
Tekla Tedds	Project: Fluor Eli Lilly LP1 Unloading Rack				Job Ref.	
	Section				Sheet no./rev.: 1	
	Calc. by: SJS	Date: 9/12/2025	Chk'd by: RMF	Date: 02/20/2026	App'd by	Date

WIND LOADING

In accordance with ASCE7-10

Using the directional design method

Tedds calculation version 2.1.19



Building data

Type of roof	Gable free
Length of building	b = **126.00** ft
Width of building	d = **45.00** ft
Height to eaves	H = **25.00** ft
Pitch of roof	α_0 = **5.0** deg
Mean height	h = **25.98** ft
Wind flow	Clear

General wind load requirements

Basic wind speed	V = **120.0** mph
Risk category	IV
Wind directionality factor (Table 26.6-1)	K_d = **0.85**
Exposure category (cl 26.7.3)	C
Enclosure classification (cl.26.10)	Open buildings
Internal pressure coef +ve (Table 26.11-1)	GC_{pi_p} = **0.00**
Internal pressure coef –ve (Table 26.11-1)	GC_{pi_n} = **0.00**
Gust effect factor	G_f = **0.85**

Topography

Topography factor not significant	K_{zt} = 1.0

Velocity pressure

Velocity pressure coefficient (T.27.3-1)	K_z = **0.95**
Velocity pressure	$q_h = 0.00256 \times K_z \times K_{zt} \times K_d \times V^2 \times 1psf/mph^2$ = **29.7** psf

Peak velocity pressure for internal pressure

Peak velocity pressure – internal (as roof press.)	q_i = **29.70** psf

Pressures and forces

Net pressure	$p = q_h \times G \times C_N$
Net force	$F_w = p \times A_{ref}$
Minimum design wind loading (cl.27.1.5)	p_{min_r} = **16** lb/ft^2

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Roof load case 1 - Wind 0 - Loadcase A

Zone	Ref. height (ft)	Ext pressure coefficient c_N	Peak velocity pressure q_h (psf)	Net pressure p (psf)	Area A_{ref} (ft^2)	Net force F_w (kips)
1 (+ve)	25.98	1.20	29.70	30.30	2845.83	86.21
2 (+ve)	25.98	0.30	29.70	7.57	2845.83	21.55

Total vertical net force $F_{w,v}$ = **107.36** kips

Total horizontal net force $F_{w,h}$ = **5.64** kips

Minimum loading

Projected vertical area of roof $A_{vert_r_0} = b \times d/2 \times \tan(\alpha_0)$ = **248.03** ft^2

Minimum overall horizontal loading $F_{w,total_min} = p_{min_r} \times A_{vert_r_0}$ = **3.97** kips

Roof load case 2 - Wind 0 - Loadcase B

Zone	Ref. height (ft)	Ext pressure coefficient c_N	Peak velocity pressure q_h (psf)	Net pressure p (psf)	Area A_{ref} (ft^2)	Net force F_w (kips)
1 (+ve)	25.98	-1.10	29.70	-27.77	2845.83	-79.03
2 (+ve)	25.98	-0.10	29.70	-2.52	2845.83	-7.18

Total vertical net force $F_{w,v}$ = **-85.89** kips

Total horizontal net force $F_{w,h}$ = **-6.26** kips

Minimum loading

Projected vertical area of roof $A_{vert_r_0} = b \times d/2 \times \tan(\alpha_0)$ = **248.03** ft^2

Minimum overall horizontal loading $F_{w,total_min} = p_{min_r} \times A_{vert_r_0}$ = **3.97** kips

Roof load case 3 - Wind 90 - Loadcase A

Zone	Ref. height (ft)	Ext pressure coefficient c_N	Peak velocity pressure q_h (psf)	Net pressure p (psf)	Area A_{ref} (ft^2)	Net force F_w (kips)
1 (+ve)	25.98	-0.80	29.70	-20.20	1173.76	-23.71
2 (+ve)	25.98	-0.60	29.70	-15.15	1173.76	-17.78
3 (+ve)	25.98	-0.30	29.70	-7.57	3344.14	-25.33

Total vertical net force $F_{w,v}$ = **-66.56** kips

Total horizontal net force $F_{w,h}$ = **0.00** kips

Minimum loading

Projected vertical area of roof $A_{vert_r_90}$ = **0.00** ft^2

Minimum overall horizontal loading $F_{w,total_min} = p_{min_r} \times A_{vert_r_90}$ = **0.00** kips

Roof load case 4 - Wind 90 - Loadcase B

Zone	Ref. height (ft)	Ext pressure coefficient c_N	Peak velocity pressure q_h (psf)	Net pressure p (psf)	Area A_{ref} (ft^2)	Net force F_w (kips)
1 (+ve)	25.98	0.80	29.70	20.20	1173.76	23.71
2 (+ve)	25.98	0.50	29.70	12.62	1173.76	14.82
3 (+ve)	25.98	0.30	29.70	7.57	3344.14	25.33

Total vertical net force $F_{w,v}$ = **63.61** kips

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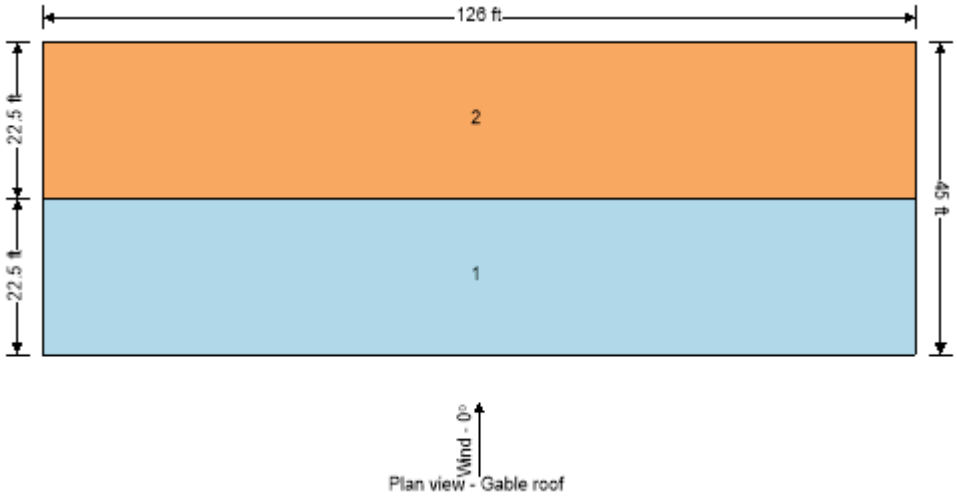
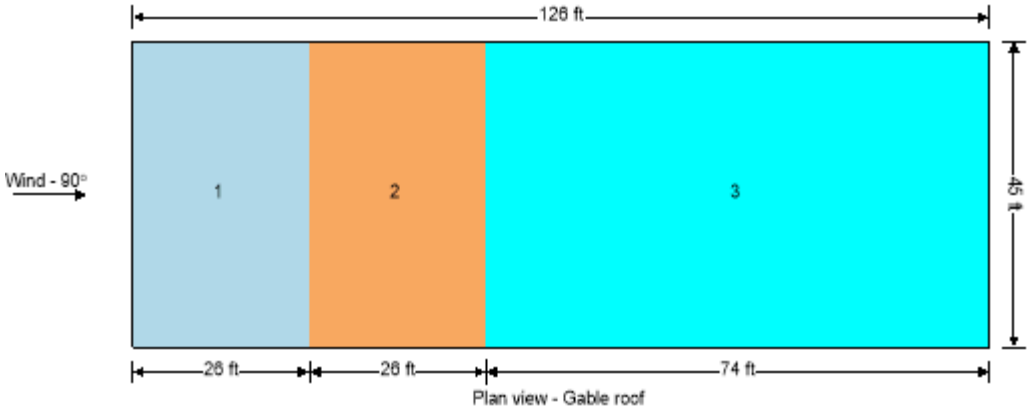
Total horizontal net force $F_{w,h}$ = **0.00** kips

Minimum loading

Projected vertical area of roof $A_{vert_r_90}$ = **0.00** ft²

Minimum overall horizontal loading $F_{w,total_min} = p_{min_r} \times A_{vert_r_90}$ = **0.00** kips





Wind Load Calculations

ASCE Wind Load Design For Petrochemical
and Other Industrial Facilities, 2nd edition

Open Pipe Rack:

F=qz*G*Cf*Ae

qz =	29.7	psf	(from wind calculation - at 33 ft above ground)
G =	0.85		(for simplicity will not reduce qz for lower heights)

Force coefficients:

Cf_str =	1.8	for structural members	Section 5.2.3
Cf_pipe =	0.7	for pipes	Section 5.2.3
Cf_ct =	2.0	for cable trays	Section 5.2.4
Cf_stair =	2.0	for stair	Section 5.3.2, Table 5-1

Frame dimensions:

W =	13	ft	width of bent (transverse to piping)
L =	15	ft	bent spacing (longitudinal to piping)

Largest Pipe:			(equation used:)
Diameter =	8	inches	
A_pipe =	1.97	ft^2/ft	=diameter/12+10%*W
Wind load =	**35**	**lbs/ft**	=ROUNDUP(qz*G*Cf_pipe*A_pipe,0)

Largest Cable Tray on Pipe Rack:			(equation used:)
H_tray =	6	inches	height of tray
A_ct =	1.8	ft^2/ft	=H_tray/12+10%*W
Wind load =	**91**	**lbs/ft**	=ROUNDUP(qz*G*Cf_ct*A_ct,0)

All Structural Members:			(equation used:)
Wind load =	**46**	**psf**	=ROUNDUP(qz*G*Cf_str,0)

By: SJS Date: 11/24/25

Checked by: RMF Date: 02/20/26

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Wind on Stairs

Stringers

Assume 12" deep x2 → 2 sf / ft of stairs

2 sf x 46 psf = 92 lb/ft of stairs

↳ from ASCE Wind Loads for Petrochemical Facilities standard, 2nd ed.

Railings

1 1/2" pipe → 1.9 O.D. → use 2" pipe

Using ASCE wind standard → 26 lb/ft

2 railings per side x 2 sides = 26 x 4 = 104 lb/ft

verticals: $\frac{3.5 \text{ ft} + 11 \times 2 \text{ sides}}{4 \text{ ft average centers}}$ = 1.75 ft x 26 lb/ft

= 46 lb/ft (avg.)

150 lb/ft of stairs of wind force

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SEISMIC FORCES

In accordance with ASCE 7-10

Tedds calculation version 3.1.05

Site parameters

Site class	C
Mapped acceleration parameters (Section 11.4.1)	
at short period	S_S = **0.151**
at 1 sec period	S_1 = **0.083**
Site coefficient	
at short period (Table 11.4-1)	F_a = **1.200**
at 1 sec period (Table 11.4-2)	F_v = **1.700**

Spectral response acceleration parameters

at short period (Eq. 11.4-1)	$S_{MS} = F_a \times S_S$ = **0.181**
at 1 sec period (Eq. 11.4-2)	$S_{M1} = F_v \times S_1$ = **0.141**

Design spectral acceleration parameters (Sect 11.4.4)

at short period (Eq. 11.4-3)	$S_{DS} = 2 / 3 \times S_{MS}$ = **0.121**
at 1 sec period (Eq. 11.4-4)	$S_{D1} = 2 / 3 \times S_{M1}$ = **0.094**

Seismic design category

Risk category (Table 1.5-1)	IV
Seismic design category based on short period response acceleration (Table 11.6-1)	A
Seismic design category based on 1 sec period response acceleration (Table 11.6-2)	C
Seismic design category	C

Approximate fundamental period

Height above base to highest level of building	h_n = **32** ft
From Table 12.8-2:	
Structure type	All other systems
Building period parameter C_t	C_t = **0.02**
Building period parameter x	x = **0.75**
Approximate fundamental period (Eq 12.8-7)	$T_a = C_t \times (h_n)^x \times 1sec / (1ft)^x$= **0.269** sec
Building fundamental period (Sect 12.8.2)	$T = T_a$ = **0.269** sec
Long-period transition period	T_L = **12** sec

Seismic response coefficient

Seismic force-resisting system (Table 12.2-1)	H. STEEL SYSTEMS NOT SPECIFICALLY DETAILED 1. Steel system not specially detailed for seismic resistance, ex
Response modification factor (Table 12.2-1)	R = **3**
Seismic importance factor (Table 1.5-2)	I_e = **1.500**
Seismic response coefficient (Sect 12.8.1.1)	
Calculated (Eq 12.8-2)	$C_{s_calc} = S_{DS} / (R / I_e)$ = **0.0604**

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Maximum (Eq 12.8-3) $C_{s_max} = S_{D1} / ((T / 1\ sec) \times (R / I_e)) =$ **0.1748**

Minimum (Eq 12.8-5) $C_{s_min} = \max(0.044 \times S_{DS} \times I_e, 0.01) =$ **0.0100**

Seismic response coefficient $C_s =$ **0.0604**

Seismic base shear (Sect 12.8.1)

Effective seismic weight of the structure $W =$ **2.0** kips

Seismic response coefficient $C_s =$ **0.0604**

Seismic base shear (Eq 12.8-1) $V = C_s \times W =$ **0.1** kips

Vertical distribution of seismic forces (Sect 12.8.3)

Vertical distribution factor (Eq 12.8-12) $C_{vx} = w_x \times h_x^k / \Sigma(w_i \times h_i^k)$

Lateral force induced at level i (Eq 12.8-11) $F_x = C_{vx} \times V$

Vertical force distribution table

Level	Height from base to Level i (ft), h_x	Portion of effective seismic weight assigned to Level i (kips), w_x	Distribution exponent related to building period, k	Vertical distribution factor, C_{vx}	Lateral force induced at Level i (kips), F_x
1	13.0;	1.0;	1.00;	0.289;	0.0
2	32.0;	1.0;	1.00;	0.711;	0.1

Ice Loads

ASCE 7-10, Section 10.8 Design Procedure

t=	1	inches	Fig 10.4-2
Vc=	40	mph	Fig 10.4-2
Kzt=	1		(no topographic effects)
I[ice]=	1.25		Risk Cat. IV
z=	35	ft	
fz=	1.01		=(z/33)^0.1
td=	2.51	inches	=2*t*Ii*fz*Kzt^0.35

Ice Load Calculations (see ASCE 7-10, Section 10.4):

td= 2.51 inches

Shape	d	bf	Dc (in)	Ai (sq in)	lb/ft
Pipes			6	67.3	**26.2**
W10x22	10.2	5.75	11.7	112.4	**43.7**
W12x30	12.3	6.52	13.9	129.8	**50.5**
W12x40	11.9	8.01	14.3	133.2	**51.8**
WT6x17.5	6.2	6.6	9.1	91.4	**35.5**
WT4x6.5	4	4	5.7	64.6	**25.1**
Railings (1.5" ID)			1.9	34.9	**13.6**
C12x20.7	12	3	15.0	138.4	**53.8**
HSS4x4	4	4	5.7	64.6	**25.1**

C-C Spacing of Pipes	9.5	inches	
Area Ice load for Pipes:	**33.0**	psf	=F15*12/B25

Ice Load Calculations for Cable Trays:

W (Cable Tray width)	144	inches	
Max ice Thickness	2.51	inches	
Ice Density	56	pcf	
Ice Load	**140.8**	plf	=B29*B30*B31/144

Ice Load Calculation for Metal Roof, Cable Trays, Platform Grating, Stairs

As =	1.00	sf	
Vi =	0.66	ASCE Eq. 10.4-2	
Weight	36.87	psf	
0.6	**22.12**	psf	(0.6 for horizontal plates, ASCE 10.4)

Ice on Stairs	277	lb/ft	(see Ice on Stair hand calculation)

Location:	Length	Total Weight (lbs)	Pt Load at End of each Stringer (kips)
South Stair - Upper	14.67	4,064	1.02
South Stair - Lower	9.17	2,540	0.64
North Stair - Each Flight	13	3,601	0.90

By: SJS Date: 11/21/25

Checked by: RMF Date: 02/20/2026

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Ice on Stairs

South Stair - Lower Flight Ice Loading

Grating: 22.12 psf x 3.75' width = 83 lb/ft of stair

Railings: 13.6 x 4 horiz. + (13.6 x 3.5 ft x 6 vert.) / 9.17 ft = 85.6 lb/ft of stair

Stringers: C12x20.7 is 53.8 lb/ft x 2 stringers = 107.6 lb/ft of stair

total = 277 lb/ft

Eli Lilly Unloading Rack - Gangway Loading Assumptions

Per the supplier, the typical weight of a gangway is 550 lbs. A conservative value of 1,000 lbs dead load has been used in the RISA-3D framing analysis model loading.

Per LP1Y-255-DST-K215-05, the design live load of the gangway is to be 500 lbs. This LL has been applied in the RISA-3D framing analysis model loading.

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Company : Hanson Professional Services
Designer : S. Steiner
Job Number : 25i0007
Model Name : Eli Lilly K215 Unloading Rack

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Model Settings

Number of Reported Sections	5
Number of Internal Sections	200
Member Area Load Mesh Size (in^2)	144
Consider Shear Deformation	Yes
Consider Torsional Warping	Yes
Approximate Mesh Size (in)	24
Transfer Forces Between Intersecting Wood Walls	Yes
Increase Wood Wall Nailing Capacity for Wind Loads	Yes
Include P-Delta for Walls	Yes
Optimize Masonry and Wood Walls	Yes
Maximum Number of Iterations	3
Single	No
Multiple (Optimum)	Yes
Maximum	No

Global Axis corresponding to vertical direction	Y
Convert Existing Data	Yes
Default Global Plane for z-axis	XZ
Plate Local Axis Orientation	Global

Hot Rolled Steel	AISC 14th (360-10): LRFD
Stiffness Adjustment	Yes (Iterative)
Notional Annex	None
Connections	AISC 14th (360-10): LRFD
Cold Formed Steel	AISI S100-10: LRFD
Stiffness Adjustment	Yes (Iterative)
Wood	AWC NDS-18 / SDPWS-15 ASD
Temperature	< 100F
Concrete	ACI 318-14
Masonry	TMS 402-16: ASD
Aluminum	AA ADM1-15: ASD
Structure Type	Building
Stiffness Adjustment	Yes (Iterative)
Stainless	AISC 14th (360-10): ASD
Stiffness Adjustment	Yes (Iterative)

Analysis Methodology	Exact Integration Method
Parme Beta Factor	0.65

Compression Stress Block	Rectangular Stress Block
Analyze using Cracked Sections	Yes
Leave room for horizontal rebar splices (2*d bar spacing)	Yes
List forces which were ignored for design in the Detail Report	Yes

Column Min Steel	1
Column Max Steel	8
Rebar Material Spec	ASTM A615
Warn if beam-column framing arrangement is not understood	Yes
Number of Shear Regions	4
Region 2 & 3 Spacing Increase Increment (in)	4

Code	ASCE 7-10
Risk Category	IV
Drift Cat	Other
Base Elevation (ft)	0
Include the weight of the structure in base shear calcs	Yes
S_1 (g)	0.083
SD_1 (g)	0.094
SD_S (g)	0.121
T_L (sec)	12
T Z (sec)	
T X (sec)	
C_tZ	0.02
C_tX	0.02
$C_tExp.$ Z	0.75
$C_tExp.$ X	0.75
R Z	3
R X	3
Ω_0Z	3
Ω_0X	3
C_dZ	3
C_dX	3
ρ Z	1
ρ X	1

Model Settings (Continued)

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Designer : S. Steiner
Job Number : 25i0007
Model Name : Eli Lilly K215 Unloading Rack

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Node Boundary Conditions

	Node Label	X [k/in]	Y [k/in]	Z [k/in]	X Rot [k-ft/rad]	Y Rot [k-ft/rad]	Z Rot [k-ft/rad]
1	3	Reaction	Reaction	Reaction	Reaction	Reaction	Reaction
2	4	Reaction	Reaction	Reaction	Reaction	Reaction	Reaction
3	1	Reaction	Reaction	Reaction	Reaction	Reaction	Reaction
4	2	Reaction	Reaction	Reaction	Reaction	Reaction	Reaction
5	5	Reaction	Reaction	Reaction	Reaction	Reaction	Reaction
6	6	Reaction	Reaction	Reaction	Reaction	Reaction	Reaction
7	7	Reaction	Reaction	Reaction	Reaction	Reaction	Reaction
8	8	Reaction	Reaction	Reaction	Reaction	Reaction	Reaction
9	9	Reaction	Reaction	Reaction	Reaction	Reaction	Reaction
10	10	Reaction	Reaction	Reaction	Reaction	Reaction	Reaction
11	11	Reaction	Reaction	Reaction	Reaction	Reaction	Reaction
12	12	Reaction	Reaction	Reaction	Reaction	Reaction	Reaction
13	15	Reaction	Reaction	Reaction	Reaction	Reaction	Reaction
14	16	Reaction	Reaction	Reaction	Reaction	Reaction	Reaction
15	17	Reaction	Reaction	Reaction	Reaction	Reaction	Reaction
16	18	Reaction	Reaction	Reaction	Reaction	Reaction	Reaction
17	19	Reaction	Reaction	Reaction	Reaction	Reaction	Reaction
18	20	Reaction	Reaction	Reaction	Reaction	Reaction	Reaction
19	21	Reaction	Reaction	Reaction	Reaction	Reaction	Reaction
20	22	Reaction	Reaction	Reaction	Reaction	Reaction	Reaction
21	23	Reaction	Reaction	Reaction	Reaction	Reaction	Reaction
22	24	Reaction	Reaction	Reaction	Reaction	Reaction	Reaction
23	25	Reaction	Reaction	Reaction	Reaction	Reaction	Reaction
24	26	Reaction	Reaction	Reaction	Reaction	Reaction	Reaction
25	13	Reaction	Reaction	Reaction	Reaction	Reaction	Reaction
26	14	Reaction	Reaction	Reaction	Reaction	Reaction	Reaction
27	385	Reaction	Reaction	Reaction	Reaction	Reaction	Reaction
28	386	Reaction	Reaction	Reaction	Reaction	Reaction	Reaction
29	387	Reaction	Reaction	Reaction	Reaction	Reaction	Reaction
30	388	Reaction	Reaction	Reaction	Reaction	Reaction	Reaction
31	412	Reaction	Reaction	Reaction	Reaction	Reaction	Reaction
32	411	Reaction	Reaction	Reaction	Reaction	Reaction	Reaction
33	410	Reaction	Reaction	Reaction	Reaction	Reaction	Reaction

Hot Rolled Steel Properties

	Label	E [ksi]	G [ksi]	Nu	Therm. Coeff. [1e^{6}°F^{-1}]	Density [k/ft^{3}]	Yield [ksi]	Ry	Fu [ksi]	Rt
1	A992	29000	11154	0.3	0.65	0.49	50	1.1	65	1.1
2	A36 Gr.36	29000	11154	0.3	0.65	0.49	36	1.5	58	1.2
3	A572 Gr.50	29000	11154	0.3	0.65	0.49	50	1.1	65	1.1
4	A500 Gr.B RND	29000	11154	0.3	0.65	0.527	42	1.4	58	1.3
5	A500 Gr.C Rect	29000	11154	0.3	0.65	0.527	50	1.4	62	1.3
6	A500 Gr.B Rect	29000	11154	0.3	0.65	0.527	46	1.4	58	1.3
7	A53 Gr.B	29000	11154	0.3	0.65	0.49	35	1.6	60	1.2
8	A1085	29000	11154	0.3	0.65	0.49	50	1.25	65	1.15
9	A913 Gr.65	29000	11154	0.3	0.65	0.49	65	1.1	80	1.1

Cold Formed Steel Properties

	Label	E [ksi]	G [ksi]	Nu	Therm. Coeff. [1e^{6}°F^{-1}]	Density [k/ft^{3}]	Yield [ksi]	Fu [ksi]
1	A653 SS Gr33	29500	11346	0.3	0.65	0.49	33	45
2	A653 SS Gr50/1	29500	11346	0.3	0.65	0.49	50	65

Hot Rolled Steel Section Sets

	Label	Shape	Type	Design List	Material	Design Rule	Area [in^{2}]	Iyy [in^{4}]	Izz [in^{4}]	J [in^{4}]
1	Beams 1	W12X30	Beam	Wide Flange	A992	Typical	8.79	20.3	238	0.457
2	Beams 2	W12X30	Beam	Wide Flange	A992	Typical	8.79	20.3	238	0.457
3	Columns	W12X45	Column	Wide Flange	A992	Columns	13.1	50	348	1.26
4	Purlins - Not Used	W8X18	Beam	Wide Flange	A992	Typical	5.26	7.97	61.9	0.172
5	Roof Girders	W12X40	Beam	Wide Flange	A992	Typical	11.7	44.1	307	0.906
6	Lateral Braces	WT6X17.5	VBrace	W Tee	A992	Typical	5.17	12.2	16	0.369
7	Grating Support Beams	W10X22	Beam	Wide Flange	A992	Typical	6.49	11.4	118	0.239
8	Horizontal Braces	WT4X6.5	HBrace	W Tee	A992	Typical	1.92	1.36	2.89	0.043
9	Beams at Catwalk	W10X33	Beam	Wide Flange	A992	Typical	9.71	36.6	171	0.583
10	Pipe Support Beams	W10X22	Beam	Wide Flange	A992	Typical	6.49	11.4	118	0.239
11	Vertical Braces	L3X3X4	VBrace	Single Angle	A992	Typical	1.44	1.23	1.23	0.031
12	Stair Posts	HSS4X4X4	Column	SquareTube	A500 Gr.B Rect	Typical	3.37	7.8	7.8	12.8

Cold Formed Steel Section Sets

	Label	Shape	Type	Design List	Material	Design Rule	Area [in^{2}]	Iyy [in^{4}]	Izz [in^{4}]	J [in^{4}]
1	Purlins	12CS4X105	Beam	CS B-to-B	A653 SS Gr33	Typical	2.2	4.27	47.5	0.008

Member Primary Data

	Label	I Node	J Node	Rotate(deg)	Section/Shape	Type	Design List	Material	Design Rule
1	M312	376	287		Roof Girders	Beam	Wide Flange	A992	Typical
2	M290	372	254		Roof Girders	Beam	Wide Flange	A992	Typical
3	M193	370	280		Roof Girders	Beam	Wide Flange	A992	Typical
4	105	368	249		Roof Girders	Beam	Wide Flange	A992	Typical
5	M170	372	255		Roof Girders	Beam	Wide Flange	A992	Typical
6	107	368	250		Roof Girders	Beam	Wide Flange	A992	Typical
7	M195	370	282		Roof Girders	Beam	Wide Flange	A992	Typical
8	M298	376	285		Roof Girders	Beam	Wide Flange	A992	Typical
9	M43	371	248		Roof Girders	Beam	Wide Flange	A992	Typical

Member Primary Data (Continued)

	Label	I Node	J Node	Rotate(deg)	Section/Shape	Type	Design List	Material	Design Rule
10	M336	375	284		Roof Girders	Beam	Wide Flange	A992	Typical
11	M145	374	253		Roof Girders	Beam	Wide Flange	A992	Typical
12	M143	374	252		Roof Girders	Beam	Wide Flange	A992	Typical
13	M319	373	283		Roof Girders	Beam	Wide Flange	A992	Typical
14	M316	373	281		Roof Girders	Beam	Wide Flange	A992	Typical
15	M291	369	288		Roof Girders	Beam	Wide Flange	A992	Typical
16	M335	375	289		Roof Girders	Beam	Wide Flange	A992	Typical
17	M294	369	290		Roof Girders	Beam	Wide Flange	A992	Typical
18	M45	371	251		Roof Girders	Beam	Wide Flange	A992	Typical
19	M311	355	353	355	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
20	M314	329	333	355	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
21	M315	330	331	5	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
22	M318	352	351	5	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
23	M320	353	350	355	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
24	M322	333	332	355	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
25	M328	332	343	355	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
26	M325	351	364	5	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
27	M327	350	363	355	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
28	M330	344	335	5	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
29	M331	360	357	5	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
30	M333	366	354	355	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
31	M310	356	352	5	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
32	M323	331	340	5	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
33	M309	328	330	5	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
34	M297	262	258	355	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
35	M306	367	356	5	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
36	M305	346	328	5	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
37	M304	337	341	5	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
38	M303	336	342	355	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
39	M302	276	277	355	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
40	M301	325	326	355	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
41	M300	266	265	355	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
42	M334	345	334	355	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
43	M296	361	365	355	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
44	M293	362	367	5	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
45	M256	261	260	5	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
46	M255	271	270	5	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
47	M254	275	273	5	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
48	M253	323	324	5	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
49	M308	347	329	355	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
50	M338	259	256	5	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
51	M307	365	355	355	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
52	M340	267	269	355	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
53	M313	287	283	355	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
54	M321	283	293	355	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
55	M299	285	281	5	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
56	M329	299	289	5	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
57	M324	297	288	5	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
58	M326	294	290	355	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
59	M317	281	295	5	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
60	M292	288	285	5	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
61	M360	348	361	355	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
62	M361	349	362	5	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
63	M363	280	286	355	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
64	M365	282	296	5	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
65	M295	290	287	355	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
66	M177	255	253	5	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
67	M332	298	284	355	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
68	M50	248	249	355	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
69	M175	254	252	355	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
70	M341	321	322	5	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
71	M342	272	278	5	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
72	M343	364	360	5	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
73	M344	363	366	355	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
74	M345	263	257	355	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
75	M350	327	320	355	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
76	M351	274	279	355	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
77	M339	268	264	5	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
78	M52	251	250	5	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
79	M426	181	178		Horizontal Braces	HBrace	W Tee	A992	Typical
80	M425	175	181		Horizontal Braces	HBrace	W Tee	A992	Typical
81	M424	149	175		Horizontal Braces	HBrace	W Tee	A992	Typical
82	M423	150	149		Horizontal Braces	HBrace	W Tee	A992	Typical
83	M422	153	150		Horizontal Braces	HBrace	W Tee	A992	Typical
84	M421	154	153		Horizontal Braces	HBrace	W Tee	A992	Typical
85	M416	166	163		Horizontal Braces	HBrace	W Tee	A992	Typical
86	M418	170	159		Horizontal Braces	HBrace	W Tee	A992	Typical
87	M417	163	168		Horizontal Braces	HBrace	W Tee	A992	Typical
88	M427	178	177		Horizontal Braces	HBrace	W Tee	A992	Typical
89	M409	174	170		Horizontal Braces	HBrace	W Tee	A992	Typical
90	M407	168	174		Horizontal Braces	HBrace	W Tee	A992	Typical
91	M389	165	166		Horizontal Braces	HBrace	W Tee	A992	Typical
92	M419	157	154		Horizontal Braces	HBrace	W Tee	A992	Typical
93	M490	221	226		Horizontal Braces	HBrace	W Tee	A992	Typical
94	M489	204	221		Horizontal Braces	HBrace	W Tee	A992	Typical
95	M475	234	208		Horizontal Braces	HBrace	W Tee	A992	Typical
96	M474	216	234		Horizontal Braces	HBrace	W Tee	A992	Typical
97	M463	217	223		Horizontal Braces	HBrace	W Tee	A992	Typical

HANSON™

Company : Hanson Professional Services
Designer : S. Steiner
Job Number : 25i0007
Model Name : Eli Lilly K215 Unloading Rack

2/19/2026
2:59:44 PM
Checked By : RMF

Member Primary Data (Continued)

	Label	I Node	J Node	Rotate(deg)	Section/Shape	Type	Design List	Material	Design Rule
98	M454	219	213		Horizontal Braces	HBrace	W Tee	A992	Typical
99	M452	207	220		Horizontal Braces	HBrace	W Tee	A992	Typical
100	M451	222	204		Horizontal Braces	HBrace	W Tee	A992	Typical
101	M449	219	229		Horizontal Braces	HBrace	W Tee	A992	Typical
102	M447	226	212		Horizontal Braces	HBrace	W Tee	A992	Typical
103	M445	208	222		Horizontal Braces	HBrace	W Tee	A992	Typical
104	M443	212	228		Horizontal Braces	HBrace	W Tee	A992	Typical
105	M442	213	230		Horizontal Braces	HBrace	W Tee	A992	Typical
106	M453	230	217		Horizontal Braces	HBrace	W Tee	A992	Typical
107	M269	122	123		Grating Support Beams	Beam	Wide Flange	A992	Typical
108	M258	83	89		Grating Support Beams	Beam	Wide Flange	A992	Typical
109	M260	81	79		Grating Support Beams	Beam	Wide Flange	A992	Typical
110	M265	80	87		Grating Support Beams	Beam	Wide Flange	A992	Typical
111	M266	90	116		Grating Support Beams	Beam	Wide Flange	A992	Typical
112	M267	92	119		Grating Support Beams	Beam	Wide Flange	A992	Typical
113	M268	120	121		Grating Support Beams	Beam	Wide Flange	A992	Typical
114	M270	124	125		Grating Support Beams	Beam	Wide Flange	A992	Typical
115	M271	126	127		Grating Support Beams	Beam	Wide Flange	A992	Typical
116	M272	132	118		Grating Support Beams	Beam	Wide Flange	A992	Typical
117	M273	142	133		Grating Support Beams	Beam	Wide Flange	A992	Typical
118	M275	145	91		Grating Support Beams	Beam	Wide Flange	A992	Typical
119	M276	105	138		Grating Support Beams	Beam	Wide Flange	A992	Typical
120	M277	136	137		Grating Support Beams	Beam	Wide Flange	A992	Typical
121	M278	129	117		Grating Support Beams	Beam	Wide Flange	A992	Typical
122	M279	143	144		Grating Support Beams	Beam	Wide Flange	A992	Typical
123	M380	131	128		Grating Support Beams	Beam	Wide Flange	A992	Typical
124	M382	139	93		Grating Support Beams	Beam	Wide Flange	A992	Typical
125	M383	134	135		Grating Support Beams	Beam	Wide Flange	A992	Typical
126	M385	140	141		Grating Support Beams	Beam	Wide Flange	A992	Typical
127	M379	85	84		Grating Support Beams	Beam	Wide Flange	A992	Typical
128	M386	94	95		Grating Support Beams	Beam	Wide Flange	A992	Typical
129	M388	96	97		Grating Support Beams	Beam	Wide Flange	A992	Typical
130	M280	173	172		Grating Support Beams	Beam	Wide Flange	A992	Typical
131	M274	160	176		Grating Support Beams	Beam	Wide Flange	A992	Typical
132	M234	242	23		Columns	Column	Wide Flange	A992	Columns
133	M357	318	14		Columns	Column	Wide Flange	A992	Columns
134	M358	305	11		Columns	Column	Wide Flange	A992	Columns
135	M359	304	12		Columns	Column	Wide Flange	A992	Columns
136	M362	303	9		Columns	Column	Wide Flange	A992	Columns
137	M356	317	13		Columns	Column	Wide Flange	A992	Columns
138	M364	302	10		Columns	Column	Wide Flange	A992	Columns
139	M500	300	8		Columns	Column	Wide Flange	A992	Columns
140	M501	312	5		Columns	Column	Wide Flange	A992	Columns
141	M502	314	6		Columns	Column	Wide Flange	A992	Columns
142	M503	313	3		Columns	Column	Wide Flange	A992	Columns
143	M504	319	4		Columns	Column	Wide Flange	A992	Columns
144	M505	309	1		Columns	Column	Wide Flange	A992	Columns
145	M499	301	7		Columns	Column	Wide Flange	A992	Columns
146	M506	306	2		Columns	Column	Wide Flange	A992	Columns
147	M355	316	16		Columns	Column	Wide Flange	A992	Columns
148	M354	315	15		Columns	Column	Wide Flange	A992	Columns
149	M235	243	24		Columns	Column	Wide Flange	A992	Columns
150	M211	311	19		Columns	Column	Wide Flange	A992	Columns
151	M212	310	20		Columns	Column	Wide Flange	A992	Columns
152	M245	244	25		Columns	Column	Wide Flange	A992	Columns
153	M223	247	21		Columns	Column	Wide Flange	A992	Columns
154	M224	246	22		Columns	Column	Wide Flange	A992	Columns
155	M246	245	26		Columns	Column	Wide Flange	A992	Columns
156	M352	308	17		Columns	Column	Wide Flange	A992	Columns
157	M353	307	18		Columns	Column	Wide Flange	A992	Columns
158	M462	34	33	180	W12X30	Beam	Wide Flange	A992	Typical
159	M497	101	102		Beams at Catwalk	Beam	Wide Flange	A992	Typical
160	M496	100	101		Beams at Catwalk	Beam	Wide Flange	A992	Typical
161	M495	99	100		Beams at Catwalk	Beam	Wide Flange	A992	Typical
162	M494	98	99		Beams at Catwalk	Beam	Wide Flange	A992	Typical
163	M498	146	114		Beams 1	Beam	Wide Flange	A992	Typical
164	M479	220	224		Beams 2	Beam	Wide Flange	A992	Typical
165	M483	230	223		Beams 2	Beam	Wide Flange	A992	Typical
166	M481	207	219		Beams 2	Beam	Wide Flange	A992	Typical
167	M482	219	230		Beams 2	Beam	Wide Flange	A992	Typical
168	M478	229	220		Beams 2	Beam	Wide Flange	A992	Typical
169	M484	223	216		Beams 2	Beam	Wide Flange	A992	Typical
170	M485	215	237		Beams 2	Beam	Wide Flange	A992	Typical
171	M286	234	232		Beams 2	Beam	Wide Flange	A992	Typical
172	M480	224	215		Beams 2	Beam	Wide Flange	A992	Typical
173	M48	179	181		Beams 2	Beam	Wide Flange	A992	Typical
174	M392	76	63		Beams 2	Beam	Wide Flange	A992	Typical
175	M124	148	150		Beams 2	Beam	Wide Flange	A992	Typical
176	M99	175	148		Beams 2	Beam	Wide Flange	A992	Typical
177	M98	147	149		Beams 2	Beam	Wide Flange	A992	Typical
178	M73	181	147		Beams 2	Beam	Wide Flange	A992	Typical
179	M74	180	175		Beams 2	Beam	Wide Flange	A992	Typical
180	M287	222	233		Beams 2	Beam	Wide Flange	A992	Typical
181	128	82	178		Beams 2	Beam	Wide Flange	A992	Typical
182	M123	149	151		Beams 2	Beam	Wide Flange	A992	Typical
183	M404	237	234		Beams 2	Beam	Wide Flange	A992	Typical
184	M288	221	218		Beams 2	Beam	Wide Flange	A992	Typical
185	M289	212	211		Beams 2	Beam	Wide Flange	A992	Typical

Member Primary Data (Continued)

	Label	I Node	J Node	Rotate(deg)	Section/Shape	Type	Design List	Material	Design Rule
186	M402	233	221		Beams 2	Beam	Wide Flange	A992	Typical
187	M403	218	212		Beams 2	Beam	Wide Flange	A992	Typical
188	M149	150	152		Beams 2	Beam	Wide Flange	A992	Typical
189	M148	151	153		Beams 2	Beam	Wide Flange	A992	Typical
190	M405	232	222		Beams 2	Beam	Wide Flange	A992	Typical
191	127	177	179		Beams 2	Beam	Wide Flange	A992	Typical
192	M282	204	225		Beams 2	Beam	Wide Flange	A992	Typical
193	M391	66	65		Beams 2	Beam	Wide Flange	A992	Typical
194	M198	155	157		Beams 2	Beam	Wide Flange	A992	Typical
195	M174	152	154		Beams 2	Beam	Wide Flange	A992	Typical
196	M173	153	155		Beams 2	Beam	Wide Flange	A992	Typical
197	M49	178	180		Beams 2	Beam	Wide Flange	A992	Typical
198	M251	162	164		Beams 2	Beam	Wide Flange	A992	Typical
199	M250	163	165		Beams 2	Beam	Wide Flange	A992	Typical
200	M240	161	162		Beams 2	Beam	Wide Flange	A992	Typical
201	M239	174	163		Beams 2	Beam	Wide Flange	A992	Typical
202	M229	158	161		Beams 2	Beam	Wide Flange	A992	Typical
203	M284	226	227		Beams 2	Beam	Wide Flange	A992	Typical
204	M281	202	204		Beams 2	Beam	Wide Flange	A992	Typical
205	M264	208	202		Beams 2	Beam	Wide Flange	A992	Typical
206	M262	216	238		Beams 2	Beam	Wide Flange	A992	Typical
207	M398	61	59		Beams 2	Beam	Wide Flange	A992	Typical
208	M397	62	60		Beams 2	Beam	Wide Flange	A992	Typical
209	M395	63	61		Beams 2	Beam	Wide Flange	A992	Typical
210	M394	65	62		Beams 2	Beam	Wide Flange	A992	Typical
211	M199	154	156		Beams 2	Beam	Wide Flange	A992	Typical
212	M217	157	159		Beams 2	Beam	Wide Flange	A992	Typical
213	M228	159	174		Beams 2	Beam	Wide Flange	A992	Typical
214	M283	225	226		Beams 2	Beam	Wide Flange	A992	Typical
215	M285	227	228		Beams 2	Beam	Wide Flange	A992	Typical
216	M263	238	208		Beams 2	Beam	Wide Flange	A992	Typical
217	M218	156	158		Beams 2	Beam	Wide Flange	A992	Typical
218	M215	159	158		Beams 1	Beam	Wide Flange	A992	Typical
219	M384	67	68		Beams 1	Beam	Wide Flange	A992	Typical
220	M366	50	49		Beams 1	Beam	Wide Flange	A992	Typical
221	M369	52	64		Beams 1	Beam	Wide Flange	A992	Typical
222	M372	74	75		Beams 1	Beam	Wide Flange	A992	Typical
223	M375	51	73		Beams 1	Beam	Wide Flange	A992	Typical
224	M381	69	70		Beams 1	Beam	Wide Flange	A992	Typical
225	M488	214	217		W10X22	Beam	Wide Flange	A992	Typical
226	M390	63	65		Beams 1	Beam	Wide Flange	A992	Typical
227	M346	48	47		Beams 1	Beam	Wide Flange	A992	Typical
228	M468	202	222		Beams 1	Beam	Wide Flange	A992	Typical
229	M378	71	72		Beams 1	Beam	Wide Flange	A992	Typical
230	M347	46	45		Beams 1	Beam	Wide Flange	A992	Typical
231	M466	238	234		Beams 1	Beam	Wide Flange	A992	Typical
232	M467	208	232		Beams 1	Beam	Wide Flange	A992	Typical
233	M261	171	170		Beams 1	Beam	Wide Flange	A992	Typical
234	M465	216	237		Beams 1	Beam	Wide Flange	A992	Typical
235	M401	53	54		Pipe Support Beams	Beam	Wide Flange	A992	Typical
236	M400	55	56		Pipe Support Beams	Beam	Wide Flange	A992	Typical
237	M399	57	58		Pipe Support Beams	Beam	Wide Flange	A992	Typical
238	M40	181	180		Beams 1	Beam	Wide Flange	A992	Typical
239	M436	39	36		Beams 2	Beam	Wide Flange	A992	Typical
240	M396	59	60		Beams 1	Beam	Wide Flange	A992	Typical
241	M456	130	88		W10X22	Beam	Wide Flange	A992	Typical
242	M455	169	130		W10X22	Beam	Wide Flange	A992	Typical
243	M393	61	62		Beams 1	Beam	Wide Flange	A992	Typical
244	M387	76	66		Beams 1	Beam	Wide Flange	A992	Typical
245	M227	174	161		Beams 1	Beam	Wide Flange	A992	Typical
246	M238	163	162		Beams 1	Beam	Wide Flange	A992	Typical
247	M487	210	213		W10X22	Beam	Wide Flange	A992	Typical
248	M477	223	215		Beams 1	Beam	Wide Flange	A992	Typical
249	M476	230	224		Beams 1	Beam	Wide Flange	A992	Typical
250	M473	228	211		Beams 1	Beam	Wide Flange	A992	Typical
251	M472	227	212		Beams 1	Beam	Wide Flange	A992	Typical
252	M471	226	218		Beams 1	Beam	Wide Flange	A992	Typical
253	M470	225	221		Beams 1	Beam	Wide Flange	A992	Typical
254	M469	204	233		Beams 1	Beam	Wide Flange	A992	Typical
255	M446	31	38		Beams 2	Beam	Wide Flange	A992	Typical
256	M115	151	150		Beams 1	Beam	Wide Flange	A992	Typical
257	M190	157	156		Beams 1	Beam	Wide Flange	A992	Typical
258	M415	219	220		Beams 1	Beam	Wide Flange	A992	Typical
259	M414	207	229		Beams 1	Beam	Wide Flange	A992	Typical
260	M486	205	209		W10X22	Beam	Wide Flange	A992	Typical
261	M140	153	152		Beams 1	Beam	Wide Flange	A992	Typical
262	90	179	178		Beams 1	Beam	Wide Flange	A992	Typical
263	M165	155	154		Beams 1	Beam	Wide Flange	A992	Typical
264	M461	35	30		Beams 2	Beam	Wide Flange	A992	Typical
265	M259	169	168		Beams 1	Beam	Wide Flange	A992	Typical
266	M257	167	166		Beams 1	Beam	Wide Flange	A992	Typical
267	M460	40	35		Beams 2	Beam	Wide Flange	A992	Typical
268	M459	36	40		Beams 2	Beam	Wide Flange	A992	Typical
269	M249	165	164		Beams 1	Beam	Wide Flange	A992	Typical
270	M458	32	39		C12X30	Beam	Channel	A992	Typical
271	M457	32	37		Beams 2	Beam	Wide Flange	A992	Typical
272	M65	147	175		Beams 1	Beam	Wide Flange	A992	Typical
273	M90	149	148		Beams 1	Beam	Wide Flange	A992	Typical

HANSON™

Company : Hanson Professional Services
Designer : S. Steiner
Job Number : 25i0007
Model Name : Eli Lilly K215 Unloading Rack

2/19/2026
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Checked By : RMF

Member Primary Data (Continued)

	Label	I Node	J Node	Rotate(deg)	Section/Shape	Type	Design List	Material	Design Rule
274	M337	146	88		W10X22	Beam	Wide Flange	A992	Typical
275	125	177	82		Beams 1	Beam	Wide Flange	A992	Typical
276	M430	377	292		Roof Girders	Beam	Wide Flange	A992	Typical
277	M431	377	291		Roof Girders	Beam	Wide Flange	A992	Typical
278	M432	291	280	355	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
279	M433	339	336	355	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
280	M508	358	348	355	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
281	M509	359	349	5	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
282	M510	338	337	5	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
283	M511	292	282	5	Purlins	Beam	CS B-to-B	A653 SS Gr33	Typical
284	M512	201	200		W10X22	Beam	Wide Flange	A992	Typical
285	M513	199	196		W10X22	Beam	Wide Flange	A992	Typical
286	M514	195	194		W10X22	Beam	Wide Flange	A992	Typical
287	M515	193	192		W10X22	Beam	Wide Flange	A992	Typical
288	M516	191	190		W10X22	Beam	Wide Flange	A992	Typical
289	M517	203	188		W10X22	Beam	Wide Flange	A992	Typical
290	M518	187	186		W10X22	Beam	Wide Flange	A992	Typical
291	M519	185	184		W10X22	Beam	Wide Flange	A992	Typical
292	M520	30	41		Beams 2	Beam	Wide Flange	A992	Typical
293	M521	77	78		Beams 2	Beam	Wide Flange	A992	Typical
294	M522	33	30		Horizontal Braces	HBrace	W Tee	A992	Typical
295	M523	30	40		Horizontal Braces	HBrace	W Tee	A992	Typical
296	M525	392	388		HSS4X4X2	Column	Tube	A992	Typical
297	M526	390	386		HSS4X4X2	Column	Tube	A992	Typical
298	M527	391	387		HSS4X4X2	Column	Tube	A992	Typical
299	M528	389	385		HSS4X4X2	Column	Tube	A992	Typical
300	M529	88	169		Horizontal Braces	HBrace	W Tee	A992	Typical
301	M530	392	390		Purlins - Not Used	Beam	Wide Flange	A992	Typical
302	M531	390	389		Purlins - Not Used	Beam	Wide Flange	A992	Typical
303	M532	389	391		Purlins - Not Used	Beam	Wide Flange	A992	Typical
304	M533	391	392		Purlins - Not Used	Beam	Wide Flange	A992	Typical
305	M534	392	395		Vertical Braces	VBrace	Single Angle	A992	Typical
306	M535	395	388		Vertical Braces	VBrace	Single Angle	A992	Typical
307	M536	390	396		Vertical Braces	VBrace	Single Angle	A992	Typical
308	M537	396	386		Vertical Braces	VBrace	Single Angle	A992	Typical
309	M538	391	396		Vertical Braces	VBrace	Single Angle	A992	Typical
310	M539	396	387		Vertical Braces	VBrace	Single Angle	A992	Typical
311	M540	392	397		Vertical Braces	VBrace	Single Angle	A992	Typical
312	M541	397	388		Vertical Braces	VBrace	Single Angle	A992	Typical
313	M464	32	412		Stair Posts	Column	SquareTube	A500 Gr.B Rect	Typical
314	M507	35	411		Stair Posts	Column	SquareTube	A500 Gr.B Rect	Typical
315	M524	40	410		Stair Posts	Column	SquareTube	A500 Gr.B Rect	Typical
316	M542	35	410		Vertical Braces	VBrace	Single Angle	A992	Typical
317	M543	55	130		Horizontal Braces	HBrace	W Tee	A992	Typical
318	M544	88	413		Horizontal Braces	HBrace	W Tee	A992	Typical
319	M545	202	153		Lateral Braces	VBrace	W Tee	A992	Typical
320	M546	150	13	90	Lateral Braces	VBrace	W Tee	A992	Typical
321	M547	153	12	90	Lateral Braces	VBrace	W Tee	A992	Typical
322	M548	208	151		Lateral Braces	VBrace	W Tee	A992	Typical
323	M549	151	14	90	Lateral Braces	VBrace	W Tee	A992	Typical
324	M550	152	11	90	Lateral Braces	VBrace	W Tee	A992	Typical
325	M551	222	152		Lateral Braces	VBrace	W Tee	A992	Typical
326	M552	232	150		Lateral Braces	VBrace	W Tee	A992	Typical
327	M367	418	419		W10X22	Beam	Wide Flange	A992	Typical
328	M368	418	420		W10X22	Beam	Wide Flange	A992	Typical
329	M559	150	9	90	Lateral Braces	VBrace	W Tee	A992	Typical
330	M560	151	10	90	Lateral Braces	VBrace	W Tee	A992	Typical
331	M561	148	11	90	Lateral Braces	VBrace	W Tee	A992	Typical
332	M562	222	148		Lateral Braces	VBrace	W Tee	A992	Typical
333	M563	149	12	90	Lateral Braces	VBrace	W Tee	A992	Typical
334	M564	233	150		Lateral Braces	VBrace	W Tee	A992	Typical
335	M565	204	151		Lateral Braces	VBrace	W Tee	A992	Typical
336	M566	202	149		Lateral Braces	VBrace	W Tee	A992	Typical
337	M370	433	435		W10X22	Beam	Wide Flange	A992	Typical
338	M371	433	434		W10X22	Beam	Wide Flange	A992	Typical
339	M373	438	439		W10X22	Beam	Wide Flange	A992	Typical
340	M374	436	437		W10X22	Beam	Wide Flange	A992	Typical
341	M376	439	441		W10X22	Beam	Wide Flange	A992	Typical
342	M377	437	440		W10X22	Beam	Wide Flange	A992	Typical
343	M406	441	443		W10X22	Beam	Wide Flange	A992	Typical
344	M408	440	442		W10X22	Beam	Wide Flange	A992	Typical
345	M410	443	445		W10X22	Beam	Wide Flange	A992	Typical
346	M411	442	444		W10X22	Beam	Wide Flange	A992	Typical

Node Loads and Enforced Displacements (BLC 1 : Dead)

	Node Label	L, D, M	Direction	Magnitude [(k, k-ft), (in, rad), (k*s²/ft, k*s²*ft)]
1	86	L	Y	-0.15
2	176	L	MZ	-4
3	160	L	MZ	4
4	137	L	MZ	-4
5	136	L	MZ	4
6	389	L	Y	-0.19
7	390	L	Y	-0.19
8	391	L	Y	-0.19
9	392	L	Y	-0.19
10	383	L	Y	-0.21
11	88	L	Y	-0.21
12	34	L	Y	-0.21

Node Loads and Enforced Displacements (BLC 1 : Dead) (Continued)

	Node Label	L, D, M	Direction	Magnitude [(k, k-ft), (in, rad), (k*s²/ft, k*s²*ft)]
13	31	L	Y	-0.21
14	30	L	Y	-0.13
15	35	L	Y	-0.13

Node Loads and Enforced Displacements (BLC 2 : Live)

	Node Label	L, D, M	Direction	Magnitude [(k, k-ft), (in, rad), (k*s²/ft, k*s²*ft)]
1	389	L	Y	-1.22
2	390	L	Y	-1.22
3	391	L	Y	-1.22
4	392	L	Y	-1.22

Node Loads and Enforced Displacements (BLC 5 : Framing Wind East-West)

	Node Label	L, D, M	Direction	Magnitude [(k, k-ft), (in, rad), (k*s²/ft, k*s²*ft)]
1	390	L	X	0.5
2	389	L	X	0.5
3	391	L	X	0.5
4	392	L	X	0.5
5	30	L	X	0.35
6	35	L	X	0.35
7	34	L	X	0.55
8	31	L	X	0.55
9	383	L	X	0.55
10	88	L	X	0.55
11	136	L	X	2
12	160	L	X	2
13	137	L	X	2
14	176	L	X	2

Node Loads and Enforced Displacements (BLC 6 : Framing Wind North-South)

	Node Label	L, D, M	Direction	Magnitude [(k, k-ft), (in, rad), (k*s²/ft, k*s²*ft)]
1	392	L	Z	0.37
2	391	L	Z	0.37
3	390	L	Z	0.37
4	389	L	Z	0.37
5	30	L	Z	0.3
6	35	L	Z	0.3
7	34	L	Z	0.41
8	31	L	Z	0.41
9	383	L	Z	0.41
10	88	L	Z	0.41
11	393	L	Z	0.37
12	394	L	Z	0.37

Node Loads and Enforced Displacements (BLC 8 : Pipe Wind East-West)

	Node Label	L, D, M	Direction	Magnitude [(k, k-ft), (in, rad), (k*s²/ft, k*s²*ft)]
1	209	L	X	0.91
2	220	L	X	0.91
3	213	L	X	0.91
4	224	L	X	0.91
5	217	L	X	0.91
6	215	L	X	0.91
7	237	L	X	1.365
8	234	L	X	1.365
9	232	L	X	1.365
10	222	L	X	1.365
11	233	L	X	1.365
12	221	L	X	1.365
13	218	L	X	1.365
14	212	L	X	1.365
15	211	L	X	1.365
16	58	L	X	0.35
17	62	L	X	0.35
18	56	L	X	0.35
19	65	L	X	0.35
20	54	L	X	0.35
21	66	L	X	0.35
22	68	L	X	0.525
23	70	L	X	0.525
24	72	L	X	0.525
25	73	L	X	0.525
26	75	L	X	0.525
27	64	L	X	0.525
28	49	L	X	0.525
29	47	L	X	0.525
30	45	L	X	0.525
31	211	L	X	1.365
32	212	L	X	1.365
33	222	L	X	1.365
34	232	L	X	1.365
35	233	L	X	1.365
36	233	L	X	1.365
37	221	L	X	1.365
38	221	L	X	1.365

Node Loads and Enforced Displacements (BLC 8 : Pipe Wind East-West) (Continued)

	Node Label	L, D, M	Direction	Magnitude [(k, k-ft), (in, rad), (k*s²/ft, k*s²*ft)]
39	222	L	X	1.365
40	222	L	X	1.365
41	222	L	X	1.365
42	233	L	X	1.365
43	233	L	X	1.365
44	233	L	X	1.365

Node Loads and Enforced Displacements (BLC 10 : Ice Dead Load)

	Node Label	L, D, M	Direction	Magnitude [(k, k-ft), (in, rad), (k*s²/ft, k*s²*ft)]
1	35	L	Y	-0.64
2	30	L	Y	-0.64
3	31	L	Y	-1.02
4	34	L	Y	-1.02
5	88	L	Y	-1.02
6	383	L	Y	-1.02
7	389	L	Y	-0.9
8	390	L	Y	-0.9
9	391	L	Y	-0.9
10	392	L	Y	-0.9

Node Loads and Enforced Displacements (BLC 11 : Snow)

	Node Label	L, D, M	Direction	Magnitude [(k, k-ft), (in, rad), (k*s²/ft, k*s²*ft)]
1	35	L	Y	-0.18
2	30	L	Y	-0.18
3	31	L	Y	-0.28
4	34	L	Y	-0.28
5	88	L	Y	-0.28
6	383	L	Y	-0.28
7	389	L	Y	-0.25
8	390	L	Y	-0.25
9	391	L	Y	-0.25
10	392	L	Y	-0.25

Node Loads and Enforced Displacements (BLC 15 : Earthquake Load East-West)

	Node Label	L, D, M	Direction	Magnitude [(k, k-ft), (in, rad), (k*s²/ft, k*s²*ft)]
1	164	L	X	-0.4
2	162	L	X	-0.76
3	161	L	X	-0.71
4	158	L	X	-0.47
5	156	L	X	-0.39
6	154	L	X	-0.54
7	152	L	X	-0.54
8	150	L	X	-0.54
9	148	L	X	-0.54
10	175	L	X	-0.54
11	180	L	X	-0.54
12	178	L	X	-0.54
13	82	L	X	-0.27
14	229	L	X	-0.81
15	220	L	X	-1.53
16	224	L	X	-1.44
17	215	L	X	-0.96
18	237	L	X	-0.78
19	234	L	X	-1.08
20	232	L	X	-1.08
21	222	L	X	-1.08
22	233	L	X	-1.08
23	221	L	X	-1.08
24	218	L	X	-1.08
25	212	L	X	-1.08
26	211	L	X	-0.54
27	150	L	X	-0.54
28	152	L	X	-0.39
29	222	L	X	-1.08
30	232	L	X	-0.78
31	233	L	X	-1.08
32	175	L	X	-0.27
33	221	L	X	-0.54
34	148	L	X	-0.54
35	150	L	X	-0.54
36	150	L	X	-0.54
37	148	L	X	-0.27
38	148	L	X	-0.54
39	222	L	X	-1.08
40	222	L	X	-1.08
41	233	L	X	-0.54
42	233	L	X	-1.08

Member Point Loads (BLC 1 : Dead)

	Member Label	Direction	Magnitude [k, k-ft]	Location [(ft, %)]
1	M49	y	-1	%50
2	M48	y	-1	%50
3	M174	y	-1	%50

Member Point Loads (BLC 1 : Dead) (Continued)

	Member Label	Direction	Magnitude [k, k-ft]	Location [(ft, %)]
4	M173	y	-1	%50

Member Point Loads (BLC 2 : Live)

	Member Label	Direction	Magnitude [k, k-ft]	Location [(ft, %)]
1	M446	y	-1.22	3.8
2	M337	y	-1.22	3.8
3	M174	y	-0.5	%50
4	M173	y	-0.5	%50
5	M49	y	-0.5	%50
6	M48	y	-0.5	%50
7	M337	y	-1.22	0.5
8	M446	y	-1.22	0.5
9	M461	y	-1.22	%10
10	M461	y	-1.22	%90

Member Distributed Loads (BLC 1 : Dead)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
1	127	Y	-15	-15	0	%100
2	128	Y	-15	-15	0	%100
3	M48	Y	-15	-15	0	%100
4	M49	Y	-15	-15	0	%100
5	M73	Y	-15	-15	0	%100
6	M74	Y	-15	-15	0	%100
7	M98	Y	-15	-15	0	%100
8	M99	Y	-15	-15	0	%100
9	M123	Y	-15	-15	0	%100
10	M124	Y	-15	-15	0	%100
11	M148	Y	-15	-15	0	%100
12	M149	Y	-15	-15	0	%100
13	M173	Y	-15	-15	0	%100
14	M174	Y	-15	-15	0	%100
15	M198	Y	-15	-15	0	%100
16	M199	Y	-15	-15	0	%100
17	M217	Y	-15	-15	0	%100
18	M218	Y	-15	-15	0	%100
19	M228	Y	-15	-15	0	%100
20	M239	Y	-15	-15	0	%100
21	M250	Y	-15	-15	0	%100

Member Distributed Loads (BLC 5 : Framing Wind East-West)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
1	127	X	75	75	0	%100
2	M48	X	75	75	0	%100
3	M73	X	75	75	0	%100
4	M98	X	75	75	0	%100
5	M123	X	75	75	0	%100
6	M148	X	75	75	0	%100
7	M173	X	75	75	0	%100
8	M198	X	75	75	0	%100
9	M217	X	75	75	0	%100
10	M228	X	75	75	0	%100
11	M239	X	75	75	0	%100
12	M456	X	75	75	0	%100
13	M457	X	75	75	0	%100
14	M460	X	75	75	0	%100
15	M495	X	75	75	0	%100
16	M496	X	75	75	0	%100
17	M497	X	75	75	0	%100
18	M218	X	75	75	0	%100
19	M199	X	75	75	0	%100
20	M174	X	75	75	0	%100
21	M149	X	75	75	0	%100
22	M124	X	75	75	0	%100
23	M99	X	75	75	0	%100
24	M74	X	75	75	0	%100
25	M49	X	75	75	0	%100
26	128	X	75	75	0	%100

Member Distributed Loads (BLC 6 : Framing Wind North-South)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
1	125	Z	75	75	0	%100
2	M215	Z	75	75	0	%100
3	M455	Z	75	75	0	%100
4	M459	Z	75	75	0	%100

Member Distributed Loads (BLC 10 : Ice Dead Load)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
1	M399	Y	-43.7	-43.7	0	%100
2	M400	Y	-43.7	-43.7	0	%100
3	M401	Y	-43.7	-43.7	0	%100
4	M90	Y	-50.5	-50.5	0	%100
5	M115	Y	-50.5	-50.5	0	%100

Member Distributed Loads (BLC 10 : Ice Dead Load) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
6	M165	Y	-50.5	-50.5	0	%100
7	M215	Y	-50.5	-50.5	0	%100
8	M227	Y	-50.5	-50.5	0	%100
9	M238	Y	-50.5	-50.5	0	%100
10	M259	Y	-50.5	-50.5	0	%100
11	M261	Y	-50.5	-50.5	0	%100
12	M346	Y	-50.5	-50.5	0	%100
13	M347	Y	-50.5	-50.5	0	%100
14	M366	Y	-50.5	-50.5	0	%100
15	M369	Y	-50.5	-50.5	0	%100
16	M372	Y	-50.5	-50.5	0	%100
17	M375	Y	-50.5	-50.5	0	%100
18	M381	Y	-50.5	-50.5	0	%100
19	M414	Y	-50.5	-50.5	0	%100
20	M467	Y	-50.5	-50.5	0	%100
21	M471	Y	-50.5	-50.5	0	%100
22	125	Y	-50.5	-50.5	0	%100
23	M476	Y	-50.5	-50.5	0	%100
24	M415	Y	-50.5	-50.5	0	%100
25	M473	Y	-50.5	-50.5	0	%100
26	M468	Y	-50.5	-50.5	0	%100
27	M390	Y	-50.5	-50.5	0	%100
28	M466	Y	-50.5	-50.5	0	%100
29	M387	Y	-50.5	-50.5	0	%100
30	M396	Y	-50.5	-50.5	0	%100
31	M470	Y	-50.5	-50.5	0	%100
32	M257	Y	-50.5	-50.5	0	%100
33	M465	Y	-50.5	-50.5	0	%100
34	M249	Y	-50.5	-50.5	0	%100
35	M384	Y	-50.5	-50.5	0	%100
36	M190	Y	-50.5	-50.5	0	%100
37	M469	Y	-50.5	-50.5	0	%100
38	M393	Y	-50.5	-50.5	0	%100
39	M378	Y	-50.5	-50.5	0	%100
40	M65	Y	-50.5	-50.5	0	%100
41	M477	Y	-50.5	-50.5	0	%100
42	M472	Y	-50.5	-50.5	0	%100
43	M140	Y	-50.5	-50.5	0	%100
44	M40	Y	-50.5	-50.5	0	%100
45	90	Y	-50.5	-50.5	0	%100
46	128	Y	-50.5	-50.5	0	%100
47	M48	Y	-50.5	-50.5	0	%100
48	M49	Y	-50.5	-50.5	0	%100
49	M74	Y	-50.5	-50.5	0	%100
50	M99	Y	-50.5	-50.5	0	%100
51	M148	Y	-50.5	-50.5	0	%100
52	M174	Y	-50.5	-50.5	0	%100
53	M198	Y	-50.5	-50.5	0	%100
54	M199	Y	-50.5	-50.5	0	%100
55	M228	Y	-50.5	-50.5	0	%100
56	M392	Y	-50.5	-50.5	0	%100
57	M397	Y	-50.5	-50.5	0	%100
58	M398	Y	-50.5	-50.5	0	%100
59	M281	Y	-50.5	-50.5	0	%100
60	M284	Y	-50.5	-50.5	0	%100
61	M285	Y	-50.5	-50.5	0	%100
62	M404	Y	-50.5	-50.5	0	%100
63	M403	Y	-50.5	-50.5	0	%100
64	M436	Y	-50.5	-50.5	0	%100
65	M457	Y	-50.5	-50.5	0	%100
66	M458	Y	-50.5	-50.5	0	%100
67	M459	Y	-50.5	-50.5	0	%100
68	M460	Y	-50.5	-50.5	0	%100
69	M461	Y	-50.5	-50.5	0	%100
70	M478	Y	-50.5	-50.5	0	%100
71	M481	Y	-50.5	-50.5	0	%100
72	M483	Y	-50.5	-50.5	0	%100
73	M482	Y	-50.5	-50.5	0	%100
74	M484	Y	-50.5	-50.5	0	%100
75	M485	Y	-50.5	-50.5	0	%100
76	M479	Y	-50.5	-50.5	0	%100
77	M262	Y	-50.5	-50.5	0	%100
78	M282	Y	-50.5	-50.5	0	%100
79	M287	Y	-50.5	-50.5	0	%100
80	M480	Y	-50.5	-50.5	0	%100
81	M394	Y	-50.5	-50.5	0	%100
82	M263	Y	-50.5	-50.5	0	%100
83	M264	Y	-50.5	-50.5	0	%100
84	M446	Y	-50.5	-50.5	0	%100
85	M251	Y	-50.5	-50.5	0	%100
86	M250	Y	-50.5	-50.5	0	%100
87	M405	Y	-50.5	-50.5	0	%100
88	M229	Y	-50.5	-50.5	0	%100
89	M395	Y	-50.5	-50.5	0	%100
90	M218	Y	-50.5	-50.5	0	%100
91	M217	Y	-50.5	-50.5	0	%100
92	M289	Y	-50.5	-50.5	0	%100
93	M173	Y	-50.5	-50.5	0	%100

Member Distributed Loads (BLC 10 : Ice Dead Load) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
94	M286	Y	-50.5	-50.5	0	%100
95	M123	Y	-50.5	-50.5	0	%100
96	M283	Y	-50.5	-50.5	0	%100
97	M124	Y	-50.5	-50.5	0	%100
98	M402	Y	-50.5	-50.5	0	%100
99	M98	Y	-50.5	-50.5	0	%100
100	M240	Y	-50.5	-50.5	0	%100
101	M288	Y	-50.5	-50.5	0	%100
102	M73	Y	-50.5	-50.5	0	%100
103	M149	Y	-50.5	-50.5	0	%100
104	M239	Y	-50.5	-50.5	0	%100
105	127	Y	-50.5	-50.5	0	%100
106	M391	Y	-50.5	-50.5	0	%100
107	M520	Y	-50.5	-50.5	0	%100
108	M460	Y	-50.5	-50.5	0	%100
109	M457	Y	-50.5	-50.5	8.2	18.8
110	M459	Y	-50.5	-50.5	1.34	%100
111	M335	Y	-50.5	-50.5	0	%100
112	M336	Y	-50.5	-50.5	0	%100
113	M193	Y	-50.5	-50.5	0	%100
114	M195	Y	-50.5	-50.5	0	%100
115	M520	Y	-50.5	-50.5	0	%100
116	M460	Y	-50.5	-50.5	0	%100
117	M457	Y	-50.5	-50.5	8.2	18.8
118	M459	Y	-50.5	-50.5	1.34	%100
119	M235	Y	-51.8	-51.8	0	%100
120	M234	Y	-51.8	-51.8	0	%100
121	M211	Y	-51.8	-51.8	0	%100
122	M245	Y	-51.8	-51.8	0	%100
123	M212	Y	-51.8	-51.8	0	%100
124	M246	Y	-51.8	-51.8	0	%100
125	M224	Y	-51.8	-51.8	0	%100
126	M223	Y	-51.8	-51.8	0	%100
127	M353	Y	-51.8	-51.8	0	26.351
128	M352	Y	-51.8	-51.8	0	26.351
129	M354	Y	-51.8	-51.8	0	26.351
130	M355	Y	-51.8	-51.8	0	26.351
131	M356	Y	-51.8	-51.8	0	26.351
132	M357	Y	-51.8	-51.8	0	26.351
133	M358	Y	-51.8	-51.8	0	26.351
134	M359	Y	-51.8	-51.8	0	26.351
135	M362	Y	-51.8	-51.8	0	26.351
136	M364	Y	-51.8	-51.8	0	26.351
137	M499	Y	-51.8	-51.8	0	26.351
138	M500	Y	-51.8	-51.8	0	26.351
139	M501	Y	-51.8	-51.8	0	26.351
140	M502	Y	-51.8	-51.8	0	26.351
141	M503	Y	-51.8	-51.8	0	26.351
142	M504	Y	-51.8	-51.8	0	26.351
143	M505	Y	-51.8	-51.8	0	26.351
144	M506	Y	-51.8	-51.8	0	26.351
145	M456	Y	-43.7	-43.7	0	%100
146	M486	Y	-43.7	-43.7	0	%100
147	M337	Y	-43.7	-43.7	0	%100
148	M455	Y	-43.7	-43.7	0	%100
149	M488	Y	-43.7	-43.7	0	%100
150	M487	Y	-43.7	-43.7	0	%100
151	M455	Y	-43.7	-43.7	0	%100
152	M456	Y	-43.7	-43.7	0	%100
153	M462	Y	-43.7	-43.7	0	%100
154	M462	Y	-43.7	-43.7	0	%100
155	M462	Y	-43.7	-43.7	0	%100
156	M455	Y	-43.7	-43.7	0	%100
157	M456	Y	-43.7	-43.7	0	%100
158	M462	Y	-43.7	-43.7	0	%100
159	M280	Y	-44	-44	0	%100
160	M274	Y	-44	-44	0	%100
161	M379	Y	-44	-44	0	%100
162	M268	Y	-44	-44	0	%100
163	M266	Y	-44	-44	0	%100
164	M267	Y	-44	-44	0	%100
165	M271	Y	-44	-44	0	%100
166	M269	Y	-44	-44	0	%100
167	M270	Y	-44	-44	0	%100
168	M278	Y	-44	-44	0	%100
169	M272	Y	-44	-44	0	%100
170	M380	Y	-44	-44	0	%100
171	M279	Y	-44	-44	0	%100
172	M273	Y	-44	-44	0	%100
173	M275	Y	-44	-44	0	%100
174	M382	Y	-44	-44	0	%100
175	M277	Y	-44	-44	0	%100
176	M276	Y	-44	-44	0	%100
177	M386	Y	-44	-44	0	%100
178	M383	Y	-44	-44	0	%100
179	M385	Y	-44	-44	0	%100
180	M260	Y	-44	-44	0	%100
181	M258	Y	-44	-44	0	%100

Member Distributed Loads (BLC 10 : Ice Dead Load) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
182	M265	Y	-44	-44	0	%100
183	M388	Y	-44	-44	0	%100
184	M514	Y	-44	-44	0	%100
185	M513	Y	-44	-44	0	%100
186	M512	Y	-44	-44	0	%100
187	M522	Y	-30.5	-30.5	0	%100
188	M523	Y	-30.5	-30.5	0	%100
189	M529	Y	-30.5	-30.5	0	%100
190	M464	Y	-25.1	-25.1	0	%100
191	M507	Y	-25.1	-25.1	0	%100
192	M524	Y	-25.1	-25.1	0	%100
193	M545	Y	-35.7	-35.37	0	%100
194	M546	Y	-35.7	-35.37	0	%100
195	M547	Y	-35.7	-35.37	0	%100
196	M548	Y	-35.7	-35.37	0	%100
197	M549	Y	-35.7	-35.37	0	%100
198	M550	Y	-35.7	-35.37	0	%100
199	M551	Y	-35.7	-35.37	0	%100
200	M552	Y	-35.7	-35.37	0	%100
201	M559	Y	-35.7	-35.37	0	%100
202	M559	Y	-35.7	-35.37	0	%100
203	M560	Y	-35.7	-35.37	0	%100
204	M560	Y	-35.7	-35.37	0	%100
205	M561	Y	-35.7	-35.37	0	%100
206	M561	Y	-35.7	-35.37	0	%100
207	M562	Y	-35.7	-35.37	0	%100
208	M562	Y	-35.7	-35.37	0	%100
209	M563	Y	-35.7	-35.37	0	%100
210	M563	Y	-35.7	-35.37	0	%100
211	M564	Y	-35.7	-35.37	0	%100
212	M564	Y	-35.7	-35.37	0	%100
213	M565	Y	-35.7	-35.37	0	%100
214	M565	Y	-35.7	-35.37	0	%100
215	M566	Y	-35.7	-35.37	0	%100
216	M566	Y	-35.7	-35.37	0	%100
217	M368	Y	-44	-44	0	%100
218	M367	Y	-44	-44	0	%100
219	M370	Y	-44	-44	0	%100
220	M371	Y	-44	-44	0	%100
221	M373	Y	-44	-44	0	%100
222	M374	Y	-44	-44	0	%100
223	M376	Y	-44	-44	0	%100
224	M377	Y	-44	-44	0	%100
225	M406	Y	-44	-44	0	%100
226	M408	Y	-44	-44	0	%100
227	M410	Y	-44	-44	0	%100
228	M411	Y	-44	-44	0	%100
229	M409	Y	-25.1	-25.1	0	%100
230	M418	Y	-25.1	-25.1	0	%100
231	M453	Y	-25.1	-25.1	0	%100
232	M463	Y	-25.1	-25.1	0	%100
233	M442	Y	-25.1	-25.1	0	%100
234	M454	Y	-25.1	-25.1	0	%100
235	M449	Y	-25.1	-25.1	0	%100
236	M452	Y	-25.1	-25.1	0	%100
237	M389	Y	-25.1	-25.1	0	%100
238	M416	Y	-25.1	-25.1	0	%100
239	M417	Y	-25.1	-25.1	0	%100
240	M407	Y	-25.1	-25.1	0	%100
241	M421	Y	-25.1	-25.1	0	%100
242	M489	Y	-25.1	-25.1	0	%100
243	M426	Y	-25.1	-25.1	0	%100
244	M443	Y	-25.1	-25.1	0	%100
245	M451	Y	-25.1	-25.1	0	%100
246	M419	Y	-25.1	-25.1	0	%100
247	M422	Y	-25.1	-25.1	0	%100
248	M447	Y	-25.1	-25.1	0	%100
249	M475	Y	-25.1	-25.1	0	%100
250	M425	Y	-25.1	-25.1	0	%100
251	M423	Y	-25.1	-25.1	0	%100
252	M427	Y	-25.1	-25.1	0	%100
253	M490	Y	-25.1	-25.1	0	%100
254	M445	Y	-25.1	-25.1	0	%100
255	M474	Y	-25.1	-25.1	0	%100
256	M424	Y	-25.1	-25.1	0	%100
257	M543	Y	-25.1	-25.1	0	%100
258	M544	Y	-25.1	-25.1	0	%100

Member Distributed Loads (BLC 16 : Framing Thermal)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
1	M195	T	100	100	0	%100
2	M338	T	100	100	0	%100
3	M149	T	100	100	0	%100
4	M325	T	100	100	0	%100
5	M312	T	100	100	0	%100
6	M260	T	100	100	0	%100
7	M427	T	100	100	0	%100
8	M302	T	100	100	0	%100

Member Distributed Loads (BLC 16 : Framing Thermal) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
9	M318	T	100	100	0	%100
10	M193	T	100	100	0	%100
11	M474	T	100	100	0	%100
12	M294	T	100	100	0	%100
13	M290	T	100	100	0	%100
14	M324	T	100	100	0	%100
15	105	T	100	100	0	%100
16	M307	T	100	100	0	%100
17	M284	T	100	100	0	%100
18	M332	T	100	100	0	%100
19	M298	T	100	100	0	%100
20	M365	T	100	100	0	%100
21	M526	T	100	100	0	%100
22	M361	T	100	100	0	%100
23	M170	T	100	100	0	%100
24	M483	T	100	100	0	%100
25	107	T	100	100	0	%100
26	M43	T	100	100	0	%100
27	M319	T	100	100	0	%100
28	M254	T	100	100	0	%100
29	M460	T	100	100	0	%100
30	M336	T	100	100	0	%100
31	M537	T	100	100	0	%100
32	M470	T	100	100	0	%100
33	M145	T	100	100	0	%100
34	M282	T	100	100	0	%100
35	M143	T	100	100	0	%100
36	M329	T	100	100	0	%100
37	M316	T	100	100	0	%100
38	M291	T	100	100	0	%100
39	M475	T	100	100	0	%100
40	M335	T	100	100	0	%100
41	M45	T	100	100	0	%100
42	M520	T	100	100	0	%100
43	M364	T	100	100	0	%100
44	M310	T	100	100	0	%100
45	M229	T	100	100	0	%100
46	M311	T	100	100	0	%100
47	M52	T	100	100	0	%100
48	M292	T	100	100	0	%100
49	M250	T	100	100	0	%100
50	M314	T	100	100	0	%100
51	M534	T	100	100	0	%100
52	M403	T	100	100	0	%100
53	M301	T	100	100	0	%100
54	M256	T	100	100	0	%100
55	M464	T	100	100	0	%100
56	M425	T	100	100	0	%100
57	M315	T	100	100	0	%100
58	M334	T	100	100	0	%100
59	M407	T	100	100	0	%100
60	M320	T	100	100	0	%100
61	M322	T	100	100	0	%100
62	M304	T	100	100	0	%100
63	M328	T	100	100	0	%100
64	M381	T	100	100	0	%100
65	M313	T	100	100	0	%100
66	M388	T	100	100	0	%100
67	M300	T	100	100	0	%100
68	M347	T	100	100	0	%100
69	M327	T	100	100	0	%100
70	M330	T	100	100	0	%100
71	M331	T	100	100	0	%100
72	M333	T	100	100	0	%100
73	M323	T	100	100	0	%100
74	M259	T	100	100	0	%100
75	M309	T	100	100	0	%100
76	M514	T	100	100	0	%100
77	M326	T	100	100	0	%100
78	M297	T	100	100	0	%100
79	M306	T	100	100	0	%100
80	M529	T	100	100	0	%100
81	M415	T	100	100	0	%100
82	M253	T	100	100	0	%100
83	M305	T	100	100	0	%100
84	M276	T	100	100	0	%100
85	M303	T	100	100	0	%100
86	M519	T	100	100	0	%100
87	M296	T	100	100	0	%100
88	M299	T	100	100	0	%100
89	M293	T	100	100	0	%100
90	M255	T	100	100	0	%100
91	M308	T	100	100	0	%100
92	M50	T	100	100	0	%100
93	M340	T	100	100	0	%100
94	M321	T	100	100	0	%100
95	M286	T	100	100	0	%100
96	M317	T	100	100	0	%100

Member Distributed Loads (BLC 16 : Framing Thermal) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
97	M471	T	100	100	0	%100
98	M465	T	100	100	0	%100
99	M360	T	100	100	0	%100
100	M363	T	100	100	0	%100
101	M295	T	100	100	0	%100
102	M177	T	100	100	0	%100
103	M175	T	100	100	0	%100
104	M341	T	100	100	0	%100
105	M342	T	100	100	0	%100
106	M218	T	100	100	0	%100
107	M343	T	100	100	0	%100
108	M459	T	100	100	0	%100
109	M344	T	100	100	0	%100
110	M345	T	100	100	0	%100
111	M375	T	100	100	0	%100
112	M350	T	100	100	0	%100
113	M351	T	100	100	0	%100
114	M339	T	100	100	0	%100
115	M523	T	100	100	0	%100
116	M426	T	100	100	0	%100
117	M424	T	100	100	0	%100
118	M423	T	100	100	0	%100
119	M507	T	100	100	0	%100
120	M422	T	100	100	0	%100
121	M421	T	100	100	0	%100
122	M123	T	100	100	0	%100
123	M416	T	100	100	0	%100
124	M528	T	100	100	0	%100
125	M418	T	100	100	0	%100
126	M417	T	100	100	0	%100
127	M409	T	100	100	0	%100
128	M389	T	100	100	0	%100
129	M419	T	100	100	0	%100
130	M506	T	100	100	0	%100
131	M490	T	100	100	0	%100
132	M489	T	100	100	0	%100
133	M261	T	100	100	0	%100
134	M463	T	100	100	0	%100
135	M454	T	100	100	0	%100
136	M393	T	100	100	0	%100
137	M452	T	100	100	0	%100
138	M451	T	100	100	0	%100
139	M509	T	100	100	0	%100
140	M449	T	100	100	0	%100
141	M479	T	100	100	0	%100
142	M447	T	100	100	0	%100
143	M445	T	100	100	0	%100
144	M443	T	100	100	0	%100
145	M442	T	100	100	0	%100
146	M397	T	100	100	0	%100
147	128	T	100	100	0	%100
148	M468	T	100	100	0	%100
149	M453	T	100	100	0	%100
150	M269	T	100	100	0	%100
151	M258	T	100	100	0	%100
152	M265	T	100	100	0	%100
153	M266	T	100	100	0	%100
154	125	T	100	100	0	%100
155	M267	T	100	100	0	%100
156	M268	T	100	100	0	%100
157	M270	T	100	100	0	%100
158	M271	T	100	100	0	%100
159	M272	T	100	100	0	%100
160	M273	T	100	100	0	%100
161	M505	T	100	100	0	%100
162	M275	T	100	100	0	%100
163	M277	T	100	100	0	%100
164	M278	T	100	100	0	%100
165	M372	T	100	100	0	%100
166	M279	T	100	100	0	%100
167	M380	T	100	100	0	%100
168	M477	T	100	100	0	%100
169	M382	T	100	100	0	%100
170	M383	T	100	100	0	%100
171	M385	T	100	100	0	%100
172	M379	T	100	100	0	%100
173	M228	T	100	100	0	%100
174	M386	T	100	100	0	%100
175	M280	T	100	100	0	%100
176	M274	T	100	100	0	%100
177	M234	T	100	100	0	%100
178	M357	T	100	100	0	%100
179	M458	T	100	100	0	%100
180	M394	T	100	100	0	%100
181	M358	T	100	100	0	%100
182	M476	T	100	100	0	%100
183	M359	T	100	100	0	%100
184	127	T	100	100	0	%100

Member Distributed Loads (BLC 16 : Framing Thermal) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
185	M485	T	100	100	0	%100
186	M362	T	100	100	0	%100
187	M356	T	100	100	0	%100
188	M500	T	100	100	0	%100
189	M432	T	100	100	0	%100
190	M501	T	100	100	0	%100
191	M249	T	100	100	0	%100
192	M502	T	100	100	0	%100
193	M503	T	100	100	0	%100
194	M199	T	100	100	0	%100
195	M504	T	100	100	0	%100
196	M499	T	100	100	0	%100
197	M355	T	100	100	0	%100
198	M354	T	100	100	0	%100
199	M235	T	100	100	0	%100
200	M211	T	100	100	0	%100
201	M212	T	100	100	0	%100
202	M245	T	100	100	0	%100
203	M223	T	100	100	0	%100
204	M224	T	100	100	0	%100
205	M436	T	100	100	0	%100
206	M246	T	100	100	0	%100
207	M352	T	100	100	0	%100
208	M369	T	100	100	0	%100
209	M353	T	100	100	0	%100
210	M462	T	100	100	0	%100
211	M497	T	100	100	0	%100
212	M496	T	100	100	0	%100
213	M392	T	100	100	0	%100
214	M495	T	100	100	0	%100
215	M494	T	100	100	0	%100
216	M498	T	100	100	0	%100
217	M430	T	100	100	0	%100
218	M400	T	100	100	0	%100
219	M481	T	100	100	0	%100
220	M238	T	100	100	0	%100
221	M482	T	100	100	0	%100
222	M478	T	100	100	0	%100
223	M484	T	100	100	0	%100
224	M473	T	100	100	0	%100
225	M480	T	100	100	0	%100
226	M48	T	100	100	0	%100
227	M124	T	100	100	0	%100
228	M99	T	100	100	0	%100
229	M98	T	100	100	0	%100
230	M73	T	100	100	0	%100
231	M74	T	100	100	0	%100
232	M287	T	100	100	0	%100
233	M404	T	100	100	0	%100
234	M288	T	100	100	0	%100
235	M289	T	100	100	0	%100
236	M402	T	100	100	0	%100
237	M190	T	100	100	0	%100
238	M148	T	100	100	0	%100
239	M405	T	100	100	0	%100
240	M391	T	100	100	0	%100
241	M198	T	100	100	0	%100
242	M174	T	100	100	0	%100
243	M173	T	100	100	0	%100
244	M49	T	100	100	0	%100
245	M283	T	100	100	0	%100
246	M251	T	100	100	0	%100
247	M240	T	100	100	0	%100
248	M239	T	100	100	0	%100
249	M281	T	100	100	0	%100
250	M264	T	100	100	0	%100
251	M262	T	100	100	0	%100
252	M398	T	100	100	0	%100
253	M395	T	100	100	0	%100
254	M217	T	100	100	0	%100
255	M285	T	100	100	0	%100
256	M263	T	100	100	0	%100
257	M540	T	100	100	0	%100
258	M215	T	100	100	0	%100
259	M543	T	100	100	0	%100
260	M384	T	100	100	0	%100
261	M366	T	100	100	0	%100
262	M488	T	100	100	0	%100
263	M390	T	100	100	0	%100
264	M346	T	100	100	0	%100
265	M378	T	100	100	0	%100
266	M466	T	100	100	0	%100
267	M467	T	100	100	0	%100
268	M401	T	100	100	0	%100
269	M40	T	100	100	0	%100
270	M399	T	100	100	0	%100
271	M396	T	100	100	0	%100
272	M227	T	100	100	0	%100

Member Distributed Loads (BLC 16 : Framing Thermal) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
273	M456	T	100	100	0	%100
274	M455	T	100	100	0	%100
275	M387	T	100	100	0	%100
276	M487	T	100	100	0	%100
277	M472	T	100	100	0	%100
278	M469	T	100	100	0	%100
279	M446	T	100	100	0	%100
280	M115	T	100	100	0	%100
281	M414	T	100	100	0	%100
282	M486	T	100	100	0	%100
283	M140	T	100	100	0	%100
284	90	T	100	100	0	%100
285	M535	T	100	100	0	%100
286	M165	T	100	100	0	%100
287	M461	T	100	100	0	%100
288	M257	T	100	100	0	%100
289	M457	T	100	100	0	%100
290	M65	T	100	100	0	%100
291	M90	T	100	100	0	%100
292	M337	T	100	100	0	%100
293	M431	T	100	100	0	%100
294	M433	T	100	100	0	%100
295	M508	T	100	100	0	%100
296	M510	T	100	100	0	%100
297	M511	T	100	100	0	%100
298	M512	T	100	100	0	%100
299	M513	T	100	100	0	%100
300	M515	T	100	100	0	%100
301	M516	T	100	100	0	%100
302	M517	T	100	100	0	%100
303	M518	T	100	100	0	%100
304	M521	T	100	100	0	%100
305	M522	T	100	100	0	%100
306	M525	T	100	100	0	%100
307	M527	T	100	100	0	%100
308	M530	T	100	100	0	%100
309	M531	T	100	100	0	%100
310	M532	T	100	100	0	%100
311	M533	T	100	100	0	%100
312	M536	T	100	100	0	%100
313	M538	T	100	100	0	%100
314	M539	T	100	100	0	%100
315	M541	T	100	100	0	%100
316	M524	T	100	100	0	%100
317	M542	T	100	100	0	%100
318	M544	T	100	100	0	%100
319	M545	T	100	100	0	%100
320	M546	T	100	100	0	%100
321	M547	T	100	100	0	%100
322	M548	T	100	100	0	%100
323	M549	T	100	100	0	%100
324	M550	T	100	100	0	%100
325	M551	T	100	100	0	%100
326	M552	T	100	100	0	%100
327	M559	T	100	100	0	%100
328	M560	T	100	100	0	%100
329	M561	T	100	100	0	%100
330	M562	T	100	100	0	%100
331	M563	T	100	100	0	%100
332	M564	T	100	100	0	%100
333	M565	T	100	100	0	%100
334	M566	T	100	100	0	%100
335	M368	T	100	100	0	%100
336	M367	T	100	100	0	%100
337	M370	T	100	100	0	%100
338	M371	T	100	100	0	%100
339	M373	T	100	100	0	%100
340	M374	T	100	100	0	%100
341	M376	T	100	100	0	%100
342	M377	T	100	100	0	%100
343	M406	T	100	100	0	%100
344	M408	T	100	100	0	%100
345	M410	T	100	100	0	%100
346	M411	T	100	100	0	%100

Member Distributed Loads (BLC 17 : BLC 1 Transient Area Loads)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
1	M461	Y	-18.35	-18.35	0	4.64
2	M459	Y	-18.35	-18.35	5.09	9.73
3	M462	Y	-19.532	-19.532	0	2.049
4	M462	Y	-19.532	-19.532	2.049	4.097
5	M462	Y	-19.532	-19.532	4.097	6.146
6	M462	Y	-19.532	-19.532	6.146	8.194
7	M462	Y	-19.532	-18.164	8.194	10.243
8	M462	Y	-18.164	-16.795	10.243	12.291
9	M462	Y	-16.795	-16.795	12.291	14.34
10	M457	Y	-5.081	-13.973	6.75	9
11	M457	Y	-13.973	-18.418	9	11.25

Member Distributed Loads (BLC 17 : BLC 1 Transient Area Loads) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
12	M457	Y	-18.418	-18.418	11.25	13.5
13	M457	Y	-18.418	-18.418	13.5	15.75
14	M457	Y	-18.418	-21.618	15.75	18
15	M457	Y	-21.618	-20.373	18	20.25
16	M457	Y	-20.373	-11.481	20.25	22.5
17	M384	Y	-430	-430	0	13
18	M366	Y	-600	-600	1.443e-15	13
19	M369	Y	-600	-600	5.329e-15	13
20	M372	Y	-600	-600	4.441e-15	13
21	M375	Y	-600	-600	6.55e-15	13
22	M381	Y	-600	-600	0	13
23	M390	Y	-400	-400	5.551e-16	13
24	M346	Y	-600	-600	0	13
25	M378	Y	-600	-600	0	13
26	M347	Y	-300	-300	5.551e-16	13
27	M401	Y	-400	-400	5.551e-16	13
28	M400	Y	-400	-400	5.551e-16	13
29	M399	Y	-450	-450	5.551e-16	13
30	M396	Y	-225	-225	2.22e-16	13
31	M393	Y	-425	-425	5.551e-16	13
32	M387	Y	-330	-330	5.551e-16	13
33	M455	Y	-15.85	-15.85	0	4.33
34	M337	Y	-15.85	-15.85	0	4.33
35	M269	Y	-93.75	-93.75	2.22e-16	13
36	M258	Y	-93.75	-93.75	2.22e-16	13
37	M260	Y	-93.75	-93.75	2.22e-16	13
38	M265	Y	-93.75	-93.75	2.22e-16	13
39	M266	Y	-93.75	-93.75	2.22e-16	13
40	M267	Y	-93.75	-93.75	2.22e-16	13
41	M268	Y	-93.75	-93.75	2.22e-16	13
42	M270	Y	-93.75	-93.75	2.22e-16	13
43	M271	Y	-93.75	-93.75	2.22e-16	13
44	M272	Y	-93.75	-93.75	2.22e-16	13
45	M273	Y	-93.75	-93.75	2.22e-16	13
46	M275	Y	-93.75	-93.75	2.22e-16	13
47	M276	Y	-93.75	-93.75	2.22e-16	13
48	M277	Y	-93.75	-93.75	2.22e-16	13
49	M278	Y	-93.75	-93.75	2.22e-16	13
50	M279	Y	-93.75	-93.75	2.22e-16	13
51	M380	Y	-93.75	-93.75	2.22e-16	13
52	M382	Y	-93.75	-93.75	2.22e-16	13
53	M383	Y	-93.75	-93.75	2.22e-16	13
54	M385	Y	-93.75	-93.75	2.22e-16	13
55	M379	Y	-93.75	-93.75	2.22e-16	13
56	M386	Y	-93.75	-93.75	2.22e-16	13
57	M388	Y	-81.25	-81.25	5.551e-16	13
58	M280	Y	-93.75	-93.75	2.22e-16	13
59	M274	Y	-93.75	-93.75	2.22e-16	13
60	M215	Y	-40.625	-40.625	5.551e-16	13
61	M40	Y	-93.75	-93.75	2.22e-16	13
62	M115	Y	-93.75	-93.75	2.22e-16	13
63	M190	Y	-87.5	-87.5	5.551e-16	13
64	M140	Y	-93.75	-93.75	2.22e-16	13
65	90	Y	-93.75	-93.75	2.22e-16	13
66	M165	Y	-93.75	-93.75	2.22e-16	13
67	M65	Y	-93.75	-93.75	2.22e-16	13
68	M90	Y	-93.75	-93.75	2.22e-16	13
69	125	Y	-46.875	-46.875	6.661e-16	13
70	M497	Y	-15	-15	1	9
71	M496	Y	-15	-15	1	9
72	M495	Y	-15	-15	1	9
73	M494	Y	-21.632	-10.249	1	4
74	M494	Y	-10.249	-5.674	4	7
75	M494	Y	-5.674	-7.907	7	10
76	M498	Y	-19.308	-19.308	0	3
77	M239	Y	-8.063	-13.313	0	2.857
78	M239	Y	-13.313	-15.938	2.857	5.714
79	M239	Y	-15.938	-10.143	5.714	8.571
80	M239	Y	-10.143	-4.348	8.571	11.429
81	M239	Y	-4.348	-8.244	11.429	14.286
82	M239	Y	-8.244	-12.14	14.286	17.143
83	M239	Y	-12.14	-12.14	17.143	20
84	M228	Y	-13.333	-13.333	1	19
85	M215	Y	-10	-10	0	3
86	M261	Y	-20	-20	0	3
87	M227	Y	-20	-20	0	3
88	M238	Y	-11.383	-11.383	0	3
89	M259	Y	-17.925	-17.925	2.776e-15	3
90	M498	Y	-75	-75	4.663e-15	3
91	M215	Y	-36.058	-70.673	0	2.167
92	M215	Y	-70.673	-79.327	2.167	4.333
93	M215	Y	-79.327	-87.981	4.333	6.5
94	M215	Y	-87.981	-87.981	6.5	8.667
95	M215	Y	-87.981	-70.673	8.667	10.833
96	M215	Y	-70.673	-70.673	10.833	13
97	M261	Y	-72.115	-141.346	0	2.167
98	M261	Y	-141.346	-158.654	2.167	4.333
99	M261	Y	-158.654	-175.962	4.333	6.5

Member Distributed Loads (BLC 17 : BLC 1 Transient Area Loads) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
100	M261	Y	-175.962	-175.962	6.5	8.667
101	M261	Y	-175.962	-141.346	8.667	10.833
102	M261	Y	-141.346	-141.346	10.833	13
103	M227	Y	-72.115	-141.346	0	2.167
104	M227	Y	-141.346	-158.654	2.167	4.333
105	M227	Y	-158.654	-175.962	4.333	6.5
106	M227	Y	-175.962	-175.962	6.5	8.667
107	M227	Y	-175.962	-141.346	8.667	10.833
108	M227	Y	-141.346	-141.346	10.833	13
109	M238	Y	-124.932	-139.962	0	2.167
110	M238	Y	-139.962	-155.868	2.167	4.333
111	M238	Y	-155.868	-176.209	4.333	6.5
112	M238	Y	-176.209	-169.284	6.5	8.667
113	M238	Y	-169.284	-149.976	8.667	10.833
114	M238	Y	-149.976	-149.108	10.833	13
115	M259	Y	-35.015	-107.777	0	2.167
116	M259	Y	-107.777	-140.318	2.167	4.333
117	M259	Y	-140.318	-172.154	4.333	6.5
118	M259	Y	-172.154	-174.382	6.5	8.667
119	M259	Y	-174.382	-141.152	8.667	10.833
120	M259	Y	-141.152	-141.59	10.833	13
121	M257	Y	-170.876	-170.305	0	2.167
122	M257	Y	-170.305	-169.517	2.167	4.333
123	M257	Y	-169.517	-168.608	4.333	6.5
124	M257	Y	-168.608	-167.838	6.5	8.667
125	M257	Y	-167.838	-167.308	8.667	10.833
126	M257	Y	-167.308	-166.972	10.833	13
127	M249	Y	-84.375	-84.375	2.22e-16	13
128	M488	Y	-400	-400	9.992e-16	13
129	M468	Y	-300	-300	6.661e-16	13
130	M466	Y	-300	-300	6.661e-16	13
131	M467	Y	-300	-300	6.661e-16	13
132	M465	Y	-280	-280	5.551e-16	13
133	M487	Y	-400	-400	9.992e-16	13
134	M477	Y	-330	-330	9.992e-16	13
135	M476	Y	-400	-400	9.992e-16	13
136	M473	Y	-150	-150	6.661e-16	13
137	M472	Y	-300	-300	6.661e-16	13
138	M471	Y	-300	-300	6.661e-16	13
139	M470	Y	-300	-300	6.661e-16	13
140	M469	Y	-300	-300	6.661e-16	13
141	M415	Y	-425	-425	2.22e-16	13
142	M414	Y	-225	-225	5.551e-16	13
143	M486	Y	-450	-450	5.551e-16	13
144	M512	Y	-300	-300	6.661e-16	13
145	M513	Y	-300	-300	6.661e-16	13
146	M514	Y	-300	-300	6.661e-16	13
147	M515	Y	-300	-300	6.661e-16	13
148	M516	Y	-300	-300	6.661e-16	13
149	M517	Y	-300	-300	6.661e-16	13
150	M518	Y	-300	-300	6.661e-16	13
151	M519	Y	-300	-300	6.661e-16	13
152	M340	Y	-13.05	-13.05	4.219e-15	15
153	M50	Y	-6.72	-6.72	4.219e-15	15
154	M344	Y	-7.653	-8.951	0	2.143
155	M344	Y	-8.951	-8.733	2.143	4.286
156	M344	Y	-8.733	-8.109	4.286	6.429
157	M344	Y	-8.109	-8.119	6.429	8.571
158	M344	Y	-8.119	-8.758	8.571	10.714
159	M344	Y	-8.758	-8.964	10.714	12.857
160	M344	Y	-8.964	-7.634	12.857	15
161	M345	Y	-13.32	-13.32	3.442e-15	15
162	M350	Y	-11.998	-12.979	0	2.143
163	M350	Y	-12.979	-13.449	2.143	4.286
164	M350	Y	-13.449	-14.092	4.286	6.429
165	M350	Y	-14.092	-14.083	6.429	8.571
166	M350	Y	-14.083	-13.426	8.571	10.714
167	M350	Y	-13.426	-12.954	10.714	12.857
168	M350	Y	-12.954	-11.983	12.857	15
169	M351	Y	-12.9	-12.9	3.997e-15	15
170	M297	Y	-13.32	-13.32	2.82e-14	15
171	M302	Y	-12.9	-12.9	0	15
172	M301	Y	-11.998	-12.979	0	2.143
173	M301	Y	-12.979	-13.449	2.143	4.286
174	M301	Y	-13.449	-14.092	4.286	6.429
175	M301	Y	-14.092	-14.083	6.429	8.571
176	M301	Y	-14.083	-13.426	8.571	10.714
177	M301	Y	-13.426	-12.954	10.714	12.857
178	M301	Y	-12.954	-11.983	12.857	15
179	M300	Y	-13.05	-13.05	1.921e-14	15
180	M296	Y	-7.653	-8.951	0	2.143
181	M296	Y	-8.951	-8.733	2.143	4.286
182	M296	Y	-8.733	-8.109	4.286	6.429
183	M296	Y	-8.109	-8.119	6.429	8.571
184	M296	Y	-8.119	-8.758	8.571	10.714
185	M296	Y	-8.758	-8.964	10.714	12.857
186	M296	Y	-8.964	-7.634	12.857	15
187	M175	Y	-6.72	-6.72	9.881e-15	15

Member Distributed Loads (BLC 17 : BLC 1 Transient Area Loads) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
188	M303	Y	-8.224	-11.494	0	2.143
189	M303	Y	-11.494	-12.208	2.143	4.286
190	M303	Y	-12.208	-10.85	4.286	6.429
191	M303	Y	-10.85	-10.856	6.429	8.571
192	M303	Y	-10.856	-12.214	8.571	10.714
193	M303	Y	-12.214	-11.492	10.714	12.857
194	M303	Y	-11.492	-8.219	12.857	15
195	M360	Y	-6.178	-8.563	0	2.143
196	M360	Y	-8.563	-8.455	2.143	4.286
197	M360	Y	-8.455	-7.17	4.286	6.429
198	M360	Y	-7.17	-7.169	6.429	8.571
199	M360	Y	-7.169	-8.456	8.571	10.714
200	M360	Y	-8.456	-8.566	10.714	12.857
201	M360	Y	-8.566	-6.179	12.857	15
202	M363	Y	-5.401	-5.462	0	2.143
203	M363	Y	-5.462	-5.671	2.143	4.286
204	M363	Y	-5.671	-6.424	4.286	6.429
205	M363	Y	-6.424	-6.423	6.429	8.571
206	M363	Y	-6.423	-5.663	8.571	10.714
207	M363	Y	-5.663	-5.453	10.714	12.857
208	M363	Y	-5.453	-5.4	12.857	15
209	M311	Y	-6.178	-8.563	0	2.143
210	M311	Y	-8.563	-8.455	2.143	4.286
211	M311	Y	-8.455	-7.17	4.286	6.429
212	M311	Y	-7.17	-7.169	6.429	8.571
213	M311	Y	-7.169	-8.456	8.571	10.714
214	M311	Y	-8.456	-8.566	10.714	12.857
215	M311	Y	-8.566	-6.179	12.857	15
216	M314	Y	-8.224	-11.494	0	2.143
217	M314	Y	-11.494	-12.208	2.143	4.286
218	M314	Y	-12.208	-10.85	4.286	6.429
219	M314	Y	-10.85	-10.856	6.429	8.571
220	M314	Y	-10.856	-12.214	8.571	10.714
221	M314	Y	-12.214	-11.492	10.714	12.857
222	M314	Y	-11.492	-8.219	12.857	15
223	M320	Y	-6.178	-8.563	0	2.143
224	M320	Y	-8.563	-8.455	2.143	4.286
225	M320	Y	-8.455	-7.17	4.286	6.429
226	M320	Y	-7.17	-7.169	6.429	8.571
227	M320	Y	-7.169	-8.456	8.571	10.714
228	M320	Y	-8.456	-8.566	10.714	12.857
229	M320	Y	-8.566	-6.179	12.857	15
230	M322	Y	-8.224	-11.494	0	2.143
231	M322	Y	-11.494	-12.208	2.143	4.286
232	M322	Y	-12.208	-10.85	4.286	6.429
233	M322	Y	-10.85	-10.856	6.429	8.571
234	M322	Y	-10.856	-12.214	8.571	10.714
235	M322	Y	-12.214	-11.492	10.714	12.857
236	M322	Y	-11.492	-8.219	12.857	15
237	M328	Y	-8.224	-11.494	0	2.143
238	M328	Y	-11.494	-12.208	2.143	4.286
239	M328	Y	-12.208	-10.85	4.286	6.429
240	M328	Y	-10.85	-10.856	6.429	8.571
241	M328	Y	-10.856	-12.214	8.571	10.714
242	M328	Y	-12.214	-11.492	10.714	12.857
243	M328	Y	-11.492	-8.219	12.857	15
244	M327	Y	-6.178	-8.563	0	2.143
245	M327	Y	-8.563	-8.455	2.143	4.286
246	M327	Y	-8.455	-7.17	4.286	6.429
247	M327	Y	-7.17	-7.169	6.429	8.571
248	M327	Y	-7.169	-8.456	8.571	10.714
249	M327	Y	-8.456	-8.566	10.714	12.857
250	M327	Y	-8.566	-6.179	12.857	15
251	M308	Y	-8.224	-11.494	0	2.143
252	M308	Y	-11.494	-12.208	2.143	4.286
253	M308	Y	-12.208	-10.85	4.286	6.429
254	M308	Y	-10.85	-10.856	6.429	8.571
255	M308	Y	-10.856	-12.214	8.571	10.714
256	M308	Y	-12.214	-11.492	10.714	12.857
257	M308	Y	-11.492	-8.219	12.857	15
258	M307	Y	-6.178	-8.563	0	2.143
259	M307	Y	-8.563	-8.455	2.143	4.286
260	M307	Y	-8.455	-7.17	4.286	6.429
261	M307	Y	-7.17	-7.169	6.429	8.571
262	M307	Y	-7.169	-8.456	8.571	10.714
263	M307	Y	-8.456	-8.566	10.714	12.857
264	M307	Y	-8.566	-6.179	12.857	15
265	M313	Y	-5.401	-5.462	0	2.143
266	M313	Y	-5.462	-5.671	2.143	4.286
267	M313	Y	-5.671	-6.424	4.286	6.429
268	M313	Y	-6.424	-6.423	6.429	8.571
269	M313	Y	-6.423	-5.663	8.571	10.714
270	M313	Y	-5.663	-5.453	10.714	12.857
271	M313	Y	-5.453	-5.4	12.857	15
272	M321	Y	-5.401	-5.462	0	2.143
273	M321	Y	-5.462	-5.671	2.143	4.286
274	M321	Y	-5.671	-6.424	4.286	6.429
275	M321	Y	-6.424	-6.423	6.429	8.571

Member Distributed Loads (BLC 17 : BLC 1 Transient Area Loads) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
276	M321	Y	-6.423	-5.663	8.571	10.714
277	M321	Y	-5.663	-5.453	10.714	12.857
278	M321	Y	-5.453	-5.4	12.857	15
279	M326	Y	-5.401	-5.462	0	2.143
280	M326	Y	-5.462	-5.671	2.143	4.286
281	M326	Y	-5.671	-6.424	4.286	6.429
282	M326	Y	-6.424	-6.423	6.429	8.571
283	M326	Y	-6.423	-5.663	8.571	10.714
284	M326	Y	-5.663	-5.453	10.714	12.857
285	M326	Y	-5.453	-5.4	12.857	15
286	M295	Y	-5.401	-5.462	0	2.143
287	M295	Y	-5.462	-5.671	2.143	4.286
288	M295	Y	-5.671	-6.424	4.286	6.429
289	M295	Y	-6.424	-6.423	6.429	8.571
290	M295	Y	-6.423	-5.663	8.571	10.714
291	M295	Y	-5.663	-5.453	10.714	12.857
292	M295	Y	-5.453	-5.4	12.857	15
293	M333	Y	-6.178	-8.563	0	2.143
294	M333	Y	-8.563	-8.455	2.143	4.286
295	M333	Y	-8.455	-7.17	4.286	6.429
296	M333	Y	-7.17	-7.169	6.429	8.571
297	M333	Y	-7.169	-8.456	8.571	10.714
298	M333	Y	-8.456	-8.566	10.714	12.857
299	M333	Y	-8.566	-6.179	12.857	15
300	M334	Y	-8.224	-11.494	0	2.143
301	M334	Y	-11.494	-12.208	2.143	4.286
302	M334	Y	-12.208	-10.85	4.286	6.429
303	M334	Y	-10.85	-10.856	6.429	8.571
304	M334	Y	-10.856	-12.214	8.571	10.714
305	M334	Y	-12.214	-11.492	10.714	12.857
306	M334	Y	-11.492	-8.219	12.857	15
307	M332	Y	-5.401	-5.462	0	2.143
308	M332	Y	-5.462	-5.671	2.143	4.286
309	M332	Y	-5.671	-6.424	4.286	6.429
310	M332	Y	-6.424	-6.423	6.429	8.571
311	M332	Y	-6.423	-5.663	8.571	10.714
312	M332	Y	-5.663	-5.453	10.714	12.857
313	M332	Y	-5.453	-5.4	12.857	15
314	M304	Y	-8.219	-11.492	0	2.143
315	M304	Y	-11.492	-12.214	2.143	4.286
316	M304	Y	-12.214	-10.856	4.286	6.429
317	M304	Y	-10.856	-10.85	6.429	8.571
318	M304	Y	-10.85	-12.208	8.571	10.714
319	M304	Y	-12.208	-11.494	10.714	12.857
320	M304	Y	-11.494	-8.224	12.857	15
321	M361	Y	-6.179	-8.566	0	2.143
322	M361	Y	-8.566	-8.456	2.143	4.286
323	M361	Y	-8.456	-7.169	4.286	6.429
324	M361	Y	-7.169	-7.17	6.429	8.571
325	M361	Y	-7.17	-8.455	8.571	10.714
326	M361	Y	-8.455	-8.563	10.714	12.857
327	M361	Y	-8.563	-6.178	12.857	15
328	M365	Y	-5.4	-5.453	0	2.143
329	M365	Y	-5.453	-5.663	2.143	4.286
330	M365	Y	-5.663	-6.423	4.286	6.429
331	M365	Y	-6.423	-6.424	6.429	8.571
332	M365	Y	-6.424	-5.671	8.571	10.714
333	M365	Y	-5.671	-5.462	10.714	12.857
334	M365	Y	-5.462	-5.401	12.857	15
335	M338	Y	-13.32	-13.32	0	15
336	M341	Y	-11.983	-12.954	0	2.143
337	M341	Y	-12.954	-13.426	2.143	4.286
338	M341	Y	-13.426	-14.083	4.286	6.429
339	M341	Y	-14.083	-14.092	6.429	8.571
340	M341	Y	-14.092	-13.449	8.571	10.714
341	M341	Y	-13.449	-12.979	10.714	12.857
342	M341	Y	-12.979	-11.998	12.857	15
343	M342	Y	-12.9	-12.9	4.552e-15	15
344	M343	Y	-7.634	-8.964	0	2.143
345	M343	Y	-8.964	-8.758	2.143	4.286
346	M343	Y	-8.758	-8.119	4.286	6.429
347	M343	Y	-8.119	-8.109	6.429	8.571
348	M343	Y	-8.109	-8.733	8.571	10.714
349	M343	Y	-8.733	-8.951	10.714	12.857
350	M343	Y	-8.951	-7.653	12.857	15
351	M339	Y	-13.05	-13.05	2.887e-15	15
352	M52	Y	-6.72	-6.72	2.109e-15	15
353	M315	Y	-8.224	-11.494	0	2.143
354	M315	Y	-11.494	-12.208	2.143	4.286
355	M315	Y	-12.208	-10.85	4.286	6.429
356	M315	Y	-10.85	-10.856	6.429	8.571
357	M315	Y	-10.856	-12.214	8.571	10.714
358	M315	Y	-12.214	-11.492	10.714	12.857
359	M315	Y	-11.492	-8.219	12.857	15
360	M318	Y	-6.178	-8.563	0	2.143
361	M318	Y	-8.563	-8.455	2.143	4.286
362	M318	Y	-8.455	-7.17	4.286	6.429
363	M318	Y	-7.17	-7.169	6.429	8.571

Member Distributed Loads (BLC 17 : BLC 1 Transient Area Loads) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
364	M318	Y	-7.169	-8.456	8.571	10.714
365	M318	Y	-8.456	-8.566	10.714	12.857
366	M318	Y	-8.566	-6.179	12.857	15
367	M325	Y	-6.178	-8.563	0	2.143
368	M325	Y	-8.563	-8.455	2.143	4.286
369	M325	Y	-8.455	-7.17	4.286	6.429
370	M325	Y	-7.17	-7.169	6.429	8.571
371	M325	Y	-7.169	-8.456	8.571	10.714
372	M325	Y	-8.456	-8.566	10.714	12.857
373	M325	Y	-8.566	-6.179	12.857	15
374	M310	Y	-6.178	-8.563	0	2.143
375	M310	Y	-8.563	-8.455	2.143	4.286
376	M310	Y	-8.455	-7.17	4.286	6.429
377	M310	Y	-7.17	-7.169	6.429	8.571
378	M310	Y	-7.169	-8.456	8.571	10.714
379	M310	Y	-8.456	-8.566	10.714	12.857
380	M310	Y	-8.566	-6.179	12.857	15
381	M323	Y	-8.224	-11.494	0	2.143
382	M323	Y	-11.494	-12.208	2.143	4.286
383	M323	Y	-12.208	-10.85	4.286	6.429
384	M323	Y	-10.85	-10.856	6.429	8.571
385	M323	Y	-10.856	-12.214	8.571	10.714
386	M323	Y	-12.214	-11.492	10.714	12.857
387	M323	Y	-11.492	-8.219	12.857	15
388	M309	Y	-8.224	-11.494	0	2.143
389	M309	Y	-11.494	-12.208	2.143	4.286
390	M309	Y	-12.208	-10.85	4.286	6.429
391	M309	Y	-10.85	-10.856	6.429	8.571
392	M309	Y	-10.856	-12.214	8.571	10.714
393	M309	Y	-12.214	-11.492	10.714	12.857
394	M309	Y	-11.492	-8.219	12.857	15
395	M306	Y	-6.178	-8.563	0	2.143
396	M306	Y	-8.563	-8.455	2.143	4.286
397	M306	Y	-8.455	-7.17	4.286	6.429
398	M306	Y	-7.17	-7.169	6.429	8.571
399	M306	Y	-7.169	-8.456	8.571	10.714
400	M306	Y	-8.456	-8.566	10.714	12.857
401	M306	Y	-8.566	-6.179	12.857	15
402	M305	Y	-8.224	-11.494	0	2.143
403	M305	Y	-11.494	-12.208	2.143	4.286
404	M305	Y	-12.208	-10.85	4.286	6.429
405	M305	Y	-10.85	-10.856	6.429	8.571
406	M305	Y	-10.856	-12.214	8.571	10.714
407	M305	Y	-12.214	-11.492	10.714	12.857
408	M305	Y	-11.492	-8.219	12.857	15
409	M299	Y	-5.401	-5.462	0	2.143
410	M299	Y	-5.462	-5.671	2.143	4.286
411	M299	Y	-5.671	-6.424	4.286	6.429
412	M299	Y	-6.424	-6.423	6.429	8.571
413	M299	Y	-6.423	-5.663	8.571	10.714
414	M299	Y	-5.663	-5.453	10.714	12.857
415	M299	Y	-5.453	-5.4	12.857	15
416	M324	Y	-5.401	-5.462	0	2.143
417	M324	Y	-5.462	-5.671	2.143	4.286
418	M324	Y	-5.671	-6.424	4.286	6.429
419	M324	Y	-6.424	-6.423	6.429	8.571
420	M324	Y	-6.423	-5.663	8.571	10.714
421	M324	Y	-5.663	-5.453	10.714	12.857
422	M324	Y	-5.453	-5.4	12.857	15
423	M317	Y	-5.401	-5.462	0	2.143
424	M317	Y	-5.462	-5.671	2.143	4.286
425	M317	Y	-5.671	-6.424	4.286	6.429
426	M317	Y	-6.424	-6.423	6.429	8.571
427	M317	Y	-6.423	-5.663	8.571	10.714
428	M317	Y	-5.663	-5.453	10.714	12.857
429	M317	Y	-5.453	-5.4	12.857	15
430	M292	Y	-5.401	-5.462	0	2.143
431	M292	Y	-5.462	-5.671	2.143	4.286
432	M292	Y	-5.671	-6.424	4.286	6.429
433	M292	Y	-6.424	-6.423	6.429	8.571
434	M292	Y	-6.423	-5.663	8.571	10.714
435	M292	Y	-5.663	-5.453	10.714	12.857
436	M292	Y	-5.453	-5.4	12.857	15
437	M293	Y	-7.634	-8.964	0	2.143
438	M293	Y	-8.964	-8.758	2.143	4.286
439	M293	Y	-8.758	-8.119	4.286	6.429
440	M293	Y	-8.119	-8.109	6.429	8.571
441	M293	Y	-8.109	-8.733	8.571	10.714
442	M293	Y	-8.733	-8.951	10.714	12.857
443	M293	Y	-8.951	-7.653	12.857	15
444	M256	Y	-13.32	-13.32	1.221e-15	15
445	M255	Y	-13.05	-13.05	1.943e-14	15
446	M254	Y	-12.9	-12.9	2.032e-14	15
447	M253	Y	-11.983	-12.954	0	2.143
448	M253	Y	-12.954	-13.426	2.143	4.286
449	M253	Y	-13.426	-14.083	4.286	6.429
450	M253	Y	-14.083	-14.092	6.429	8.571
451	M253	Y	-14.092	-13.449	8.571	10.714

Member Distributed Loads (BLC 17 : BLC 1 Transient Area Loads) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
452	M253	Y	-13.449	-12.979	10.714	12.857
453	M253	Y	-12.979	-11.998	12.857	15
454	M177	Y	-6.72	-6.72	9.659e-15	15
455	M330	Y	-8.224	-11.494	0	2.143
456	M330	Y	-11.494	-12.208	2.143	4.286
457	M330	Y	-12.208	-10.85	4.286	6.429
458	M330	Y	-10.85	-10.856	6.429	8.571
459	M330	Y	-10.856	-12.214	8.571	10.714
460	M330	Y	-12.214	-11.492	10.714	12.857
461	M330	Y	-11.492	-8.219	12.857	15
462	M331	Y	-6.178	-8.563	0	2.143
463	M331	Y	-8.563	-8.455	2.143	4.286
464	M331	Y	-8.455	-7.17	4.286	6.429
465	M331	Y	-7.17	-7.169	6.429	8.571
466	M331	Y	-7.169	-8.456	8.571	10.714
467	M331	Y	-8.456	-8.566	10.714	12.857
468	M331	Y	-8.566	-6.179	12.857	15
469	M329	Y	-5.401	-5.462	0	2.143
470	M329	Y	-5.462	-5.671	2.143	4.286
471	M329	Y	-5.671	-6.424	4.286	6.429
472	M329	Y	-6.424	-6.423	6.429	8.571
473	M329	Y	-6.423	-5.663	8.571	10.714
474	M329	Y	-5.663	-5.453	10.714	12.857
475	M329	Y	-5.453	-5.4	12.857	15

Member Distributed Loads (BLC 18 : BLC 2 Transient Area Loads)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
1	M462	Y	-195.319	-195.319	0	2.049
2	M462	Y	-195.319	-195.319	2.049	4.097
3	M462	Y	-195.319	-195.319	4.097	6.146
4	M462	Y	-195.319	-195.319	6.146	8.194
5	M462	Y	-195.319	-181.636	8.194	10.243
6	M462	Y	-181.636	-167.952	10.243	12.291
7	M462	Y	-167.952	-167.952	12.291	14.34
8	M457	Y	-50.81	-139.726	6.75	9
9	M457	Y	-139.726	-184.185	9	11.25
10	M457	Y	-184.185	-184.185	11.25	13.5
11	M457	Y	-184.185	-184.185	13.5	15.75
12	M457	Y	-184.185	-216.185	15.75	18
13	M457	Y	-216.185	-203.726	18	20.25
14	M457	Y	-203.726	-114.81	20.25	22.5
15	M461	Y	-183.5	-183.5	0	4.64
16	M459	Y	-183.5	-183.5	5.09	9.73
17	M383	Y	-562.5	-562.5	2.22e-16	13
18	M385	Y	-562.5	-562.5	2.22e-16	13
19	M386	Y	-562.5	-562.5	2.22e-16	13
20	90	Y	-281.25	-281.25	6.661e-16	13
21	125	Y	-281.25	-281.25	5.551e-16	13
22	M455	Y	-158.5	-158.5	0	4.33
23	M337	Y	-158.5	-158.5	0	4.33
24	M498	Y	-237.75	-237.75	2.554e-15	3
25	M215	Y	-750	-750	7.772e-16	3
26	M261	Y	-1500	-1500	3.22e-15	3
27	M227	Y	-1500	-1500	3.22e-15	3
28	M259	Y	-987.75	-987.75	3.22e-15	3
29	M269	Y	-562.5	-562.5	2.22e-16	13
30	M270	Y	-562.5	-562.5	2.22e-16	13
31	M271	Y	-562.5	-562.5	2.22e-16	13
32	M115	Y	-281.25	-281.25	6.661e-16	13
33	M90	Y	-281.25	-281.25	5.551e-16	13
34	M266	Y	-562.5	-562.5	2.22e-16	13
35	M267	Y	-562.5	-562.5	2.22e-16	13
36	M268	Y	-562.5	-562.5	2.22e-16	13
37	M115	Y	-281.25	-281.25	5.551e-16	13
38	M140	Y	-281.25	-281.25	6.661e-16	13
39	M379	Y	-562.5	-562.5	2.22e-16	13
40	M280	Y	-562.5	-562.5	2.22e-16	13
41	M274	Y	-562.5	-562.5	2.22e-16	13
42	M140	Y	-281.25	-281.25	5.551e-16	13
43	M165	Y	-281.25	-281.25	6.661e-16	13
44	M258	Y	-562.5	-562.5	2.22e-16	13
45	M260	Y	-562.5	-562.5	2.22e-16	13
46	M265	Y	-562.5	-562.5	2.22e-16	13
47	M190	Y	-281.25	-281.25	6.661e-16	13
48	M165	Y	-281.25	-281.25	5.551e-16	13
49	M388	Y	-487.5	-487.5	5.551e-16	13
50	M215	Y	-243.75	-243.75	5.551e-16	13
51	M190	Y	-243.75	-243.75	5.551e-16	13
52	M272	Y	-562.5	-562.5	2.22e-16	13
53	M278	Y	-562.5	-562.5	2.22e-16	13
54	M380	Y	-562.5	-562.5	2.22e-16	13
55	M65	Y	-281.25	-281.25	5.551e-16	13
56	M90	Y	-281.25	-281.25	6.661e-16	13
57	M276	Y	-562.5	-562.5	2.22e-16	13
58	M277	Y	-562.5	-562.5	2.22e-16	13
59	M382	Y	-562.5	-562.5	2.22e-16	13
60	M40	Y	-281.25	-281.25	6.661e-16	13
61	90	Y	-281.25	-281.25	5.551e-16	13

Member Distributed Loads (BLC 18 : BLC 2 Transient Area Loads) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
62	M273	Y	-562.5	-562.5	2.22e-16	13
63	M275	Y	-562.5	-562.5	2.22e-16	13
64	M279	Y	-562.5	-562.5	2.22e-16	13
65	M40	Y	-281.25	-281.25	5.551e-16	13
66	M65	Y	-281.25	-281.25	6.661e-16	13

Member Distributed Loads (BLC 19 : BLC 3 Transient Area Loads)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
1	M340	Y	-33.06	-33.06	4.219e-15	15
2	M50	Y	-17.024	-17.024	4.219e-15	15
3	M344	Y	-19.388	-22.675	0	2.143
4	M344	Y	-22.675	-22.125	2.143	4.286
5	M344	Y	-22.125	-20.542	4.286	6.429
6	M344	Y	-20.542	-20.569	6.429	8.571
7	M344	Y	-20.569	-22.188	8.571	10.714
8	M344	Y	-22.188	-22.708	10.714	12.857
9	M344	Y	-22.708	-19.338	12.857	15
10	M345	Y	-33.744	-33.744	3.442e-15	15
11	M350	Y	-30.396	-32.88	0	2.143
12	M350	Y	-32.88	-34.071	2.143	4.286
13	M350	Y	-34.071	-35.699	4.286	6.429
14	M350	Y	-35.699	-35.678	6.429	8.571
15	M350	Y	-35.678	-34.012	8.571	10.714
16	M350	Y	-34.012	-32.816	10.714	12.857
17	M350	Y	-32.816	-30.356	12.857	15
18	M351	Y	-32.68	-32.68	3.997e-15	15
19	M303	Y	-20.834	-29.117	0	2.143
20	M303	Y	-29.117	-30.927	2.143	4.286
21	M303	Y	-30.927	-27.487	4.286	6.429
22	M303	Y	-27.487	-27.502	6.429	8.571
23	M303	Y	-27.502	-30.942	8.571	10.714
24	M303	Y	-30.942	-29.114	10.714	12.857
25	M303	Y	-29.114	-20.822	12.857	15
26	M360	Y	-15.652	-21.694	0	2.143
27	M360	Y	-21.694	-21.42	2.143	4.286
28	M360	Y	-21.42	-18.164	4.286	6.429
29	M360	Y	-18.164	-18.162	6.429	8.571
30	M360	Y	-18.162	-21.423	8.571	10.714
31	M360	Y	-21.423	-21.7	10.714	12.857
32	M360	Y	-21.7	-15.654	12.857	15
33	M363	Y	-13.683	-13.837	0	2.143
34	M363	Y	-13.837	-14.366	2.143	4.286
35	M363	Y	-14.366	-16.274	4.286	6.429
36	M363	Y	-16.274	-16.271	6.429	8.571
37	M363	Y	-16.271	-14.347	8.571	10.714
38	M363	Y	-14.347	-13.814	10.714	12.857
39	M363	Y	-13.814	-13.681	12.857	15
40	M304	Y	-83.061	-116.085	0	2.143
41	M304	Y	-116.085	-123.301	2.143	4.286
42	M304	Y	-123.301	-109.588	4.286	6.429
43	M304	Y	-109.588	-109.645	6.429	8.571
44	M304	Y	-109.645	-123.361	8.571	10.714
45	M304	Y	-123.361	-116.073	10.714	12.857
46	M304	Y	-116.073	-83.012	12.857	15
47	M361	Y	-62.403	-86.489	0	2.143
48	M361	Y	-86.489	-85.399	2.143	4.286
49	M361	Y	-85.399	-72.417	4.286	6.429
50	M361	Y	-72.417	-72.409	6.429	8.571
51	M361	Y	-72.409	-85.41	8.571	10.714
52	M361	Y	-85.41	-86.516	10.714	12.857
53	M361	Y	-86.516	-62.411	12.857	15
54	M365	Y	-54.55	-55.167	0	2.143
55	M365	Y	-55.167	-57.276	2.143	4.286
56	M365	Y	-57.276	-64.882	4.286	6.429
57	M365	Y	-64.882	-64.871	6.429	8.571
58	M365	Y	-64.871	-57.197	8.571	10.714
59	M365	Y	-57.197	-55.075	10.714	12.857
60	M365	Y	-55.075	-54.544	12.857	15
61	M311	Y	-15.652	-21.694	0	2.143
62	M311	Y	-21.694	-21.42	2.143	4.286
63	M311	Y	-21.42	-18.164	4.286	6.429
64	M311	Y	-18.164	-18.162	6.429	8.571
65	M311	Y	-18.162	-21.423	8.571	10.714
66	M311	Y	-21.423	-21.7	10.714	12.857
67	M311	Y	-21.7	-15.654	12.857	15
68	M314	Y	-20.834	-29.117	0	2.143
69	M314	Y	-29.117	-30.927	2.143	4.286
70	M314	Y	-30.927	-27.487	4.286	6.429
71	M314	Y	-27.487	-27.502	6.429	8.571
72	M314	Y	-27.502	-30.942	8.571	10.714
73	M314	Y	-30.942	-29.114	10.714	12.857
74	M314	Y	-29.114	-20.822	12.857	15
75	M320	Y	-15.652	-21.694	0	2.143
76	M320	Y	-21.694	-21.42	2.143	4.286
77	M320	Y	-21.42	-18.164	4.286	6.429
78	M320	Y	-18.164	-18.162	6.429	8.571
79	M320	Y	-18.162	-21.423	8.571	10.714
80	M320	Y	-21.423	-21.7	10.714	12.857

Member Distributed Loads (BLC 19 : BLC 3 Transient Area Loads) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
81	M320	Y	-21.7	-15.654	12.857	15
82	M322	Y	-20.834	-29.117	0	2.143
83	M322	Y	-29.117	-30.927	2.143	4.286
84	M322	Y	-30.927	-27.487	4.286	6.429
85	M322	Y	-27.487	-27.502	6.429	8.571
86	M322	Y	-27.502	-30.942	8.571	10.714
87	M322	Y	-30.942	-29.114	10.714	12.857
88	M322	Y	-29.114	-20.822	12.857	15
89	M328	Y	-20.834	-29.117	0	2.143
90	M328	Y	-29.117	-30.927	2.143	4.286
91	M328	Y	-30.927	-27.487	4.286	6.429
92	M328	Y	-27.487	-27.502	6.429	8.571
93	M328	Y	-27.502	-30.942	8.571	10.714
94	M328	Y	-30.942	-29.114	10.714	12.857
95	M328	Y	-29.114	-20.822	12.857	15
96	M327	Y	-15.652	-21.694	0	2.143
97	M327	Y	-21.694	-21.42	2.143	4.286
98	M327	Y	-21.42	-18.164	4.286	6.429
99	M327	Y	-18.164	-18.162	6.429	8.571
100	M327	Y	-18.162	-21.423	8.571	10.714
101	M327	Y	-21.423	-21.7	10.714	12.857
102	M327	Y	-21.7	-15.654	12.857	15
103	M308	Y	-20.834	-29.117	0	2.143
104	M308	Y	-29.117	-30.927	2.143	4.286
105	M308	Y	-30.927	-27.487	4.286	6.429
106	M308	Y	-27.487	-27.502	6.429	8.571
107	M308	Y	-27.502	-30.942	8.571	10.714
108	M308	Y	-30.942	-29.114	10.714	12.857
109	M308	Y	-29.114	-20.822	12.857	15
110	M307	Y	-15.652	-21.694	0	2.143
111	M307	Y	-21.694	-21.42	2.143	4.286
112	M307	Y	-21.42	-18.164	4.286	6.429
113	M307	Y	-18.164	-18.162	6.429	8.571
114	M307	Y	-18.162	-21.423	8.571	10.714
115	M307	Y	-21.423	-21.7	10.714	12.857
116	M307	Y	-21.7	-15.654	12.857	15
117	M313	Y	-13.682	-13.837	0	2.143
118	M313	Y	-13.837	-14.366	2.143	4.286
119	M313	Y	-14.366	-16.274	4.286	6.429
120	M313	Y	-16.274	-16.271	6.429	8.571
121	M313	Y	-16.271	-14.347	8.571	10.714
122	M313	Y	-14.347	-13.814	10.714	12.857
123	M313	Y	-13.814	-13.681	12.857	15
124	M321	Y	-13.682	-13.837	0	2.143
125	M321	Y	-13.837	-14.366	2.143	4.286
126	M321	Y	-14.366	-16.274	4.286	6.429
127	M321	Y	-16.274	-16.271	6.429	8.571
128	M321	Y	-16.271	-14.347	8.571	10.714
129	M321	Y	-14.347	-13.814	10.714	12.857
130	M321	Y	-13.814	-13.681	12.857	15
131	M326	Y	-13.683	-13.837	0	2.143
132	M326	Y	-13.837	-14.366	2.143	4.286
133	M326	Y	-14.366	-16.274	4.286	6.429
134	M326	Y	-16.274	-16.271	6.429	8.571
135	M326	Y	-16.271	-14.347	8.571	10.714
136	M326	Y	-14.347	-13.814	10.714	12.857
137	M326	Y	-13.814	-13.681	12.857	15
138	M295	Y	-13.682	-13.837	0	2.143
139	M295	Y	-13.837	-14.366	2.143	4.286
140	M295	Y	-14.366	-16.274	4.286	6.429
141	M295	Y	-16.274	-16.271	6.429	8.571
142	M295	Y	-16.271	-14.347	8.571	10.714
143	M295	Y	-14.347	-13.814	10.714	12.857
144	M295	Y	-13.814	-13.681	12.857	15
145	M333	Y	-15.652	-21.694	0	2.143
146	M333	Y	-21.694	-21.42	2.143	4.286
147	M333	Y	-21.42	-18.164	4.286	6.429
148	M333	Y	-18.164	-18.162	6.429	8.571
149	M333	Y	-18.162	-21.423	8.571	10.714
150	M333	Y	-21.423	-21.7	10.714	12.857
151	M333	Y	-21.7	-15.654	12.857	15
152	M334	Y	-20.834	-29.117	0	2.143
153	M334	Y	-29.117	-30.927	2.143	4.286
154	M334	Y	-30.927	-27.487	4.286	6.429
155	M334	Y	-27.487	-27.502	6.429	8.571
156	M334	Y	-27.502	-30.942	8.571	10.714
157	M334	Y	-30.942	-29.114	10.714	12.857
158	M334	Y	-29.114	-20.822	12.857	15
159	M332	Y	-13.683	-13.837	0	2.143
160	M332	Y	-13.837	-14.366	2.143	4.286
161	M332	Y	-14.366	-16.274	4.286	6.429
162	M332	Y	-16.274	-16.271	6.429	8.571
163	M332	Y	-16.271	-14.347	8.571	10.714
164	M332	Y	-14.347	-13.814	10.714	12.857
165	M332	Y	-13.814	-13.681	12.857	15
166	M330	Y	-83.012	-116.073	0	2.143
167	M330	Y	-116.073	-123.361	2.143	4.286
168	M330	Y	-123.361	-109.645	4.286	6.429

Member Distributed Loads (BLC 19 : BLC 3 Transient Area Loads) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
169	M330	Y	-109.645	-109.588	6.429	8.571
170	M330	Y	-109.588	-123.301	8.571	10.714
171	M330	Y	-123.301	-116.085	10.714	12.857
172	M330	Y	-116.085	-83.061	12.857	15
173	M331	Y	-62.411	-86.516	0	2.143
174	M331	Y	-86.516	-85.41	2.143	4.286
175	M331	Y	-85.41	-72.409	4.286	6.429
176	M331	Y	-72.409	-72.417	6.429	8.571
177	M331	Y	-72.417	-85.399	8.571	10.714
178	M331	Y	-85.399	-86.489	10.714	12.857
179	M331	Y	-86.489	-62.403	12.857	15
180	M329	Y	-54.544	-55.075	0	2.143
181	M329	Y	-55.075	-57.197	2.143	4.286
182	M329	Y	-57.197	-64.871	4.286	6.429
183	M329	Y	-64.871	-64.882	6.429	8.571
184	M329	Y	-64.882	-57.276	8.571	10.714
185	M329	Y	-57.276	-55.167	10.714	12.857
186	M329	Y	-55.167	-54.55	12.857	15
187	M338	Y	-134.533	-134.533	0	15
188	M341	Y	-121.024	-130.832	0	2.143
189	M341	Y	-130.832	-135.599	2.143	4.286
190	M341	Y	-135.599	-142.241	4.286	6.429
191	M341	Y	-142.241	-142.327	6.429	8.571
192	M341	Y	-142.327	-135.837	8.571	10.714
193	M341	Y	-135.837	-131.087	10.714	12.857
194	M341	Y	-131.087	-121.183	12.857	15
195	M342	Y	-130.29	-130.29	4.552e-15	15
196	M343	Y	-77.099	-90.532	0	2.143
197	M343	Y	-90.532	-88.459	2.143	4.286
198	M343	Y	-88.459	-82.004	4.286	6.429
199	M343	Y	-82.004	-81.899	6.429	8.571
200	M343	Y	-81.899	-88.207	8.571	10.714
201	M343	Y	-88.207	-90.401	10.714	12.857
202	M343	Y	-90.401	-77.297	12.857	15
203	M339	Y	-131.805	-131.805	2.887e-15	15
204	M52	Y	-67.873	-67.873	2.109e-15	15
205	M293	Y	-77.099	-90.532	0	2.143
206	M293	Y	-90.532	-88.459	2.143	4.286
207	M293	Y	-88.459	-82.004	4.286	6.429
208	M293	Y	-82.004	-81.899	6.429	8.571
209	M293	Y	-81.899	-88.207	8.571	10.714
210	M293	Y	-88.207	-90.401	10.714	12.857
211	M293	Y	-90.401	-77.297	12.857	15
212	M256	Y	-134.533	-134.533	1.221e-15	15
213	M255	Y	-131.805	-131.805	1.943e-14	15
214	M254	Y	-130.29	-130.29	2.032e-14	15
215	M253	Y	-121.024	-130.832	0	2.143
216	M253	Y	-130.832	-135.599	2.143	4.286
217	M253	Y	-135.599	-142.241	4.286	6.429
218	M253	Y	-142.241	-142.327	6.429	8.571
219	M253	Y	-142.327	-135.837	8.571	10.714
220	M253	Y	-135.837	-131.087	10.714	12.857
221	M253	Y	-131.087	-121.183	12.857	15
222	M177	Y	-67.873	-67.873	9.659e-15	15
223	M297	Y	-33.744	-33.744	1.221e-15	15
224	M302	Y	-32.68	-32.68	2.032e-14	15
225	M301	Y	-30.356	-32.816	0	2.143
226	M301	Y	-32.816	-34.012	2.143	4.286
227	M301	Y	-34.012	-35.678	4.286	6.429
228	M301	Y	-35.678	-35.699	6.429	8.571
229	M301	Y	-35.699	-34.071	8.571	10.714
230	M301	Y	-34.071	-32.88	10.714	12.857
231	M301	Y	-32.88	-30.396	12.857	15
232	M300	Y	-33.06	-33.06	1.943e-14	15
233	M296	Y	-19.338	-22.708	0	2.143
234	M296	Y	-22.708	-22.188	2.143	4.286
235	M296	Y	-22.188	-20.569	4.286	6.429
236	M296	Y	-20.569	-20.542	6.429	8.571
237	M296	Y	-20.542	-22.125	8.571	10.714
238	M296	Y	-22.125	-22.675	10.714	12.857
239	M296	Y	-22.675	-19.388	12.857	15
240	M175	Y	-17.024	-17.024	9.659e-15	15
241	M315	Y	-83.012	-116.073	0	2.143
242	M315	Y	-116.073	-123.361	2.143	4.286
243	M315	Y	-123.361	-109.645	4.286	6.429
244	M315	Y	-109.645	-109.588	6.429	8.571
245	M315	Y	-109.588	-123.301	8.571	10.714
246	M315	Y	-123.301	-116.085	10.714	12.857
247	M315	Y	-116.085	-83.061	12.857	15
248	M318	Y	-62.411	-86.516	0	2.143
249	M318	Y	-86.516	-85.41	2.143	4.286
250	M318	Y	-85.41	-72.409	4.286	6.429
251	M318	Y	-72.409	-72.417	6.429	8.571
252	M318	Y	-72.417	-85.399	8.571	10.714
253	M318	Y	-85.399	-86.489	10.714	12.857
254	M318	Y	-86.489	-62.403	12.857	15
255	M325	Y	-62.411	-86.516	0	2.143
256	M325	Y	-86.516	-85.41	2.143	4.286

Member Distributed Loads (BLC 19 : BLC 3 Transient Area Loads) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
257	M325	Y	-85.41	-72.409	4.286	6.429
258	M325	Y	-72.409	-72.417	6.429	8.571
259	M325	Y	-72.417	-85.399	8.571	10.714
260	M325	Y	-85.399	-86.489	10.714	12.857
261	M325	Y	-86.489	-62.403	12.857	15
262	M310	Y	-62.411	-86.516	0	2.143
263	M310	Y	-86.516	-85.41	2.143	4.286
264	M310	Y	-85.41	-72.409	4.286	6.429
265	M310	Y	-72.409	-72.417	6.429	8.571
266	M310	Y	-72.417	-85.399	8.571	10.714
267	M310	Y	-85.399	-86.489	10.714	12.857
268	M310	Y	-86.489	-62.403	12.857	15
269	M323	Y	-83.012	-116.073	0	2.143
270	M323	Y	-116.073	-123.361	2.143	4.286
271	M323	Y	-123.361	-109.645	4.286	6.429
272	M323	Y	-109.645	-109.588	6.429	8.571
273	M323	Y	-109.588	-123.301	8.571	10.714
274	M323	Y	-123.301	-116.085	10.714	12.857
275	M323	Y	-116.085	-83.061	12.857	15
276	M309	Y	-83.012	-116.073	0	2.143
277	M309	Y	-116.073	-123.361	2.143	4.286
278	M309	Y	-123.361	-109.645	4.286	6.429
279	M309	Y	-109.645	-109.588	6.429	8.571
280	M309	Y	-109.588	-123.301	8.571	10.714
281	M309	Y	-123.301	-116.085	10.714	12.857
282	M309	Y	-116.085	-83.061	12.857	15
283	M306	Y	-62.411	-86.516	0	2.143
284	M306	Y	-86.516	-85.41	2.143	4.286
285	M306	Y	-85.41	-72.409	4.286	6.429
286	M306	Y	-72.409	-72.417	6.429	8.571
287	M306	Y	-72.417	-85.399	8.571	10.714
288	M306	Y	-85.399	-86.489	10.714	12.857
289	M306	Y	-86.489	-62.403	12.857	15
290	M305	Y	-83.012	-116.073	0	2.143
291	M305	Y	-116.073	-123.361	2.143	4.286
292	M305	Y	-123.361	-109.645	4.286	6.429
293	M305	Y	-109.645	-109.588	6.429	8.571
294	M305	Y	-109.588	-123.301	8.571	10.714
295	M305	Y	-123.301	-116.085	10.714	12.857
296	M305	Y	-116.085	-83.061	12.857	15
297	M299	Y	-54.544	-55.075	0	2.143
298	M299	Y	-55.075	-57.197	2.143	4.286
299	M299	Y	-57.197	-64.871	4.286	6.429
300	M299	Y	-64.871	-64.882	6.429	8.571
301	M299	Y	-64.882	-57.276	8.571	10.714
302	M299	Y	-57.276	-55.167	10.714	12.857
303	M299	Y	-55.167	-54.55	12.857	15
304	M324	Y	-54.544	-55.075	0	2.143
305	M324	Y	-55.075	-57.197	2.143	4.286
306	M324	Y	-57.197	-64.871	4.286	6.429
307	M324	Y	-64.871	-64.882	6.429	8.571
308	M324	Y	-64.882	-57.276	8.571	10.714
309	M324	Y	-57.276	-55.167	10.714	12.857
310	M324	Y	-55.167	-54.55	12.857	15
311	M317	Y	-54.544	-55.075	0	2.143
312	M317	Y	-55.075	-57.197	2.143	4.286
313	M317	Y	-57.197	-64.871	4.286	6.429
314	M317	Y	-64.871	-64.882	6.429	8.571
315	M317	Y	-64.882	-57.276	8.571	10.714
316	M317	Y	-57.276	-55.167	10.714	12.857
317	M317	Y	-55.167	-54.55	12.857	15
318	M292	Y	-54.544	-55.075	0	2.143
319	M292	Y	-55.075	-57.197	2.143	4.286
320	M292	Y	-57.197	-64.871	4.286	6.429
321	M292	Y	-64.871	-64.882	6.429	8.571
322	M292	Y	-64.882	-57.276	8.571	10.714
323	M292	Y	-57.276	-55.167	10.714	12.857
324	M292	Y	-55.167	-54.55	12.857	15
325	M432	Y	-14.175	-14.175	2.926e-5	6.5
326	M433	Y	-20.895	-23.155	0	1.3
327	M433	Y	-23.155	-38.022	1.3	2.6
328	M433	Y	-38.022	-38.021	2.6	3.9
329	M433	Y	-38.021	-23.15	3.9	5.2
330	M433	Y	-23.15	-20.887	5.2	6.5
331	M508	Y	-25.683	-16.478	0	1.3
332	M508	Y	-16.478	-19.595	1.3	2.6
333	M508	Y	-19.595	-19.601	2.6	3.9
334	M508	Y	-19.601	-16.482	3.9	5.2
335	M508	Y	-16.482	-25.671	5.2	6.5
336	M509	Y	-102.415	-65.717	0	1.3
337	M509	Y	-65.717	-78.158	1.3	2.6
338	M509	Y	-78.158	-78.16	2.6	3.9
339	M509	Y	-78.16	-65.676	3.9	5.2
340	M509	Y	-65.676	-102.284	5.2	6.5
341	M510	Y	-83.306	-92.323	0	1.3
342	M510	Y	-92.323	-151.579	1.3	2.6
343	M510	Y	-151.579	-151.575	2.6	3.9
344	M510	Y	-151.575	-92.3	3.9	5.2

Member Distributed Loads (BLC 19 : BLC 3 Transient Area Loads) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
345	M510	Y	-92.3	-83.255	5.2	6.5
346	M511	Y	-56.514	-56.514	8.777e-5	6.5

Member Distributed Loads (BLC 20 : BLC 4 Transient Area Loads)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
1	M340	Y	-87.87	-87.87	4.219e-15	15
2	M50	Y	-45.249	-45.249	4.219e-15	15
3	M344	Y	-51.531	-60.267	0	2.143
4	M344	Y	-60.267	-58.805	2.143	4.286
5	M344	Y	-58.805	-54.6	4.286	6.429
6	M344	Y	-54.6	-54.669	6.429	8.571
7	M344	Y	-54.669	-58.973	8.571	10.714
8	M344	Y	-58.973	-60.355	10.714	12.857
9	M344	Y	-60.355	-51.399	12.857	15
10	M345	Y	-89.689	-89.689	3.442e-15	15
11	M350	Y	-80.788	-87.392	0	2.143
12	M350	Y	-87.392	-90.558	2.143	4.286
13	M350	Y	-90.558	-94.884	4.286	6.429
14	M350	Y	-94.884	-94.827	6.429	8.571
15	M350	Y	-94.827	-90.399	8.571	10.714
16	M350	Y	-90.399	-87.221	10.714	12.857
17	M350	Y	-87.221	-80.683	12.857	15
18	M351	Y	-86.86	-86.86	3.997e-15	15
19	M303	Y	-55.374	-77.39	0	2.143
20	M303	Y	-77.39	-82.201	2.143	4.286
21	M303	Y	-82.201	-73.059	4.286	6.429
22	M303	Y	-73.059	-73.097	6.429	8.571
23	M303	Y	-73.097	-82.241	8.571	10.714
24	M303	Y	-82.241	-77.382	10.714	12.857
25	M303	Y	-77.382	-55.342	12.857	15
26	M360	Y	-41.602	-57.659	0	2.143
27	M360	Y	-57.659	-56.933	2.143	4.286
28	M360	Y	-56.933	-48.278	4.286	6.429
29	M360	Y	-48.278	-48.273	6.429	8.571
30	M360	Y	-48.273	-56.94	8.571	10.714
31	M360	Y	-56.94	-57.678	10.714	12.857
32	M360	Y	-57.678	-41.607	12.857	15
33	M363	Y	-36.367	-36.778	0	2.143
34	M363	Y	-36.778	-38.184	2.143	4.286
35	M363	Y	-38.184	-43.255	4.286	6.429
36	M363	Y	-43.255	-43.247	6.429	8.571
37	M363	Y	-43.247	-38.132	8.571	10.714
38	M363	Y	-38.132	-36.716	10.714	12.857
39	M363	Y	-36.716	-36.363	12.857	15
40	M304	Y	-55.374	-77.39	0	2.143
41	M304	Y	-77.39	-82.201	2.143	4.286
42	M304	Y	-82.201	-73.059	4.286	6.429
43	M304	Y	-73.059	-73.097	6.429	8.571
44	M304	Y	-73.097	-82.241	8.571	10.714
45	M304	Y	-82.241	-77.382	10.714	12.857
46	M304	Y	-77.382	-55.342	12.857	15
47	M361	Y	-41.602	-57.659	0	2.143
48	M361	Y	-57.659	-56.933	2.143	4.286
49	M361	Y	-56.933	-48.278	4.286	6.429
50	M361	Y	-48.278	-48.273	6.429	8.571
51	M361	Y	-48.273	-56.94	8.571	10.714
52	M361	Y	-56.94	-57.678	10.714	12.857
53	M361	Y	-57.678	-41.607	12.857	15
54	M365	Y	-36.367	-36.778	0	2.143
55	M365	Y	-36.778	-38.184	2.143	4.286
56	M365	Y	-38.184	-43.255	4.286	6.429
57	M365	Y	-43.255	-43.247	6.429	8.571
58	M365	Y	-43.247	-38.132	8.571	10.714
59	M365	Y	-38.132	-36.716	10.714	12.857
60	M365	Y	-36.716	-36.363	12.857	15
61	M311	Y	-41.602	-57.659	0	2.143
62	M311	Y	-57.659	-56.933	2.143	4.286
63	M311	Y	-56.933	-48.278	4.286	6.429
64	M311	Y	-48.278	-48.273	6.429	8.571
65	M311	Y	-48.273	-56.94	8.571	10.714
66	M311	Y	-56.94	-57.678	10.714	12.857
67	M311	Y	-57.678	-41.607	12.857	15
68	M314	Y	-55.374	-77.39	0	2.143
69	M314	Y	-77.39	-82.201	2.143	4.286
70	M314	Y	-82.201	-73.059	4.286	6.429
71	M314	Y	-73.059	-73.097	6.429	8.571
72	M314	Y	-73.097	-82.241	8.571	10.714
73	M314	Y	-82.241	-77.382	10.714	12.857
74	M314	Y	-77.382	-55.342	12.857	15
75	M320	Y	-41.602	-57.659	0	2.143
76	M320	Y	-57.659	-56.933	2.143	4.286
77	M320	Y	-56.933	-48.278	4.286	6.429
78	M320	Y	-48.278	-48.273	6.429	8.571
79	M320	Y	-48.273	-56.94	8.571	10.714
80	M320	Y	-56.94	-57.678	10.714	12.857
81	M320	Y	-57.678	-41.607	12.857	15
82	M322	Y	-55.374	-77.39	0	2.143
83	M322	Y	-77.39	-82.201	2.143	4.286

Member Distributed Loads (BLC 20 : BLC 4 Transient Area Loads) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
84	M322	Y	-82.201	-73.059	4.286	6.429
85	M322	Y	-73.059	-73.097	6.429	8.571
86	M322	Y	-73.097	-82.241	8.571	10.714
87	M322	Y	-82.241	-77.382	10.714	12.857
88	M322	Y	-77.382	-55.342	12.857	15
89	M328	Y	-55.374	-77.39	0	2.143
90	M328	Y	-77.39	-82.201	2.143	4.286
91	M328	Y	-82.201	-73.059	4.286	6.429
92	M328	Y	-73.059	-73.097	6.429	8.571
93	M328	Y	-73.097	-82.241	8.571	10.714
94	M328	Y	-82.241	-77.382	10.714	12.857
95	M328	Y	-77.382	-55.342	12.857	15
96	M327	Y	-41.602	-57.659	0	2.143
97	M327	Y	-57.659	-56.933	2.143	4.286
98	M327	Y	-56.933	-48.278	4.286	6.429
99	M327	Y	-48.278	-48.273	6.429	8.571
100	M327	Y	-48.273	-56.94	8.571	10.714
101	M327	Y	-56.94	-57.678	10.714	12.857
102	M327	Y	-57.678	-41.607	12.857	15
103	M308	Y	-55.374	-77.39	0	2.143
104	M308	Y	-77.39	-82.201	2.143	4.286
105	M308	Y	-82.201	-73.059	4.286	6.429
106	M308	Y	-73.059	-73.097	6.429	8.571
107	M308	Y	-73.097	-82.241	8.571	10.714
108	M308	Y	-82.241	-77.382	10.714	12.857
109	M308	Y	-77.382	-55.342	12.857	15
110	M307	Y	-41.602	-57.659	0	2.143
111	M307	Y	-57.659	-56.933	2.143	4.286
112	M307	Y	-56.933	-48.278	4.286	6.429
113	M307	Y	-48.278	-48.273	6.429	8.571
114	M307	Y	-48.273	-56.94	8.571	10.714
115	M307	Y	-56.94	-57.678	10.714	12.857
116	M307	Y	-57.678	-41.607	12.857	15
117	M313	Y	-36.367	-36.778	0	2.143
118	M313	Y	-36.778	-38.184	2.143	4.286
119	M313	Y	-38.184	-43.255	4.286	6.429
120	M313	Y	-43.255	-43.247	6.429	8.571
121	M313	Y	-43.247	-38.132	8.571	10.714
122	M313	Y	-38.132	-36.716	10.714	12.857
123	M313	Y	-36.716	-36.363	12.857	15
124	M321	Y	-36.367	-36.778	0	2.143
125	M321	Y	-36.778	-38.184	2.143	4.286
126	M321	Y	-38.184	-43.255	4.286	6.429
127	M321	Y	-43.255	-43.247	6.429	8.571
128	M321	Y	-43.247	-38.132	8.571	10.714
129	M321	Y	-38.132	-36.716	10.714	12.857
130	M321	Y	-36.716	-36.363	12.857	15
131	M326	Y	-36.367	-36.778	0	2.143
132	M326	Y	-36.778	-38.184	2.143	4.286
133	M326	Y	-38.184	-43.255	4.286	6.429
134	M326	Y	-43.255	-43.247	6.429	8.571
135	M326	Y	-43.247	-38.132	8.571	10.714
136	M326	Y	-38.132	-36.716	10.714	12.857
137	M326	Y	-36.716	-36.363	12.857	15
138	M295	Y	-36.367	-36.778	0	2.143
139	M295	Y	-36.778	-38.184	2.143	4.286
140	M295	Y	-38.184	-43.255	4.286	6.429
141	M295	Y	-43.255	-43.247	6.429	8.571
142	M295	Y	-43.247	-38.132	8.571	10.714
143	M295	Y	-38.132	-36.716	10.714	12.857
144	M295	Y	-36.716	-36.363	12.857	15
145	M333	Y	-41.602	-57.659	0	2.143
146	M333	Y	-57.659	-56.933	2.143	4.286
147	M333	Y	-56.933	-48.278	4.286	6.429
148	M333	Y	-48.278	-48.273	6.429	8.571
149	M333	Y	-48.273	-56.94	8.571	10.714
150	M333	Y	-56.94	-57.678	10.714	12.857
151	M333	Y	-57.678	-41.607	12.857	15
152	M334	Y	-55.374	-77.39	0	2.143
153	M334	Y	-77.39	-82.201	2.143	4.286
154	M334	Y	-82.201	-73.059	4.286	6.429
155	M334	Y	-73.059	-73.097	6.429	8.571
156	M334	Y	-73.097	-82.241	8.571	10.714
157	M334	Y	-82.241	-77.382	10.714	12.857
158	M334	Y	-77.382	-55.342	12.857	15
159	M332	Y	-36.367	-36.778	0	2.143
160	M332	Y	-36.778	-38.184	2.143	4.286
161	M332	Y	-38.184	-43.255	4.286	6.429
162	M332	Y	-43.255	-43.247	6.429	8.571
163	M332	Y	-43.247	-38.132	8.571	10.714
164	M332	Y	-38.132	-36.716	10.714	12.857
165	M332	Y	-36.716	-36.363	12.857	15
166	M330	Y	-55.342	-77.382	0	2.143
167	M330	Y	-77.382	-82.241	2.143	4.286
168	M330	Y	-82.241	-73.097	4.286	6.429
169	M330	Y	-73.097	-73.059	6.429	8.571
170	M330	Y	-73.059	-82.201	8.571	10.714
171	M330	Y	-82.201	-77.39	10.714	12.857

Member Distributed Loads (BLC 20 : BLC 4 Transient Area Loads) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
172	M330	Y	-77.39	-55.374	12.857	15
173	M331	Y	-41.607	-57.678	0	2.143
174	M331	Y	-57.678	-56.94	2.143	4.286
175	M331	Y	-56.94	-48.273	4.286	6.429
176	M331	Y	-48.273	-48.278	6.429	8.571
177	M331	Y	-48.278	-56.933	8.571	10.714
178	M331	Y	-56.933	-57.659	10.714	12.857
179	M331	Y	-57.659	-41.602	12.857	15
180	M329	Y	-36.363	-36.717	0	2.143
181	M329	Y	-36.717	-38.132	2.143	4.286
182	M329	Y	-38.132	-43.247	4.286	6.429
183	M329	Y	-43.247	-43.255	6.429	8.571
184	M329	Y	-43.255	-38.184	8.571	10.714
185	M329	Y	-38.184	-36.778	10.714	12.857
186	M329	Y	-36.778	-36.367	12.857	15
187	M338	Y	-89.689	-89.689	0	15
188	M341	Y	-80.683	-87.221	0	2.143
189	M341	Y	-87.221	-90.399	2.143	4.286
190	M341	Y	-90.399	-94.827	4.286	6.429
191	M341	Y	-94.827	-94.884	6.429	8.571
192	M341	Y	-94.884	-90.558	8.571	10.714
193	M341	Y	-90.558	-87.392	10.714	12.857
194	M341	Y	-87.392	-80.788	12.857	15
195	M342	Y	-86.86	-86.86	4.552e-15	15
196	M343	Y	-51.399	-60.355	0	2.143
197	M343	Y	-60.355	-58.973	2.143	4.286
198	M343	Y	-58.973	-54.669	4.286	6.429
199	M343	Y	-54.669	-54.6	6.429	8.571
200	M343	Y	-54.6	-58.805	8.571	10.714
201	M343	Y	-58.805	-60.267	10.714	12.857
202	M343	Y	-60.267	-51.531	12.857	15
203	M339	Y	-87.87	-87.87	2.887e-15	15
204	M52	Y	-45.249	-45.249	2.109e-15	15
205	M293	Y	-51.399	-60.355	0	2.143
206	M293	Y	-60.355	-58.973	2.143	4.286
207	M293	Y	-58.973	-54.669	4.286	6.429
208	M293	Y	-54.669	-54.6	6.429	8.571
209	M293	Y	-54.6	-58.805	8.571	10.714
210	M293	Y	-58.805	-60.267	10.714	12.857
211	M293	Y	-60.267	-51.531	12.857	15
212	M256	Y	-89.689	-89.689	1.221e-15	15
213	M255	Y	-87.87	-87.87	1.943e-14	15
214	M254	Y	-86.86	-86.86	2.032e-14	15
215	M253	Y	-80.683	-87.221	0	2.143
216	M253	Y	-87.221	-90.399	2.143	4.286
217	M253	Y	-90.399	-94.827	4.286	6.429
218	M253	Y	-94.827	-94.884	6.429	8.571
219	M253	Y	-94.884	-90.558	8.571	10.714
220	M253	Y	-90.558	-87.392	10.714	12.857
221	M253	Y	-87.392	-80.788	12.857	15
222	M177	Y	-45.249	-45.249	9.659e-15	15
223	M297	Y	-89.689	-89.689	1.221e-15	15
224	M302	Y	-86.86	-86.86	2.032e-14	15
225	M301	Y	-80.683	-87.221	0	2.143
226	M301	Y	-87.221	-90.399	2.143	4.286
227	M301	Y	-90.399	-94.827	4.286	6.429
228	M301	Y	-94.827	-94.884	6.429	8.571
229	M301	Y	-94.884	-90.558	8.571	10.714
230	M301	Y	-90.558	-87.392	10.714	12.857
231	M301	Y	-87.392	-80.788	12.857	15
232	M300	Y	-87.87	-87.87	1.943e-14	15
233	M296	Y	-51.399	-60.355	0	2.143
234	M296	Y	-60.355	-58.973	2.143	4.286
235	M296	Y	-58.973	-54.669	4.286	6.429
236	M296	Y	-54.669	-54.6	6.429	8.571
237	M296	Y	-54.6	-58.805	8.571	10.714
238	M296	Y	-58.805	-60.267	10.714	12.857
239	M296	Y	-60.267	-51.531	12.857	15
240	M175	Y	-45.249	-45.249	9.659e-15	15
241	M315	Y	-55.342	-77.382	0	2.143
242	M315	Y	-77.382	-82.241	2.143	4.286
243	M315	Y	-82.241	-73.097	4.286	6.429
244	M315	Y	-73.097	-73.059	6.429	8.571
245	M315	Y	-73.059	-82.201	8.571	10.714
246	M315	Y	-82.201	-77.39	10.714	12.857
247	M315	Y	-77.39	-55.374	12.857	15
248	M318	Y	-41.607	-57.678	0	2.143
249	M318	Y	-57.678	-56.94	2.143	4.286
250	M318	Y	-56.94	-48.273	4.286	6.429
251	M318	Y	-48.273	-48.278	6.429	8.571
252	M318	Y	-48.278	-56.933	8.571	10.714
253	M318	Y	-56.933	-57.659	10.714	12.857
254	M318	Y	-57.659	-41.602	12.857	15
255	M325	Y	-41.607	-57.678	0	2.143
256	M325	Y	-57.678	-56.94	2.143	4.286
257	M325	Y	-56.94	-48.273	4.286	6.429
258	M325	Y	-48.273	-48.278	6.429	8.571
259	M325	Y	-48.278	-56.933	8.571	10.714

Member Distributed Loads (BLC 20 : BLC 4 Transient Area Loads) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
260	M325	Y	-56.933	-57.659	10.714	12.857
261	M325	Y	-57.659	-41.602	12.857	15
262	M310	Y	-41.607	-57.678	0	2.143
263	M310	Y	-57.678	-56.94	2.143	4.286
264	M310	Y	-56.94	-48.273	4.286	6.429
265	M310	Y	-48.273	-48.278	6.429	8.571
266	M310	Y	-48.278	-56.933	8.571	10.714
267	M310	Y	-56.933	-57.659	10.714	12.857
268	M310	Y	-57.659	-41.602	12.857	15
269	M323	Y	-55.342	-77.382	0	2.143
270	M323	Y	-77.382	-82.241	2.143	4.286
271	M323	Y	-82.241	-73.097	4.286	6.429
272	M323	Y	-73.097	-73.059	6.429	8.571
273	M323	Y	-73.059	-82.201	8.571	10.714
274	M323	Y	-82.201	-77.39	10.714	12.857
275	M323	Y	-77.39	-55.374	12.857	15
276	M309	Y	-55.342	-77.382	0	2.143
277	M309	Y	-77.382	-82.241	2.143	4.286
278	M309	Y	-82.241	-73.097	4.286	6.429
279	M309	Y	-73.097	-73.059	6.429	8.571
280	M309	Y	-73.059	-82.201	8.571	10.714
281	M309	Y	-82.201	-77.39	10.714	12.857
282	M309	Y	-77.39	-55.374	12.857	15
283	M306	Y	-41.607	-57.678	0	2.143
284	M306	Y	-57.678	-56.94	2.143	4.286
285	M306	Y	-56.94	-48.273	4.286	6.429
286	M306	Y	-48.273	-48.278	6.429	8.571
287	M306	Y	-48.278	-56.933	8.571	10.714
288	M306	Y	-56.933	-57.659	10.714	12.857
289	M306	Y	-57.659	-41.602	12.857	15
290	M305	Y	-55.342	-77.382	0	2.143
291	M305	Y	-77.382	-82.241	2.143	4.286
292	M305	Y	-82.241	-73.097	4.286	6.429
293	M305	Y	-73.097	-73.059	6.429	8.571
294	M305	Y	-73.059	-82.201	8.571	10.714
295	M305	Y	-82.201	-77.39	10.714	12.857
296	M305	Y	-77.39	-55.374	12.857	15
297	M299	Y	-36.363	-36.716	0	2.143
298	M299	Y	-36.716	-38.132	2.143	4.286
299	M299	Y	-38.132	-43.247	4.286	6.429
300	M299	Y	-43.247	-43.255	6.429	8.571
301	M299	Y	-43.255	-38.184	8.571	10.714
302	M299	Y	-38.184	-36.778	10.714	12.857
303	M299	Y	-36.778	-36.367	12.857	15
304	M324	Y	-36.363	-36.717	0	2.143
305	M324	Y	-36.717	-38.132	2.143	4.286
306	M324	Y	-38.132	-43.247	4.286	6.429
307	M324	Y	-43.247	-43.255	6.429	8.571
308	M324	Y	-43.255	-38.184	8.571	10.714
309	M324	Y	-38.184	-36.778	10.714	12.857
310	M324	Y	-36.778	-36.367	12.857	15
311	M317	Y	-36.363	-36.716	0	2.143
312	M317	Y	-36.716	-38.132	2.143	4.286
313	M317	Y	-38.132	-43.247	4.286	6.429
314	M317	Y	-43.247	-43.255	6.429	8.571
315	M317	Y	-43.255	-38.184	8.571	10.714
316	M317	Y	-38.184	-36.778	10.714	12.857
317	M317	Y	-36.778	-36.367	12.857	15
318	M292	Y	-36.363	-36.716	0	2.143
319	M292	Y	-36.716	-38.132	2.143	4.286
320	M292	Y	-38.132	-43.247	4.286	6.429
321	M292	Y	-43.247	-43.255	6.429	8.571
322	M292	Y	-43.255	-38.184	8.571	10.714
323	M292	Y	-38.184	-36.778	10.714	12.857
324	M292	Y	-36.778	-36.367	12.857	15
325	M432	Y	-37.676	-37.676	2.926e-5	6.5
326	M433	Y	-55.536	-61.545	0	1.3
327	M433	Y	-61.545	-101.06	1.3	2.6
328	M433	Y	-101.06	-101.055	2.6	3.9
329	M433	Y	-101.055	-61.531	3.9	5.2
330	M433	Y	-61.531	-55.514	5.2	6.5
331	M508	Y	-68.264	-43.796	0	1.3
332	M508	Y	-43.796	-52.082	1.3	2.6
333	M508	Y	-52.082	-52.098	2.6	3.9
334	M508	Y	-52.098	-43.807	3.9	5.2
335	M508	Y	-43.807	-68.231	5.2	6.5
336	M509	Y	-68.277	-43.811	0	1.3
337	M509	Y	-43.811	-52.105	1.3	2.6
338	M509	Y	-52.105	-52.107	2.6	3.9
339	M509	Y	-52.107	-43.784	3.9	5.2
340	M509	Y	-43.784	-68.19	5.2	6.5
341	M510	Y	-55.537	-61.549	0	1.3
342	M510	Y	-61.549	-101.053	1.3	2.6
343	M510	Y	-101.053	-101.05	2.6	3.9
344	M510	Y	-101.05	-61.533	3.9	5.2
345	M510	Y	-61.533	-55.503	5.2	6.5
346	M511	Y	-37.676	-37.676	8.777e-5	6.5

Member Distributed Loads (BLC 21 : BLC 5 Transient Area Loads)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
1	M234	X	30.858	30.858	0	24.9
2	M357	X	30.858	30.858	2.351	27.351
3	M358	X	30.858	30.858	2.351	27.351
4	M359	X	30.858	30.858	2.351	27.351
5	M362	X	30.858	30.858	2.351	27.351
6	M356	X	30.858	30.858	2.351	27.351
7	M364	X	30.858	30.858	2.351	27.351
8	M500	X	30.858	30.858	2.351	27.351
9	M501	X	30.858	30.858	2.351	27.351
10	M502	X	30.858	30.858	2.351	27.351
11	M503	X	30.858	30.858	2.351	27.351
12	M504	X	30.858	30.858	2.351	27.351
13	M505	X	30.858	30.858	2.351	27.351
14	M499	X	30.858	30.858	2.351	27.351
15	M506	X	30.858	30.858	2.351	27.351
16	M355	X	30.858	30.858	2.351	27.351
17	M354	X	30.858	30.858	2.351	27.351
18	M235	X	30.858	30.858	0	24.9
19	M211	X	30.858	30.858	2.351	27.351
20	M212	X	30.858	30.858	2.351	27.351
21	M245	X	30.858	30.858	0	24.9
22	M223	X	30.858	30.858	0	24.9
23	M224	X	30.858	30.858	0	24.9
24	M246	X	30.858	30.858	0	24.9
25	M352	X	30.858	30.858	2.351	27.351
26	M353	X	30.858	30.858	2.351	27.351
27	M462	X	47.15	47.15	0	14.34
28	M497	X	37.298	37.298	0	10
29	M496	X	37.298	37.298	0	10
30	M495	X	37.298	37.298	0	10
31	M494	X	37.298	37.298	0	10
32	M479	X	47.15	47.15	0	20
33	M483	X	47.15	47.15	0	20
34	M481	X	47.15	47.15	0	22.5
35	M482	X	47.15	47.15	0	20
36	M478	X	47.15	47.15	0	22.5
37	M484	X	47.15	47.15	0	6.5
38	M485	X	47.15	47.15	0	6.5
39	M286	X	47.15	47.15	0	15
40	M480	X	47.15	47.15	0	20
41	M48	X	47.15	47.15	0	15
42	M392	X	47.15	47.15	0	20
43	M124	X	47.15	47.15	0	15
44	M99	X	47.15	47.15	0	15
45	M98	X	47.15	47.15	0	15
46	M73	X	47.15	47.15	0	15
47	M74	X	47.15	47.15	0	15
48	M287	X	47.15	47.15	0	15
49	128	X	47.15	47.15	0	15
50	M123	X	47.15	47.15	0	15
51	M404	X	47.15	47.15	0	15
52	M288	X	47.15	47.15	0	15
53	M289	X	47.15	47.15	0	15
54	M402	X	47.15	47.15	0	15
55	M403	X	47.15	47.15	0	15
56	M149	X	47.15	47.15	0	15
57	M148	X	47.15	47.15	0	15
58	M405	X	47.15	47.15	0	15
59	127	X	47.15	47.15	0	15
60	M282	X	47.15	47.15	0	15
61	M391	X	47.15	47.15	0	20
62	M198	X	47.15	47.15	0	15
63	M174	X	47.15	47.15	0	15
64	M173	X	47.15	47.15	0	15
65	M49	X	47.15	47.15	0	15
66	M251	X	47.15	47.15	0	22.5
67	M250	X	47.15	47.15	0	22.5
68	M240	X	47.15	47.15	0	20
69	M239	X	47.15	47.15	0	20
70	M229	X	47.15	47.15	0	20
71	M284	X	47.15	47.15	0	15
72	M281	X	47.15	47.15	0	15
73	M264	X	47.15	47.15	0	15
74	M262	X	47.15	47.15	0	15
75	M398	X	47.15	47.15	0	22.5
76	M397	X	47.15	47.15	0	22.5
77	M395	X	47.15	47.15	0	20
78	M394	X	47.15	47.15	0	20
79	M199	X	47.15	47.15	0	15
80	M217	X	47.15	47.15	0	6.5
81	M228	X	47.15	47.15	0	20
82	M283	X	47.15	47.15	0	15
83	M285	X	47.15	47.15	0	15
84	M263	X	47.15	47.15	0	15
85	M218	X	47.15	47.15	0	6.5
86	M436	X	47.15	47.15	0	22.5
87	M456	X	39.1	39.1	0	3.17
88	M460	X	47.15	47.15	0	3.67

Member Distributed Loads (BLC 21 : BLC 5 Transient Area Loads) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
89	M457	X	47.15	47.15	0	22.5
90	M521	X	47.15	47.15	0	22.5
91	M525	X	15.333	15.333	0	5
92	M526	X	15.333	15.333	0	5
93	M527	X	15.333	15.333	0	5
94	M528	X	15.333	15.333	0	5
95	M530	X	31.203	31.203	0	2.5
96	M532	X	31.203	31.203	0	2.5
97	M464	X	15.333	15.333	0	3.73
98	M507	X	15.333	15.333	0	3.73
99	M524	X	15.333	15.333	0	3.73
100	M367	X	39.1	39.1	0	15
101	M368	X	39.1	39.1	0	15
102	M370	X	39.1	39.1	0	15
103	M371	X	39.1	39.1	0	15
104	M373	X	39.1	39.1	0	15
105	M374	X	39.1	39.1	0	15
106	M376	X	39.1	39.1	0	15
107	M377	X	39.1	39.1	0	15
108	M406	X	39.1	39.1	0	15
109	M408	X	39.1	39.1	0	15
110	M410	X	39.1	39.1	0	15
111	M411	X	39.1	39.1	0	15

Member Distributed Loads (BLC 22 : BLC 6 Transient Area Loads)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
1	M269	Z	39.1	39.1	0	13
2	M258	Z	39.1	39.1	0	13
3	M260	Z	39.1	39.1	0	13
4	M265	Z	39.1	39.1	0	13
5	M266	Z	39.1	39.1	0	13
6	M267	Z	39.1	39.1	0	13
7	M268	Z	39.1	39.1	0	13
8	M270	Z	39.1	39.1	0	13
9	M271	Z	39.1	39.1	0	13
10	M272	Z	39.1	39.1	0	13
11	M273	Z	39.1	39.1	0	13
12	M275	Z	39.1	39.1	0	13
13	M276	Z	39.1	39.1	0	13
14	M277	Z	39.1	39.1	0	13
15	M278	Z	39.1	39.1	0	13
16	M279	Z	39.1	39.1	0	13
17	M380	Z	39.1	39.1	0	13
18	M382	Z	39.1	39.1	0	13
19	M383	Z	39.1	39.1	0	13
20	M385	Z	39.1	39.1	0	13
21	M379	Z	39.1	39.1	0	13
22	M386	Z	39.1	39.1	0	13
23	M388	Z	39.1	39.1	0	13
24	M280	Z	39.1	39.1	0	13
25	M274	Z	39.1	39.1	0	13
26	M234	Z	46.383	46.383	0	24.9
27	M357	Z	46.383	46.383	2.351	27.351
28	M358	Z	46.383	46.383	2.351	27.351
29	M359	Z	46.383	46.383	2.351	27.351
30	M362	Z	46.383	46.383	2.351	27.351
31	M356	Z	46.383	46.383	2.351	27.351
32	M364	Z	46.383	46.383	2.351	27.351
33	M500	Z	46.383	46.383	2.351	27.351
34	M501	Z	46.383	46.383	2.351	27.351
35	M502	Z	46.383	46.383	2.351	27.351
36	M503	Z	46.383	46.383	2.351	27.351
37	M504	Z	46.383	46.383	2.351	27.351
38	M505	Z	46.383	46.383	2.351	27.351
39	M499	Z	46.383	46.383	2.351	27.351
40	M506	Z	46.383	46.383	2.351	27.351
41	M355	Z	46.383	46.383	2.351	27.351
42	M354	Z	46.383	46.383	2.351	27.351
43	M235	Z	46.383	46.383	0	24.9
44	M211	Z	46.383	46.383	2.351	27.351
45	M212	Z	46.383	46.383	2.351	27.351
46	M245	Z	46.383	46.383	0	24.9
47	M223	Z	46.383	46.383	0	24.9
48	M224	Z	46.383	46.383	0	24.9
49	M246	Z	46.383	46.383	0	24.9
50	M352	Z	46.383	46.383	2.351	27.351
51	M353	Z	46.383	46.383	2.351	27.351
52	M498	Z	47.15	47.15	0	3
53	M215	Z	47.15	47.15	0	13
54	M384	Z	47.15	47.15	0	13
55	M366	Z	47.15	47.15	0	13
56	M369	Z	47.15	47.15	0	13
57	M372	Z	47.15	47.15	0	13
58	M375	Z	47.15	47.15	0	13
59	M381	Z	47.15	47.15	0	13
60	M488	Z	39.1	39.1	0	13
61	M390	Z	47.15	47.15	0	13
62	M346	Z	47.15	47.15	0	13

Member Distributed Loads (BLC 22 : BLC 6 Transient Area Loads) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
63	M468	Z	47.15	47.15	0	13
64	M378	Z	47.15	47.15	0	13
65	M347	Z	47.15	47.15	0	13
66	M466	Z	47.15	47.15	0	13
67	M467	Z	47.15	47.15	0	13
68	M261	Z	47.15	47.15	0	13
69	M465	Z	47.15	47.15	0	13
70	M401	Z	39.1	39.1	0	13
71	M400	Z	39.1	39.1	0	13
72	M399	Z	39.1	39.1	0	13
73	M40	Z	47.15	47.15	0	13
74	M396	Z	47.15	47.15	0	13
75	M455	Z	39.1	39.1	0	4.33
76	M393	Z	47.15	47.15	0	13
77	M387	Z	47.15	47.15	0	13
78	M227	Z	47.15	47.15	0	13
79	M238	Z	47.15	47.15	0	13
80	M487	Z	39.1	39.1	0	13
81	M477	Z	47.15	47.15	0	13
82	M476	Z	47.15	47.15	0	13
83	M473	Z	47.15	47.15	0	13
84	M472	Z	47.15	47.15	0	13
85	M471	Z	47.15	47.15	0	13
86	M470	Z	47.15	47.15	0	13
87	M469	Z	47.15	47.15	0	13
88	M446	Z	47.15	47.15	0	5.09
89	M115	Z	47.15	47.15	0	13
90	M190	Z	47.15	47.15	0	13
91	M415	Z	47.15	47.15	0	13
92	M414	Z	47.15	47.15	0	13
93	M486	Z	39.1	39.1	0	13
94	M140	Z	47.15	47.15	0	13
95	90	Z	47.15	47.15	0	13
96	M165	Z	47.15	47.15	0	13
97	M461	Z	47.15	47.15	0	4.64
98	M259	Z	47.15	47.15	0	13
99	M257	Z	47.15	47.15	0	13
100	M459	Z	47.15	47.15	0	9.73
101	M249	Z	47.15	47.15	0	13
102	M458	Z	46	46	0	5.09
103	M65	Z	47.15	47.15	0	13
104	M90	Z	47.15	47.15	0	13
105	M337	Z	39.1	39.1	0	4.33
106	125	Z	47.15	47.15	0	13
107	M512	Z	39.1	39.1	0	13
108	M513	Z	39.1	39.1	0	13
109	M514	Z	39.1	39.1	0	13
110	M515	Z	39.1	39.1	0	13
111	M516	Z	39.1	39.1	0	13
112	M517	Z	39.1	39.1	0	13
113	M518	Z	39.1	39.1	0	13
114	M519	Z	39.1	39.1	0	13
115	M520	Z	47.15	47.15	0	3.75
116	M525	Z	15.333	15.333	0	5
117	M526	Z	15.333	15.333	0	5
118	M527	Z	15.333	15.333	0	5
119	M528	Z	15.333	15.333	0	5
120	M531	Z	31.203	31.203	0	4
121	M533	Z	31.203	31.203	0	4
122	M464	Z	15.333	15.333	0	3.73
123	M507	Z	15.333	15.333	0	3.73
124	M524	Z	15.333	15.333	0	3.73

Member Distributed Loads (BLC 23 : BLC 7 Transient Area Loads)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
1	M253	Y	-112.052	-103.585	12.857	15
2	M177	Y	-58.017	-58.017	9.659e-15	15
3	M303	Y	-0.413	-49.105	6	7.8
4	M303	Y	-49.105	-101.954	7.8	9.6
5	M303	Y	-101.954	-106.077	9.6	11.4
6	M303	Y	-106.077	-101.912	11.4	13.2
7	M303	Y	-101.912	-93.65	13.2	15
8	M360	Y	-0.808	-38.511	6	7.8
9	M360	Y	-38.511	-63.524	7.8	9.6
10	M360	Y	-63.524	-64.68	9.6	11.4
11	M360	Y	-64.68	-70.446	11.4	13.2
12	M360	Y	-70.446	-81.992	13.2	15
13	M363	Y	-48.304	-48.304	7.5	15
14	M333	Y	-1.142	-54.421	6	7.8
15	M333	Y	-54.421	-89.767	7.8	9.6
16	M333	Y	-89.767	-91.401	9.6	11.4
17	M333	Y	-91.401	-99.549	11.4	13.2
18	M333	Y	-99.549	-115.865	13.2	15
19	M334	Y	-0.584	-69.391	6	7.8
20	M334	Y	-69.391	-144.074	7.8	9.6
21	M334	Y	-144.074	-149.9	9.6	11.4
22	M334	Y	-149.9	-144.015	11.4	13.2
23	M334	Y	-144.015	-132.34	13.2	15

Member Distributed Loads (BLC 23 : BLC 7 Transient Area Loads) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
24	M332	Y	-68.259	-68.259	7.5	15
25	M330	Y	-93.652	-101.936	0	1.8
26	M330	Y	-101.936	-106.078	1.8	3.6
27	M330	Y	-106.078	-101.933	3.6	5.4
28	M330	Y	-101.933	-49.101	5.4	7.2
29	M330	Y	-49.101	-0.414	7.2	9
30	M331	Y	-81.994	-70.461	0	1.8
31	M331	Y	-70.461	-64.679	1.8	3.6
32	M331	Y	-64.679	-63.509	3.6	5.4
33	M331	Y	-63.509	-38.508	5.4	7.2
34	M331	Y	-38.508	-0.809	7.2	9
35	M329	Y	-48.304	-48.304	0.0001519	7.5
36	M315	Y	-100.272	-140.207	0	2.143
37	M315	Y	-140.207	-149.011	2.143	4.286
38	M315	Y	-149.011	-132.443	4.286	6.429
39	M315	Y	-132.443	-132.374	6.429	8.571
40	M315	Y	-132.374	-148.938	8.571	10.714
41	M315	Y	-148.938	-140.222	10.714	12.857
42	M315	Y	-140.222	-100.331	12.857	15
43	M318	Y	-75.387	-104.505	0	2.143
44	M318	Y	-104.505	-103.169	2.143	4.286
45	M318	Y	-103.169	-87.464	4.286	6.429
46	M318	Y	-87.464	-87.474	6.429	8.571
47	M318	Y	-87.474	-103.156	8.571	10.714
48	M318	Y	-103.156	-104.472	10.714	12.857
49	M318	Y	-104.472	-75.378	12.857	15
50	M325	Y	-115.865	-99.549	0	1.8
51	M325	Y	-99.549	-91.401	1.8	3.6
52	M325	Y	-91.401	-89.767	3.6	5.4
53	M325	Y	-89.767	-54.421	5.4	7.2
54	M325	Y	-54.421	-1.142	7.2	9
55	M310	Y	-75.387	-104.505	0	2.143
56	M310	Y	-104.505	-103.169	2.143	4.286
57	M310	Y	-103.169	-87.464	4.286	6.429
58	M310	Y	-87.464	-87.474	6.429	8.571
59	M310	Y	-87.474	-103.156	8.571	10.714
60	M310	Y	-103.156	-104.472	10.714	12.857
61	M310	Y	-104.472	-75.378	12.857	15
62	M323	Y	-132.34	-144.015	0	1.8
63	M323	Y	-144.015	-149.9	1.8	3.6
64	M323	Y	-149.9	-144.074	3.6	5.4
65	M323	Y	-144.074	-69.391	5.4	7.2
66	M323	Y	-69.391	-0.584	7.2	9
67	M309	Y	-100.272	-140.207	0	2.143
68	M309	Y	-140.207	-149.011	2.143	4.286
69	M309	Y	-149.011	-132.443	4.286	6.429
70	M309	Y	-132.443	-132.374	6.429	8.571
71	M309	Y	-132.374	-148.938	8.571	10.714
72	M309	Y	-148.938	-140.222	10.714	12.857
73	M309	Y	-140.222	-100.331	12.857	15
74	M306	Y	-1.136	-54.419	6	7.8
75	M306	Y	-54.419	-89.79	7.8	9.6
76	M306	Y	-89.79	-91.395	9.6	11.4
77	M306	Y	-91.395	-99.55	11.4	13.2
78	M306	Y	-99.55	-115.839	13.2	15
79	M305	Y	-0.587	-69.384	6	7.8
80	M305	Y	-69.384	-144.019	7.8	9.6
81	M305	Y	-144.019	-149.904	9.6	11.4
82	M305	Y	-149.904	-144.077	11.4	13.2
83	M305	Y	-144.077	-132.328	13.2	15
84	M299	Y	-65.885	-66.526	0	2.143
85	M299	Y	-66.526	-69.09	2.143	4.286
86	M299	Y	-69.09	-78.359	4.286	6.429
87	M299	Y	-78.359	-78.372	6.429	8.571
88	M299	Y	-78.372	-69.185	8.571	10.714
89	M299	Y	-69.185	-66.637	10.714	12.857
90	M299	Y	-66.637	-65.892	12.857	15
91	M324	Y	-68.259	-68.259	7.5	15
92	M317	Y	-68.259	-68.259	0.0001518	7.5
93	M292	Y	-65.885	-66.526	0	2.143
94	M292	Y	-66.526	-69.09	2.143	4.286
95	M292	Y	-69.09	-78.359	4.286	6.429
96	M292	Y	-78.359	-78.372	6.429	8.571
97	M292	Y	-78.372	-69.185	8.571	10.714
98	M292	Y	-69.185	-66.637	10.714	12.857
99	M292	Y	-66.637	-65.892	12.857	15
100	M306	Y	-81.994	-70.461	0	1.8
101	M306	Y	-70.461	-64.679	1.8	3.6
102	M306	Y	-64.679	-63.509	3.6	5.4
103	M306	Y	-63.509	-38.508	5.4	7.2
104	M306	Y	-38.508	-0.809	7.2	9
105	M305	Y	-93.652	-101.936	0	1.8
106	M305	Y	-101.936	-106.078	1.8	3.6
107	M305	Y	-106.078	-101.933	3.6	5.4
108	M305	Y	-101.933	-49.101	5.4	7.2
109	M305	Y	-49.101	-0.414	7.2	9
110	M324	Y	-48.304	-48.304	0.0001519	7.5
111	M304	Y	-132.34	-144.015	0	1.8

Member Distributed Loads (BLC 23 : BLC 7 Transient Area Loads) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
112	M304	Y	-144.015	-149.9	1.8	3.6
113	M304	Y	-149.9	-144.074	3.6	5.4
114	M304	Y	-144.074	-69.391	5.4	7.2
115	M304	Y	-69.391	-0.584	7.2	9
116	M361	Y	-115.865	-99.549	0	1.8
117	M361	Y	-99.549	-91.401	1.8	3.6
118	M361	Y	-91.401	-89.767	3.6	5.4
119	M361	Y	-89.767	-54.421	5.4	7.2
120	M361	Y	-54.421	-1.142	7.2	9
121	M365	Y	-68.26	-68.26	0.0001519	7.5
122	M509	Y	-123.627	-79.373	0	1.3
123	M509	Y	-79.373	-94.395	1.3	2.6
124	M509	Y	-94.395	-94.366	2.6	3.9
125	M509	Y	-94.366	-79.353	3.9	5.2
126	M509	Y	-79.353	-123.685	5.2	6.5
127	M510	Y	-100.585	-111.486	0	1.3
128	M510	Y	-111.486	-183.099	1.3	2.6
129	M510	Y	-183.099	-183.108	2.6	3.9
130	M510	Y	-183.108	-111.512	3.9	5.2
131	M510	Y	-111.512	-100.625	5.2	6.5
132	M511	Y	-68.264	-68.264	8.777e-5	6.5
133	M311	Y	-75.378	-104.472	0	2.143
134	M311	Y	-104.472	-103.156	2.143	4.286
135	M311	Y	-103.156	-87.474	4.286	6.429
136	M311	Y	-87.474	-87.464	6.429	8.571
137	M311	Y	-87.464	-103.169	8.571	10.714
138	M311	Y	-103.169	-104.505	10.714	12.857
139	M311	Y	-104.505	-75.387	12.857	15
140	M314	Y	-100.331	-140.222	0	2.143
141	M314	Y	-140.222	-148.938	2.143	4.286
142	M314	Y	-148.938	-132.374	4.286	6.429
143	M314	Y	-132.374	-132.443	6.429	8.571
144	M314	Y	-132.443	-149.011	8.571	10.714
145	M314	Y	-149.011	-140.207	10.714	12.857
146	M314	Y	-140.207	-100.272	12.857	15
147	M320	Y	-75.378	-104.472	0	2.143
148	M320	Y	-104.472	-103.156	2.143	4.286
149	M320	Y	-103.156	-87.474	4.286	6.429
150	M320	Y	-87.474	-87.464	6.429	8.571
151	M320	Y	-87.464	-103.169	8.571	10.714
152	M320	Y	-103.169	-104.505	10.714	12.857
153	M320	Y	-104.505	-75.387	12.857	15
154	M322	Y	-100.331	-140.222	0	2.143
155	M322	Y	-140.222	-148.938	2.143	4.286
156	M322	Y	-148.938	-132.374	4.286	6.429
157	M322	Y	-132.374	-132.443	6.429	8.571
158	M322	Y	-132.443	-149.011	8.571	10.714
159	M322	Y	-149.011	-140.207	10.714	12.857
160	M322	Y	-140.207	-100.272	12.857	15
161	M328	Y	-132.328	-144.077	0	1.8
162	M328	Y	-144.077	-149.904	1.8	3.6
163	M328	Y	-149.904	-144.019	3.6	5.4
164	M328	Y	-144.019	-69.384	5.4	7.2
165	M328	Y	-69.384	-0.587	7.2	9
166	M327	Y	-115.839	-99.55	0	1.8
167	M327	Y	-99.55	-91.395	1.8	3.6
168	M327	Y	-91.395	-89.79	3.6	5.4
169	M327	Y	-89.79	-54.419	5.4	7.2
170	M327	Y	-54.419	-1.136	7.2	9
171	M308	Y	-0.585	-69.386	6	7.8
172	M308	Y	-69.386	-144.044	7.8	9.6
173	M308	Y	-144.044	-149.901	9.6	11.4
174	M308	Y	-149.901	-144.049	11.4	13.2
175	M308	Y	-144.049	-132.342	13.2	15
176	M307	Y	-1.143	-54.416	6	7.8
177	M307	Y	-54.416	-89.747	7.8	9.6
178	M307	Y	-89.747	-91.4	9.6	11.4
179	M307	Y	-91.4	-99.57	11.4	13.2
180	M307	Y	-99.57	-115.868	13.2	15
181	M313	Y	-65.892	-66.637	0	2.143
182	M313	Y	-66.637	-69.185	2.143	4.286
183	M313	Y	-69.185	-78.372	4.286	6.429
184	M313	Y	-78.372	-78.359	6.429	8.571
185	M313	Y	-78.359	-69.09	8.571	10.714
186	M313	Y	-69.09	-66.526	10.714	12.857
187	M313	Y	-66.526	-65.885	12.857	15
188	M321	Y	-68.259	-68.259	5.061e-5	7.5
189	M326	Y	-68.259	-68.259	7.5	15
190	M295	Y	-65.892	-66.637	0	2.143
191	M295	Y	-66.637	-69.185	2.143	4.286
192	M295	Y	-69.185	-78.372	4.286	6.429
193	M295	Y	-78.372	-78.359	6.429	8.571
194	M295	Y	-78.359	-69.09	8.571	10.714
195	M295	Y	-69.09	-66.526	10.714	12.857
196	M295	Y	-66.526	-65.885	12.857	15
197	M308	Y	-93.642	-101.956	0	1.8
198	M308	Y	-101.956	-106.08	1.8	3.6
199	M308	Y	-106.08	-101.915	3.6	5.4

Member Distributed Loads (BLC 23 : BLC 7 Transient Area Loads) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
200	M308	Y	-101.915	-49.1	5.4	7.2
201	M308	Y	-49.1	-0.416	7.2	9
202	M307	Y	-81.974	-70.446	0	1.8
203	M307	Y	-70.446	-64.676	1.8	3.6
204	M307	Y	-64.676	-63.54	3.6	5.4
205	M307	Y	-63.54	-38.509	5.4	7.2
206	M307	Y	-38.509	-0.804	7.2	9
207	M326	Y	-48.304	-48.304	5.054e-5	7.5
208	M333	Y	-81.974	-70.446	0	1.8
209	M333	Y	-70.446	-64.676	1.8	3.6
210	M333	Y	-64.676	-63.54	3.6	5.4
211	M333	Y	-63.54	-38.509	5.4	7.2
212	M333	Y	-38.509	-0.804	7.2	9
213	M334	Y	-93.642	-101.956	0	1.8
214	M334	Y	-101.956	-106.08	1.8	3.6
215	M334	Y	-106.08	-101.915	3.6	5.4
216	M334	Y	-101.915	-49.1	5.4	7.2
217	M334	Y	-49.1	-0.416	7.2	9
218	M332	Y	-48.304	-48.304	5.054e-5	7.5
219	M303	Y	-132.331	-144.044	0	1.8
220	M303	Y	-144.044	-149.902	1.8	3.6
221	M303	Y	-149.902	-144.049	3.6	5.4
222	M303	Y	-144.049	-69.391	5.4	7.2
223	M303	Y	-69.391	-0.586	7.2	9
224	M360	Y	-115.838	-99.527	0	1.8
225	M360	Y	-99.527	-91.393	1.8	3.6
226	M360	Y	-91.393	-89.808	3.6	5.4
227	M360	Y	-89.808	-54.424	5.4	7.2
228	M360	Y	-54.424	-1.135	7.2	9
229	M363	Y	-68.26	-68.26	5.054e-5	7.5
230	M432	Y	-68.264	-68.264	2.926e-5	6.5
231	M433	Y	-100.565	-111.491	0	1.3
232	M433	Y	-111.491	-183.091	1.3	2.6
233	M433	Y	-183.091	-183.096	2.6	3.9
234	M433	Y	-183.096	-111.519	3.9	5.2
235	M433	Y	-111.519	-100.627	5.2	6.5
236	M508	Y	-123.551	-79.332	0	1.3
237	M508	Y	-79.332	-94.411	1.3	2.6
238	M508	Y	-94.411	-94.408	2.6	3.9
239	M508	Y	-94.408	-79.381	3.9	5.2
240	M508	Y	-79.381	-123.71	5.2	6.5
241	M340	Y	-112.665	-112.665	4.219e-15	15
242	M50	Y	-58.017	-58.017	4.219e-15	15
243	M344	Y	-66.073	-77.273	0	2.143
244	M344	Y	-77.273	-75.398	2.143	4.286
245	M344	Y	-75.398	-70.006	4.286	6.429
246	M344	Y	-70.006	-70.095	6.429	8.571
247	M344	Y	-70.095	-75.613	8.571	10.714
248	M344	Y	-75.613	-77.385	10.714	12.857
249	M344	Y	-77.385	-65.903	12.857	15
250	M345	Y	-114.997	-114.997	3.442e-15	15
251	M350	Y	-103.585	-112.052	0	2.143
252	M350	Y	-112.052	-116.111	2.143	4.286
253	M350	Y	-116.111	-121.659	4.286	6.429
254	M350	Y	-121.659	-121.585	6.429	8.571
255	M350	Y	-121.585	-115.908	8.571	10.714
256	M350	Y	-115.908	-111.833	10.714	12.857
257	M350	Y	-111.833	-103.45	12.857	15
258	M351	Y	-111.37	-111.37	3.997e-15	15
259	M297	Y	-114.997	-114.997	2.82e-14	15
260	M302	Y	-111.37	-111.37	0	15
261	M301	Y	-103.585	-112.052	0	2.143
262	M301	Y	-112.052	-116.111	2.143	4.286
263	M301	Y	-116.111	-121.659	4.286	6.429
264	M301	Y	-121.659	-121.585	6.429	8.571
265	M301	Y	-121.585	-115.908	8.571	10.714
266	M301	Y	-115.908	-111.833	10.714	12.857
267	M301	Y	-111.833	-103.45	12.857	15
268	M300	Y	-112.665	-112.665	1.921e-14	15
269	M296	Y	-66.073	-77.273	0	2.143
270	M296	Y	-77.273	-75.398	2.143	4.286
271	M296	Y	-75.398	-70.006	4.286	6.429
272	M296	Y	-70.006	-70.095	6.429	8.571
273	M296	Y	-70.095	-75.613	8.571	10.714
274	M296	Y	-75.613	-77.385	10.714	12.857
275	M296	Y	-77.385	-65.903	12.857	15
276	M175	Y	-58.017	-58.017	9.881e-15	15
277	M330	Y	-0.586	-69.391	6	7.8
278	M330	Y	-69.391	-144.049	7.8	9.6
279	M330	Y	-144.049	-149.902	9.6	11.4
280	M330	Y	-149.902	-144.044	11.4	13.2
281	M330	Y	-144.044	-132.331	13.2	15
282	M331	Y	-1.135	-54.424	6	7.8
283	M331	Y	-54.424	-89.808	7.8	9.6
284	M331	Y	-89.808	-91.393	9.6	11.4
285	M331	Y	-91.393	-99.527	11.4	13.2
286	M331	Y	-99.527	-115.838	13.2	15
287	M329	Y	-68.259	-68.259	7.5	15

Member Distributed Loads (BLC 23 : BLC 7 Transient Area Loads) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
288	M325	Y	-0.803	-38.513	6	7.8
289	M325	Y	-38.513	-63.553	7.8	9.6
290	M325	Y	-63.553	-64.674	9.6	11.4
291	M325	Y	-64.674	-70.431	11.4	13.2
292	M325	Y	-70.431	-81.973	13.2	15
293	M323	Y	-0.414	-49.104	6	7.8
294	M323	Y	-49.104	-101.937	7.8	9.6
295	M323	Y	-101.937	-106.078	9.6	11.4
296	M323	Y	-106.078	-101.933	11.4	13.2
297	M323	Y	-101.933	-93.644	13.2	15
298	M317	Y	-48.304	-48.304	7.5	15
299	M304	Y	-0.414	-49.104	6	7.8
300	M304	Y	-49.104	-101.937	7.8	9.6
301	M304	Y	-101.937	-106.078	9.6	11.4
302	M304	Y	-106.078	-101.933	11.4	13.2
303	M304	Y	-101.933	-93.644	13.2	15
304	M361	Y	-0.803	-38.513	6	7.8
305	M361	Y	-38.513	-63.553	7.8	9.6
306	M361	Y	-63.553	-64.674	9.6	11.4
307	M361	Y	-64.674	-70.43	11.4	13.2
308	M361	Y	-70.43	-81.973	13.2	15
309	M365	Y	-48.304	-48.304	7.5	15
310	M328	Y	-0.413	-49.105	6	7.8
311	M328	Y	-49.105	-101.954	7.8	9.6
312	M328	Y	-101.954	-106.077	9.6	11.4
313	M328	Y	-106.077	-101.912	11.4	13.2
314	M328	Y	-101.912	-93.65	13.2	15
315	M327	Y	-0.808	-38.511	6	7.8
316	M327	Y	-38.511	-63.524	7.8	9.6
317	M327	Y	-63.524	-64.68	9.6	11.4
318	M327	Y	-64.68	-70.446	11.4	13.2
319	M327	Y	-70.446	-81.992	13.2	15
320	M321	Y	-48.304	-48.304	7.5	15
321	M338	Y	-114.997	-114.997	0	15
322	M341	Y	-103.45	-111.833	0	2.143
323	M341	Y	-111.833	-115.908	2.143	4.286
324	M341	Y	-115.908	-121.585	4.286	6.429
325	M341	Y	-121.585	-121.659	6.429	8.571
326	M341	Y	-121.659	-116.111	8.571	10.714
327	M341	Y	-116.111	-112.052	10.714	12.857
328	M341	Y	-112.052	-103.585	12.857	15
329	M342	Y	-111.37	-111.37	4.552e-15	15
330	M343	Y	-65.903	-77.385	0	2.143
331	M343	Y	-77.385	-75.613	2.143	4.286
332	M343	Y	-75.613	-70.095	4.286	6.429
333	M343	Y	-70.095	-70.006	6.429	8.571
334	M343	Y	-70.006	-75.398	8.571	10.714
335	M343	Y	-75.398	-77.273	10.714	12.857
336	M343	Y	-77.273	-66.073	12.857	15
337	M339	Y	-112.665	-112.665	2.887e-15	15
338	M52	Y	-58.017	-58.017	2.109e-15	15
339	M293	Y	-65.903	-77.385	0	2.143
340	M293	Y	-77.385	-75.613	2.143	4.286
341	M293	Y	-75.613	-70.095	4.286	6.429
342	M293	Y	-70.095	-70.006	6.429	8.571
343	M293	Y	-70.006	-75.398	8.571	10.714
344	M293	Y	-75.398	-77.273	10.714	12.857
345	M293	Y	-77.273	-66.073	12.857	15
346	M256	Y	-114.997	-114.997	1.221e-15	15
347	M255	Y	-112.665	-112.665	1.943e-14	15
348	M254	Y	-111.37	-111.37	2.032e-14	15
349	M253	Y	-103.45	-111.833	0	2.143
350	M253	Y	-111.833	-115.908	2.143	4.286
351	M253	Y	-115.908	-121.585	4.286	6.429
352	M253	Y	-121.585	-121.659	6.429	8.571
353	M253	Y	-121.659	-116.111	8.571	10.714
354	M253	Y	-116.111	-112.052	10.714	12.857

Member Distributed Loads (BLC 24 : BLC 10 Transient Area Loads)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
1	M462	Y	-43.205	-43.205	0	2.049
2	M462	Y	-43.205	-43.205	2.049	4.097
3	M462	Y	-43.205	-43.205	4.097	6.146
4	M462	Y	-43.205	-43.205	6.146	8.194
5	M462	Y	-43.205	-40.178	8.194	10.243
6	M462	Y	-40.178	-37.151	10.243	12.291
7	M462	Y	-37.151	-37.151	12.291	14.34
8	M457	Y	-11.239	-30.907	6.75	9
9	M457	Y	-30.907	-40.742	9	11.25
10	M457	Y	-40.742	-40.742	11.25	13.5
11	M457	Y	-40.742	-40.742	13.5	15.75
12	M457	Y	-40.742	-47.82	15.75	18
13	M457	Y	-47.82	-45.064	18	20.25
14	M457	Y	-45.064	-25.396	20.25	22.5
15	M461	Y	-40.59	-40.59	0	4.64
16	M459	Y	-40.59	-40.59	5.09	9.73
17	M384	Y	-354.75	-354.75	0	13
18	M366	Y	-495	-495	1.443e-15	13

Member Distributed Loads (BLC 24 : BLC 10 Transient Area Loads) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
19	M369	Y	-495	-495	5.329e-15	13
20	M372	Y	-495	-495	4.441e-15	13
21	M375	Y	-495	-495	6.55e-15	13
22	M381	Y	-495	-495	0	13
23	M390	Y	-330	-330	5.551e-16	13
24	M346	Y	-495	-495	0	13
25	M378	Y	-495	-495	0	13
26	M347	Y	-247.5	-247.5	5.551e-16	13
27	M401	Y	-330	-330	5.551e-16	13
28	M400	Y	-330	-330	5.551e-16	13
29	M399	Y	-371.25	-371.25	5.551e-16	13
30	M396	Y	-185.625	-185.625	2.22e-16	13
31	M393	Y	-350.625	-350.625	5.551e-16	13
32	M387	Y	-272.25	-272.25	5.551e-16	13
33	M455	Y	-35.06	-35.06	0	4.33
34	M337	Y	-35.06	-35.06	0	4.33
35	M497	Y	-33.18	-33.18	5.551e-16	10
36	M496	Y	-33.18	-33.18	5.551e-16	10
37	M495	Y	-33.18	-33.18	5.551e-16	10
38	M494	Y	-33.18	-33.18	6.83	10
39	M239	Y	-34.119	-34.119	0	2
40	M239	Y	-34.119	-34.119	2	4
41	M239	Y	-34.119	-34.119	4	6
42	M239	Y	-34.119	-34.119	6	8
43	M239	Y	-34.119	-37.25	8	10
44	M239	Y	-37.25	-27.234	10	12
45	M239	Y	-27.234	-0.939	12	14
46	M228	Y	-33.18	-33.18	1.887e-15	20
47	M269	Y	-82.95	-82.95	2.22e-16	13
48	M258	Y	-82.95	-82.95	2.22e-16	13
49	M260	Y	-82.95	-82.95	2.22e-16	13
50	M265	Y	-82.95	-82.95	2.22e-16	13
51	M266	Y	-82.95	-82.95	2.22e-16	13
52	M267	Y	-82.95	-82.95	2.22e-16	13
53	M268	Y	-82.95	-82.95	2.22e-16	13
54	M270	Y	-82.95	-82.95	2.22e-16	13
55	M271	Y	-82.95	-82.95	2.22e-16	13
56	M272	Y	-82.95	-82.95	2.22e-16	13
57	M273	Y	-82.95	-82.95	2.22e-16	13
58	M275	Y	-82.95	-82.95	2.22e-16	13
59	M276	Y	-82.95	-82.95	2.22e-16	13
60	M277	Y	-82.95	-82.95	2.22e-16	13
61	M278	Y	-82.95	-82.95	2.22e-16	13
62	M279	Y	-82.95	-82.95	2.22e-16	13
63	M380	Y	-82.95	-82.95	2.22e-16	13
64	M382	Y	-82.95	-82.95	2.22e-16	13
65	M383	Y	-82.95	-82.95	2.22e-16	13
66	M385	Y	-82.95	-82.95	2.22e-16	13
67	M379	Y	-82.95	-82.95	2.22e-16	13
68	M386	Y	-82.95	-82.95	2.22e-16	13
69	M388	Y	-71.89	-71.89	5.551e-16	13
70	M280	Y	-82.95	-82.95	2.22e-16	13
71	M274	Y	-82.95	-82.95	2.22e-16	13
72	M215	Y	-35.945	-35.945	5.551e-16	13
73	M40	Y	-82.95	-82.95	2.22e-16	13
74	M115	Y	-82.95	-82.95	2.22e-16	13
75	M190	Y	-77.42	-77.42	2.22e-16	13
76	M140	Y	-82.95	-82.95	2.22e-16	13
77	90	Y	-82.95	-82.95	2.22e-16	13
78	M165	Y	-82.95	-82.95	2.22e-16	13
79	M65	Y	-82.95	-82.95	2.22e-16	13
80	M90	Y	-82.95	-82.95	2.22e-16	13
81	125	Y	-41.475	-41.475	5.551e-16	13
82	M488	Y	-221.2	-221.2	9.992e-16	13
83	M468	Y	-165.9	-165.9	6.661e-16	13
84	M466	Y	-165.9	-165.9	6.661e-16	13
85	M467	Y	-165.9	-165.9	6.661e-16	13
86	M465	Y	-154.84	-154.84	5.551e-16	13
87	M487	Y	-221.2	-221.2	9.992e-16	13
88	M477	Y	-182.49	-182.49	9.992e-16	13
89	M476	Y	-221.2	-221.2	9.992e-16	13
90	M473	Y	-82.95	-82.95	6.661e-16	13
91	M472	Y	-165.9	-165.9	6.661e-16	13
92	M471	Y	-165.9	-165.9	6.661e-16	13
93	M470	Y	-165.9	-165.9	6.661e-16	13
94	M469	Y	-165.9	-165.9	6.661e-16	13
95	M415	Y	-235.025	-235.025	2.22e-16	13
96	M414	Y	-124.425	-124.425	5.551e-16	13
97	M486	Y	-248.85	-248.85	5.551e-16	13
98	M512	Y	-165.9	-165.9	6.661e-16	13
99	M513	Y	-165.9	-165.9	6.661e-16	13
100	M514	Y	-165.9	-165.9	6.661e-16	13
101	M515	Y	-165.9	-165.9	6.661e-16	13
102	M516	Y	-165.9	-165.9	6.661e-16	13
103	M517	Y	-165.9	-165.9	6.661e-16	13
104	M518	Y	-165.9	-165.9	6.661e-16	13
105	M519	Y	-165.9	-165.9	6.661e-16	13
106	M340	Y	-96.222	-96.222	4.219e-15	15

Member Distributed Loads (BLC 24 : BLC 10 Transient Area Loads) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
107	M50	Y	-49.55	-49.55	4.219e-15	15
108	M344	Y	-56.43	-65.996	0	2.143
109	M344	Y	-65.996	-64.394	2.143	4.286
110	M344	Y	-64.394	-59.789	4.286	6.429
111	M344	Y	-59.789	-59.865	6.429	8.571
112	M344	Y	-59.865	-64.578	8.571	10.714
113	M344	Y	-64.578	-66.091	10.714	12.857
114	M344	Y	-66.091	-56.285	12.857	15
115	M345	Y	-98.214	-98.214	3.442e-15	15
116	M350	Y	-88.467	-95.698	0	2.143
117	M350	Y	-95.698	-99.165	2.143	4.286
118	M350	Y	-99.165	-103.903	4.286	6.429
119	M350	Y	-103.903	-103.841	6.429	8.571
120	M350	Y	-103.841	-98.992	8.571	10.714
121	M350	Y	-98.992	-95.511	10.714	12.857
122	M350	Y	-95.511	-88.352	12.857	15
123	M351	Y	-95.116	-95.116	3.997e-15	15
124	M297	Y	-98.214	-98.214	2.82e-14	15
125	M302	Y	-95.116	-95.116	0	15
126	M301	Y	-88.467	-95.698	0	2.143
127	M301	Y	-95.698	-99.165	2.143	4.286
128	M301	Y	-99.165	-103.903	4.286	6.429
129	M301	Y	-103.903	-103.841	6.429	8.571
130	M301	Y	-103.841	-98.992	8.571	10.714
131	M301	Y	-98.992	-95.511	10.714	12.857
132	M301	Y	-95.511	-88.352	12.857	15
133	M300	Y	-96.222	-96.222	1.921e-14	15
134	M296	Y	-56.43	-65.996	0	2.143
135	M296	Y	-65.996	-64.394	2.143	4.286
136	M296	Y	-64.394	-59.789	4.286	6.429
137	M296	Y	-59.789	-59.865	6.429	8.571
138	M296	Y	-59.865	-64.578	8.571	10.714
139	M296	Y	-64.578	-66.091	10.714	12.857
140	M296	Y	-66.091	-56.285	12.857	15
141	M175	Y	-49.55	-49.55	9.881e-15	15
142	M303	Y	-60.637	-84.746	0	2.143
143	M303	Y	-84.746	-90.014	2.143	4.286
144	M303	Y	-90.014	-80.003	4.286	6.429
145	M303	Y	-80.003	-80.045	6.429	8.571
146	M303	Y	-80.045	-90.058	8.571	10.714
147	M303	Y	-90.058	-84.737	10.714	12.857
148	M303	Y	-84.737	-60.602	12.857	15
149	M360	Y	-45.556	-63.14	0	2.143
150	M360	Y	-63.14	-62.344	2.143	4.286
151	M360	Y	-62.344	-52.866	4.286	6.429
152	M360	Y	-52.866	-52.861	6.429	8.571
153	M360	Y	-52.861	-62.352	8.571	10.714
154	M360	Y	-62.352	-63.16	10.714	12.857
155	M360	Y	-63.16	-45.562	12.857	15
156	M363	Y	-39.823	-40.274	0	2.143
157	M363	Y	-40.274	-41.813	2.143	4.286
158	M363	Y	-41.813	-47.366	4.286	6.429
159	M363	Y	-47.366	-47.358	6.429	8.571
160	M363	Y	-47.358	-41.756	8.571	10.714
161	M363	Y	-41.756	-40.206	10.714	12.857
162	M363	Y	-40.206	-39.819	12.857	15
163	M311	Y	-45.556	-63.14	0	2.143
164	M311	Y	-63.14	-62.344	2.143	4.286
165	M311	Y	-62.344	-52.867	4.286	6.429
166	M311	Y	-52.867	-52.861	6.429	8.571
167	M311	Y	-52.861	-62.352	8.571	10.714
168	M311	Y	-62.352	-63.16	10.714	12.857
169	M311	Y	-63.16	-45.562	12.857	15
170	M314	Y	-60.637	-84.746	0	2.143
171	M314	Y	-84.746	-90.014	2.143	4.286
172	M314	Y	-90.014	-80.003	4.286	6.429
173	M314	Y	-80.003	-80.045	6.429	8.571
174	M314	Y	-80.045	-90.058	8.571	10.714
175	M314	Y	-90.058	-84.737	10.714	12.857
176	M314	Y	-84.737	-60.602	12.857	15
177	M320	Y	-45.556	-63.14	0	2.143
178	M320	Y	-63.14	-62.344	2.143	4.286
179	M320	Y	-62.344	-52.867	4.286	6.429
180	M320	Y	-52.867	-52.861	6.429	8.571
181	M320	Y	-52.861	-62.352	8.571	10.714
182	M320	Y	-62.352	-63.16	10.714	12.857
183	M320	Y	-63.16	-45.562	12.857	15
184	M322	Y	-60.637	-84.746	0	2.143
185	M322	Y	-84.746	-90.014	2.143	4.286
186	M322	Y	-90.014	-80.003	4.286	6.429
187	M322	Y	-80.003	-80.045	6.429	8.571
188	M322	Y	-80.045	-90.058	8.571	10.714
189	M322	Y	-90.058	-84.737	10.714	12.857
190	M322	Y	-84.737	-60.602	12.857	15
191	M328	Y	-60.637	-84.746	0	2.143
192	M328	Y	-84.746	-90.014	2.143	4.286
193	M328	Y	-90.014	-80.003	4.286	6.429
194	M328	Y	-80.003	-80.045	6.429	8.571

Member Distributed Loads (BLC 24 : BLC 10 Transient Area Loads) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
195	M328	Y	-80.045	-90.058	8.571	10.714
196	M328	Y	-90.058	-84.737	10.714	12.857
197	M328	Y	-84.737	-60.602	12.857	15
198	M327	Y	-45.556	-63.14	0	2.143
199	M327	Y	-63.14	-62.344	2.143	4.286
200	M327	Y	-62.344	-52.867	4.286	6.429
201	M327	Y	-52.867	-52.861	6.429	8.571
202	M327	Y	-52.861	-62.352	8.571	10.714
203	M327	Y	-62.352	-63.16	10.714	12.857
204	M327	Y	-63.16	-45.562	12.857	15
205	M308	Y	-60.637	-84.746	0	2.143
206	M308	Y	-84.746	-90.014	2.143	4.286
207	M308	Y	-90.014	-80.003	4.286	6.429
208	M308	Y	-80.003	-80.045	6.429	8.571
209	M308	Y	-80.045	-90.058	8.571	10.714
210	M308	Y	-90.058	-84.737	10.714	12.857
211	M308	Y	-84.737	-60.602	12.857	15
212	M307	Y	-45.556	-63.14	0	2.143
213	M307	Y	-63.14	-62.344	2.143	4.286
214	M307	Y	-62.344	-52.866	4.286	6.429
215	M307	Y	-52.866	-52.861	6.429	8.571
216	M307	Y	-52.861	-62.352	8.571	10.714
217	M307	Y	-62.352	-63.16	10.714	12.857
218	M307	Y	-63.16	-45.562	12.857	15
219	M313	Y	-39.823	-40.273	0	2.143
220	M313	Y	-40.273	-41.813	2.143	4.286
221	M313	Y	-41.813	-47.366	4.286	6.429
222	M313	Y	-47.366	-47.358	6.429	8.571
223	M313	Y	-47.358	-41.756	8.571	10.714
224	M313	Y	-41.756	-40.206	10.714	12.857
225	M313	Y	-40.206	-39.819	12.857	15
226	M321	Y	-39.823	-40.273	0	2.143
227	M321	Y	-40.273	-41.813	2.143	4.286
228	M321	Y	-41.813	-47.366	4.286	6.429
229	M321	Y	-47.366	-47.358	6.429	8.571
230	M321	Y	-47.358	-41.756	8.571	10.714
231	M321	Y	-41.756	-40.206	10.714	12.857
232	M321	Y	-40.206	-39.819	12.857	15
233	M326	Y	-39.823	-40.274	0	2.143
234	M326	Y	-40.274	-41.813	2.143	4.286
235	M326	Y	-41.813	-47.366	4.286	6.429
236	M326	Y	-47.366	-47.358	6.429	8.571
237	M326	Y	-47.358	-41.756	8.571	10.714
238	M326	Y	-41.756	-40.206	10.714	12.857
239	M326	Y	-40.206	-39.819	12.857	15
240	M295	Y	-39.823	-40.273	0	2.143
241	M295	Y	-40.273	-41.813	2.143	4.286
242	M295	Y	-41.813	-47.366	4.286	6.429
243	M295	Y	-47.366	-47.358	6.429	8.571
244	M295	Y	-47.358	-41.756	8.571	10.714
245	M295	Y	-41.756	-40.206	10.714	12.857
246	M295	Y	-40.206	-39.819	12.857	15
247	M333	Y	-45.556	-63.14	0	2.143
248	M333	Y	-63.14	-62.344	2.143	4.286
249	M333	Y	-62.344	-52.866	4.286	6.429
250	M333	Y	-52.866	-52.861	6.429	8.571
251	M333	Y	-52.861	-62.352	8.571	10.714
252	M333	Y	-62.352	-63.16	10.714	12.857
253	M333	Y	-63.16	-45.562	12.857	15
254	M334	Y	-60.637	-84.746	0	2.143
255	M334	Y	-84.746	-90.014	2.143	4.286
256	M334	Y	-90.014	-80.003	4.286	6.429
257	M334	Y	-80.003	-80.045	6.429	8.571
258	M334	Y	-80.045	-90.058	8.571	10.714
259	M334	Y	-90.058	-84.737	10.714	12.857
260	M334	Y	-84.737	-60.602	12.857	15
261	M332	Y	-39.823	-40.274	0	2.143
262	M332	Y	-40.274	-41.813	2.143	4.286
263	M332	Y	-41.813	-47.366	4.286	6.429
264	M332	Y	-47.366	-47.358	6.429	8.571
265	M332	Y	-47.358	-41.756	8.571	10.714
266	M332	Y	-41.756	-40.206	10.714	12.857
267	M332	Y	-40.206	-39.819	12.857	15
268	M304	Y	-60.602	-84.737	0	2.143
269	M304	Y	-84.737	-90.058	2.143	4.286
270	M304	Y	-90.058	-80.045	4.286	6.429
271	M304	Y	-80.045	-80.003	6.429	8.571
272	M304	Y	-80.003	-90.014	8.571	10.714
273	M304	Y	-90.014	-84.746	10.714	12.857
274	M304	Y	-84.746	-60.637	12.857	15
275	M361	Y	-45.562	-63.16	0	2.143
276	M361	Y	-63.16	-62.352	2.143	4.286
277	M361	Y	-62.352	-52.861	4.286	6.429
278	M361	Y	-52.861	-52.866	6.429	8.571
279	M361	Y	-52.866	-62.344	8.571	10.714
280	M361	Y	-62.344	-63.14	10.714	12.857
281	M361	Y	-63.14	-45.556	12.857	15
282	M365	Y	-39.819	-40.206	0	2.143

Member Distributed Loads (BLC 24 : BLC 10 Transient Area Loads) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
283	M365	Y	-40.206	-41.756	2.143	4.286
284	M365	Y	-41.756	-47.358	4.286	6.429
285	M365	Y	-47.358	-47.366	6.429	8.571
286	M365	Y	-47.366	-41.813	8.571	10.714
287	M365	Y	-41.813	-40.274	10.714	12.857
288	M365	Y	-40.274	-39.823	12.857	15
289	M338	Y	-98.214	-98.214	0	15
290	M341	Y	-88.352	-95.511	0	2.143
291	M341	Y	-95.511	-98.992	2.143	4.286
292	M341	Y	-98.992	-103.841	4.286	6.429
293	M341	Y	-103.841	-103.903	6.429	8.571
294	M341	Y	-103.903	-99.165	8.571	10.714
295	M341	Y	-99.165	-95.698	10.714	12.857
296	M341	Y	-95.698	-88.467	12.857	15
297	M342	Y	-95.116	-95.116	4.552e-15	15
298	M343	Y	-56.285	-66.091	0	2.143
299	M343	Y	-66.091	-64.578	2.143	4.286
300	M343	Y	-64.578	-59.865	4.286	6.429
301	M343	Y	-59.865	-59.789	6.429	8.571
302	M343	Y	-59.789	-64.394	8.571	10.714
303	M343	Y	-64.394	-65.996	10.714	12.857
304	M343	Y	-65.996	-56.43	12.857	15
305	M339	Y	-96.222	-96.222	2.887e-15	15
306	M52	Y	-49.55	-49.55	2.109e-15	15
307	M315	Y	-60.637	-84.746	0	2.143
308	M315	Y	-84.746	-90.014	2.143	4.286
309	M315	Y	-90.014	-80.003	4.286	6.429
310	M315	Y	-80.003	-80.045	6.429	8.571
311	M315	Y	-80.045	-90.058	8.571	10.714
312	M315	Y	-90.058	-84.737	10.714	12.857
313	M315	Y	-84.737	-60.602	12.857	15
314	M318	Y	-45.556	-63.14	0	2.143
315	M318	Y	-63.14	-62.344	2.143	4.286
316	M318	Y	-62.344	-52.867	4.286	6.429
317	M318	Y	-52.867	-52.861	6.429	8.571
318	M318	Y	-52.861	-62.352	8.571	10.714
319	M318	Y	-62.352	-63.16	10.714	12.857
320	M318	Y	-63.16	-45.562	12.857	15
321	M325	Y	-45.556	-63.14	0	2.143
322	M325	Y	-63.14	-62.344	2.143	4.286
323	M325	Y	-62.344	-52.867	4.286	6.429
324	M325	Y	-52.867	-52.861	6.429	8.571
325	M325	Y	-52.861	-62.352	8.571	10.714
326	M325	Y	-62.352	-63.16	10.714	12.857
327	M325	Y	-63.16	-45.562	12.857	15
328	M310	Y	-45.556	-63.14	0	2.143
329	M310	Y	-63.14	-62.344	2.143	4.286
330	M310	Y	-62.344	-52.867	4.286	6.429
331	M310	Y	-52.867	-52.861	6.429	8.571
332	M310	Y	-52.861	-62.352	8.571	10.714
333	M310	Y	-62.352	-63.16	10.714	12.857
334	M310	Y	-63.16	-45.562	12.857	15
335	M323	Y	-60.637	-84.746	0	2.143
336	M323	Y	-84.746	-90.014	2.143	4.286
337	M323	Y	-90.014	-80.003	4.286	6.429
338	M323	Y	-80.003	-80.045	6.429	8.571
339	M323	Y	-80.045	-90.058	8.571	10.714
340	M323	Y	-90.058	-84.737	10.714	12.857
341	M323	Y	-84.737	-60.602	12.857	15
342	M309	Y	-60.637	-84.746	0	2.143
343	M309	Y	-84.746	-90.014	2.143	4.286
344	M309	Y	-90.014	-80.003	4.286	6.429
345	M309	Y	-80.003	-80.045	6.429	8.571
346	M309	Y	-80.045	-90.058	8.571	10.714
347	M309	Y	-90.058	-84.737	10.714	12.857
348	M309	Y	-84.737	-60.602	12.857	15
349	M306	Y	-45.556	-63.14	0	2.143
350	M306	Y	-63.14	-62.344	2.143	4.286
351	M306	Y	-62.344	-52.866	4.286	6.429
352	M306	Y	-52.866	-52.861	6.429	8.571
353	M306	Y	-52.861	-62.352	8.571	10.714
354	M306	Y	-62.352	-63.16	10.714	12.857
355	M306	Y	-63.16	-45.562	12.857	15
356	M305	Y	-60.637	-84.746	0	2.143
357	M305	Y	-84.746	-90.014	2.143	4.286
358	M305	Y	-90.014	-80.003	4.286	6.429
359	M305	Y	-80.003	-80.045	6.429	8.571
360	M305	Y	-80.045	-90.058	8.571	10.714
361	M305	Y	-90.058	-84.737	10.714	12.857
362	M305	Y	-84.737	-60.602	12.857	15
363	M299	Y	-39.823	-40.273	0	2.143
364	M299	Y	-40.273	-41.813	2.143	4.286
365	M299	Y	-41.813	-47.366	4.286	6.429
366	M299	Y	-47.366	-47.358	6.429	8.571
367	M299	Y	-47.358	-41.756	8.571	10.714
368	M299	Y	-41.756	-40.206	10.714	12.857
369	M299	Y	-40.206	-39.819	12.857	15
370	M324	Y	-39.823	-40.274	0	2.143

Member Distributed Loads (BLC 24 : BLC 10 Transient Area Loads) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
371	M324	Y	-40.274	-41.813	2.143	4.286
372	M324	Y	-41.813	-47.366	4.286	6.429
373	M324	Y	-47.366	-47.358	6.429	8.571
374	M324	Y	-47.358	-41.756	8.571	10.714
375	M324	Y	-41.756	-40.206	10.714	12.857
376	M324	Y	-40.206	-39.819	12.857	15
377	M317	Y	-39.823	-40.273	0	2.143
378	M317	Y	-40.273	-41.813	2.143	4.286
379	M317	Y	-41.813	-47.366	4.286	6.429
380	M317	Y	-47.366	-47.358	6.429	8.571
381	M317	Y	-47.358	-41.756	8.571	10.714
382	M317	Y	-41.756	-40.206	10.714	12.857
383	M317	Y	-40.206	-39.819	12.857	15
384	M292	Y	-39.823	-40.273	0	2.143
385	M292	Y	-40.273	-41.813	2.143	4.286
386	M292	Y	-41.813	-47.366	4.286	6.429
387	M292	Y	-47.366	-47.358	6.429	8.571
388	M292	Y	-47.358	-41.756	8.571	10.714
389	M292	Y	-41.756	-40.206	10.714	12.857
390	M292	Y	-40.206	-39.819	12.857	15
391	M293	Y	-56.285	-66.091	0	2.143
392	M293	Y	-66.091	-64.578	2.143	4.286
393	M293	Y	-64.578	-59.865	4.286	6.429
394	M293	Y	-59.865	-59.789	6.429	8.571
395	M293	Y	-59.789	-64.394	8.571	10.714
396	M293	Y	-64.394	-65.996	10.714	12.857
397	M293	Y	-65.996	-56.43	12.857	15
398	M256	Y	-98.214	-98.214	1.221e-15	15
399	M255	Y	-96.222	-96.222	1.943e-14	15
400	M254	Y	-95.116	-95.116	2.032e-14	15
401	M253	Y	-88.352	-95.511	0	2.143
402	M253	Y	-95.511	-98.992	2.143	4.286
403	M253	Y	-98.992	-103.841	4.286	6.429
404	M253	Y	-103.841	-103.903	6.429	8.571
405	M253	Y	-103.903	-99.165	8.571	10.714
406	M253	Y	-99.165	-95.698	10.714	12.857
407	M253	Y	-95.698	-88.467	12.857	15
408	M177	Y	-49.55	-49.55	9.659e-15	15
409	M330	Y	-60.637	-84.746	0	2.143
410	M330	Y	-84.746	-90.014	2.143	4.286
411	M330	Y	-90.014	-80.003	4.286	6.429
412	M330	Y	-80.003	-80.045	6.429	8.571
413	M330	Y	-80.045	-90.058	8.571	10.714
414	M330	Y	-90.058	-84.737	10.714	12.857
415	M330	Y	-84.737	-60.602	12.857	15
416	M331	Y	-45.556	-63.14	0	2.143
417	M331	Y	-63.14	-62.344	2.143	4.286
418	M331	Y	-62.344	-52.866	4.286	6.429
419	M331	Y	-52.866	-52.861	6.429	8.571
420	M331	Y	-52.861	-62.352	8.571	10.714
421	M331	Y	-62.352	-63.16	10.714	12.857
422	M331	Y	-63.16	-45.562	12.857	15
423	M329	Y	-39.823	-40.274	0	2.143
424	M329	Y	-40.274	-41.813	2.143	4.286
425	M329	Y	-41.813	-47.366	4.286	6.429
426	M329	Y	-47.366	-47.358	6.429	8.571
427	M329	Y	-47.358	-41.756	8.571	10.714
428	M329	Y	-41.756	-40.206	10.714	12.857
429	M329	Y	-40.206	-39.819	12.857	15
430	M432	Y	-41.257	-41.257	2.926e-5	6.5
431	M433	Y	-60.815	-67.395	0	1.3
432	M433	Y	-67.395	-110.665	1.3	2.6
433	M433	Y	-110.665	-110.66	2.6	3.9
434	M433	Y	-110.66	-67.379	3.9	5.2
435	M433	Y	-67.379	-60.791	5.2	6.5
436	M508	Y	-74.752	-47.959	0	1.3
437	M508	Y	-47.959	-57.032	1.3	2.6
438	M508	Y	-57.032	-57.05	2.6	3.9
439	M508	Y	-57.05	-47.971	3.9	5.2
440	M508	Y	-47.971	-74.717	5.2	6.5
441	M509	Y	-74.767	-47.975	0	1.3
442	M509	Y	-47.975	-57.058	1.3	2.6
443	M509	Y	-57.058	-57.06	2.6	3.9
444	M509	Y	-57.06	-47.946	3.9	5.2
445	M509	Y	-47.946	-74.671	5.2	6.5
446	M510	Y	-60.816	-67.399	0	1.3
447	M510	Y	-67.399	-110.658	1.3	2.6
448	M510	Y	-110.658	-110.655	2.6	3.9
449	M510	Y	-110.655	-67.382	3.9	5.2
450	M510	Y	-67.382	-60.779	5.2	6.5
451	M511	Y	-41.257	-41.257	8.777e-5	6.5

Member Distributed Loads (BLC 25 : BLC 11 Transient Area Loads)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
1	M462	Y	-39.064	-39.064	0	2.049
2	M462	Y	-39.064	-39.064	2.049	4.097
3	M462	Y	-39.064	-39.064	4.097	6.146
4	M462	Y	-39.064	-39.064	6.146	8.194

Member Distributed Loads (BLC 25 : BLC 11 Transient Area Loads) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
5	M462	Y	-39.064	-36.327	8.194	10.243
6	M462	Y	-36.327	-33.59	10.243	12.291
7	M462	Y	-33.59	-33.59	12.291	14.34
8	M457	Y	-10.162	-27.945	6.75	9
9	M457	Y	-27.945	-36.837	9	11.25
10	M457	Y	-36.837	-36.837	11.25	13.5
11	M457	Y	-36.837	-36.837	13.5	15.75
12	M457	Y	-36.837	-43.237	15.75	18
13	M457	Y	-43.237	-40.745	18	20.25
14	M457	Y	-40.745	-22.962	20.25	22.5
15	M461	Y	-36.7	-36.7	0	4.64
16	M459	Y	-36.7	-36.7	5.09	9.73
17	M239	Y	-21.65	-21.65	10.792	12.378
18	M456	Y	-21.65	-21.65	0.793	2.378
19	M455	Y	-9.906	-25.756	0	1.082
20	M455	Y	-25.756	-33.681	1.082	2.165
21	M455	Y	-33.681	-25.756	2.165	3.247
22	M455	Y	-25.756	-9.906	3.247	4.33
23	M337	Y	-9.906	-25.756	0	1.082
24	M337	Y	-25.756	-33.681	1.082	2.165
25	M337	Y	-33.681	-25.756	2.165	3.247
26	M337	Y	-25.756	-9.906	3.247	4.33
27	M497	Y	-30	-30	5.551e-16	10
28	M496	Y	-30	-30	5.551e-16	10
29	M495	Y	-30	-30	5.551e-16	10
30	M494	Y	-30	-30	6.83	10
31	M239	Y	-30.849	-30.849	0	2
32	M239	Y	-30.849	-30.849	2	4
33	M239	Y	-30.849	-30.849	4	6
34	M239	Y	-30.849	-30.849	6	8
35	M239	Y	-30.849	-33.68	8	10
36	M239	Y	-33.68	-24.624	10	12
37	M239	Y	-24.624	-0.849	12	14
38	M228	Y	-30	-30	1.887e-15	20
39	M269	Y	-75	-75	2.22e-16	13
40	M258	Y	-75	-75	2.22e-16	13
41	M260	Y	-75	-75	2.22e-16	13
42	M265	Y	-75	-75	2.22e-16	13
43	M266	Y	-75	-75	2.22e-16	13
44	M267	Y	-75	-75	2.22e-16	13
45	M268	Y	-75	-75	2.22e-16	13
46	M270	Y	-75	-75	2.22e-16	13
47	M271	Y	-75	-75	2.22e-16	13
48	M272	Y	-75	-75	2.22e-16	13
49	M273	Y	-75	-75	2.22e-16	13
50	M275	Y	-75	-75	2.22e-16	13
51	M276	Y	-75	-75	2.22e-16	13
52	M277	Y	-75	-75	2.22e-16	13
53	M278	Y	-75	-75	2.22e-16	13
54	M279	Y	-75	-75	2.22e-16	13
55	M380	Y	-75	-75	2.22e-16	13
56	M382	Y	-75	-75	2.22e-16	13
57	M383	Y	-75	-75	2.22e-16	13
58	M385	Y	-75	-75	2.22e-16	13
59	M379	Y	-75	-75	2.22e-16	13
60	M386	Y	-75	-75	2.22e-16	13
61	M388	Y	-65	-65	5.551e-16	13
62	M280	Y	-75	-75	2.22e-16	13
63	M274	Y	-75	-75	2.22e-16	13
64	M215	Y	-32.5	-32.5	5.551e-16	13
65	M40	Y	-75	-75	2.22e-16	13
66	M115	Y	-75	-75	2.22e-16	13
67	M190	Y	-70	-70	5.551e-16	13
68	M140	Y	-75	-75	2.22e-16	13
69	90	Y	-75	-75	2.22e-16	13
70	M165	Y	-75	-75	2.22e-16	13
71	M65	Y	-75	-75	2.22e-16	13
72	M90	Y	-75	-75	2.22e-16	13
73	125	Y	-37.5	-37.5	6.661e-16	13
74	M340	Y	-104.4	-104.4	4.219e-15	15
75	M50	Y	-53.761	-53.761	4.219e-15	15
76	M344	Y	-61.226	-71.605	0	2.143
77	M344	Y	-71.605	-69.867	2.143	4.286
78	M344	Y	-69.867	-64.871	4.286	6.429
79	M344	Y	-64.871	-64.953	6.429	8.571
80	M344	Y	-64.953	-70.066	8.571	10.714
81	M344	Y	-70.066	-71.708	10.714	12.857
82	M344	Y	-71.708	-61.068	12.857	15
83	M345	Y	-106.561	-106.561	3.442e-15	15
84	M350	Y	-95.986	-103.832	0	2.143
85	M350	Y	-103.832	-107.594	2.143	4.286
86	M350	Y	-107.594	-112.734	4.286	6.429
87	M350	Y	-112.734	-112.666	6.429	8.571
88	M350	Y	-112.666	-107.405	8.571	10.714
89	M350	Y	-107.405	-103.629	10.714	12.857
90	M350	Y	-103.629	-95.861	12.857	15
91	M351	Y	-103.2	-103.2	3.997e-15	15
92	M297	Y	-106.561	-106.561	2.82e-14	15

Member Distributed Loads (BLC 25 : BLC 11 Transient Area Loads) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
93	M302	Y	-103.2	-103.2	0	15
94	M301	Y	-95.986	-103.832	0	2.143
95	M301	Y	-103.832	-107.594	2.143	4.286
96	M301	Y	-107.594	-112.734	4.286	6.429
97	M301	Y	-112.734	-112.666	6.429	8.571
98	M301	Y	-112.666	-107.405	8.571	10.714
99	M301	Y	-107.405	-103.629	10.714	12.857
100	M301	Y	-103.629	-95.861	12.857	15
101	M300	Y	-104.4	-104.4	1.921e-14	15
102	M296	Y	-61.226	-71.605	0	2.143
103	M296	Y	-71.605	-69.867	2.143	4.286
104	M296	Y	-69.867	-64.871	4.286	6.429
105	M296	Y	-64.871	-64.953	6.429	8.571
106	M296	Y	-64.953	-70.066	8.571	10.714
107	M296	Y	-70.066	-71.708	10.714	12.857
108	M296	Y	-71.708	-61.068	12.857	15
109	M175	Y	-53.761	-53.761	9.881e-15	15
110	M303	Y	-65.791	-91.949	0	2.143
111	M303	Y	-91.949	-97.664	2.143	4.286
112	M303	Y	-97.664	-86.802	4.286	6.429
113	M303	Y	-86.802	-86.848	6.429	8.571
114	M303	Y	-86.848	-97.712	8.571	10.714
115	M303	Y	-97.712	-91.939	10.714	12.857
116	M303	Y	-91.939	-65.752	12.857	15
117	M360	Y	-49.428	-68.506	0	2.143
118	M360	Y	-68.506	-67.643	2.143	4.286
119	M360	Y	-67.643	-57.36	4.286	6.429
120	M360	Y	-57.36	-57.354	6.429	8.571
121	M360	Y	-57.354	-67.652	8.571	10.714
122	M360	Y	-67.652	-68.528	10.714	12.857
123	M360	Y	-68.528	-49.434	12.857	15
124	M363	Y	-43.208	-43.696	0	2.143
125	M363	Y	-43.696	-45.367	2.143	4.286
126	M363	Y	-45.367	-51.392	4.286	6.429
127	M363	Y	-51.392	-51.383	6.429	8.571
128	M363	Y	-51.383	-45.305	8.571	10.714
129	M363	Y	-45.305	-43.623	10.714	12.857
130	M363	Y	-43.623	-43.203	12.857	15
131	M311	Y	-49.428	-68.506	0	2.143
132	M311	Y	-68.506	-67.643	2.143	4.286
133	M311	Y	-67.643	-57.36	4.286	6.429
134	M311	Y	-57.36	-57.354	6.429	8.571
135	M311	Y	-57.354	-67.652	8.571	10.714
136	M311	Y	-67.652	-68.528	10.714	12.857
137	M311	Y	-68.528	-49.434	12.857	15
138	M314	Y	-65.791	-91.949	0	2.143
139	M314	Y	-91.949	-97.664	2.143	4.286
140	M314	Y	-97.664	-86.802	4.286	6.429
141	M314	Y	-86.802	-86.848	6.429	8.571
142	M314	Y	-86.848	-97.712	8.571	10.714
143	M314	Y	-97.712	-91.939	10.714	12.857
144	M314	Y	-91.939	-65.752	12.857	15
145	M320	Y	-49.428	-68.506	0	2.143
146	M320	Y	-68.506	-67.643	2.143	4.286
147	M320	Y	-67.643	-57.36	4.286	6.429
148	M320	Y	-57.36	-57.354	6.429	8.571
149	M320	Y	-57.354	-67.652	8.571	10.714
150	M320	Y	-67.652	-68.528	10.714	12.857
151	M320	Y	-68.528	-49.434	12.857	15
152	M322	Y	-65.791	-91.949	0	2.143
153	M322	Y	-91.949	-97.664	2.143	4.286
154	M322	Y	-97.664	-86.802	4.286	6.429
155	M322	Y	-86.802	-86.848	6.429	8.571
156	M322	Y	-86.848	-97.712	8.571	10.714
157	M322	Y	-97.712	-91.939	10.714	12.857
158	M322	Y	-91.939	-65.752	12.857	15
159	M328	Y	-65.791	-91.949	0	2.143
160	M328	Y	-91.949	-97.664	2.143	4.286
161	M328	Y	-97.664	-86.802	4.286	6.429
162	M328	Y	-86.802	-86.848	6.429	8.571
163	M328	Y	-86.848	-97.712	8.571	10.714
164	M328	Y	-97.712	-91.939	10.714	12.857
165	M328	Y	-91.939	-65.752	12.857	15
166	M327	Y	-49.428	-68.506	0	2.143
167	M327	Y	-68.506	-67.643	2.143	4.286
168	M327	Y	-67.643	-57.36	4.286	6.429
169	M327	Y	-57.36	-57.354	6.429	8.571
170	M327	Y	-57.354	-67.652	8.571	10.714
171	M327	Y	-67.652	-68.528	10.714	12.857
172	M327	Y	-68.528	-49.434	12.857	15
173	M308	Y	-65.791	-91.949	0	2.143
174	M308	Y	-91.949	-97.664	2.143	4.286
175	M308	Y	-97.664	-86.802	4.286	6.429
176	M308	Y	-86.802	-86.848	6.429	8.571
177	M308	Y	-86.848	-97.712	8.571	10.714
178	M308	Y	-97.712	-91.939	10.714	12.857
179	M308	Y	-91.939	-65.752	12.857	15
180	M307	Y	-49.428	-68.506	0	2.143

Member Distributed Loads (BLC 25 : BLC 11 Transient Area Loads) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
181	M307	Y	-68.506	-67.643	2.143	4.286
182	M307	Y	-67.643	-57.36	4.286	6.429
183	M307	Y	-57.36	-57.354	6.429	8.571
184	M307	Y	-57.354	-67.652	8.571	10.714
185	M307	Y	-67.652	-68.528	10.714	12.857
186	M307	Y	-68.528	-49.434	12.857	15
187	M313	Y	-43.208	-43.696	0	2.143
188	M313	Y	-43.696	-45.367	2.143	4.286
189	M313	Y	-45.367	-51.392	4.286	6.429
190	M313	Y	-51.392	-51.383	6.429	8.571
191	M313	Y	-51.383	-45.305	8.571	10.714
192	M313	Y	-45.305	-43.623	10.714	12.857
193	M313	Y	-43.623	-43.203	12.857	15
194	M321	Y	-43.208	-43.696	0	2.143
195	M321	Y	-43.696	-45.367	2.143	4.286
196	M321	Y	-45.367	-51.392	4.286	6.429
197	M321	Y	-51.392	-51.383	6.429	8.571
198	M321	Y	-51.383	-45.305	8.571	10.714
199	M321	Y	-45.305	-43.623	10.714	12.857
200	M321	Y	-43.623	-43.203	12.857	15
201	M326	Y	-43.208	-43.696	0	2.143
202	M326	Y	-43.696	-45.367	2.143	4.286
203	M326	Y	-45.367	-51.392	4.286	6.429
204	M326	Y	-51.392	-51.383	6.429	8.571
205	M326	Y	-51.383	-45.305	8.571	10.714
206	M326	Y	-45.305	-43.623	10.714	12.857
207	M326	Y	-43.623	-43.203	12.857	15
208	M295	Y	-43.208	-43.696	0	2.143
209	M295	Y	-43.696	-45.367	2.143	4.286
210	M295	Y	-45.367	-51.392	4.286	6.429
211	M295	Y	-51.392	-51.383	6.429	8.571
212	M295	Y	-51.383	-45.305	8.571	10.714
213	M295	Y	-45.305	-43.623	10.714	12.857
214	M295	Y	-43.623	-43.203	12.857	15
215	M333	Y	-49.428	-68.506	0	2.143
216	M333	Y	-68.506	-67.643	2.143	4.286
217	M333	Y	-67.643	-57.36	4.286	6.429
218	M333	Y	-57.36	-57.354	6.429	8.571
219	M333	Y	-57.354	-67.652	8.571	10.714
220	M333	Y	-67.652	-68.528	10.714	12.857
221	M333	Y	-68.528	-49.434	12.857	15
222	M334	Y	-65.791	-91.949	0	2.143
223	M334	Y	-91.949	-97.664	2.143	4.286
224	M334	Y	-97.664	-86.802	4.286	6.429
225	M334	Y	-86.802	-86.848	6.429	8.571
226	M334	Y	-86.848	-97.712	8.571	10.714
227	M334	Y	-97.712	-91.939	10.714	12.857
228	M334	Y	-91.939	-65.752	12.857	15
229	M332	Y	-43.208	-43.696	0	2.143
230	M332	Y	-43.696	-45.367	2.143	4.286
231	M332	Y	-45.367	-51.392	4.286	6.429
232	M332	Y	-51.392	-51.383	6.429	8.571
233	M332	Y	-51.383	-45.305	8.571	10.714
234	M332	Y	-45.305	-43.623	10.714	12.857
235	M332	Y	-43.623	-43.203	12.857	15
236	M304	Y	-65.752	-91.939	0	2.143
237	M304	Y	-91.939	-97.712	2.143	4.286
238	M304	Y	-97.712	-86.848	4.286	6.429
239	M304	Y	-86.848	-86.802	6.429	8.571
240	M304	Y	-86.802	-97.664	8.571	10.714
241	M304	Y	-97.664	-91.949	10.714	12.857
242	M304	Y	-91.949	-65.791	12.857	15
243	M361	Y	-49.434	-68.528	0	2.143
244	M361	Y	-68.528	-67.651	2.143	4.286
245	M361	Y	-67.651	-57.354	4.286	6.429
246	M361	Y	-57.354	-57.36	6.429	8.571
247	M361	Y	-57.36	-67.643	8.571	10.714
248	M361	Y	-67.643	-68.506	10.714	12.857
249	M361	Y	-68.506	-49.428	12.857	15
250	M365	Y	-43.203	-43.624	0	2.143
251	M365	Y	-43.624	-45.305	2.143	4.286
252	M365	Y	-45.305	-51.383	4.286	6.429
253	M365	Y	-51.383	-51.392	6.429	8.571
254	M365	Y	-51.392	-45.367	8.571	10.714
255	M365	Y	-45.367	-43.696	10.714	12.857
256	M365	Y	-43.696	-43.208	12.857	15
257	M315	Y	-65.791	-91.949	0	2.143
258	M315	Y	-91.949	-97.664	2.143	4.286
259	M315	Y	-97.664	-86.802	4.286	6.429
260	M315	Y	-86.802	-86.848	6.429	8.571
261	M315	Y	-86.848	-97.712	8.571	10.714
262	M315	Y	-97.712	-91.939	10.714	12.857
263	M315	Y	-91.939	-65.752	12.857	15
264	M318	Y	-49.428	-68.506	0	2.143
265	M318	Y	-68.506	-67.643	2.143	4.286
266	M318	Y	-67.643	-57.36	4.286	6.429
267	M318	Y	-57.36	-57.354	6.429	8.571
268	M318	Y	-57.354	-67.652	8.571	10.714

Member Distributed Loads (BLC 25 : BLC 11 Transient Area Loads) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
269	M318	Y	-67.652	-68.528	10.714	12.857
270	M318	Y	-68.528	-49.434	12.857	15
271	M325	Y	-49.428	-68.506	0	2.143
272	M325	Y	-68.506	-67.643	2.143	4.286
273	M325	Y	-67.643	-57.36	4.286	6.429
274	M325	Y	-57.36	-57.354	6.429	8.571
275	M325	Y	-57.354	-67.652	8.571	10.714
276	M325	Y	-67.652	-68.528	10.714	12.857
277	M325	Y	-68.528	-49.434	12.857	15
278	M310	Y	-49.428	-68.506	0	2.143
279	M310	Y	-68.506	-67.643	2.143	4.286
280	M310	Y	-67.643	-57.36	4.286	6.429
281	M310	Y	-57.36	-57.354	6.429	8.571
282	M310	Y	-57.354	-67.652	8.571	10.714
283	M310	Y	-67.652	-68.528	10.714	12.857
284	M310	Y	-68.528	-49.434	12.857	15
285	M323	Y	-65.791	-91.949	0	2.143
286	M323	Y	-91.949	-97.664	2.143	4.286
287	M323	Y	-97.664	-86.802	4.286	6.429
288	M323	Y	-86.802	-86.848	6.429	8.571
289	M323	Y	-86.848	-97.712	8.571	10.714
290	M323	Y	-97.712	-91.939	10.714	12.857
291	M323	Y	-91.939	-65.752	12.857	15
292	M309	Y	-65.791	-91.949	0	2.143
293	M309	Y	-91.949	-97.664	2.143	4.286
294	M309	Y	-97.664	-86.802	4.286	6.429
295	M309	Y	-86.802	-86.848	6.429	8.571
296	M309	Y	-86.848	-97.712	8.571	10.714
297	M309	Y	-97.712	-91.939	10.714	12.857
298	M309	Y	-91.939	-65.752	12.857	15
299	M306	Y	-49.428	-68.506	0	2.143
300	M306	Y	-68.506	-67.643	2.143	4.286
301	M306	Y	-67.643	-57.36	4.286	6.429
302	M306	Y	-57.36	-57.354	6.429	8.571
303	M306	Y	-57.354	-67.652	8.571	10.714
304	M306	Y	-67.652	-68.528	10.714	12.857
305	M306	Y	-68.528	-49.434	12.857	15
306	M305	Y	-65.791	-91.949	0	2.143
307	M305	Y	-91.949	-97.664	2.143	4.286
308	M305	Y	-97.664	-86.802	4.286	6.429
309	M305	Y	-86.802	-86.848	6.429	8.571
310	M305	Y	-86.848	-97.712	8.571	10.714
311	M305	Y	-97.712	-91.939	10.714	12.857
312	M305	Y	-91.939	-65.752	12.857	15
313	M299	Y	-43.208	-43.696	0	2.143
314	M299	Y	-43.696	-45.367	2.143	4.286
315	M299	Y	-45.367	-51.392	4.286	6.429
316	M299	Y	-51.392	-51.383	6.429	8.571
317	M299	Y	-51.383	-45.305	8.571	10.714
318	M299	Y	-45.305	-43.623	10.714	12.857
319	M299	Y	-43.623	-43.203	12.857	15
320	M324	Y	-43.208	-43.696	0	2.143
321	M324	Y	-43.696	-45.367	2.143	4.286
322	M324	Y	-45.367	-51.392	4.286	6.429
323	M324	Y	-51.392	-51.383	6.429	8.571
324	M324	Y	-51.383	-45.305	8.571	10.714
325	M324	Y	-45.305	-43.623	10.714	12.857
326	M324	Y	-43.623	-43.203	12.857	15
327	M317	Y	-43.208	-43.696	0	2.143
328	M317	Y	-43.696	-45.367	2.143	4.286
329	M317	Y	-45.367	-51.392	4.286	6.429
330	M317	Y	-51.392	-51.383	6.429	8.571
331	M317	Y	-51.383	-45.305	8.571	10.714
332	M317	Y	-45.305	-43.623	10.714	12.857
333	M317	Y	-43.623	-43.203	12.857	15
334	M292	Y	-43.208	-43.696	0	2.143
335	M292	Y	-43.696	-45.367	2.143	4.286
336	M292	Y	-45.367	-51.392	4.286	6.429
337	M292	Y	-51.392	-51.383	6.429	8.571
338	M292	Y	-51.383	-45.305	8.571	10.714
339	M292	Y	-45.305	-43.623	10.714	12.857
340	M292	Y	-43.623	-43.203	12.857	15
341	M338	Y	-106.561	-106.561	0	15
342	M341	Y	-95.861	-103.629	0	2.143
343	M341	Y	-103.629	-107.405	2.143	4.286
344	M341	Y	-107.405	-112.666	4.286	6.429
345	M341	Y	-112.666	-112.734	6.429	8.571
346	M341	Y	-112.734	-107.594	8.571	10.714
347	M341	Y	-107.594	-103.832	10.714	12.857
348	M341	Y	-103.832	-95.986	12.857	15
349	M342	Y	-103.2	-103.2	4.552e-15	15
350	M343	Y	-61.068	-71.708	0	2.143
351	M343	Y	-71.708	-70.066	2.143	4.286
352	M343	Y	-70.066	-64.953	4.286	6.429
353	M343	Y	-64.953	-64.871	6.429	8.571
354	M343	Y	-64.871	-69.867	8.571	10.714
355	M343	Y	-69.867	-71.605	10.714	12.857
356	M343	Y	-71.605	-61.226	12.857	15

Member Distributed Loads (BLC 25 : BLC 11 Transient Area Loads) (Continued)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
357	M339	Y	-104.4	-104.4	2.887e-15	15
358	M52	Y	-53.761	-53.761	2.109e-15	15
359	M293	Y	-61.068	-71.708	0	2.143
360	M293	Y	-71.708	-70.066	2.143	4.286
361	M293	Y	-70.066	-64.953	4.286	6.429
362	M293	Y	-64.953	-64.871	6.429	8.571
363	M293	Y	-64.871	-69.867	8.571	10.714
364	M293	Y	-69.867	-71.605	10.714	12.857
365	M293	Y	-71.605	-61.226	12.857	15
366	M256	Y	-106.561	-106.561	1.221e-15	15
367	M255	Y	-104.4	-104.4	1.943e-14	15
368	M254	Y	-103.2	-103.2	2.032e-14	15
369	M253	Y	-95.861	-103.629	0	2.143
370	M253	Y	-103.629	-107.405	2.143	4.286
371	M253	Y	-107.405	-112.666	4.286	6.429
372	M253	Y	-112.666	-112.734	6.429	8.571
373	M253	Y	-112.734	-107.594	8.571	10.714
374	M253	Y	-107.594	-103.832	10.714	12.857
375	M253	Y	-103.832	-95.986	12.857	15
376	M177	Y	-53.761	-53.761	9.659e-15	15
377	M330	Y	-65.791	-91.949	0	2.143
378	M330	Y	-91.949	-97.664	2.143	4.286
379	M330	Y	-97.664	-86.802	4.286	6.429
380	M330	Y	-86.802	-86.848	6.429	8.571
381	M330	Y	-86.848	-97.712	8.571	10.714
382	M330	Y	-97.712	-91.939	10.714	12.857
383	M330	Y	-91.939	-65.752	12.857	15
384	M331	Y	-49.428	-68.506	0	2.143
385	M331	Y	-68.506	-67.643	2.143	4.286
386	M331	Y	-67.643	-57.36	4.286	6.429
387	M331	Y	-57.36	-57.354	6.429	8.571
388	M331	Y	-57.354	-67.652	8.571	10.714
389	M331	Y	-67.652	-68.528	10.714	12.857
390	M331	Y	-68.528	-49.434	12.857	15
391	M329	Y	-43.208	-43.696	0	2.143
392	M329	Y	-43.696	-45.367	2.143	4.286
393	M329	Y	-45.367	-51.392	4.286	6.429
394	M329	Y	-51.392	-51.383	6.429	8.571
395	M329	Y	-51.383	-45.305	8.571	10.714
396	M329	Y	-45.305	-43.623	10.714	12.857
397	M329	Y	-43.623	-43.203	12.857	15
398	M509	Y	-81.067	-52.048	0	1.3
399	M509	Y	-52.048	-61.899	1.3	2.6
400	M509	Y	-61.899	-61.879	2.6	3.9
401	M509	Y	-61.879	-52.035	3.9	5.2
402	M509	Y	-52.035	-81.105	5.2	6.5
403	M510	Y	-65.957	-73.106	0	1.3
404	M510	Y	-73.106	-120.065	1.3	2.6
405	M510	Y	-120.065	-120.071	2.6	3.9
406	M510	Y	-120.071	-73.123	3.9	5.2
407	M510	Y	-73.123	-65.983	5.2	6.5
408	M511	Y	-44.763	-44.763	8.777e-5	6.5
409	M432	Y	-44.763	-44.763	2.926e-5	6.5
410	M433	Y	-65.944	-73.109	0	1.3
411	M433	Y	-73.109	-120.059	1.3	2.6
412	M433	Y	-120.059	-120.063	2.6	3.9
413	M433	Y	-120.063	-73.127	3.9	5.2
414	M433	Y	-73.127	-65.985	5.2	6.5
415	M508	Y	-81.017	-52.021	0	1.3
416	M508	Y	-52.021	-61.909	1.3	2.6
417	M508	Y	-61.909	-61.907	2.6	3.9
418	M508	Y	-61.907	-52.053	3.9	5.2
419	M508	Y	-52.053	-81.121	5.2	6.5

Member Distributed Loads (BLC 26 : BLC 13 Transient Area Loads)

	Member Label	Direction	Start Magnitude [lb/ft, F, psf, k-ft/ft]	End Magnitude [lb/ft, F, psf, k-ft/ft]	Start Location [(ft, %)]	End Location [(ft, %)]
1	M372	Z	240	240	3.442e-15	13
2	M375	Z	240	240	6.55e-15	13
3	M381	Z	240	240	0	13
4	M390	Z	160	160	5.551e-16	13
5	M346	Z	240	240	0	13
6	M378	Z	240	240	0	13
7	M347	Z	120	120	5.551e-16	13
8	M401	Z	160	160	5.551e-16	13
9	M400	Z	160	160	5.551e-16	13
10	M399	Z	180	180	5.551e-16	13
11	M396	Z	90	90	5.551e-16	13
12	M393	Z	170	170	5.551e-16	13
13	M387	Z	132	132	9.992e-16	13
14	M384	Z	172	172	0	13
15	M366	Z	240	240	4.996e-15	13
16	M369	Z	240	240	1.776e-15	13

HANSON™

Company : Hanson Professional Services
Designer : S. Steiner
Job Number : 25i0007
Model Name : Eli Lilly K215 Unloading Rack

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Member Area Loads (BLC 1 : Dead)

	Node A	Node B	Node C	Node D	Direction	Load Direction	A Magnitude [psf]	B Magnitude [psf]	C Magnitude [psf]	D Magnitude [psf]	Exclude Braces
1	30	35	40	37	Y	B-C	-10	-10	-10	-10	Yes
2	31	34	33	37	Y	A-B	-10	-10	-10	-10	Yes
3	59	46	45	60	Y	A-B	-40	-40	-40	-40	Yes
4	88	130	169	146	Y	A-B	-10	-10	-10	-10	Yes
5	159	158	82	177	Y	B-C	-15	-15	-15	-15	Yes
6	159	158	82	177	Y	B-C	-10	-10	-10	-10	Yes
7	163	98	102	159	Y	Two Way	-10	-10	-10	-10	Yes
8	165	159	158	164	Y	A-B	-15	-15	-15	-15	Yes
9	207	229	211	228	Y	B-C	-40	-40	-40	-40	Yes
10	248	371	368	249	Y	A-B	-3	-3	-3	-3	Yes
11	254	372	374	252	Y	A-B	-3	-3	-3	-3	Yes
12	280	370	372	286	Y	A-B	-3	-3	-3	-3	Yes
13	294	374	371	293	Y	A-B	-3	-3	-3	-3	Yes
14	298	368	375	284	Y	A-B	-3	-3	-3	-3	Yes
15	370	282	296	372	Y	A-B	-3	-3	-3	-3	Yes
16	371	251	250	368	Y	A-B	-3	-3	-3	-3	Yes
17	371	295	297	374	Y	A-B	-3	-3	-3	-3	Yes
18	372	255	253	374	Y	A-B	-3	-3	-3	-3	Yes
19	375	289	299	368	Y	A-B	-3	-3	-3	-3	Yes

Member Area Loads (BLC 2 : Live)

	Node A	Node B	Node C	Node D	Direction	Load Direction	A Magnitude [psf]	B Magnitude [psf]	C Magnitude [psf]	D Magnitude [psf]	Exclude Braces
1	31	34	33	37	Y	A-B	-100	-100	-100	-100	Yes
2	35	40	37	30	Y	A-B	-100	-100	-100	-100	Yes
3	82	178	179	177	Y	A-B	-150	-150	-150	-150	Yes
4	88	130	169	146	Y	A-B	-100	-100	-100	-100	Yes
5	146	114	102	159	Y	B-C	-150	-150	-150	-150	Yes
6	148	150	151	149	Y	A-B	-150	-150	-150	-150	Yes
7	150	152	153	151	Y	A-B	-150	-150	-150	-150	Yes
8	152	154	155	153	Y	A-B	-150	-150	-150	-150	Yes
9	154	156	157	155	Y	A-B	-150	-150	-150	-150	Yes
10	156	158	159	157	Y	A-B	-150	-150	-150	-150	Yes
11	175	148	149	147	Y	A-B	-150	-150	-150	-150	Yes
12	178	180	181	179	Y	A-B	-150	-150	-150	-150	Yes
13	180	175	147	181	Y	A-B	-150	-150	-150	-150	Yes

Member Area Loads (BLC 3 : Roof Wind East-West)

	Node A	Node B	Node C	Node D	Direction	Load Direction	A Magnitude [psf]	B Magnitude [psf]	C Magnitude [psf]	D Magnitude [psf]	Exclude Braces
1	248	371	368	249	Y	A-B	-7.6	-7.6	-7.6	-7.6	Yes
2	280	370	372	286	Y	A-B	-7.6	-7.6	-7.6	-7.6	Yes
3	282	370	372	296	Y	A-B	-30.3	-30.3	-30.3	-30.3	Yes
4	294	374	371	293	Y	A-B	-7.6	-7.6	-7.6	-7.6	Yes
5	298	368	375	284	Y	A-B	-7.6	-7.6	-7.6	-7.6	Yes
6	368	299	289	375	Y	A-B	-30.3	-30.3	-30.3	-30.3	Yes
7	371	251	250	368	Y	A-B	-30.3	-30.3	-30.3	-30.3	Yes
8	372	255	253	374	Y	A-B	-30.3	-30.3	-30.3	-30.3	Yes
9	372	254	252	374	Y	A-B	-7.6	-7.6	-7.6	-7.6	Yes
10	374	297	295	371	Y	A-B	-30.3	-30.3	-30.3	-30.3	Yes
11	291	377	370	280	Y	A-B	-7.6	-7.6	-7.6	-7.6	Yes
12	377	292	282	370	Y	A-B	-30.3	-30.3	-30.3	-30.3	Yes

Member Area Loads (BLC 4 : Roof Wind North-South)

	Node A	Node B	Node C	Node D	Direction	Load Direction	A Magnitude [psf]	B Magnitude [psf]	C Magnitude [psf]	D Magnitude [psf]	Exclude Braces
1	248	371	368	249	Y	A-B	-20.2	-20.2	-20.2	-20.2	Yes
2	280	370	372	286	Y	A-B	-20.2	-20.2	-20.2	-20.2	Yes
3	282	370	372	296	Y	A-B	-20.2	-20.2	-20.2	-20.2	Yes
4	294	374	371	293	Y	A-B	-20.2	-20.2	-20.2	-20.2	Yes
5	298	368	375	284	Y	A-B	-20.2	-20.2	-20.2	-20.2	Yes
6	368	299	289	375	Y	A-B	-20.2	-20.2	-20.2	-20.2	Yes
7	371	251	250	368	Y	A-B	-20.2	-20.2	-20.2	-20.2	Yes
8	372	255	253	374	Y	A-B	-20.2	-20.2	-20.2	-20.2	Yes
9	372	254	252	374	Y	A-B	-20.2	-20.2	-20.2	-20.2	Yes
10	374	297	295	371	Y	A-B	-20.2	-20.2	-20.2	-20.2	Yes
11	291	377	370	280	Y	A-B	-20.2	-20.2	-20.2	-20.2	Yes
12	377	292	282	370	Y	A-B	-20.2	-20.2	-20.2	-20.2	Yes

Member Area Loads (BLC 5 : Framing Wind East-West)

	Node A	Node B	Node C	Node D	Direction	Load Direction	A Magnitude [psf]	B Magnitude [psf]	C Magnitude [psf]	D Magnitude [psf]	Exclude Braces
1	240	241	29	28	X	Open Structure	46	46	46	46	Yes

Member Area Loads (BLC 6 : Framing Wind North-South)

	Node A	Node B	Node C	Node D	Direction	Load Direction	A Magnitude [psf]	B Magnitude [psf]	C Magnitude [psf]	D Magnitude [psf]	Exclude Braces
1	239	240	28	27	Z	Open Structure	46	46	46	46	Yes

Member Area Loads (BLC 7 : Roof Live)

	Node A	Node B	Node C	Node D	Direction	Load Direction	A Magnitude [psf]	B Magnitude [psf]	C Magnitude [psf]	D Magnitude [psf]	Exclude Braces
1	248	371	368	249	Y	A-B	-25.9	-25.9	-25.9	-25.9	Yes
2	254	372	374	252	Y	A-B	-25.9	-25.9	-25.9	-25.9	Yes
3	289	375	408	399	Y	A-B	-36.6	-36.6	-36.6	-36.6	Yes

HANSON™

Company : Hanson Professional Services
Designer : S. Steiner
Job Number : 25i0007
Model Name : Eli Lilly K215 Unloading Rack

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Member Area Loads (BLC 7 : Roof Live) (Continued)

	Node A	Node B	Node C	Node D	Direction	Load Direction	A Magnitude [psf]	B Magnitude [psf]	C Magnitude [psf]	D Magnitude [psf]	Exclude Braces
4	295	371	406	401	Y	A-B	-25.9	-25.9	-25.9	-25.9	Yes
5	296	372	409	404	Y	A-B	-25.9	-25.9	-25.9	-25.9	Yes
6	371	293	400	406	Y	A-B	-25.9	-25.9	-25.9	-25.9	Yes
7	371	251	250	368	Y	A-B	-25.9	-25.9	-25.9	-25.9	Yes
8	372	255	253	374	Y	A-B	-25.9	-25.9	-25.9	-25.9	Yes
9	372	286	405	409	Y	A-B	-25.9	-25.9	-25.9	-25.9	Yes
10	375	284	398	408	Y	A-B	-36.6	-36.6	-36.6	-36.6	Yes
11	399	408	368	299	Y	A-B	-25.9	-25.9	-25.9	-25.9	Yes
12	401	406	407	402	Y	A-B	-36.6	-36.6	-36.6	-36.6	Yes
13	402	407	374	297	Y	A-B	-25.9	-25.9	-25.9	-25.9	Yes
14	404	409	377	292	Y	A-B	-36.6	-36.6	-36.6	-36.6	Yes
15	406	400	403	407	Y	A-B	-36.6	-36.6	-36.6	-36.6	Yes
16	407	403	294	374	Y	A-B	-25.9	-25.9	-25.9	-25.9	Yes
17	408	398	298	368	Y	A-B	-25.9	-25.9	-25.9	-25.9	Yes
18	409	405	291	377	Y	A-B	-36.6	-36.6	-36.6	-36.6	Yes

Member Area Loads (BLC 10 : Ice Dead Load)

	Node A	Node B	Node C	Node D	Direction	Load Direction	A Magnitude [psf]	B Magnitude [psf]	C Magnitude [psf]	D Magnitude [psf]	Exclude Braces
1	31	34	33	37	Y	A-B	-22.12	-22.12	-22.12	-22.12	Yes
2	37	30	35	40	Y	A-B	-22.12	-22.12	-22.12	-22.12	Yes
3	59	46	45	60	Y	A-B	-33	-33	-33	-33	Yes
4	88	130	169	146	Y	A-B	-22.12	-22.12	-22.12	-22.12	Yes
5	146	114	102	159	Y	A-B	-22.12	-22.12	-22.12	-22.12	Yes
6	159	177	82	158	Y	A-B	-22.12	-22.12	-22.12	-22.12	Yes
7	207	229	211	228	Y	B-C	-22.12	-22.12	-22.12	-22.12	Yes
8	248	371	368	249	Y	A-B	-22.12	-22.12	-22.12	-22.12	Yes
9	254	372	374	252	Y	A-B	-22.12	-22.12	-22.12	-22.12	Yes
10	280	370	372	286	Y	A-B	-22.12	-22.12	-22.12	-22.12	Yes
11	294	374	371	293	Y	A-B	-22.12	-22.12	-22.12	-22.12	Yes
12	298	368	375	284	Y	A-B	-22.12	-22.12	-22.12	-22.12	Yes
13	370	282	296	372	Y	A-B	-22.12	-22.12	-22.12	-22.12	Yes
14	371	251	250	368	Y	A-B	-22.12	-22.12	-22.12	-22.12	Yes
15	371	295	297	374	Y	A-B	-22.12	-22.12	-22.12	-22.12	Yes
16	372	255	253	374	Y	A-B	-22.12	-22.12	-22.12	-22.12	Yes
17	375	289	299	368	Y	A-B	-22.12	-22.12	-22.12	-22.12	Yes
18	291	377	370	280	Y	A-B	-22.12	-22.12	-22.12	-22.12	Yes
19	377	292	282	370	Y	A-B	-22.12	-22.12	-22.12	-22.12	Yes

Member Area Loads (BLC 11 : Snow)

	Node A	Node B	Node C	Node D	Direction	Load Direction	A Magnitude [psf]	B Magnitude [psf]	C Magnitude [psf]	D Magnitude [psf]	Exclude Braces
1	37	31	34	33	Y	B-C	-20	-20	-20	-20	Yes
2	40	35	30	37	Y	A-B	-20	-20	-20	-20	Yes
3	88	130	169	146	Y	Two Way	-20	-20	-20	-20	Yes
4	146	159	102	114	Y	B-C	-20	-20	-20	-20	Yes
5	159	158	82	177	Y	B-C	-20	-20	-20	-20	Yes
6	248	371	368	249	Y	A-B	-24	-24	-24	-24	Yes
7	254	372	374	252	Y	A-B	-24	-24	-24	-24	Yes
8	280	370	372	286	Y	A-B	-24	-24	-24	-24	Yes
9	294	374	371	293	Y	A-B	-24	-24	-24	-24	Yes
10	298	368	375	284	Y	A-B	-24	-24	-24	-24	Yes
11	370	282	296	372	Y	A-B	-24	-24	-24	-24	Yes
12	371	295	297	374	Y	A-B	-24	-24	-24	-24	Yes
13	371	251	250	368	Y	A-B	-24	-24	-24	-24	Yes
14	372	255	253	374	Y	A-B	-24	-24	-24	-24	Yes
15	375	289	299	368	Y	A-B	-24	-24	-24	-24	Yes
16	282	370	377	292	Y	A-B	-24	-24	-24	-24	Yes
17	370	280	291	377	Y	A-B	-24	-24	-24	-24	Yes

Member Area Loads (BLC 13 : Thermal Pipe)

	Node A	Node B	Node C	Node D	Direction	Load Direction	A Magnitude [psf]	B Magnitude [psf]	C Magnitude [psf]	D Magnitude [psf]	Exclude Braces
1	45	46	59	60	Z	B-C	16	16	16	16	Yes

Basic Load Cases

	BLC Description	Category	Y Gravity	Nodal	Point	Distributed	Area(Member)
1	Dead	DL		15	4	21	19
2	Live	LL		4	10		13
3	Roof Wind East-West	WLX					12
4	Roof Wind North-South	WLZ					12
5	Framing Wind East-West	WLX		14		26	1
6	Framing Wind North-South	WLZ		12		4	1
7	Roof Live	RLL					18
8	Pipe Wind East-West	WLX		44			
9	Pipe Wind North-South	WLZ					
10	Ice Dead Load	IL		10		258	19
11	Snow	SL		10			17
12	Self-Wt	DL	-1				
13	Thermal Pipe	TL					1
14	Earthquake Load North-South	ELZ					
15	Earthquake Load East-West	ELX		42			
16	Framing Thermal	TL				346	
17	BLC 1 Transient Area Loads	None				475	
18	BLC 2 Transient Area Loads	None				66	
19	BLC 3 Transient Area Loads	None				346	

Basic Load Cases (Continued)

	BLC Description	Category	Y Gravity	Nodal	Point	Distributed	Area(Member)
20	BLC 4 Transient Area Loads	None				346	
21	BLC 5 Transient Area Loads	None				111	
22	BLC 6 Transient Area Loads	None				124	
23	BLC 7 Transient Area Loads	None				354	
24	BLC 10 Transient Area Loads	None				451	
25	BLC 11 Transient Area Loads	None				419	
26	BLC 13 Transient Area Loads	None				16	

Load Combinations

	Description	Solve	P-Delta	BLC	Factor	BLC	Factor	BLC	Factor	BLC	Factor	BLC	Factor	BLC	Factor	BLC	Factor
1	Deflection 1	Yes	Y	DL	1												
2	Deflection 2	Yes	Y	LL	1												
3	Deflection 3 (DL+LL)	Yes	Y	DL	1	LL	1										
4	Deflection 3 (DL+RLL)	Yes	Y	DL	1	RLL	1										
5	Wind +X Deflection	Yes	Y	DL	1	WLX	0.4										
6	Wind -X Deflection	Yes	Y	DL	1	WLX	-0.4										
7	Wind +Z Deflection	Yes	Y	DL	1	WLZ	0.4										
8	Wind -Z Deflection	Yes	Y	DL	1	WLZ	-0.4										
9	SERVICE DL		Y	DL	1												
10	SERVICE LL		Y	LL	1												
11	SERVICE SL		Y	SL	1												
12	SERVICE RLL		Y	RLL	1												
13	SERVICE WLX		Y	WLX	0.6												
14	SERVICE WLZ		Y	WLZ	0.6												
15	SERVICE ICE DL		Y	IL	1												
16	SERVICE ELX		Y	ELX	0.7												
17	SERVICE ELZ		Y	ELZ	0.7												
18	WLX Load Combinations		Y														
19	ASCE Strength 1	Yes	Y	DL	1.4												
20	D-T	Yes	Y	DL	1.4	TL	-1.2										
21	D+T	Yes	Y	DL	1.4	TL	1.2										
22	D0.5L-T	Yes	Y	DL	1.2	TL	-1.2	LL	0.5								
23	D0.5L+T	Yes	Y	DL	1.2	TL	1.2	LL	0.5								
24	D1.6L-T	Yes	Y	DL	1.2	TL	-1	LL	1.6								
25	D1.6L+T	Yes	Y	DL	1.2	TL	1	LL	1.6								
26	ASCE Strength 2 (a)	Yes	Y	DL	1.2	LL	1.6	LLS	1.6	RLL	0.5						
27	ASCE Strength 2 (b)	Yes	Y	DL	1.2	LL	1.6	LLS	1.6	SL	0.5	SLN	0.5				
28	ASCE Strength 2 (c)	Yes	Y	DL	1.2	LL	1.6	LLS	1.6								
29	ASCE Strength 2 + Ice	Yes	Y	DL	1.2	LL	1.6	LLS	1.6	SL	0.5	SLN	0.5	IL	0.2		
30	ASCE Strength 3 (a)	Yes	Y	DL	1.2	RLL	1.6	LL	0.5	LLS	1						
31	ASCE Strength 3 (c)	Yes	Y	DL	1.2	SL	1.6	SLN	1.6	LL	0.5	LLS	1				
32	ASCE Strength 3 (b) (a)	Yes	Y	DL	1.2	RLL	1.6	WLX	0.5								
33	ASCE Strength 3 (b) (b)	Yes	Y	DL	1.2	RLL	1.6	WLX	-0.5								
34	ASCE Strength 3 (d) (a)	Yes	Y	DL	1.2	SL	1.6	SLN	1.6	WLX	0.5						
35	ASCE Strength 3 (d) (b)	Yes	Y	DL	1.2	SL	1.6	SLN	1.6	WLX	-0.5						
36	ASCE Strength 3 (f) (a)	Yes	Y	DL	1.2	WLX	0.5										
37	ASCE Strength 3 (f) (b)	Yes	Y	DL	1.2	WLX	-0.5										
38	ASCE Strength 4 (a) (a)	Yes	Y	DL	1.2	WLX	1	LL	0.5	LLS	1	RLL	0.5				
39	ASCE Strength 4 (a) (b)	Yes	Y	DL	1.2	WLX	-1	LL	0.5	LLS	1	RLL	0.5				
40	ASCE Strength 4 (b) (a)	Yes	Y	DL	1.2	WLX	1	LL	0.5	LLS	1	SL	0.5	SLN	0.5		
41	ASCE Strength 4 (b) (b)	Yes	Y	DL	1.2	WLX	-1	LL	0.5	LLS	1	SL	0.5	SLN	0.5		
42	ASCE Strength 4 (c) (a)	Yes	Y	DL	1.2	WLX	1	LL	0.5	LLS	1						
43	ASCE Strength 4 (c) (b)	Yes	Y	DL	1.2	WLX	-1	LL	0.5	LLS	1						
44	ASCE Strength 4 + Ice	Yes	Y	DL	1.2	WLX	0.21	LL	0.5	LLS	1	SL	0.5	SLN	0.5	IL	1
45	ASCE Strength 4 + Ice	Yes	Y	DL	1.2	WLX	-0.21	LL	0.5	LLS	1	SL	0.5	SLN	0.5	IL	1
46	ASCE Strength 6 (a)	Yes	Y	DL	0.9	WLX	1										
47	ASCE Strength 6 (b)	Yes	Y	DL	0.9	WLX	-1										
48	ASCE Strength 6 + Ice	Yes	Y	DL	0.9	WLX	0.21	IL	1								
49	ASCE Strength 6 + Ice	Yes	Y	DL	0.9	WLX	-0.21	IL	1								
50	Thermal with Wind	Yes	Y														
51	DWT	Yes	Y	DL	0.9	WLZ	1	TL	1								
52	DW-T	Yes	Y	DL	0.9	WLZ	1	TL	-1								
53	DWT	Yes	Y	DL	0.9	WLZ	-1	TL	1								
54	DW-T	Yes	Y	DL	0.9	WLZ	-1	TL	-1								
55	Ice-T	Yes	Y	DL	0.9	WLZ	0.21	IL	1	TL	-1						
56	Ice+T	Yes	Y	DL	0.9	WLZ	-0.21	IL	1	TL	-1						
57	DWT	Yes	Y	DL	0.9	WLX	1	TL	1								
58	DW-T	Yes	Y	DL	0.9	WLX	1	TL	-1								
59	DWT	Yes	Y	DL	0.9	WLX	-1	TL	1								
60	DW-T	Yes	Y	DL	0.9	WLX	-1	TL	-1								
61	Ice-T	Yes	Y	DL	0.9	WLX	0.21	IL	1	TL	-1						
62	Ice+T	Yes	Y	DL	0.9	WLX	-0.21	IL	1	TL	-1						
63	WLZ Load Combinations	Yes	Y														
64	ASCE Strength 3 (b) (a)	Yes	Y	DL	1.2	RLL	1.6	WLZ	0.5								
65	ASCE Strength 3 (b) (b)	Yes	Y	DL	1.2	RLL	1.6	WLZ	-0.5								
66	ASCE Strength 3 (d) (a)	Yes	Y	DL	1.2	SL	1.6	SLN	1.6	WLX	0.5						
67	ASCE Strength 3 (d) (b)	Yes	Y	DL	1.2	SL	1.6	SLN	1.6	WLX	-0.5						
68	ASCE Strength 3 (f) (a)	Yes	Y	DL	1.2	WLZ	0.5										
69	ASCE Strength 3 (f) (b)	Yes	Y	DL	1.2	WLZ	-0.5										
70	ASCE Strength 4 (a) (a)	Yes	Y	DL	1.2	WLZ	1	LL	0.5	LLS	1	RLL	0.5				
71	ASCE Strength 4 (a) (b)	Yes	Y	DL	1.2	WLZ	-1	LL	0.5	LLS	1	RLL	0.5				
72	ASCE Strength 4 (b) (a)	Yes	Y	DL	1.2	WLZ	1	LL	0.5	LLS	1	SL	0.5	SLN	0.5		
73	ASCE Strength 4 (b) (b)	Yes	Y	DL	1.2	WLZ	-1	LL	0.5	LLS	1	SL	0.5	SLN	0.5		
74	ASCE Strength 4 (c) (a)	Yes	Y	DL	1.2	WLZ	1	LL	0.5	LLS	1						
75	ASCE Strength 4 (c) (b)	Yes	Y	DL	1.2	WLZ	-1	LL	0.5	LLS	1						
76	ASCE Strength 4 + Ice	Yes	Y	DL	1.2	WLZ	0.21	LL	0.5	LLS	1	SL	0.5	SLN	0.5		
77	ASCE Strength 4 + Ice	Yes	Y	DL	1.2	WLZ	-0.21	LL	0.5	LLS	1	SL	0.5	SLN	0.5		
78	ASCE Strength 6 (a)	Yes	Y	DL	0.9	WLZ	1										

HANSON™

Company : Hanson Professional Services
Designer : S. Steiner
Job Number : 25i0007
Model Name : Eli Lilly K215 Unloading Rack

2/19/2026
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Checked By : RMF

Load Combinations (Continued)

	Description	Solve	P-Delta	BLC	Factor	BLC	Factor	BLC	Factor	BLC	Factor	BLC	Factor	BLC	Factor	BLC	Factor
79	ASCE Strength 6 (b)	Yes	Y	DL	0.9	WLZ	-1										
80	ASCE Strength 6 + Ice	Yes	Y	DL	0.9	WLZ	0.21	IL	1								
81	ASCE Strength 6 + Ice	Yes	Y	DL	0.9	WLZ	-0.21	IL	1								
82	ASCE Strength 5 (a)	Yes	Y	DL	1.2	ELX	1	LL	0.5	LLS	1	SL	0.2	SLN	0.2		
83	ASCE Strength 5 (b)	Yes	Y	DL	1.2	ELX+Z	1	LL	0.5	LLS	1	SL	0.2	SLN	0.2		
84	ASCE Strength 5 (c)	Yes	Y	DL	1.2	ELX-Z	1	LL	0.5	LLS	1	SL	0.2	SLN	0.2		
85	ASCE Strength 5 (d)	Yes	Y	DL	1.2	ELZ	1	LL	0.5	LLS	1	SL	0.2	SLN	0.2		
86	ASCE Strength 5 (e)	Yes	Y	DL	1.2	ELZ+X	1	LL	0.5	LLS	1	SL	0.2	SLN	0.2		
87	ASCE Strength 5 (f)	Yes	Y	DL	1.2	ELZ-X	1	LL	0.5	LLS	1	SL	0.2	SLN	0.2		
88	ASCE Strength 5 (g)	Yes	Y	DL	1.2	ELX	-1	LL	0.5	LLS	1	SL	0.2	SLN	0.2		
89	ASCE Strength 5 (h)	Yes	Y	DL	1.2	ELX+Z	-1	LL	0.5	LLS	1	SL	0.2	SLN	0.2		
90	ASCE Strength 5 (i)	Yes	Y	DL	1.2	ELX-Z	-1	LL	0.5	LLS	1	SL	0.2	SLN	0.2		
91	ASCE Strength 5 (j)	Yes	Y	DL	1.2	ELZ	-1	LL	0.5	LLS	1	SL	0.2	SLN	0.2		
92	ASCE Strength 5 (k)	Yes	Y	DL	1.2	ELZ+X	-1	LL	0.5	LLS	1	SL	0.2	SLN	0.2		
93	ASCE Strength 5 (l)	Yes	Y	DL	1.2	ELZ-X	-1	LL	0.5	LLS	1	SL	0.2	SLN	0.2		
94	ASCE Strength 7 (a)	Yes	Y	DL	0.9	ELX	1										
95	ASCE Strength 7 (b)	Yes	Y	DL	0.9	ELX+Z	1										
96	ASCE Strength 7 (c)	Yes	Y	DL	0.9	ELX-Z	1										
97	ASCE Strength 7 (d)	Yes	Y	DL	0.9	ELZ	1										
98	ASCE Strength 7 (e)	Yes	Y	DL	0.9	ELZ+X	1										
99	ASCE Strength 7 (f)	Yes	Y	DL	0.9	ELZ-X	1										
100	ASCE Strength 7 (g)	Yes	Y	DL	0.9	ELX	-1										
101	ASCE Strength 7 (h)	Yes	Y	DL	0.9	ELX+Z	-1										
102	ASCE Strength 7 (i)	Yes	Y	DL	0.9	ELX-Z	-1										
103	ASCE Strength 7 (j)	Yes	Y	DL	0.9	ELZ	-1										
104	ASCE Strength 7 (k)	Yes	Y	DL	0.9	ELZ+X	-1										
105	ASCE Strength 7 (l)	Yes	Y	DL	0.9	ELZ-X	-1										
106	ASD Load Combinations		Y														
107	ASCE ASD 1		Y	DL	1												
108	ASCE ASD 2		Y	DL	1	LL	1	LLS	1								
109	ASCE ASD 3 (a)		Y	DL	1	RLL	1										
110	ASCE ASD 3 (b)		Y	DL	1	SL	1	SLN	1								
111	ASCE ASD 4 (a)		Y	DL	1	LL	0.75	LLS	0.75	RLL	0.75						
112	ASCE ASD 4 (b)		Y	DL	1	LL	0.75	LLS	0.75	SL	0.75	SLN	0.75				
113	WLX Combinations		Y		1												
114	ASCE ASD 5 (a) (a)		Y	DL	1	WLX	0.6										
115	ASCE ASD 5 (a) (b)		Y	DL	1	WLX	-0.6										
116	ASCE ASD 6 (a) (a)		Y	DL	1	WLX	0.45	LL	0.75	LLS	0.75	RLL	0.75				
117	ASCE ASD 6 (a) (b)		Y	DL	1	WLX	-0.45	LL	0.75	LLS	0.75	RLL	0.75				
118	ASCE ASD 6 (c) (a)		Y	DL	1	WLX	0.45	LL	0.75	LLS	0.75	SL	0.75	SLN	0.75		
119	ASCE ASD 6 (c) (b)		Y	DL	1	WLX	-0.45	LL	0.75	LLS	0.75	SL	0.75	SLN	0.75		
120	ASCE ASD 6 (e) (a)		Y	DL	1	WLX	0.45	LL	0.75	LLS	0.75						
121	ASCE ASD 6 (e) (b)		Y	DL	1	WLX	-0.45	LL	0.75	LLS	0.75						
122	ASCE ASD 7 (a)		Y	DL	0.6	WLX	0.6										
123	ASCE ASD 7 (b)		Y	DL	0.6	WLX	-0.6										
124	WLZ Combinations		Y		1												
125	ASCE ASD 5 (a) (a)		Y	DL	1	WLZ	0.6										
126	ASCE ASD 5 (a) (b)		Y	DL	1	WLZ	-0.6										
127	ASCE ASD 6 (a) (a)		Y	DL	1	WLZ	0.45	LL	0.75	LLS	0.75	RLL	0.75				
128	ASCE ASD 6 (a) (b)		Y	DL	1	WLZ	-0.45	LL	0.75	LLS	0.75	RLL	0.75				
129	ASCE ASD 6 (c) (a)		Y	DL	1	WLZ	0.45	LL	0.75	LLS	0.75	SL	0.75	SLN	0.75		
130	ASCE ASD 6 (c) (b)		Y	DL	1	WLZ	-0.45	LL	0.75	LLS	0.75	SL	0.75	SLN	0.75		
131	ASCE ASD 6 (e) (a)		Y	DL	1	WLZ	0.45	LL	0.75	LLS	0.75						
132	ASCE ASD 6 (e) (b)		Y	DL	1	WLZ	-0.45	LL	0.75	LLS	0.75						
133	ASCE ASD 7 (a)		Y	DL	0.6	WLZ	0.6										
134	ASCE ASD 7 (b)		Y	DL	0.6	WLZ	-0.6										
135	Seismic Combinations		Y														
136	ASCE ASD 5 (b) (a)		Y	DL	1	ELX	0.7										
137	ASCE ASD 5 (b) (b)		Y	DL	1	ELX+Z	0.7										
138	ASCE ASD 5 (b) (c)		Y	DL	1	ELX-Z	0.7										
139	ASCE ASD 5 (b) (d)		Y	DL	1	ELZ	0.7										
140	ASCE ASD 5 (b) (e)		Y	DL	1	ELZ+X	0.7										
141	ASCE ASD 5 (b) (f)		Y	DL	1	ELZ-X	0.7										
142	ASCE ASD 5 (b) (g)		Y	DL	1	ELX	-0.7										
143	ASCE ASD 5 (b) (h)		Y	DL	1	ELX+Z	-0.7										
144	ASCE ASD 5 (b) (i)		Y	DL	1	ELX-Z	-0.7										
145	ASCE ASD 5 (b) (j)		Y	DL	1	ELZ	-0.7										
146	ASCE ASD 5 (b) (k)		Y	DL	1	ELZ+X	-0.7										
147	ASCE ASD 5 (b) (l)		Y	DL	1	ELZ-X	-0.7										
148	ASCE ASD 6 (b) (a)		Y	DL	1	ELX	0.525	LL	0.75	LLS	0.75	RLL	0.75				
149	ASCE ASD 6 (b) (b)		Y	DL	1	ELX+Z	0.525	LL	0.75	LLS	0.75	RLL	0.75				
150	ASCE ASD 6 (b) (c)		Y	DL	1	ELX-Z	0.525	LL	0.75	LLS	0.75	RLL	0.75				
151	ASCE ASD 6 (b) (d)		Y	DL	1	ELZ	0.525	LL	0.75	LLS	0.75	RLL	0.75				
152	ASCE ASD 6 (b) (e)		Y	DL	1	ELZ+X	0.525	LL	0.75	LLS	0.75	RLL	0.75				
153	ASCE ASD 6 (b) (f)		Y	DL	1	ELZ-X	0.525	LL	0.75	LLS	0.75	RLL	0.75				
154	ASCE ASD 6 (b) (g)		Y	DL	1	ELX	-0.525	LL	0.75	LLS	0.75	RLL	0.75				
155	ASCE ASD 6 (b) (h)		Y	DL	1	ELX+Z	-0.525	LL	0.75	LLS	0.75	RLL	0.75				
156	ASCE ASD 6 (b) (i)		Y	DL	1	ELX-Z	-0.525	LL	0.75	LLS	0.75	RLL	0.75				
157	ASCE ASD 6 (b) (j)		Y	DL	1	ELZ	-0.525	LL	0.75	LLS	0.75	RLL	0.75				
158	ASCE ASD 6 (b) (k)		Y	DL	1	ELZ+X	-0.525	LL	0.75	LLS	0.75	RLL	0.75				
159	ASCE ASD 6 (b) (l)		Y	DL	1	ELZ-X	-0.525	LL	0.75	LLS	0.75	RLL	0.75				
160	ASCE ASD 6 (d) (a)		Y	DL	1	ELX	0.525	LL	0.75	LLS	0.75	SL	0.75	SLN	0.75		
161	ASCE ASD 6 (d) (b)		Y	DL	1	ELX+Z	0.525	LL	0.75	LLS	0.75	SL	0.75	SLN	0.75		
162	ASCE ASD 6 (d) (c)		Y	DL	1	ELX-Z	0.525	LL	0.75	LLS	0.75	SL	0.75	SLN	0.75		
163	ASCE ASD 6 (d) (d)		Y	DL	1	ELZ	0.525	LL	0.75	LLS	0.75	SL	0.75	SLN	0.75		
164	ASCE ASD 6 (d) (e)		Y	DL	1	ELZ+X	0.525	LL	0.75	LLS	0.75	SL	0.75	SLN	0.75		
165	ASCE ASD 6 (d) (f)		Y	DL	1	ELZ-X	0.525	LL	0.75	LLS	0.75	SL	0.75	SLN	0.75		
166	ASCE ASD 6 (d) (g)		Y	DL	1	ELX	-0.525	LL	0.75	LLS	0.75	SL	0.75	SLN	0.75		

HANSON™

Company : Hanson Professional Services
Designer : S. Steiner
Job Number : 25i0007
Model Name : Eli Lilly K215 Unloading Rack

2/19/2026
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Checked By : RMF

Load Combinations (Continued)

	Description	Solve	P-Delta	BLC	Factor	BLC	Factor	BLC	Factor	BLC	Factor	BLC	Factor	BLC	Factor	BLC	Factor
167	ASCE ASD 6 (d) (h)		Y	DL	1	ELX+Z	-0.525	LL	0.75	LLS	0.75	SL	0.75	SLN	0.75		
168	ASCE ASD 6 (d) (i)		Y	DL	1	ELX-Z	-0.525	LL	0.75	LLS	0.75	SL	0.75	SLN	0.75		
169	ASCE ASD 6 (d) (j)		Y	DL	1	ELZ	-0.525	LL	0.75	LLS	0.75	SL	0.75	SLN	0.75		
170	ASCE ASD 6 (d) (k)		Y	DL	1	ELZ+X	-0.525	LL	0.75	LLS	0.75	SL	0.75	SLN	0.75		
171	ASCE ASD 6 (d) (l)		Y	DL	1	ELZ-X	-0.525	LL	0.75	LLS	0.75	SL	0.75	SLN	0.75		
172	ASCE ASD 6 (f) (a)		Y	DL	1	ELX	0.525	LL	0.75	LLS	0.75						
173	ASCE ASD 6 (f) (b)		Y	DL	1	ELX+Z	0.525	LL	0.75	LLS	0.75						
174	ASCE ASD 6 (f) (c)		Y	DL	1	ELX-Z	0.525	LL	0.75	LLS	0.75						
175	ASCE ASD 6 (f) (d)		Y	DL	1	ELZ	0.525	LL	0.75	LLS	0.75						
176	ASCE ASD 6 (f) (e)		Y	DL	1	ELZ+X	0.525	LL	0.75	LLS	0.75						
177	ASCE ASD 6 (f) (f)		Y	DL	1	ELZ-X	0.525	LL	0.75	LLS	0.75						
178	ASCE ASD 6 (f) (g)		Y	DL	1	ELX	-0.525	LL	0.75	LLS	0.75						
179	ASCE ASD 6 (f) (h)		Y	DL	1	ELX+Z	-0.525	LL	0.75	LLS	0.75						
180	ASCE ASD 6 (f) (i)		Y	DL	1	ELX-Z	-0.525	LL	0.75	LLS	0.75						
181	ASCE ASD 6 (f) (j)		Y	DL	1	ELZ	-0.525	LL	0.75	LLS	0.75						
182	ASCE ASD 6 (f) (k)		Y	DL	1	ELZ+X	-0.525	LL	0.75	LLS	0.75						
183	ASCE ASD 6 (f) (l)		Y	DL	1	ELZ-X	-0.525	LL	0.75	LLS	0.75						
184	ASCE ASD 8 (a)		Y	DL	0.6	ELX	0.7										
185	ASCE ASD 8 (b)		Y	DL	0.6	ELX+Z	0.7										
186	ASCE ASD 8 (c)		Y	DL	0.6	ELX-Z	0.7										
187	ASCE ASD 8 (d)		Y	DL	0.6	ELZ	0.7										
188	ASCE ASD 8 (e)		Y	DL	0.6	ELZ+X	0.7										
189	ASCE ASD 8 (f)		Y	DL	0.6	ELZ-X	0.7										
190	ASCE ASD 8 (g)		Y	DL	0.6	ELX	-0.7										
191	ASCE ASD 8 (h)		Y	DL	0.6	ELX+Z	-0.7										
192	ASCE ASD 8 (i)		Y	DL	0.6	ELX-Z	-0.7										
193	ASCE ASD 8 (j)		Y	DL	0.6	ELZ	-0.7										
194	ASCE ASD 8 (k)		Y	DL	0.6	ELZ+X	-0.7										
195	ASCE ASD 8 (l)		Y	DL	0.6	ELZ-X	-0.7										

Load Combination Design

	Description	CD	Service	Hot Rolled	Cold Formed	Wood	Concrete	Masonry	Aluminum	Stainless	Connection
1	Deflection 1		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
2	Deflection 2		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
3	Deflection 3 (DL+LL)		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
4	Deflection 3 (DL+RLL)		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
5	Wind +X Deflection		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
6	Wind -X Deflection		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
7	Wind +Z Deflection		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
8	Wind -Z Deflection		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
9	SERVICE DL		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
10	SERVICE LL		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
11	SERVICE SL		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
12	SERVICE RLL		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
13	SERVICE WLX		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
14	SERVICE WLZ		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
15	SERVICE ICE DL		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
16	SERVICE ELX		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
17	SERVICE ELZ		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
18	WLX Load Combinations			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
19	ASCE Strength 1			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
20	D-T			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
21	D+T			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
22	D0.5L-T			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
23	D0.5L+T			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
24	D1.6L-T			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
25	D1.6L+T			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
26	ASCE Strength 2 (a)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
27	ASCE Strength 2 (b)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
28	ASCE Strength 2 (c)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
29	ASCE Strength 2 + Ice			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
30	ASCE Strength 3 (a)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
31	ASCE Strength 3 (c)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
32	ASCE Strength 3 (b) (a)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
33	ASCE Strength 3 (b) (b)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
34	ASCE Strength 3 (d) (a)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
35	ASCE Strength 3 (d) (b)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
36	ASCE Strength 3 (f) (a)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
37	ASCE Strength 3 (f) (b)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
38	ASCE Strength 4 (a) (a)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
39	ASCE Strength 4 (a) (b)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
40	ASCE Strength 4 (b) (a)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
41	ASCE Strength 4 (b) (b)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
42	ASCE Strength 4 (c) (a)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
43	ASCE Strength 4 (c) (b)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
44	ASCE Strength 4 + Ice			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
45	ASCE Strength 4 + Ice			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
46	ASCE Strength 6 (a)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
47	ASCE Strength 6 (b)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
48	ASCE Strength 6 + Ice			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
49	ASCE Strength 6 + Ice			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
50	Thermal with Wind			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
51	DWT			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
52	DW-T			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
53	DWT			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
54	DW-T			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
55	Ice-T			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
56	Ice+T			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Load Combination Design (Continued)

	Description	CD	Service	Hot Rolled	Cold Formed	Wood	Concrete	Masonry	Aluminum	Stainless	Connection
57	DWT			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
58	DW-T			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
59	DWT			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
60	DW-T			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
61	Ice-T			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
62	Ice+T			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
63	WLZ Load Combinations			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
64	ASCE Strength 3 (b) (a)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
65	ASCE Strength 3 (b) (b)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
66	ASCE Strength 3 (d) (a)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
67	ASCE Strength 3 (d) (b)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
68	ASCE Strength 3 (f) (a)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
69	ASCE Strength 3 (f) (b)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
70	ASCE Strength 4 (a) (a)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
71	ASCE Strength 4 (a) (b)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
72	ASCE Strength 4 (b) (a)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
73	ASCE Strength 4 (b) (b)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
74	ASCE Strength 4 (c) (a)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
75	ASCE Strength 4 (c) (b)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
76	ASCE Strength 4 + Ice			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
77	ASCE Strength 4 + Ice			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
78	ASCE Strength 6 (a)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
79	ASCE Strength 6 (b)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
80	ASCE Strength 6 + Ice			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
81	ASCE Strength 6 + Ice			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
82	ASCE Strength 5 (a)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
83	ASCE Strength 5 (b)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
84	ASCE Strength 5 (c)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
85	ASCE Strength 5 (d)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
86	ASCE Strength 5 (e)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
87	ASCE Strength 5 (f)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
88	ASCE Strength 5 (g)			Yes	Yes		Yes	Yes	Yes	Yes	Yes
89	ASCE Strength 5 (h)			Yes	Yes		Yes	Yes	Yes	Yes	Yes
90	ASCE Strength 5 (i)			Yes	Yes		Yes	Yes	Yes	Yes	Yes
91	ASCE Strength 5 (j)			Yes	Yes		Yes	Yes	Yes	Yes	Yes
92	ASCE Strength 5 (k)			Yes	Yes		Yes	Yes	Yes	Yes	Yes
93	ASCE Strength 5 (l)			Yes	Yes		Yes	Yes	Yes	Yes	Yes
94	ASCE Strength 7 (a)			Yes	Yes		Yes	Yes	Yes	Yes	Yes
95	ASCE Strength 7 (b)			Yes	Yes		Yes	Yes	Yes	Yes	Yes
96	ASCE Strength 7 (c)			Yes	Yes		Yes	Yes	Yes	Yes	Yes
97	ASCE Strength 7 (d)			Yes	Yes		Yes	Yes	Yes	Yes	Yes
98	ASCE Strength 7 (e)			Yes	Yes		Yes	Yes	Yes	Yes	Yes
99	ASCE Strength 7 (f)			Yes	Yes		Yes	Yes	Yes	Yes	Yes
100	ASCE Strength 7 (g)			Yes	Yes		Yes	Yes	Yes	Yes	Yes
101	ASCE Strength 7 (h)			Yes	Yes		Yes	Yes	Yes	Yes	Yes
102	ASCE Strength 7 (i)			Yes	Yes		Yes	Yes	Yes	Yes	Yes
103	ASCE Strength 7 (j)			Yes	Yes		Yes	Yes	Yes	Yes	Yes
104	ASCE Strength 7 (k)			Yes	Yes		Yes	Yes	Yes	Yes	Yes
105	ASCE Strength 7 (l)			Yes	Yes		Yes	Yes	Yes	Yes	Yes
106	ASD Load Combinations			Yes	Yes		Yes	Yes	Yes	Yes	Yes
107	ASCE ASD 1			Yes	Yes		Yes	Yes	Yes	Yes	Yes
108	ASCE ASD 2			Yes	Yes		Yes	Yes	Yes	Yes	Yes
109	ASCE ASD 3 (a)			Yes	Yes		Yes	Yes	Yes	Yes	Yes
110	ASCE ASD 3 (b)			Yes	Yes		Yes	Yes	Yes	Yes	Yes
111	ASCE ASD 4 (a)			Yes	Yes		Yes	Yes	Yes	Yes	Yes
112	ASCE ASD 4 (b)			Yes	Yes		Yes	Yes	Yes	Yes	Yes
113	WLX Combinations	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
114	ASCE ASD 5 (a) (a)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
115	ASCE ASD 5 (a) (b)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
116	ASCE ASD 6 (a) (a)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
117	ASCE ASD 6 (a) (b)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
118	ASCE ASD 6 (c) (a)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
119	ASCE ASD 6 (c) (b)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
120	ASCE ASD 6 (e) (a)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
121	ASCE ASD 6 (e) (b)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
122	ASCE ASD 7 (a)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
123	ASCE ASD 7 (b)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
124	WLZ Combinations	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
125	ASCE ASD 5 (a) (a)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
126	ASCE ASD 5 (a) (b)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
127	ASCE ASD 6 (a) (a)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
128	ASCE ASD 6 (a) (b)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
129	ASCE ASD 6 (c) (a)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
130	ASCE ASD 6 (c) (b)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
131	ASCE ASD 6 (e) (a)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
132	ASCE ASD 6 (e) (b)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
133	ASCE ASD 7 (a)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
134	ASCE ASD 7 (b)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
135	Seismic Combinations	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
136	ASCE ASD 5 (b) (a)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
137	ASCE ASD 5 (b) (b)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
138	ASCE ASD 5 (b) (c)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
139	ASCE ASD 5 (b) (d)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
140	ASCE ASD 5 (b) (e)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
141	ASCE ASD 5 (b) (f)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
142	ASCE ASD 5 (b) (g)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
143	ASCE ASD 5 (b) (h)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
144	ASCE ASD 5 (b) (i)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Load Combination Design (Continued)

	Description	CD	Service	Hot Rolled	Cold Formed	Wood	Concrete	Masonry	Aluminum	Stainless	Connection
145	ASCE ASD 5 (b) (j)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
146	ASCE ASD 5 (b) (k)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
147	ASCE ASD 5 (b) (l)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
148	ASCE ASD 6 (b) (a)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
149	ASCE ASD 6 (b) (b)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
150	ASCE ASD 6 (b) (c)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
151	ASCE ASD 6 (b) (d)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
152	ASCE ASD 6 (b) (e)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
153	ASCE ASD 6 (b) (f)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
154	ASCE ASD 6 (b) (g)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
155	ASCE ASD 6 (b) (h)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
156	ASCE ASD 6 (b) (i)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
157	ASCE ASD 6 (b) (j)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
158	ASCE ASD 6 (b) (k)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
159	ASCE ASD 6 (b) (l)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
160	ASCE ASD 6 (d) (a)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
161	ASCE ASD 6 (d) (b)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
162	ASCE ASD 6 (d) (c)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
163	ASCE ASD 6 (d) (d)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
164	ASCE ASD 6 (d) (e)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
165	ASCE ASD 6 (d) (f)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
166	ASCE ASD 6 (d) (g)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
167	ASCE ASD 6 (d) (h)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
168	ASCE ASD 6 (d) (i)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
169	ASCE ASD 6 (d) (j)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
170	ASCE ASD 6 (d) (k)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
171	ASCE ASD 6 (d) (l)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
172	ASCE ASD 6 (f) (a)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
173	ASCE ASD 6 (f) (b)	1.6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
174	ASCE ASD 6 (f) (c)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
175	ASCE ASD 6 (f) (d)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
176	ASCE ASD 6 (f) (e)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
177	ASCE ASD 6 (f) (f)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
178	ASCE ASD 6 (f) (g)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
179	ASCE ASD 6 (f) (h)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
180	ASCE ASD 6 (f) (i)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
181	ASCE ASD 6 (f) (j)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
182	ASCE ASD 6 (f) (k)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
183	ASCE ASD 6 (f) (l)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
184	ASCE ASD 8 (a)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
185	ASCE ASD 8 (b)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
186	ASCE ASD 8 (c)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
187	ASCE ASD 8 (d)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
188	ASCE ASD 8 (e)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
189	ASCE ASD 8 (f)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
190	ASCE ASD 8 (g)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
191	ASCE ASD 8 (h)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
192	ASCE ASD 8 (i)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
193	ASCE ASD 8 (j)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
194	ASCE ASD 8 (k)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
195	ASCE ASD 8 (l)			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Envelope Node Reactions

	Node Label		X [k]	LC	Y [k]	LC	Z [k]	LC	MX [k-ft]	LC	MY [k-ft]	LC	MZ [k-ft]	LC
1	3	max	6.743	60	54.415	40	0.364	54	0	105	0.01	53	0	105
2		min	-6.203	57	-7.761	59	-0.375	51	0	1	-0.01	72	0	1
3	4	max	6.279	59	53.797	45	0.365	54	0	105	0.01	51	0	105
4		min	-6.669	58	0	50	-0.376	51	0	1	-0.01	73	0	1
5	1	max	6.063	60	31.361	40	0.462	53	0	105	0.041	54	0	105
6		min	-5.829	57	-7.363	59	-0.442	52	0	1	-0.041	52	0	1
7	2	max	5.714	59	28.541	41	0.463	53	0	105	0.041	52	0	105
8		min	-6.131	58	-5.45	57	-0.443	52	0	1	-0.041	54	0	1
9	5	max	7.072	60	55.557	44	0.545	54	0	105	0.01	53	0	105
10		min	-6.575	57	-7.725	59	-0.547	51	0	1	-0.01	72	0	1
11	6	max	6.563	59	52.32	45	0.545	54	0	105	0.01	51	0	105
12		min	-7.055	58	0	50	-0.546	51	0	1	-0.01	73	0	1
13	7	max	7.272	60	48.808	40	0.741	54	0	105	0.01	53	0	105
14		min	-6.717	57	-6.824	59	-0.731	51	0	1	-0.01	52	0	1
15	8	max	6.753	59	48.465	45	0.742	54	0	105	0.01	51	0	105
16		min	-7.207	58	-2.026	57	-0.732	51	0	1	-0.01	73	0	1
17	9	max	7.453	60	56.026	25	34.728	53	0	105	0.017	40	0.02	40
18		min	-6.609	57	-21.964	60	-31.856	52	0	1	-0.017	60	-0.02	60
19	10	max	6.576	59	56.448	25	35.166	53	0	105	0.017	40	0.02	40
20		min	-7.551	58	-18.255	58	-32.237	52	0	1	-0.017	59	-0.02	59
21	11	max	6.716	60	83.451	24	25.163	54	0	105	0.01	53	0.039	57
22		min	-7.131	57	-46.561	59	-25.08	51	0	1	-0.01	52	-0.039	41
23	12	max	7.179	59	83.61	24	25.843	54	0	105	0.01	51	0.039	58
24		min	-6.613	58	-41.459	57	-25.722	51	0	1	-0.01	54	-0.039	41
25	15	max	6.69	60	54.086	44	0.811	54	0	105	0.01	53	0	105
26		min	-6.182	57	-6.976	59	-0.801	51	0	1	-0.01	52	0	1
27	16	max	6.285	59	53.928	45	0.813	54	0	105	0.01	51	0	105
28		min	-6.658	58	0	50	-0.803	51	0	1	-0.01	54	0	1
29	17	max	5.997	60	36.958	44	0.895	54	0	105	0.01	53	0	105
30		min	-5.779	57	-4.906	59	-0.896	51	0	1	-0.01	52	0	1
31	18	max	5.669	59	35.1	45	0.891	54	0	105	0.009	51	0	105
32		min	-6.058	58	-1.918	57	-0.893	51	0	1	-0.009	54	0	1
33	19	max	6.377	60	32.176	40	0.684	54	0	105	0.015	54	0	105
34		min	-6.032	57	-3.834	59	-0.681	51	0	1	-0.015	51	0	1

Envelope Node Reactions (Continued)

	Node Label		X [k]	LC	Y [k]	LC	Z [k]	LC	MX [k-ft]	LC	MY [k-ft]	LC	MZ [k-ft]	LC
35	20	max	6	59	34.465	45	0.738	54	0	105	0.01	51	0	105
36		min	-6.331	58	-2.421	57	-0.734	51	0	1	-0.01	54	0	1
37	21	max	6.709	60	37.238	44	0.948	54	0	105	0.008	53	0	105
38		min	-6.387	57	0	50	-0.954	51	0	1	-0.009	72	0	1
39	22	max	6.335	59	47.139	45	0.986	54	0	105	0.007	72	0	105
40		min	-6.638	58	0	50	-0.991	51	0	1	-0.007	54	0	1
41	23	max	6.623	60	38.39	44	1.217	54	0	105	0.008	53	0	105
42		min	-6.334	57	0	50	-1.222	51	0	1	-0.008	52	0	1
43	24	max	6.985	59	50.237	45	1.561	53	0	105	0.006	52	0	105
44		min	-7.316	58	0	50	-1.567	52	0	1	-0.006	53	0	1
45	25	max	6.332	60	23.726	38	1.6	54	0	105	0.01	53	0	105
46		min	-6.057	57	-2.603	60	-1.604	51	0	1	-0.01	52	0	1
47	26	max	8.472	59	33.478	45	3.195	54	0	105	0.009	51	0	105
48		min	-8.734	58	-0.809	57	-3.203	51	0	1	-0.009	54	0	1
49	13	max	7.376	60	69.804	25	39.893	54	0	105	0.015	60	0.019	40
50		min	-6.194	57	-35.702	60	-42.895	51	0	1	-0.016	40	-0.018	60
51	14	max	6.166	59	69.372	25	40.188	54	0	105	0.015	59	0.019	40
52		min	-7.355	58	-34.01	54	-43.203	51	0	1	-0.016	40	-0.018	59
53	385	max	0.202	23	3.001	72	0.12	20	0	105	0	59	0	105
54		min	-0.196	20	-1.598	53	-0.13	23	0	1	0	58	0	1
55	386	max	1.384	60	3.001	72	0.119	20	0	54	0	59	0	105
56		min	-1.389	57	-1.597	53	-0.129	23	0	51	0	58	0	1
57	387	max	0.201	23	3.001	73	1.116	53	0	105	0	58	0	58
58		min	-0.195	20	-1.598	51	-1.111	52	0	1	0	59	0	59
59	388	max	1.383	60	3.001	73	1.116	53	0	54	0	57	0	58
60		min	-1.388	57	-1.597	51	-1.11	52	0	51	0	60	0	59
61	412	max	0.02	60	5.485	29	0.028	53	0	105	0	105	0	105
62		min	-0.02	57	0	50	-0.028	52	0	1	0	1	0	1
63	411	max	0.061	59	3.887	24	0.03	54	0	105	1.154	23	0	105
64		min	-0.006	54	-2.977	59	-0.029	52	0	1	-1.147	20	0	1
65	410	max	0.014	24	6.581	45	3.174	58	0	105	1.155	23	0	54
66		min	-0.053	58	-2.422	58	-3.237	59	0	1	-1.148	20	-0.001	25
67	Totals:	max	157.904	59	1107.276	44	125.002	54						
68		min	-157.904	58	0	50	-125.002	51						

Envelope Node Displacements

	Node Label		X [in]	LC	Y [in]	LC	Z [in]	LC	X Rotation [rad]	LC	Y Rotation [rad]	LC	Z Rotation [rad]	LC
1	17	max	0	57	0	59	0	51	0	105	0	52	0	105
2		min	0	60	0	44	0	54	0	1	0	53	0	1
3	8	max	0	58	0	57	0	51	0	105	0	73	0	105
4		min	0	59	0	45	0	54	0	1	0	51	0	1
5	9	max	0	57	0	60	0	52	0	105	0	60	0	60
6		min	0	60	0	25	0	53	0	1	0	40	0	40
7	25	max	0	57	0	60	0	51	0	105	0	52	0	105
8		min	0	60	0	38	0	54	0	1	0	53	0	1
9	11	max	0	57	0	59	0	51	0	105	0	52	0	41
10		min	0	60	0	24	0	54	0	1	0	53	0	57
11	12	max	0	58	0	57	0	51	0	105	0	54	0	41
12		min	0	59	0	24	0	54	0	1	0	51	0	58
13	13	max	0	57	0	60	0	51	0	105	0	40	0	60
14		min	0	60	0	25	0	54	0	1	0	60	0	40
15	24	max	0	58	0	63	0	52	0	105	0	53	0	105
16		min	0	59	0	45	0	53	0	1	0	52	0	1
17	23	max	0	57	0	63	0	51	0	105	0	52	0	105
18		min	0	60	0	44	0	54	0	1	0	53	0	1
19	14	max	0	58	0	54	0	51	0	105	0	40	0	59
20		min	0	59	0	25	0	54	0	1	0	59	0	40
21	15	max	0	57	0	59	0	51	0	105	0	52	0	105
22		min	0	60	0	44	0	54	0	1	0	53	0	1
23	22	max	0	58	0	63	0	51	0	105	0	54	0	105
24		min	0	59	0	45	0	54	0	1	0	72	0	1
25	21	max	0	57	0	63	0	51	0	105	0	72	0	105
26		min	0	60	0	44	0	54	0	1	0	53	0	1
27	16	max	0	58	0	63	0	51	0	105	0	54	0	105
28		min	0	59	0	45	0	54	0	1	0	51	0	1
29	20	max	0	58	0	57	0	51	0	105	0	54	0	105
30		min	0	59	0	45	0	54	0	1	0	51	0	1
31	18	max	0	58	0	57	0	51	0	105	0	54	0	105
32		min	0	59	0	45	0	54	0	1	0	51	0	1
33	26	max	0	58	0	57	0	51	0	105	0	54	0	105
34		min	0	59	0	45	0	54	0	1	0	51	0	1
35	7	max	0	57	0	59	0	51	0	105	0	52	0	105
36		min	0	60	0	40	0	54	0	1	0	53	0	1
37	10	max	0	58	0	58	0	52	0	105	0	59	0	59
38		min	0	59	0	25	0	53	0	1	0	40	0	40
39	6	max	0	58	0	63	0	51	0	105	0	73	0	105
40		min	0	59	0	45	0	54	0	1	0	51	0	1
41	2	max	0	58	0	57	0	52	0	105	0	54	0	105
42		min	0	59	0	41	0	53	0	1	0	52	0	1
43	27	max	0	105	0	105	0	105	0	105	0	105	0	105
44		min	0	1	0	1	0	1	0	1	0	1	0	1
45	1	max	0	57	0	59	0	52	0	105	0	52	0	105
46		min	0	60	0	40	0	53	0	1	0	54	0	1
47	3	max	0	57	0	59	0	51	0	105	0	72	0	105
48		min	0	60	0	40	0	54	0	1	0	53	0	1
49	28	max	0	105	0	105	0	105	0	105	0	105	0	105
50		min	0	1	0	1	0	1	0	1	0	1	0	1
51	29	max	0	105	0	105	0	105	0	105	0	105	0	105

Envelope Node Displacements (Continued)

	Node Label		X [in]	LC	Y [in]	LC	Z [in]	LC	X Rotation [rad]	LC	Y Rotation [rad]	LC	Z Rotation [rad]	LC
52		min	0	1	0	1	0	1	0	1	0	1	0	1
53	4	max	0	58	0	63	0	51	0	105	0	73	0	105
54		min	0	59	0	45	0	54	0	1	0	51	0	1
55	5	max	0	57	0	59	0	51	0	105	0	72	0	105
56		min	0	60	0	44	0	54	0	1	0	53	0	1
57	19	max	0	57	0	59	0	51	0	105	0	51	0	105
58		min	0	60	0	40	0	54	0	1	0	54	0	1
59	30	max	0.647	57	0	63	0.238	51	0	63	6.039e-3	60	2.468e-4	23
60		min	-0.649	60	-0.436	29	-0.238	54	-6.963e-3	29	-6.071e-3	57	-2.453e-4	20
61	31	max	1.331	57	0	63	0.156	51	3.799e-3	29	1.854e-3	21	2.468e-4	23
62		min	-1.321	60	-0.682	29	-0.155	54	0	50	-1.852e-3	22	-2.453e-4	20
63	32	max	0.615	58	0.034	21	0.092	51	8.127e-3	21	0	105	2.468e-4	23
64		min	-0.617	59	-0.036	22	-0.092	54	-8.088e-3	22	0	1	-2.453e-4	20
65	33	max	0.562	58	0.027	21	0.472	21	8.997e-3	21	4.815e-3	20	1.578e-3	29
66		min	-0.563	59	-0.059	24	-0.47	22	-9.009e-3	22	-4.838e-3	23	0	50
67	34	max	1.361	57	0	63	0.354	51	2.092e-3	29	4.933e-3	20	5.676e-3	29
68		min	-1.35	60	-0.439	29	-0.353	54	0	50	-4.955e-3	23	0	50
69	35	max	0.611	57	0.035	21	0.035	60	5.953e-5	45	4.317e-3	20	5.529e-3	24
70		min	-0.613	60	-0.036	22	-0.036	57	-3.445e-5	58	-4.341e-3	23	-1.589e-3	51
71	36	max	0.552	58	0.033	23	0.551	21	1.056e-2	21	3.58e-4	54	1.14e-2	59
72		min	-0.552	59	-0.037	20	-0.549	22	-1.052e-2	20	-3.587e-4	51	-1.139e-2	58
73	37	max	0.592	58	0.021	57	0.267	51	4.985e-3	21	4.567e-3	20	0	63
74		min	-0.593	59	-0.097	24	-0.267	54	-4.982e-3	22	-4.593e-3	23	-1.317e-4	45
75	38	max	1.371	57	0	63	0.422	51	1.513e-3	45	4.081e-3	57	1.185e-2	59
76		min	-1.36	60	-0.346	45	-0.42	54	0	50	-4.045e-3	60	-1.182e-2	58
77	39	max	0.575	58	0.031	23	0.359	51	8.127e-3	21	2.366e-4	53	1.21e-2	59
78		min	-0.577	59	-0.039	20	-0.358	54	-8.088e-3	22	-2.431e-4	52	-1.207e-2	58
79	40	max	0.629	58	0.034	21	0.015	59	4.894e-5	59	4.317e-3	20	5.531e-3	24
80		min	-0.63	59	-0.036	22	-0.015	58	-3.854e-5	58	-4.342e-3	23	-1.586e-3	51
81	41	max	0.676	57	0.004	57	0.437	21	0	63	4.804e-3	60	2.627e-3	29
82		min	-0.678	60	-0.196	24	-0.436	54	-3.301e-3	29	-4.817e-3	57	0	50
83	44	max	1.052	58	0.061	23	0.681	21	6.596e-3	21	4.631e-4	53	9.739e-3	59
84		min	-1.055	59	-0.076	20	-0.678	22	-6.58e-3	20	-4.758e-4	52	-9.702e-3	58
85	45	max	1.165	57	0.075	21	0.488	20	4.173e-3	20	3.507e-3	52	8.657e-3	60
86		min	-1.136	60	-0.083	22	-0.493	21	-4.18e-3	23	-3.502e-3	54	-8.967e-3	40
87	46	max	1.162	58	0.075	21	0.488	22	4.181e-3	22	3.497e-3	54	8.679e-3	59
88		min	-1.137	59	-0.083	22	-0.493	21	-4.176e-3	21	-3.496e-3	52	-8.961e-3	40
89	47	max	1.267	57	0.071	21	0.355	20	3.653e-3	22	8.405e-4	72	9.789e-3	41
90		min	-1.266	60	-0.088	22	-0.359	21	-3.66e-3	23	-8.367e-4	53	-9.651e-3	57
91	48	max	1.265	58	0.071	21	0.355	22	3.66e-3	22	8.332e-4	73	9.816e-3	41
92		min	-1.268	59	-0.088	22	-0.36	21	-3.655e-3	23	-8.298e-4	51	-9.624e-3	58
93	49	max	1.345	57	0.071	21	0.242	52	2.993e-3	22	8.398e-4	72	9.981e-3	60
94		min	-1.314	60	-0.088	22	-0.245	53	-2.998e-3	23	-8.37e-4	53	-1.039e-2	40
95	50	max	1.342	58	0.071	21	0.243	52	2.998e-3	22	8.32e-4	73	1.002e-2	59
96		min	-1.316	59	-0.088	22	-0.246	53	-2.996e-3	23	-8.295e-4	51	-1.037e-2	40
97	51	max	1.419	58	0.088	21	0.11	51	6.693e-4	22	8.219e-4	54	1.122e-2	41
98		min	-1.426	41	-0.102	22	-0.116	54	-5.961e-4	23	-8.223e-4	51	-1.095e-2	58
99	52	max	1.391	58	0.072	21	0.137	52	2.352e-3	22	8.31e-4	73	1.089e-2	41
100		min	-1.397	59	-0.087	22	-0.139	53	-2.356e-3	23	-8.298e-4	51	-1.065e-2	58
101	53	max	1.666	58	0	63	0.497	51	1.822e-5	44	1.015e-4	58	9.12e-3	59
102		min	-1.681	59	-0.392	62	-0.494	54	0	50	-1.086e-4	43	-9.102e-3	58
103	54	max	1.682	57	0	63	0.492	21	1.018e-5	62	1.013e-4	58	9.089e-3	60
104		min	-1.668	60	-0.391	61	-0.488	20	-3.969e-6	2	-1.088e-4	43	-9.128e-3	57
105	55	max	1.879	40	0	63	0.667	21	2.452e-4	29	2.779e-3	29	8.962e-3	59
106		min	-1.72	59	-0.453	44	-0.664	20	0	50	0	50	-8.931e-3	58
107	56	max	1.879	40	0	63	0.677	21	1.868e-6	61	1.709e-4	41	8.952e-3	60
108		min	-1.706	60	-0.392	61	-0.673	20	-9.181e-7	2	-1.711e-4	58	-8.978e-3	57
109	57	max	1.862	58	0	63	0.865	21	0	63	2.853e-4	60	8.405e-3	59
110		min	-1.884	59	-0.592	45	-0.862	20	-2.435e-5	45	-3.021e-4	57	-8.38e-3	58
111	58	max	1.877	57	0	2	0.873	21	0	63	2.881e-4	60	8.473e-3	60
112		min	-1.866	60	-0.59	44	-0.869	20	-1.837e-5	44	-3.052e-4	57	-8.476e-3	57
113	59	max	1.107	58	0.074	23	0.97	21	4.577e-3	23	8.062e-4	54	8.048e-3	59
114		min	-1.108	59	-0.084	20	-0.967	20	-4.569e-3	20	-8.078e-4	51	-8.036e-3	58
115	60	max	1.099	57	0.075	23	0.976	21	5.097e-3	23	8.809e-4	52	8.166e-3	60
116		min	-1.101	60	-0.083	20	-0.972	20	-5.079e-3	22	-8.823e-4	53	-8.153e-3	57
117	61	max	1.175	58	0.07	23	0.759	21	5.035e-3	23	5.329e-4	53	8.763e-3	59
118		min	-1.179	59	-0.088	20	-0.756	20	-5.028e-3	20	-5.474e-4	52	-8.724e-3	58
119	62	max	1.181	57	0.072	23	0.768	21	4.304e-3	23	7.141e-4	52	8.781e-3	60
120		min	-1.179	60	-0.087	20	-0.765	20	-4.288e-3	22	-7.065e-4	53	-8.799e-3	57
121	63	max	1.216	58	0.07	21	0.575	21	3.632e-3	21	6.094e-4	54	9.161e-3	59
122		min	-1.218	59	-0.087	22	-0.572	22	-3.626e-3	22	-6.452e-4	72	-9.139e-3	58
123	64	max	1.394	57	0.072	21	0.136	52	2.35e-3	22	8.383e-4	52	1.087e-2	41
124		min	-1.396	60	-0.087	22	-0.138	53	-2.355e-3	23	-8.374e-4	53	-1.069e-2	57
125	65	max	1.219	57	0.072	23	0.584	21	3.577e-3	23	7.576e-4	72	9.123e-3	60
126		min	-1.217	60	-0.086	20	-0.581	20	-3.562e-3	22	-7.299e-4	53	-9.157e-3	57
127	66	max	1.196	57	0.073	23	0.407	51	2.889e-3	23	1.269e-3	51	9.055e-3	60
128		min	-1.192	60	-0.084	20	-0.405	54	-2.853e-3	22	-1.265e-3	54	-9.098e-3	57
129	67	max	1.139	58	0.074	21	0.418	23	1.92e-3	21	8.16e-4	54	8.459e-3	59
130		min	-1.113	59	-0.085	22	-0.413	22	-1.908e-3	20	-8.168e-4	51	-8.766e-3	40
131	68	max	1.142	57	0.073	21	0.419	23	1.931e-3	21	8.594e-4	52	8.429e-3	60
132		min	-1.111	60	-0.085	22	-0.414	22	-1.918e-3	20	-8.6e-4	53	-8.782e-3	40
133	69	max	1.25	58	0.071	21	0.303	51	1.104e-3	21	8.21e-4	54	9.654e-3	41
134		min	-1.252	59	-0.088	22	-0.299	54	-1.094e-3	20	-8.225e-4	51	-9.476e-3	58
135	70	max	1.247	57	0.071	21	0.301	51	1.111e-3	21	8.512e-4	52	9.63e-3	41
136		min	-1.246	41	-0.088	22	-0.296	54	-1.097e-3	20	-8.517e-4	53	-9.469e-3	57
137	71	max	1.361	58	0.072	21	0.199	51	4.539e-4	53	8.217e-4	54	1.019e-2	59
138		min	-1.332	59	-0.088	22	-0.194	54	-4.437e-4	52	-8.228e-4	72	-1.058e-2	40
139	72	max	1.364	57	0.072	21	0.197	51	4.547e-4	53	8.41e-4	52	1.019e-2	60

Envelope Node Displacements (Continued)

	Node Label		X [in]	LC	Y [in]	LC	Z [in]	LC	X Rotation [rad]	LC	Y Rotation [rad]	LC	Z Rotation [rad]	LC
140		min	-1.334	60	-0.088	22	-0.191	54	-4.427e-4	52	-8.411e-4	53	-1.06e-2	40
141	73	max	1.422	57	0.088	21	0.109	51	6.683e-4	22	8.321e-4	52	1.119e-2	41
142		min	-1.425	60	-0.102	22	-0.115	54	-5.932e-4	23	-8.324e-4	53	-1.102e-2	57
143	74	max	1.472	58	0.075	21	0.081	51	1.748e-3	20	8.266e-4	54	1.118e-2	59
144		min	-1.443	59	-0.089	22	-0.085	54	-1.751e-3	23	-8.267e-4	51	-1.156e-2	40
145	75	max	1.472	40	0.075	21	0.081	51	1.749e-3	22	8.344e-4	52	1.115e-2	60
146		min	-1.44	60	-0.089	22	-0.085	54	-1.746e-3	23	-8.345e-4	53	-1.159e-2	40
147	76	max	1.191	58	0.073	21	0.419	51	2.95e-3	21	8.246e-4	54	9.079e-3	59
148		min	-1.193	59	-0.085	22	-0.416	54	-2.922e-3	22	-8.329e-4	51	-9.066e-3	58
149	77	max	1.402	58	0.092	23	0.858	21	2.14e-3	53	6.915e-4	53	6.334e-3	41
150		min	-1.407	59	-0.113	20	-0.855	20	-2.145e-3	52	-7.104e-4	52	-6.229e-3	38
151	78	max	1.316	58	0.097	23	1.068	21	2.245e-3	59	1.046e-3	54	5.816e-3	41
152		min	-1.318	59	-0.108	20	-1.064	20	-2.299e-3	58	-1.048e-3	51	-5.755e-3	38
153	79	max	1.766	57	0.07	59	0.338	23	4.703e-3	29	2.101e-3	40	4.303e-3	41
154		min	-1.766	41	-0.395	24	-0.335	20	0	50	-1.826e-3	60	-4.068e-3	58
155	80	max	1.704	58	0.073	57	0.41	21	0	63	3.95e-3	41	3.849e-3	60
156		min	-1.671	59	-0.391	24	-0.406	22	-4.797e-3	29	-3.584e-3	58	-4.147e-3	40
157	81	max	1.776	58	0.072	57	0.337	21	4.702e-3	29	2.106e-3	40	4.089e-3	39
158		min	-1.766	41	-0.395	24	-0.334	22	0	50	-1.841e-3	60	-4.273e-3	40
159	82	max	1.539	57	0.117	21	0.671	20	2.141e-3	20	5.469e-3	52	4.051e-3	41
160		min	-1.5	60	-0.129	22	-0.673	23	-2.017e-3	23	-5.462e-3	54	-4.083e-3	38
161	83	max	1.805	58	0.061	57	0.373	21	0	63	1.199e-3	41	3.965e-3	39
162		min	-1.779	59	-0.503	24	-0.37	22	-5.139e-5	45	-7.96e-4	58	-4.21e-3	40
163	84	max	1.998	57	0.062	59	0.266	23	0	63	6.153e-3	60	2.472e-3	41
164		min	-2	60	-0.428	24	-0.261	20	-5.291e-3	29	-6.483e-3	40	-2.378e-3	58
165	85	max	2.014	58	0.053	57	0.266	51	0	63	6.625e-3	60	2.584e-3	60
166		min	-2	41	-0.428	24	-0.264	54	-5.291e-3	29	-6.94e-3	40	-2.615e-3	40
167	86	max	1.535	40	0.138	57	0.838	52	2.142e-3	20	3.101e-3	52	7.65e-4	40
168		min	-1.473	41	-0.15	60	-0.836	53	-2.015e-3	23	-3.096e-3	54	-5.693e-4	60
169	87	max	1.698	57	0.07	59	0.411	23	0	63	3.957e-3	41	4.04e-3	41
170		min	-1.675	60	-0.391	24	-0.407	20	-4.794e-3	29	-3.6e-3	58	-3.933e-3	38
171	88	max	1.747	58	0	63	0.824	21	1.121e-3	41	5.021e-4	59	6.237e-3	29
172		min	-1.799	59	-1.038	29	-0.832	22	-1.079e-3	57	-7.789e-4	40	-3.198e-4	58
173	89	max	1.794	57	0.055	59	0.375	23	7.332e-6	60	1.196e-3	41	4.171e-3	41
174		min	-1.785	60	-0.503	24	-0.371	20	-4.858e-5	40	-7.923e-4	58	-3.996e-3	58
175	90	max	2.067	58	0.072	57	0.106	51	1.528e-4	51	9.42e-4	41	4.931e-3	39
176		min	-2.058	59	-0.512	24	-0.104	54	-1.449e-4	54	-4.978e-4	57	-5.189e-3	40
177	91	max	1.957	57	0.072	59	0.357	20	4.739e-3	29	3.183e-3	57	4.733e-3	41
178		min	-1.921	60	-0.395	24	-0.36	23	0	50	-3.587e-3	41	-4.621e-3	38
179	92	max	2.026	58	0.089	57	0.075	51	4.753e-3	29	2.386e-3	40	5.112e-3	39
180		min	-2.033	41	-0.407	24	-0.074	54	0	50	-1.99e-3	60	-5.292e-3	40
181	93	max	2.068	57	0.061	59	0.426	20	0	63	5.627e-3	41	2.597e-3	41
182		min	-2.033	60	-0.427	24	-0.429	23	-5.302e-3	29	-5.259e-3	57	-2.618e-3	58
183	94	max	1.798	58	0.073	57	0.566	22	0	63	1.792e-3	60	4.251e-3	39
184		min	-1.794	59	-0.393	24	-0.568	21	-4.677e-3	29	-2.14e-3	40	-4.412e-3	40
185	95	max	1.801	57	0.071	59	0.566	20	0	63	1.795e-3	59	4.448e-3	41
186		min	-1.793	41	-0.393	24	-0.568	23	-4.678e-3	29	-2.138e-3	40	-4.219e-3	58
187	96	max	1.548	58	0.112	21	0.477	21	2.167e-5	60	9.351e-4	57	3.907e-3	60
188		min	-1.53	59	-0.138	22	-0.474	22	-3.154e-5	29	-1.396e-3	43	-4.159e-3	40
189	97	max	1.552	57	0.112	21	0.479	23	1.224e-5	53	9.341e-4	57	4.058e-3	41
190		min	-1.528	60	-0.137	22	-0.475	20	-7.929e-5	29	-1.395e-3	43	-3.963e-3	38
191	98	max	1.513	58	0.138	59	0.89	23	1.035e-3	53	6.359e-4	53	6.626e-5	60
192		min	-1.518	59	-0.162	58	-0.884	20	-1.037e-3	52	-6.506e-4	52	-2.655e-4	38
193	99	max	1.713	58	0	63	0.796	23	2.638e-4	29	8.888e-4	53	3.647e-3	29
194		min	-1.791	41	-0.709	24	-0.791	20	-2.085e-5	58	-8.732e-4	52	-1.944e-5	58
195	100	max	1.57	58	0.14	59	0.702	23	1.351e-3	53	7.204e-4	54	5.742e-5	60
196		min	-1.573	59	-0.163	58	-0.697	20	-1.353e-3	52	-7.527e-4	72	-3.125e-4	40
197	101	max	1.708	58	0	63	0.609	23	2.09e-5	44	8.498e-4	54	1.248e-3	27
198		min	-1.76	41	-0.587	24	-0.604	54	0	50	-8.589e-4	51	-3.371e-4	21
199	102	max	1.549	58	0.144	59	0.529	51	1.556e-3	53	1.184e-3	54	8.753e-5	60
200		min	-1.55	59	-0.162	58	-0.526	54	-1.605e-3	52	-1.192e-3	51	-4.081e-4	29
201	103	max	1.816	40	0.036	57	0.671	60	2.264e-3	40	1.882e-4	40	1.116e-4	39
202		min	-1.804	41	-0.752	24	-0.672	57	-2.144e-3	59	-1.976e-4	41	-1.128e-4	38
203	104	max	2.127	40	0.034	59	0.323	58	2.202e-3	59	1.105e-4	88	1.227e-4	59
204		min	-2.108	41	-0.757	24	-0.332	59	-2.233e-3	57	-8.98e-5	94	-1.23e-4	58
205	105	max	2.015	58	0.054	57	0.497	22	5.304e-3	29	6.617e-3	40	2.618e-3	60
206		min	-2.01	59	-0.427	24	-0.498	21	0	50	-6.151e-3	59	-2.641e-3	40
207	106	max	0	105	0	105	0	105	0	105	0	105	0	105
208		min	0	1	0	1	0	1	0	1	0	1	0	1
209	107	max	2.066	40	0.043	57	0.24	57	2.246e-3	59	1.008e-4	41	1.206e-4	59
210		min	-2.053	41	-0.765	24	-0.239	60	-2.242e-3	58	-8.86e-5	57	-1.223e-4	57
211	108	max	2.168	40	0.056	59	0.654	58	1.118e-2	59	2.952e-6	57	1.369e-4	59
212		min	-2.15	41	-0.738	24	-0.656	59	-1.12e-2	57	-2.278e-4	45	-1.377e-4	57
213	109	max	1.455	38	0	63	0.999	21	0	63	2.93e-4	60	1.525e-4	41
214		min	-1.468	41	-0.367	56	-0.996	20	-2.889e-5	45	-3.165e-4	57	-1.676e-4	61
215	110	max	2.172	40	0.064	59	0.451	59	1.132e-2	58	2.354e-4	44	1.343e-4	60
216		min	-2.154	41	-0.74	24	-0.461	58	-1.139e-2	60	-9.243e-5	94	-1.347e-4	58
217	111	max	2.147	40	0.04	59	0.207	60	2.165e-3	58	2.835e-5	100	1.196e-4	59
218		min	-2.131	41	-0.764	24	-0.219	57	-2.322e-3	59	-4.789e-5	82	-1.206e-4	57
219	112	max	2.03	40	0.028	57	0.437	60	2.204e-3	57	1.181e-4	57	1.232e-4	60
220		min	-2.015	41	-0.757	24	-0.443	57	-2.176e-3	59	-1.326e-4	41	-1.244e-4	57
221	113	max	1.689	38	0	63	0.625	51	1.718e-5	45	1.993e-4	59	2.413e-3	29
222		min	-1.759	41	-0.512	24	-0.622	54	-1.191e-6	2	-1.984e-4	57	0	50
223	114	max	1.69	58	0	63	0.826	23	0	63	1.275e-3	41	2.46e-3	29
224		min	-1.776	41	-0.53	24	-0.82	20	-5.352e-3	29	-1.083e-3	58	-8.67e-5	58
225	115	max	1.788	40	0.036	57	0.478	57	2.189e-3	60	2.107e-4	41	1.05e-4	59
226		min	-1.776	41	-0.756	24	-0.486	60	-2.27e-3	58	-2.063e-4	57	-1.061e-4	58
227	116	max	2.074	57	0.068	59	0.104	51	1.533e-4	51	9.406e-4	41	5.149e-3	41

Envelope Node Displacements (Continued)

	Node Label		X [in]	LC	Y [in]	LC	Z [in]	LC	X Rotation [rad]	LC	Y Rotation [rad]	LC	Z Rotation [rad]	LC
228		min	-2.055	60	-0.512	24	-0.103	54	-1.434e-4	54	-4.962e-4	57	-4.969e-3	38
229	117	max	2.101	40	0.078	59	0.164	52	0	63	2.522e-3	41	5.45e-3	41
230		min	-2.06	60	-0.395	24	-0.166	53	-4.763e-3	29	-2.183e-3	57	-5.354e-3	38
231	118	max	2.136	57	0.064	59	0.194	52	5.064e-5	54	9.702e-4	88	5.358e-3	41
232		min	-2.108	41	-0.504	24	-0.196	53	-5.494e-5	51	-5.255e-4	94	-5.187e-3	38
233	119	max	2.034	57	0.087	59	0.074	51	4.753e-3	29	2.387e-3	40	5.331e-3	41
234		min	-2.033	41	-0.407	24	-0.073	54	0	50	-1.995e-3	59	-5.082e-3	38
235	120	max	1.989	40	0.079	57	0.136	51	0	63	3.664e-3	41	4.751e-3	60
236		min	-1.96	59	-0.398	24	-0.134	54	-4.724e-3	29	-3.229e-3	57	-5.085e-3	40
237	121	max	1.989	40	0.078	59	0.135	51	0	63	3.669e-3	41	4.966e-3	41
238		min	-1.959	60	-0.398	24	-0.133	54	-4.721e-3	29	-3.224e-3	57	-4.857e-3	38
239	122	max	2.147	40	0.069	57	0.078	52	1.268e-4	52	1.278e-4	59	5.301e-3	39
240		min	-2.143	59	-0.512	24	-0.079	53	-1.272e-4	53	-7.699e-4	44	-5.555e-3	40
241	123	max	2.16	57	0.073	59	0.077	52	1.244e-4	52	1.261e-4	59	5.528e-3	41
242		min	-2.131	41	-0.512	24	-0.078	53	-1.265e-4	53	-7.694e-4	44	-5.357e-3	38
243	124	max	2.111	40	0.078	57	0.107	52	4.736e-3	29	2.466e-3	57	5.306e-3	39
244		min	-2.084	59	-0.399	24	-0.108	53	0	50	-2.778e-3	41	-5.635e-3	40
245	125	max	2.111	57	0.082	59	0.105	52	4.734e-3	29	2.447e-3	57	5.535e-3	41
246		min	-2.07	60	-0.399	24	-0.107	53	0	50	-2.77e-3	41	-5.439e-3	38
247	126	max	2.064	58	0.087	57	0.05	52	0	63	2.936e-3	59	5.296e-3	39
248		min	-2.073	59	-0.407	24	-0.05	53	-4.745e-3	29	-3.296e-3	40	-5.475e-3	40
249	127	max	2.078	57	0.091	59	0.048	52	0	63	2.95e-3	59	5.521e-3	41
250		min	-2.071	41	-0.407	24	-0.049	53	-4.744e-3	29	-3.298e-3	40	-5.276e-3	38
251	128	max	2.042	57	0.076	59	0.224	52	4.74e-3	29	3.607e-3	40	5.266e-3	41
252		min	-2.036	41	-0.394	24	-0.226	53	0	50	-3.207e-3	59	-5.022e-3	58
253	129	max	2.101	40	0.072	57	0.165	52	0	63	2.529e-3	41	5.221e-3	39
254		min	-2.071	59	-0.395	24	-0.167	53	-4.765e-3	29	-2.202e-3	57	-5.55e-3	40
255	130	max	1.771	58	0	63	0.794	21	1.127e-3	41	5.648e-4	59	4.481e-3	45
256		min	-1.82	59	-1.013	29	-0.802	22	-1.076e-3	57	-7.491e-4	40	0	50
257	131	max	2.031	58	0.07	57	0.225	52	4.742e-3	29	3.609e-3	40	5.042e-3	39
258		min	-2.036	41	-0.394	24	-0.227	53	0	50	-3.197e-3	59	-5.221e-3	40
259	132	max	2.127	40	0.057	57	0.195	52	5.027e-5	54	9.722e-4	88	5.131e-3	39
260		min	-2.117	59	-0.504	24	-0.197	53	-5.721e-5	51	-5.275e-4	94	-5.386e-3	40
261	133	max	2.039	57	0.057	59	0.323	20	2.451e-6	60	4.279e-4	100	4.88e-3	41
262		min	-2.015	41	-0.504	24	-0.326	23	-2.215e-5	44	-8.515e-4	82	-4.698e-3	38
263	134	max	1.828	58	0.061	57	0.602	22	7.299e-5	29	7.362e-4	58	4.144e-3	39
264		min	-1.815	59	-0.5	24	-0.603	21	0	50	-1.161e-3	41	-4.346e-3	40
265	135	max	1.831	57	0.056	59	0.601	20	7.08e-5	29	7.332e-4	58	4.315e-3	41
266		min	-1.812	60	-0.5	24	-0.603	23	-3.262e-6	59	-1.158e-3	41	-4.166e-3	38
267	136	max	2.172	58	0.033	57	0.462	22	2.911e-6	41	6.789e-4	41	1.06e-3	20
268		min	-2.159	59	-0.552	24	-0.463	23	-5.311e-6	48	-3.837e-4	57	-3.625e-3	29
269	137	max	2.179	57	0.041	59	0.461	20	4.981e-6	49	1.461e-3	40	3.627e-3	29
270		min	-2.152	60	-0.552	24	-0.464	23	-1.407e-6	58	-1.124e-3	59	-1.059e-3	20
271	138	max	2.019	57	0.061	59	0.496	20	5.303e-3	29	6.162e-3	40	2.594e-3	41
272		min	-2.008	41	-0.427	24	-0.499	23	0	50	-5.707e-3	59	-2.5e-3	58
273	139	max	2.065	40	0.054	57	0.427	22	0	63	5.17e-3	41	2.649e-3	60
274		min	-2.038	59	-0.427	24	-0.429	23	-5.305e-3	29	-4.808e-3	57	-2.795e-3	40
275	140	max	1.732	58	0.074	57	0.637	22	4.822e-3	29	3.476e-3	57	4.037e-3	39
276		min	-1.708	59	-0.387	24	-0.638	21	0	50	-3.901e-3	41	-4.28e-3	40
277	141	max	1.735	57	0.071	59	0.636	20	4.819e-3	29	3.476e-3	58	4.183e-3	41
278		min	-1.706	60	-0.387	24	-0.638	23	0	50	-3.909e-3	41	-4.124e-3	38
279	142	max	2.03	40	0.059	57	0.323	22	2.906e-6	53	4.29e-4	100	4.665e-3	39
280		min	-2.022	59	-0.504	24	-0.325	23	-1.842e-5	34	-8.525e-4	82	-4.922e-3	40
281	143	max	1.985	58	0.072	57	0.289	22	0	63	2.132e-3	59	4.808e-3	39
282		min	-1.989	41	-0.394	24	-0.291	23	-4.764e-3	29	-2.531e-3	40	-4.989e-3	40
283	144	max	1.993	57	0.072	59	0.288	20	0	63	2.141e-3	59	5.027e-3	41
284		min	-1.989	41	-0.394	24	-0.291	23	-4.763e-3	29	-2.528e-3	40	-4.781e-3	58
285	145	max	1.957	40	0.071	57	0.358	22	4.742e-3	29	3.186e-3	57	4.527e-3	60
286		min	-1.928	59	-0.395	24	-0.36	23	0	50	-3.58e-3	41	-4.854e-3	40
287	146	max	1.714	58	0	63	0.826	21	0	63	7.99e-4	59	6.496e-3	29
288		min	-1.777	41	-0.73	24	-0.823	22	-4.765e-3	29	-7.548e-4	40	-1.589e-4	58
289	147	max	1.841	58	0.113	21	0.255	52	2.289e-3	20	1.296e-3	73	4.952e-3	39
290		min	-1.863	41	-0.135	22	-0.257	53	-2.237e-3	23	-1.294e-3	51	-5.057e-3	40
291	148	max	1.975	40	0.118	21	0.134	52	2.418e-3	22	1.301e-3	52	5.542e-3	41
292		min	-1.919	60	-0.138	22	-0.135	53	-2.373e-3	23	-1.301e-3	53	-5.521e-3	38
293	149	max	1.975	40	0.118	21	0.135	52	2.416e-3	20	1.289e-3	54	5.321e-3	60
294		min	-1.923	59	-0.138	22	-0.137	53	-2.384e-3	23	-1.289e-3	51	-5.715e-3	40
295	150	max	1.894	57	0.139	21	0.043	51	1.143e-3	22	1.298e-3	52	5.514e-3	41
296		min	-1.913	41	-0.158	22	-0.043	54	-1.039e-3	23	-1.298e-3	53	-5.194e-3	38
297	151	max	1.888	58	0.139	21	0.044	51	1.143e-3	22	1.282e-3	54	5.292e-3	39
298		min	-1.913	41	-0.158	22	-0.044	54	-1.04e-3	23	-1.282e-3	51	-5.396e-3	40
299	152	max	1.814	40	0.114	21	0.165	51	5.308e-4	54	1.312e-3	52	4.784e-3	41
300		min	-1.764	60	-0.136	22	-0.163	54	-5.859e-4	51	-1.312e-3	53	-4.745e-3	38
301	153	max	1.815	40	0.114	21	0.167	51	5.523e-4	54	1.281e-3	54	4.591e-3	60
302		min	-1.762	59	-0.136	22	-0.164	54	-6.104e-4	51	-1.283e-3	72	-4.982e-3	40
303	154	max	1.642	57	0.112	21	0.302	23	3.382e-4	54	1.327e-3	52	4.435e-3	41
304		min	-1.657	41	-0.136	22	-0.298	20	-4.161e-4	64	-1.328e-3	53	-4.141e-3	58
305	155	max	1.645	58	0.112	21	0.301	21	3.376e-4	54	1.28e-3	54	4.213e-3	39
306		min	-1.656	41	-0.136	22	-0.297	22	-4.213e-4	64	-1.283e-3	51	-4.336e-3	40
307	156	max	1.505	57	0.115	21	0.447	23	1.98e-4	54	1.34e-3	52	3.908e-3	41
308		min	-1.464	60	-0.132	22	-0.444	20	-4.424e-4	32	-1.341e-3	53	-3.875e-3	38
309	157	max	1.506	40	0.116	21	0.446	21	2.614e-4	60	1.273e-3	54	3.738e-3	60
310		min	-1.467	59	-0.131	22	-0.443	22	-4.512e-4	32	-1.274e-3	51	-4.084e-3	40
311	158	max	1.578	57	0.115	23	0.51	23	1.313e-3	53	1.98e-3	51	4.209e-3	41
312		min	-1.572	60	-0.13	20	-0.506	20	-1.36e-3	52	-1.973e-3	54	-4.069e-3	58
313	159	max	1.573	58	0.115	21	0.508	21	1.629e-3	53	1.286e-3	54	4.075e-3	60
314		min	-1.574	59	-0.131	22	-0.505	22	-1.679e-3	52	-1.299e-3	51	-4.234e-3	40
315	160	max	2.187	58	0.033	53	0.233	51	4.127e-5	53	3.248e-4	94	1.064e-3	21

HANSON™

Company : Hanson Professional Services
Designer : S. Steiner
Job Number : 25i0007
Model Name : Eli Lilly K215 Unloading Rack

2/19/2026
2:59:44 PM
Checked By : RMF

Envelope Node Displacements (Continued)

	Node Label		X [in]	LC	Y [in]	LC	Z [in]	LC	X Rotation [rad]	LC	Y Rotation [rad]	LC	Z Rotation [rad]	LC
316		min	-2.154	41	-0.553	24	-0.231	54	-3.391e-5	52	-6.31e-4	88	-3.626e-3	29
317	161	max	1.598	57	0.114	23	0.702	23	1.119e-3	53	1.182e-3	72	4.018e-3	41
318		min	-1.593	60	-0.133	20	-0.698	20	-1.118e-3	52	-1.138e-3	53	-3.921e-3	58
319	162	max	1.542	57	0.113	23	0.894	23	9.574e-4	53	1.114e-3	52	3.919e-3	41
320		min	-1.54	60	-0.133	20	-0.89	20	-9.556e-4	52	-1.102e-3	53	-3.829e-3	58
321	163	max	1.536	58	0.111	23	0.891	21	1.058e-3	53	8.311e-4	53	3.884e-3	41
322		min	-1.541	59	-0.135	20	-0.888	20	-1.061e-3	52	-8.538e-4	52	-3.86e-3	38
323	164	max	1.432	57	0.117	23	1.107	21	7.856e-4	53	1.374e-3	52	3.592e-3	41
324		min	-1.435	60	-0.129	20	-1.104	20	-7.888e-4	52	-1.376e-3	53	-3.533e-3	38
325	165	max	1.44	58	0.117	23	1.104	21	1.081e-3	53	1.257e-3	54	3.576e-3	41
326		min	-1.442	59	-0.129	20	-1.101	20	-1.134e-3	58	-1.26e-3	51	-3.554e-3	38
327	166	max	1.492	57	0	2	1	21	0	63	3.875e-4	60	3.754e-3	41
328		min	-1.493	60	-0.352	61	-0.997	20	-2.429e-5	44	-4.081e-4	57	-3.676e-3	58
329	167	max	1.496	58	0	63	0.998	21	0	63	3.874e-4	60	3.73e-3	41
330		min	-1.499	59	-0.338	20	-0.994	20	-3.444e-5	45	-4.069e-4	57	-3.707e-3	38
331	168	max	1.738	57	0.025	59	0.798	23	2.402e-6	55	2.294e-4	41	3.967e-3	41
332		min	-1.789	41	-0.257	20	-0.794	20	-1.422e-6	2	-2.424e-4	40	-3.879e-3	58
333	169	max	1.737	58	0	63	0.796	21	3.432e-4	29	4.147e-4	59	3.623e-3	29
334		min	-1.791	41	-0.811	24	-0.792	22	-2.752e-5	58	-6.57e-4	40	-5.704e-5	58
335	170	max	1.731	57	0.016	59	0.606	23	1.277e-5	62	8.758e-5	58	4.109e-3	41
336		min	-1.777	60	-0.281	61	-0.602	20	-6.19e-6	2	-9.556e-5	43	-3.996e-3	58
337	171	max	1.732	58	0	63	0.604	21	2.45e-5	44	8.892e-5	58	4.001e-3	60
338		min	-1.781	59	-0.609	24	-0.601	22	0	50	-9.971e-5	43	-4.139e-3	40
339	172	max	2.083	40	0.063	59	0.198	51	5.315e-3	29	5.05e-3	58	2.647e-3	41
340		min	-2.057	60	-0.428	24	-0.195	54	0	50	-5.315e-3	41	-2.664e-3	58
341	173	max	2.085	58	0.055	53	0.2	51	5.318e-3	29	4.575e-3	58	2.788e-3	60
342		min	-2.043	59	-0.428	24	-0.198	54	0	50	-4.86e-3	41	-2.938e-3	40
343	174	max	1.594	58	0.111	21	0.7	21	1.42e-3	53	9.504e-4	54	3.927e-3	60
344		min	-1.597	59	-0.135	22	-0.697	22	-1.423e-3	52	-1.006e-3	72	-4.044e-3	40
345	175	max	1.846	57	0.113	21	0.254	52	2.288e-3	22	1.307e-3	52	5.174e-3	41
346		min	-1.864	41	-0.135	22	-0.257	23	-2.232e-3	23	-1.306e-3	53	-4.869e-3	58
347	176	max	2.173	40	0.044	59	0.231	51	3.837e-5	53	1.393e-3	60	3.628e-3	29
348		min	-2.165	60	-0.553	24	-0.228	54	-3.382e-5	52	-1.69e-3	40	-1.063e-3	21
349	177	max	1.537	58	0.117	21	0.672	22	2.146e-3	20	5.454e-3	54	3.931e-3	39
350		min	-1.502	59	-0.129	22	-0.673	21	-2.012e-3	23	-5.452e-3	52	-4.215e-3	40
351	178	max	1.672	57	0.112	21	0.531	20	2.168e-3	20	1.311e-3	72	4.58e-3	41
352		min	-1.682	41	-0.136	22	-0.533	23	-2.071e-3	23	-1.305e-3	53	-4.282e-3	58
353	179	max	1.668	58	0.112	21	0.531	22	2.173e-3	20	1.299e-3	73	4.358e-3	39
354		min	-1.682	41	-0.136	22	-0.533	21	-2.068e-3	23	-1.294e-3	51	-4.478e-3	40
355	180	max	1.782	40	0.112	21	0.391	20	2.215e-3	20	1.31e-3	72	4.586e-3	41
356		min	-1.731	60	-0.136	22	-0.394	23	-2.146e-3	23	-1.305e-3	53	-4.544e-3	38
357	181	max	1.783	40	0.112	21	0.392	22	2.22e-3	20	1.298e-3	73	4.398e-3	60
358		min	-1.735	59	-0.136	22	-0.394	23	-2.146e-3	23	-1.294e-3	51	-4.786e-3	40
359	184	max	2.127	57	0.145	23	0.623	22	9.631e-5	44	6.383e-4	58	2.684e-3	45
360		min	-2.107	60	-0.313	20	-0.627	21	-1.064e-5	59	-1.075e-3	41	-5.301e-7	2
361	185	max	2.124	58	0.145	23	0.624	20	9.965e-5	45	6.581e-4	58	0	63
362		min	-2.109	59	-0.313	20	-0.627	23	0	50	-1.07e-3	41	-2.682e-3	44
363	186	max	2.26	57	0.144	59	0.483	22	6.072e-6	48	9.598e-4	44	2.689e-3	45
364		min	-2.241	41	-0.317	20	-0.486	21	-1.549e-6	82	-2.254e-4	94	0	50
365	187	max	2.259	40	0.14	23	0.483	20	1.548e-6	82	9.602e-4	44	1.809e-7	2
366		min	-2.241	41	-0.317	20	-0.486	23	-6.07e-6	48	-2.26e-4	94	-2.687e-3	44
367	188	max	2.384	40	0.143	59	0.342	22	8.939e-6	60	5.471e-4	57	2.687e-3	45
368		min	-2.366	41	-0.316	20	-0.344	21	-3.931e-5	34	-9.921e-4	45	0	50
369	189	max	1.918	38	0	63	0.815	51	4.665e-6	62	1.64e-4	60	7.08e-5	41
370		min	-1.93	41	-0.657	56	-0.811	54	-6.518e-6	27	-1.783e-4	40	-1.794e-4	61
371	190	max	2.558	40	0.145	23	0.212	52	6.054e-5	54	1.088e-3	40	2.688e-3	45
372		min	-2.54	41	-0.317	20	-0.212	53	-6.55e-5	51	-6.292e-4	59	0	50
373	191	max	2.558	40	0.145	23	0.213	52	6.006e-5	54	1.079e-3	40	2.013e-7	2
374		min	-2.54	41	-0.317	20	-0.214	53	-6.839e-5	51	-6.202e-4	59	-2.685e-3	44
375	192	max	2.613	40	0.158	23	0.092	52	1.149e-4	52	1.646e-4	60	2.687e-3	45
376		min	-2.595	41	-0.328	20	-0.092	53	-1.156e-4	53	-9.491e-4	44	0	50
377	193	max	2.613	40	0.158	23	0.094	52	1.181e-4	52	1.591e-4	60	2.83e-7	2
378		min	-2.595	41	-0.328	20	-0.093	53	-1.168e-4	53	-9.479e-4	44	-2.687e-3	44
379	194	max	2.472	40	0.155	23	0.12	51	1.491e-4	51	1.182e-3	41	2.899e-3	45
380		min	-2.455	41	-0.327	20	-0.119	54	-1.363e-4	54	-7.397e-4	57	-4.191e-7	2
381	195	max	2.472	40	0.155	23	0.121	51	1.482e-4	51	1.18e-3	41	0	63
382		min	-2.455	41	-0.327	20	-0.12	54	-1.376e-4	54	-7.104e-4	57	-2.9e-3	44
383	196	max	2.266	40	0.145	59	0.244	51	4.943e-5	53	7.424e-4	60	2.9e-3	45
384		min	-2.248	41	-0.317	20	-0.241	54	-4.417e-5	52	-1.175e-3	40	-4.84e-7	2
385	197	max	1.859	38	0	63	0.988	21	5.531e-4	52	2.234e-4	60	2.826e-4	41
386		min	-1.875	41	-0.844	56	-0.984	22	-5.932e-4	73	-2.375e-4	40	-2.681e-4	58
387	198	max	1.947	38	0	63	0.656	51	2.183e-5	45	1.276e-4	60	7.768e-5	41
388		min	-1.954	41	-0.654	61	-0.653	54	-1.191e-6	2	-1.405e-4	40	-1.89e-4	61
389	199	max	2.266	40	0.142	23	0.245	51	5.299e-5	53	7.404e-4	60	0	63
390		min	-2.248	41	-0.317	20	-0.243	54	-4.41e-5	52	-1.173e-3	40	-2.897e-3	44
391	200	max	2.065	57	0.143	23	0.376	21	1.542e-5	60	1.13e-3	41	2.894e-3	45
392		min	-2.047	41	-0.314	20	-0.37	22	-7.383e-5	40	-7.052e-4	58	-5.119e-7	2
393	201	max	2.067	58	0.143	23	0.375	21	0	63	1.124e-3	41	0	63
394		min	-2.048	59	-0.314	20	-0.371	22	-7.346e-5	45	-7.254e-4	58	-2.891e-3	44
395	202	max	2.434	58	0.234	21	0.059	51	5.576e-4	51	1.242e-3	54	1.686e-3	39
396		min	-2.479	41	-0.26	22	-0.058	54	-5.401e-4	54	-1.249e-3	70	-1.8e-3	42
397	203	max	2.384	40	0.141	23	0.342	20	6.085e-6	53	5.184e-4	57	1.656e-7	2
398		min	-2.366	41	-0.316	20	-0.344	23	-3.164e-5	34	-1.025e-3	45	-2.686e-3	44
399	204	max	2.572	40	0.217	21	0.154	52	7.929e-4	23	1.238e-3	71	1.756e-3	41
400		min	-2.492	59	-0.243	22	-0.154	53	-8.145e-4	20	-1.238e-3	74	-1.95e-3	42
401	205	max	1.897	58	0	63	0.987	21	0	63	5.02e-4	41	1.169e-3	59
402		min	-1.901	59	-0.66	62	-0.985	22	-4.385e-5	45	-5.184e-4	57	-1.591e-3	38
403	206	max	2.017	40	0.191	23	0.37	51	1.402e-3	64	1.045e-4	60	2.718e-4	40

Envelope Node Displacements (Continued)

	Node Label		X [in]	LC	Y [in]	LC	Z [in]	LC	X Rotation [rad]	LC	Y Rotation [rad]	LC	Z Rotation [rad]	LC
404		min	-2.064	41	-0.258	20	-0.365	54	-4.428e-4	54	-1.122e-4	40	-2.832e-4	39
405	207	max	1.789	58	0.216	23	1.092	21	7.603e-4	58	1.304e-3	54	1.156e-3	59
406		min	-1.791	59	-0.233	20	-1.091	22	-7.642e-4	59	-1.311e-3	72	-1.462e-3	38
407	208	max	2.293	40	0.212	21	0.184	51	7.077e-4	64	1.242e-3	54	1.408e-3	41
408		min	-2.21	59	-0.242	22	-0.181	54	-4.121e-4	54	-1.305e-3	70	-1.628e-3	38
409	209	max	1.896	57	0	2	0.988	23	0	63	5.025e-4	41	1.619e-3	41
410		min	-1.897	60	-0.658	61	-0.984	20	-3.384e-5	44	-5.171e-4	57	-1.166e-3	57
411	210	max	1.951	58	0.032	57	0.789	21	5.868e-6	49	1.759e-4	60	1.136e-3	59
412		min	-1.954	59	-0.487	62	-0.785	22	-1.305e-5	27	-1.95e-4	40	-1.648e-3	38
413	211	max	1.945	40	0.216	21	0.694	22	1.529e-3	59	1.319e-3	52	1.214e-3	41
414		min	-1.888	60	-0.232	22	-0.698	21	-2.84e-3	32	-1.332e-3	71	-1.183e-3	38
415	212	max	2.086	57	0.209	21	0.553	22	1.256e-3	59	1.281e-3	72	1.493e-3	39
416		min	-2.117	41	-0.24	22	-0.557	21	-2.043e-3	32	-1.267e-3	71	-1.296e-3	40
417	213	max	1.954	57	0.028	59	0.79	21	3.84e-6	61	1.756e-4	60	1.673e-3	41
418		min	-1.951	60	-0.512	61	-0.784	20	-1.464e-6	2	-1.987e-4	40	-1.146e-3	57
419	214	max	1.976	58	0.035	57	0.601	21	2.926e-5	44	3.733e-5	60	1.096e-3	59
420		min	-1.978	59	-0.484	62	-0.597	22	0	50	-5.244e-5	40	-1.424e-3	42
421	215	max	1.979	57	0.214	23	0.508	21	2.071e-3	32	1.295e-3	70	1.302e-3	43
422		min	-1.971	60	-0.234	20	-0.502	20	-7.948e-4	59	-1.288e-3	54	-1.12e-3	57
423	216	max	1.882	40	0.214	21	0.446	21	1.949e-3	64	1.254e-3	54	9.733e-4	41
424		min	-1.82	59	-0.235	22	-0.441	22	-4.03e-4	54	-1.275e-3	70	-1.154e-3	42
425	217	max	1.982	57	0.03	59	0.602	21	1.778e-5	43	3.09e-5	60	1.459e-3	41
426		min	-1.976	60	-0.51	61	-0.596	20	-6.144e-6	2	-5.416e-5	40	-1.112e-3	58
427	218	max	2.222	40	0.208	21	0.413	22	9.436e-4	53	1.272e-3	52	1.44e-3	41
428		min	-2.145	60	-0.241	22	-0.415	21	-1.156e-3	64	-1.267e-3	71	-1.327e-3	38
429	219	max	1.92	58	0.209	23	0.883	21	6.898e-4	58	1.295e-3	54	1.183e-3	59
430		min	-1.924	59	-0.24	20	-0.878	22	-6.662e-4	59	-1.306e-3	72	-1.721e-3	38
431	220	max	1.923	57	0.211	23	0.884	21	5.792e-4	20	1.33e-3	52	1.749e-3	41
432		min	-1.922	60	-0.239	20	-0.878	20	-5.59e-4	23	-1.335e-3	53	-1.18e-3	58
433	221	max	2.332	57	0.21	21	0.271	22	7.948e-4	59	1.256e-3	52	1.592e-3	43
434		min	-2.369	41	-0.239	22	-0.273	21	-9.404e-4	61	-1.263e-3	71	-1.42e-3	40
435	222	max	2.444	57	0.234	21	0.058	51	5.577e-4	51	1.266e-3	70	1.823e-3	43
436		min	-2.481	41	-0.26	22	-0.057	54	-5.402e-4	54	-1.264e-3	53	-1.655e-3	40
437	223	max	1.972	58	0.213	21	0.507	21	1.957e-3	64	1.259e-3	54	1.107e-3	41
438		min	-1.973	59	-0.235	22	-0.503	22	-5.745e-4	53	-1.279e-3	70	-1.273e-3	42
439	224	max	1.968	57	0.211	23	0.696	21	6.004e-4	20	1.328e-3	52	1.623e-3	41
440		min	-1.964	60	-0.238	20	-0.69	20	-5.805e-4	23	-1.335e-3	73	-1.1e-3	57
441	225	max	2.326	58	0.21	21	0.273	52	7.882e-4	53	1.243e-3	71	1.449e-3	39
442		min	-2.368	41	-0.239	22	-0.274	53	-9.278e-4	55	-1.232e-3	74	-1.559e-3	42
443	226	max	2.227	40	0.209	21	0.413	20	9.366e-4	53	1.248e-3	71	1.275e-3	41
444		min	-2.149	59	-0.24	22	-0.415	23	-1.156e-3	64	-1.25e-3	72	-1.494e-3	38
445	227	max	2.085	58	0.208	21	0.554	20	1.227e-3	53	1.249e-3	71	1.336e-3	39
446		min	-2.116	41	-0.241	22	-0.557	23	-2.023e-3	64	-1.262e-3	72	-1.471e-3	40
447	228	max	1.944	40	0.216	21	0.694	20	1.41e-3	53	1.315e-3	71	1.133e-3	41
448		min	-1.888	59	-0.232	22	-0.698	23	-2.776e-3	64	-1.29e-3	74	-1.251e-3	38
449	229	max	1.783	57	0.216	23	1.093	23	5.417e-4	54	1.336e-3	52	1.49e-3	41
450		min	-1.787	60	-0.232	20	-1.091	20	-5.518e-4	51	-1.343e-3	73	-1.162e-3	57
451	230	max	1.963	58	0.208	21	0.695	21	7.938e-4	58	1.301e-3	54	1.088e-3	59
452		min	-1.965	59	-0.239	22	-0.691	22	-7.727e-4	59	-1.31e-3	72	-1.575e-3	38
453	231	max	2.291	40	0.195	23	0.247	51	7.063e-4	64	7.74e-5	60	3.516e-4	38
454		min	-2.202	41	-0.259	20	-0.242	54	-4.118e-4	54	-8.189e-5	40	-3.353e-4	41
455	232	max	2.289	40	0.212	21	0.182	51	7.049e-4	64	1.321e-3	70	1.566e-3	41
456		min	-2.212	60	-0.242	22	-0.18	54	-4.115e-4	54	-1.268e-3	73	-1.454e-3	38
457	233	max	2.577	40	0.217	21	0.152	52	7.97e-4	23	1.261e-3	52	1.879e-3	43
458		min	-2.492	60	-0.243	22	-0.152	53	-8.163e-4	20	-1.261e-3	53	-1.792e-3	38
459	234	max	2.029	57	0.209	21	0.306	51	1.463e-3	32	1.31e-3	70	1.39e-3	39
460		min	-2.065	41	-0.24	22	-0.303	54	-5.572e-4	60	-1.272e-3	73	-1.193e-3	40
461	235	max	2.068	40	0.191	23	0.571	52	1.231e-3	53	9.892e-5	58	3.02e-4	40
462		min	-2.117	41	-0.258	20	-0.571	53	-2.023e-3	64	-9.536e-5	41	-3.135e-4	39
463	236	max	2.224	40	0.191	23	0.453	52	9.401e-4	53	7.785e-5	58	3.084e-4	38
464		min	-2.136	41	-0.258	20	-0.453	53	-1.156e-3	64	-7.725e-5	82	-2.915e-4	41
465	237	max	1.883	40	0.214	21	0.447	21	2.096e-3	32	1.287e-3	70	1.125e-3	41
466		min	-1.817	60	-0.235	22	-0.441	22	-6.927e-4	60	-1.28e-3	75	-1.033e-3	38
467	238	max	2.035	58	0.208	21	0.307	51	1.401e-3	64	1.24e-3	73	1.231e-3	39
468		min	-2.064	41	-0.241	22	-0.304	54	-4.378e-4	54	-1.297e-3	70	-1.366e-3	40
469	239	max	0	105	0	105	0	105	0	105	0	105	0	105
470		min	0	1	0	1	0	1	0	1	0	1	0	1
471	240	max	0	105	0	105	0	105	0	105	0	105	0	105
472		min	0	1	0	1	0	1	0	1	0	1	0	1
473	241	max	0	105	0	105	0	105	0	105	0	105	0	105
474		min	0	1	0	1	0	1	0	1	0	1	0	1
475	242	max	1.937	57	0.22	23	0.877	21	5.792e-4	20	1.33e-3	52	1.749e-3	41
476		min	-1.942	60	-0.248	20	-0.871	20	-5.59e-4	23	-1.335e-3	53	-1.18e-3	58
477	243	max	1.94	58	0.218	23	0.877	21	6.898e-4	58	1.295e-3	54	1.183e-3	59
478		min	-1.938	59	-0.249	20	-0.872	22	-6.662e-4	59	-1.306e-3	72	-1.721e-3	38
479	244	max	1.797	57	0.226	23	1.087	23	5.408e-4	54	1.336e-3	52	1.49e-3	41
480		min	-1.804	60	-0.242	20	-1.084	20	-5.509e-4	51	-1.343e-3	73	-1.163e-3	57
481	245	max	1.806	58	0.225	23	1.084	21	7.603e-4	58	1.304e-3	54	1.156e-3	59
482		min	-1.805	59	-0.242	20	-1.084	22	-7.642e-4	59	-1.311e-3	72	-1.462e-3	38
483	246	max	1.981	58	0.218	21	0.688	21	7.938e-4	58	1.301e-3	54	1.088e-3	59
484		min	-1.978	59	-0.249	22	-0.683	22	-7.727e-4	59	-1.31e-3	72	-1.575e-3	38
485	247	max	1.981	57	0.221	23	0.689	21	6.004e-4	20	1.328e-3	52	1.623e-3	41
486		min	-1.983	60	-0.247	20	-0.683	20	-5.805e-4	23	-1.335e-3	73	-1.101e-3	57
487	248	max	2.393	58	0.412	53	0.404	52	9.823e-4	53	5.412e-4	32	8.266e-3	64
488		min	-2.305	59	-1.283	64	-0.39	53	-1.631e-3	64	-3.659e-4	60	-1.408e-3	53
489	249	max	2.273	58	0.407	53	0.531	20	1.454e-3	53	1.124e-3	32	8.301e-3	64
490		min	-2.259	59	-1.29	64	-0.523	23	-3.04e-3	64	-4.254e-4	60	-1.383e-3	54
491	250	max	2.183	57	0.771	59	0.529	22	1.639e-3	59	2.695e-4	54	3.481e-3	59

Envelope Node Displacements (Continued)

	Node Label		X [in]	LC	Y [in]	LC	Z [in]	LC	X Rotation [rad]	LC	Y Rotation [rad]	LC	Z Rotation [rad]	LC
492		min	-2.226	60	-1.464	32	-0.521	21	-3.139e-3	32	-1.038e-3	64	-9.309e-3	32
493	251	max	2.304	57	0.766	59	0.397	52	9.904e-4	53	1.811e-4	58	3.454e-3	60
494		min	-2.272	60	-1.471	32	-0.384	53	-1.635e-3	32	-4.472e-4	33	-9.342e-3	32
495	252	max	2.46	58	0.422	53	0.186	59	9.675e-4	64	2.595e-4	60	8.268e-3	64
496		min	-2.369	59	-1.286	64	-0.195	58	-4.556e-4	54	-3.382e-4	38	-1.41e-3	53
497	253	max	2.372	57	0.779	59	0.154	23	9.656e-4	64	2.799e-4	33	3.508e-3	59
498		min	-2.341	60	-1.475	32	-0.172	56	-4.586e-4	54	-1.789e-4	58	-9.369e-3	32
499	254	max	2.22	58	0.407	53	0.301	59	2.173e-3	64	3.289e-4	60	8.3e-3	64
500		min	-2.204	59	-1.29	64	-0.301	58	-6.6e-4	54	-9.135e-4	32	-1.382e-3	53
501	255	max	2.124	57	0.761	59	0.3	23	2.279e-3	32	9.015e-4	64	3.434e-3	60
502		min	-2.166	60	-1.459	32	-0.302	20	-8.69e-4	60	-3.146e-4	54	-9.285e-3	32
503	256	max	2.175	57	0.589	59	0.522	22	1.631e-3	59	2.1e-4	54	3.405e-3	59
504		min	-2.207	60	-0.968	32	-0.527	21	-3.121e-3	32	-8.483e-4	64	-9.068e-3	32
505	257	max	2.231	58	0.337	53	0.523	20	1.449e-3	53	9.548e-4	32	8.078e-3	64
506		min	-2.229	59	-0.848	64	-0.528	23	-3.024e-3	64	-4.074e-4	60	-1.343e-3	54
507	258	max	2.419	58	0.35	53	0.175	59	9.837e-4	64	2.787e-4	60	8.045e-3	64
508		min	-2.339	59	-0.845	64	-0.181	58	-4.607e-4	54	-4.872e-4	32	-1.37e-3	53
509	259	max	2.296	57	0.585	59	0.391	20	9.955e-4	53	1.235e-4	54	3.378e-3	60
510		min	-2.252	60	-0.973	32	-0.38	53	-1.652e-3	32	-5.724e-4	64	-9.102e-3	32
511	260	max	2.364	57	0.595	59	0.159	23	9.818e-4	64	4.159e-4	64	3.432e-3	59
512		min	-2.322	60	-0.975	32	-0.169	20	-4.638e-4	54	-1.456e-4	54	-9.128e-3	32
513	261	max	2.115	57	0.581	59	0.308	21	2.261e-3	32	7.105e-4	64	3.358e-3	60
514		min	-2.147	60	-0.964	32	-0.3	22	-8.61e-4	60	-2.535e-4	54	-9.044e-3	32
515	262	max	2.178	58	0.336	53	0.307	51	2.157e-3	64	3.097e-4	60	8.077e-3	64
516		min	-2.174	59	-0.848	64	-0.303	54	-6.549e-4	54	-7.434e-4	32	-1.341e-3	53
517	263	max	2.352	58	0.34	53	0.394	52	9.873e-4	53	7.104e-4	32	8.043e-3	64
518		min	-2.275	59	-0.843	64	-0.385	53	-1.647e-3	64	-3.839e-4	60	-1.368e-3	53
519	264	max	2.165	57	0.418	59	0.517	22	1.62e-3	59	1.289e-4	54	3.054e-3	59
520		min	-2.187	60	-0.558	58	-0.531	21	-3.095e-3	32	-5.634e-4	64	-7.904e-3	32
521	265	max	2.38	58	0.284	53	0.17	51	1.009e-3	64	3.169e-4	60	6.966e-3	64
522		min	-2.31	59	-0.461	55	-0.174	54	-4.69e-4	54	-7.604e-4	32	-1.189e-3	53
523	266	max	2.138	58	0.272	53	0.324	51	2.131e-3	64	2.715e-4	60	6.998e-3	64
524		min	-2.144	59	-0.472	55	-0.315	54	-6.466e-4	54	-4.71e-4	32	-1.161e-3	53
525	267	max	2.312	58	0.274	53	0.387	20	9.944e-4	53	9.696e-4	32	6.964e-3	64
526		min	-2.245	59	-0.458	55	-0.382	23	-1.671e-3	64	-4.098e-4	60	-1.187e-3	53
527	268	max	2.286	57	0.416	59	0.385	22	1.003e-3	53	2.045e-4	54	3.027e-3	60
528		min	-2.233	60	-0.56	58	-0.381	21	-1.678e-3	32	-8.584e-4	64	-7.938e-3	32
529	269	max	2.191	58	0.272	53	0.517	22	1.442e-3	53	6.967e-4	32	6.999e-3	64
530		min	-2.199	59	-0.472	55	-0.531	21	-3.e-3	64	-3.816e-4	60	-1.162e-3	54
531	270	max	2.355	57	0.423	59	0.168	51	1.007e-3	64	7.182e-4	64	3.081e-3	59
532		min	-2.303	60	-0.566	58	-0.171	54	-4.721e-4	54	-2.439e-4	54	-7.964e-3	32
533	271	max	2.105	57	0.414	59	0.322	51	2.234e-3	32	4.09e-4	64	3.007e-3	60
534		min	-2.127	60	-0.553	58	-0.312	54	-8.488e-4	60	-1.552e-4	54	-7.88e-3	32
535	272	max	2.272	57	0.28	59	0.379	22	1.007e-3	53	2.513e-4	54	2.156e-3	60
536		min	-2.211	60	-0.333	58	-0.386	21	-1.696e-3	32	-1.054e-3	64	-5.061e-3	32
537	273	max	2.341	57	0.285	59	0.191	51	1.026e-3	64	9.363e-4	64	2.21e-3	59
538		min	-2.281	60	-0.336	58	-0.187	54	-4.778e-4	54	-3.124e-4	54	-5.087e-3	32
539	274	max	2.276	40	0.225	53	0.379	20	9.985e-4	53	1.142e-3	32	4.297e-3	64
540		min	-2.215	59	-0.293	20	-0.386	21	-1.687e-3	64	-4.083e-4	60	-7.368e-4	53
541	275	max	2.091	57	0.279	59	0.334	51	2.214e-3	32	2.468e-4	33	2.136e-3	60
542		min	-2.108	41	-0.33	58	-0.321	54	-8.391e-4	60	-1.809e-4	58	-5.004e-3	32
543	276	max	2.101	58	0.224	53	0.335	51	2.113e-3	64	2.599e-4	60	4.331e-3	64
544		min	-2.114	59	-0.299	55	-0.323	54	-6.408e-4	54	-3.174e-4	38	-7.104e-4	53
545	277	max	2.346	40	0.234	53	0.192	51	1.028e-3	64	3.285e-4	60	4.299e-3	64
546		min	-2.281	59	-0.295	55	-0.189	54	-4.748e-4	54	-9.51e-4	32	-7.389e-4	53
547	278	max	2.151	57	0.281	59	0.513	22	1.612e-3	59	1.868e-4	58	2.183e-3	59
548		min	-2.166	60	-0.332	58	-0.534	21	-3.077e-3	32	-4.274e-4	33	-5.027e-3	32
549	279	max	2.154	58	0.224	53	0.514	22	1.438e-3	53	5.272e-4	32	4.332e-3	64
550		min	-2.17	59	-0.299	55	-0.534	21	-2.984e-3	64	-3.831e-4	60	-7.122e-4	54
551	280	max	1.915	58	0.245	21	0.462	51	3.019e-3	64	4.75e-4	60	5.467e-4	43
552		min	-1.864	59	-0.266	22	-0.444	54	-7.002e-4	54	-9.056e-4	32	-6.286e-4	38
553	281	max	2.389	57	0.242	21	0.239	22	7.941e-4	59	1.303e-4	54	7.639e-4	39
554		min	-2.414	41	-0.27	22	-0.245	21	-1.069e-3	32	-5.492e-4	64	-6.678e-4	42
555	282	max	1.919	57	0.245	21	0.46	51	3.21e-3	32	8.137e-4	64	5.393e-4	39
556		min	-1.862	60	-0.266	22	-0.443	54	-1.088e-3	60	-3.031e-4	54	-5.004e-4	42
557	283	max	2.383	58	0.241	21	0.24	52	7.782e-4	53	6.028e-4	32	7.986e-4	43
558		min	-2.414	41	-0.27	22	-0.245	21	-1.055e-3	64	-2.282e-4	60	-8.67e-4	38
559	284	max	1.985	58	0.248	21	0.649	22	1.719e-3	53	1.097e-3	32	5.983e-4	43
560		min	-1.936	59	-0.263	22	-0.68	21	-4.159e-3	64	-5.122e-4	60	-6.721e-4	38
561	285	max	2.632	40	0.248	21	0.128	52	6.19e-4	23	2.491e-5	100	8.818e-4	39
562		min	-2.558	60	-0.274	22	-0.129	53	-6.672e-4	61	-1.104e-4	32	-8.246e-4	57
563	286	max	2.091	58	0.227	23	0.338	51	2.116e-3	64	2.832e-4	60	3.211e-3	64
564		min	-2.109	41	-0.282	20	-0.325	54	-6.42e-4	54	-3.523e-4	38	-5.206e-4	53
565	287	max	2.629	40	0.248	21	0.131	52	6.113e-4	23	1.25e-4	32	9.522e-4	43
566		min	-2.56	59	-0.274	22	-0.132	53	-6.548e-4	22	-5.651e-5	60	-1.061e-3	38
567	288	max	2.507	57	0.265	21	0.078	51	5.817e-4	51	3.633e-4	64	8.643e-4	39
568		min	-2.533	41	-0.291	22	-0.076	54	-5.28e-4	54	-1.242e-4	53	-7.709e-4	42
569	289	max	1.989	57	0.247	21	0.649	22	2.045e-3	59	2.197e-4	54	5.93e-4	39
570		min	-1.935	60	-0.263	22	-0.68	21	-4.324e-3	32	-9.423e-4	64	-5.62e-4	42
571	290	max	2.499	58	0.265	21	0.079	51	5.804e-4	51	1.112e-4	59	9.13e-4	59
572		min	-2.532	41	-0.292	22	-0.078	54	-5.268e-4	54	-3.581e-4	32	-9.846e-4	38
573	291	max	2.021	58	0.244	21	0.516	21	2.983e-3	64	4.744e-4	60	5.985e-4	43
574		min	-2.022	59	-0.266	22	-0.494	54	-6.718e-4	54	-9.422e-4	32	-6.09e-4	38
575	292	max	2.028	57	0.245	23	0.518	21	3.191e-3	32	8.453e-4	64	6.162e-4	39
576		min	-2.021	60	-0.265	20	-0.493	22	-1.089e-3	60	-2.971e-4	54	-5.798e-4	57
577	293	max	2.274	40	0.228	21	0.378	22	9.982e-4	53	1.111e-3	32	3.177e-3	64
578		min	-2.207	59	-0.282	22	-0.388	21	-1.685e-3	64	-3.956e-4	60	-5.47e-4	53
579	294	max	2.345	40	0.232	23	0.199	51	1.025e-3	64	3.052e-4	60	3.179e-3	64

Envelope Node Displacements (Continued)

	Node Label		X [in]	LC	Y [in]	LC	Z [in]	LC	X Rotation [rad]	LC	Y Rotation [rad]	LC	Z Rotation [rad]	LC
580		min	-2.273	59	-0.283	20	-0.194	54	-4.735e-4	54	-9.093e-4	32	-5.491e-4	53
581	295	max	2.27	40	0.252	59	0.377	22	1.006e-3	53	2.488e-4	54	1.789e-3	60
582		min	-2.204	60	-0.287	58	-0.388	21	-1.695e-3	32	-1.028e-3	64	-3.852e-3	32
583	296	max	2.086	57	0.251	59	0.336	51	2.216e-3	32	2.776e-4	33	1.769e-3	60
584		min	-2.11	41	-0.285	58	-0.323	54	-8.397e-4	60	-1.821e-4	58	-3.795e-3	32
585	297	max	2.341	40	0.256	59	0.198	51	1.023e-3	64	8.992e-4	64	1.843e-3	59
586		min	-2.274	60	-0.29	58	-0.192	54	-4.765e-4	54	-2.968e-4	54	-3.879e-3	32
587	298	max	2.145	58	0.227	23	0.513	22	1.438e-3	53	5.597e-4	32	3.212e-3	64
588		min	-2.164	41	-0.282	20	-0.535	21	-2.986e-3	64	-3.958e-4	60	-5.223e-4	54
589	299	max	2.145	57	0.252	59	0.513	22	1.611e-3	59	1.839e-4	58	1.816e-3	59
590		min	-2.165	41	-0.287	58	-0.535	21	-3.079e-3	32	-4.516e-4	33	-3.818e-3	32
591	300	max	2.372	58	0.242	21	0.238	22	7.777e-4	53	6.189e-4	32	7.984e-4	59
592		min	-2.415	41	-0.272	22	-0.246	21	-1.056e-3	64	-2.371e-4	60	-8.982e-4	38
593	301	max	2.377	57	0.243	21	0.238	22	7.937e-4	59	1.415e-4	54	7.604e-4	39
594		min	-2.416	41	-0.271	22	-0.246	21	-1.071e-3	32	-5.663e-4	64	-6.318e-4	42
595	302	max	2.631	40	0.249	21	0.13	52	6.092e-4	23	1.317e-4	32	9.509e-4	43
596		min	-2.548	59	-0.276	22	-0.131	53	-6.529e-4	22	-7.838e-5	60	-1.093e-3	38
597	303	max	2.633	40	0.249	21	0.128	52	6.17e-4	23	4.432e-5	60	8.792e-4	39
598		min	-2.546	60	-0.276	22	-0.129	53	-6.659e-4	61	-1.154e-4	32	-7.907e-4	57
599	304	max	2.487	58	0.266	21	0.083	51	5.802e-4	51	1.158e-4	59	9.137e-4	59
600		min	-2.534	41	-0.293	22	-0.08	54	-5.262e-4	54	-3.599e-4	32	-1.016e-3	38
601	305	max	2.495	57	0.266	21	0.082	51	5.814e-4	51	3.651e-4	64	8.608e-4	39
602		min	-2.534	41	-0.293	22	-0.079	54	-5.275e-4	54	-1.288e-4	53	-7.349e-4	42
603	306	max	1.982	40	0.248	21	0.647	22	1.721e-3	53	1.206e-3	32	5.972e-4	43
604		min	-1.924	59	-0.264	22	-0.683	21	-4.168e-3	64	-5.241e-4	60	-6.876e-4	38
605	307	max	1.916	40	0.246	21	0.471	51	3.027e-3	64	4.838e-4	60	5.462e-4	43
606		min	-1.852	59	-0.267	22	-0.451	54	-7.024e-4	54	-9.886e-4	32	-6.5e-4	38
607	308	max	1.917	40	0.246	21	0.471	51	3.218e-3	32	9.027e-4	64	5.361e-4	39
608		min	-1.849	60	-0.267	22	-0.45	54	-1.09e-3	60	-3.27e-4	54	-4.753e-4	42
609	309	max	1.983	40	0.248	21	0.647	22	2.048e-3	59	2.4e-4	54	5.911e-4	39
610		min	-1.923	60	-0.265	22	-0.683	21	-4.334e-3	32	-1.054e-3	64	-5.436e-4	42
611	310	max	2.009	58	0.245	21	0.521	51	2.991e-3	64	5.099e-4	60	5.983e-4	43
612		min	-2.01	59	-0.267	22	-0.501	54	-6.753e-4	54	-1.031e-3	32	-6.156e-4	38
613	311	max	2.016	57	0.246	23	0.521	21	3.2e-3	32	9.384e-4	64	6.156e-4	39
614		min	-2.009	60	-0.267	20	-0.499	54	-1.094e-3	60	-3.412e-4	54	-5.727e-4	57
615	312	max	2.275	40	0.24	21	0.375	22	1.005e-3	53	2.43e-4	54	1.139e-3	60
616		min	-2.194	60	-0.273	22	-0.39	21	-1.691e-3	32	-9.759e-4	64	-1.86e-3	38
617	313	max	2.136	57	0.241	21	0.511	22	1.616e-3	59	1.232e-4	54	1.166e-3	59
618		min	-2.167	41	-0.273	22	-0.536	21	-3.096e-3	32	-5.974e-4	64	-1.841e-3	38
619	314	max	2.273	40	0.241	21	0.375	22	9.969e-4	53	1.051e-3	32	1.333e-3	33
620		min	-2.196	59	-0.273	22	-0.39	21	-1.681e-3	64	-3.774e-4	60	-4.568e-4	57
621	315	max	2.076	57	0.241	21	0.342	51	2.233e-3	32	4.306e-4	64	1.119e-3	60
622		min	-2.111	41	-0.273	22	-0.327	54	-8.454e-4	60	-1.682e-4	54	-1.794e-3	38
623	316	max	2.078	58	0.24	21	0.342	51	2.133e-3	64	3.378e-4	60	1.346e-3	33
624		min	-2.111	41	-0.273	22	-0.328	54	-6.476e-4	54	-5.214e-4	32	-3.893e-4	57
625	317	max	2.346	40	0.244	21	0.206	51	1.014e-3	64	7.995e-4	64	1.193e-3	59
626		min	-2.264	60	-0.274	22	-0.199	54	-4.732e-4	54	-2.572e-4	54	-1.915e-3	38
627	318	max	2.344	40	0.244	21	0.207	51	1.016e-3	64	2.68e-4	60	1.366e-3	33
628		min	-2.261	59	-0.274	22	-0.2	54	-4.701e-4	54	-8.098e-4	32	-5.228e-4	57
629	319	max	2.132	58	0.24	21	0.511	22	1.442e-3	53	7.584e-4	32	1.372e-3	33
630		min	-2.167	41	-0.273	22	-0.536	21	-3.003e-3	64	-4.272e-4	60	-4.421e-4	57
631	320	max	2.122	58	0.245	21	0.509	22	1.578e-3	53	7.555e-4	32	7.291e-4	33
632		min	-2.168	41	-0.271	22	-0.538	21	-3.975e-3	64	-4.232e-4	60	-2.567e-4	57
633	321	max	2.276	40	0.245	21	0.373	22	1.134e-3	53	2.386e-4	54	6.081e-4	60
634		min	-2.186	41	-0.271	22	-0.392	21	-2.492e-3	32	-9.357e-4	64	-1.005e-3	38
635	322	max	2.126	57	0.245	21	0.509	22	1.703e-3	59	1.228e-4	54	6.199e-4	59
636		min	-2.168	41	-0.272	22	-0.538	21	-4.043e-3	32	-5.95e-4	64	-9.925e-4	38
637	323	max	2.066	57	0.245	21	0.347	51	2.972e-3	32	4.198e-4	64	5.949e-4	60
638		min	-2.112	41	-0.272	22	-0.331	54	-1.006e-3	60	-1.562e-4	54	-9.678e-4	38
639	324	max	2.347	40	0.248	21	0.213	51	1.556e-3	64	7.562e-4	64	6.342e-4	59
640		min	-2.255	41	-0.273	22	-0.204	54	-6.441e-4	54	-2.378e-4	54	-1.031e-3	38
641	325	max	2.068	58	0.245	21	0.348	51	2.914e-3	64	3.411e-4	60	7.142e-4	33
642		min	-2.112	41	-0.271	22	-0.332	54	-8.917e-4	54	-5.198e-4	32	-2.269e-4	57
643	326	max	2.345	40	0.248	21	0.214	51	1.558e-3	64	2.475e-4	60	7.267e-4	33
644		min	-2.255	41	-0.273	22	-0.205	54	-6.417e-4	54	-7.665e-4	32	-2.998e-4	57
645	327	max	2.274	40	0.245	21	0.374	22	1.128e-3	53	9.98e-4	32	7.084e-4	33
646		min	-2.186	41	-0.272	22	-0.392	21	-2.484e-3	64	-3.467e-4	60	-2.64e-4	57
647	328	max	2.483	40	0.265	21	0.086	51	6.751e-4	64	2.971e-4	64	5.57e-4	33
648		min	-2.535	41	-0.296	22	-0.082	54	-5.637e-4	54	-1.116e-4	53	-1.543e-4	57
649	329	max	2.482	40	0.266	21	0.087	51	6.757e-4	64	1.142e-4	59	4.521e-4	59
650		min	-2.535	41	-0.296	22	-0.083	54	-5.628e-4	54	-2.999e-4	32	-6.939e-4	32
651	330	max	2.634	40	0.249	21	0.127	52	5.699e-4	59	5.876e-5	60	5.618e-4	33
652		min	-2.54	41	-0.279	22	-0.13	53	-6.617e-4	61	-1.05e-4	32	-1.679e-4	57
653	331	max	2.365	40	0.242	21	0.236	22	8.224e-4	59	1.402e-4	54	5.425e-4	33
654		min	-2.416	41	-0.275	22	-0.247	21	-1.397e-3	32	-4.953e-4	64	-1.216e-4	58
655	332	max	2.364	40	0.242	21	0.237	22	8.107e-4	53	5.438e-4	32	4.105e-4	59
656		min	-2.416	41	-0.275	22	-0.248	21	-1.385e-3	64	-2.269e-4	60	-6.716e-4	32
657	333	max	2.633	40	0.248	21	0.129	52	5.601e-4	59	1.223e-4	32	4.655e-4	59
658		min	-2.54	41	-0.279	22	-0.131	53	-6.509e-4	61	-9.369e-5	60	-7.287e-4	32
659	334	max	1.983	40	0.248	21	0.643	22	1.781e-3	53	1.113e-3	32	2.824e-4	59
660		min	-1.909	59	-0.267	22	-0.686	21	-4.646e-3	64	-5.327e-4	60	-4.258e-4	38
661	335	max	1.983	40	0.248	21	0.643	22	1.967e-3	59	2.078e-4	54	3.154e-4	33
662		min	-1.908	60	-0.267	22	-0.686	21	-4.742e-3	32	-9.384e-4	64	-1.5e-4	57
663	336	max	1.917	40	0.246	21	0.484	51	3.484e-3	64	4.813e-4	60	2.839e-4	59
664		min	-1.837	59	-0.27	22	-0.46	54	-8.752e-4	54	-9.195e-4	32	-4.844e-4	32
665	337	max	1.917	40	0.246	21	0.484	51	3.609e-3	32	8.148e-4	64	3.807e-4	33
666		min	-1.835	41	-0.27	22	-0.46	54	-1.129e-3	60	-2.907e-4	54	-1.015e-4	58
667	338	max	2.001	57	0.247	23	0.534	51	3.671e-3	32	8.682e-4	64	2.819e-4	39

Envelope Node Displacements (Continued)

	Node Label		X [in]	LC	Y [in]	LC	Z [in]	LC	X Rotation [rad]	LC	Y Rotation [rad]	LC	Z Rotation [rad]	LC
668		min	-1.994	60	-0.269	20	-0.51	54	-1.167e-3	60	-3.26e-4	54	-2.111e-4	57
669	339	max	1.994	58	0.246	21	0.534	51	3.529e-3	64	5.287e-4	60	2.655e-4	43
670		min	-1.994	59	-0.269	22	-0.511	54	-8.796e-4	54	-9.779e-4	32	-3.144e-4	38
671	340	max	2.277	40	0.246	21	0.373	22	1.19e-3	53	2.198e-4	54	4.323e-4	60
672		min	-2.186	41	-0.271	22	-0.392	21	-2.835e-3	32	-8.534e-4	64	-7.28e-4	38
673	341	max	2.066	40	0.246	21	0.35	51	3.293e-3	32	3.997e-4	33	4.215e-4	60
674		min	-2.112	41	-0.271	22	-0.333	54	-1.075e-3	60	-1.447e-4	54	-7.015e-4	38
675	342	max	2.064	58	0.246	21	0.35	51	3.253e-3	64	3.345e-4	60	5.159e-4	33
676		min	-2.112	41	-0.271	22	-0.334	54	-9.975e-4	54	-4.998e-4	32	-1.605e-4	57
677	343	max	2.274	40	0.246	21	0.373	22	1.184e-3	53	9.174e-4	32	5.128e-4	33
678		min	-2.186	41	-0.271	22	-0.392	21	-2.829e-3	64	-3.313e-4	60	-1.867e-4	57
679	344	max	2.122	57	0.246	21	0.508	22	1.74e-3	59	1.147e-4	54	4.386e-4	59
680		min	-2.168	41	-0.271	22	-0.539	21	-4.453e-3	32	-5.634e-4	33	-7.184e-4	38
681	345	max	2.117	40	0.246	21	0.508	22	1.637e-3	53	7.199e-4	32	5.266e-4	33
682		min	-2.168	41	-0.271	22	-0.539	21	-4.396e-3	64	-4.153e-4	60	-1.82e-4	57
683	346	max	2.348	40	0.25	21	0.216	51	1.788e-3	64	6.858e-4	64	4.484e-4	59
684		min	-2.256	41	-0.272	22	-0.206	54	-7.176e-4	54	-2.169e-4	54	-7.444e-4	38
685	347	max	2.345	40	0.249	21	0.216	51	1.789e-3	64	2.372e-4	60	5.255e-4	33
686		min	-2.256	41	-0.272	22	-0.207	54	-7.155e-4	54	-7.014e-4	32	-2.116e-4	57
687	348	max	1.917	40	0.247	21	0.501	51	4.276e-3	64	4.154e-4	60	2.176e-5	54
688		min	-1.835	41	-0.275	22	-0.472	54	-1.177e-3	54	-4.688e-4	38	-9.948e-5	64
689	349	max	1.917	40	0.247	21	0.501	51	4.277e-3	32	3.811e-4	39	1.078e-4	32
690		min	-1.835	41	-0.275	22	-0.471	54	-1.181e-3	60	-3.389e-4	58	-2.621e-5	59
691	350	max	2.365	40	0.242	21	0.234	22	8.694e-4	53	1.399e-4	32	2.507e-5	54
692		min	-2.416	41	-0.28	22	-0.249	21	-1.964e-3	64	-1.085e-4	60	-1.48e-4	64
693	351	max	2.365	40	0.242	21	0.234	22	8.734e-4	59	5.436e-5	58	1.508e-4	32
694		min	-2.417	41	-0.28	22	-0.249	21	-1.968e-3	32	-1.063e-4	33	-5.051e-5	59
695	352	max	2.633	40	0.249	21	0.127	52	5.269e-4	59	3.634e-5	100	1.845e-4	32
696		min	-2.54	41	-0.285	22	-0.129	53	-6.571e-4	61	-4.528e-5	82	-8.757e-5	60
697	353	max	2.633	40	0.249	21	0.127	52	5.253e-4	59	4.051e-5	88	3.173e-5	53
698		min	-2.54	41	-0.285	22	-0.13	53	-6.553e-4	61	-3.611e-5	82	-1.44e-4	64
699	354	max	1.983	40	0.249	21	0.637	22	1.881e-3	53	5.827e-4	38	1.275e-5	79
700		min	-1.904	41	-0.272	22	-0.689	21	-5.469e-3	64	-5.228e-4	60	-7.109e-5	64
701	355	max	2.483	40	0.266	21	0.092	51	9.67e-4	64	6.611e-5	59	2.35e-5	54
702		min	-2.535	41	-0.302	22	-0.087	54	-6.293e-4	54	-8.3e-5	38	-1.482e-4	64
703	356	max	2.483	40	0.266	21	0.092	51	9.669e-4	64	5.741e-5	33	1.636e-4	32
704		min	-2.535	41	-0.302	22	-0.087	54	-6.294e-4	54	-4.034e-5	57	-7.789e-5	60
705	357	max	1.983	40	0.25	21	0.637	22	1.882e-3	53	4.524e-4	58	8.799e-5	32
706		min	-1.904	41	-0.272	22	-0.689	21	-5.469e-3	64	-5.e-4	39	-3.64e-5	59
707	358	max	1.983	38	0.249	23	0.553	51	4.461e-3	64	3.936e-4	60	3.292e-5	42
708		min	-1.987	41	-0.273	20	-0.523	54	-1.234e-3	54	-4.534e-4	38	-4.028e-5	33
709	359	max	1.983	38	0.249	23	0.552	51	4.481e-3	32	3.511e-4	39	6.234e-5	38
710		min	-1.987	41	-0.273	20	-0.523	54	-1.276e-3	60	-3.023e-4	58	-3.84e-5	60
711	360	max	2.119	40	0.251	21	0.503	22	1.989e-3	53	2.274e-4	58	9.145e-5	38
712		min	-2.168	41	-0.273	22	-0.542	21	-6.909e-3	64	-2.75e-4	39	-1.052e-4	43
713	361	max	2.064	40	0.251	21	0.364	51	5.28e-3	64	2.362e-4	60	8.538e-5	42
714		min	-2.112	41	-0.273	22	-0.343	54	-1.632e-3	54	-2.893e-4	38	-8.049e-5	39
715	362	max	2.065	40	0.251	21	0.364	51	5.28e-3	64	2.147e-4	39	8.352e-5	38
716		min	-2.112	41	-0.273	22	-0.343	54	-1.632e-3	54	-1.718e-4	42	-9.732e-5	43
717	363	max	2.275	40	0.251	21	0.369	22	1.517e-3	53	2.328e-4	38	9.383e-5	42
718		min	-2.186	41	-0.273	22	-0.395	21	-4.881e-3	64	-1.903e-4	60	-7.941e-5	39
719	364	max	2.276	40	0.251	21	0.369	22	1.518e-3	53	1.209e-4	58	9.178e-5	38
720		min	-2.186	41	-0.273	22	-0.395	21	-4.881e-3	64	-1.651e-4	33	-9.599e-5	43
721	365	max	2.346	40	0.254	21	0.228	51	3.169e-3	64	1.437e-4	59	1.066e-4	42
722		min	-2.256	41	-0.274	22	-0.215	54	-1.154e-3	54	-1.838e-4	38	-9.276e-5	39
723	366	max	2.118	40	0.251	21	0.503	22	1.987e-3	53	3.581e-4	38	9.239e-5	42
724		min	-2.168	41	-0.273	22	-0.542	21	-6.909e-3	64	-2.991e-4	60	-8.749e-5	39
725	367	max	2.347	40	0.254	21	0.228	51	3.169e-3	64	1.227e-4	39	1.063e-4	38
726		min	-2.256	41	-0.274	22	-0.215	54	-1.155e-3	54	-9.923e-5	57	-1.109e-4	43
727	368	max	2.119	40	0.252	21	0.502	22	2.057e-3	53	2.737e-4	58	1.027e-4	40
728		min	-2.168	41	-0.273	22	-0.542	21	-7.408e-3	64	-2.639e-4	41	-1.073e-4	59
729	369	max	2.483	40	0.266	21	0.093	51	1.025e-3	64	5.002e-5	59	5.463e-5	58
730		min	-2.535	41	-0.302	22	-0.087	54	-6.426e-4	54	-5.645e-5	57	-6.005e-5	39
731	370	max	1.917	40	0.248	21	0.503	51	4.435e-3	64	3.823e-4	59	1.699e-5	38
732		min	-1.835	41	-0.275	22	-0.473	54	-1.237e-3	54	-3.978e-4	57	-1.374e-5	41
733	371	max	2.276	40	0.252	21	0.368	22	1.584e-3	53	1.614e-4	58	1.038e-4	57
734		min	-2.186	41	-0.273	22	-0.396	21	-5.295e-3	64	-1.529e-4	41	-9.832e-5	41
735	372	max	2.064	40	0.252	21	0.366	51	5.681e-3	64	2.07e-4	59	9.451e-5	40
736		min	-2.112	41	-0.273	22	-0.344	54	-1.757e-3	54	-2.199e-4	57	-9.928e-5	60
737	373	max	2.365	40	0.243	21	0.234	22	8.836e-4	59	8.067e-5	58	2.727e-5	40
738		min	-2.416	41	-0.281	22	-0.249	21	-2.082e-3	32	-7.312e-5	41	-3.155e-5	39
739	374	max	2.346	40	0.255	21	0.229	51	3.449e-3	64	1.257e-4	59	1.19e-4	58
740		min	-2.256	41	-0.274	22	-0.216	54	-1.243e-3	54	-1.347e-4	57	-1.136e-4	41
741	375	max	1.983	40	0.25	21	0.636	22	1.901e-3	53	5.124e-4	58	3.551e-5	38
742		min	-1.904	41	-0.273	22	-0.689	21	-5.636e-3	64	-5.024e-4	41	-3.285e-5	41
743	376	max	2.633	40	0.249	21	0.126	52	5.185e-4	59	3.826e-5	100	7.559e-5	57
744		min	-2.54	41	-0.285	22	-0.129	53	-6.562e-4	61	-4.074e-5	82	-6.897e-5	41
745	377	max	1.983	38	0.249	23	0.554	51	4.647e-3	64	3.451e-4	59	5.249e-5	40
746		min	-1.987	41	-0.273	20	-0.524	54	-1.304e-3	54	-3.618e-4	57	-4.184e-5	60
747	383	max	1.719	58	0	63	0.826	21	0	63	6.604e-4	59	6.497e-3	29
748		min	-1.777	41	-0.775	24	-0.823	22	-3.928e-3	29	-7.394e-4	40	-2.027e-4	58
749	385	max	0	20	0	53	0	23	0	105	0	58	0	105
750		min	0	23	0	72	0	20	0	1	0	59	0	1
751	386	max	0	57	0	53	0	23	0	51	0	58	0	105
752		min	0	60	0	72	0	20	0	54	0	59	0	1
753	387	max	0	20	0	51	0	52	0	105	0	59	0	59
754		min	0	23	0	73	0	53	0	1	0	58	0	58
755	388	max	0	57	0	51	0	52	0	51	0	60	0	59

Envelope Node Displacements (Continued)

	Node Label		X [in]	LC	Y [in]	LC	Z [in]	LC	X Rotation [rad]	LC	Y Rotation [rad]	LC	Z Rotation [rad]	LC
756		min	0	60	0	73	0	53	0	54	0	57	0	58
757	389	max	0.039	58	0.046	21	0.029	51	4.779e-4	51	6.658e-7	58	6.507e-4	59
758		min	-0.039	59	-0.048	22	-0.029	54	-4.764e-4	54	-6.658e-7	59	-6.497e-4	58
759	390	max	0.007	58	0.046	21	0.029	51	4.779e-4	51	1.99e-7	58	1.256e-4	59
760		min	-0.008	59	-0.048	22	-0.029	54	-4.764e-4	54	-2.005e-7	59	-1.247e-4	58
761	391	max	0.039	58	0.046	21	0.009	51	1.511e-4	51	1.628e-7	51	6.507e-4	59
762		min	-0.039	59	-0.048	22	-0.009	54	-1.496e-4	54	-1.611e-7	54	-6.497e-4	58
763	392	max	0.007	58	0.047	21	0.009	51	1.511e-4	51	0	58	1.256e-4	59
764		min	-0.008	59	-0.049	22	-0.009	54	-1.496e-4	54	0	59	-1.247e-4	58
765	393	max	1.536	40	0.132	59	0.855	52	2.144e-3	20	2.509e-3	54	8.776e-4	58
766		min	-1.473	41	-0.146	58	-0.853	53	-2.013e-3	23	-2.506e-3	52	-1.008e-3	41
767	394	max	1.536	40	0.132	57	0.855	52	2.142e-3	20	2.522e-3	52	1.041e-3	40
768		min	-1.473	41	-0.143	60	-0.853	53	-2.015e-3	23	-2.517e-3	54	-8.579e-4	60
769	395	max	0.037	20	0.023	21	0.007	51	1.51e-4	51	1.628e-7	51	6.498e-4	59
770		min	-0.037	23	-0.024	22	-0.007	54	-1.495e-4	54	-1.611e-7	54	-6.491e-4	58
771	396	max	0.037	20	0.023	21	0.024	23	4.781e-4	51	6.658e-7	58	6.5e-4	59
772		min	-0.037	23	-0.024	22	-0.023	54	-4.766e-4	54	-6.658e-7	59	-6.49e-4	58
773	397	max	0.006	58	0.023	21	0.024	23	4.778e-4	51	1.99e-7	58	1.254e-4	59
774		min	-0.006	59	-0.024	22	-0.023	54	-4.768e-4	54	-2.005e-7	59	-1.245e-4	58
775	398	max	2.037	58	0.233	53	0.581	22	3.645e-4	64	8.432e-4	58	9.881e-4	41
776		min	-2.056	39	-0.294	55	-0.608	21	-3.987e-5	54	-1.525e-3	41	-6.163e-3	64
777	399	max	2.178	38	0.278	59	0.581	22	4.206e-4	66	1.017e-3	58	6.5e-3	32
778		min	-2.074	60	-0.321	58	-0.608	21	-1.355e-4	59	-1.508e-3	43	-2.744e-4	100
779	400	max	2.291	58	0.226	53	0.308	22	6.249e-5	53	7.125e-4	57	9.191e-4	43
780		min	-2.327	39	-0.288	55	-0.316	21	-3.088e-4	64	-1.257e-3	43	-1.649e-2	64
781	401	max	2.525	38	0.27	59	0.308	22	1.113e-4	60	5.376e-4	57	1.824e-2	32
782		min	-2.316	60	-0.317	58	-0.316	21	-3.682e-4	34	-1.513e-3	45	-2.692e-4	100
783	402	max	2.63	38	0.288	59	0.138	51	3.652e-4	66	1.567e-3	45	1.992e-2	32
784		min	-2.404	41	-0.332	58	-0.134	54	-1.998e-4	60	-8.115e-4	57	-3.577e-4	20
785	403	max	2.388	58	0.245	53	0.139	51	3.322e-4	64	1.532e-3	43	1.01e-3	43
786		min	-2.421	39	-0.303	20	-0.136	54	-1.939e-4	54	-9.864e-4	57	-1.814e-2	64
787	404	max	2.157	38	0.276	59	0.398	51	1.416e-4	60	1.582e-3	43	1.218e-2	32
788		min	-1.99	60	-0.32	58	-0.383	54	-4.026e-4	34	-1.107e-3	58	-4.234e-4	20
789	405	max	1.979	58	0.235	53	0.4	51	4.58e-5	53	1.599e-3	41	7.777e-4	43
790		min	-1.997	39	-0.295	55	-0.385	54	-3.461e-4	64	-9.329e-4	58	-1.108e-2	64
791	406	max	0	105	0	105	0	105	0	105	0	105	0	105
792		min	0	1	0	1	0	1	0	1	0	1	0	1
793	407	max	0	105	0	105	0	105	0	105	0	105	0	105
794		min	0	1	0	1	0	1	0	1	0	1	0	1
795	408	max	0	105	0	105	0	105	0	105	0	105	0	105
796		min	0	1	0	1	0	1	0	1	0	1	0	1
797	409	max	0	105	0	105	0	105	0	105	0	105	0	105
798		min	0	1	0	1	0	1	0	1	0	1	0	1
799	410	max	0	58	0	58	0	59	0	105	0	20	0	25
800		min	0	24	0	45	0	58	0	1	0	23	0	54
801	411	max	0	54	0	59	0	52	0	105	0	20	0	105
802		min	0	59	0	24	0	54	0	1	0	23	0	1
803	412	max	0	57	0	63	0	52	0	105	0	105	0	105
804		min	0	60	0	29	0	53	0	1	0	1	0	1
805	413	max	1.839	40	0	63	0.696	21	0	63	3.538e-3	59	8.899e-3	59
806		min	-1.644	59	-0.404	44	-0.693	20	-2.658e-3	44	-3.813e-3	40	-8.865e-3	58
807	414	max	0.923	40	0.103	21	0.046	51	3.303e-4	51	4.694e-4	41	1.16e-2	60
808		min	-0.925	60	-0.113	22	-0.046	54	-3.234e-4	54	-2.486e-4	57	-1.172e-2	57
809	415	max	0.929	58	0.103	21	0.047	51	3.337e-4	51	4.679e-4	41	1.16e-2	59
810		min	-0.915	41	-0.113	22	-0.046	54	-3.278e-4	54	-2.468e-4	57	-1.169e-2	58
811	416	max	2.102	40	0.176	21	0.114	51	1.41e-4	51	1.239e-3	41	3.941e-3	41
812		min	-2.085	41	-0.202	22	-0.112	54	-1.344e-4	54	-7.081e-4	57	-3.983e-3	40
813	417	max	2.106	57	0.176	21	0.112	51	1.417e-4	51	1.242e-3	41	3.958e-3	41
814		min	-2.086	41	-0.202	22	-0.111	54	-1.335e-4	54	-7.26e-4	57	-3.981e-3	40
815	418	max	1.409	57	0.084	21	0.309	51	5.063e-4	53	1.006e-3	52	8.128e-3	41
816		min	-1.411	41	-0.103	22	-0.304	54	-5.057e-4	52	-1.007e-3	53	-7.951e-3	38
817	419	max	1.545	40	0.086	21	0.192	51	4.799e-4	54	9.942e-4	52	8.607e-3	41
818		min	-1.508	60	-0.103	22	-0.188	54	-5.173e-4	51	-9.943e-4	53	-8.968e-3	40
819	420	max	1.29	57	0.087	21	0.444	23	8.104e-4	21	1.016e-3	52	7.083e-3	41
820		min	-1.255	60	-0.1	22	-0.438	22	-8.214e-4	20	-1.017e-3	53	-7.412e-3	40
821	429	max	2.212	40	0.181	21	0.086	52	1.217e-4	52	1.605e-4	60	4.414e-3	41
822		min	-2.194	41	-0.205	22	-0.086	53	-1.186e-4	53	-1.009e-3	44	-4.441e-3	40
823	430	max	2.214	40	0.181	21	0.085	52	1.185e-4	52	1.653e-4	60	4.432e-3	41
824		min	-2.196	41	-0.205	22	-0.085	53	-1.168e-4	53	-1.01e-3	44	-4.47e-3	38
825	431	max	0.964	40	0.103	21	0.034	52	2.51e-4	52	6.255e-5	59	1.211e-2	59
826		min	-0.96	59	-0.112	22	-0.034	53	-2.532e-4	53	-3.83e-4	44	-1.22e-2	58
827	432	max	0.971	57	0.103	21	0.033	52	2.464e-4	52	6.431e-5	59	1.209e-2	60
828		min	-0.956	41	-0.112	22	-0.034	53	-2.491e-4	53	-3.835e-4	44	-1.222e-2	57
829	433	max	1.411	58	0.084	21	0.311	51	4.899e-4	53	9.706e-4	54	8.131e-3	41
830		min	-1.414	59	-0.103	22	-0.307	54	-4.93e-4	52	-9.723e-4	51	-7.935e-3	38
831	434	max	1.543	40	0.086	21	0.195	51	4.936e-4	54	9.713e-4	54	8.63e-3	41
832		min	-1.506	59	-0.103	22	-0.19	54	-5.339e-4	51	-9.727e-4	72	-8.97e-3	40
833	435	max	1.287	58	0.087	21	0.443	21	7.976e-4	21	9.646e-4	54	7.108e-3	41
834		min	-1.257	59	-0.1	22	-0.438	22	-8.302e-4	58	-9.656e-4	51	-7.419e-3	40
835	436	max	1.672	40	0.089	21	0.058	52	2.707e-3	22	9.864e-4	52	9.524e-3	39
836		min	-1.632	60	-0.105	22	-0.061	53	-2.68e-3	23	-9.865e-4	53	-9.897e-3	40
837	437	max	1.577	57	0.085	21	0.174	52	2.901e-3	22	9.91e-4	52	9.224e-3	41
838		min	-1.583	41	-0.103	22	-0.177	53	-2.88e-3	23	-9.899e-4	53	-9.032e-3	38
839	438	max	1.67	40	0.089	21	0.059	52	2.704e-3	20	9.772e-4	54	9.529e-3	39
840		min	-1.635	59	-0.105	22	-0.062	53	-2.69e-3	23	-9.773e-4	51	-9.88e-3	40
841	439	max	1.573	58	0.085	21	0.175	52	2.901e-3	20	9.824e-4	73	9.233e-3	41
842		min	-1.584	41	-0.103	22	-0.177	53	-2.884e-3	23	-9.81e-4	51	-9.021e-3	38
843	440	max	1.521	40	0.084	21	0.289	52	3.169e-3	22	9.928e-4	72	8.399e-3	41

Envelope Node Displacements (Continued)

	Node Label		X [in]	LC	Y [in]	LC	Z [in]	LC	X Rotation [rad]	LC	Y Rotation [rad]	LC	Z Rotation [rad]	LC
844		min	-1.484	60	-0.103	22	-0.292	53	-3.148e-3	23	-9.895e-4	53	-8.753e-3	40
845	441	max	1.519	40	0.084	21	0.29	52	3.174e-3	22	9.836e-4	73	8.425e-3	41
846		min	-1.487	59	-0.103	22	-0.292	53	-3.145e-3	23	-9.806e-4	51	-8.758e-3	40
847	442	max	1.432	57	0.084	21	0.421	20	3.465e-3	22	9.936e-4	72	8.282e-3	41
848		min	-1.431	41	-0.103	22	-0.426	23	-3.446e-3	23	-9.891e-4	53	-8.098e-3	38
849	443	max	1.429	58	0.084	21	0.422	22	3.473e-3	22	9.849e-4	73	8.289e-3	41
850		min	-1.433	59	-0.103	22	-0.426	21	-3.439e-3	23	-9.809e-4	51	-8.086e-3	38
851	444	max	1.317	57	0.088	21	0.56	20	3.645e-3	22	4.145e-3	52	7.284e-3	41
852		min	-1.283	60	-0.098	22	-0.565	23	-3.627e-3	23	-4.14e-3	54	-7.597e-3	40
853	445	max	1.314	58	0.089	21	0.561	22	3.655e-3	22	4.134e-3	54	7.305e-3	41
854		min	-1.285	59	-0.098	22	-0.565	21	-3.62e-3	21	-4.132e-3	52	-7.606e-3	40
855	428	max	1.688	57	0	63	0.796	23	1.713e-4	29	2.388e-4	41	4.198e-3	29
856		min	-1.79	41	-0.565	24	-0.791	20	-1.306e-5	58	-2.484e-4	57	0	50

Envelope AISC 14TH (360-10): LRFD Member Steel Code Checks

	Member	Shape	Code Check	Loc[ft]	LC	Shear Check	Loc[ft]	Dir	LC	phi*Pnc [k]	phi*Pnt [k]	phi*Mn y-y [k-ft]	phi*Mn z-z [k-ft]	Cb	Eqn
1	M312	W12X40	0.089	6.492	33	0.067	6.492	y	64	457.832	526.5	63	213.75	1.948	H1-1b
2	M290	W12X40	0.316	6.567	64	0.08	6.451	y	64	329.91	526.5	63	213.75	2.15	H1-1b
3	M193	W12X40	0.068	6.492	33	0.051	4.858	y	64	457.832	526.5	63	213.75	1.965	H1-1b
4	105	W12X40	0.319	6.567	64	0.082	6.451	y	64	329.91	526.5	63	213.75	2.15	H1-1b
5	M170	W12X40	0.34	6.567	32	0.093	6.451	y	32	329.91	526.5	63	213.75	2.076	H1-1b
6	107	W12X40	0.343	6.567	32	0.096	6.451	y	32	329.91	526.5	63	213.75	2.07	H1-1b
7	M195	W12X40	0.088	6.492	32	0.058	4.984	y	32	457.832	526.5	63	213.75	1.971	H1-1b
8	M298	W12X40	0.117	6.492	38	0.081	6.492	y	32	457.832	526.5	63	213.75	1.885	H1-1b
9	M43	W12X40	0.305	6.567	64	0.079	6.451	y	64	329.91	526.5	63	213.75	2.16	H1-1b
10	M336	W12X40	0.063	6.492	33	0.038	4.817	y	64	457.832	526.5	63	213.75	1.932	H1-1b
11	M145	W12X40	0.327	6.567	32	0.092	6.451	y	32	329.91	526.5	63	213.75	2.056	H1-1b
12	M143	W12X40	0.303	6.567	64	0.078	6.567	y	64	329.91	526.5	63	213.75	2.159	H1-1b
13	M319	W12X40	0.087	6.492	33	0.069	5.11	y	64	457.832	526.5	63	213.75	1.956	H1-1b
14	M316	W12X40	0.104	6.492	38	0.079	5.152	y	32	457.832	526.5	63	213.75	1.917	H1-1b
15	M291	W12X40	0.111	6.492	38	0.079	6.157	y	32	457.832	526.5	63	213.75	1.906	H1-1b
16	M335	W12X40	0.082	6.492	38	0.046	4.942	y	32	457.832	526.5	63	213.75	1.799	H1-1b
17	M294	W12X40	0.089	6.492	33	0.067	6.073	y	64	457.832	526.5	63	213.75	1.946	H1-1b
18	M45	W12X40	0.329	6.567	32	0.095	6.451	y	32	329.91	526.5	63	213.75	2.063	H1-1b
19	M426	WT4X6.5	0.658	9.925	57	0.014	19.849	y	41	11.507	86.4	4.013	3.483	1	H1-1a
20	M425	WT4X6.5	0.286	9.925	59	0.012	9.925	y	62	11.507	86.4	4.013	3.483	1	H1-1b
21	M424	WT4X6.5	0.393	9.925	57	0.012	9.925	y	61	11.507	86.4	4.013	3.483	1	H1-1a
22	M423	WT4X6.5	0.221	9.925	57	0.011	9.925	y	48	11.507	86.4	4.013	3.483	1	H1-1b
23	M422	WT4X6.5	0.449	9.925	59	0.012	19.849	y	40	11.507	86.4	4.013	3.483	1	H1-1a
24	M421	WT4X6.5	0.864	9.925	58	0.018	0	y	41	11.507	86.4	4.013	3.483	1	H1-1a
25	M416	WT4X6.5	0.224	8.596	44	0.011	17.192	y	45	7.119	86.4	4.013	5.838	1	H1-1b
26	M418	WT4X6.5	0.423	8.201	41	0.011	16.401	y	44	7.811	86.4	4.013	5.838	1	H1-1a
27	M417	WT4X6.5	0.437	8.201	41	0.011	16.401	y	44	7.811	86.4	4.013	5.838	1	H1-1a
28	M427	WT4X6.5	0.505	9.925	60	0.014	19.849	y	40	11.507	86.4	4.013	3.483	1	H1-1a
29	M409	WT4X6.5	0.272	8.201	45	0.011	16.401	y	45	7.811	86.4	4.013	4.032	1	H1-1b
30	M407	WT4X6.5	0.446	8.201	41	0.011	16.401	y	44	7.811	86.4	4.013	5.838	1	H1-1a
31	M389	WT4X6.5	0.453	8.596	41	0.011	17.192	y	45	7.119	86.4	4.013	5.838	1	H1-1a
32	M419	WT4X6.5	0.533	9.925	60	0.014	0	y	40	11.507	86.4	4.013	3.483	1	H1-1a
33	M490	WT4X6.5	0.375	9.925	45	0.013	19.849	y	45	5.359	86.4	4.013	3.483	1	H1-1b
34	M489	WT4X6.5	0.378	9.925	44	0.013	19.849	y	44	5.359	86.4	4.013	3.483	1	H1-1b
35	M475	WT4X6.5	0.383	9.925	44	0.013	19.849	y	44	5.359	86.4	4.013	3.483	1	H1-1b
36	M474	WT4X6.5	0.371	9.925	45	0.013	19.849	y	45	5.359	86.4	4.013	3.483	1	H1-1b
37	M463	WT4X6.5	0.461	8.201	41	0.011	16.401	y	44	7.811	86.4	4.013	5.838	1	H1-1a
38	M454	WT4X6.5	0.23	8.201	59	0.011	16.401	y	44	7.811	86.4	4.013	5.838	1	H1-1a
39	M452	WT4X6.5	0.549	0	38	0.015	0	y	44	6.714	86.4	4.013	2.797	1	H1-1b
40	M451	WT4X6.5	0.359	9.925	44	0.013	19.849	y	44	5.359	86.4	4.013	3.483	1	H1-1b
41	M449	WT4X6.5	0.553	25.986	41	0.015	25.986	y	45	6.714	86.4	4.013	2.797	1	H1-1b
42	M447	WT4X6.5	0.364	9.925	44	0.013	19.849	y	44	5.359	86.4	4.013	3.483	1	H1-1b
43	M445	WT4X6.5	0.389	9.925	41	0.013	19.849	y	45	5.359	86.4	4.013	5.352	1	H1-1a
44	M443	WT4X6.5	0.37	9.925	45	0.013	19.849	y	45	5.359	86.4	4.013	3.483	1	H1-1b
45	M442	WT4X6.5	0.21	8.201	45	0.011	16.401	y	45	7.811	86.4	4.013	5.838	1	H1-1b
46	M453	WT4X6.5	0.195	8.201	44	0.011	16.401	y	44	7.811	86.4	4.013	5.838	1	H1-1b
47	M269	W10X22	0.318	6.5	29	0.097	0	y	29	105.827	292.05	22.875	72.708	1.135	H1-1b
48	M258	W10X22	0.318	6.5	29	0.097	13	y	29	105.827	292.05	22.875	72.689	1.135	H1-1b
49	M260	W10X22	0.32	6.5	29	0.098	13	y	29	105.827	292.05	22.875	72.8	1.136	H1-1b
50	M265	W10X22	0.32	6.5	29	0.098	13	y	29	105.827	292.05	22.875	72.8	1.136	H1-1b
51	M266	W10X22	0.318	6.5	29	0.097	0	y	29	105.827	292.05	22.875	72.696	1.135	H1-1b
52	M267	W10X22	0.32	6.5	29	0.098	13	y	29	105.827	292.05	22.875	72.8	1.136	H1-1b
53	M268	W10X22	0.32	6.5	29	0.098	13	y	29	105.827	292.05	22.875	72.8	1.136	H1-1b
54	M270	W10X22	0.32	6.5	29	0.098	13	y	29	105.827	292.05	22.875	72.8	1.136	H1-1b
55	M271	W10X22	0.32	6.5	29	0.098	13	y	29	105.827	292.05	22.875	72.8	1.136	H1-1b
56	M272	W10X22	0.319	6.5	29	0.097	0	y	29	105.827	292.05	22.875	72.72	1.135	H1-1b
57	M273	W10X22	0.318	6.5	29	0.097	0	y	29	105.827	292.05	22.875	72.71	1.135	H1-1b
58	M275	W10X22	0.32	6.5	29	0.098	13	y	29	105.827	292.05	22.875	72.8	1.136	H1-1b
59	M276	W10X22	0.32	6.5	29	0.098	13	y	29	105.827	292.05	22.875	72.8	1.136	H1-1b
60	M277	W10X22	0.33	0	40	0.097	0	y	29	105.827	292.05	22.875	84.609	1.321	H1-1b
61	M278	W10X22	0.32	6.5	29	0.098	13	y	29	105.827	292.05	22.875	72.8	1.136	H1-1b
62	M279	W10X22	0.32	6.5	29	0.098	13	y	29	105.827	292.05	22.875	72.8	1.136	H1-1b
63	M380	W10X22	0.32	6.5	29	0.098	13	y	29	105.827	292.05	22.875	72.8	1.136	H1-1b
64	M382	W10X22	0.32	6.5	29	0.098	13	y	29	105.827	292.05	22.875	72.8	1.136	H1-1b
65	M383	W10X22	0.318	6.5	29	0.097	13	y	29	105.827	292.05	22.875	72.69	1.135	H1-1b
66	M385	W10X22	0.32	6.5	29	0.098	13	y	29	105.827	292.05	22.875	72.8	1.136	H1-1b
67	M379	W10X22	0.32	6.5	29	0.098	13	y	29	105.827	292.05	22.875	72.8	1.136	H1-1b
68	M386	W10X22	0.32	6.5	29	0.098	13	y	29	105.827	292.05	22.875	72.8	1.136	H1-1b
69	M388	W10X22	0.279	6.5	29	0.085	13	y	29	105.827	292.05	22.875	72.8	1.136	H1-1b
70	M280	W10X22	0.32	6.5	29	0.098	13	y	29	105.827	292.05	22.875	72.8	1.136	H1-1b
71	M274	W10X22	0.338	0	40	0.097	0	y	29	105.827	292.05	22.875	84.583	1.32	H1-1b
72	M234	W12X45	0.409	16.515	60	0.055	24.9	y	57	383.249	589.5	71.25	185.165	1.403	H1-1b

HANSON™

Company : Hanson Professional Services
Designer : S. Steiner
Job Number : 25i0007
Model Name : Eli Lilly K215 Unloading Rack

2/19/2026
2:59:44 PM
Checked By : RMF

Envelope AISC 14TH (360-10): LRFD Member Steel Code Checks (Continued)

	Member	Shape	Code Check	Loc[ft]	LC	Shear Check	Loc[ft]	Dir	LC	phi*Pnc [k]	phi*Pnt [k]	phi*Mn y-y [k-ft]	phi*Mn z-z [k-ft]	Cb	Eqn
73	M357	W12X45	0.388	14.373	41	0.088	3.349	y	32	375.229	589.5	71.25	240.75	2.477	H1-1b
74	M358	W12X45	0.449	14.373	58	0.059	27.351	y	60	375.229	589.5	71.25	240.75	1.183	H1-1b
75	M359	W12X45	0.446	14.373	60	0.059	18.839	y	57	375.229	589.5	71.25	240.75	2.454	H1-1b
76	M362	W12X45	0.452	14.373	58	0.061	18.839	y	58	375.229	589.5	71.25	240.75	1.104	H1-1b
77	M356	W12X45	0.413	14.373	40	0.087	2.233	y	32	375.229	589.5	71.25	240.75	1.09	H1-1b
78	M364	W12X45	0.439	14.373	60	0.06	18.839	y	60	375.229	589.5	71.25	240.75	2.421	H1-1b
79	M500	W12X45	0.418	14.373	60	0.06	18.839	y	60	375.229	589.5	71.25	240.75	2.488	H1-1b
80	M501	W12X45	0.422	14.373	57	0.092	3.349	y	64	375.229	589.5	71.25	240.75	1.059	H1-1b
81	M502	W12X45	0.404	14.373	59	0.092	3.349	y	64	375.229	589.5	71.25	240.75	2.282	H1-1b
82	M503	W12X45	0.407	14.373	57	0.09	3.349	y	64	375.229	589.5	71.25	240.75	1.058	H1-1b
83	M504	W12X45	0.385	14.373	59	0.09	3.349	y	64	375.229	589.5	71.25	240.75	2.311	H1-1b
84	M505	W12X45	0.365	14.373	57	0.051	18.839	y	58	375.229	589.5	71.25	240.75	1.058	H1-1b
85	M499	W12X45	0.422	14.373	58	0.06	18.839	y	58	375.229	589.5	71.25	240.75	1.101	H1-1b
86	M506	W12X45	0.345	14.373	59	0.051	18.839	y	60	375.229	589.5	71.25	240.75	2.303	H1-1b
87	M355	W12X45	0.377	14.373	41	0.086	3.349	y	32	375.229	589.5	71.25	240.75	2.435	H1-1b
88	M354	W12X45	0.389	14.373	40	0.085	3.349	y	64	375.229	589.5	71.25	240.75	1.086	H1-1b
89	M235	W12X45	0.481	16.515	59	0.062	24.9	y	58	383.249	589.5	71.25	183.88	1.394	H1-1b
90	M211	W12X45	0.357	14.373	57	0.052	27.351	y	57	375.229	589.5	71.25	240.75	1.098	H1-1b
91	M212	W12X45	0.357	14.373	59	0.052	27.351	y	58	375.229	589.5	71.25	240.75	2.3	H1-1b
92	M245	W12X45	0.437	16.388	60	0.052	24.9	y	60	383.249	589.5	71.25	183.728	1.392	H1-1b
93	M223	W12X45	0.397	11.815	60	0.056	24.9	y	60	383.249	589.5	71.25	188.226	1.426	H1-1b
94	M224	W12X45	0.447	11.815	59	0.055	24.9	y	59	383.249	589.5	71.25	188	1.425	H1-1b
95	M246	W12X45	0.407	14.102	59	0.072	24.9	y	59	383.249	589.5	71.25	181.095	1.372	H1-1b
96	M352	W12X45	0.35	14.373	40	0.051	18.839	y	58	375.229	589.5	71.25	240.75	1.089	H1-1b
97	M353	W12X45	0.324	14.373	41	0.051	18.839	y	60	375.229	589.5	71.25	240.75	2.447	H1-1b
98	M462	W12X30	0.175	10.682	59	0.036	0	y	29	246.99	395.55	35.85	122.886	1.129	H1-1b
99	M497	W10X33	0.033	5	40	0.007	10	y	31	330.46	436.95	52.5	145.5	1.138	H1-1b
100	M496	W10X33	0.033	5	40	0.007	10	y	31	330.46	436.95	52.5	145.5	1.138	H1-1b
101	M495	W10X33	0.033	5	40	0.01	10	y	45	330.46	436.95	52.5	145.5	1.138	H1-1b
102	M494	W10X33	0.097	6.837	41	0.013	10	y	27	330.46	436.95	52.5	145.5	1.135	H1-1b
103	M498	W12X30	0.06	0	52	0.015	0	y	45	363.642	395.55	35.85	161.625	1.203	H1-1b
104	M479	W12X30	0.203	10	45	0.038	0	y	45	212.122	395.55	35.85	161.625	1.612	H1-1b
105	M483	W12X30	0.186	10	44	0.035	20	y	44	212.122	395.55	35.85	161.625	1.606	H1-1b
106	M481	W12X30	0.244	11.25	44	0.04	0	y	45	199.96	395.55	35.85	161.625	1.607	H1-1b
107	M482	W12X30	0.186	10	44	0.035	20	y	44	212.122	395.55	35.85	161.625	1.606	H1-1b
108	M478	W12X30	0.244	11.25	45	0.04	22.5	y	44	199.96	395.55	35.85	161.625	1.607	H1-1b
109	M484	W12X30	0.013	6.5	54	0.003	6.5	y	45	319.199	395.55	35.85	161.625	1.136	H1-1b*
110	M485	W12X30	0.013	6.5	54	0.003	6.5	y	45	319.199	395.55	35.85	161.625	1.136	H1-1b*
111	M286	W12X30	0.152	7.5	44	0.033	15	y	44	141.544	395.55	35.85	135.114	1.287	H1-1b
112	M480	W12X30	0.203	10	45	0.038	20	y	45	212.122	395.55	35.85	161.625	1.612	H1-1b
113	M48	W12X30	0.529	7.5	29	0.134	0	y	29	141.544	395.55	35.85	121.28	1.155	H1-1b
114	M392	W12X30	0.223	10	45	0.039	0	y	45	212.122	395.55	35.85	161.625	1.613	H1-1b
115	M124	W12X30	0.468	7.5	29	0.117	15	y	29	141.544	395.55	35.85	119.292	1.136	H1-1b
116	M99	W12X30	0.466	7.5	29	0.117	0	y	29	141.544	395.55	35.85	119.303	1.136	H1-1b
117	M98	W12X30	0.469	7.5	29	0.117	15	y	29	141.544	395.55	35.85	119.304	1.136	H1-1b
118	M73	W12X30	0.469	7.5	29	0.117	0	y	29	141.544	395.55	35.85	119.3	1.136	H1-1b
119	M74	W12X30	0.466	7.5	29	0.117	15	y	29	141.544	395.55	35.85	119.299	1.136	H1-1b
120	M287	W12X30	0.143	7.5	44	0.031	15	y	44	141.544	395.55	35.85	134.91	1.285	H1-1b
121	128	W12X30	0.466	7.5	29	0.117	15	y	29	141.544	395.55	35.85	119.291	1.136	H1-1b
122	M123	W12X30	0.468	7.5	29	0.117	0	y	29	141.544	395.55	35.85	119.292	1.136	H1-1b
123	M404	W12X30	0.147	7.5	45	0.032	0	y	44	141.544	395.55	35.85	135.117	1.287	H1-1b
124	M288	W12X30	0.14	7.5	45	0.031	15	y	44	141.544	395.55	35.85	134.914	1.285	H1-1b
125	M289	W12X30	0.138	7.5	45	0.031	15	y	44	141.544	395.55	35.85	134.916	1.285	H1-1b
126	M402	W12X30	0.143	7.5	44	0.031	0	y	44	141.544	395.55	35.85	134.911	1.285	H1-1b
127	M403	W12X30	0.142	7.5	44	0.031	0	y	44	141.544	395.55	35.85	134.913	1.285	H1-1b
128	M149	W12X30	0.468	7.5	29	0.117	0	y	29	141.544	395.55	35.85	119.286	1.136	H1-1b
129	M148	W12X30	0.468	7.5	29	0.117	15	y	29	141.544	395.55	35.85	119.287	1.136	H1-1b
130	M405	W12X30	0.149	7.5	45	0.033	0	y	44	141.544	395.55	35.85	135.114	1.287	H1-1b
131	127	W12X30	0.467	7.5	29	0.117	0	y	29	141.544	395.55	35.85	119.291	1.136	H1-1b
132	M282	W12X30	0.141	7.5	44	0.031	15	y	45	141.544	395.55	35.85	134.914	1.285	H1-1b
133	M391	W12X30	0.223	10	45	0.039	20	y	45	212.122	395.55	35.85	161.625	1.613	H1-1b
134	M198	W12X30	0.468	7.5	29	0.117	15	y	29	141.544	395.55	35.85	119.291	1.136	H1-1b
135	M174	W12X30	0.527	7.5	29	0.134	0	y	29	141.544	395.55	35.85	121.281	1.155	H1-1b
136	M173	W12X30	0.529	7.5	29	0.134	15	y	29	141.544	395.55	35.85	121.281	1.155	H1-1b
137	M49	W12X30	0.527	7.5	29	0.134	15	y	29	141.544	395.55	35.85	121.28	1.155	H1-1b
138	M251	W12X30	0.124	11.25	45	0.023	0	y	45	199.96	395.55	35.85	161.625	1.548	H1-1b
139	M250	W12X30	0.111	11.25	44	0.022	22.5	y	45	199.96	395.55	35.85	161.625	1.509	H1-1b
140	M240	W12X30	0.13	10	45	0.019	20	y	44	212.122	395.55	35.85	161.625	1.535	H1-1b
141	M239	W12X30	0.431	10	29	0.102	20	y	29	212.122	395.55	35.85	161.625	1.638	H1-1b
142	M229	W12X30	0.151	10	45	0.024	0	y	45	212.122	395.55	35.85	161.625	1.569	H1-1b
143	M284	W12X30	0.139	7.5	45	0.031	15	y	45	141.544	395.55	35.85	134.915	1.285	H1-1b
144	M281	W12X30	0.14	7.5	44	0.031	0	y	45	141.544	395.55	35.85	134.914	1.285	H1-1b
145	M264	W12X30	0.151	7.5	45	0.033	15	y	45	141.544	395.55	35.85	135.114	1.287	H1-1b
146	M262	W12X30	0.147	7.5	45	0.033	15	y	45	141.544	395.55	35.85	135.118	1.287	H1-1b
147	M398	W12X30	0.28	11.25	45	0.043	22.5	y	45	199.96	395.55	35.85	161.625	1.613	H1-1b
148	M397	W12X30	0.279	11.25	45	0.043	22.5	y	45	199.96	395.55	35.85	161.625	1.613	H1-1b
149	M395	W12X30	0.384	13.163	29	0.053	20	y	44	209.566	395.55	35.85	161.625	1.022	H1-1b
150	M394	W12X30	0.299	10	44	0.039	20	y	44	212.122	395.55	35.85	161.625	1.613	H1-1b
151	M199	W12X30	0.466	7.5	29	0.117	0	y	29	141.544	395.55	35.85	119.29	1.136	H1-1b
152	M217	W12X30	0.067	3.25	24	0.035	0	y	29	319.199	395.55	35.85	161.625	1.31	H1-1b
153	M228	W12X30	0.305	10	29	0.054	0	y	29	212.122	395.55	35.85	161.625	1.624	H1-1b
154	M283	W12X30	0.141	7.5	45	0.031	0	y	45	141.544	395.55	35.85	134.914	1.285	H1-1b
155	M285	W12X30	0.139	7.5	45	0.031	0	y	45	141.544	395.55	35.85	134.917	1.285	H1-1b
156	M263	W12X30	0.147	7.5	44	0.033	0	y	45	141.544	395.55	35.85	135.116	1.287	H1-1b
157	M218	W12X30	0.068	3.25	24	0.035	6.5	y	29	319.199	395.55	35.85	161.625	1.31	H1-1b
158	M215	W12X30	0.443	0	41	0.141	0	y	41	183.057	395.55	35.85	161.625	2.316	H1-1b
159	M384	W12X30	0.24	6.5	56	0.065	13	y	45	183.057	395.55	35.85	132.691	1.136	H1-1b
160	M366	W12X30	0.327	6.5	56	0.088	13	y	44	183.057	395.55	35.85	132.691	1.136	H1-1b

Envelope AISC 14TH (360-10): LRFD Member Steel Code Checks (Continued)

	Member	Shape	Code Check	Loc[ft]	LC	Shear Check	Loc[ft]	Dir	LC	phi*Pnc [k]	phi*Pnt [k]	phi*Mn y-y [k-ft]	phi*Mn z-z [k-ft]	Cb	Eqn
161	M369	W12X30	0.327	6.5	56	0.088	13	y	44	183.057	395.55	35.85	132.691	1.136	H1-1b
162	M372	W12X30	0.327	6.5	56	0.088	13	y	45	183.057	395.55	35.85	132.691	1.136	H1-1b
163	M375	W12X30	0.327	6.5	56	0.088	13	y	44	183.057	395.55	35.85	132.691	1.136	H1-1b
164	M381	W12X30	0.327	6.5	56	0.088	13	y	44	183.057	395.55	35.85	132.691	1.136	H1-1b
165	M488	W10X22	0.224	6.5	44	0.068	13	y	44	105.827	292.05	22.875	72.801	1.136	H1-1b
166	M390	W12X30	0.225	6.5	56	0.061	13	y	44	183.057	395.55	35.85	132.691	1.136	H1-1b
167	M346	W12X30	0.327	6.5	56	0.088	13	y	44	183.057	395.55	35.85	132.691	1.136	H1-1b
168	M468	W12X30	0.2	0	43	0.069	0	y	39	183.057	395.55	35.85	161.625	2.306	H1-1b
169	M378	W12X30	0.327	6.5	56	0.088	13	y	44	183.057	395.55	35.85	132.691	1.136	H1-1b
170	M347	W12X30	0.174	6.5	56	0.047	13	y	45	183.057	395.55	35.85	132.691	1.136	H1-1b
171	M466	W12X30	0.163	13	38	0.057	0	y	39	183.057	395.55	35.85	161.625	2.308	H1-1b
172	M467	W12X30	0.187	13	38	0.063	13	y	38	183.057	395.55	35.85	161.625	2.303	H1-1b
173	M261	W12X30	0.102	3.051	29	0.088	0	y	29	183.057	395.55	35.85	139.364	1.194	H1-1b
174	M465	W12X30	0.132	13	38	0.051	13	y	38	183.057	395.55	35.85	161.625	2.329	H1-1b
175	M401	W10X22	0.374	6.5	56	0.078	13	y	44	105.827	292.05	22.875	72.8	1.136	H1-1b
176	M400	W10X22	0.375	6.5	56	0.078	13	y	44	105.827	292.05	22.875	72.8	1.136	H1-1b
177	M399	W10X22	0.418	6.5	56	0.087	13	y	45	105.827	292.05	22.875	72.8	1.136	H1-1b
178	M40	W12X30	0.499	13	40	0.151	13	y	40	183.057	395.55	35.85	161.625	2.28	H1-1b
179	M436	W12X30	0.188	8.265	25	0.036	0	y	44	154.899	395.55	35.85	161.625	1.593	H1-1b
180	M396	W12X30	0.135	6.5	56	0.037	13	y	44	183.057	395.55	35.85	132.691	1.136	H1-1b
181	M456	W10X22	0.053	3.17	29	0.02	1.585	z	29	270.324	292.05	22.875	97.5	1.129	H1-1b
182	M455	W10X22	0.114	0	45	0.039	0	y	45	258.302	292.05	22.875	97.5	1.787	H1-1b
183	M393	W12X30	0.237	6.5	56	0.064	13	y	45	183.057	395.55	35.85	132.691	1.136	H1-1b
184	M387	W12X30	0.189	6.5	56	0.051	13	y	45	183.057	395.55	35.85	132.691	1.136	H1-1b
185	M227	W12X30	0.433	0	41	0.144	0	y	41	183.057	395.55	35.85	161.625	2.357	H1-1b
186	M238	W12X30	0.409	0	41	0.117	0	y	41	183.057	395.55	35.85	161.625	2.299	H1-1b
187	M487	W10X22	0.224	6.5	44	0.068	13	y	44	105.827	292.05	22.875	72.801	1.136	H1-1b
188	M477	W12X30	0.145	0	41	0.057	13	y	38	183.057	395.55	35.85	161.625	2.31	H1-1b
189	M476	W12X30	0.161	13	40	0.067	13	y	40	183.057	395.55	35.85	161.625	2.212	H1-1b
190	M473	W12X30	0.132	13	38	0.043	13	y	38	183.057	395.55	35.85	161.625	2.306	H1-1b
191	M472	W12X30	0.172	13	38	0.06	0	y	39	183.057	395.55	35.85	161.625	2.305	H1-1b
192	M471	W12X30	0.177	13	38	0.059	13	y	38	183.057	395.55	35.85	161.625	2.308	H1-1b
193	M470	W12X30	0.175	0	43	0.063	0	y	39	183.057	395.55	35.85	161.625	2.308	H1-1b
194	M469	W12X30	0.212	13	38	0.071	13	y	38	183.057	395.55	35.85	161.625	2.303	H1-1b
195	M446	W12X30	0.062	3.74	59	0.053	5.09	y	29	340.538	395.55	35.85	161.625	1.301	H1-1b
196	M115	W12X30	0.571	0	41	0.17	0	y	41	183.057	395.55	35.85	161.625	2.281	H1-1b
197	M190	W12X30	0.431	13	40	0.132	13	y	40	183.057	395.55	35.85	161.625	2.283	H1-1b
198	M415	W12X30	0.175	13	40	0.072	13	y	40	183.057	395.55	35.85	161.625	2.198	H1-1b
199	M414	W12X30	0.146	0	41	0.052	0	y	41	183.057	395.55	35.85	161.625	2.219	H1-1b
200	M486	W10X22	0.264	6.5	45	0.078	0	y	45	105.827	292.05	22.875	73.551	1.148	H1-1b
201	M140	W12X30	0.518	13	40	0.156	13	y	40	183.057	395.55	35.85	161.625	2.28	H1-1b
202	90	W12X30	0.478	0	41	0.145	0	y	41	183.057	395.55	35.85	161.625	2.283	H1-1b
203	M165	W12X30	0.465	0	41	0.142	0	y	41	183.057	395.55	35.85	161.625	2.283	H1-1b
204	M461	W12X30	0.012	2.32	24	0.05	4.64	y	29	346.42	395.55	35.85	161.625	1.065	H1-1b
205	M259	W12X30	0.101	0	73	0.072	0	y	29	183.057	395.55	35.85	131.995	1.13	H1-1b
206	M257	W12X30	0.066	6.5	70	0.02	0	y	45	183.057	395.55	35.85	132.686	1.136	H1-1b
207	M460	W12X30	0.034	0	54	0.005	0	z	60	357.433	395.55	35.85	161.625	1.132	H1-1b
208	M459	W12X30	0.153	5.113	29	0.079	0	y	25	256.893	395.55	35.85	161.625	1.274	H1-1b
209	M249	W12X30	0.368	0	41	0.101	0	y	41	183.057	395.55	35.85	161.625	2.284	H1-1b
210	M458	C12X30	0.013	2.545	70	0.001	5.09	z	53	247.936	396.45	12.308	126.75	1.136	H1-1b
211	M457	W12X30	0.623	10.446	29	0.087	22.5	y	29	62.908	395.55	35.85	68.136	1.141	H1-1b
212	M65	W12X30	0.538	0	41	0.161	0	y	41	183.057	395.55	35.85	161.625	2.282	H1-1b
213	M90	W12X30	0.594	13	40	0.176	13	y	40	183.057	395.55	35.85	161.625	2.283	H1-1b
214	M337	W10X22	0.054	0	45	0.062	0	y	29	258.302	292.05	22.875	97.5	3	H1-1b
215	125	W12X30	0.434	13	40	0.125	13	y	40	183.057	395.55	35.85	161.625	2.283	H1-1b
216	M430	W12X40	0.069	6.492	38	0.028	4.817	y	32	457.832	526.5	63	213.75	1.602	H1-1b
217	M431	W12X40	0.054	6.492	57	0.023	5.152	y	33	457.832	526.5	63	213.75	1.108	H1-1b
218	M512	W10X22	0.175	7.163	45	0.053	0	y	45	105.827	292.05	22.875	72.82	1.137	H1-1b
219	M513	W10X22	0.175	5.77	44	0.053	0	y	45	105.827	292.05	22.875	72.82	1.137	H1-1b
220	M514	W10X22	0.175	5.77	45	0.053	0	y	45	105.827	292.05	22.875	72.82	1.137	H1-1b
221	M515	W10X22	0.162	7.23	44	0.049	0	y	45	105.827	292.05	22.875	72.82	1.137	H1-1b
222	M516	W10X22	0.162	7.296	44	0.049	0	y	45	105.827	292.05	22.875	72.82	1.137	H1-1b
223	M517	W10X22	0.162	5.77	45	0.049	0	y	45	105.827	292.05	22.875	72.82	1.137	H1-1b
224	M518	W10X22	0.162	5.77	44	0.049	0	y	45	105.827	292.05	22.875	72.82	1.137	H1-1b
225	M519	W10X22	0.162	7.23	45	0.049	0	y	45	105.827	292.05	22.875	72.82	1.137	H1-1b
226	M520	W12X30	0.007	3.75	59	0.015	3.75	z	29	356.614	395.55	35.85	161.625	1.136	H1-1b*
227	M521	W12X30	0.118	11.25	38	0.005	22.5	y	21	62.908	395.55	35.85	67.842	1.136	H1-1b
228	M522	WT4X6.5	0.127	5.247	58	0.015	5.247	y	25	50.143	86.4	4.013	5.838	1	H1-1b*
229	M523	WT4X6.5	0.111	5.916	59	0.008	5.916	y	44	45.634	86.4	4.013	5.838	1	H1-1b*
230	M525	HSS4X4X2	0.036	5	29	0.002	5	y	60	71.647	79.65	8.7	8.7	1	H1-1b*
231	M526	HSS4X4X2	0.05	2.5	25	0.006	2.5	z	53	71.647	79.65	8.7	8.7	1	H1-1b
232	M527	HSS4X4X2	0.066	2.5	25	0.009	5	y	23	71.647	79.65	8.7	8.7	1.316	H1-1b
233	M528	HSS4X4X2	0.103	2.5	23	0.009	5	y	23	71.647	79.65	8.7	8.7	1.316	H1-1b
234	M529	WT4X6.5	0.025	2.683	45	0.006	5.366	y	44	49.357	86.4	4.013	5.838	1	H1-1b
235	M530	W8X18	0.002	2.5	54	0.001	2.5	y	21	226.64	236.7	17.475	63.75	1.136	H1-1b*
236	M531	W8X18	0.004	2	54	0.001	4	y	21	211.794	236.7	17.475	63.75	1.136	H1-1b
237	M532	W8X18	0.002	2.5	54	0.001	2.5	y	21	226.64	236.7	17.475	63.75	1.136	H1-1b*
238	M533	W8X18	0.004	2	52	0.001	4	y	21	211.794	236.7	17.475	63.75	1.136	H1-1b
239	M534	L3X3X4	0.052	2.407	59	0.001	4.717	y	19	32.259	64.8	2.345	4.42	1.136	H2-1
240	M535	L3X3X4	0.054	2.407	57	0.001	4.717	y	72	32.259	64.8	2.345	4.42	1.136	H2-1
241	M536	L3X3X4	0.052	2.407	59	0.001	4.717	y	20	32.259	64.8	2.345	4.42	1.136	H2-1
242	M537	L3X3X4	0.054	2.407	57	0.001	4.717	y	21	32.259	64.8	2.345	4.42	1.136	H2-1
243	M538	L3X3X4	0.036	1.822	51	0	3.536	y	21	43.041	64.8	2.345	4.723	1.136	H2-1
244	M539	L3X3X4	0.038	1.822	53	0.001	3.536	y	21	43.041	64.8	2.345	4.723	1.136	H2-1
245	M540	L3X3X4	0.036	1.822	51	0	3.536	y	21	43.041	64.8	2.345	4.723	1.136	H2-1
246	M541	L3X3X4	0.038	1.822	53	0.001	3.536	y	41	43.041	64.8	2.345	4.723	1.136	H2-1
247	M464	HSS4X4X4	0.042	3.73	29	0.001	3.73	y	59	131.626	139.518	16.181	16.181	1	H1-1b*
248	M507	HSS4X4X4	0.03	3.73	24	0.084	3.73	y	23	131.626	139.518	16.181	16.181	1	H1-1b*

HANSON™

Company : Hanson Professional Services
Designer : S. Steiner
Job Number : 25i0007
Model Name : Eli Lilly K215 Unloading Rack

2/19/2026
2:59:44 PM
Checked By : RMF

Envelope AISC 14TH (360-10): LRFD Member Steel Code Checks (Continued)

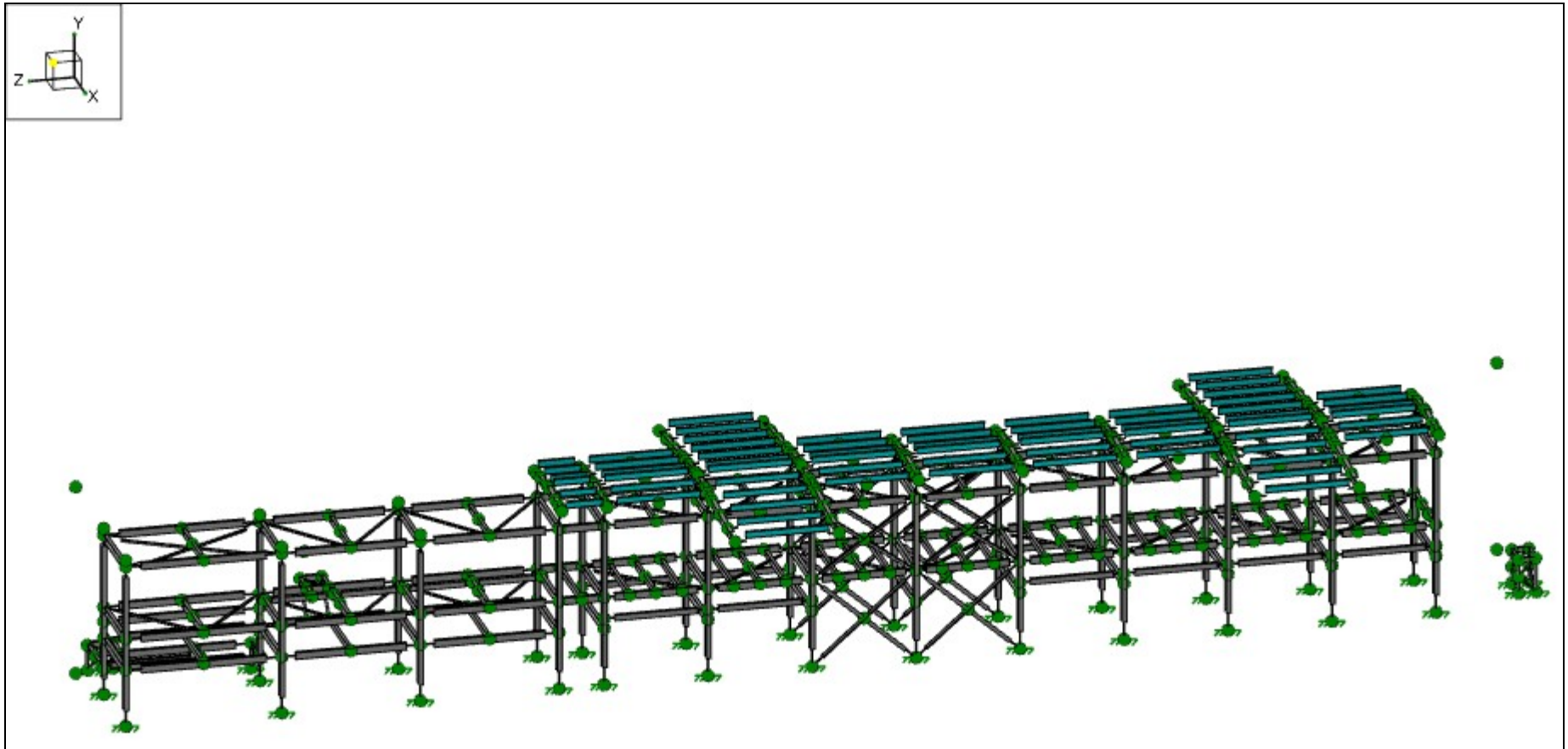
	Member	Shape	Code Check	Loc[ft]	LC	Shear Check	Loc[ft]	Dir	LC	phi*Pnc [k]	phi*Pnt [k]	phi*Mn y-y [k-ft]	phi*Mn z-z [k-ft]	Cb	Eqn
249	M524	HSS4X4X4	0.05	3.73	29	0.084	3.73	y	23	131.626	139.518	16.181	16.181	1	H1-1b*
250	M542	L3X3X4	0.172	2.696	59	0.006	5.233	y	25	27.716	64.8	2.345	4.299	1.136	H2-1
251	M543	WT4X6.5	0.029	3.228	44	0.005	6.391	y	45	42.325	86.4	4.013	5.838	1	H1-1b
252	M544	WT4X6.5	0.103	6.391	29	0.005	6.391	y	45	42.325	86.4	4.013	5.838	1	H1-1b*
253	M545	WT6X17.5	0.089	18.484	52	0.007	9.242	y	44	73.473	232.65	21.488	19.394	1	H1-1b*
254	M546	WT6X17.5	0.905	9.958	51	0.014	9.958	z	40	63.29	232.65	21.488	19.394	1	H1-1a
255	M547	WT6X17.5	0.294	12.904	54	0.015	9.958	z	41	63.29	232.65	21.488	19.394	1	H1-1a
256	M548	WT6X17.5	0.118	18.484	53	0.007	9.242	y	45	73.473	232.65	21.488	19.394	1	H1-1b*
257	M549	WT6X17.5	0.912	9.958	51	0.014	9.958	z	40	63.29	232.65	21.488	19.394	1	H1-1a
258	M550	WT6X17.5	0.285	12.904	54	0.015	9.958	z	57	63.29	232.65	21.488	19.394	1	H1-1a
259	M551	WT6X17.5	0.092	18.484	52	0.007	9.242	y	44	73.473	232.65	21.488	19.394	1	H1-1b*
260	M552	WT6X17.5	0.116	18.484	53	0.007	9.242	y	45	73.473	232.65	21.488	19.394	1	H1-1b*
261	M367	W10X22	0.064	7.5	59	0.008	15	y	44	79.488	292.05	22.875	61.034	1.136	H1-1b
262	M368	W10X22	0.065	7.5	59	0.008	15	y	45	79.488	292.05	22.875	61.034	1.136	H1-1b
263	M559	WT6X17.5	0.754	9.958	53	0.015	9.958	z	40	63.29	232.65	21.488	11.616	1	H1-1a
264	M560	WT6X17.5	0.763	9.958	53	0.015	9.958	z	40	63.29	232.65	21.488	11.616	1	H1-1a
265	M561	WT6X17.5	0.316	9.958	51	0.015	9.958	z	57	63.29	232.65	21.488	11.616	1	H1-1a
266	M562	WT6X17.5	0.097	9.242	56	0.011	9.242	y	44	73.473	232.65	21.488	11.616	1	H1-1b
267	M563	WT6X17.5	0.32	9.958	51	0.015	9.958	z	41	63.29	232.65	21.488	11.616	1	H1-1a
268	M564	WT6X17.5	0.112	18.484	51	0.01	9.242	y	45	73.473	232.65	21.488	19.394	1	H1-1b*
269	M565	WT6X17.5	0.112	18.484	51	0.01	9.242	y	45	73.473	232.65	21.488	19.394	1	H1-1b*
270	M566	WT6X17.5	0.099	9.242	56	0.01	9.242	y	44	73.473	232.65	21.488	11.616	1	H1-1b
271	M370	W10X22	0.064	7.5	57	0.008	15	y	45	79.488	292.05	22.875	61.034	1.136	H1-1b
272	M371	W10X22	0.063	7.5	57	0.008	15	y	44	79.488	292.05	22.875	61.034	1.136	H1-1b
273	M373	W10X22	0.069	7.5	57	0.008	15	y	44	79.488	292.05	22.875	61.034	1.136	H1-1b
274	M374	W10X22	0.069	7.5	59	0.008	15	y	44	79.488	292.05	22.875	61.034	1.136	H1-1b
275	M376	W10X22	0.073	7.5	57	0.008	15	y	45	79.488	292.05	22.875	61.034	1.136	H1-1b
276	M377	W10X22	0.074	7.5	59	0.008	15	y	45	79.488	292.05	22.875	61.034	1.136	H1-1b
277	M406	W10X22	0.071	7.5	57	0.008	15	y	44	79.488	292.05	22.875	61.034	1.136	H1-1b
278	M408	W10X22	0.072	7.5	59	0.008	15	y	44	79.488	292.05	22.875	61.034	1.136	H1-1b
279	M410	W10X22	0.063	7.5	57	0.008	15	y	45	79.488	292.05	22.875	61.034	1.136	H1-1b
280	M411	W10X22	0.064	7.5	59	0.008	15	y	45	79.488	292.05	22.875	61.034	1.136	H1-1b

Envelope AISI S100-10: LRFD Member Cold Formed Steel Code Checks

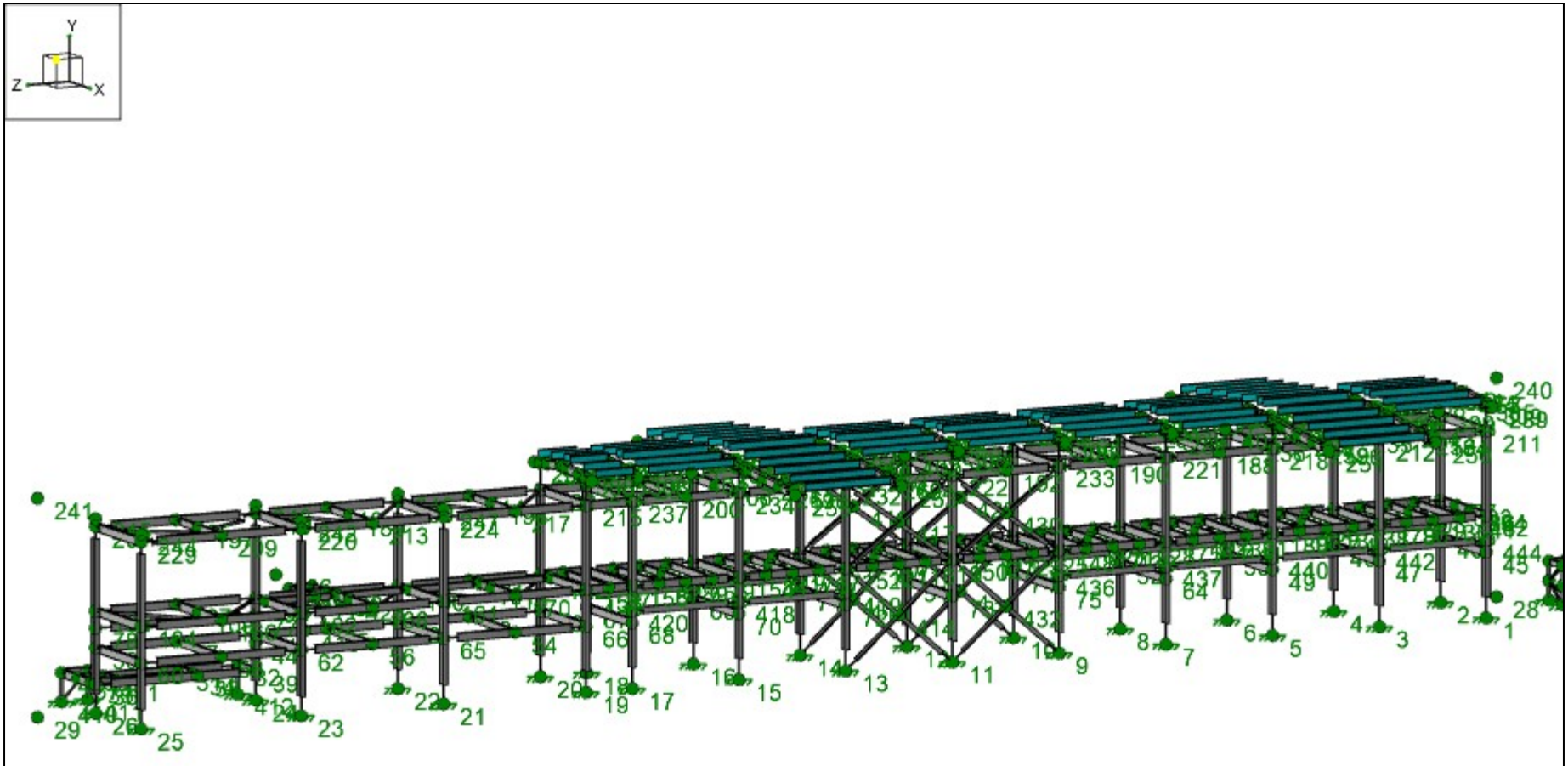
	Member	Shape	Code Check	Loc[ft]	LC	Shear Check	Loc[ft]	Dir	LC	phi*Pn[k]	phi*Tn[k]	phi*Mnyy[k-ft]	phi*Mnzz[k-ft]	Cb	Cmyy	Cmzz	Eqn
1	M311	12CS4X105	0.509	7.5	64	0.107	15	y	64	22.444	65.34	3.373	14.796	1.139	1	1	C5.2.2-1
2	M314	12CS4X105	0.691	7.5	64	0.15	15	y	64	22.444	65.34	3.373	15.113	1.206	1	1	C5.2.2-1
3	M315	12CS4X105	0.734	7.5	32	0.16	0	y	32	22.444	65.34	3.423	15.125	1.209	1	1	C5.2.2-1
4	M318	12CS4X105	0.537	7.5	32	0.114	0	y	32	22.444	65.34	3.423	14.821	1.144	1	1	C5.2.2-1
5	M320	12CS4X105	0.51	7.5	64	0.107	15	y	64	22.444	65.34	3.373	14.79	1.138	1	1	C5.2.2-1
6	M322	12CS4X105	0.691	7.5	64	0.15	15	y	64	22.444	65.34	3.373	15.107	1.205	1	1	C5.2.2-1
7	M328	12CS4X105	0.596	7.041	64	0.144	0	y	64	22.444	65.34	3.373	15.107	1.205	1	1	C5.2.2-1
8	M325	12CS4X105	0.463	7.117	32	0.108	0	y	32	22.444	65.34	3.423	14.834	1.147	1	1	C5.2.2-1
9	M327	12CS4X105	0.436	7.117	64	0.101	0	y	64	22.444	65.34	3.373	14.8	1.14	1	1	C5.2.2-1
10	M330	12CS4X105	0.659	7.883	32	0.154	15	y	32	22.444	65.34	3.423	14.939	1.169	1	1	C5.2.2-1
11	M331	12CS4X105	0.47	7.883	32	0.108	15	y	32	22.444	65.34	3.423	14.782	1.137	1	1	C5.2.2-1
12	M333	12CS4X105	0.445	7.883	64	0.101	15	y	64	22.444	65.34	3.373	14.753	1.131	1	1	C5.1.2-2
13	M310	12CS4X105	0.536	7.5	32	0.114	0	y	32	22.444	65.34	3.423	14.827	1.146	1	1	C5.2.2-1
14	M323	12CS4X105	0.639	7.117	32	0.154	0	y	32	22.444	65.34	3.423	15.125	1.209	1	1	C5.2.2-1
15	M309	12CS4X105	0.734	7.5	32	0.16	0	y	32	22.444	65.34	3.423	15.129	1.21	1	1	C5.2.2-1
16	M297	12CS4X105	0.662	7.5	64	0.138	15	y	64	22.444	65.34	3.373	14.802	1.141	1	1	C5.2.2-1
17	M306	12CS4X105	0.46	7.883	32	0.108	15	y	32	22.444	65.34	3.423	14.856	1.152	1	1	C5.2.2-1
18	M305	12CS4X105	0.638	7.883	32	0.154	15	y	32	22.444	65.34	3.423	15.143	1.213	1	1	C5.2.2-1
19	M304	12CS4X105	0.651	7.117	32	0.154	0	y	32	22.444	65.34	3.423	15.006	1.183	1	1	C5.2.2-1
20	M303	12CS4X105	0.607	7.041	64	0.144	0	y	64	22.444	65.34	3.373	14.994	1.18	1	1	C5.2.2-1
21	M302	12CS4X105	0.625	7.5	64	0.134	15	y	64	22.444	65.34	3.373	14.908	1.162	1	1	C5.2.2-1
22	M301	12CS4X105	0.636	7.5	64	0.138	0	y	64	22.444	65.34	3.373	15.139	1.212	1	1	C5.2.2-1
23	M300	12CS4X105	0.655	7.5	64	0.136	15	y	64	22.444	65.34	3.373	14.775	1.135	1	1	C5.1.2-2
24	M334	12CS4X105	0.615	7.959	64	0.144	15	y	64	22.444	65.34	3.373	14.932	1.167	1	1	C5.2.2-1
25	M296	12CS4X105	0.42	7.5	64	0.09	15	y	64	22.444	65.34	3.373	14.854	1.151	1	1	C5.2.2-1
26	M293	12CS4X105	0.45	7.5	32	0.097	0	y	32	22.444	65.34	3.423	14.894	1.159	1	1	C5.2.2-1
27	M256	12CS4X105	0.717	7.5	32	0.15	15	y	32	22.444	65.34	3.423	14.802	1.141	1	1	C5.2.2-1
28	M255	12CS4X105	0.711	7.5	32	0.148	15	y	32	22.444	65.34	3.423	14.774	1.135	1	1	C5.1.2-2
29	M254	12CS4X105	0.679	7.5	32	0.146	15	y	32	22.444	65.34	3.423	14.903	1.161	1	1	C5.2.2-1
30	M253	12CS4X105	0.689	7.5	32	0.15	15	y	32	22.444	65.34	3.423	15.146	1.213	1	1	C5.2.2-1
31	M308	12CS4X105	0.595	7.959	64	0.144	15	y	64	22.444	65.34	3.373	15.126	1.209	1	1	C5.2.2-1
32	M338	12CS4X105	0.717	7.5	32	0.15	15	y	32	22.444	65.34	3.423	14.803	1.141	1	1	C5.2.2-1
33	M307	12CS4X105	0.433	7.883	64	0.101	15	y	64	22.444	65.34	3.373	14.823	1.145	1	1	C5.2.2-1
34	M340	12CS4X105	0.656	7.5	64	0.136	15	y	64	22.444	65.34	3.373	14.773	1.135	1	1	C5.1.2-2
35	M313	12CS4X105	0.369	7.5	64	0.081	0	y	64	22.444	65.34	3.373	15.204	1.226	1	1	C5.2.2-1
36	M321	12CS4X105	0.314	7.117	64	0.09	0	y	64	22.444	65.34	3.373	15.251	1.237	1	1	C5.2.2-1
37	M299	12CS4X105	0.393	7.5	32	0.086	15	y	32	22.444	65.34	3.423	15.19	1.223	1	1	C5.2.2-1
38	M329	12CS4X105	0.354	7.883	32	0.086	15	y	32	22.444	65.34	3.423	14.943	1.169	1	1	C5.2.2-1
39	M324	12CS4X105	0.337	7.806	32	0.099	15	y	32	22.444	65.34	3.423	15.179	1.245	1	1	C5.2.2-1
40	M326	12CS4X105	0.311	7.883	64	0.092	15	y	64	22.444	65.34	3.373	15.316	1.253	1	1	C5.2.2-1
41	M317	12CS4X105	0.338	7.194	32	0.098	0	y	32	22.444	65.34	3.423	15.22	1.232	1	1	C5.2.2-1
42	M292	12CS4X105	0.392	7.5	32	0.086	15	y	32	22.444	65.34	3.423	15.2	1.226	1	1	C5.2.2-1
43	M360	12CS4X105	0.437	7.117	64	0.101	0	y	64	22.444	65.34	3.373	14.787	1.138	1	1	C5.2.2-1
44	M361	12CS4X105	0.465	7.117	32	0.108	0	y	32	22.444	65.34	3.423	14.817	1.144	1	1	C5.2.2-1
45	M363	12CS4X105	0.322	7.117	64	0.085	0	y	64	22.444	65.34	3.373	15.075	1.198	1	1	C5.2.2-1
46	M365	12CS4X105	0.346	7.117	32	0.092	0	y	32	22.444	65.34	3.423	15.063	1.195	1	1	C5.2.2-1
47	M295	12CS4X105	0.368	7.5	64	0.081	0	y	64	22.444	65.34	3.373	15.22	1.23	1	1	C5.2.2-1
48	M177	12CS4X105	0.384	7.5	32	0.078	15	y	32	22.444	65.34	3.423	14.753	1.131	1	1	C5.1.2-2
49	M332	12CS4X105	0.329	7.883	64	0.08	15	y	64	22.444	65.34	3.373	14.952	1.171	1	1	C5.2.2-1
50	M50	12CS4X105	0.355	7.5	64	0.072	15	y	64	22.444	65.34	3.373	14.753	1.131	1	1	C5.1.2-2
51	M175	12CS4X105	0.355	7.5	64	0.072	15	y	64	22.444	65.34	3.373	14.753	1.131	1	1	C5.1.2-2
52	M341	12CS4X105	0.694	7.5	32	0.15	15	y	32	22.444	65.34	3.423	15.097	1.202	1	1	C5.2.2-1
53	M342	12CS4X105	0.682	7.5	32	0.146	15	y	32	22.444	65.34	3.423	14.881	1.157	1	1	C5.2.2-1

Envelope AISI S100-10: LRFD Member Cold Formed Steel Code Checks (Continued)

	Member	Shape	Code Check	Loc[ft]	LC	Shear Check	Loc[ft]	Dir	LC	phi*Pn[k]	phi*Tn[k]	phi*Mnyy[k-ft]	phi*Mnzz[k-ft]	Cb	Cmyy	Cmzz	Eqn
54	M343	12CS4X105	0.454	7.5	32	0.097	0	y	32	22.444	65.34	3.423	14.854	1.151	1	1	C5.2.2-1
55	M344	12CS4X105	0.424	7.5	64	0.09	15	y	64	22.444	65.34	3.373	14.815	1.143	1	1	C5.2.2-1
56	M345	12CS4X105	0.661	7.5	64	0.138	15	y	64	22.444	65.34	3.373	14.803	1.141	1	1	C5.2.2-1
57	M350	12CS4X105	0.64	7.5	64	0.138	0	y	64	22.444	65.34	3.373	15.089	1.201	1	1	C5.2.2-1
58	M351	12CS4X105	0.628	7.5	64	0.134	15	y	64	22.444	65.34	3.373	14.885	1.157	1	1	C5.2.2-1
59	M339	12CS4X105	0.712	7.5	32	0.148	15	y	32	22.444	65.34	3.423	14.772	1.135	1	1	C5.1.2-2
60	M52	12CS4X105	0.384	7.5	32	0.078	15	y	32	22.444	65.34	3.423	14.753	1.131	1	1	C5.1.2-2
61	M432	12CS4X105	0.044	3.25	64	0.032	0	y	64	36.359	65.34	3.583	18.432	2.155	1	1	C5.2.2-1
62	M433	12CS4X105	0.105	3.25	64	0.063	6.5	y	64	36.359	65.34	3.583	18.432	1.348	1	1	C5.2.2-1
63	M508	12CS4X105	0.072	3.25	64	0.044	6.5	y	64	36.359	65.34	3.583	18.432	1.161	1	1	C5.2.2-1
64	M509	12CS4X105	0.085	3.25	32	0.047	0	y	32	36.359	65.34	3.423	15.471	1.172	1	1	C5.2.2-1
65	M510	12CS4X105	0.123	3.25	32	0.068	6.5	y	32	36.359	65.34	3.423	14.986	1.434	1	1	C5.2.2-1
66	M511	12CS4X105	0.052	3.25	32	0.035	6.5	y	32	36.359	65.34	3.423	14.752	2.225	1	1	C5.2.2-1



HANSON	Hanson Professional Services	Eli Lilly K215 Unloading Rack	SK-3
	S. Steiner		CAL_20260209_UnloadingRackWithCanopy.r3d
	25i0007		



HANSON	Hanson Professional Services	Eli Lilly K215 Unloading Rack	SK-1
	S. Steiner		Feb 17, 2026 at 07:11 PM
	25i0007		CAL_20260209_UnloadingRackWithCanopy.r3d

Foundation Loads / Column Reactions by Load Case

LC No.	LC Description	Node Label	Shear E-W [k]	Axial [k]	Shear N-S [k]	Moment N-S [k-ft]	Torsion [k-ft]	Moment E-W [k-ft]
1	SERVICE DL	1	-0.257	8.980	0.005	0.000	0.000	0.000
2	SERVICE LL	1	-0.112	7.403	0.001	0.000	0.000	0.000
3	SERVICE SL	1	-0.015	2.465	0.017	0.000	0.000	0.000
4	SERVICE RLL	1	0.000	2.092	0.022	0.000	0.000	0.000
5	SERVICE WLX	1	-2.925	8.623	0.007	0.000	-0.001	0.000
6	SERVICE WLZ	1	0.005	0.735	-0.174	0.000	-0.010	0.000
7	SERVICE ICE DL	1	-0.317	10.305	0.016	0.000	0.000	0.000
8	SERVICE ELX	1	0.436	-1.285	0.001	0.000	0.000	0.000
9	SERVICE ELZ	1	0.000	0.000	0.000	0.000	0.000	0.000
1	SERVICE DL	2	0.132	8.530	0.005	0.000	0.000	0.000
2	SERVICE LL	2	-0.136	7.222	0.003	0.000	0.000	0.000
3	SERVICE SL	2	-0.018	2.440	0.017	0.000	0.000	0.000
4	SERVICE RLL	2	0.001	2.092	0.022	0.000	0.000	0.000
5	SERVICE WLX	2	-2.956	-7.223	0.010	0.000	-0.001	0.000
6	SERVICE WLZ	2	0.006	0.757	-0.174	0.000	0.010	0.000
7	SERVICE ICE DL	2	0.030	9.009	0.016	0.000	0.000	0.000
8	SERVICE ELX	2	0.434	1.285	-0.001	0.000	0.000	0.000
9	SERVICE ELZ	2	0.000	0.000	0.000	0.000	0.000	0.000
1	SERVICE DL	3	-0.300	16.513	0.005	0.000	0.000	0.000
2	SERVICE LL	3	0.240	14.776	0.001	0.000	0.000	0.000
3	SERVICE SL	3	0.030	7.486	0.016	0.000	0.000	0.000
4	SERVICE RLL	3	-0.002	6.149	0.022	0.000	0.000	0.000
5	SERVICE WLX	3	-3.451	13.493	0.007	0.000	-0.001	0.000
6	SERVICE WLZ	3	-0.005	2.805	-0.179	0.000	-0.009	0.000
7	SERVICE ICE DL	3	-0.126	17.463	0.016	0.000	0.000	0.000
8	SERVICE ELX	3	0.588	-1.811	0.001	0.000	0.000	0.000
9	SERVICE ELZ	3	0.000	0.000	0.000	0.000	0.000	0.000

Foundation Loads / Column Reactions by Load Case

LC No.	LC Description	Node Label	Shear E-W [k]	Axial [k]	Shear N-S [k]	Moment N-S [k-ft]	Torsion [k-ft]	Moment E-W [k-ft]
1	SERVICE DL	4	0.445	17.076	0.005	0.000	0.000	0.000
2	SERVICE LL	4	0.192	14.974	0.001	0.000	0.000	0.000
3	SERVICE SL	4	0.022	7.498	0.017	0.000	0.000	0.000
4	SERVICE RLL	4	-0.003	6.136	0.022	0.000	0.000	0.000
5	SERVICE WLX	4	-3.524	-8.243	0.007	0.000	-0.001	0.000
6	SERVICE WLZ	4	-0.006	2.792	-0.179	0.000	0.009	0.000
7	SERVICE ICE DL	4	0.510	19.576	0.018	0.000	0.000	0.000
8	SERVICE ELX	4	0.589	1.811	-0.001	0.000	0.000	0.000
9	SERVICE ELZ	4	0.000	0.000	0.000	0.000	0.000	0.000
1	SERVICE DL	5	-0.428	17.027	0.005	0.000	0.000	0.000
2	SERVICE LL	5	-0.166	14.922	0.001	0.000	0.000	0.000
3	SERVICE SL	5	-0.018	7.493	0.016	0.000	0.000	0.000
4	SERVICE RLL	5	0.004	6.139	0.021	0.000	0.000	0.000
5	SERVICE WLX	5	-3.675	13.667	0.013	0.000	-0.001	0.000
6	SERVICE WLZ	5	0.011	2.785	-0.189	0.000	-0.009	0.000
7	SERVICE ICE DL	5	-0.459	19.425	0.016	0.000	0.000	0.000
8	SERVICE ELX	5	0.772	-2.285	0.000	0.000	0.000	0.000
9	SERVICE ELZ	5	0.000	0.000	0.000	0.000	0.000	0.000
1	SERVICE DL	6	0.317	16.562	0.005	0.000	0.000	0.000
2	SERVICE LL	6	-0.215	14.828	0.000	0.000	0.000	0.000
3	SERVICE SL	6	-0.026	7.491	0.016	0.000	0.000	0.000
4	SERVICE RLL	6	0.002	6.146	0.021	0.000	0.000	0.000
5	SERVICE WLX	6	-3.737	-8.415	0.001	0.000	-0.001	0.000
6	SERVICE WLZ	6	0.011	2.812	-0.189	0.000	0.009	0.000
7	SERVICE ICE DL	6	0.177	17.615	0.018	0.000	0.000	0.000
8	SERVICE ELX	6	0.769	2.285	0.000	0.000	0.000	0.000
9	SERVICE ELZ	6	0.000	0.000	0.000	0.000	0.000	0.000

Foundation Loads / Column Reactions by Load Case

LC No.	LC Description	Node Label	Shear E-W [k]	Axial [k]	Shear N-S [k]	Moment N-S [k-ft]	Torsion [k-ft]	Moment E-W [k-ft]
1	SERVICE DL	7	-0.292	14.954	0.004	0.000	0.000	0.000
2	SERVICE LL	7	0.221	14.557	0.001	0.000	0.000	0.000
3	SERVICE SL	7	0.028	4.888	0.013	0.000	0.000	0.000
4	SERVICE RLL	7	-0.002	4.334	0.017	0.000	0.000	0.000
5	SERVICE WLX	7	-3.836	12.284	0.015	0.000	-0.001	0.000
6	SERVICE WLZ	7	-0.014	1.504	-0.205	0.000	-0.009	0.000
7	SERVICE ICE DL	7	-0.145	15.151	0.014	0.000	0.000	0.000
8	SERVICE ELX	7	1.074	-3.237	-0.002	0.000	0.000	0.000
9	SERVICE ELZ	7	0.000	0.000	0.000	0.000	0.000	0.000
1	SERVICE DL	8	0.453	15.507	0.004	0.000	0.000	0.000
2	SERVICE LL	8	0.172	14.693	-0.002	0.000	0.000	0.000
3	SERVICE SL	8	0.022	4.923	0.013	0.000	0.000	0.000
4	SERVICE RLL	8	-0.002	4.356	0.017	0.000	0.000	0.000
5	SERVICE WLX	8	-3.916	-9.485	-0.005	0.000	-0.001	0.000
6	SERVICE WLZ	8	-0.015	1.481	-0.205	0.000	0.009	0.000
7	SERVICE ICE DL	8	0.491	17.120	0.016	0.000	0.000	0.000
8	SERVICE ELX	8	1.074	3.237	0.004	0.000	0.000	0.000
9	SERVICE ELZ	8	0.000	0.000	0.000	0.000	0.000	0.000
1	SERVICE DL	9	-0.448	15.793	1.622	0.000	0.000	0.000
2	SERVICE LL	9	-0.178	14.471	1.569	0.000	0.000	0.000
3	SERVICE SL	9	-0.023	4.778	0.449	0.000	0.000	0.000
4	SERVICE RLL	9	0.001	4.507	0.394	0.000	0.000	0.000
5	SERVICE WLX	9	-4.012	13.813	0.981	0.000	0.004	0.005
6	SERVICE WLZ	9	0.016	-9.731	-5.433	0.000	-0.009	0.000
7	SERVICE ICE DL	9	-0.489	19.213	2.253	0.000	0.000	0.000
8	SERVICE ELX	9	1.470	-5.279	-0.519	0.000	-0.001	-0.002
9	SERVICE ELZ	9	0.000	0.000	0.000	0.000	0.000	0.000

Foundation Loads / Column Reactions by Load Case

LC No.	LC Description	Node Label	Shear E-W [k]	Axial [k]	Shear N-S [k]	Moment N-S [k-ft]	Torsion [k-ft]	Moment E-W [k-ft]
1	SERVICE DL	10	0.297	15.421	1.655	0.000	0.000	0.000
2	SERVICE LL	10	-0.227	14.438	1.604	0.000	0.000	0.000
3	SERVICE SL	10	-0.030	4.758	0.457	0.000	0.000	0.000
4	SERVICE RLL	10	0.001	4.481	0.396	0.000	0.000	0.000
5	SERVICE WLX	10	-4.110	-11.312	-0.900	0.000	0.004	0.005
6	SERVICE WLZ	10	0.016	-9.993	-5.592	0.000	0.009	0.000
7	SERVICE ICE DL	10	0.147	17.829	2.374	0.000	0.000	0.000
8	SERVICE ELX	10	1.470	5.331	0.559	0.000	-0.001	-0.002
9	SERVICE ELZ	10	0.000	0.000	0.000	0.000	0.000	0.000
1	SERVICE DL	11	-0.279	16.363	0.077	0.000	0.000	0.000
2	SERVICE LL	11	0.235	14.849	-0.021	0.000	0.000	0.000
3	SERVICE SL	11	0.033	5.158	0.117	0.000	0.000	0.000
4	SERVICE RLL	11	0.002	4.364	0.067	0.000	0.000	0.000
5	SERVICE WLX	11	-3.880	13.414	-0.554	0.000	0.000	0.010
6	SERVICE WLZ	11	0.005	1.464	-10.485	0.000	-0.009	0.000
7	SERVICE ICE DL	11	-0.118	19.293	-0.163	0.000	0.000	-0.001
8	SERVICE ELX	11	1.546	-4.645	-0.052	0.000	0.000	-0.004
9	SERVICE ELZ	11	0.000	0.000	0.000	0.000	0.000	0.000
1	SERVICE DL	12	0.467	16.671	0.087	0.000	0.000	0.000
2	SERVICE LL	12	0.186	15.083	0.051	0.000	0.000	0.000
3	SERVICE SL	12	0.027	5.203	0.132	0.000	0.000	0.000
4	SERVICE RLL	12	0.002	4.380	0.070	0.000	0.000	0.000
5	SERVICE WLX	12	-3.965	-10.375	0.480	0.000	0.000	0.010
6	SERVICE WLZ	12	0.005	1.422	-10.916	0.000	0.009	0.000
7	SERVICE ICE DL	12	0.519	20.187	-0.173	0.000	0.000	-0.001
8	SERVICE ELX	12	1.537	4.542	-0.019	0.000	0.000	-0.004
9	SERVICE ELZ	12	0.000	0.000	0.000	0.000	0.000	0.000

Foundation Loads / Column Reactions by Load Case

LC No.	LC Description	Node Label	Shear E-W [k]	Axial [k]	Shear N-S [k]	Moment N-S [k-ft]	Torsion [k-ft]	Moment E-W [k-ft]
1	SERVICE DL	13	-0.434	17.346	-1.710	0.000	0.000	0.000
2	SERVICE LL	13	-0.172	14.844	-1.630	0.000	0.000	0.000
3	SERVICE SL	13	-0.020	7.370	-0.595	0.000	0.000	0.000
4	SERVICE RLL	13	0.002	6.124	-0.479	0.000	0.000	0.000
5	SERVICE WLX	13	-3.700	14.948	-1.650	0.000	-0.003	0.005
6	SERVICE WLZ	13	-0.016	14.043	-5.756	0.000	-0.009	0.000
7	SERVICE ICE DL	13	-0.473	20.462	-2.086	0.000	0.000	0.000
8	SERVICE ELX	13	1.141	-3.665	0.450	0.000	0.001	-0.002
9	SERVICE ELZ	13	0.000	0.000	0.000	0.000	0.000	0.000
1	SERVICE DL	14	0.311	16.997	-1.734	0.000	0.000	0.000
2	SERVICE LL	14	-0.221	14.565	-1.572	0.000	0.000	0.000
3	SERVICE SL	14	-0.028	7.338	-0.587	0.000	0.000	0.000
4	SERVICE RLL	14	0.001	6.132	-0.481	0.000	0.000	0.000
5	SERVICE WLX	14	-3.773	-9.635	1.121	0.000	-0.003	0.005
6	SERVICE WLZ	14	-0.016	14.361	-5.946	0.000	0.009	0.000
7	SERVICE ICE DL	14	0.162	19.190	-2.219	0.000	0.000	0.000
8	SERVICE ELX	14	1.136	3.716	-0.417	0.000	0.001	-0.002
9	SERVICE ELZ	14	0.000	0.000	0.000	0.000	0.000	0.000
1	SERVICE DL	15	-0.301	16.517	-0.005	0.000	0.000	0.000
2	SERVICE LL	15	0.240	14.777	-0.001	0.000	0.000	0.000
3	SERVICE SL	15	0.029	7.485	-0.017	0.000	0.000	0.000
4	SERVICE RLL	15	-0.003	6.147	-0.022	0.000	0.000	0.000
5	SERVICE WLX	15	-3.390	12.911	-0.013	0.000	0.001	0.000
6	SERVICE WLZ	15	0.020	2.779	-0.209	0.000	-0.009	0.000
7	SERVICE ICE DL	15	-0.127	17.749	-0.016	0.000	0.000	0.000
8	SERVICE ELX	15	0.817	-2.396	0.000	0.000	0.000	0.000
9	SERVICE ELZ	15	0.000	0.000	0.000	0.000	0.000	0.000

Foundation Loads / Column Reactions by Load Case

LC No.	LC Description	Node Label	Shear E-W [k]	Axial [k]	Shear N-S [k]	Moment N-S [k-ft]	Torsion [k-ft]	Moment E-W [k-ft]
1	SERVICE DL	16	0.444	17.072	-0.005	0.000	0.000	0.000
2	SERVICE LL	16	0.191	14.973	0.000	0.000	0.000	0.000
3	SERVICE SL	16	0.022	7.499	-0.017	0.000	0.000	0.000
4	SERVICE RLL	16	-0.004	6.138	-0.022	0.000	0.000	0.000
5	SERVICE WLX	16	-3.460	-7.661	-0.002	0.000	0.001	0.000
6	SERVICE WLZ	16	0.021	2.819	-0.211	0.000	0.009	0.000
7	SERVICE ICE DL	16	0.508	19.863	-0.018	0.000	0.000	0.000
8	SERVICE ELX	16	0.819	2.396	-0.001	0.000	0.000	0.000
9	SERVICE ELZ	16	0.000	0.000	0.000	0.000	0.000	0.000
1	SERVICE DL	17	-0.335	11.485	-0.005	0.000	0.000	0.000
2	SERVICE LL	17	-0.101	10.568	-0.001	0.000	0.000	0.000
3	SERVICE SL	17	-0.014	3.529	-0.017	0.000	0.000	0.000
4	SERVICE RLL	17	0.000	3.071	-0.023	0.000	0.000	0.000
5	SERVICE WLX	17	-2.864	8.416	-0.009	0.000	0.001	0.000
6	SERVICE WLZ	17	-0.021	1.102	-0.186	0.000	-0.009	0.000
7	SERVICE ICE DL	17	-0.381	13.152	-0.017	0.000	0.000	0.000
8	SERVICE ELX	17	0.639	-1.860	0.000	0.000	0.000	0.000
9	SERVICE ELZ	17	0.000	0.000	0.000	0.000	0.000	0.000
1	SERVICE DL	18	0.208	11.087	-0.005	0.000	0.000	0.000
2	SERVICE LL	18	-0.149	10.394	-0.002	0.000	0.000	0.000
3	SERVICE SL	18	-0.020	3.502	-0.018	0.000	0.000	0.000
4	SERVICE RLL	18	0.000	3.066	-0.023	0.000	0.000	0.000
5	SERVICE WLX	18	-2.892	-6.409	-0.007	0.000	0.001	0.000
6	SERVICE WLZ	18	-0.021	1.037	-0.183	0.000	0.009	0.000
7	SERVICE ICE DL	18	0.091	11.834	-0.017	0.000	0.000	0.000
8	SERVICE ELX	18	0.636	1.860	0.000	0.000	0.000	0.000
9	SERVICE ELZ	18	0.000	0.000	0.000	0.000	0.000	0.000

Foundation Loads / Column Reactions by Load Case

LC No.	LC Description	Node Label	Shear E-W [k]	Axial [k]	Shear N-S [k]	Moment N-S [k-ft]	Torsion [k-ft]	Moment E-W [k-ft]
1	SERVICE DL	19	-0.211	12.313	0.000	0.000	0.000	0.000
2	SERVICE LL	19	0.040	3.603	0.000	0.000	0.000	0.000
3	SERVICE SL	19	0.007	1.115	0.004	0.000	0.000	0.000
4	SERVICE RLL	19	0.000	0.978	0.005	0.000	0.000	0.000
5	SERVICE WLX	19	-3.279	8.751	-0.012	0.000	0.000	0.000
6	SERVICE WLZ	19	0.014	0.289	-0.257	0.000	-0.010	0.000
7	SERVICE ICE DL	19	-0.185	11.118	0.003	0.000	0.000	0.000
8	SERVICE ELX	19	0.633	-2.030	0.003	0.000	0.000	0.000
9	SERVICE ELZ	19	0.000	0.000	0.000	0.000	0.000	0.000
1	SERVICE DL	20	0.207	12.778	0.000	0.000	0.000	0.000
2	SERVICE LL	20	0.007	7.235	-0.001	0.000	0.000	0.000
3	SERVICE SL	20	0.003	1.611	0.004	0.000	0.000	0.000
4	SERVICE RLL	20	0.000	0.975	0.005	0.000	0.000	0.000
5	SERVICE WLX	20	-3.317	-8.145	-0.051	0.000	0.000	0.000
6	SERVICE WLZ	20	0.014	0.358	-0.316	0.000	0.008	0.000
7	SERVICE ICE DL	20	0.195	11.651	0.002	0.000	0.000	0.000
8	SERVICE ELX	20	0.633	2.030	-0.003	0.000	0.000	0.000
9	SERVICE ELZ	20	0.000	0.000	0.000	0.000	0.000	0.000
1	SERVICE DL	21	-0.245	16.744	-0.003	0.000	0.000	0.000
2	SERVICE LL	21	0.020	0.717	0.000	0.000	0.000	0.000
3	SERVICE SL	21	0.002	0.075	-0.007	0.000	0.000	0.000
4	SERVICE RLL	21	0.000	-0.001	-0.009	0.000	0.000	0.000
5	SERVICE WLX	21	-3.602	7.798	-0.015	0.000	0.000	0.000
6	SERVICE WLZ	21	0.006	-0.022	-0.246	0.000	-0.010	0.000
7	SERVICE ICE DL	21	-0.229	14.019	-0.008	0.000	0.000	0.000
8	SERVICE ELX	21	0.650	-1.986	0.002	0.000	0.000	0.000
9	SERVICE ELZ	21	0.000	0.000	0.000	0.000	0.000	0.000

Foundation Loads / Column Reactions by Load Case

LC No.	LC Description	Node Label	Shear E-W [k]	Axial [k]	Shear N-S [k]	Moment N-S [k-ft]	Torsion [k-ft]	Moment E-W [k-ft]
1	SERVICE DL	22	0.254	18.268	-0.003	0.000	0.000	0.000
2	SERVICE LL	22	-0.001	9.169	-0.001	0.000	0.000	0.000
3	SERVICE SL	22	-0.001	1.365	-0.007	0.000	0.000	0.000
4	SERVICE RLL	22	0.000	0.001	-0.009	0.000	0.000	0.000
5	SERVICE WLX	22	-3.630	-7.798	-0.062	0.000	0.000	0.000
6	SERVICE WLZ	22	0.006	0.022	-0.300	0.000	0.008	0.000
7	SERVICE ICE DL	22	0.227	17.166	-0.010	0.000	0.000	0.000
8	SERVICE ELX	22	0.651	1.986	-0.002	0.000	0.000	0.000
9	SERVICE ELZ	22	0.000	0.000	0.000	0.000	0.000	0.000
1	SERVICE DL	23	-0.271	17.492	-0.003	0.000	0.000	0.000
2	SERVICE LL	23	-0.057	0.007	-0.001	0.000	0.000	0.000
3	SERVICE SL	23	-0.011	-0.034	-0.006	0.000	0.000	0.000
4	SERVICE RLL	23	0.000	-0.001	-0.008	0.000	0.000	0.000
5	SERVICE WLX	23	-3.512	7.571	-0.008	0.000	0.000	0.000
6	SERVICE WLZ	23	0.020	0.003	-0.235	0.000	-0.010	0.000
7	SERVICE ICE DL	23	-0.261	14.596	-0.007	0.000	0.000	0.000
8	SERVICE ELX	23	0.631	-1.997	0.002	0.000	0.000	0.000
9	SERVICE ELZ	23	0.000	0.000	0.000	0.000	0.000	0.000
1	SERVICE DL	24	0.256	20.072	-0.002	0.000	0.000	0.000
2	SERVICE LL	24	-0.066	5.653	0.001	0.000	0.000	0.000
3	SERVICE SL	24	-0.013	1.080	-0.005	0.000	0.000	0.000
4	SERVICE RLL	24	0.000	0.001	-0.006	0.000	0.000	0.000
5	SERVICE WLX	24	-4.076	-7.571	-0.393	0.000	0.000	0.000
6	SERVICE WLZ	24	0.007	-0.003	-0.510	0.000	0.008	0.000
7	SERVICE ICE DL	24	0.214	20.065	-0.007	0.000	0.000	0.000
8	SERVICE ELX	24	0.634	1.997	-0.001	0.000	0.000	0.000
9	SERVICE ELZ	24	0.000	0.000	0.000	0.000	0.000	0.000

Foundation Loads / Column Reactions by Load Case

LC No.	LC Description	Node Label	Shear E-W [k]	Axial [k]	Shear N-S [k]	Moment N-S [k-ft]	Torsion [k-ft]	Moment E-W [k-ft]
1	SERVICE DL	25	-0.144	10.017	-0.002	0.000	0.000	0.000
2	SERVICE LL	25	0.028	-0.072	-0.001	0.000	0.000	0.000
3	SERVICE SL	25	0.006	-0.017	-0.006	0.000	0.000	0.000
4	SERVICE RLL	25	0.000	0.000	-0.007	0.000	0.000	0.000
5	SERVICE WLX	25	-3.189	6.515	-0.005	0.000	0.000	0.000
6	SERVICE WLZ	25	-0.040	0.054	-0.229	0.000	-0.010	0.000
7	SERVICE ICE DL	25	-0.121	8.930	-0.006	0.000	0.000	0.000
8	SERVICE ELX	25	0.562	-1.738	0.001	0.000	0.000	0.000
9	SERVICE ELZ	25	0.000	0.000	0.000	0.000	0.000	0.000
1	SERVICE DL	26	0.153	12.040	-0.002	0.000	0.000	0.000
2	SERVICE LL	26	0.028	3.974	0.001	0.000	0.000	0.000
3	SERVICE SL	26	0.006	0.784	-0.006	0.000	0.000	0.000
4	SERVICE RLL	26	0.000	0.000	-0.008	0.000	0.000	0.000
5	SERVICE WLX	26	-4.543	-6.515	-0.399	0.000	0.000	0.000
6	SERVICE WLZ	26	-0.027	-0.054	-0.511	0.000	0.008	0.000
7	SERVICE ICE DL	26	0.160	14.134	-0.009	0.000	0.000	0.000
8	SERVICE ELX	26	0.562	1.738	0.000	0.000	0.000	0.000
9	SERVICE ELZ	26	0.000	0.000	0.000	0.000	0.000	0.000

Project Title: Eli Lilly K215 Unloading Rack
Engineer:
Project ID: 25I0007
Project Descr:

Concrete Column

Project File: CAL_20251010_LillyUnloadingRackCalculations.ec6

LIC# : KW-06015723, Build:20.26.02.06 HANSON ENGINEERS, INC. (c) ENERCALC, LLC 1982-2026

DESCRIPTION: Typical Concrete Pier (Transverse Moment Frames)

Note - ACI 318-11 was not available for code checks in Enercalc, so ACI 318-14 was used instead. There are no significant changes betwen the editions that would affect the design of these elements.

Code References

Governing Code : IBC 2015, CBC 2016
Referenced Design Standard(s) : ACI 318-14
Load Combinations Used : ASCE 7-10

General Information

f'c : Concrete 28 day streng	=	5.0 ksi
E =	=	3,122.0 ksi
Density	=	150.0 pcf
β	=	0.0
fy - Main Rebar	=	60.0 ksi
E - Main Rebar	=	29,000.0 ksi
Allow. Reinforcing Limits		*ASTM A615 Bars Used*
Min. Reinf.	=	1.0 %
Max. Reinf.	=	8.0 %

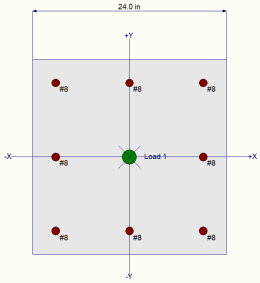
Overall Column Height = 7.110 ft
End Fixity Top Free, Bottom Fixed

Brace condition for deflection (buckling) along colun
X-X (width) axis :
Unbraced Length for buckling ABOUT X-X Axis = 7.110 ft, K = 1.0
Y-Y (depth) axis :
Unbraced Length for buckling ABOUT Y-Y Axis = 7.110 ft, K = 1.0

Column Cross Section

Column Dimensions : 24.0in Square Column, Column Edge to Rebar Edge Cover = 2.375in

Column Reinforcing : general



Rebar Sizes & Locations

Total bars = 8

X & Y distances measured from lower-left corner.

Bar Size #	X in	Y in	Bar Size #	X in	Y in	Bar Size #	X in	Y in	Bar Size #	X in	Y in
# 8	2.875	2.875	# 8	2.875	21.125	# 8	21.125	21.125	# 8	21.125	2.875
# 8	12.000	21.125	# 8	12.000	2.875	# 8	2.875	12.000	# 8	21.125	12.000

Applied Loads

Entered loads are factored per load combinations specified by user.

Column self weight included : 4,266.0 lbs * Dead Load Factor
AXIAL LOADS . . .
All max downward: Axial Load at 7.110 ft above base, D = 40.10, LR = 6.20, L = 15.0, S = 7.50, W = 13.70, E = 3.30 k
BENDING LOADS . . .
Lat. Point Load at 7.110 ft creating Mx-x, W = 2.90, E = 1.10 k

Project Title: Eli Lilly K215 Unloading Rack
Engineer:
Project ID: 25I0007
Project Descr:

Concrete Column

Project File: CAL_20251010_LillyUnloadingRackCalculations.ec6

LIC# : KW-06015723, Build:20.26.02.06 HANSON ENGINEERS, INC. (c) ENERCALC, LLC 1982-2026

DESCRIPTION: Typical Concrete Pier (Transverse Moment Frames)

Note - ACI 318-11 was not available for code checks in Enercalc, so ACI 318-14 was used instead. There are no significant changes betwen the editions that would affect the design of these elements.

DESIGN SUMMARY

Load Combination	0.0
Location of max.above base	0.0 ft

Maximum Stress Ratio **0.0 : 1**

Ratio = (Pu^2+Mu^2)^.5 / (PhiPn^2+PhiMn^2)^.5

Pu =	0.0 k	φ * Pn =	0.0 k
Mu-x =	0.0 k-ft	φ * Mn-x =	0.0 k-ft
Mu-y =	0.0 k-ft	φ* Mn-y =	0.0 k-ft
Mu Angle =	deg	φ =	
Mu at Angle =	k-ft	φMn at Angle =	k-ft

Pn & Mn values located at Pu-Mu vector intersection with capacity curv

Column Capacities . .

Pnmax : Nominal Max. Compressive Axial Capaci	0.0 k
Pnmin : Nominal Min. Tension Axial Capacity	k
φ Pn, max : Usable Compressive Axial Capacity	0.0 k
φ Pn, min : Usable Tension Axial Capacity	k

Maximum SERVICE Load Reactions .

Top along Y-Y	0.0 k	Bottom along Y-Y	0.0 k
Top along X-X	0.0 k	Bottom along X-X	0.0 k

Maximum SERVICE Load Deflections . .

Along Y-Y	0.0 in	at	0.0 ft above base
for load combination :	0.0		
Along X-X	0.0 in	at	0.0 ft above base
for load combination :	0.0		

General Section Informatio β = 0.0 θ = 0.0

ρ : % Reinforcing	0.0 %	Rebar < Min of 1.0 %
Reinforcing Area	0.0 in^2	
Concrete Area	0.0 in^2	

Governing Load Combination Results

Governing Factored Load Combination	Moment X-X	Moment Y-Y	Dist. from base ft	Axial Load k Pu	φ * Pn	Bending Analysis k-ft δx	δx * Mux	δy	δy * Muy	Alpha (deg)	δ Mu	φ Mn	Utilization Ratio

Maximum Reactions

Note: Only non-zero reactions are listed.

Load Combination	X-X Axis Reaction k @ Base	@ Top	Y-Y Axis Reaction @ Base	@ Top	Axial Reaction @ Base	Mx - End Moments k-ft @ Base	@ Top	My - End Moments @ Base	@ Top

Maximum Moment Reactions

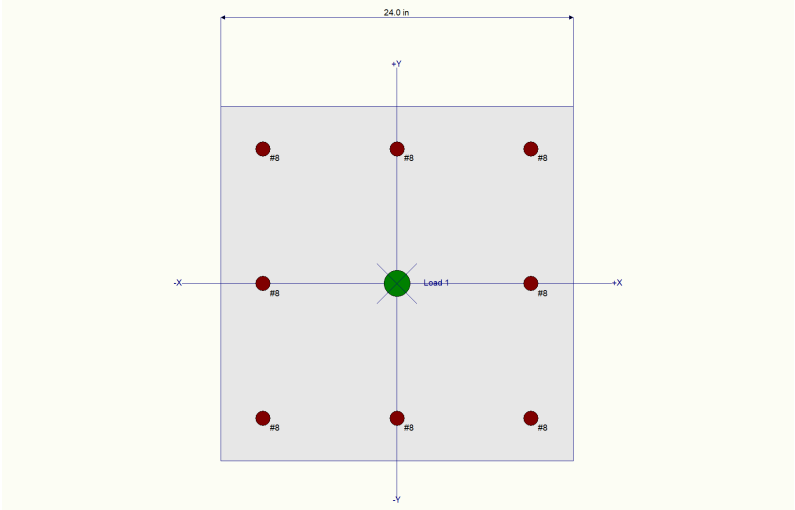
Note: Only non-zero reactions are listed.

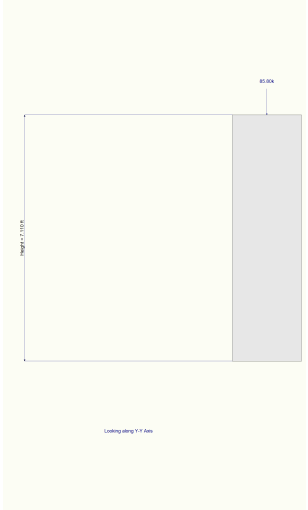
Load Combination	Moment About X-X Axis @ Base	@ Top	Moment About Y-Y Axis @ Base	@ Top

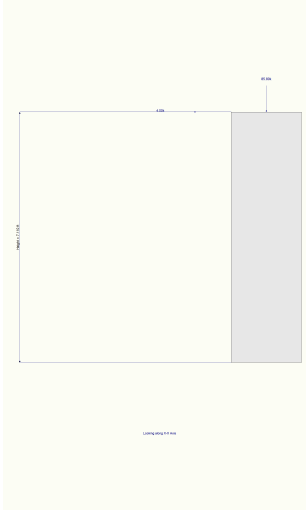
Maximum Deflections for Load Combinations

Load Combination	Max. X-X Deflection	Distance	Max. Y-Y Deflection	Distance

Sketches







Project Title: Eli Lilly K215 Unloading Rack
Engineer:
Project ID: 25I0007
Project Descr:

Concrete Column

Project File: CAL_20251010_LillyUnloadingRackCalculations.ec6

LIC# : KW-06015723, Build:20.26.02.06 HANSON ENGINEERS, INC. (c) ENERCALC, LLC 1982-2026

DESCRIPTION: Concrete Pier at Lateral Braces (Longitudinal Braced Frames)

Note - ACI 318-11 was not available for code checks in Enercalc, so ACI 318-14 was used instead. There are no significant changes betwen the editions that would affect the design of these elements.

Code References

Governing Code : IBC 2015, CBC 2016
Referenced Design Standard(s) : ACI 318-14
Load Combinations Used : ASCE 7-10

General Information

f'c : Concrete 28 day streng	=	5.0 ksi
E =	=	3,122.0 ksi
Density	=	150.0 pcf
β	=	0.0
fy - Main Rebar	=	60.0 ksi
E - Main Rebar	=	29,000.0 ksi
Allow. Reinforcing Limits		*ASTM A615 Bars Used*
Min. Reinf.	=	1.0 %
Max. Reinf.	=	8.0 %

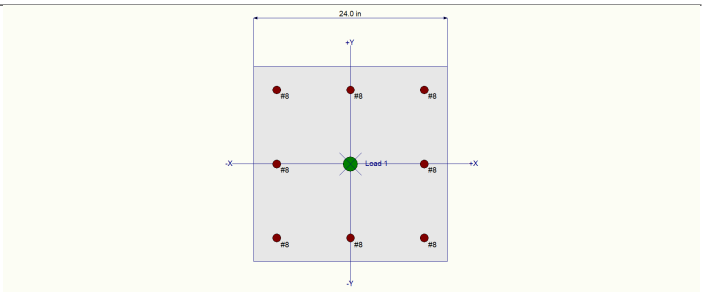
Overall Column Height = 7.110 ft
End Fixity Top Free, Bottom Fixed

Brace condition for deflection (buckling) along colun
X-X (width) axis :
Unbraced Length for buckling ABOUT X-X Axis = 7.110 ft, K = 1.0
Y-Y (depth) axis :
Unbraced Length for buckling ABOUT Y-Y Axis = 7.110 ft, K = 1.0

Column Cross Section

Column Dimensions : 24.0in Square Column, Column Edge to Rebar Edge Cover = 2.375in

Column Reinforcing : general



Rebar Sizes & Locations

Total bars = 8

X & Y distances measured from lower-left corner.

Bar Size #	X in	Y in	Bar Size #	X in	Y in	Bar Size #	X in	Y in	Bar Size #	X in	Y in
# 8	2.875	2.875	# 8	2.875	21.125	# 8	21.125	21.125	# 8	21.125	2.875
# 8	12.000	21.125	# 8	12.000	2.875	# 8	2.875	12.000	# 8	21.125	12.000

Applied Loads

Entered loads are factored per load combinations specified by user.

Column self weight included : 4,266.0 lbs * Dead Load Factor
AXIAL LOADS . . .
All max downward: Axial Load at 7.110 ft above base, D = 37.80, LR = 6.20, L = 15.10, S = 7.40, W = 15.0, E = 5.30 k
BENDING LOADS . . .
Transverse Shear (RISA X/EW Dir): Lat. Point Load at 7.110 ft creating My-y, W = 4.10 k
Longitudinal Shear (RISA Z/NS Dir): Lat. Point Load at 7.110 ft creating Mx-x, W = 15.0, E = 0.60 k

Project Title: Eli Lilly K215 Unloading Rack
Engineer:
Project ID: 25I0007
Project Descr:

Concrete Column

Project File: CAL_20251010_LillyUnloadingRackCalculations.ec6

LIC# : KW-06015723, Build:20.26.02.06 HANSON ENGINEERS, INC. (c) ENERCALC, LLC 1982-2026

DESCRIPTION: Concrete Pier at Lateral Braces (Longitudinal Braced Frames)

Note - ACI 318-11 was not available for code checks in Enercalc, so ACI 318-14 was used instead. There are no significant changes betwen the editions that would affect the design of these elements.

DESIGN SUMMARY

Load Combination	0.0
Location of max.above base	0.0 ft

Maximum Stress Ratio **0.0 : 1**

Ratio = (Pu^2+Mu^2)^.5 / (PhiPn^2+PhiMn^2)^.5

Pu =	0.0 k	φ * Pn =	0.0 k
Mu-x =	0.0 k-ft	φ* Mn-x =	0.0 k-ft
Mu-y =	0.0 k-ft	φ* Mn-y =	0.0 k-ft
Mu Angle =	deg	φ =	
Mu at Angle =	k-ft	φMn at Angle =	k-ft

Pn & Mn values located at Pu-Mu vector intersection with capacity curv

Column Capacities . .

Pnmax : Nominal Max. Compressive Axial Capaci	0.0 k
Pnmin : Nominal Min. Tension Axial Capacity	k
φ Pn, max : Usable Compressive Axial Capacity	0.0 k
φ Pn, min : Usable Tension Axial Capacity	k

Maximum SERVICE Load Reactions .

Top along Y-Y	0.0 k	Bottom along Y-Y	0.0 k
Top along X-X	0.0 k	Bottom along X-X	0.0 k

Maximum SERVICE Load Deflections . .

Along Y-Y	0.0 in	at	0.0 ft above base
for load combination :	0.0		
Along X-X	0.0 in	at	0.0 ft above base
for load combination :	0.0		

General Section Informatio β = 0.0 θ = 0.0

ρ : % Reinforcing	0.0 %	Rebar < Min of 1.0 %
Reinforcing Area	0.0 in^2	
Concrete Area	0.0 in^2	

Governing Load Combination Results

Governing Factored Load Combination	Moment X-X	Moment Y-Y	Dist. from base ft	Axial Load k Pu	φ * Pn	Bending Analysis k-ft δx	δx * Mux	δy	δy * Muy	Alpha (deg)	δ Mu	φ Mn	Utilization Ratio

Maximum Reactions

Note: Only non-zero reactions are listed.

Load Combination	X-X Axis Reaction k @ Base	@ Top	Y-Y Axis Reaction @ Base	@ Top	Axial Reaction @ Base	Mx - End Moments k-ft @ Base	@ Top	My - End Moments @ Base	@ Top

Maximum Moment Reactions

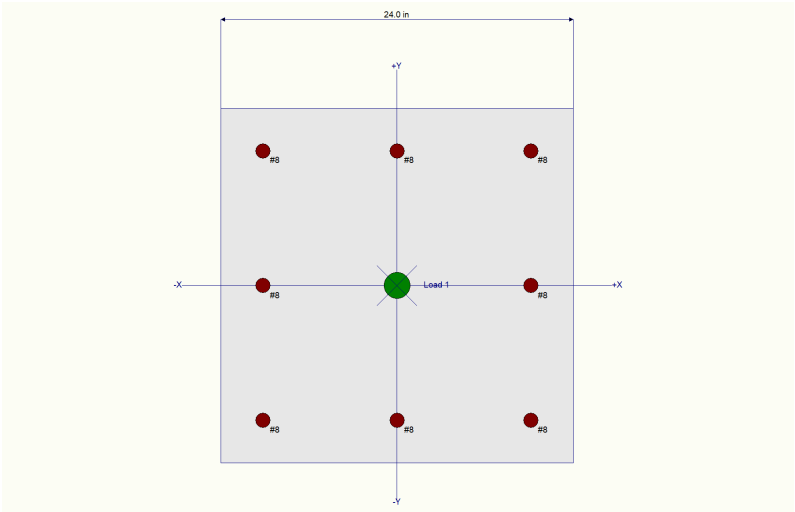
Note: Only non-zero reactions are listed.

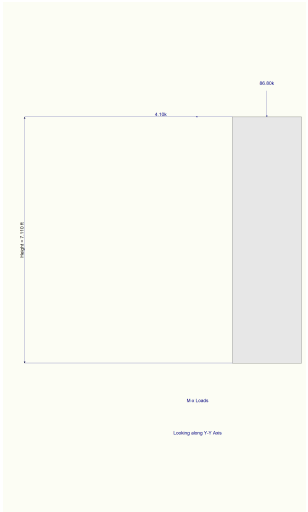
Load Combination	Moment About X-X Axis @ Base	@ Top	Moment About Y-Y Axis @ Base	@ Top

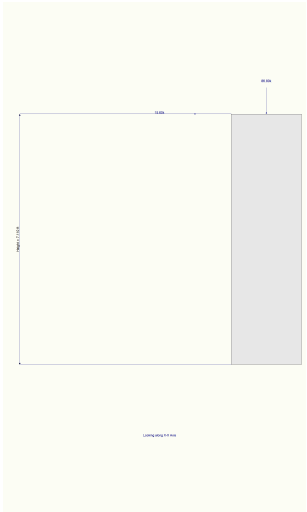
Maximum Deflections for Load Combinations

Load Combination	Max. X-X Deflection	Distance	Max. Y-Y Deflection	Distance

Sketches







Project Title: Eli Lilly K215 Unloading Rack
Engineer:
Project ID: 25I0007
Project Descr:

Concrete Beam

Project File: CAL_20251010_LillyUnloadingRackCalculations.ec6

LIC# : KW-06015723, Build:20.26.02.06 HANSON ENGINEERS, INC. (c) ENERCALC, LLC 1982-2026

DESCRIPTION: Pile Cap at Lateral Braces

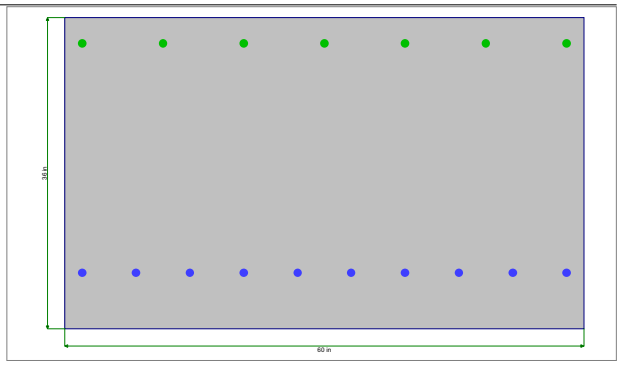
Note - ACI 318-11 was not available for code checks in Enercalc, so ACI 318-14 was used instead. There are no significant changes betwen the editions that would affect the design of these elements.

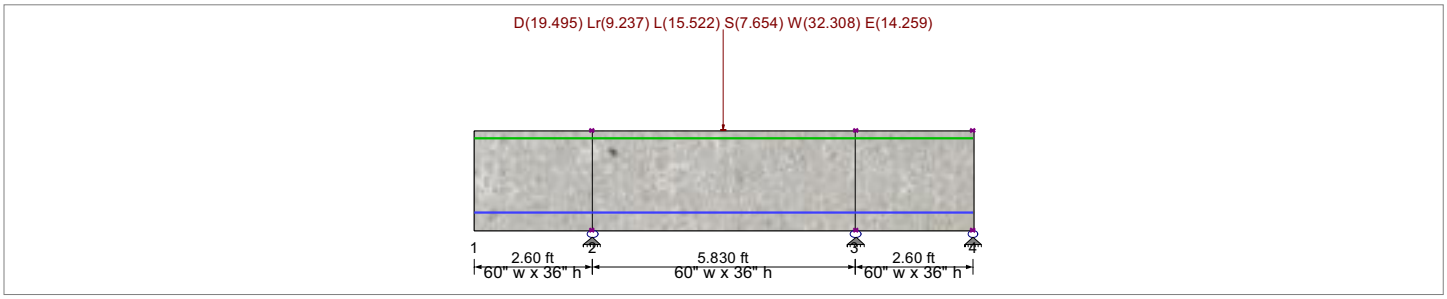
Code References

Governing Code : IBC 2015, CBC 2016
Referenced Design Standard(s) : ACI 318-14
Load Combination Set : ASCE 7-10

General Information

f'c	=	5.0 ksi	ϕ Phi Values	Flexure :	0.90
fr =7.5 * λ * f'c$^{1/2}$	=	530.33 psi		Shear :	0.750
ψ Density	=	145.0 pcf	β_1	=	0.80
λ LtWt Factor	=	1.0			
Elastic Modulus	=	3,122.0 ksi	Fy - Stirrups		40.0 ksi
fy - Main Rebar	=	60.0	E - Stirrups	=	29,000.0 ksi
E - Main Rebar	=	29,000.0 ksi	Stirrup Bar Size #		4
			Number of Resisting Legs Per Stirrup	=	2





Cross Section & Reinforcing Details

Rectangular Section, Width = 60.0 in, Height = 36.0 in

Span #1 Reinforcing....
10-#8 at 6.50 in from Bottom, from 0.0 to 2.60 ft in this span
7-#8 at 3.0 in from Top, from 0.0 to 2.60 ft in this span

Span #2 Reinforcing....
10-#8 at 6.50 in from Bottom, from 0.0 to 5.830 ft in this span
7-#8 at 3.0 in from Top, from 0.0 to 5.830 ft in this span

Span #3 Reinforcing....
10-#8 at 6.50 in from Bottom, from 0.0 to 2.60 ft in this span
7-#8 at 3.0 in from Top, from 0.0 to 2.60 ft in this span

Load for Span Number 2
Point Load : D = 19.495, Lr = 9.237, L = 15.522, S = 7.654, W = 32.308, E = 14.259 k @ 2.915 ft, (Reactions from Pier)

DESIGN SUMMARY

Design OK

Maximum Bending Stress Ratio =	**0.072** : 1	Design Limits	
Section used for this span	**Typical Section**	Tensile Strain >= Min Strain Limit	(Section 9.3.3.1)
Mu : Applied	73.495 k-ft	0.02245 in / in >=	0.00400 in / in OK
Mn * Phi : Allowable	1,020.63 k-ft	As Provided >= As,min	(Section 9.6.1.2)
Location of maximum on span	2.915 ft	7.900 in^2 >=	0.878 in^2 OK
Span # where maximum occurs	Span # 2		

Maximum Deflection

Max Downward Transient Deflection	0.000 in	Ratio =	0 <360.0	
Max Upward Transient Deflection	0.000 in	Ratio =	0 <360.0	
Max Downward Total Deflection	0.000 in	Ratio =	0 <180.0	
Max Upward Total Deflection	0.000 in	Ratio =	0 <180.0	Span: 3 : +D+0.750Lr+0.750L+0.450W

Project Title: Eli Lilly K215 Unloading Rack
Engineer:
Project ID: 25I0007
Project Descr:

Concrete Beam

Project File: CAL_20251010_LillyUnloadingRackCalculations.ec6

LIC# : KW-06015723, Build:20.26.02.06 HANSON ENGINEERS, INC. (c) ENERCALC, LLC 1982-2026

DESCRIPTION: Pile Cap at Lateral Braces

Note - ACI 318-11 was not available for code checks in Enercalc, so ACI 318-14 was used instead. There are no significant changes betwen the editions that would affect the design of these elements.

Cross Section Strength & Inertia

Top & Bottom references are for tension side of sect

Cross Section	Bar Layout Description	Phi*Mn (k-ft) Bottom	Phi*Mn (k-ft) Top	Moment of Inertia (in^4) I gross	Moment of Inertia (in^4) Icr - Bottom	Moment of Inertia (in^4) Icr - Top
Section 1	10- #8 @ d=29.5",7- #8 @ d=3",	1,020.63	953.64	233280.00	44,743.26	41,548.26
Section 2	10- #8 @ d=29.5",7- #8 @ d=3",	1,020.63	953.64	233280.00	44,743.26	41,548.26
Section 3	10- #8 @ d=29.5",7- #8 @ d=3",	1,020.63	953.64	233280.00	44,743.26	41,548.26

Vertical Reactions

Support notation : Far left is #1

Load Combination	Support 1	Support 2	Support 3	Support 4
Max Upward from all Load Conditions	19.480	48.417	2.235	
Max Upward from Load Combinations	19.480	48.417	2.235	
Max Downward from all Load Conditions (Resisting Uplift)	-2.847	-7.076	-15.295	
Max Downward from Load Combinations (Resisting Uplift)	-2.847	-7.076	-15.295	
+D+L	12.968	32.231	-10.182	
+D+Lr	10.640	26.446	-8.354	
+D+S	10.054	24.989	-7.894	
+D+0.750Lr+0.750L	14.096	35.036	-11.068	
+D+0.750L+0.750S	13.657	33.943	-10.722	
+D+0.60W	14.398	35.786	-11.305	
+D-0.60W	0.041	0.101	-0.032	
+D+0.70E	10.916	27.131	-8.571	
+D-0.70E	3.523	8.757	-2.766	
+D+0.750Lr+0.750L+0.450W	19.480	48.417	-15.295	
+D+0.750Lr+0.750L-0.450W	8.712	21.654	-6.840	
+D+0.750L+0.750S+0.450W	19.041	47.325	-14.950	
+D+0.750L+0.750S-0.450W	8.273	20.561	-6.495	
+D+0.750L+0.750S+0.5250E	16.429	40.833	-12.899	
+D+0.750L+0.750S-0.5250E	10.884	27.052	-8.546	
+0.60D+0.60W	11.511	28.609	-9.037	
+0.60D-0.60W	-2.847	-7.076	2.235	
+0.60D+0.70E	8.028	19.953	-6.303	
+0.60D-0.70E	0.635	1.579	-0.499	

Shear Stirrup Requirements

Entire Beam Span Length : Vu < Phi*Vc / 2, Req'd Vs = Not Reqd per 9.6.3.1, Stirrups are not required.

Overall Maximum Deflections

Span	Load Combination	Max. "-" Defl (in)	Location in Span (ft)	Load Combination	Max. "+" Defl (in)	Location in Span (ft)
1	+D+0.750Lr+0.750L+0.450W	0.0000	2.717	+D+0.750Lr+0.750L+0.450W	-0.0005	0.000
2	+D+0.750Lr+0.750L+0.450W	0.0003	2.682	+D+0.750Lr+0.750L+0.450W	-0.0000	5.882
3		0.0000	2.682	+D+0.750Lr+0.750L+0.450W	-0.0000	1.092

Project Title: Eli Lilly K215 Unloading Rack
Engineer:
Project ID: 25I0007
Project Descr:

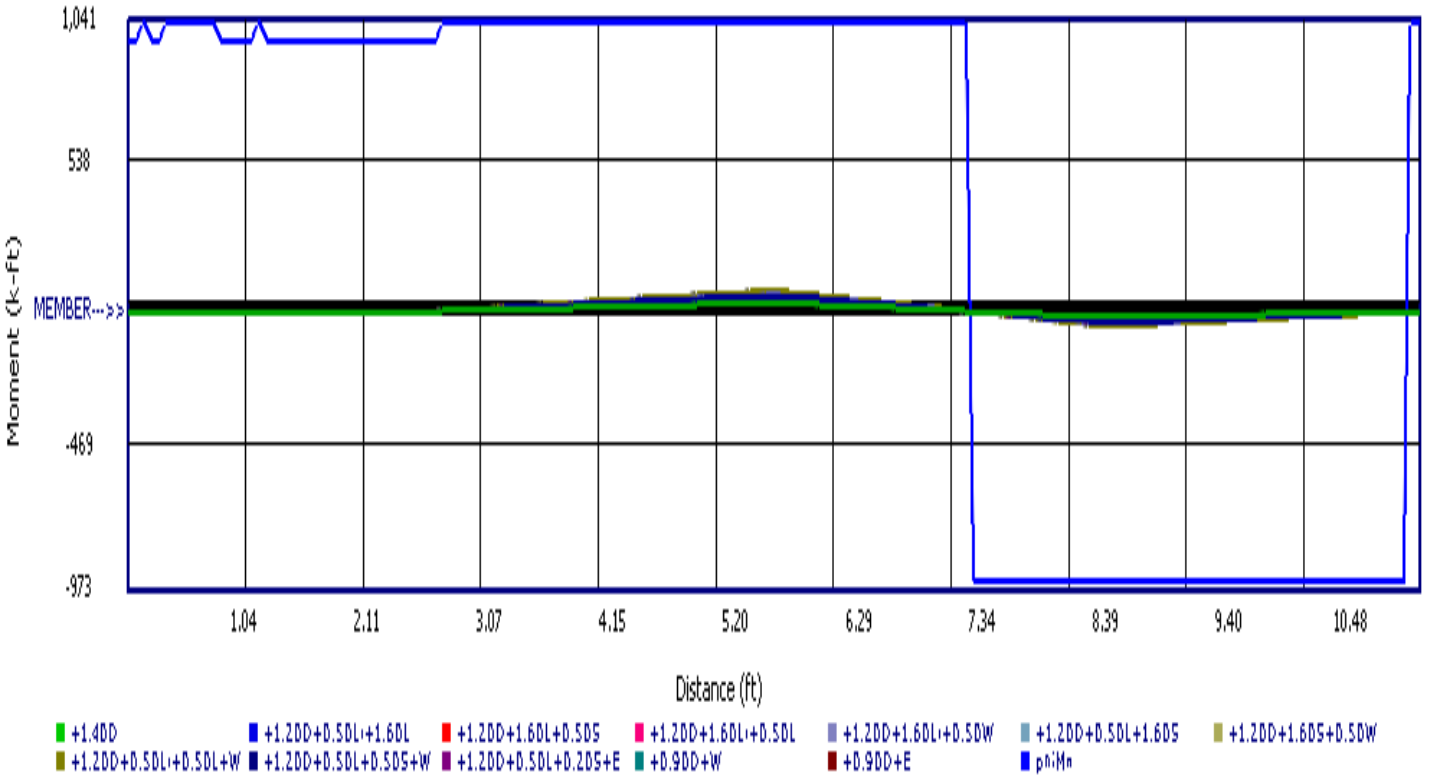
Concrete Beam

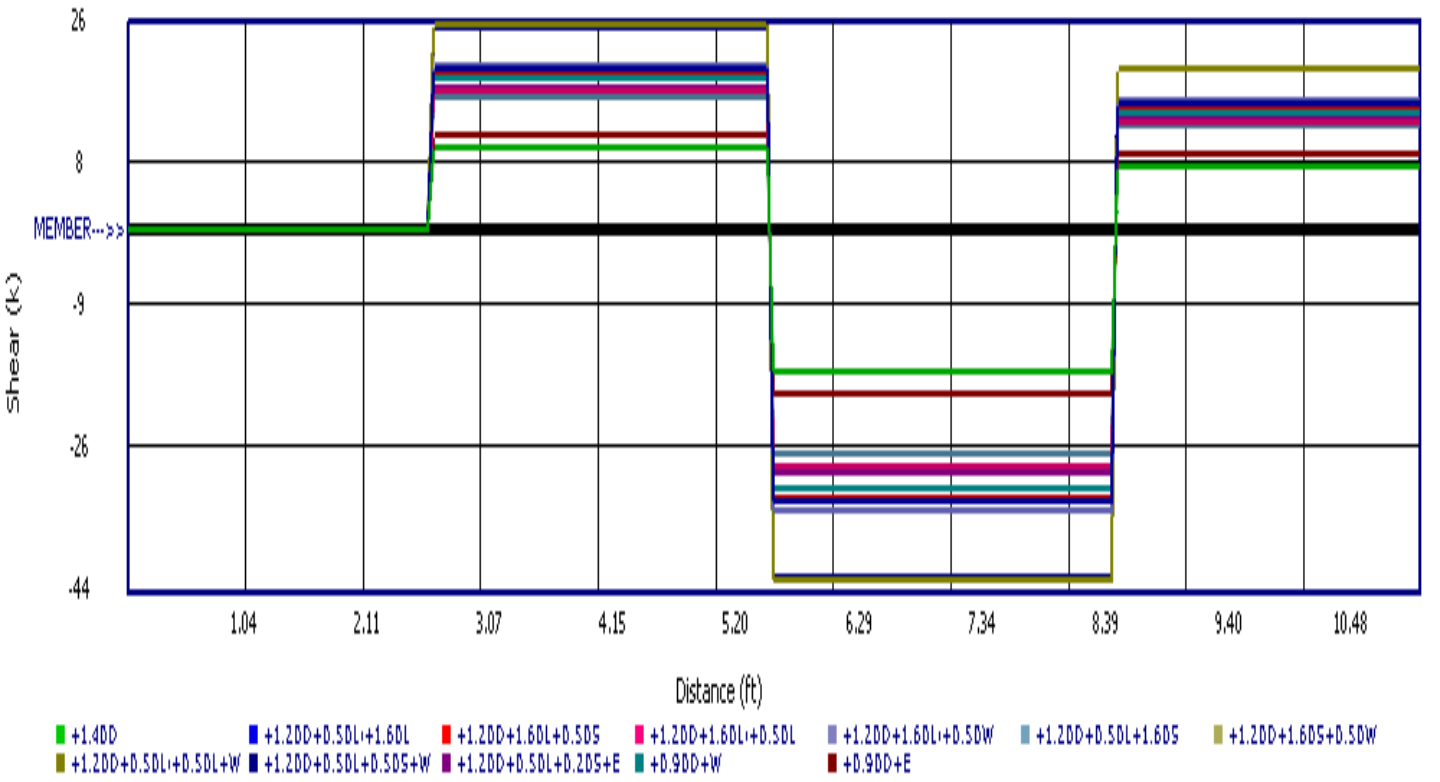
Project File: CAL_20251010_LillyUnloadingRackCalculations.ec6

LIC# : KW-06015723, Build:20.26.02.06 HANSON ENGINEERS, INC. (c) ENERCALC, LLC 1982-2026

DESCRIPTION: Pile Cap at Lateral Braces

Note - ACI 318-11 was not available for code checks in Enercalc, so ACI 318-14 was used instead. There are no significant changes betwen the editions that would affect the design of these elements.





Project Title: Eli Lilly K215 Unloading Rack
Engineer:
Project ID: 25I0007
Project Descr:

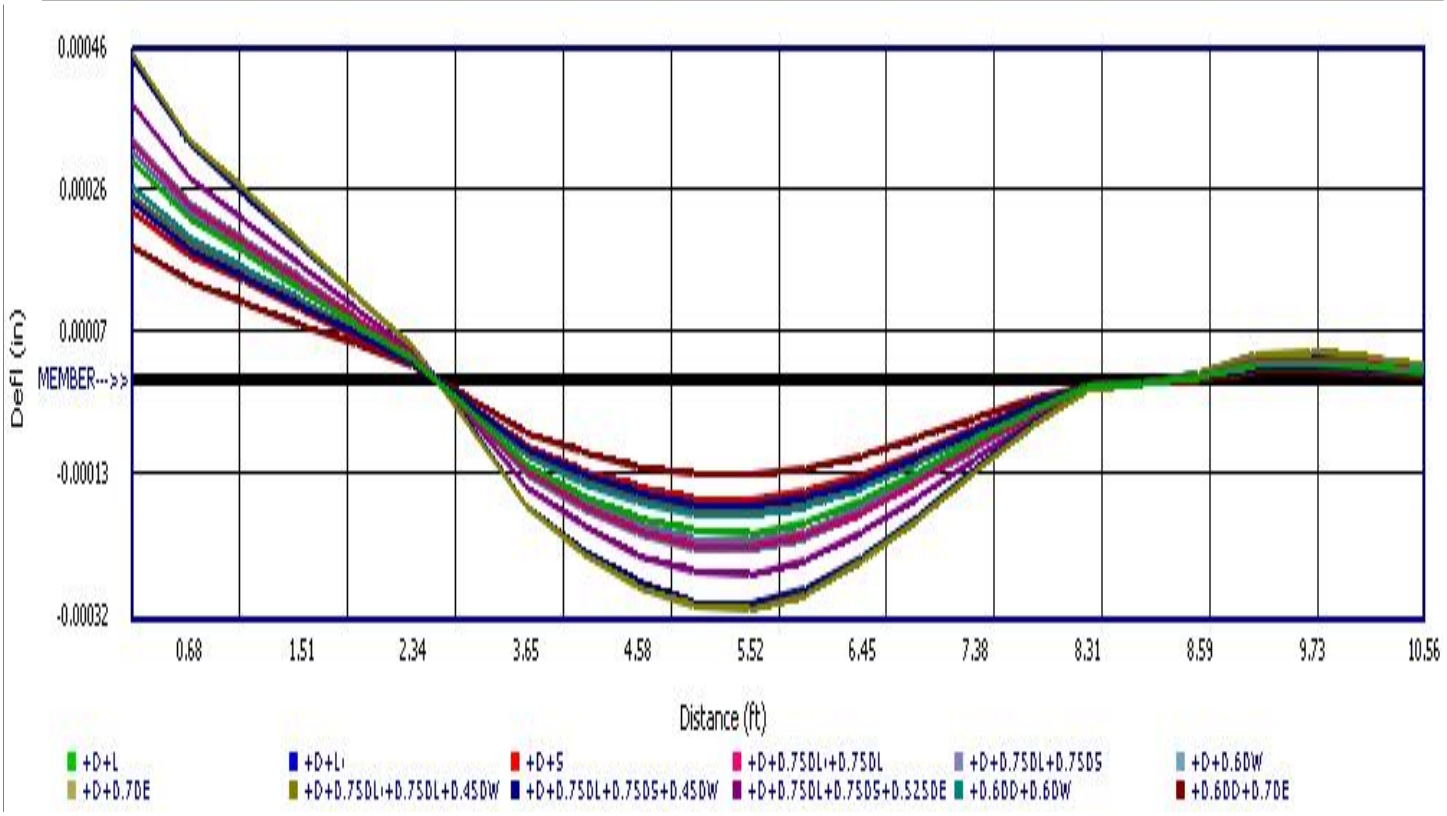
Concrete Beam

Project File: CAL_20251010_LillyUnloadingRackCalculations.ec6

LIC# : KW-06015723, Build:20.26.02.06 HANSON ENGINEERS, INC. (c) ENERCALC, LLC 1982-2026

DESCRIPTION: Pile Cap at Lateral Braces

Note - ACI 318-11 was not available for code checks in Enercalc, so ACI 318-14 was used instead. There are no significant changes betwen the editions that would affect the design of these elements.



Project Title: Eli Lilly K215 Unloading Rack
Engineer:
Project ID: 25I0007
Project Descr:

Concrete Column

Project File: CAL_20251010_LillyUnloadingRackCalculations.ec6

LIC# : KW-06015723, Build:20.26.02.06 HANSON ENGINEERS, INC. (c) ENERCALC, LLC 1982-2026

DESCRIPTION: Tie Beam at Lateral Braces

Note - ACI 318-11 was not available for code checks in Enercalc, so ACI 318-14 was used instead. There are no significant changes betwen the editions that would affect the design of these elements.

Code References

Governing Code : IBC 2015, CBC 2016
Referenced Design Standard(s) : ACI 318-14
Load Combinations Used : ASCE 7-10

General Information

f'c : Concrete 28 day streng	=	5.0 ksi
E =	=	3,122.0 ksi
Density	=	150.0 pcf
β	=	0.0
fy - Main Rebar	=	60.0 ksi
E - Main Rebar	=	29,000.0 ksi
Allow. Reinforcing Limits		*ASTM A615 Bars Used*
Min. Reinf.	=	1.0 %
Max. Reinf.	=	8.0 %

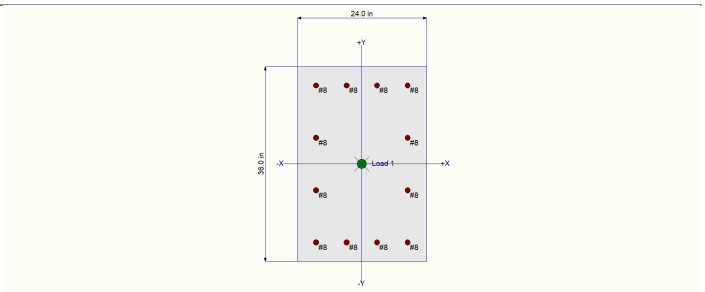
Overall Column Height = 10.0 ft
End Fixity Top & Bottom Pinned

Brace condition for deflection (buckling) along colun
X-X (width) axis :
Unbraced Length for buckling ABOUT X-X Axis = 10.0 ft, K = 1.0
Y-Y (depth) axis :
Unbraced Length for buckling ABOUT Y-Y Axis = 10.0 ft, K = 1.0

Column Cross Section

Column Dimensions : 36.0in high x 24.0in Wide, Column Edge to Rebar Edge Cover = 3.0in

Column Reinforcing : 4 - #8 bars @ corners,, 2 - #8 bars top & bottom between corner bars, 2 - #8 bars left & right between corner



Applied Loads

Entered loads are factored per load combinations specified by user.

Column self weight not internally calculated
AXIAL LOADS . . .
Axial Load at 10.0 ft above base, W = 1.747, E = 0.790 k

DESIGN SUMMARY

Load Combination 0.0
Location of max.above base 0.0 ft

Maximum Stress Ratio **0.0 : 1**
Ratio = (Pu^2+Mu^2)^.5 / (PhiPn^2+PhiMn^2)^.5

Pu =	0.0 k	φ * Pn =	0.0 k
Mu-x =	0.0 k-ft	φ * Mn-x =	0.0 k-ft
Mu-y =	0.0 k-ft	φ* Mn-y =	0.0 k-ft
Mu Angle =	deg	φ =	
Mu at Angle =	k-ft	φMn at Angle =	k-ft

Pn & Mn values located at Pu-Mu vector intersection with capacity curv

Column Capacities . .

Pnmax : Nominal Max. Compressive Axial Capaci	0.0 k
Pnmin : Nominal Min. Tension Axial Capacity	k
φ Pn, max : Usable Compressive Axial Capacity	0.0 k
φ Pn, min : Usable Tension Axial Capacity	k

Maximum SERVICE Load Reactions .

Top along Y-Y	0.0 k	Bottom along Y-Y	0.0 k
Top along X-X	0.0 k	Bottom along X-X	0.0 k

Maximum SERVICE Load Deflections . .

Along Y-Y 0.0 in at 0.0 ft above base
for load combination : 0.0
Along X-X 0.0 in at 0.0 ft above base
for load combination : 0.0

General Section Informatio β = 0.0 θ = 0.0
ρ : % Reinforcing 0.0 % Rebar < Min of 1.0 %
Reinforcing Area 0.0 in^2
Concrete Area 0.0 in^2

Governing Load Combination Results

Governing Factored Load Combination	Moment X-X	Moment Y-Y	Dist. from base ft	Axial Load Pu k	φ * Pn	δx	δx * Mux	δy	δy * Muy	Bending Analysis k-ft Alpha (deg)	δ Mu	φ Mn	Utilization Ratio

Project Title: Eli Lilly K215 Unloading Rack
Engineer:
Project ID: 25I0007
Project Descr:

Concrete Column

Project File: CAL_20251010_LillyUnloadingRackCalculations.ec6

LIC# : KW-06015723, Build:20.26.02.06 HANSON ENGINEERS, INC. (c) ENERCALC, LLC 1982-2026

DESCRIPTION: Tie Beam at Lateral Braces

Note - ACI 318-11 was not available for code checks in Enercalc, so ACI 318-14 was used instead. There are no significant changes betwen the editions that would affect the design of these elements.

Maximum Reactions

Note: Only non-zero reactions are listed.

Load Combination	X-X Axis Reaction k @ Base	@ Top	Y-Y Axis Reaction @ Base	@ Top	Axial Reaction @ Base	Mx - End Moments k-ft @ Base	@ Top	My - End Moments @ Base	@ Top

Maximum Moment Reactions

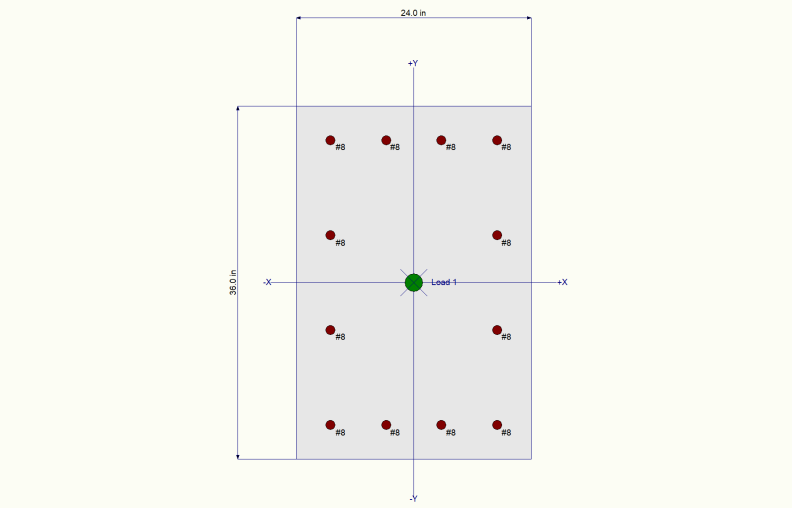
Note: Only non-zero reactions are listed.

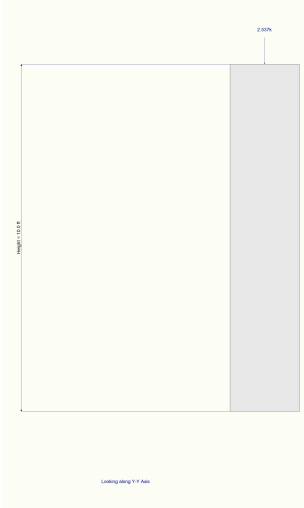
Load Combination	Moment About X-X Axis @ Base	@ Top	Moment About Y-Y Axis @ Base	@ Top

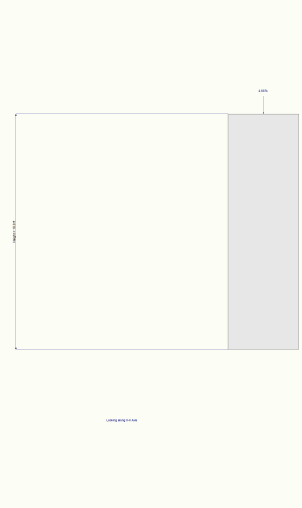
Maximum Deflections for Load Combinations

Load Combination	Max. X-X Deflection	Distance	Max. Y-Y Deflection	Distance

Sketches







Project Title: Eli Lilly K215 Unloading Rack
Engineer:
Project ID: 25I0007
Project Descr:

Steel Beam

Project File: CAL_20251010_LillyUnloadingRackCalculations.ec6

LIC# : KW-06015723, Build:20.26.02.06 HANSON ENGINEERS, INC. (c) ENERCALC, LLC 1982-2026

DESCRIPTION: Typical Guardrail Rail

Note - ACI 318-11 was not available for code checks in Enercalc, so ACI 318-14 was used instead. There are no significant changes betwen the editions that would affect the design of these elements.

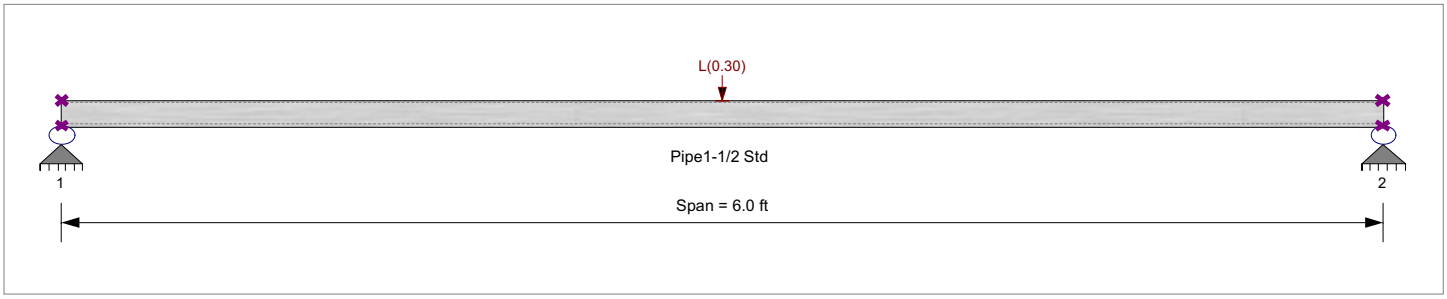
Code References

Governing Code : IBC 2015, CBC 2016
Referenced Design Standard(s) : AISC 360-10
Load Combination Set : ASCE 7-10

Material Properties

Analysis Method : Load Resistance Factor Design
Beam Bracing : Completely Unbraced
Bending Axis : Major Axis Bending

Fy : Steel Yield : 35.0 ksi
E: Modulus : 29,000.0 ksi



Applied Loads

Service loads entered. Load Factors will be applied for calculations.

Beam self weight NOT internally calculated and added
Load(s) for Span Number 1
Point Load : L = 0.30 k @ 3.0 ft, (Live Load)

DESIGN SUMMARY

Design OK

Maximum Bending Stress Ratio =	0.652 : 1	Maximum Shear Stress Ratio =	0.034 : 1
Section used for this span	Pipe1-1/2 Std	Section used for this span	Pipe1-1/2 Std
Mu : Applied	0.720 k-ft	Vu : Applied	0.240 k
Mn * Phi : Allowable	1.105 k-ft	Vn * Phi : Allowable	7.088 k
Load Combination	+1.60L	Load Combination	+1.60L
		Location of maximum on span	0.000 ft
Span # where maximum occurs	Span # 1	Span # where maximum occurs	Span # 1

Maximum Deflection

Max Downward Transient Deflection	0 in	Ratio =	0 <360	n/a
Max Upward Transient Deflection	0 in	Ratio =	0 <360	n/a
Max Downward Total Deflection	0.276 in	Ratio =	261 >=180	Span: 1 : L Only
Max Upward Total Deflection	0 in	Ratio =	0 <180	n/a

Maximum Forces & Stresses for Load Combinations

Load Combination / Segment Length	Span #	Max Stress Ratios M	Max Stress Ratios V	Summary of Moment Values max Mu +	max Mu -	Mu Max	Mnx	Phi*Mnx	Cb	Rm	Summary of Shear Values VuMax	Vnx	Phi*Vnx
Dsgn. L = 6.00 ft	1		0.000				1.23	1.11	1.00		-0.00	7.88	7.09
+1.60L													
Dsgn. L = 6.00 ft	1	0.652	0.034	0.72		0.72	1.23	1.11	1.32	1.00	0.24	7.88	7.09
+0.50L													
Dsgn. L = 6.00 ft	1	0.204	0.011	0.22		0.22	1.23	1.11	1.32	1.00	0.08	7.88	7.09

Project Title: Eli Lilly K215 Unloading Rack
Engineer:
Project ID: 25I0007
Project Descr:

Steel Beam

Project File: CAL_20251010_LillyUnloadingRackCalculations.ec6

LIC# : KW-06015723, Build:20.26.02.06 HANSON ENGINEERS, INC. (c) ENERCALC, LLC 1982-2026

DESCRIPTION: Typical Guardrail Rail

Note - ACI 318-11 was not available for code checks in Enercalc, so ACI 318-14 was used instead. There are no significant changes betwen the editions that would affect the design of these elements.

Steel Section Properties : Pipe1-1/2 Std

Depth	=	1.900 in	I xx	=	0.29 in^4	J	=	0.586 in^4
			S xx		0.31 in^3			
Diameter	=	1.900 in	R xx	=	0.626 in			
Wall Thick	=	0.145 in	Zx	=	0.421 in^3			
Area	=	0.750 in^2	I yy	=	0.293 in^4			
Weight	=	2.742 plf	S yy	=	0.309 in^3			
			R yy	=	0.626 in			
Ycg	=	0.950 in						

Project Title: Eli Lilly K215 Unloading Rack
Engineer:
Project ID: 25I0007
Project Descr:

Steel Column

Project File: CAL_20251010_LillyUnloadingRackCalculations.ec6

LIC# : KW-06015723, Build:20.26.02.06 HANSON ENGINEERS, INC. (c) ENERCALC, LLC 1982-2026

DESCRIPTION: Typical Guardrail Post

Note - ACI 318-11 was not available for code checks in Enercalc, so ACI 318-14 was used instead. There are no significant changes betwen the editions that would affect the design of these elements.

Code References

Governing Code : IBC 2015, CBC 2016
Referenced Design Standard(s) : AISC 360-10
Load Combinations Used : ASCE 7-10

General Information

Steel Section Name : **Pipe1-1/2 x-Strong**
Analysis Method : Load Resistance Factor
Steel Stress Grade
Fy : Steel Yield 35.0 ksi
E : Elastic Bending Modulus 29,000.0 ksi

Overall Column Height 3.50 ft
Top & Bottom Fixity Top Free, Bottom Fixed
Brace condition :
Unbraced Length for buckling ABOUT X-X Axis = 3.50 ft, K = 2.1
Unbraced Length for buckling ABOUT Y-Y Axis = 3.50 ft, K = 2.1

Applied Loads

Service loads entered. Load Factors will be applied for calculations.

Column self weight included : 12.796 lbs * Dead Load Factor
BENDING LOADS . . .
Wind: Lat. Uniform Load creating Mx-x, W = 0.0060 k/ft
300 lb conc. load spread across two posts: Lat. Point Load at 3.50 ft creating Mx-x, L = 0.150 k

DESIGN SUMMARY

Bending & Shear Check Results

PASS Max. Axial+Bending Stress Ratio = **0.5836** : 1
Load Combination +1.20D+1.60L
Location of max.above base 0.0 ft
At maximum location values are . . .
Pu 0.01536 k
0.9 * Pn 10.806 k
Mu-x -0.840 k-ft
0.9 * Mn-x : 1.441 k-ft
Mu-y 0.0 k-ft
0.9 * Mn-y : 1.441 k-ft

PASS Maximum Shear Stress Ratio **0.03048** : 1
Load Combination +1.20D+1.60L
Location of max.above base 0.0 ft
At maximum location values are . . .
Vu : Applied 0.240 k
Vn * Phi : Allowable 7.875 k

Maximum Load Reactions . .
Top along X-X 0.0 k
Bottom along X-X 0.0 k
Top along Y-Y 0.0 k
Bottom along Y-Y 0.150 k

Maximum Load Deflections . . .
Along Y-Y 0.3416 in at 3.50 ft above base
for load combination : +D+L
Along X-X 0.0 in at 0.0 ft above base
for load combination :

Load Combination Results

Load Combination	Maximum Axial + Bending Stress Ratios: Stress Ratio	Status	Location	Cbx	Cby	KxLx/Rx	KyLy/Ry	Maximum Shear Ratios: Stress Ratio	Status	Location
+1.40D	0.002	PASS	0.00 ft	1.67	1.00	144.59	0.00	0.000	PASS	0.00 ft
+1.20D+1.60L	0.584	PASS	0.00 ft	1.67	1.00	144.59	0.00	0.030	PASS	0.00 ft
+1.20D+0.50L	0.183	PASS	0.00 ft	1.67	1.00	144.59	0.00	0.010	PASS	0.00 ft
+1.20D+0.50W	0.013	PASS	0.00 ft	1.67	1.00	144.59	0.00	0.001	PASS	0.00 ft
+1.20D-0.50W	0.013	PASS	0.00 ft	1.67	1.00	144.59	0.00	0.001	PASS	0.00 ft
+1.20D+0.50L+W	0.208	PASS	0.00 ft	1.67	1.00	144.59	0.00	0.012	PASS	0.00 ft
+1.20D+0.50L-W	0.157	PASS	0.00 ft	1.67	1.00	144.59	0.00	0.010	PASS	3.50 ft
+0.90D+W	0.026	PASS	0.00 ft	1.67	1.00	144.59	0.00	0.003	PASS	0.00 ft
+0.90D-W	0.026	PASS	0.00 ft	1.67	1.00	144.59	0.00	0.003	PASS	0.00 ft
+0.90D	0.001	PASS	0.00 ft	1.67	1.00	144.59	0.00	0.000	PASS	0.00 ft

Maximum Reactions

Note: Only non-zero reactions are listed.

Load Combination	Axial (k) @ Base	Rx (k) @ Base	Rx (k) @ Top	Ry (k) @ Base	Ry (k) @ Top	Mx (k-ft) @ Base	Mx (k-ft) @ Top	My (k-ft) @ Base	My (k-ft) @ Top
D Only	0.013								
+D+L	0.013			-0.150		0.525			
+D+0.750L	0.013			-0.113		0.394			
+D+0.60W	0.013			-0.013		0.022			

Project Title: Eli Lilly K215 Unloading Rack
Engineer:
Project ID: 25I0007
Project Descr:

Steel Column

Project File: CAL_20251010_LillyUnloadingRackCalculations.ec6

LIC# : KW-06015723, Build:20.26.02.06 HANSON ENGINEERS, INC. (c) ENERCALC, LLC 1982-2026

DESCRIPTION: Typical Guardrail Post

Note - ACI 318-11 was not available for code checks in Enercalc, so ACI 318-14 was used instead. There are no significant changes betwen the editions that would affect the design of these elements.

Maximum Reactions

Note: Only non-zero reactions are listed.

Load Combination	Axial (k) @ Base	Rx (k) @ Base	Rx (k) @ Top	Ry (k) @ Base	Ry (k) @ Top	Mx (k-ft) @ Base	Mx (k-ft) @ Top	My (k-ft) @ Base	My (k-ft) @ Top
+D+0.750L+0.450W	0.013			-0.122		0.410			
+0.60D+0.60W	0.008			-0.013		0.022			
+0.60D	0.008								
L Only				-0.150		0.525			
W Only				-0.021		0.037			

Extreme Reactions

Item	Extreme Value	Axial (k) @ Base	Rx (k) @ Base	Rx (k) @ Top	Ry (k) @ Base	Ry (k) @ Top	Mx (k-ft) @ Base	Mx (k-ft) @ Top	My (k-ft) @ Base	My (k-ft) @ Top
Axial, Base Max		0.013								
Axial, Base Min										
Rx, Base Max										
Rx, Base Min										
Ry, Base Max										
Ry, Base Min					-0.150					
Mx, Base Max							0.525			
Mx, Base Min										
My, Base Max										
My, Base Min										

Maximum Deflections for Load Combinations

Load Combination	Max. Deflection in X dir	Distance	Max. Deflection in Y dir	Distance
D Only	0.0000 in	0.000 ft	0.000 in	0.000 ft
+D+L	0.0000 in	0.000 ft	0.342 in	3.500 ft
+D+0.750L	0.0000 in	0.000 ft	0.256 in	3.500 ft
+D+0.60W	0.0000 in	0.000 ft	0.011 in	3.500 ft
+D+0.750L+0.450W	0.0000 in	0.000 ft	0.264 in	3.500 ft
+0.60D+0.60W	0.0000 in	0.000 ft	0.011 in	3.500 ft
+0.60D	0.0000 in	0.000 ft	0.000 in	0.000 ft
L Only	0.0000 in	0.000 ft	0.342 in	3.500 ft
W Only	0.0000 in	0.000 ft	0.018 in	3.477 ft

Steel Section Properties : Pipe1-1/2 x-Strong

Depth	=	1.900 in	I xx	=	0.37 in^4	J	=	0.744 in^4
			S xx	=	0.39 in^3			
Diameter	=	1.900 in	R xx	=	0.610 in			
Wall Thick	=	0.200 in	Zx	=	0.549 in^3			
Area	=	1.000 in^2	I yy	=	0.372 in^4			
Weight	=	3.656 plf	S yy	=	0.392 in^3			
			R yy	=	0.610 in			
Ycg	=	0.000 in						

Project Title: Eli Lilly K215 Unloading Rack
Engineer:
Project ID: 25I0007
Project Descr:

Steel Column

Project File: CAL_20251010_LillyUnloadingRackCalculations.ec6

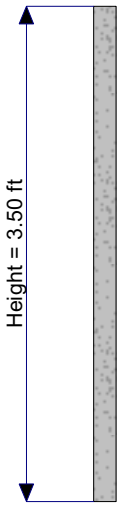
LIC# : KW-06015723, Build:20.26.02.06 HANSON ENGINEERS, INC. (c) ENERCALC, LLC 1982-2026

DESCRIPTION: Typical Guardrail Post

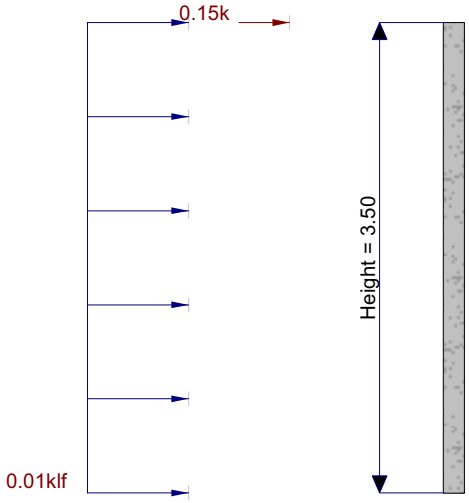
Note - ACI 318-11 was not available for code checks in Enercalc, so ACI 318-14 was used instead. There are no significant changes betwen the editions that would affect the design of these elements.

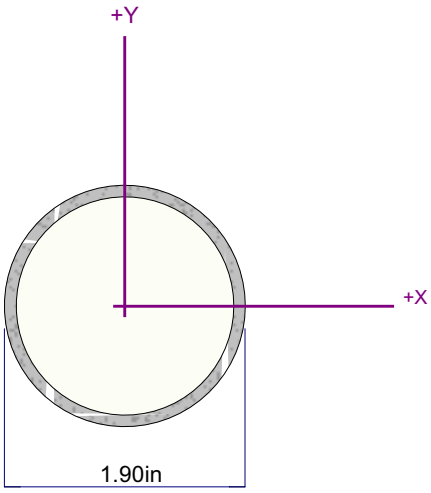
Sketches

My Loads - Looking along Y-Y Axis



Mx Loads - Looking along X-X Axis





Eli Lilly Unloading Rack K215
Platform Guardrail OSHA Live Load Check

$kip := 1000\ lb$

OSHA concentrated LL: $LL_conc := 300\ lb$

Assume concentrated LL is resisted by two adjacent guardrail posts: $LL_conc_single_post := \frac{LL_conc}{2} = 150\ lb$

$post_height := 3.5\ ft$

$LL_factor := 1.6$

$M_u := post_height \cdot LL_conc_single_post \cdot LL_factor = 840\ lb \cdot ft$ at base of post

$d_{angle_toe_to_bolt_center} := 2\ in$ $\#_{bolts} := 2$ per post

$$P_{bolt} := \frac{M_u}{d_{angle_toe_to_bolt_center} \cdot \#_{bolts}} = 2520\ lb$$

$F_{nt_A325} := 90\ ksi$ $d_{bolts} := 0.75\ in$

$$\phi R_{n_bolt} := 0.75 \cdot F_{nt_A325} \cdot d_{bolts}^{\ 2} \cdot \frac{\pi}{4} = 29.821\ kip$$

Therefore bolt strength OK

STRUCTURAL MTO

Estimate of Structural Steel Column Material Takeoff

Type	Count	Length	Volume	Weight (lbs)	Weight (tons)
W12X40	26	640 ft	51 CF	25,100 lb	12.5 Tons
SUBTOTAL		640 ft	51 CF	25,100 lb	12.5 Tons

Estimate of Concrete Pier Material Takeoff

Type	Count	Length	Volume
24" SQ. CONCRETE PIER	26	185 ft	27 CY
SUBTOTAL		185 ft	27 CY

Estimate of Concrete Foundation Material Takeoff

Type	Count	Volume
DOUBLE PIER PILE CAP	2	18 CY
SINGLE PIER PILE CAP	22	132 CY
STAIR PAD	1	1 CY
TIE BEAM	12	23 CY
SUBTOTAL: 37		174 CY

Estimate of Concrete Slab Material Takeoff

Type	Count	Area	Volume
Concrete Slab - 8"	5	5,300 SF	131 CY
SUBTOTAL		5,300 SF	131 CY

Estimate of Structural Steel Framing Material Takeoff

Type	Count	Length	Volume	Weight (lbs)	Weight (tons)
C10X20	1	14.3 ft	0.6 CF	310 lbf	0.2 Tons
C12X20.7	1	4.7 ft	0.2 CF	100 lbf	0.0 Tons
W6X15	1	11.3 ft	0.3 CF	160 lbf	0.1 Tons
W8X24	2	6.0 ft	0.3 CF	140 lbf	0.1 Tons
W10X22	93	1,167.3 ft	48.6 CF	23,830 lbf	11.9 Tons
W12X30	83	1,259.5 ft	73.4 CF	35,970 lbf	18.0 Tons
WT4X6.5	46	947.7 ft	11.6 CF	5,710 lbf	2.9 Tons
WT6X9.5	8	149.4 ft	2.7 CF	1,300 lbf	0.6 Tons
SUBTOTAL: 235		3,560.1 ft	137.8 CF	67,520 lbf	33.8 Tons

Estimate of Cold-Formed Framing Material Takeoff

Type	Count	Length	Volume	Weight (lbs)	Weight (tons)
C8	67	998.2 ft	2.7 CF	1,340 lbf	0.7 Tons
SUBTOTAL: 67		998.2 ft	2.7 CF	1,340 lbf	0.7 Tons

Estimate of Grating Material Takeoff

Type	Area (sf)
1 1/4" Grating	1,890
SUBTOTAL	1,890

Estimate of Railing Material Takeoff

Unit Description	Count	Total Length
Railing	35	380 ft
SUBTOTAL		380 ft

ADDITIONAL ITEMS:	QUANTITY
1. STEEL CONNECTIONS	4,700 LBS
2. METAL ROOF PANELS	3,200 SF
3. STAIRS	2 FLIGHTS
4. LADDERS	2
5. SWING GATES:	7

NOTES:
1. MATERIAL TAKEOFFS DO NOT INCLUDE A CONTINGENCY.
2. ALL STEEL (INCLUDING NON-STRUCTURAL STEEL) TO BE GALV.

V1- VOID - DO NOT RESUBMIT

Authorization to proceed does not relieve Contractor/Supplier of its responsibility or liability under the Contract and or Purchase Order.

By dac54131 at Dec 03, 2025

PRELIMINARY
NOT FOR CONSTRUCTION

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
A		ISSUED FOR REVIEW		10-OCT-2025

REVISIONS

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FLUOR®
EFDEE Engineering PC
PROJECT NO. B1LG
NODE:

REV	DATE	DES	ENG	CHK	APP	PM APP
A	10-OCT-2025	RKN	SJS	FCB	RMF	SS

Lilly

STRUCTURAL MTO

SYSTEM TYPE: NA			EDIFICE/BUILDING: UNLOADING RACK	LEVEL/FLOOR:
SUB DISCIPLINE	LOCATION CODE: LP1	LOCATION/GRID	CCI CODE	PROJECT NAME: LP1-306-1181
CRITICAL QUALITY		ENTERPRISE GROUP: API	DRAWING INTENTION	DRAWING SCALE: NONE
CRITICAL BUSINESS		EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA	MERIDIAN REV: 1
CRITICAL HSE		VENDOR: NA	DOCUMENT ID: LP1Y-215-MTO-K215-01	
NON-CRITICAL		ENGINEERING AGENCY DOC ID: NA		

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Reviewed by Y.D. Byeon_Process

HANSON

Document Title:	K215 PIPING SPECIALTY ITEM LIST
Document Number:	LP1Y-250-LST-K215-01
Security Class:	Restricted

PLEASE SEE FLUOR PIPING MATERIAL ENGINEERING COMMENTS BELOW. RECOMMEND B1 STATUS. - LC, FLUOR, 21APR26

PIPING SPECIALTY ITEM LIST

Lilly - Lebanon Plant 1 (B1LG)

K215 - Lilly Rail Offloading Rack

Hanson Project Number: 25i0007

B1 - DATA COMPLETE - WITH COMMENTS - PROCEED - RESUBMIT
Authorization to proceed does not relieve Contractor/Supplier of its responsibility or liability under the Contract and or Purchase Order.
By: Ricardo DaCosta dac54131 April 22, 2026

A	IFR	MW	QR	SS	4/7/2026
Revision	**Issue Status**	**Originator**	**Reviewer**	**Approver**	**Date**

DISCLAIMER: REFERENCED PCS FILES ARE NOT EXACTLY IDENTICAL TO THOSE ISSUED BY FLUOR PIPING ENGINEERING. THE ONES LISTED AND ATTACHED TO THIS LIST ARE FOR REFERENCE AND MAY DEVIATE SLIGHTLY FROM THE FLUOR ISSUED PCS FILES. (LC, FLUOR, 21APR26)

K215 PIPING SPECIALTY ITEM LIST

Revision	Scope	Hygenic (YES/NO)	Impact (YES/NO)	CWP	Node	P&ID	Item Size	Item Length	Item Tag Number	Client Property Tag	SP Item Type	Service Code	Service Description	Line Number	Pipe Schedule	Item Description	Piping Cutsheet				Elastomer	Internal Surface Finish (uin Max)	Vender Package Number	Vender Package Description	Remarks
0	BY CONTRACTOR - RAIL	NO	YES	TBD	TFS	LP1Y-DI-00000063	3	N/A	LP1Y BCU-0631A1	BY OWNER	BREAKAWAY COUPLING	MTBE	Methyl Tert Butyl Ether	LP1Y 063101	315	BREAKAWAY COUPLING 150# 316 SS RF	PCS-BCU-001	MANNTEK	INDUSTRIAL SAFETY BREAKAWAY	SS	VITON	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	YES	TBD	TFS	LP1Y-DI-00000062	1	N/A	LP1Y CCU-0623N1	BY OWNER	CAMLOCK CONNECTION	DMF	Dimethylformamide	LP1Y 062301	315	HOSE CONNECTION 316 SS SW ADAPTER CAMLOCK WITH DUST CAP	PCS-CCU-029	DIXON	SOCKETWELD ADAPTER WITH DUST CAP	SS	PTFE	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	YES	TBD	TFS	LP1Y-DI-00000062	1	N/A	LP1Y CCU-0621N2	BY OWNER	CAMLOCK CONNECTION	N2	Nitrogen	LP1Y 061035	315	HOSE CONNECTION 316 SS SW ADAPTER CAMLOCK WITH DUST CAP	PCS-CCU-029	DIXON	SOCKETWELD ADAPTER WITH DUST CAP	SS	PTFE	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	YES	TBD	TFS	LP1Y-DI-00000063	0.75	N/A	LP1Y CCU-0631N3	BY OWNER	CAMLOCK CONNECTION	MTBE	Methyl Tert Butyl Ether	LP1Y 063101	315	HOSE CONNECTION 316 SS SW ADAPTER CAMLOCK WITH DUST CAP	PCS-CCU-029	DIXON	SOCKETWELD ADAPTER WITH DUST CAP	SS	PTFE	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	YES	TBD	TFS	LP1Y-DI-00000063	0.75	N/A	LP1Y CCU-0631N4	BY OWNER	CAMLOCK CONNECTION	N2	Nitrogen	LP1Y 061064	315	HOSE CONNECTION 316 SS SW ADAPTER CAMLOCK WITH DUST CAP	PCS-CCU-029	DIXON	SOCKETWELD ADAPTER WITH DUST CAP	SS	PTFE	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	YES	TBD	TFS	LP1Y-DI-00000063	1	N/A	LP1Y CCU-0633N2	BY OWNER	CAMLOCK CONNECTION	ACN	Acetonitrile	LP1Y 063301	315	HOSE CONNECTION 316 SS SW ADAPTER CAMLOCK WITH DUST CAP	PCS-CCU-029	DIXON	SOCKETWELD ADAPTER WITH DUST CAP	SS	PTFE	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	YES	TBD	TFS	LP1Y-DI-00000063	1	N/A	LP1Y CCU-0631N2	BY OWNER	CAMLOCK CONNECTION	N2	Nitrogen	LP1Y 061002	315	HOSE CONNECTION 316 SS SW ADAPTER CAMLOCK WITH DUST CAP	PCS-CCU-029	DIXON	SOCKETWELD ADAPTER WITH DUST CAP	SS	PTFE	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	YES	TBD	TFS	LP1Y-DI-00000064	1	N/A	LP1Y CCU-0641N2	BY OWNER	CAMLOCK CONNECTION	N2	Nitrogen	LP1Y 061003	315	HOSE CONNECTION 316 SS SW ADAPTER CAMLOCK WITH DUST CAP	PCS-CCU-029	DIXON	SOCKETWELD ADAPTER WITH DUST CAP	SS	PTFE	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	YES	TBD	TFS	LP1Y-DI-00000065	1	N/A	LP1Y CCU-0651N2	BY OWNER	CAMLOCK CONNECTION	N2	Nitrogen	LP1Y 061004	315	HOSE CONNECTION 316 SS SW ADAPTER CAMLOCK WITH DUST CAP	PCS-CCU-029	DIXON	SOCKETWELD ADAPTER WITH DUST CAP	SS	PTFE	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	YES	TBD	TFS	LP1Y-DI-00000063	3	N/A	LP1Y CCU-0631A2	BY OWNER	CAMLOCK CONNECTION	MTBE	Methyl Tert Butyl Ether	LP1Y 063101	315	HOSE CONNECTION 316 SS BW COUPLER CAMLOCK WITH DUST PLUG	PCS-CCU-030	PT COUPLING	DWB-COUPLER WITH W- ADAPTER DUST PLUG	SS	PTFE	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	YES	TBD	TFS	LP1Y-DI-00000063	0.5	N/A	LP1Y CCU-0631N1	BY OWNER	CAMLOCK CONNECTION	MTBE	Methyl Tert Butyl Ether	LP1Y 063101	315	HOSE CONNECTION 316 SS SW ADAPTER CAMLOCK WITH DUST CAP	PCS-CCU-032	PT COUPLING	AWS-ADAPTER WITH V- COUPLER DUST CAP	SS	PTFE	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	YES	TBD	TFS	LP1Y-DI-00000063	3	10'-0"	LP1Y FLX-0631A1	BY OWNER	FLEXIBLE HOSE	MTBE	Methyl Tert Butyl Ether	LP1Y 063101	315	FLEXIBLE HOSE 360 PSIG 316 SS DRY DISCONNECT COUPLER PTFE-LINED WHITE ANTISTATIC W DUST PLUG SEL	PCS-FLX-054	AFLEX	BIOFLEX ULTRA	SS	TFE	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	YES	TBD	TFS	LP1Y-DI-00000062	1	50'-0"	LP1Y FLX-0621N1	BY OWNER	FLEXIBLE HOSE	N2	Nitrogen	LP1Y 061035	315	FLEXIBLE HOSE COUPLER CAMLOCK X COUPLER CAMLOCK PFA-LINED WHITE WITH DUST PLUGS	PCS-FLX-056	SWAGELOCK	U SERIES	SS	N/A	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	YES	TBD	TFS	LP1Y-DI-00000062	1	50'-0"	LP1Y FLX-0620N1	BY OWNER	FLEXIBLE HOSE	N2	Nitrogen	LP1Y 061035	315	FLEXIBLE HOSE 2" 150# FLG X COUPLER CAMLOCK PFA-LINED WHITE WITH DUST PLUG	PCS-FLX-056	SWAGELOCK	U SERIES	SS	N/A	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	YES	TBD	TFS	LP1Y-DI-00000063	0.75	50'-0"	LP1Y FLX-0631N2	BY OWNER	FLEXIBLE HOSE	N2	Nitrogen	LP1Y 061064	315	FLEXIBLE HOSE COUPLER CAMLOCK X COUPLER CAMLOCK PFA-LINED WHITE WITH DUST PLUGS	PCS-FLX-056	SWAGELOCK	U SERIES	SS	N/A	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	YES	TBD	TFS	LP1Y-DI-00000063	1	50'-0"	LP1Y FLX-0631N1	BY OWNER	FLEXIBLE HOSE	N2	Nitrogen	LP1Y 061002	315	FLEXIBLE HOSE COUPLER CAMLOCK X COUPLER CAMLOCK PFA-LINED WHITE WITH DUST PLUGS	PCS-FLX-056	SWAGELOCK	U SERIES	SS	N/A	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	YES	TBD	TFS	LP1Y-DI-00000063	1	50'-0"	FLX-0630N1	BY OWNER	FLEXIBLE HOSE	N2	Nitrogen	LP1Y 061006	315	FLEXIBLE HOSE 2" 150# FLG X COUPLER CAMLOCK PFA-LINED WHITE WITH DUST PLUG	PCS-FLX-056	SWAGELOCK	U SERIES	SS	N/A	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	YES	TBD	TFS	LP1Y-DI-00000064	1	50'-0"	LP1Y FLX-0641N1	BY OWNER	FLEXIBLE HOSE	N2	Nitrogen	LP1Y 061003	315	FLEXIBLE HOSE COUPLER CAMLOCK X COUPLER CAMLOCK PFA-LINED WHITE WITH DUST PLUGS	PCS-FLX-056	SWAGELOCK	U SERIES	SS	PTFE	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	YES	TBD	TFS	LP1Y-DI-00000064	1	50'-0"	LP1Y FLX-0640N1	BY OWNER	FLEXIBLE HOSE	N2	Nitrogen	LP1Y 061007	315	FLEXIBLE HOSE 2" 150# FLG X COUPLER CAMLOCK PFA-LINED WHITE WITH DUST PLUG	PCS-FLX-056	SWAGELOCK	U SERIES	SS	N/A	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	YES	TBD	TFS	LP1Y-DI-00000065	1	50'-0"	LP1Y FLX-0651N1	BY OWNER	FLEXIBLE HOSE	N2	Nitrogen	N2 E 061004	315	FLEXIBLE HOSE COUPLER CAMLOCK X COUPLER CAMLOCK PFA-LINED WHITE WITH DUST PLUGS	PCS-FLX-056	SWAGELOCK	U SERIES	SS	PTFE	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	YES	TBD	TFS	LP1Y-DI-00000065	1	50'-0"	LP1Y FLX-0650N1	BY OWNER	FLEXIBLE HOSE	N2	Nitrogen	LP1Y 061008	315	FLEXIBLE HOSE 2" 150# FLG X COUPLER CAMLOCK PFA-LINED WHITE WITH DUST PLUG	PCS-FLX-056	SWAGELOCK	U SERIES	SS	N/A	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	NO	TBD	TFS	LP1Y-DI-00000061	3	N/A	LP1Y PC-0610V1	BY OWNER	PIPE COVER	RV	Relief Vent	LP1Y 061041	315	PIPE COVER PVC PE	PCS-PC-001	SL&C	WSL SERIES	NM	N/A	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	NO	TBD	TFS	LP1Y-DI-00000061	3	N/A	LP1Y PC-0610V2	BY OWNER	PIPE COVER	RV	Relief Vent	LP1Y 061042	315	PIPE COVER PVC PE	PCS-PC-001	SL&C	WSL SERIES	NM	N/A	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	NO	TBD	TFS	LP1Y-DI-00000061	3	N/A	LP1Y PC-0610V3	BY OWNER	PIPE COVER	RV	Relief Vent	LP1Y 061054	315	PIPE COVER PVC PE	PCS-PC-001	SL&C	WSL SERIES	NM	N/A	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	NO	TBD	TFS	LP1Y-DI-00000061	3	N/A	LP1Y PC-0610V4	BY OWNER	PIPE COVER	RV	Relief Vent	LP1Y 061057	315	PIPE COVER PVC PE	PCS-PC-001	SL&C	WSL SERIES	NM	N/A	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	YES	TBD	TFS	LP1Y-DI-00000063	3	N/A	LP1Y QCU-0631A2	BY OWNER	QUICK CONNECT UNIT	MTBE	Methyl Tert Butyl Ether	LP1Y 063101	315	HOSE CONNECTION 150# 316 SS RF X DRY DISCONNECT ADAPTER WITH DUST CAP SEL 6	PCS-QCU-045	MANNTEK	DRY DISCONNECT COUPLING	SS	PTFE	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	YES	TBD	TFS	LP1Y-DI-00000063	3	N/A	LP1Y QCU-0631A1	BY OWNER	QUICK CONNECT UNIT	MTBE	Methyl Tert Butyl Ether	LP1Y 063101	315	HOSE CONNECTION 316 SS FNPT X DRY DISCONNECT ADAPTER WITH DUST CAP SEL 6	PCS-QCU-046	MANNTEK	DRY DISCONNECT COUPLING	SS	PTFE	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	YES	TBD	TFS	LP1Y-DI-00000063	3	N/A	LP1Y EI-0631A1	BY OWNER	RAILCAR ADAPTER	MTBE	Methyl Tert Butyl Ether	LP1Y 063101	315	RAIL TANK CAR 316 SS 5" SWIVEL X 3" COUPLER ADAPTER	PCS-RCA-001	DIXON	RTCA30 (316 SS)	SS	PTFE	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	YES	TBD	TFS	LP1Y-DI-00000062	3	N/A	LP1Y SG-0623A	BY OWNER	SIGHT GLASS	DMF	Dimethylformamide	LP1Y 062301	315	SIGHT GLASS FULL VIEW 150# 316 SS RF PFA LINED	PCS-SG-016					NA	NA	NA	
0	BY CONTRACTOR - RAIL	NO	YES	TBD	TFS	LP1Y-DI-00000063	4	N/A	LP1Y SG-0631A	BY OWNER	SIGHT GLASS	MTBE	Methyl Tert Butyl Ether	LP1Y 063101	315	SIGHT GLASS FULL VIEW 150# 316 SS RF PFA LINED	PCS-SG-016					NA	NA	NA	FULL BORE SIGHT GLASS TO BE LOCATED MINIMUM TO THE SUCTION FLANGE
0	BY CONTRACTOR - RAIL	NO	YES	TBD	TFS	LP1Y-DI-00000063	3	N/A	LP1Y SG-0633A	BY OWNER	SIGHT GLASS	ACN	Acetonitrile	LP1Y 063301	315	SIGHT GLASS FULL VIEW 150# 316 SS RF PFA LINED	PCS-SG-016					NA	NA	NA	
0	BY CONTRACTOR - RAIL	NO	NO	TBD	TFS	LP1Y-DI-00000061	0.75	N/A	LP1Y STR-0610U3	BY OWNER	STRAINER	S(130)	Steam (130 psig)	LP1Y 061059	2	STRAINER Y-TYPE 600# STL FNPT 304 SCR .045 PERF	PCS-STR-005	KECKLEY	STYLE 587	CS	N/A	N/A	N/A	N/A	USE 3/8" BALL VALVE FOR BLOWDOWN
0	BY CONTRACTOR - RAIL	NO	NO	TBD	TFS	LP1Y-DI-00000061	0.75	N/A	LP1Y STR-0610U4	BY OWNER	STRAINER	S(130)	Steam (130 psig)	LP1Y 061061	2	STRAINER Y-TYPE 600# STL FNPT 304 SCR .045 PERF	PCS-STR-005	KECKLEY	STYLE 587	CS	N/A	N/A	N/A	N/A	USE 3/8" BALL VALVE FOR BLOWDOWN
0	BY CONTRACTOR - RAIL	NO	NO	TBD	TFS	LP1Y-DI-00000062	0.75	N/A	LP1Y STR-0620U2	BY OWNER	STRAINER	S(130)	Steam (130 psig)	LP1Y 062001	2	STRAINER Y-TYPE 600# STL FNPT 304 SCR .045 PERF	PCS-STR-005	KECKLEY	STYLE 587	CS	N/A	N/A	N/A	N/A	USE 3/8" BALL VALVE FOR BLOWDOWN
0	BY CONTRACTOR - RAIL	NO	NO	TBD	TFS	LP1Y-DI-00000063	0.75	N/A	LP1Y STR-0630U2	BY OWNER	STRAINER	S(130)	Steam (130 psig)	LP1Y 063001	2	STRAINER Y-TYPE 600# STL FNPT 304 SCR .045 PERF	PCS-STR-005	KECKLEY	STYLE 587	CS	N/A	N/A	N/A	N/A	USE 3/8" BALL VALVE FOR BLOWDOWN
0	BY CONTRACTOR - RAIL	NO	NO	TBD	TFS	LP1Y-DI-00000064	0.75	N/A	LP1Y STR-0640U1	BY OWNER	STRAINER	S(130)	Steam (130 psig)	LP1Y 064003	2	STRAINER Y-TYPE 600# STL FNPT 304 SCR .045 PERF	PCS-STR-005	KECKLEY	STYLE 587	CS	N/A	N/A	N/A	N/A	USE 3/8" BALL VALVE FOR BLOWDOWN
0	BY CONTRACTOR - RAIL	NO	NO	TBD	TFS	LP1Y-DI-00000065	0.75	N/A	LP1Y STR-0650U1	BY OWNER	STRAINER	S(130)	Steam (130 psig)	LP1Y 065003	2	STRAINER Y-TYPE 600# STL FNPT 304 SCR .045 PERF	PCS-STR-005	KECKLEY	STYLE 587	CS	N/A	N/A	N/A	N/A	USE 3/8" BALL VALVE FOR BLOWDONE
0	BY CONTRACTOR - RAIL	NO	NO	TBD	TFS	LP1Y-DI-00000061	0.75	N/A	LP1Y STT-0610U3	BY OWNER	STEAM TRAP	S(130)	Steam (130 psig)	LP1Y 061059	2	STEAM TRAP INVERTED BUCKET 400 PSIG 304L SS FNPT	PCS-STT-003	ARMSTRONG	2000 SERIES	SS	N/A	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	NO	TBD	TFS	LP1Y-DI-00000061	0.75	N/A	LP1Y STT-0610U4	BY OWNER	STEAM TRAP	S(130)	Steam (130 psig)	LP1Y 061061	2	STEAM TRAP INVERTED BUCKET 400 PSIG 304L SS FNPT	PCS-STT-003	ARMSTRONG	2000 SERIES	SS	N/A	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	NO	TBD	TFS	LP1Y-DI-00000062	0.75	N/A	LP1Y STT-0620U1	BY OWNER	STEAM TRAP	S(130)	Steam (130 psig)	LP1Y 062001	2	STEAM TRAP INVERTED BUCKET 400 PSIG 304L SS FNPT	PCS-STT-003	ARMSTRONG	2000 SERIES	SS	N/A	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	NO	TBD	TFS	LP1Y-DI-00000063	0.75	N/A	LP1Y STT-0630U1	BY OWNER	STEAM TRAP	S(130)	Steam (130 psig)	LP1Y 063001	2	STEAM TRAP INVERTED BUCKET 400 PSIG 304L SS FNPT	PCS-STT-003	ARMSTRONG	2000 SERIES	SS	N/A	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	NO	TBD	TFS	LP1Y-DI-00000064	0.75	N/A	LP1Y STT-0640U1	BY OWNER	STEAM TRAP	S(130)	Steam (130 psig)	LP1Y 064003	2	STEAM TRAP INVERTED BUCKET 400 PSIG 304L SS FNPT	PCS-STT-003	ARMSTRONG	2000 SERIES	SS	N/A	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	NO	TBD	TFS	LP1Y-DI-00000065	0.75	N/A	LP1Y STT-0650U1	BY OWNER	STEAM TRAP	S(130)	Steam (130 psig)	LP1Y 065003	2	STEAM TRAP INVERTED BUCKET 400 PSIG 304L SS FNPT	PCS-STT-003	ARMSTRONG	2000 SERIES	SS	N/A	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	NO	TBD	TFS	LP1Y-DI-00000061	1.25	N/A	LP1Y WH-0610	BY OWNER	WATER HEATER	WPC	Water, Potable - Cold	LP1Y 061027	36	WATER HEATER 316 SS MNPT	PCS-WH-003	EEMAX	SAFEADVANTAGE	SS	N/A	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	NO	TBD	TFS	LP1Y-DI-00000061	1	N/A	LP1Y WHA-~~0610E1~~	BY OWNER	WATER HAMMER ARRESTER	WPC	Water, Potable - Cold	LP1Y 061027	36	WATER HAMMER ARRESTER 150 PSIG STAINLESS STEEL MNPT	PCS-WHA-002	JAY R. SMITH	5000 SERIES	SS	N/A	N/A	N/A	N/A	
0	BY CONTRACTOR - RAIL	NO	NO	TBD	TFS	LP1Y-DI-00000061	1	N/A	LP1Y WHA-~~0610E2~~	BY OWNER	WATER HAMMER ARRESTER	WPC	Water, Potable - Cold	LP1Y 061028	36	WATER HAMMER ARRESTER 150 PSIG STAINLESS STEEL MNPT	PCS-WHA-002	JAY R. SMITH	5000 SERIES	SS	N/A	N/A	N/A	N/A	

MTBE was deleted from the list in the cutsheet. Please confirm whether it is BCU-001 or other BCU.

SPECIFY KALREZ SEAL MATERIAL. (LC, FLUOR, 21APR26)

SPECIFY 'SEL 6' TO MATCH QCU END. (LC, FLUOR, 21APR26)

Please confirm if it means 2" flanged 1" hose or 2" flex hose.

SHOWN AS NPS 2 ON P&IDS. (LC, FLUOR, 21APR26)

MTBE is marked in red in the cutsheets. Please confirm if QCU-045/046 is applicable for MTBE service.

5", if it is the column for maximum size.

PURCHASE PCS-SG-018 INSTEAD - SWISS FLUID FULL VIEW SIGHT GLASS. PCS-SG-016 SHALL ONLY BE INSTALLED IN A VERTICAL ORIENTATION.

0610E1

0610E2

CONFIRM WH LOCATION IS NOT WITHIN HAC CLASS 1 DIV 2 ZONE. (LC, FLUOR, 21APR26)

CATALOG REFERENCE SUMMARY

Cutsheet: Tag Number	Specialty Item Type	Item Description	Page
PCS-BCU-001	BREAKAWAY COUPLING	BREAKAWAY COUPLING 150# 316 SS RF	4
PCS-CCU-029	CAMLOCK CONNECTION	HOSE CONNECTION 316 SS SW ADAPTER CAMLOCK WITH DUST CAP	12
PCS-CCU-030	CAMLOCK CONNECTION	HOSE CONNECTION 316 SS BW ADAPTER CAMLOCK WITH DUST CAP	20
PCS-CCU-032	CAMLOCK CONNECTION	HOSE CONNECTION 316 SS SW ADAPTER CAMLOCK WITH DUST CAP	28
PCS-FLX-054	FLEXIBLE HOSE	FLEXIBLE HOSE 360 PSIG 316 SS DRY DISCONNECT COUPLER PTFE-LINED WHITE ANTISTATIC W DUST PLUG SEL	35
PCS-FLX-056	FLEXIBLE HOSE	FLEXIBLE HOSE "VARIES" X COUPLER CAMLOCK PFA-LINED WHITE WITH DUST PLUGS	56
PCS-PC-001	PIPE COVER	PIPE COVER PVC PE	65
PCS-QCU-045	QUICK CONNECT UNIT	HOSE CONNECTION 150# 316 SS RF X DRY DISCONNECT ADAPTER WITH DUST CAP SEL 6	71
PCS-QCU-046	QUICK CONNECT UNIT	HOSE CONNECTION 316 SS FNPT X DRY DISCONNECT ADAPTER WITH DUST CAP SEL 6	81
PCS-RCA-001	RAILCAR ADAPTER	RAIL TANK CAR 316 SS 5" SWIVEL X 3' COUPLER ADAPTER	91
~~PCS-SG-016~~ USE PCS-SG-018 INSTEAD. (LC, FLUOR, 21APR26)	SIGHT GLASS	SIGHT GLASS FULL VIEW 150# 316SS RF PFA LINED	94
PCS-STR-005	STRAINER	STRAINER Y-TYPE 600# STL FNPT 304 SCR .045 PERF	96
PCS-STT-003	STEAM TRAP	STEAM TRAP INVERTED BUCKET 400 PSIG 304L SS FNPT	102
PCS-WH-003	WATER HEATER	WATER HEATER 316 SS MNPT	105
PCS-WHA-002	WATER HAMMER ARRESTER	WATER HAMMER ARRESTER 150 PSIG STAINLESS STEEL MNPT	110

=== PIPING SPECIALTY ITEM LIST ===

CUT SHEET NUMBER:	PCS-BCU-001
DESCRIPTION:	Breakaway Coupling
VENDOR:	MannTek
MODEL:	316 SS Industrial Safety Breakaway Coupling with Flanged Ends

NOTE: USED AS INDICATED ON P&ID (PS 315).

APPLICABLE SERVICES
Acetonitrile
Alcohol, Isopropyl
Dichloromethane
Dimethyl Sulfoxide
Dimethylformamide
General Solvent
Methyl Tert Butyl Ether
Please confirm whether MTBE is applicable to BCU-001 or it needs other type of BCU.
Sodium Hydroxide, Industrial 50%
Sulfuric Acid

CUT SHEET BEGINS ON NEXT PAGE

MannTek Safety Break-away Couplings - Industrial Version

Application:

- loading rack, bulk loading / unloading, road tankers and rail tankers

Features:

- designed to be installed between a fixed point (pipe, pump, manifold) and a hose
- minimizes spillage and damage associated with pull-away incidents
- coupling automatically senses excessive load, closes valves and disconnects, release is executed when force causes bolts to break
- high flow rate / low pressure drop
- working pressure: stainless steel **360 PSI** and aluminum **230 PSI** at ambient temperature **70°F (21°C)**
- ~~FKM (FPM) seal standard,~~ additional seal materials are available
- optional non-closure design available, contact Dixon®

SPECIFY KALREZ SEAL MATERIAL. (LC, FLUOR, 21APR26)

Female NPT x Female NPT

Size	DN Size	*316 Stainless Steel* Part #	*Aluminum* Part #
1"	25	***SBC100SS***	***SBC100AL***
2"	50	***SBC200SS***	***SBC200AL***
3"	80	***SBC300SS***	***SBC300AL***
4"	100	***SBC400SS***	***SBC400AL***
5"	125	***SBC500SS***	***SBC500AL***
6"	150	***SBC600SS***	***SBC600AL***

Male NPT x Male NPT

Size	DN Size	*316 Stainless Steel* Part #	*Aluminum* Part #
2"	50	***SBC200SSMNPT***	***SBC200ALMNPT***
3"	80	***SBC300SSMNPT***	***SBC300ALMNPT***
4"	100	***SBC400SSMNPT***	***SBC400ALMNPT***
5"	125	***SBC500SSMNPT***	***SBC500ALMNPT***
6"	150	***SBC600SSMNPT***	***SBC600ALMNPT***

150# Flange x 150# Flange

Size	DN Size	*316 Stainless Steel* Part #	*Aluminum* Part #
1"	25	***SBC100SSFL***	**---**
2"	50	***SBC200SSFL***	***SBC200ALFL***
3"	80	***SBC300SSFL***	***SBC300ALFL***
4"	100	***SBC400SSFL***	***SBC400ALFL***
5"	125	***SBC500SSFL***	***SBC500ALFL***
6"	150	***SBC600SSFL***	***SBC600ALFL***
8"	200	***SBC800SSFL***	**---**

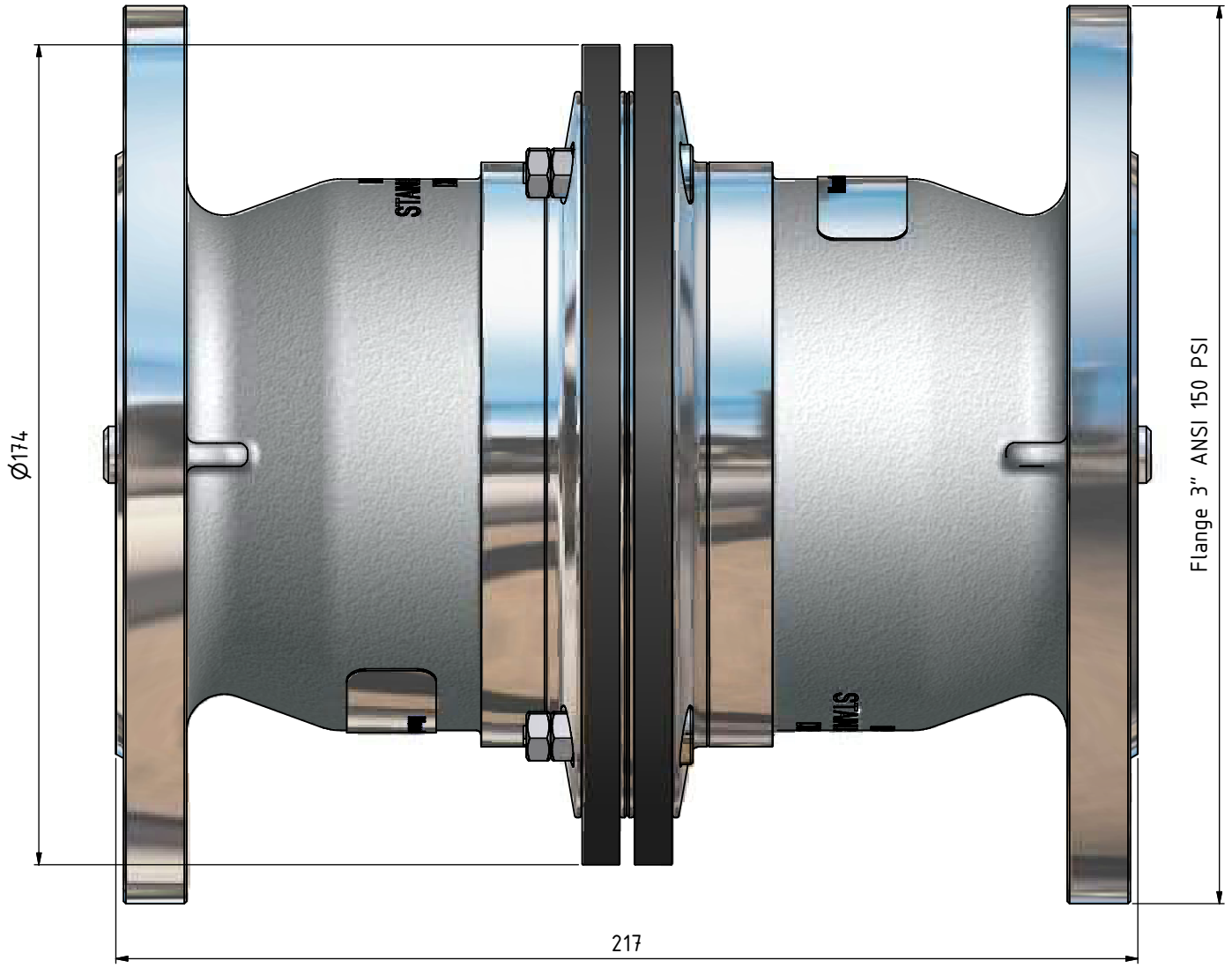
- For flange dimensions, diagrams and additional information please reference dixonvalve.com.

TTMA Flanged Breakaway Coupling

Features:

- minimizes spillage and damage associated with drive away and pull away incidents
- installs between an API coupler and loading arm
- coupling automatically senses excessive load, closes the valves and disconnects, release is executed when force causes the bolts to break
- high flow rate / low pressure drop
- **150 PSI** at ambient temperature **70°F (21°C)**
- FKM/FPM is standard seal

Notice. This drawing contains poprietary information and shall not be used or reproduced or its contents disclosed in whole or in part, without the prior written concent of the Mann Teknik AB



Dimensions are in metric.

Weight: 11,5 kg

Detail No:	Material	Ämne/Blank	Ytbeh./Treatment	Vikt/Weight

Mann Tek

Designed/drawn by	Scale:	Dimension with no tolerances acc to	
P. Nyberg	1:1	Machined	SS-ISO 2768-1 medium
Checked		Weld Forge	SS-ISO 2768-1 coarse
G. Kopplin	Date	Surface roughness	ISO 1302
Approved	2009-10-20	Form & position tol	SS ISO 1101
T. Mann			

Industrial Break-Away 3"

Flange 3" ANSI 150 PSI

Data nr./Comp No.	Blad/Sheet
N461D4401	1 / 1

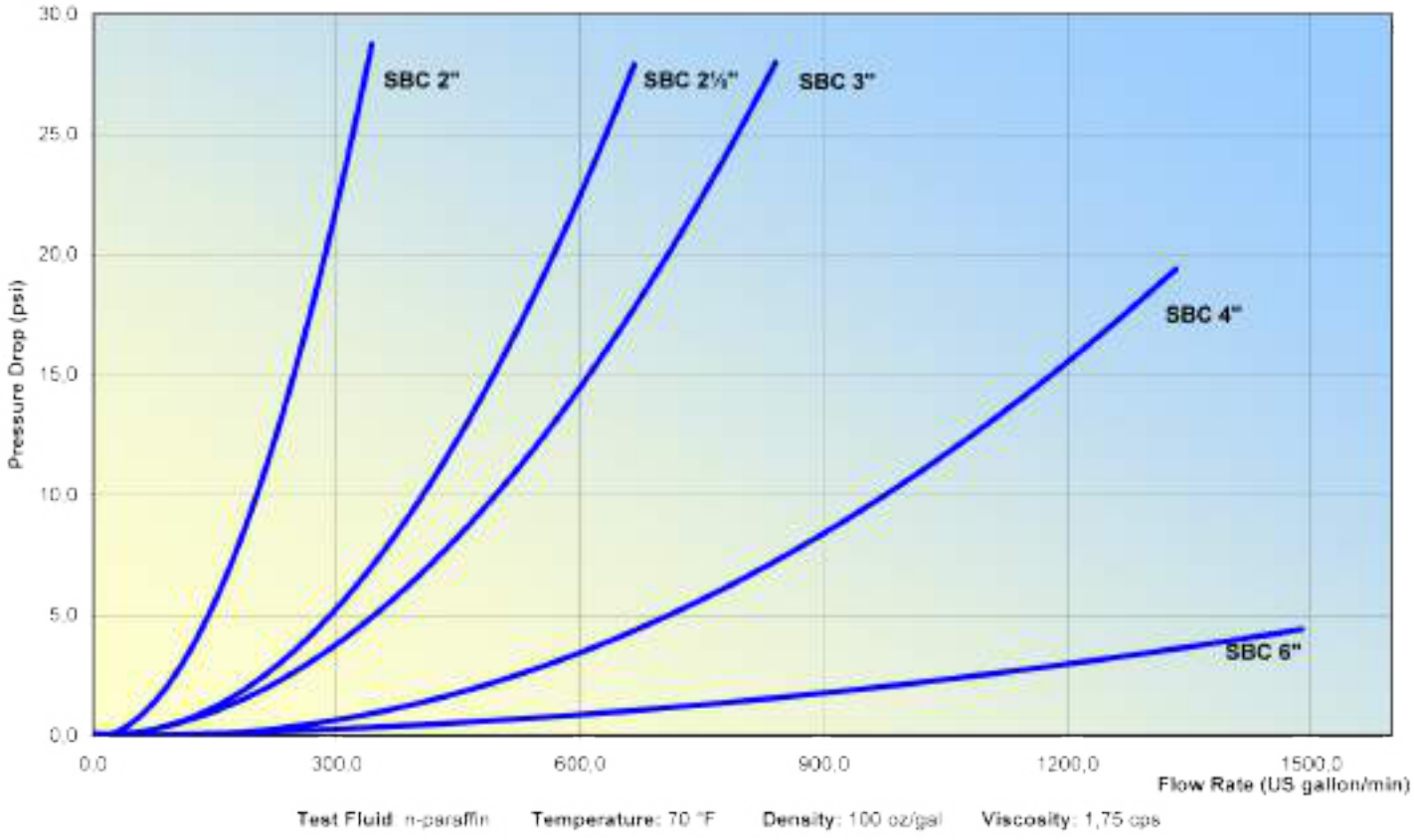
Drawing No.	Rev.
N461D4401-1	42

NO.	Revision comprises	Date	Sign.	Appr by

The Right Connection®

Safety Break-away Couplings Breaking Bolt Series

Flow Diagram (pressure drop)



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800 High Street
Chestertown, MD 21620
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fx: 800.283.4966
dixonvalve.com

MannTek

Configurator MannTek Products

Part Number: N461D4401

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https://configurator.manntek.se/SBC/INDUSTRIAL/4/SS/FLANGE/3_ASA150_B/FLANGE/3_ASA150_B/VITON/NONE/Product11.html

Product Group

Safety Breakaway Couplings
The couplings are designed to be the weakest link of the hose line or loading arm. Suitable for a wide range of products including petroleum based products, chemicals and foodstuffs.

Three types can be distinguished:
- SBC Breaking Bolt Type 'Industrial Version'
- SBC Breaking Bolt Type 'Marine Version'
- SBC-C Cable Release Type

1) SBC Breaking Bolt Type 'Industrial Version'

Used to protect hose assemblies and loading arms against excessive loads, e.g. due to drive/pull away incidents or rapid removal of the mobile tanker in emergency situations.
Typical applications include container discharge, fuel bunkering truck-to-ship, ship-to-ship, terminal-to-ship, loading/unloading of tank trucks, rail tankers and ships.

Designed with breaking bolts; activation at a defined tensile (pulling) force on the transfer line being applied, either straight or at an angle.

Installation between a hose assembly or loading arm and a fixed point, e.g. pipe system, rail tanker or road tanker connection.
Release by pull at any angle up to 90 degrees.

SBC 'Industrial Type' have thre external breaking bolts. Breaking forces (release forces) are calculated on the basis of the working pressure of the system in

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Question.Standard breaking forces for SBC are calculated to protect a mating hose with a burst pressure 4 times WP. To protect hoses with other burst pressure specification or adjacent equipment which may be weaker than the hose please contact your sales representative who will provide a design guideline or can forward your special request to the MannTek technical department.

If the transfer line is subjected to inadvertent and strong tensile loads, e.g. due to a premature movement or the drifting of ships / barges, the coupling will separate and instantly seal the two open ends. It can be reassembled on site with a set of spare parts.
2) SBC Breaking Bolt Type 'Marine Version'

Used to protect hose assemblies against excessive loads, e.g. due to pull away incidents or rapid removal of the vessel in emergency situations.
Typical applications are ship to rig and ship to ship media transfers. Typical applications include container discharge, fuel bunkering truck-to-ship, ship-to-ship, terminal-to-ship.

Designed with breaking bolts; activation at a defined tensile (pulling) force on the transfer line being applied applied axially. In comparison to the 'Industrial Version' the coupling offers an increased resistance to torsion and bending loads to prevent unwanted separation.

SBC 'Marine-Version' have thre external breaking bolts. Breaking forces (release forces) are calculated on the basis of the working pressure of the system in question.
Standard breaking forces for SBC are calculated to protect a mating hose with a burst pressure 4 times WP.

To protect hoses with other burst pressure specification or adjacent equipment which may be weaker than the hose please contact your sales representative who will provide a design guideline or can forward your special request to the MannTek technical department.

If the transfer line is subjected to inadvertent and strong tensile loads, e.g. due to a premature movement or the drifting of ships / barges, the coupling will separate and instantly seal the two open ends. It can be reassembled on site with a set of spare parts.
3) SBC-C Cable Release Type

Used to protect hose assemblies against excessive loads. Special type, recommended in transfer systems with the combination of high operating pressure and weak piping structure.

Compared with standard Safety Break Couplings with breaking bolts SBC-C are designed with a clamp system and an external collar with an attached cable.
The tensile force (pulling force) is not transmitted via the hose or loading arm, but by an integrated cable which is attached to a fixed point (pipework, rail tanker or road tanker).
The length of the cable is adjusted slightly shorter than the maximum length of the loading line.

Release forces are almost independent from the internal pressure (working pressure).

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After the admitted travel limit of the cable is reached (angle up to 90 degrees), only a small pull is sufficient to actuate the separation.

If the cable reaches the admitted travel limit and a defined pulling force, the external colllar is pulled off and clamps are actuated; the coupling will separate and instantly seal the two open ends. There are no loose components which can get lost. After separation the coupling is checked and can be reconnected without spare parts or special tools.

Type of Coupling

SBC Breaking Bolt Type - Industrial Version
Used to protect hose assemblies and loading arms against excessive loads, e.g. due to drive/pull away incidents or rapid removal of the mobile tanker in emergency situations.
Typical applications include container discharge, fuel bunkering truck-to-ship, ship-to-ship, terminal-to-ship, loading/unloading of tank trucks, rail tankers and ships.

Designed with breaking bolts; activation at a defined tensile (pulling) force on the transfer line being applied, either straight or at an angle.

Installation between a hose assembly or loading arm and a fixed point, e.g. pipe system, rail tanker or road tanker connection.
Release by pull at any angle up to 90 degrees.

SBC 'Industrial Type' have thre external breaking bolts. Breaking forces (release forces) are calculated on the basis of the working pressure of the system in Question.
Standard breaking forces for SBC are calculated to protect a mating hose with a burst pressure 4 times WP.
To protect hoses with other burst pressure specification or adjacent equipment which may be weaker than the hose please contact your sales representative who will provide a design guideline or can forward your special request to the MannTek technical department.

If the transfer line is subjected to inadvertent and strong tensile loads, e.g. due to a premature movement or the drifting of ships / barges, the coupling will separate and instantly seal the two open ends. It can be reassembled on site with a set of spare parts.
Certificates: SBC fulfill the requirements of European Pressure Equipment Directive category 2 (for higher demand on request).
Suitable for the use in potentially explosive atmosphere (ATEX) zone 1.
Approved for the transport of dangerous goods on road (ADR), by rail (RID) and by ship (IMDG).

Lloyd’s Register 0038 RET0271208, TDT-UW-21/16, EAC TC No. RU Д-SE.МЮ62.B.02151

Nominal Diameter

DN 80 (3")
Similar to 2½", but where higher loading rates are required. Especially on rail tankers, marine tankers, ISO containers, Aviation fuel bunkering and related activities.

Body Material

Stainless Steel 1.4408 (AISI 316)

Maximum Working Pressure	25 bar
Test Pressure	37,5 bar

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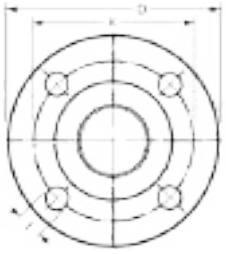
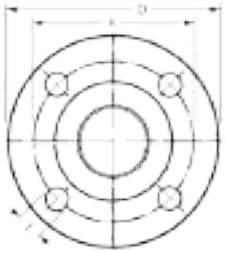
Minimum Burst Pressure 125 bar

Connection 1	**Flanged Connection**
Connection 1 - Detail	**3" ANSI Cl.150** Dimension: D: (190,5 mm, 7 ½"), k: (152,4 mm, 6"), 4 holes, l: (19 mm, ¾")



Infographic

Connection Detail	**Technical Drawing**
Connection 2	**Other Flanged Connection**
Connection 2 - Detail	**3" ANSI Cl.150** Dimension: D: (190,5 mm, 7 ½"), k: (152,4 mm, 6"), 4 holes, l: (19 mm, ¾")



Infographic

Connection Detail	**Technical Drawing**
O-Rings	**Viton® (FPM/FKM) Standard** FKM/FPM (Fluorocarbon Rubber): FKM is known especially for its non-flammability, low gas permeability and excellent resistance to ozone, weathering and aging. The operating temperatures of the Fluorocarbon Rubber range between -20° C and +200° C). FKM is often used with mineral based oils and greases at high temperatures, aliphatic, aromatic and chlorinated hydrocarbons and oxidizing acids.
Options: Special Types	**No Special Types required**

PDF Documents

Homepage SBC (Safety Breakaway Couplings)

=== PIPING SPECIALTY ITEM LIST ===

CUT SHEET NUMBER:	PCS-CCU-029
DESCRIPTION:	Camlock Connection
VENDOR:	Dixon
MODEL:	316 SS SW Adapter with Type H Dust Cap

SELECTION:

TO BE USED WHEN 316 STAINLESS STEEL CAMLOCK ADAPTER IS NEEDED IN SOCKETWELD PIPE SCHEDULE METHOD OF CONSTRUCTION.

CUT SHEET BEGINS ON NEXT PAGE

EZ Boss-Lock Coupler Butt Weld to Schedule 40 Pipe / Socket Weld to Nominal Tubing

Feature

- Nitrile rubber gaskets standard, other gaskets available on pages 75-77

Size	Bore	316 Stainless Steel Part #
3/4"	.765	RDWBPST075EZ
1"	1.015	RDWBPST100EZ
1-1/2"	1.515	RDWBPST150EZ
2"	2.015	RDWBPST200EZ
3"	3.015	RDWBPST300EZ
4"	4.015	RDWBPST400EZ
6"	6.020	RDWBPST600EZ

Adapter Socket Weld to Schedule 40 Pipe

Feature

- Not all configurations are stocked

Size	Bore	356T6 Aluminum Part #	316 Stainless Steel Part #
3/4"	1.065	---	75AWSPSS
1"	1.330	---	100AWSPSS
1-1/2"	1.915	150AWSPAL	150AWSPSS
2"	2.390	200AWSPAL	200AWSPSS
3"	3.530	300AWSPAL	300AWSPSS
4"	4.530	400AWSPAL	400AWSPSS
5"	5.593	500AWSPAL	500AWSPSS
6"	6.655	600AWSPAL	600AWSPSS
8" DIX	8.711	800AWSPAL[1]	---
8" BL	8.711	801AWSPAL[1]	---

[1] Dixon® and Boss-Lock cam & groove couplings do not interchange in the 8" size.
NOTE: Final weld preparation, as with all weld fittings, is the responsibility of the welder.

Adapter Butt Weld to Schedule 40 Pipe/Socket Weld to Nominal Tubing

Size	Bore	356T6 Aluminum Part #	316 Stainless Steel Part #
3/4"	.765	---	75AWBPSTSS
1"	1.015	---	100AWBPSTSS
1-1/2"	1.515	150AWBPSTAL	150AWBPSTSS
2"	2.015	200AWBPSTAL	200AWBPSTSS
3"	3.015	300AWBPSTAL	300AWBPSTSS
4"	4.015	400AWBPSTAL	400AWBPSTSS
5"	5.015	500AWBPSTAL	500AWBPSTSS
6"	6.020	600AWBPSTAL	600AWBPSTSS
8" BL	8.035	801AWBPSTAL[1]	---

[1] Dixon and Boss-Lock cam & groove couplings do not interchange in the 8" size

EZ Boss-Lock Type H Dust Caps

356T6 aluminum

brass

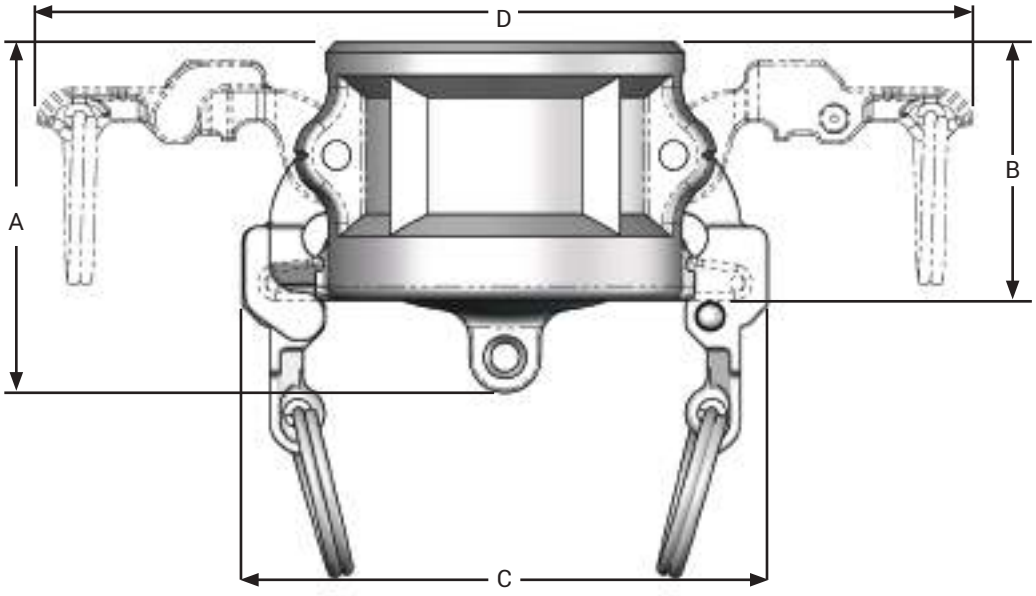
316 stainless steel

Features

- Nitrile rubber gasket standard, other gaskets available on pages 75-77
- Dust caps and dust plugs are not to be used in pressure applications for safety and environmental reasons

Size	356T6 Aluminum Part #	Brass Part #	316 Stainless Steel Part #
3/4"	---	---	RH075EZ
1"	AH100EZ	BH100EZ	RH100EZ
1-1/4"	---	---	RH125EZ
1-1/2"	AH150EZ	BH150EZ	RH150EZ
2"	AH200EZ	BH200EZ	RH200EZ
2-1/2"	AH250EZ	---	---
3"	AH300EZ	BH300EZ	RH300EZ
4"	AH400EZ	---	RH400EZ
6"	AH600EZ	---	RH600EZ

NOTE: Under no circumstances should the EZ Boss-Lock cam arms be used on any fitting not specifically designed for their use. See pages 5 and 73 for further specifications.



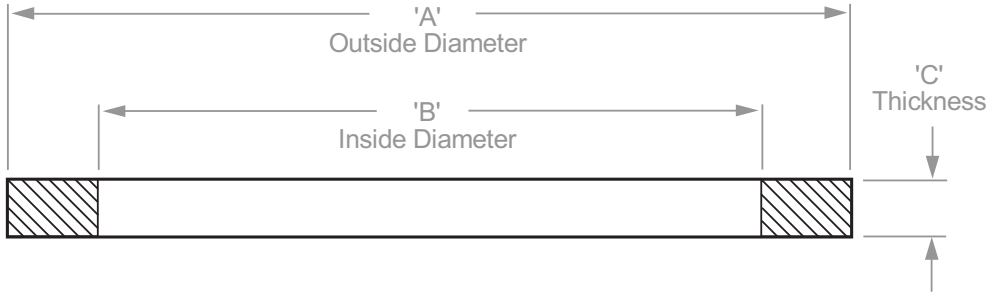
356T6 Aluminum, Brass, and Iron Dimensions

	Size	1"	1-1/2"	2"	2-1/2"	3"	4"	6"
A	overall length	2-3/16	2-15/16	3-1/16	3-7/32	3-1/8	3-11/32	3-29/32
B	coupler length	1-17/32	1-29/32	2-3/16	2-1/4	2-5/16	2-5/16	2-5/8
C	distance across closed cam arms	3-7/32	4-7/32	4-19/32	5-1/16	5-27/32	6-15/16	9-31/32
D	distance across open cam arms	6-11/32	7-25/32	8-5/32	8-21/32	9-29/32	11	16-25/32

316 Stainless Steel Dimensions

	Size	3/4"	1"	1-1/4"	1-1/2"	2"	3"	4"	6"
A	overall length	1-15/16	2-1/4	2-3/4	2-13/16	3-1/16	3-1/8	3-19/32	3-3/4
B	coupler length	1-9/32	1-9/16	1-29/32	1-29/32	2-1/4	2-9/32	2-5/16	2-19/32
C	distance across closed cam arms	3-3/32	3-1/4	3-29/32	4-7/32	4-19/32	5-27/32	6-15/16	9-31/32
D	distance across open cam arms	6-7/32	6-		7-25/32	8-5/32	9-29/32	11	16-25/32

Cam & Groove Gasket Dimensions



Features

- Nominal dimensions for elastomeric cam & groove gaskets are listed
- I.D.s of the PTFE envelope gaskets and the PTFE accordion gaskets are 1/16" smaller than the I.D.s for the elastomeric gaskets shown below
- Thickness of PTFE accordion gaskets may vary from chart below

Size	Part #	Dimensions		
		A	B	C
1/2"	50-G [1]	1-1/32"	11/16"	.156"
3/4"	75-G	1-3/8"	7/8"	.218"
1"	100-G	1-9/16"	1-1/16"	.250"
1-1/4"	125-G	1-15/16"	1-23/64"	.250"
1-1/2"	150-G	2-3/16"	1-5/8"	.250"
2"	200-G	2-5/8"	2"	.250"
2-1/2"	250-G	3-1/8"	2-3/8"	.250"
3"	300-G	3-23/32"	3"	.250"
4"	400-G	4-7/8"	4"	.250"
5"	500-G [1]	5-15/16"	4-7/8"	.250"
6"	600-G	7-1/16"	6"	.250"
8"	800-G [1]	9-5/16"	8-1/8"	.343"

[1] No interchange standard exists for the 1/2", 5", and 8" fittings. Generally, these sizes do not interchange with other manufacturers' products. As such, the gaskets in these sizes are not suitable for use in other manufacturers' couplings.

Cam & Groove Gasket Color Codes

Gasket Code [1]	Material	Color	Color code stripes (appear on the O.D. of the gasket or filler)
-BU	Nitrile rubber (standard in all metal fittings)	black	1 blue
-BF	Nitrile rubber (formulated for fuel service)	black	1 green
-VI	FKM-A	green	none
-VIA	FKM-A	black	1 yellow
-THK	Extra thick nitrile rubber	black	2 blue
-EPR	Ethylene Propylene	black	1 white
-NE	Neoprene	black	1 red
-WNE	White Neoprene, FDA approved material	white	none
-WB	White nitrile rubber, FDA approved material	white	blue
-SIL	Silicone	red	none
-TF	PTFE (TFE) envelope with nitrile rubber filler	white / black	1 blue
TFVI	PTFE (TFE) envelope with FKM-A filler	white / black	1 yellow
TFWB	PTFE (TFE) envelope with white nitrile rubber filler, FDA approved materials	white / white	blue
TFEP	PTFE (TFE) envelope with Ethylene Propylene filler	white / black	1 white stripe
-TES	PTFE (FEP) encapsulated Silicone core, FDA approved materials	translucent / red	none
-TEV	PTFE (FEP) encapsulated FKM-A core	translucent / black	none
TFACC	PTFE (TFE) accordion, FDA approved materials	white	none

[1] Part number suffix

NOTE: Since other manufacturers' coding systems vary, this chart should not be used as a reference in identifying gaskets of unknown origin.

Fuel Gaskets

Size	Material	Part #
1-1/2"	nitrile rubber	150-G-BF
2"	nitrile rubber	200-G-BF
3"	nitrile rubber	300-G-BF
4"	nitrile rubber	400-G-BF
3"	extra thick nitrile rubber	300GTHKBF
4"	extra thick nitrile rubber	400GTHKBF

Cam & Groove Envelope Gaskets

Feature

- Maximum operating temperatures for specific chemicals may be lower; contact Dixon® for specific recommendations

Approval

- PTFE white nitrile rubber filler (TFWB) contains FDA approved materials

envelope

Size	PTFE (TFE) with Nitrile Rubber filler Part #	PTFE (TFE) with FKM-A filler Part #
1/2"	50-G-TF	50-G-TF-VI
3/4"	75-G-TF	75-G-TF-VI
1"	100-G-TF	100GTFVI
1-1/4"	125-G-TF	125GTFVI
1-1/2"	150-G-TF	150GTFVI
2"	200-G-TF	200GTFVI
2-1/2"	250-G-TF	250GTFVI
3"	300-G-TF	300GTFVI
4"	400-G-TF	400GTFVI
6"	600-G-TF	600GTFVI

Size	PTFE (TFE) with white Nitrile Rubber filler Part #	PTFE (TFE) with Ethylene Propylene filler Part #
1-1/2"	---	150GTFEP
2"	200GTFWB	200GTFEP
3"	300GTFWB	300GTFEP
4"	400GTFWB	---
5"	500GTFWB	---

NOTE: The torque (force) required to close the handles on a pair of cam & groove fittings will vary according to the gasket chosen. Due to the nature of the PTFE material, the various PTFE gasket designs will require the most torque, increasing as you go from envelope to encapsulated, and finally, accordion.

Abrasion Resistant Inserts

Features
- Extends the life of adapters when transferring proppant to storage tanks
- Av ailable for multiple adapters and multiple coupler materials

Description	UHMW Polyethylene Part #
to be used with 400-A-AL, 400-A-BR, 400-A-MI, 400-A-SS	400-A-INSERT
to be used with 400-C-AL	400-C-INSERTAL
to be used with 400-C-MI	400-C-INSERTMI

400-A-INSERT

400-C-INSERTAL

Clip for Boss-Lock Couplings

Feature
- .072 stainless steel wire

Stainless Steel Part #
AC7

Clip Assembly

Features
- F or use with Boss-Lock couplings
- Includes pull ring, chain, and AC3 clip

Carbon Steel Part #
BLCA152CS

Jack Chain with S-Hook

Length	Material	Part #	Optional Qty
6"	brass	CH-B-6	10
12"	brass	CH-B-12	10
24"	brass	CH-B-24	---
12"	carbon steel	CH-C-12	10
6"	stainless	CH-SS-6	10
12"	stainless	CH-SS-12	10
24"	stainless	CH-SS-24	10

Locking Cam Arm Velcro® Straps

Application
- Use with cam & groove couplers to prevent spills, secure from damage, and hold in place during transport

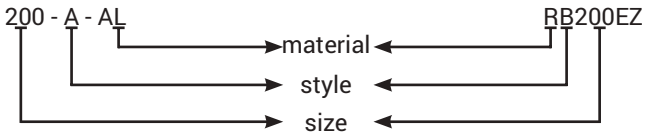
Features
- Used on the female end of a cam & groove coupler
- High visibility orange hook and loop with metal buckle provides strength, durability, and long life

Specification
- T emperature range **-70°F to 200°F (-57°C to 93°C)**

Length	Used On	Part #
12"	fits up to 1-1/2" coupler	CVS12
18"	fits up to 3" coupler	CVS18
24"	fits up to 5" coupler	CVS24
36"	fits up to 8" coupler	CVS36

coupler sold separately

Ordering System



Size represents the coupler or adapter and hose or pipe end (i.e. 200 = 2"). If the coupling is a jump size, the coupler or adapter is the first size (i.e. 4030-C is 4" coupler to 3" hose shank).

Operation of Cam & Groove

1. To make connection, slide the adapter into the coupler and, with normal hand pressure, press the cam levers down.

2. Uncoupling is as quick and simple as coupling. Just lift the cam levers and remove the adapter.

Specifications on Dixon® Cam & Groove

Specifications

- Dixon, Boss-Lock, and EZ Boss-Lock cam & groove couplers and adapters are produced to interchange with all product produced to Commercial Item Description A-A-59326D.
- No standard exists for the 1/2", 5", and 8" fittings, and generally these sizes do not interchange with other manufacturers' products.
- Dust caps and dust plugs are **not** to be used in pressure applications for safety and environmental reasons.
- *Designed for use with liquids*; consult Dixon for specific recommendations.

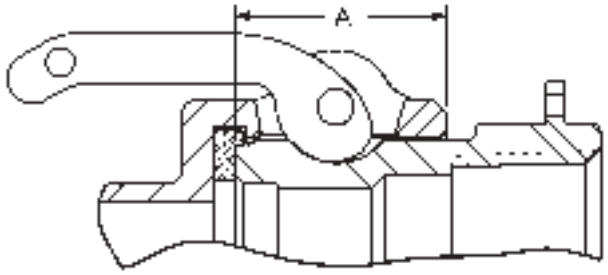
Pressure Ratings

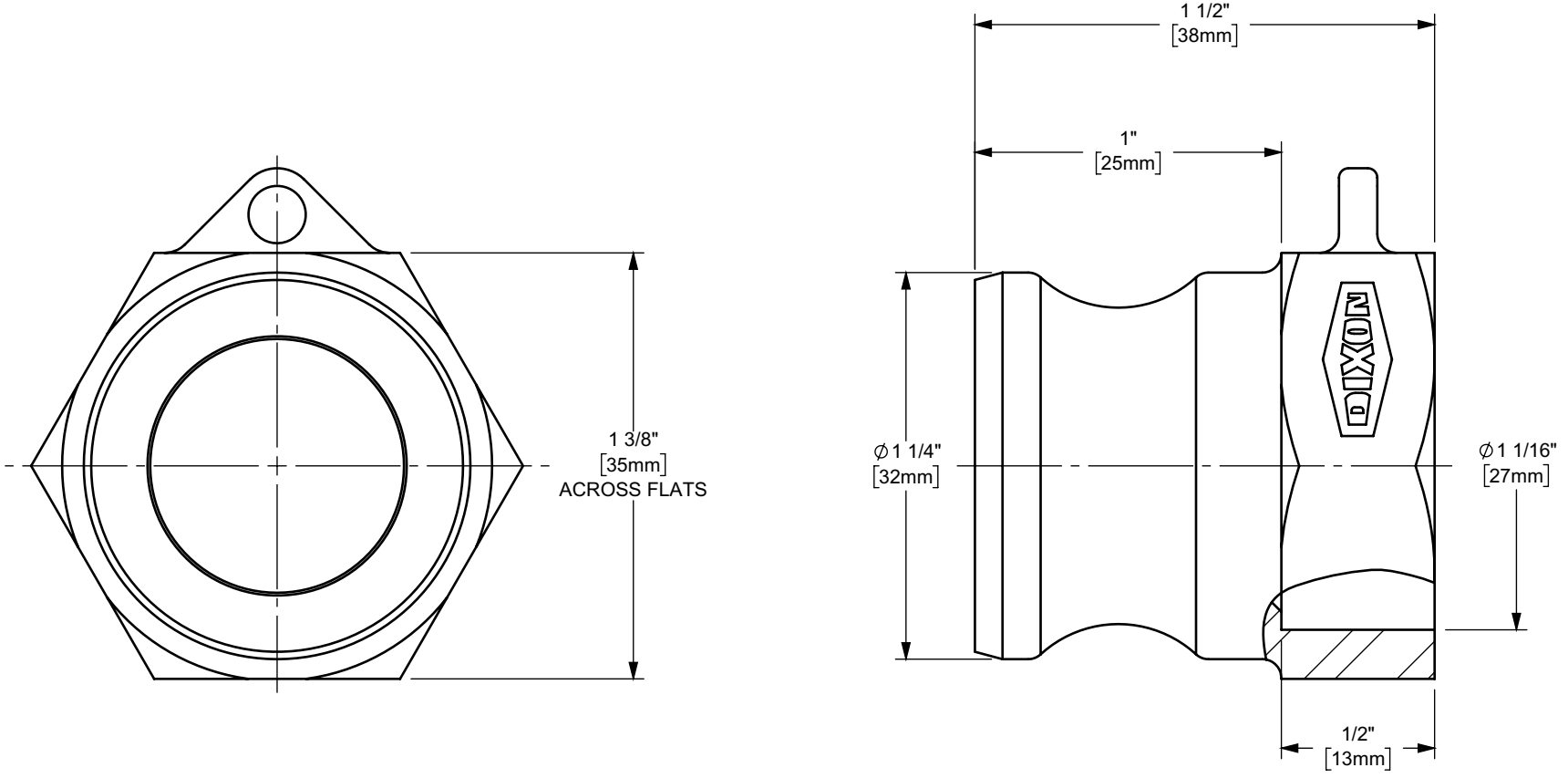
Recommendations based on the use of mating Dixon fittings at ambient temperature **70°F (21°C)** with standard nitrile rubber seal installed. For use at elevated temperature or other unusual operating conditions, consult Dixon. Dixon, Boss-Lock, and EZ Boss-Lock couplers' and adapters' maximum working pressures are as follows:

Size	1/2"	3/4" thru 2"	2-1/2"	3"	4"	5" and 6"	8"
PSI	**150**	**250**	**150**	**125**	**100**	**75**	**50**
PSI (working pressure with King Crimp® ferrule system)	**150**	**250**	**150**	**150**	**150**	**75**	**50**

Cam & Groove Nominal Take-Up Lengths

Coupling Size	Dimension A
1/2"	0.97"
3/4"	0.97"
1"	1.20"
1-1/4"	1.44"
1-1/2"	1.50"
2"	1.81"
2-1/2"	1.82"
3"	1.78"
4"	1.84"
5"	2.00"
6"	2.13"
8"	2.06"
8"	3.22"





DIXON
The Right Connection™
dixonvalve.com

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LE THIS DRAWING

DIXON U.S.A.
THE RIGHT CONNECTION ™

TITLE: 3/4" DIXON™ / BOSS-LOCK™ ADAPTER x SOCKET TO PIPE CAM AND GROOVE FITTING

MATERIAL: STAINLESS STEEL

PART NUMBER: **75AWSPSS**

=== PIPING SPECIALTY ITEM LIST ===

CUT SHEET NUMBER:	PCS-CCU-030
DESCRIPTION:	Camlock Connection
VENDOR:	PT Coupling
MODEL:	316 SS BW Coupler with Dust Plug

SELECTION:

TO BE USED WHEN 316 STAINLESS STEEL CAMLOCK COUPLER IS NEEDED IN BUTTWELD PIPE SCHEDULE METHOD OF CONSTRUCTION.

MANUFACTURER MUST BE PT COUPLING FOR PRESSURE REQUIREMENTS.

CUT SHEET BEGINS ON NEXT PAGE

Special Cam & Groove - Weld Fittings

DWB COUPLER
COUPLER X BUTT WELD
Made in the U.S.A.

All Stainless Steel castings are 316 Stainless Steel and all components are 300 series Stainless Steel.

HBS cam arms are made of 300 series Stainless Steel.

HB cam arms are made of Brass.

Other alloys and sizes are available via special quote.

Ordering Information:
- *Coupler Size*
- *Material*
- *Pipe Size & Schedule Number or Tube O.D.*
- *Wall Thickness*
- *Butt Weld*

Example:
- *Coupler Size: 3"*
- *Material: Aluminum*
- *Pipe Size: 3"*
- *Butt Weld*

All couplings are available with STA-LOK® II cam arms.

DWB line drawings & dimensions are on page 166.

Replacement gaskets are available on pages 66-68.

All registered trademarks are the property of their respective owners.

BUTT WELD FOR SCHEDULE 40 PIPE

		Aluminum	Stainless Steel	Hastelloy®C	Carpenter-20	Carbon Steel
Size	Part Name	Item #	Item #	Item #	Item #	Item #
1/2"	E05DWB	☎	64620107	☎	☎	☎
3/4"	07DWB	☎	64620711	☎	☎	☎
1"	10DWB	☎	64621013	☎	75621025	☎
1-1/4"	12DWB	60621225	64621211	☎	☎	☎
1-1/2"	15DWB	60621515	64621512	76621505	☎	74621505
1-1/2" X 1"	15X10DWB	☎	64621517	☎	☎	☎
2"	20DWB	60622013	64622011	76622006	75622011	☎
2-1/2"	25DWB	☎	64622512	☎	☎	☎
3"	30DWB	60623013	64623016	76623002	75623025	74623025
4"	40DWB	60624018	64624015	76624025	☎	74624025
5"	50DWB	60625015	64625016	☎	☎	☎
6"	60DWB	60626014	64626012	76626001	☎	74626012
8"	80DWB	60628011	64628028	☎	☎	☎
10"	100DWB	60629015	64629011	☎	☎	☎

BUTT WELD FOR SCHEDULE 10 PIPE

		Aluminum	Stainless Steel	Hastelloy®C	Carpenter-20	Carbon Steel
Size	Part Name	Item #	Item #	Item #	Item #	Item #
1/2"	E05DWB	☎	64620109	☎	☎	☎
3/4"	07DWB	☎	64620713	☎	☎	☎
1"	10DWB	☎	64621010	☎	☎	☎
1-1/4"	12DWB	☎	64621214	☎	☎	☎
1-1/2"	15DWB	☎	64621515	76621506	☎	☎
2"	20DWB	60622019	64622017	76622005	75622012	74622005
2-1/2"	25DWB	☎	64622515	☎	☎	☎
3"	30DWB	60623016	64623013	☎	☎	☎
4"	40DWB	60624013	64624012	☎	☎	☎
5"	50DWB	60625011	64625012	☎	☎	☎
6"	60DWB	60626016	64626013	76626003	☎	☎
8"	80DWB	60628013	64628027	☎	☎	☎

BUTT WELD FOR .065 TUBE

		Aluminum	Stainless Steel	Hastelloy®C	Carpenter-20	Carbon Steel
Size	Part Name	Item #	Item #	Item #	Item #	Item #
1/2"	E05DWB	☎	64620110	☎	☎	☎
3/4"	07DWB	☎	64620712	☎	☎	☎
1"	10DWB	☎	64621014	76621014	☎	☎
1-1/4"	12DWB	☎	64621213	☎	☎	☎
1-1/2"	15DWB	60621514	64621516	76621514	☎	☎
1-1/2"X 2"	15X20DWB	☎	64621520	☎	☎	☎
2"	20DWB	60622014	64622016	76622014	☎	☎
2-1/2"	25DWB	60622512	64622514	☎	☎	☎
3"	30DWB	60623010	64623017	☎	☎	☎
4"	40DWB	60624011	64624016	☎	☎	☎
5"	50DWB	60625014	64625014	☎	☎	☎
6"	60DWB	60626011	64626019	☎	☎	☎
8"	80DWB	60628010	☎	☎	☎	☎
10"	100DWB	☎	64629010	☎	☎	☎

Accessories & Replacement Parts

STANDARD CAM ASSEMBLY HB-HBS-HS

1 cam arm, 1 ring & 1 pin

	Brass		*300 Series Stainless Steel		**316 Stainless Steel	
Coupler Size	Part Name	Item #	Part Name	Item #	Part Name	Item #
1/2 & 3/4"	HB10	5000505A	HBS10	5001505A	HS10	5001005A
1"	HB11	5000510A	HBS11	5001510A	HS11	5001010A
1-1/4"	HB12	5000512A	HBS12	5001512A	HS12	5001012A
1-1/2"	HB15	5000515A	HBS15	5001515A	HS15	5001015A
2 & 2-1/2"	HB20/25	5000520A	HBS20/25	5001520A	HS20/25	5001020A
The above assemblies include (1) cam arm, (1) ring and (1) pin						
3,4, & 5"	HB30/40/50	5000530A	HBS30/40/50	5001530A	HS30/40/50	5001030A
The above assembly includes (1) cam arm, (1) ring and (1) 2" long pin						
3,4, & 5"	HBV/30/40/50	5000630A	HBSV30/40/50	5001730A	☎	☎
The above assembly includes (1) cam arm, (1) ring and (1) 1-5/8" pin						
6, 8, 10, & 12"	HB60/80	5000560A	HBS60/80	5001560A	HS60/80	5001060A
10 & 12"	HB100	5000590A	HBS100	☎	HS100	☎
The above assembly includes (1) cam arm, (1) ring and (1) pin						

**All cams, assemblies, rings and pins are 300 series Stainless.Steel*
***All cams, assemblies, rings and pins are 316 Stainless Steel*

STANDARD CAM ASSEMBLY SETS HB-HBS

2 cam arms, 2 rings, & 2 pins

	Brass		*300 Series Stainless Steel	
Coupler Size	Part Name	Item #	Part Name	Item #
1/2 & 3/4"	HB10	5000505S	HBS10	5001505S
1"	HB11	5000510S	HBS11	5001510S
1-1/4"	HB12	5000512S	HBS12	5001512S
1-1/2"	HB15	5000515S	HBS15	5001515S
2 & 2-1/2"	HB20/25	5000520S	HBS20/25	5001520S
3,4, & 5"	HB30/40/50	5000530S	HBS30/40/50	5001530S
6,8,10 & 12"	HB60/80	5000560S	HBS60/80	☎

**All cams, assemblies, rings and pins are 300 series Stainless Steel.*

STA-LOK® II CAM ASSEMBLY

1 CAM ARM, PIN, PULL RING & LOCK PLATE

STA-LOK® II is a passive locking feature that automatically locks the handle into place, if it is fully closed. To open the handles there are manual quick release levers. Proper use of this mechanism can help prevent spills by keeping the cam arms from opening under adverse conditions, such as high vibrations, pulsing, wear and hazardous materials. **STA-LOK® II couplers meet the Coast Guard ASTM 1122-87 specifications for Quick Disconnect Couplers.**

	Brass		***316 Stainless Steel	
Coupler Size	Part Name	Item #	Part Name	Item #
1"	SLB11	5006010A	SLS11	5009010A
1-1/4"	SLB12	5006012A	SLS12	5009012A
1-1/2"	SLB15	5006015A	SLS15	5009015A
2 & 2-1/2"	SLB20	5006020A	SLS20	5009020A
3,4, & 5"	SLB30	5006030A	SLS30	5009030A
6,8,10 & 12"	SLB60	5006060A	SLS60	5009060A

****All cams arms are 316 Stainless Steel, rings and pins are 300 series Stainless Steel.*
Call for special quote on 316 Stainless Steel components

Accessories & Replacement Parts - Coupling Gaskets

COUPLING GASKET

Made in the U.S.A.

Buna N Standard
1 Blue Stripe

Buna N Fuel Resistant
2 Blue Stripes

Neoprene Gasket
1 White Stripe

	Buna N Standard		Buna Fuel Resistant		Neoprene		Viton®	
Size	Part Name	Item #	Part Name	Item #	Part Name	Item #	Part Name	Item #
1/2"	B05Y	5500205	☎	☎	N05Y	5500405	V05Y	5500805
3/4"	B01	5500207	B01F	5500207F	N01	5500407	V01	5500807
1"	B02	5500210	B02F	5500210F	N02	5500410	V02	5500810
1-1/4"	B03	5500212	B03F	5500212F	N03	5500412	V03	5500812
1-1/2"	B04	5500215	B04F	5500215F	N04	5500415	V04	5500815
2"	B05	5500220	B05F	5500220F	N05	5500420	V05	5500820
2-1/2"	B06	5500225	B06F	5500225F	N06	5500425	V06	5500825
3"	B07	5500230	B07F	5500230F	N07	5500430	V07	5500830
4"	B08	5500240	B08F	5500240F	N08	5500440	V08	5500840
5"	B09	5500250	B09F	5500250F	N09	5500450	V09	5500850
6"	B010	5500260	B010F	5500260F	N010	5500460	V010	5500860
8"	B011	5500280	☎	☎	N011	5500480	V011	5500880
10"	B012	5500290	☎	☎	☎	☎	V012	5500890
12"	B013	5500292	☎	☎	☎	☎	☎	☎

VITON® 1/2" - 6"
Green

VITON® 8" & 10"*
1 Green Stripe

TEFLON® ENVELOPED COUPLING GASKET

Made in the U.S.A.

	Teflon® Enveloped Viton® Filler		Teflon® Enveloped Neoprene Filler		Teflon® Enveloped Buna Filler	
Size	Part Name	Item #	Part Name	Item #	Part Name	Item #
1/2"	TOFV05Y	5503905	TOFN05Y	5501605	TOFB05Y	5503705
3/4"	TOFV1	5503907	TOFN1	5501607	TOFB1	5503707
1"	TOFV2	5503910	TOFN2	5501610	TOFB2	5503710
1-1/4"	TOFV3	5503912	TOFN3	5501612	TOFB3	5503712
1-1/2"	TOFV4	5503915	TOFN4	5501615	TOFB4	5503715
2"	TOFV5	5503920	TOFN5	5501620	TOFB5	5503720
2-1/2"	☎	☎	TOFN6	5501625	TOFB6	5503725
3"	TOFV7	5503930	TOFN7	5501630	TOFB7	5503730
4"	TOFV8	5503940	TOFN8	5501640	☎	☎
6"	TOFV10	5503960	TOFN10	5501660	☎	☎

All registered trademarks are the property of their respective owners.

General Dimensions

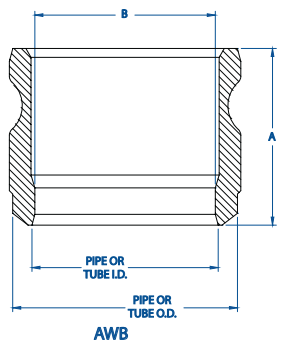
PT COUPLING CO. GENERAL SOCKET WELD DIMENSIONS

PART AWB (ADAPTER X BUTT WELD)

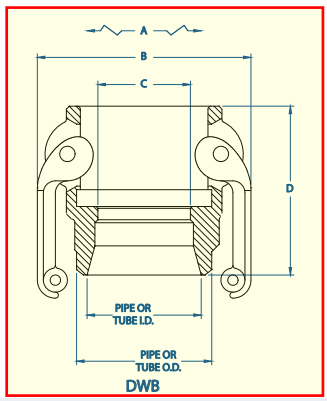
DIM	DESCRIPTION	E05	07	10	12	15	20	25	30	40	50	60	80	100
A	OVERALL LENGTH	1.43	1.40	1.74	1.99	2.06	2.50	2.44	2.75	2.88	2.70	3.00	5.90	6.73
B	INSIDE DIAMETER	.53	.75	.88	1.06	1.25	1.75	2.14	2.81	3.75	4.75	5.84	7.80	9.90

PART DWB (COUPLER X BUTT WELD)

DIM	DESCRIPTION	E05	07	10	12	15	20	25	30	40	50	60	80	100
A	O.D. WITH CAM ARMS EXTENDED	4.21	4.51	5.10	6.86	7.16	7.54	8.03	9.56	10.66	11.67	16.26	18.23	20.66
B	OUTSIDE DIAMETER	1.82	2.11	2.44	3.26	3.56	3.94	4.43	5.46	6.56	7.57	10.16	12.13	14.56
C	INSIDE DIAMETER	.56	.78	.97	1.25	1.54	1.88	2.38	2.88	3.50	4.50	5.50	7.50	9.40
D	OVERALL LENGTH	2.00	2.00	2.51	2.89	2.81	3.20	3.63	3.82	4.00	4.20	4.54	4.80	5.50



AWB



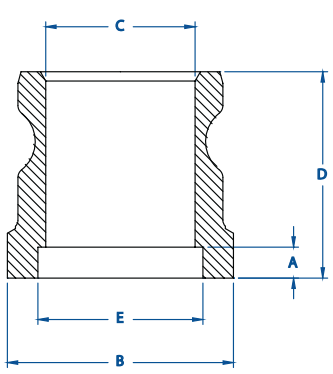
DWB

PART AWS (ADAPTER X SOCKET WELD)

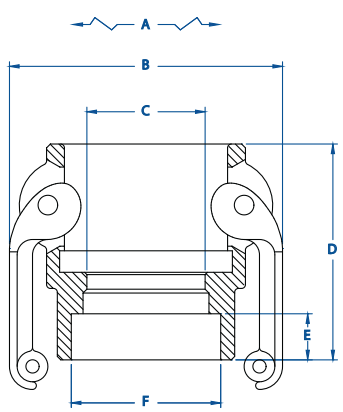
DIM	DESCRIPTION		E05	07	10	12	15	20	25	30	40	50	60	80	100
A	SOCKET DEPTH		.30	.30	.30	.30	.40	.40	.40	.40	.60	.60	.60	.90	.90
B	OUTSIDE DIAMETER		1.00	1.25	1.45	2.00	2.25	2.74	3.25	3.75	5.01	6.00	7.00	10.30	12.80
C	INSIDE DIAMETER		.53	.75	.88	1.06	1.35	1.81	2.14	2.81	3.00	4.78	5.84	7.86	9.77
D	OVERALL LENGTH		1.00	1.13	1.44	2.00	2.13	2.50	2.56	2.75	2.97	2.94	2.90	4.50	4.60
E	SOCKET DIAMETER	PIPE	.88	1.10	1.36	1.71	1.95	2.44	2.94	3.57	4.58	5.66	6.72	8.74	10.89
		TUBE	.52	.77	1.02	1.27	1.52	2.02	2.52	3.02	4.03	5.03	6.03	8.04	10.05

PART DWS (COUPLER X SOCKET WELD)

DIM	DESCRIPTION		E05	07	10	12	15	20	25	30	40	50	60	80	100
A	O.D. WITH CAM ARMS EXTENDED		4.21	4.51	5.10	6.86	7.16	7.54	8.03	9.56	10.66	11.67	16.26	18.23	20.66
B	OUTSIDE DIAMETER		1.82	2.11	2.44	3.26	3.56	3.94	4.43	5.46	6.56	7.57	10.16	12.13	14.40
C	INSIDE DIAMETER		.56	.88	.97	1.25	1.50	1.88	2.38	2.88	3.60	4.50	5.50	7.50	9.40
D	OVERALL LENGTH		2.00	2.06	2.50	2.70	2.81	3.10	3.40	3.50	3.90	4.20	4.54	4.80	5.50
E	SOCKET DEPTH		.50	.50	.50	.50	.50	.70	.70	.70	.70	.90	.90	.90	.90
F	SOCKET DIAMETER	PIPE	0.88	1.10	1.36	1.71	1.95	2.44	2.94	3.57	4.58	5.66	6.72	8.74	10.89
		TUBE	0.52	.77	1.02	1.27	1.52	2.02	2.52	3.02	4.03	5.03	6.03	8.04	10.05



AWS



DWS

Basic Standard Parts - W-Adapter

W-ADAPTER
DUST PLUG
Made in the U.S.A.

All Stainless Steel cam and groove fittings are 316 Stainless Steel.

W-Adapter line drawings & dimensions are on page 167.

PT Coupling maintains inventories at all of their branch warehouse locations.

		Aluminum	Brass	Stainless Steel
Size	Part Name	Item #	Item #	Item #
1/2"	E05W	1001201	1201201	1401201
3/4"	07W	1001207	1201207	1401207
1"	10W	1001210	1201210	1401210
1-1/4"	12W	1001212	1201212	1401212
1-1/2"	15W	1001215	1201215	1401215
2"	20W	1001220	1201220	1401220
2-1/2"	25W	1001225	1201225	1401225
3"	30W	1001230	1201230	1401230
4"	40W	1001240	1201240	1401240
5"	50W	1001250	1201250	1401250
6"	60W	1001260	1201260	1401260
8"	80W	1001280	1201280	1401280
10"	100W	1001290	1201290	1401290
12"	120W	1001292	☎	☎

		Ductile Iron	Hard Coat Alum	Carbon Steel
Size	Part Name	Item #	Item #	Item #
1/2"	E05W	1801201	2001201	2401201
3/4"	07W	1801207	2001207	2401207
1"	10W	1801210	2001210	2401210
1-1/4"	12W	1801212	2001212	☎
1-1/2"	15W	1801215	2001215	2401215
2"	20W	1801220	2001220	2401220
2-1/2"	25W	1801225	2001225	☎
3"	30W	1801230	2001230	2401230
4"	40W	1801240	2001240	2401240
5"	50W	1801250	2001250	☎
6"	60W	1801260	2001260	☎
8"	80W	1801280	2001280	☎
10"	100W	1801290	☎	☎
12"	120W	☎	☎	☎

		Polypropylene	Food Grade Poly	Nylon
Size	Part Name	Item #	Item #	Item #
3/4"	07W	2701207	2771207	2901207
1"	10W	2701210	2771210	2901210
1-1/4"	12W	2701212	2771212	☎
1-1/2"	15W	2701215	2771215	2901215
2"	20W	2701220	2771220	2901220
3"	30W	2701230	2771230	2901230
4"	40W	2701240	2771240	☎

Accessories & Replacement Parts - SECURE-LOK™

A locking system that secures your couplings. Prevents tampering of cam & groove couplings. Made of 12 Gauge 316 Stainless Steel for strength & corrosion resistance. Prevents removal of coupler cam pins by concealing coupler cam pin area. Can be used for lockout/tagout system. Tag hole size .20" Lock hole size .32"

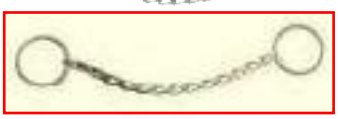
		Stainless Steel	Carbon Steel
Size	Part Name	Item #	Item #
2"	SECURE-LOK™ 20	5007020	☎
3"	SECURE-LOK™ 30	5007030	☎
4"	SECURE-LOK™ 40	5007740	☎

SECURE-LOK™

Made in the U.S.A.

		SS ChainW/SS Hooks	SS Chain W/SS Rings
Length	Part Name	Item #	Item #
6"	S56	5101060A	5101061A
8"	S51	5101080A	5101081A
12"	S52	5101092A	5101093A

SECURITY CHAINS
FOR DUST CAPS & DUST PLUGS
Made in the U.S.A.



		Brass Chain W/SS Hooks	Brass Chain W/Zinc Rings
Length	Part Name	Item #	Item #
6"	C51	5100560A	5100560B
12"	C52	5100592A	5105592A

Length	Width	Part Name	Item #	Description
12"	1"	CAM STRAP 112	700001	Fits up to 1-1/2" couplings
18"	2"	CAM STRAP 218	700002	Fits up to 3" couplings
24"	2"	CAM STRAP 224	700004	Fits up to 5" couplings
36"	3"	CAM STRAP 236	700003	Fits up to 8" couplings

HOOK & LOOP FASTENER CAM STRAPS

Size	Part Name	Item #
1"	10SBL	77041
2"	20SBL	77015
3"	30SBL	77013
4"	40SBL	77014

SAFETY LOCK™
Eliminate Spills. Visually Confirm Ear Locking
Safety Lock™ is an alternative product to the Hook & Loop Fastener Straps with an easy-to-install and easy-to-remove, heavy duty "clip" that clicks into place over the ears.

Recommended Operating Conditions

PT Coupling Company does not recommend cam and groove fittings for any compressed gas service, which includes air and steam service.

Material	Gasket	Size	Max. Allowable Working Pressure (PSI)*	Heavy Duty Max. Allowable Working Pressure (PSI)*
Metal Couplings	Standard gaskets are Buna any variation must be specified	½", ¾ x ½", ¾", 1", 1¼", 1½", 2"	250	500
		2½" 3"	200	400
		4"	150	300
		5", 6"	75	150
		8", 10", 12"	50	N/A
Non-Metal Couplings	Standard gaskets are Buna any variation must be specified	¾", 1", 1¼", 1½", 2"	100	N/A
		3", 4"	50	N/A

**Note: Above maximum allowable working pressures are based on operating temperatures less than 100°F. If operating temperatures exceed 100°F, use temperature derating factor from table below.*

Locate the correct factor from the table below and multiply this by the appropriate MAWP (psi) above for the elevated temperature working pressure.

Example: 2" Alum. at 250°F = 250 psi x .84 = 210 psi at elevated temperature.

Temperature Derating Factors										
Temp (deg F)	**Alum.**	**Brass**	**Stain. Steel**	**Ductile Iron**	**Carbon Steel**	**Hast. C®**	**Carp 20®**	**Monel®**	**Poly**	**Nylon**
***Min.**	**-150 F**	**-150 F**	**-150 F**	**-20 F**	**-20 F**	**-150 F**	**-150 F**	**-150 F**	**-15F**	**-30 F**
100	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
150	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.80	0.90
200	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	NR	0.60
250	0.84	1.00	1.00	1.00	1.00	1.00	1.00	1.00	NR	NR
300	NR	0.97	1.00	1.00	1.00	1.00	1.00	1.00	NR	NR
350	NR	0.95	1.00	1.00	1.00	1.00	1.00	1.00	NR	NR
400	NR	0.92	1.00	1.00	1.00	1.00	1.00	1.00	NR	NR
450	NR	0.90	1.00	1.00	1.00	1.00	1.00	1.00	NR	NR
*500	NR	NR	1.00	1.00	1.00	1.00	1.00	1.00	NR	NR

NR = Not Recommended

**These temperatures represent the extremes that can be handled by current gasket materials.*

MATERIALS AVAILABLE FOR SPECIAL USES ARE:	
Monel®	Aluminum Anodized
Carpenter 20®	Teflon® Lined
Hastelloy C®. Hastelloy C22®	Others Per Request
Tantaline	

These listings are not exclusive—please contact us for parts engineered to fit your needs..

All registered trademarks are the property of their respective owners.

All PT Quick Connect/Disconnect Couplings in every metal and size through 6"parts are manufactured to interchange with all products produced to Commercial Item Description **A-A-59326**. All "RFQ's" identified as Government will be quoted with the required components and castings to meet the **A-A-59326** Commercial Item Description. Couplings **NOT** interchangeable in the industry are sizes 8" through 12". No standards were established by the government for these sizes. Quality is controlled in every phase of manufacture as is required for compliance to **A-A-59326**. PT Coupling's quality management system is certified and conforms to the requirements of ISO9001.

=== PIPING SPECIALTY ITEM LIST ===

CUT SHEET NUMBER:	PCS-CCU-032
DESCRIPTION:	Camlock Connection
VENDOR:	PT Coupling
MODEL:	316 SS SW Adapter with Dust Cap

SELECTION:

PT COUPLING CAMLOCK ADAPTER TO BE USED FOR 1/2" CONNECTIONS. DIXON DOES NOT OFFER 1/2" SW CAMLOCK ADAPTERS.

TO BE USED WHEN 316 STAINLESS STEEL CAMLOCK ADAPTER IS NEEDED IN SOCKETWELD PIPE SCHEDULE METHOD OF CONSTRUCTION.

CUT SHEET BEGINS ON NEXT PAGE

Special Cam & Groove - Weld Fittings

SOCKET WELD FOR PIPE

		Aluminum	Stainless Steel	Hastelloy®C	Carpenter 20	Carbon Steel
Size	Part Name	Item #	Item #	Item #	Item #	Item #
1/2"	E05AWS	☎	64600101	☎	72600101	☎
3/4"	07AWS	☎	64600701	☎	75600701	☎
1"	10AWS	60601002	64601004	76601001	☎	74601001
1"X1-1/2"	10X15AWS	☎	64601006	☎	☎	☎
1-1/4"	12AWS	60601201	64601201	☎	☎	☎
1-1/2"	15AWS	60601502	64601503	76601501	☎	74601501
2"	20AWS	60602004	64602001	76602004	75602013	74602002
2"X1-1/2"	20X15AWS	☎	64602005	☎	☎	☎
2-1/2"	25AWS	60602502	64602504	☎	☎	74602501
3"	30AWS	60603005	64603003	76603003	☎	☎
4"	40AWS	60604005B	64604003	☎	75604001	74604099
4" X 3"	40X30AWS	60604032	☎	☎	☎	☎
5"	50AWS	60605004	64605001	☎	☎	74605003
5" X 4"	50X40AWS	606050401	☎	☎	☎	☎
6"	60AWS	60606001	64606003	☎	☎	74606002
6" X 5"	60X50AWS	60606050	☎	☎	☎	☎
8"	80AWS	60608004	64608001	☎	☎	74608001
10"	100AWS	60609002	64609001	☎	☎	☎

SOCKET WELD FOR TUBE

		Aluminum	Stainless Steel	Hastelloy®C	Carpenter 20	Carbon Steel
Size	Part Name	Item #	Item #	Item #	Item #	Item #
3/4"	07AWS	☎	64600703	☎	☎	☎
1"	10AWS	60601001	64601001	☎	☎	74601002
1-1/4"	12AWS	60601202	☎	☎	☎	☎
1-1/2"	15AWS	60601501	64601504	☎	☎	74601502
2"	20AWS	60602003	64602002	☎	☎	74602003
2" X 1-1/2"	20X15AWS	60602015	☎	☎	☎	☎
2-1/2"	25AWS	60602501	64602503	☎	☎	74602502
3"	30AWS	60603001	64603005	☎	☎	74603002
4"	40AWS	60604007A	64604004	☎	☎	74604001
4" x 3"	40X30AWS	☎	☎	☎	☎	☎
5"	50AWS	60605001	64605004	☎	☎	74605001
5" X 4"	50X40AWS	☎	☎	☎	☎	☎
6"	60AWS	60606002	64606001	☎	☎	74606001
6" x 5"	60X50AWS	☎	☎	☎	☎	☎
8"	80AWS	60608002	64608002	☎	☎	☎
10"	100AWS	☎	☎	☎	☎	☎

AWS-ADAPTER
ADAPTER X
SOCKET WELD
Made in the U.S.A.

All Stainless Steel castings are 316 Stainless Steel.

Other alloys and sizes are available via special quote.

Ordering Information:
- *Adapter Size*
- *Material*
- *Pipe Size & Schedule Number or Tube O.D.*
- *Wall Thickness*
- *Socket Weld*

Example:
- *Adapter Size: 3"*
- *Material: Aluminum*
- *Pipe Size: 3"*
- *Sch. Number: Sch 40*
- *Socket Weld*

AWS line drawings & dimensions are on page 170.

All registered trademarks are the property of their respective owners.

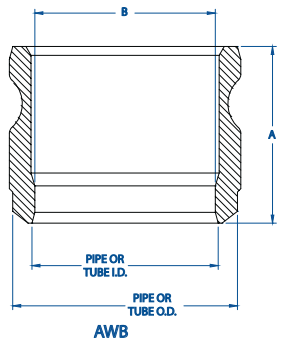
PT COUPLING CO. GENERAL SOCKET WELD DIMENSIONS

PART AWB (ADAPTER X BUTT WELD)

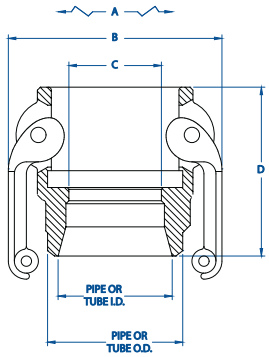
DIM	DESCRIPTION	E05	07	10	12	15	20	25	30	40	50	60	80	100
A	OVERALL LENGTH	1.43	1.40	1.74	1.99	2.06	2.50	2.44	2.75	2.88	2.70	3.00	5.90	6.73
B	INSIDE DIAMETER	.53	.75	.88	1.06	1.25	1.75	2.14	2.81	3.75	4.75	5.84	7.80	9.90

PART DWB (COUPLER X BUTT WELD)

DIM	DESCRIPTION	E05	07	10	12	15	20	25	30	40	50	60	80	100
A	O.D. WITH CAM ARMS EXTENDED	4.21	4.51	5.10	6.86	7.16	7.54	8.03	9.56	10.66	11.67	16.26	18.23	20.66
B	OUTSIDE DIAMETER	1.82	2.11	2.44	3.26	3.56	3.94	4.43	5.46	6.56	7.57	10.16	12.13	14.56
C	INSIDE DIAMETER	.56	.78	.97	1.25	1.54	1.88	2.38	2.88	3.50	4.50	5.50	7.50	9.40
D	OVERALL LENGTH	2.00	2.00	2.51	2.89	2.81	3.20	3.63	3.82	4.00	4.20	4.54	4.80	5.50



AWB



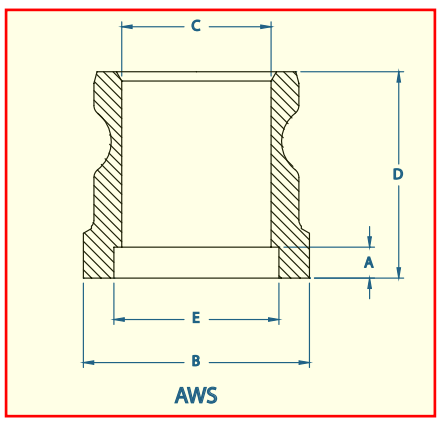
DWB

PART AWS (ADAPTER X SOCKET WELD)

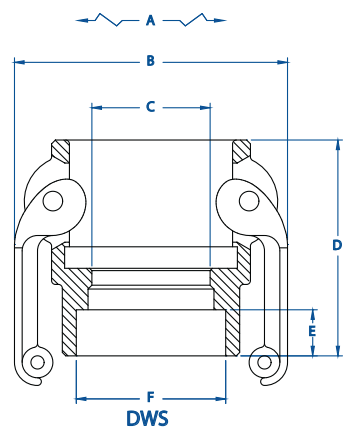
DIM	DESCRIPTION		E05	07	10	12	15	20	25	30	40	50	60	80	100
A	SOCKET DEPTH		.30	.30	.30	.30	.40	.40	.40	.40	.60	.60	.60	.90	.90
B	OUTSIDE DIAMETER		1.00	1.25	1.45	2.00	2.25	2.74	3.25	3.75	5.01	6.00	7.00	10.30	12.80
C	INSIDE DIAMETER		.53	.75	.88	1.06	1.35	1.81	2.14	2.81	3.00	4.78	5.84	7.86	9.77
D	OVERALL LENGTH		1.00	1.13	1.44	2.00	2.13	2.50	2.56	2.75	2.97	2.94	2.90	4.50	4.60
E	SOCKET DIAMETER	PIPE	.88	1.10	1.36	1.71	1.95	2.44	2.94	3.57	4.58	5.66	6.72	8.74	10.89
		TUBE	.52	.77	1.02	1.27	1.52	2.02	2.52	3.02	4.03	5.03	6.03	8.04	10.05

PART DWS (COUPLER X SOCKET WELD)

DIM	DESCRIPTION		E05	07	10	12	15	20	25	30	40	50	60	80	100
A	O.D. WITH CAM ARMS EXTENDED		4.21	4.51	5.10	6.86	7.16	7.54	8.03	9.56	10.66	11.67	16.26	18.23	20.66
B	OUTSIDE DIAMETER		1.82	2.11	2.44	3.26	3.56	3.94	4.43	5.46	6.56	7.57	10.16	12.13	14.40
C	INSIDE DIAMETER		.56	.88	.97	1.25	1.50	1.88	2.38	2.88	3.60	4.50	5.50	7.50	9.40
D	OVERALL LENGTH		2.00	2.06	2.50	2.70	2.81	3.10	3.40	3.50	3.90	4.20	4.54	4.80	5.50
E	SOCKET DEPTH		.50	.50	.50	.50	.50	.70	.70	.70	.70	.90	.90	.90	.90
F	SOCKET DIAMETER	PIPE	0.88	1.10	1.36	1.71	1.95	2.44	2.94	3.57	4.58	5.66	6.72	8.74	10.89
		TUBE	0.52	.77	1.02	1.27	1.52	2.02	2.52	3.02	4.03	5.03	6.03	8.04	10.05



AWS



DWS

Basic Standard Parts - V-Coupler

V-COUPLER
DUST CAP
Made in the U.S.A.

HBS cam arms are made of 300 series Stainless Steel.

HB cam arms are made of Brass.

8, 10 and 12 inch couplers are 4-cam designs with the STA-LOK® positive locking feature.

All Stainless Steel cam and groove fittings are 316 Stainless Steel with 300 series Stainless Steel components, unless otherwise noted.

Please call for special quote on components in 316 Stainless Steel.

V-Coupler line drawings & dimensions are on page 167.

PT Coupling maintains inventories at all of their branch warehouse locations.

		Aluminum/HBS	Aluminum/HB	Brass/HB	Brass/HBS
Size	Part Name	Item #	Item #	Item #	Item #
1/2"	E05V	1001101	1071101	1201101	1291101
3/4"	07V	1001107	1071107	1201107	1291107
1"	10V	1001110	1071110	1201110	1291110
1-1/4"	12V	1001112	1071112	1201112	1291112
1-1/2"	15V	1001115	1071115	1201115	1291115
2"	20V	1001120	1071120	1201120	1291120
2-1/2"	25V	1001125	1071125	1201125	1291125
3"	30V	1001130	1071130	1201130	1291130
4"	40V	1001140	1071140	1201140	1291140
5"	50V	1001150	1071150	1201150	☎
6"	60V	1001160	1071160	1201160	☎
8"	80V	1001180	1071180	1201180	☎
10"	100V	1001190	1071190	1201190	☎
12"	120V	1001192	1071192	1201192	☎

		Stainless Steel	Ductile Iron	Hard Coat Alum	Carbon Steel
Size	Part Name	Item #	Item #	Item #	Item #
1/2"	E05V	1401101	1801101	2001101	☎
3/4"	07V	1401107	1801107	2001107	2401107
1"	10V	1401110	1801110	2001110	2401110
1-1/4"	12V	1401112	1801112	2001112	☎
1-1/2"	15V	1401115	1801115	2001115	2401115
2"	20V	1401120	1801120	2001120	2401120
2-1/2"	25V	1401125	1801125	2001125	2401125
3"	30V	1401130	1801130	2001130	2401130
4"	40V	1401140	1801140	2001140	2401140
5"	50V	1401150	1801150	2001150	☎
6"	60V	1401160	1801160	2001160	2401160
8"	80V	1401180	1801180	2001180	2401180
10"	100V	1401190	1801190	☎	☎
12"	120V	1401192	1801192	☎	☎

		Polypropylene	Food Grade Poly	Nylon
Size	Part Name	Item #	Item #	Item #
3/4"	07V	2701107	2771107	2901107
1"	10V	2701110	2771110	2901110
1-1/4"	12V	2701112	2771112	☎
1-1/2"	15V	2701115	2771115	2901115
2"	20V	2701120	2771120	2901120
3"	30V	2701130	2771130	2901130
4"	40V	2701140	2771140	2901140

Toll-Free: (800) 654-0320 • Toll-Free Fax: (800) 828-2649 • info@ptcoupling.com • www.ptcoupling.com

Accessories & Replacement Parts - SECURE-LOK™

A locking system that secures your couplings. Prevents tampering of cam & groove couplings. Made of 12 Gauge 316 Stainless Steel for strength & corrosion resistance. Prevents removal of coupler cam pins by concealing coupler cam pin area. Can be used for lockout/tagout system. Tag hole size .20" Lock hole size .32"

		Stainless Steel	Carbon Steel
Size	Part Name	Item #	Item #
2"	SECURE-LOK™ 20	5007020	☎
3"	SECURE-LOK™ 30	5007030	☎
4"	SECURE-LOK™ 40	5007740	☎

SECURE-LOK™

Made in the U.S.A.

		SS ChainW/SS Hooks	SS Chain W/SS Rings
Length	Part Name	Item #	Item #
6"	S56	5101060A	5101061A
8"	S51	5101080A	5101081A
12"	S52	5101092A	5101093A

SECURITY CHAINS
FOR DUST CAPS & DUST PLUGS
Made in the U.S.A.

		Brass Chain W/SS Hooks	Brass Chain W/Zinc Rings
Length	Part Name	Item #	Item #
6"	C51	5100560A	5100560B
12"	C52	5100592A	5105592A

Length	Width	Part Name	Item #	Description
12"	1"	CAM STRAP 112	700001	Fits up to 1-1/2" couplings
18"	2"	CAM STRAP 218	700002	Fits up to 3" couplings
24"	2"	CAM STRAP 224	700004	Fits up to 5" couplings
36"	3"	CAM STRAP 236	700003	Fits up to 8" couplings

HOOK & LOOP FASTENER CAM STRAPS

Size	Part Name	Item #
1"	10SBL	77041
2"	20SBL	77015
3"	30SBL	77013
4"	40SBL	77014

SAFETY LOCK™
Eliminate Spills. Visually Confirm Ear Locking
Safety Lock™ is an alternative product to the Hook & Loop Fastener Straps with an easy-to-install and easy-to-remove, heavy duty "clip" that clicks into place over the ears.

Toll-Free: (800) 654-0320 • Toll-Free Fax: (800) 826-2649 • info@ptcoupling.com • www.ptcoupling.com

Accessories & Replacement Parts - Coupling Gaskets

COUPLING GASKET

Made in the U.S.A.

Buna N Standard
1 Blue Stripe

Buna N Fuel Resistant
2 Blue Stripes

Neoprene Gasket
1 White Stripe

	Buna N Standard		Buna Fuel Resistant		Neoprene		Viton®	
Size	Part Name	Item #	Part Name	Item #	Part Name	Item #	Part Name	Item #
1/2"	B05Y	5500205	☎	☎	N05Y	5500405	V05Y	5500805
3/4"	B01	5500207	B01F	5500207F	N01	5500407	V01	5500807
1"	B02	5500210	B02F	5500210F	N02	5500410	V02	5500810
1-1/4"	B03	5500212	B03F	5500212F	N03	5500412	V03	5500812
1-1/2"	B04	5500215	B04F	5500215F	N04	5500415	V04	5500815
2"	B05	5500220	B05F	5500220F	N05	5500420	V05	5500820
2-1/2"	B06	5500225	B06F	5500225F	N06	5500425	V06	5500825
3"	B07	5500230	B07F	5500230F	N07	5500430	V07	5500830
4"	B08	5500240	B08F	5500240F	N08	5500440	V08	5500840
5"	B09	5500250	B09F	5500250F	N09	5500450	V09	5500850
6"	B010	5500260	B010F	5500260F	N010	5500460	V010	5500860
8"	B011	5500280	☎	☎	N011	5500480	V011	5500880
10"	B012	5500290	☎	☎	☎	☎	V012	5500890
12"	B013	5500292	☎	☎	☎	☎	☎	☎

VITON® 1/2" - 6"
Green

VITON® 8" & 10"*
1 Green Stripe

TEFLON® ENVELOPED COUPLING GASKET

Made in the U.S.A.

	Teflon® Enveloped Viton® Filler		Teflon® Enveloped Neoprene Filler		Teflon® Enveloped Buna Filler	
Size	Part Name	Item #	Part Name	Item #	Part Name	Item #
1/2"	TOFV05Y	5503905	TOFN05Y	5501605	TOFB05Y	5503705
3/4"	TOFV1	5503907	TOFN1	5501607	TOFB1	5503707
1"	TOFV2	5503910	TOFN2	5501610	TOFB2	5503710
1-1/4"	TOFV3	5503912	TOFN3	5501612	TOFB3	5503712
1-1/2"	TOFV4	5503915	TOFN4	5501615	TOFB4	5503715
2"	TOFV5	5503920	TOFN5	5501620	TOFB5	5503720
2-1/2"	☎	☎	TOFN6	5501625	TOFB6	5503725
3"	TOFV7	5503930	TOFN7	5501630	TOFB7	5503730
4"	TOFV8	5503940	TOFN8	5501640	☎	☎
6"	TOFV10	5503960	TOFN10	5501660	☎	☎

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Recommended Operating Conditions

PT Coupling Company does not recommend cam and groove fittings for any compressed gas service, which includes air and steam service.

Material	Gasket	Size	Max. Allowable Working Pressure (PSI)*	Heavy Duty Max. Allowable Working Pressure (PSI)*
Metal Couplings	Standard gaskets are Buna any variation must be specified	½", ¾ x ½", ¾", 1", 1¼", 1½", 2"	250	500
		2½" 3"	200	400
		4"	150	300
		5", 6"	75	150
		8", 10", 12"	50	N/A
Non-Metal Couplings	Standard gaskets are Buna any variation must be specified	¾", 1", 1¼", 1½", 2"	100	N/A
		3", 4"	50	N/A

**Note: Above maximum allowable working pressures are based on operating temperatures less than 100°F. If operating temperatures exceed 100°F, use temperature derating factor from table below.*

Locate the correct factor from the table below and multiply this by the appropriate MAWP (psi) above for the elevated temperature working pressure.

Example: 2" Alum. at 250°F = 250 psi x .84 = 210 psi at elevated temperature.

Temperature Derating Factors										
Temp (deg F)	**Alum.**	**Brass**	**Stain. Steel**	**Ductile Iron**	**Carbon Steel**	**Hast. C®**	**Carp 20®**	**Monel®**	**Poly**	**Nylon**
***Min.**	**-150 F**	**-150 F**	**-150 F**	**-20 F**	**-20 F**	**-150 F**	**-150 F**	**-150 F**	**-15F**	**-30 F**
100	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
150	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.80	0.90
200	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	NR	0.60
250	0.84	1.00	1.00	1.00	1.00	1.00	1.00	1.00	NR	NR
300	NR	0.97	1.00	1.00	1.00	1.00	1.00	1.00	NR	NR
350	NR	0.95	1.00	1.00	1.00	1.00	1.00	1.00	NR	NR
400	NR	0.92	1.00	1.00	1.00	1.00	1.00	1.00	NR	NR
450	NR	0.90	1.00	1.00	1.00	1.00	1.00	1.00	NR	NR
*500	NR	NR	1.00	1.00	1.00	1.00	1.00	1.00	NR	NR

NR = Not Recommended

**These temperatures represent the extremes that can be handled by current gasket materials.*

MATERIALS AVAILABLE FOR SPECIAL USES ARE:	
Monel®	Aluminum Anodized
Carpenter 20®	Teflon® Lined
Hastelloy C®. Hastelloy C22®	Others Per Request
Tantaline	

These listings are not exclusive—please contact us for parts engineered to fit your needs..

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All PT Quick Connect/Disconnect Couplings in every metal and size through 6"parts are manufactured to interchange with all products produced to Commercial Item Description **A-A-59326**. All "RFQ's" identified as Government will be quoted with the required components and castings to meet the **A-A-59326** Commercial Item Description. Couplings **NOT** interchangeable in the industry are sizes 8" through 12". No standards were established by the government for these sizes. Quality is controlled in every phase of manufacture as is required for compliance to **A-A-59326**. PT Coupling's quality management system is certified and conforms to the requirements of ISO9001.

=== PIPING SPECIALTY ITEM LIST ===

CUT SHEET NUMBER:	PCS-FLX-054
DESCRIPTION:	Flexible Hose Assembly
VENDOR:	Aflex
MODEL:	Bioflex Ultra 316 SS White Antistatic Hose with MannTek Dry Disconnect Couplers and Dust Plugs

NOTE:

USED FOR THE FOLLOWING SERVICES
Acetonitrile
Alcohol, Isopropyl
Dichloromethane
Dimethyl Sulfoxide
Dimethylformamide
General Solvent
Methyl Tert Butyl Ether
Sodium Hydroxide, Industrial 50%
Sulfuric Acid

CUT SHEET BEGINS ON NEXT PAGE

Bioflex Ultra Hose Assembly Part Number System

If required, Bioflex Ultra Hose Assembly can be defined by an individual Part Number, made up of **7** entries as below:

1

Hose Size	Size Part No.
3/8"	06
1/2"	08
5/8"	10
3/4"	12
7/8"	14
1"	16
1 3/8"	22
1 1/2"	24
1 7/8"	30
2"	32
2 1/2"	40
3"	48

2

Hose Type	Type Part No.
Bioflex Ultra GP (Natural PTFE Liner)	BFXU/GP
Bioflex Ultra AS (Antistatic PTFE Liner)	BFXU/AS

3

Braid and Cover	
Tube Only	TO
Stainless Steel Braid	SS
Polypropylene Braid	PB
Blue EPDM Rubber Cover (on SS)	RC
Transparent Silicone Rubber (on SS)	SI
Black Fireproof, Antistatic Rubber (on SS)	BK
RC-300 Rubber Covered End Protection Systems (Pg 16) see *note below	

***Note** - *If one of the rubber end protection systems is required, for one or both ends, please define the requirement in writing in addition to Part Number.*

4

External Protection Systems	
No External Protection System	OO
SS Wire Protection Coil	PC
Rubber Anti-Scuff Rings	SR
'Safeguard' HDPE Spiral Wrap	SG

5

Length
The overall hose length between the sealing faces at each end is given as the **Length Part No** either in decimal Metres followed by 'm' or inches followed by 'in'

6 & 7

Assembled End Fitting Description * All Components in Stainless Steel	End Fitting Part No.
JIC Female	02
Fixed Male Pipe, NPT Thread	03
Fixed Male Pipe, BSPT Thread	03/B
Fixed Female Pipe, NPT Thread	06
JIC-to-NPT Male Union	08
JIC-to-Female Union	08F
Straight Sanitary Tri Clamp, 1.984" Diameter 0.870" Exit Diameter (Standard) 1.370" Exit Diameter (Step Up)	- 10 10/S
Straight Mini Sanitary Tri Clamp, 0.984" Diameter 0.370" Exit Diameter (Standard) 0.625" Exit Diameter (Step-Up)	- 11 11/S
*ANSI 150# Swivelling Flange Non-Lined	12
DIN PN 10/16 Swivelling Flange Non-Lined	12/PN
*ANSI 150# Swivelling Flange, PTFE Lined	12L
DIN PN 10/16 Swivelling Flange, PTFE Lined	12L/PN
Cam and Groove, Locking Arm Swivelling Female, Non-Lined	16
Cam and Groove, Locking Arm Female, PTFE Lined	16L
Cam and Groove Male, Non-Lined	17
Cam and Groove, Male PTFE Lined	17L
DIN 11851 Female, PTFE Lined	23L
DIN 11851 Male, PTFE Lined	24L
SMS Female, PTFE Lined	26L
RJT Female, Non-Lined	27
BSPP Cone Seat Female	33
BSP Lug Nut Female	34

Notes - ***For Flange only:*** *Add '**/ZP**' for Carbon Steel Zinc Plated, or add '**/EC**' for Epoxy coated*

Elbows: *For all types of Elbow fittings add '**/90°**' for 90° elbows*

Triclamps: *For 'Hot Formed' PTFE Lined Triclamps add '**/HF**'*

Additional Requirements:

- Any additional requirements which are not included in the Part Number must be written out in full in the Order, including any special labelling or colour coding.

- When purchasing Triclovers/Sanitary Fittings please specify the surface finish required.

Example:

A 3/4" bore [illegible] an outer Safegard Sle[illegible]

End (1) - a 3/[illegible]

End (2) - a C[illegible]

and a Lengt[illegible]

48 - BFXU/AS - RC - OO - Xin* - 03 - 03

SPECIFY WHITE RUBBER COVER AND HOSE TAGGING REQUIREMENTS

*SEE LAST PAGE FOR HOSE LENGTH DETAILS

Hose Assembly Part No. =	**12**	**-**	**BFXU/AS**	**-**	**RC**	**-**	**SG**	**-**	**54in**	**-**	**12L**	**-**	**16L**
Entry No.	**1**		**2**		**3**		**4**		**5**		**6**		**7**

Bioflex Ultra Hose Design and Comparative Properties

Bioflex Ultra Hose Design Options

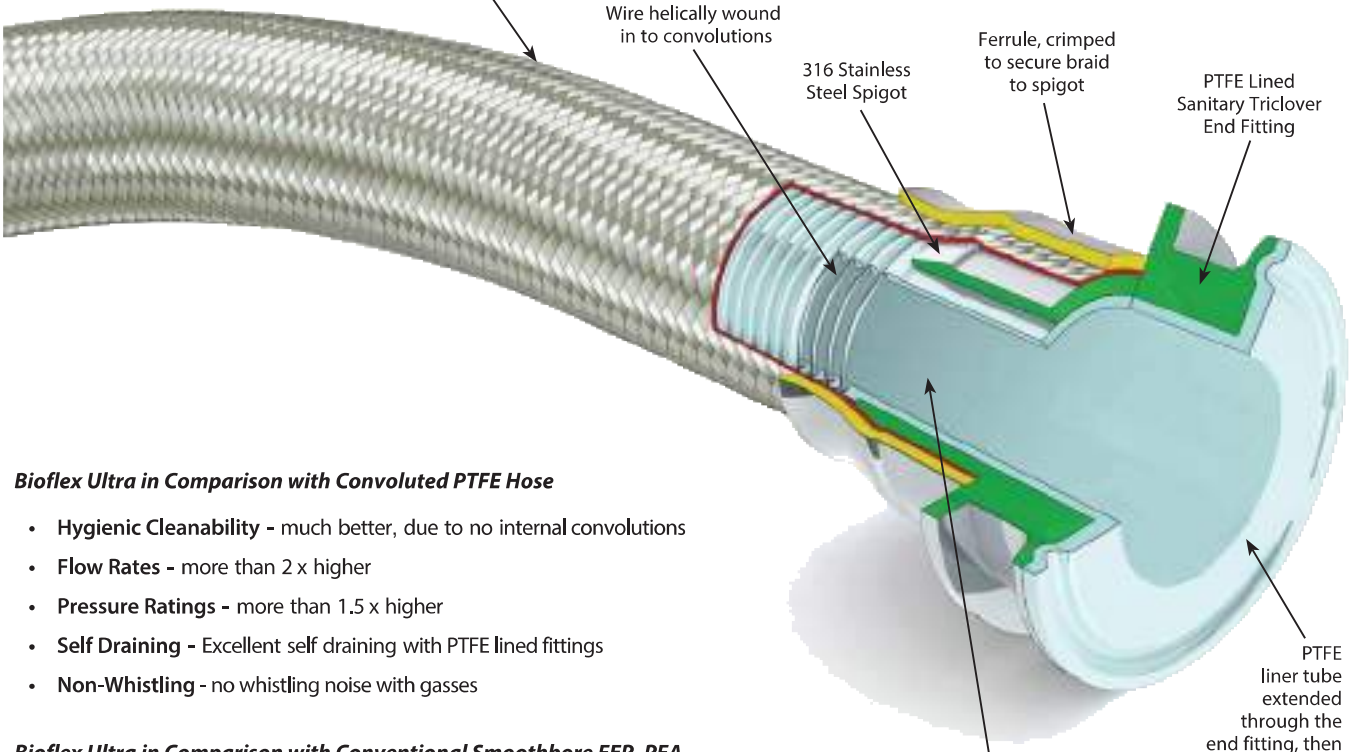
Bioflex Ultra Hose Grades are made up by combining the Design Options, which are defined by 2 letters as shown below, and fully described on the indicated pages.

For example, "Bioflex Ultra AS, PB, SG" defines a hose with an Antistatic PTFE liner (AS) and a Polypropylene braid (PB) and an outer "Safegard" (SG) spiral HDPE protection sleeve.

PTFE Liner Tube Options -	**GP** (Natural PTFE) and **AS** (Antistatic PTFE) -	Page 15
Braid Options -	**TO** (Tube Only, No Braid), **SS** (316 SS Wire) and **PB** (Polypropylene) -	Page 16
Rubber Cover Options -	**RC** (Blue EPDM), **BK** (Black EPDM) and **SI** (Silicone Rubber) - (WHITE EPDM)	Page 17
External Protection Options -	**SR** (Scuff Rings), **SG** (Safegard Spiral) and **PC** (SS Wire Coil) -	Page 18

Note: Rubber covered hose is marked in accordance with EN 16643

BIOFLEX ULTRA GP, SS WITH A PTFE LINED SANITARY TRICLOVER FITTING



Bioflex Ultra in Comparison with Convoluted PTFE Hose

- **Hygienic Cleanability -** much better, due to no internal convolutions
- **Flow Rates -** more than 2 x higher
- **Pressure Ratings -** more than 1.5 x higher
- **Self Draining -** Excellent self draining with PTFE lined fittings
- **Non-Whistling** - no whistling noise with gasses

Bioflex Ultra in Comparison with Conventional Smoothbore FEP, PFA or PTFE Lined Rubber Hose

- **No Adhesives -** No toxic adhesives used in the Bioflex Ultra construction, eliminating the possibilities of Process Fluid contamination in service
- **Flexibility -** Much better flexibility
- **Internal Shape Control** - No distortions in bore when flexed

Bioflex Ultra in Comparison with Silicone Rubber Hose

- **Hygienic Cleanability -** much better, due to the non-stick PTFE Liner
- **Chemical Resistance -** considerably improved, particularly to strong oxidising acids and bases
- **Temperature & Pressure Ratings -** much higher temperature and pressure capability
- **Steam Resistance -** permanently resistant to steam sterilising (unlike silicone hose, which has a limited life)

Bioflex Ultra Specifications: Temperatures, Pressures & Flow Rates

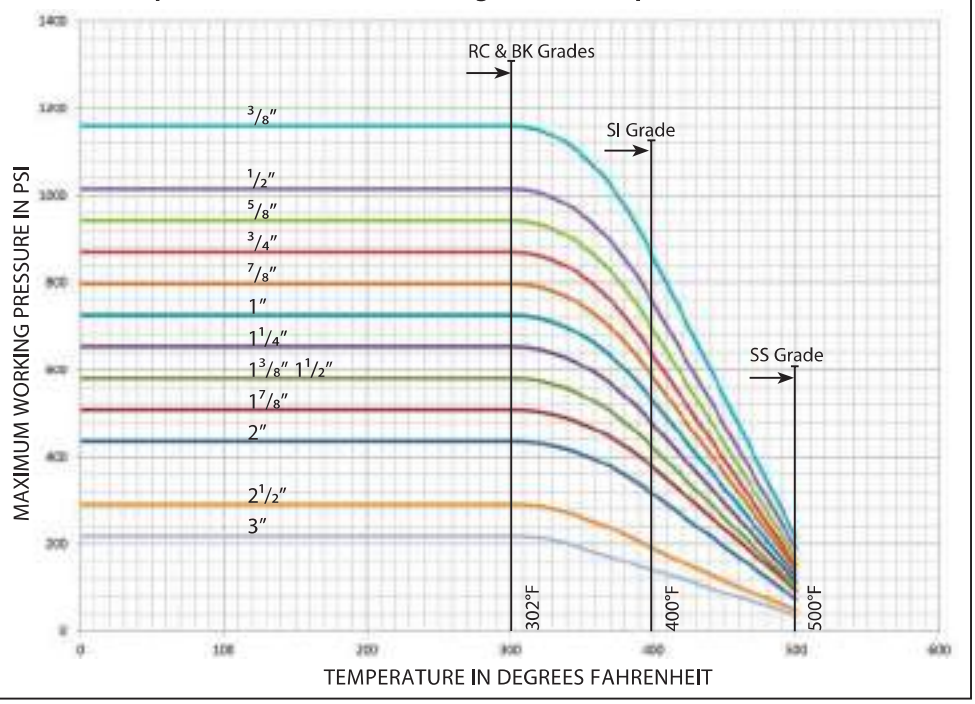
Temperature vs Pressure

Due to its extremely strong construction, Bioflex Ultra has outstanding resistance to temperature and pressure, much higher than that of conventional convoluted PTFE lined hose.

Maximum Working Pressure (MWP) Variation with Temperature:

MWP for SS grade hose, also for RC, BK and SI grades are as given in the graph, BUT only within the temperature range for the particular grade of hose, as given below:

Temperature & Maximum Working Pressure Graph for Bioflex Ultra



Min/Max Operating Temperatures:

Bioflex Ultra GP or AS	SS	-99.4˚F to +500˚F
Bioflex Ultra GP or AS	SS,RC & SS, BK	-40˚F to +302˚F
Bioflex Ultra GP or AS	SS, SI	-99.4˚F to +399.2˚F

Bioflex Ultra GP or AS PB - MWPs for PB are as given in the graph up to 176˚F, then reduced by 50% between 176˚F and 212˚F (the maximum for PB).

Temperature vs Vacuum

All sizes and grades of Bioflex Ultra hose are usable at vacuum to -0.9bar up to 392˚F (212˚F for TO grades). Above this, the vacuum resistance should be reduced 2% for every degree above 392˚F (212˚F for TO grades).

Flow Rates

In practice, flow rates will vary with hose flexing, fluid viscosity, end fitting design and other parameters, but Bioflex Ultra hose flow rates are always around 2 times better than convoluted PTFE hose, when comparing hose assemblies with PTFE lined (non bore-restricted) end fittings.

Bioflex Ultra Hose - Flow Rate Calculation

If it is required to determine the flow rate of a particular hose assembly, or if it is required to determine the pressure required to generate a certain flow rate, then this can sometimes be approximately calculated by the Bioflex Ultra supplier.

It should be noted that calculations can only be made for fluids with a viscosity equal to water, and for hose assemblies with PTFE lined end fittings (no bore restrictions at the ends of the hose).

The following information should be given to the supplier:

To calculate the Flow Rate in Cubic Metres per Hour:

- Pressure in Bars at the Entry into the Hose Assembly
- Pressure in Bars at the Exit from the Hose Assembly > (Subtracted to calculate Pressure Drop over the Hose Length
- The hose configuration (roughly straight, or 33% Bends, or 66% Bends, or 100% Tightly Coiled)

OR To Calculate the Pressure Drop in bars over the length of the Hose Assembly:

- Required Flow Rate in Cubic Metres per Hour
- The hose configuration (roughly straight, or 33% Bends, or 66% Bends, or 100% Tightly Coiled)

Bioflex Ultra Sizes, Grades, Bend Radius and Dimensions

Nominal Hose Bore Size		Actual Bore Size		Bioflex Ultra Grade (Braid & Cover)	Helical Wire	O/D of Tube, Braid or Rubber		Minimum Bend Radius		† Maximum Continuous Hose Length	
in	mm	in	mm			in	mm	in	mm	Ft	Mtrs
3/8	9.5	0.375	9.5	TO	-	0.47	12.0	1 3/8	35	100	30
		0.375	9.5	SS	-	0.50	12.8	3/4	19	100	30
		0.375	9.5	RC/BK/SI	-	0.68	17.4	3/4	19	100	30
1/2	15	0.530	13.5	TO	√	0.654	16.4	2 3/8	60	100	30
		0.530	13.5	SS	√	0.700	17.8	1 1/2	38	100	30
		0.530	13.5	PB	√	0.800	20.5	1 1/2	38	100	30
		0.530	13.5	RC/BK/SI	√	0.845	21.4	1 1/2	38	100	30
5/8	16	0.650	16.5	TO	√	0.780	20.1	2 1/2	64	100	30
		0.650	16.5	SS	√	0.850	21.6	1 3/4	45	100	30
		0.650	16.5	PB	√	0.955	24.3	1 3/4	45	100	30
		0.650	16.5	RC/BK/SI	√	0.990	25.2	1 3/4	45	100	30
3/4	20	0.780	19.8	TO	√	0.920	23.4	3	75	100	30
		0.780	19.8	SS	√	0.980	24.9	2	50	100	30
		0.780	19.8	PB	√	1.085	27.6	2	50	100	30
		0.780	19.8	RC/BK/SI	√	1.120	28.5	2	50	100	30
* 7/8	22	0.875	22.2	TO	√	1.040	26.4	3 1/2	90	100	30
		0.875	22.2	SS	√	1.090	27.7	2 1/8	55	100	30
		0.875	22.2	PB	√	1.240	31.5	2 1/8	55	100	30
		0.875	22.2	RC/BK/SI	√	1.260	32.0	2 1/8	55	100	30
1	25	1.030	26.1	TO	√	1.200	30.5	4 3/4	110	100	30
		1.030	26.1	SS	√	1.255	31.9	2 3/4	70	100	30
		1.030	26.1	PB	√	1.455	37.0	2 3/4	70	100	30
		1.030	26.1	RC/BK/SI	√	1.455	37.0	2 3/4	70	100	30
1 1/4	32	1.280	32.5	TO	√	1.490	37.8	5 1/2	140	100	30
		1.280	32.5	SS	√	1.555	39.5	4	100	100	30
		1.280	32.5	PB	√	1.740	44.2	4	100	100	30
		1.280	32.5	RC/BK/SI	√	1.755	44.6	4	100	100	30
* 1 3/8	35	1.375	34.9	TO	√	1.590	40.3	6 1/2	160	100	30
		1.375	34.9	SS	√	1.665	42.3	4	100	100	30
		1.375	34.9	PB	√	1.850	47.0	4	100	100	30
		1.375	34.9	RC/BK/SI	√	1.865	47.4	4	100	100	30
1 1/2	40	1.530	38.8	TO	√	1.750	44.5	7	180	100	30
		1.530	38.8	SS	√	1.840	46.7	5 1/2	140	100	30
		1.530	38.8	PB	√	2.020	51.4	5 1/2	140	100	30
		1.530	38.8	RC/BK/SI	√	2.035	51.7	5 1/2	140	100	30
* 1 7/8	48	1.875	47.6	TO	√	2.145	54.4	11	280	100	30
		1.875	47.6	SS	√	2.220	56.4	6 5/8	170	100	30
		1.875	47.6	PB	√	2.410	61.3	6 5/8	170	100	30
		1.875	47.6	RC/BK/SI	√	2.410	61.3	6 5/8	170	100	30
2	50	2.030	51.5	TO	√	2.320	58.9	12	300	100	30
		2.030	51.5	SS	√	2.390	60.7	8	200	100	30
		2.030	51.5	PB	√	2.575	65.4	8	200	100	30
		2.030	51.5	RC/BK/SI	√	2.580	65.6	8	200	100	30
2 1/2	65	2.508	63.7	TO	√	2.882	73.2	17 3/4	450	60	18
		2.508	63.7	SS	√	2.965	75.3	11 7/8	300	60	18
		2.508	63.7	PB	√	3.181	80.8	11 7/8	300	60	18
		2.508	63.7	RC/BK/SI	√	3.169	80.5	11 7/8	300	60	18
3	80	3.024	76.8	TO	√	3.394	86.2	20 3/4	525	50	15
		3.024	76.8	SS	√	3.457	87.8	13 3/4	350	50	15
		3.024	76.8	PB	√	3.732	94.8	13 3/4	350	50	15
		3.024	76.8	RC/BK/SI	√	3.654	92.8	13 3/4	350	50	15

*The 7/8", 1 3/8" and 1 7/8" hose sizes are only suitable for use with PTFE lined sanitary clamp (or triclover) end fittings and PTFE lined I-Line end fittings.

† Longer lengths may be available to special order

Bioflex Ultra Sizes, Grades, Pressures & Weights

Nominal Hose Bore Size		Actual Bore Size		Bioflex Ultra Grade (Braid & Cover)	Helical Wire	**Maximum Working Pressure of Hose		Burst Pressure		Weight per Unit Length	
in	mm	in	mm			Bar	psi	Bar	psi	Kg/Mtr	lb/ft
3/8	9.5	0.375	9.5	TO	-	5	72	20	290	.06	.04
		0.375	9.5	SS	-	80	1160	500	7200	.14	.09
		0.375	9.5	RC/BK/SI	-	80	1160	500	7200	.22	.15
1/2	15	0.530	13.5	TO	√	5	72	20	290	.15	.10
		0.530	13.5	SS	√	70	1015	400	5800	.29	.19
		0.530	13.5	PB	√	35	500	140	2000	.22	.15
		0.530	13.5	RC/BK/SI	√	70	1015	400	5800	.39	.26
5/8	16	0.650	16.5	TO	√	5	72	20	290	.17	.11
		0.650	16.5	SS	√	65	940	380	5500	.35	.23
		0.650	16.5	PB	√	33	480	130	1900	.25	.17
		0.650	16.5	RC/BK/SI	√	65	940	380	5500	.47	.31
3/4	20	0.780	19.8	TO	√	5	72	20	290	.20	.13
		0.780	19.8	SS	√	60	870	300	4350	.40	.27
		0.780	19.8	PB	√	30	440	120	1750	.28	.19
		0.780	19.8	RC/BK/SI	√	60	870	300	4350	.55	.37
* 7/8	22	0.875	22.2	TO	√	4	60	16	230	.34	.23
		0.875	22.2	SS	√	55	800	220	3200	.60	.40
		0.875	22.2	PB	√	27.5	400	110	1600	.44	.30
		0.875	22.2	RC/BK/SI	√	55	800	220	3200	.82	.55
1	25	1.030	26.1	TO	√	4	60	16	230	.36	.24
		1.030	26.1	SS	√	50	720	200	2900	.63	.42
		1.030	26.1	PB	√	25	360	100	1450	.47	.31
		1.030	26.1	RC/BK/SI	√	50	720	200	2900	.92	.62
1 1/4	32	1.280	32.5	TO	√	3	43	12	175	.45	.30
		1.280	32.5	SS	√	45	650	180	2600	.85	.57
		1.280	32.5	PB	√	23	330	90	1300	.72	.48
		1.280	32.5	RC/BK/SI	√	45	650	180	2600	1.15	.77
* 1 3/8	35	1.375	34.9	TO	√	2	29	8	115	.68	.46
		1.375	34.9	SS	√	40	580	160	2320	1.14	.77
		1.375	34.9	PB	√	20	290	80	1160	1.00	.67
		1.375	34.9	RC/BK/SI	√	40	580	160	2320	1.51	1.01
1 1/2	40	1.530	38.8	TO	√	2	29	8	116	.66	.44
		1.530	38.8	SS	√	40	580	160	2320	1.10	.74
		1.530	38.8	PB	√	20	290	80	1160	.90	.60
		1.530	38.8	RC/BK/SI	√	40	580	160	2320	1.55	1.04
* 1 7/8	48	1.875	47.6	TO	√	2	29	8	115	1.12	.75
		1.875	47.6	SS	√	35	500	140	2000	1.70	1.14
		1.875	47.6	PB	√	18	250	72	1040	1.40	.94
		1.875	47.6	RC/BK/SI	√	35	500	140	2000	2.22	1.49
2	50	2.030	51.5	TO	√	2	29	8	116	1.25	.84
		2.030	51.5	SS	√	30	430	120	1750	1.90	1.27
		2.030	51.5	PB	√	15	215	60	870	1.60	1.07
		2.030	51.5	RC/BK/SI	√	30	430	120	1750	2.56	1.71
2 1/2	65	2.508	63.7	TO	√	4	60	16	230	1.99	1.33
		2.508	63.7	SS	√	20	290	80	1160	2.58	1.73
		2.508	63.7	PB	√	12	174	48	696	2.38	1.59
		2.508	63.7	RC/BK/SI	√	20	290	80	1160	3.59	2.41
3	80	3.024	76.8	TO	√	3	43	12	175	2.45	1.64
		3.024	76.8	SS	√	15	215	60	870	3.13	2.10
		3.024	76.8	PB	√	10	145	40	580	3.02	2.02
		3.024	76.8	RC/BK/SI	√	15	215	60	870	4.30	2.96

*The 7/8", 1 3/8" and 1 7/8" hose sizes are only suitable for use with PTFE lined sanitary clamp (or triclover) end fittings and PTFE lined I-Line end fittings.

** The Maximum Working Pressure of a hose assembly is limited to the lowest of the MWP's of either of the two end fittings, or of the hose itself as listed above. The MWP of the hose reduces as the operating temperature increases as specified in the Graph on the 'Bioflex Ultra Specification' page 7.

Bioflex Ultra Hose Liners

*GP - General Purpose Liner

Purpose

Bioflex Ultra GP is the 'General Purpose' grade, for use in all applications where fluids or gases are being conveyed which do not generate a risk of static charge development (see "AS").

Design & Approvals

A full list of Approvals is given on Page 11.

Bioflex Ultra GP is a virgin PTFE liner, manufactured from hose grade PTFE which conforms to the requirements of:

FDA 21 CFR 177.1550

Bioflex Ultra GP hose liner tube has also been tested, and complies with USP Class VI, at 37˚C (99˚F), 70˚C (158˚F) and at 121˚C (250˚F) - see page 11.

Bioflex Ultra GP and AS PTFE Liner Tubes have also been tested for Leachables and Extractables in accordance with BPSA recommendations, and were found to be satisfactory.

EC - ELECTRICAL CONTINUITY
(Also known as 'Electrically Bonded')

All Bioflex Ultra hose assemblies are electrically continuous, except TO grade hose assemblies. Electrical Continuity requires that the hose assembly supplied is electrically continuous, or conductive, between metal end fittings at each end of the hose (whether GP or AS grade).

The requirements for this are specified in the German Document BRG 132 and EN 16643, when tested in accordance with EN ISO 8031, which requires that the resistance between end fittings shall be <100 Ohms per assembly. For hose assemblies which meet this requirement a Grade "M" marking is applied in accordance with EN 16643.

Grade TO (Tube Only) hose assemblies are not normally EC (Electrically Continuous), which means that the electrical resistance between end fittings will usually be more than 100 ohms. However, if EC or non-EC are specific requirements for the TO hose assembly, then it is possible to ensure either, but only if the requirement is discussed in detail with Aflex Hose at the enquiry stage.

*AS - Anti-Static PTFE Liner

Purpose

Bioflex Ultra AS is an essential requirement in applications where there is the risk of an electrostatic charge build-up on the inside surface of the PTFE tube which may then discharge through the tube wall. Media passing through which create such a risk are fluids which have a Conductance of less than 10^8 S/m (Siemens per Metre), or 10^4 pS/m such as fuels, solvents, freons, some WFI (ultra-pure "Water for Injection") and non-polar organics which are being transferred at a medium to high flow velocity.

All twin or multi phase media, and any non-mixing media, such as powder in air, or water droplets in steam, in gases or in oil, also colloidal fluids constitute a particular hazard for static charge generation, and always require grade AS.

If in doubt, either order AS grade, or consult Aflex Hose.

Design & Approvals

Bioflex Ultra AS is an anti-static PTFE liner manufactured from FDA 21 CFR 177.1550 approved PTFE, mixed with less than 2.5% of 'high purity' Carbon Black material to FDA requirement 21 CFR 178.3297. Bioflex Ultra AS grade also meets the requirements of EC Directive 2007/19/EC. The carbon is encapsulated by the PTFE, and in normal, non-abrasive applications will not come loose to contaminate any fluid passing through.

Bioflex Ultra AS also conforms to USP Class VI, at 37˚C (99˚F), 70˚C (158˚F) and at 121˚C (250˚F) - see page 11.

Antistatic Hose Assemblies

When "AS" (Antistatic) grade hose is specified, then the hose or hose assembly supplied will be tested in accordance with EN ISO 8031 and meet the Antistatic requirements of EN 16643. This requires, for an antistatic liner or antistatic cover, that the resistance between an appropriately placed foam electrode and a metallic end fitting will be between 10^3 to 10^8 ohms per assembly. For hose assemblies which meet these requirements an appropriate Grade "Ω" marking is applied in accordance with EN 16643.

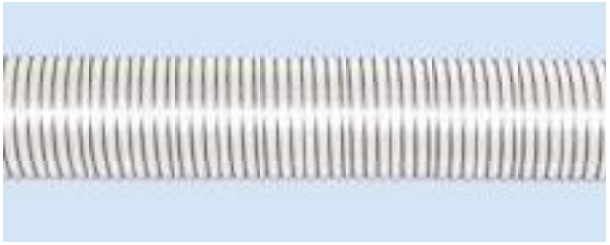
Note: When in service, at least one end fitting must be connected to earth, to permit dissipation of the static charge from the end fitting.

EXCEPTIONS -

The $^3/_8$" hose size is not changed from the original Bioflex design. The hose liner tubes do not therefore, have increased bore size, and does not include a helical wire reinforcement as shown.

Bioflex Ultra Hose Braids

TO - Tube Only (no braid)



Purpose

TO grade hose (available in both GP and AS) is a lightweight hose, used in applications where working pressures are low and where there is no need for the physical protection offered by an external braid. TO grade tube is vacuum resistant to -0.9bar up to 100˚C.

Note: *2½" & 3" TO cannot be ordered as an assembly.*

SS - Stainless Steel Braid

Purpose

Stainless Steel braided hose is the general purpose product, and can be used in applications involving high temperatures and working pressures. High tensile AISI 316 stainless steel wire is used, to give maximum pressure resistance and external protection to the hose.

PB - Polypropylene Braid

Purpose

Polypropylene braided hose is often preferred to SS in applications involving frequent handling and movement of the hose, and where temperatures are within the range -30˚C and +100˚C (-22˚F to +212˚F). PB braid is lighter in weight, and any broken strands will not cut the operator's hands. In addition, PB braid is not prone to "chloride stress corrosion", and has generally good chemical resistance.

To ensure Electrical Continuity (EC) between end fittings 2 strands of Monel wire earthing strips are criss-crossed on to the liner, underneath the Polypropylene Braid. These are folded back underneath the ferrule at both end fittings, to make an electrical contact.

Note: Prolonged exposure to sunlight eventually results in UV degradation of PB braid.

Note: Previously, Bioflex HB (Hastelloy Braid) and KYB (Kynar Braid) were available, particularly for chlorine applications. It has been found however, that customers prefer the Corroflon product for such applications - please refer to the Corroflon brochure.

Note: Previously, Bioflex HB (Hastelloy Braid) and KYB (Kynar Braid) were available, particularly for chlorine applications. It has been found however, that customers prefer the Corroflon product for such applications - please refer to the Corroflon brochure.

Bioflex Ultra Rubber Covers

RC - Blue EPDM Rubber Covered

Purpose

For the most rugged applications where the hose may be subjected to rough treatment and severe external abrasion. Also for hygienic applications, where the external smoothness and cleanability of the hose is of prime importance.

Design

An SS braided hose assembly has a smooth finish, blue EPDM external rubber cover extruded over and vulcanised directly onto the braid. EPDM has excellent chemical resistance, and the hose has a temperature range from -40˚C, -40˚F up to +150˚C, +302˚F. Alternative colours of rubber, and strips with alternative text titles are available to special order.

Specifications

The Blue EPDM rubber cover has been tested and conforms to the requirements of USP Class VI.

Limitations

Rubber covers are only applicable to SS braided hose.

SI - Platinum Cured Silicone Rubber Cover

Purpose

As for RC hose, but where the hose may be required to withstand temperatures from -73˚C, -100˚F up to +204˚C, +400˚F. SI grade hose is semi-transparent, allowing visual monitoring of the braid.

Design

An SS braided hose assembly has a platinum cured silicone rubber cover extruded directly onto the braid, with a very smooth external surface finish to aid cleaning.

Specifications

The Silicone rubber cover has been tested and conforms to the requirements of USP Class VI.

BK - Black EPDM Rubber Covered

Purpose

Bioflex Ultra BK grade is designed to provide an alternative Rubber Cover which has the additional properties of making the hose Fireproof, and also making the rubber cover Anti-static.

Design

Bioflex Ultra BK grade hose is Fireproof to BS5173 Section103.13 Part 6.2 and 6.3. When end fittings are assembled to the hose, the assembly is Fire Resistant, but can be made Fireproof by adding DBK-300 rubber cover end protection. Bioflex Ultra BK meets the requirements of EN 16643 flame resistance.

Specifications

The Black EPDM Rubber Cover is anti-static in accordance with specification EN 16643 which specifies a resistance of 10^3 to 10^8 Ohms per hose assembly, and meets the requirements for 'Ω' marking.

Note: When in service at least one end fitting must be connected to Earth.

RC-300 - Rubber Covered 300mm Long End Protection

Purpose

In applications where excessive flexing of the hose at the end fitting occurs, it is sometimes necessary to 'stiffen' the hose in this area, to prevent kinking.

Design

A layer of rubber is applied and vulcanised directly to the ferrule, and 300mm (12 inch) along the hose from the fitting. This can be done either on an SS braided hose (RC-300) or on a rubber covered hose to form a 300mm (12 inch) long double layer of rubber at the end (DRC-300).

The rubber used is normally Blue EPDM, but if the hose is BK, or SI covered, then the same type and colour of rubber would be used (DBK-300 or DSI-300).

Limitations

Cannot be applied to PB braided hose. If required for a PB hose, consult Aflex Hose for an alternative "EPR" system. (EPR includes a 300mm (12 inch) length of loose rubber hose jubilee clipped to the ferrule).

Bioflex Ultra Standard & Streamline Tagging and Colour Coding Systems

Standard Labelling

All Bioflex Ultra hose assemblies are labelled with the following information:

Manufacturer's Name (Aflex Hose Ltd)

Hose Size and Grade

EN16643 and year of standard publication

EN16643 electrical property grade

Max. Working Pressure and test pressure

Working temperature range*

Unique Serial Number

Month & Year of Manufacture

Aflex Hose Telephone Number

CE Mark (if applicable)

**Note any restrictions on working pressure resulting from elevated temperatures.*

This information is normally laser-etched on to a loose stainless steel Ring mounted on the hose.

In some cases, at the discretion of Aflex Hose, the information may be etched on to a thin stainless steel plate which is clamped to the hose, or on to the end fitting ferrule at one end. This may be necessary for example, if the customer requires additional information which may not fit on to a Ring.

Customers may specify which labelling system they require, and may request additional information on the label.

Streamline Tagging

This system is only applicable to the Silicone (SI) rubber covered grades of Bioflex Ultra that have a stainless steel (SS) braid.

A label and/or Colour Code is placed around the silicone cover of the hose and then encapsulated by a transparent silicone that is formed into a thin streamlined cover.

Note: ½" size, Colour Code only, no text.

Colour Coding

This system is applicable to all grades of Bioflex Ultra hose.

A coloured PTFE spiral strip is wound on to the hose.

It can be left loose, or it can be encapsulated under a transparent, heat-shrunk polyolefin sleeve.

INCLUDE FOLLOWING DETAILS IN HOSE LABEL:
- MANUFACTURER
- HOSE TYPE / MODEL
- MONTH & YEAR OF MANUFACTURE
- P.O. NUMBER
- WETTED PARTS MATERIAL OF CONSTRUCTION (MOC)
- MAXIMUM WORKING PRESSURE AND TEMPERATURE OF ASSEMBLY
- FULL ITEM LOCATION TAG (E.G. LP150FLX1000A)
- APPROVAL / CERTIFICATION MARKS (E.G. FDA, NSF, USP, ETC.)

Bioflex Ultra NPT & BSPT Fixed Male and NPT Fixed Female Fitting

End Fitting Specifications

- NPT Taper Threads to American National Standard Pipe Taper Thread design to ANSI/AMSE B1.20.1.
- BSPT Threads to British Standard Pipe Taper Thread design to BS21

Alternatives - Parallel Threads, Metric Threads and Others.

End Fitting Materials

- Fittings in Grade 316L SS
- Ferrules, most in Grade 304 SS, some sizes in Grade 316L SS
- Available in other materials to special order

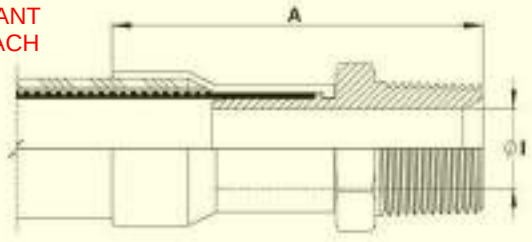
Alternatives - Hydraulic design BSPT Male Fittings in Zinc Plated Carbon Steel, to special order.

Surface Finish :

- All surface finishes are to ASME BPE-SF-O (No finish required).
- If a specified finish on a particular surface is required, please state on the enquiry and order.

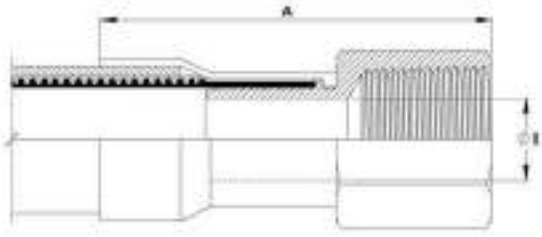
Fixed Male NPT or BSPT

MALE NPT BOTH HOSE ENDS.
USE GASOILA SOFT-SET THREAD SEALANT WITH TEFLON OR EQUIVALENT TO ATTACH TO DRY DISCONNECT COUPLINGS.



Nominal Hose Size		NPT or BSPT Thread Size	*Fitting Length A (Bioflex Ultra RC)		Fitting Inside Diameter I		Weight of Fitting	
in	mm	in	in	mm	in	mm	Kg	Lbs
1/2	13	1/2	3.30	84	0.38	9.53	0.10	0.22
3/4	20	3/4	3.74	95	0.63	15.88	0.18	0.40
1	25	1	4.13	105	0.80	20.24	0.29	0.64
1 1/4	32	1 1/4	4.90	124	1.00	25.40	0.45	0.99
1 1/2	40	1 1/2	5.19	132	1.25	31.75	0.60	1.32
2	50	2	5.78	142	1.75	44.45	0.84	1.85
** 2 1/2	65	2 1/2	5.31	135	2.25	57.15	1.84	4.04
** 3	80	3	5.39	137	2.63	66.7	2.49	5.47

Fixed Female NPT



Nominal Hose Size		NPT Thread Size	*Fitting Length A (Bioflex Ultra RC)		Fitting Inside Diameter I		Weight of Fitting	
in	mm	in	in	mm	in	mm	Kg	Lbs
1/2	13	1/2	3.42	87	0.38	9.53	0.18	0.40
3/4	20	3/4	3.66	93	0.63	15.88	0.22	0.49
1	25	1	4.13	105	0.80	20.24	0.33	0.73
1 1/2	40	1 1/2	4.80	122	1.25	31.75	0.75	1.65
2	50	2	4.96	126	1.75	44.45	1.06	2.34

**Fitting Lengths listed are for Bioflex Ultra RC, SI and BK hose grades. Shorter lengths apply for other hose grades.*
*** 2 1/2" and 3" sizes are only available as Direct Crimp Fittings*

3" (Ø 119 mm) Technical Information

Tank Unit (Adapter) and Hose Unit (Coupler)

According to NATO STANAG 3756

Material	Maximum Working Pressure	Test Pressure	Minimum Burst Pressure
Aluminium	10 bar / 145 psi	15 bar / 218 psi	50 bar / 726 psi
Brass/Gunmetal	16 bar / 232 psi	24 bar / 348 psi	80 bar / 1160 psi
Stainless Steel	25 bar / 363 psi	37,5 bar / 544 psi	125 bar / 1813 psi
Titan	25 bar / 363 psi	37,5 bar / 544 psi	125 bar / 1813 psi
Hastelloy	25 bar / 363 psi	37,5 bar / 544 psi	125 bar / 1813 psi
PEEK	6 bar / 87 psi	9 bar / 131 psi	30 bar / 435 psi

Connections
3" in BSP, NPT and flanged inlet.

Applications
For road, rail tanker loading/unloading, filling of tanks and containers, refuelling and loading/unloading bunker boats (marine tankers) and small ships.
Also suitable for other applications where high loading rates are required and product contamination and spillage needs to be minimized.

Media
Petroleum products: gasoline, diesel, oil etc.
Chemical products: ethylene oxide, propylene oxide, acrylonitrile, butadiene, ammonia, vinyl chloride, toluene, xylene, sulphuric acid, phenol etc.
Gas: vapor recovery/balance systems
Dry powder.

Material of Body
Aluminium, brass/gunmetal, stainless steel, Hastelloy and PEEK. Other materials on request.

Seals
Standard seals FPM (Viton®), EPDM, FFKM (Chemraz®, Kalrez®), NBR. Other on request.

High Flow Rates / Low Pressure Drop
Allows maximum product transfer with minimal losses.

Recommended Maximum Flow Rates
2000 litres/minute (fuel)

Selectivity
- avoid mixing products
To avoid product contamination caused by connecting a hose unit to the wrong tank unit, selective versions of the hose and tank units are available.
Each unit has a number of selective positions, designated by a coded part number according to the coupling size – specify when placing order.

Interchangeability
Compatible with couplings of other manufacturers.

Hose Unit with Integrated Swivel
All hose units are supplied with an integrated swivel.

3" (Ø 119 mm) Hose Unit (Coupler) – Female Thread

Connection [1] Inch/DN	Body Material [2]	Seal [3]		Weight ≈		Mann Tek Code No.
		O-Ring	Thread Seal	kg	lbs	
F 3" BSP	Al	SPECIFY TEFLON (PTFE) O-RING	PUR	3,8	8.4	S414B1101B
F 3" NPT			—	3,9	8.6	S415B1101
F 3" BSP	GM		PUR	8,4	18.5	S414B2201B
F 3" NPT			—	9,0	19.8	S415B2201
F 3" BSP	SS		PTFE	8,4	18.5	S414B4401A
F 3" NPT			—	8,7	19.2	S415B4401A
F 3" BSP	Hastelloy		PTFE			S414A7701B
F 3" NPT			—	8,1	17.7	S415A7701
F 3" BSP	PVDF/Hastelloy		PTFE			S414A8701B
F 3" BSP	PEEK		PTFE			S414A9901B
F 3" NPT			—			S415A9901

[1] F = Female thread, BSP = EN ISO 228, NPT = ANSI B1.20.1
[2] Material: Al = Aluminium, GM = Gunmetal, SS = Stainless Steel
[3] Standard seal FPM/FKM. Alternative materials, e.g. EPDM, Chemraz®, Kalrez®, NBR or HNBR on request

According to NATO STANAG 3756

(QTY = 2)
CONNECT DRY DISCONNECT COUPLER UNITS TO BOTH MALE THREADED ENDS OF HOSE ASSEMBLY.

3" (Ø 119 mm) Coupler – Flanged Inlet

Flange [1]	Body Material [2]	Seal [3]	Weight ≈		Mann Tek Code No.
		O-Ring	kg	lbs	
undrilled Ø 210 mm	Al	Standard: FPM/FKM (Viton®) Other on request.			S420B1101
DN 80 PN 10 / 16 Type A					S436B1101
3" ASA 150			5,5	12.1	S461B1101
DIN 28459 'TW 1' / DN 80					S465B1101
DIN 28459 'TW 3' / DN 100					S466B1101
3" TTMA					S467B1101
4" TTMA			5,5	12.1	S468B1101
undrilled Ø 210 mm	GM				S320B2201
DN 80 PN 10/16 Type B					S436B2201
DN 80 PN 25/40 Type B					S437B2201
3" ASA 150					S461B2201
3" ASA 300					S462B2201
DIN 28459 'TW 1' / DN 80					S465B2201
DIN 28459 'TW 3' / DN 100					S466B2201
3" TTMA					S467B2201
4" TTMA					S468B2201
undrilled Ø 210 mm	SS				S420B4401
o. Bohrungen Ø 210 mm**)					S420B4401F
DN 80 PN 10/ 6 Type B			12,7	28.0	S436B4401
DN 80 PN 10/16 Type B**)					S436B4401F
DN 80 PN 25/40 Type B					S437B4401
DN 80 PN 25/40 Type B**)					S437B4401F
DN 80 PN 25/40 Type E					S435B4401F
3" ASA 150			13,2	29.1	S461B4401
3" ASA 150**)			13,2	29.1	S461B4401F
3" ASA 300					S462B4401
3" ASA 300**)			14,6	32.2	S462B4401F
DIN 28459 'TW 1' / DN 80					S465B4401
DIN 28459 'TW 3' / DN 100					S466B4401
3" TTMA					S467B4401
4" TTMA					S468B4401
undrilled Ø 210 mm	PEEK				S420B9901
DN 80 PN 10/16 Type B					S436B9901
3" ASA 150					S461B9901

According to NATO STANAG 3756

*) Type E (with spigot), EN 1092-1, see page 44

**) Flange with standard thickness

[1] PN 10/16/25/40 = EN 1092 (types see page 44), ASA = ANSI B16.5 (150 o. 300 psi), DIN 28459 = old standard TW, 10 bar. TTMA = Truck Trailer Manufacturers Association

[2] Material: Al = Aluminium, GM = Gunmetal, SS = Stainless Steel

[3] Standard seal FPM/FKM. Alternative materials, e.g. EPDM, Chemraz®, Kalrez®, NBR or HNBR on request

Viton® and Teflon® are registred trademarks of DuPont, DuPont Elastomers.

O-Ring Materials

Designation	Trade name	ISO 1629	ASTM 1418	Temp. Range °C		Field of Application
Nitrile Butadiene Rubber / Nitrile Rubber	Buna N® Europrene N® Hycar® Nipol N® Perbunan ®	NBR	NBR	-55	110	Petroleum based products / aliphatic hydrocarbons (gasoline, diesel, oils, petroleum, propane, butane). Animal and vegetal oils and fats. Flame retardant liquids.. Silicone oils and greases. Water up to 80°C.
Ethylene-Propylene Diene Rubber	Dutral Keltan® Vistalon® Buna AP®	EPDM	EPDM	-55	120	Water, hot water, vapour, brake fluids, detergents. Alcohols, ketons, engine coolants, phosphate based flame retardants, organic and inorganic acids and alkalis. Not suitable for petroleum based products.
Fluoroelastomer	Fluorel® Technoflon® Viton®	FPM	FKM	-20	200	Petroleum based products / aliphatic hydrocarbons (gasoline, diesel, oils, petroleum, propane, butane) - also with high additive content. Phosphate based flame retardants. Silicone oils and greases, acids and alkalis.
Fluorosilicone Elastomer	——	MFQ	FVMQ	-60	200	Additional to the properties of silicone elastomers, fluorosilicone elastomers show a consideably higher resistance to oils, fuels and solvents - especially to aromatic and aromatic and chlorinated hydrocarbons and alcohols.
Silicone	Silastic® Silopren®	MVQ	VMQ	-60	200	Very good heat and cold resistance, very good resistance to oxygene, ozone, UV rays and wheather influence. High electrical insulating properties, physiologically safe.
Hydrogenated Nitrile Butadiene Rubber	Therban® Tornac® Zetpol®	HNBR	HNBR	-35	120	Very good resistance to petroleum based products / aliphatic hydrocarbons like NBR, but better resistance to high temperature, ageing and wheather influence. Excellent physical properties also with high temperatures.
Butyl Rubber	Exxon Butyl® Polysar Butyl®	IIR	IIR	-55	100	For O-Rings this material has largely been replaced by EPDM. Butyl is resistant to the same fluid types as EPDM and -except for resistance to gas permeation- it is somewhat inferior to EPDM for O-Ring use. Excellent weather resistance and gas permeation resistance. Not suitable for petroleum based products.
Perfluorinated Elastomer	CHEMRAZ® Kalrez® PERLAST®	FFPM	FFPM	-40	260	Best chemical resistance of all elastomers, including organic acids, acetic acid, benzoic acid, formic acid.
Chloroprene Rubber	Baypren® Neoprene®	CR	CR	-40	120	Resistant to refrigerants, ammonia, carbon dioxide, freon (R12, R13, R21, R22, R113 - R115), silicone oils, water, oxygen (low pressure), alkalis, coustic soda, alcohols, chlorine, ozone, castor oil and vegetable oils. Properties of CR are similar to NBR but slightly lower resistance to petroleum based producs, better resistance to ageing, UV light and ozone.
Polyester / Polyether Urethane Rubber	Polyurethane Adiprene® Urepan® Desmopan®	PU PUR AU EU	AU EU	-40	100	Very good resistance to petroleum based products, also with aromatic content. Good resistance to ozone and ageing. Very good mechanical properties. Not resistant to esters, concentrated acids and lyes, water above +50° C.
Polytetrafluoroethylene	Teflon®	PTFE	PTFE	-200	260	Universal chemical resistance except liquid alkaline metals and some fluorine compounds. Very good anti friction properties, low wear. High temperature resistance. Physiologically safe. Use for statical seals only.
Fluorinated Ethylene Propylene	Teflon FEP®	FEP/MVQ	FEP/VMQ	-60	200	Universal chemical resistance except liquid alkaline metals and some fluorine compounds. Very good anti friction properties, low wear. High temperature resistance. Physiologically safe.
		FEP/FPM	FEP/FKM	-20	200	
Perflouralkoxy	Teflon PFA®	PFA/MVQ	PFA/VMQ	-60	250	Universal chemical resistance except liquid alkaline metals and some fluorine compounds. Very good anti friction properties, low wear. High temperature resistance. Physiologically safe.
		PFA/FPM	PFA/FKM	-20	250	
Tetrafluoroethylene-Propylene Copolymer Elastomer	Aflas®	——	TFE / P	-25	200	Petroleum based products, oils and greases, brake fluids, fuels, alcohols, heat transfer media, oils. amines, acids, alkalis.

***Note**: Specifications in this chart are based on the information given by our suppliers together with published guides. This is not always a guarantee for a proper function. If in doubt, please ask our sales team, statiing medium and operating conditions.*

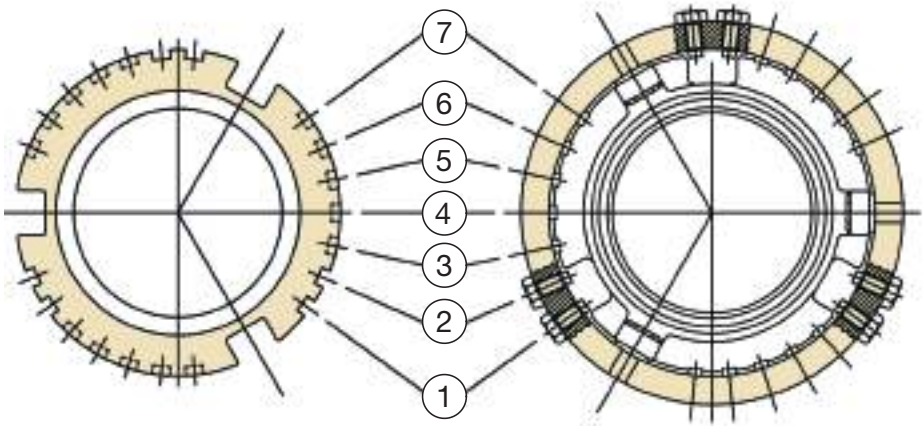
Selectivity Coding Table

Item Tag	Service Description	SEL #
LP1Y FLX-1751A1	Dimethylformamide	SEL 1
LP1Y FLX-1762A2	Dimethyl Sulfoxide	SEL 2
LP1Y FLX-1771A1	Alcohol, Isopropyl	SEL 3
LP1Y FLX-1772A1	Dichloromethane	SEL 4
LP1Y FLX-1773A1	Acetonitrile	SEL 5
LP1Y FLX-1774A1	Methyl Tert Butyl Ether	SEL 6
LP1Y FLX-1781A1	Sulfuric Acid	SEL 7
LP1Y FLX-1782A1	Sodium Hydroxide, Industrial 50%	SEL 12
LP1Y FLX-3201A1	General Solvent (Solvent 1)	SEL 13
LP1Y FLX-3202A1	General Solvent (Solvent 2)	SEL 14
LP1Y FLX-3203A1	General Solvent (Solvent 3)	SEL 15
LP1Y FLX-3204A1	General Solvent (Solvent 4)	SEL 16
LP1Y FLX-3211A1	General Solvent (Solvent 5)	SEL 17
LP1Y FLX-3221A1	General Solvent (Solvent 6)	SEL 23
LP1Y FLX-3222A1	General Solvent (Solvent 7)	SEL 24

Selectivity System for 2½" (Ø 105 mm)

Coupling diameter:
Ø 105 mm (2½")

Tank Unit **Hose Unit**



With 7 Positions

1	2	3	4	5	6	7

Selectivity System:
NATO STANAG 3756. Interchangable with other manufacturers.

Each unit can be provided with selectivity according to two systems – with 7 or 21 positions (see chart).

When ordering, please add "/SEL" and the chosen selectivity to the Mann Tek Code No.
Examples: S312A4401**/SEL 1** or S312A4401**/SEL 12** (A)

With 21 Positions

12	13	14	15	16	17	23	24	25	26	27
34	35	36	37	45	46	47	56	57	67	

Order Number Breakdown – 1/2

1. Type of Coupling

A = API Adapter
AV = Tank Unit EN 13081 (special version)
B = Ball Valve
C = Dust Cap
CG = Dust Cap LPG
D = Swivel
E = Tank Unit with pressure valves
F = Hose Unit (ISO 45)
G = Tank Unit (ISO 45)
GS = Tank Unit (ISO 45) with selectivity
H = Sampling Vent & Drain Unit
I = Dust Plug ISO 45
K = Dust Cap ISO 45
L = Tank Unit LPG
LC = Tank Unit Cryogenic
M = Hose Unit LPG
MC = Hose Unit Cryogenic
N = Break Away Pin
NC = Break Away Pin Cryogenic
O = Break Away Wire
P = Dust Plug
R = Pressure Cap
RG = Pressure Cap LPG
S = Hose Unit (STANAG 3756)
SN = Hose Unit int. Break Away Pin
SO = Hose Unit int. Break Away Wire
T = Tank Unit (STANAG 3756)
U = Filter / Sight Glass
V = Dust Plug LPG
WA = Hose Fittings

2. Socket Diameter / Nominal Diameter

0 = 50 mm or ¾"
1 = 56 mm or 1", 1¼"
2 = 70 mm or 1½", 2"
3 = 105 mm or 2½"
4 = 119 mm or 3"
5 = 164 mm or 4"
V = 5"
6 = 238 mm or 6"
8 = 272 mm or 8"
10 = 10"
12 = 12"

3. Connection (Thread, Flange etc.)

01 = ¾" BSP (female)
02 = ¾" NPT (female)
03 = 1" BSP (female)
04 = 1" NPT (female)
05 = 1¼" BSP (female)
06 = 1¼" NPT (female)
07 = 1½" BSP (female)
08 = 1½" NPT (female)
09 = 1¾" BSP (female)
10 = 2" BSP (female)
11 = 2" NPT (female)
12 = 2½" BSP (female)
13 = 2½" NPT (female)
14 = 3" BSP (female)
15 = 3" NPT (female)
16 = 4" BSP (female)
17 = 4" NPT (female)
18 = Flange undrilled Ø 156
19 = Flange undrilled Ø 165
20 = Flange undrilled Ø 210
21 = Flange undrilled Ø 230
22 = Flange undrilled Ø 254
23 = Flange DN 25 PN 10/16
24 = Flange DN 25 PN 25/40
25 = Flange DN 32 PN 10/16
26 = Flange DN 32 PN 25/40
27 = Flange DN 40 PN 10/16
28 = Flange DN 40 PN 25/40
29 = Flange DN 50 PN 25/40*
30 = Flange DN 50 PN 10/16
31 = Flange DN 50 PN 25/40
32 = Flange DN 65 PN 25/40*
33 = Flange DN 65 PN 10/16
34 = Flange DN 65 PN 25/40
35 = Flange DN 80 PN 25/40*
36 = Flange DN 80 PN 10/16
37 = Flange DN 80 PN 25/40
38 = Flange DN 100 PN 25/40*
39 = Flange DN 100 PN 10/16
40 = Flange DN 100 PN 25/40
41 = Flange DN 125 PN 6
42 = Flange DN 125 PN 10/16
43 = Flange DN 125 PN 25/40
44 = Flange DN 150 PN 6
45 = Flange DN 150 PN 10/16
46 = Flange DN 150 PN 25/40
47 = Flange DN 20 PN 10/16
48 = Flange DN 20 PN 25/40
49 = Flange ¾" ANSI Class 150
50 = Flange ¾" ANSI Class 300
51 = Flange 1" ANSI Class 150
52 = Flange 1" ANSI Class 300
53 = Flange 1¼" ANSI Class 150
54 = Flange 1¼" ANSI Class 300
55 = Flange 1½" ANSI Class 150
56 = Flange 1½" ANSI Class 300
57 = Flange 2" ANSI Class 150
58 = Flange 2" ANSI Class 300
59 = Flange 2½" ANSI Class 150
60 = Flange 2½" ANSI Class 300
61 = Flange 3" ANSI Class 150
62 = Flange 3" ANSI Class 300
63 = Flange 4" ANSI Class 150
64 = Flange 4" ANSI Class 300
65 = Flange TW 1 (3" - DN 80)
66 = Flange TW 3 (4" - DN 100)
67 = Flange 3" T.T.M.A.
68 = Flange 4" T.T.M.A.
69 = ¾" BSP (male)
70 = ¾" NPT (male)
71 = 1" BSP (male)
72 = 1" NPT (male)
73 = 1¼" BSP (male)
74 = 1¼" NPT (male)
75 = 1½" BSP (male)
76 = 1½" NPT (male)
77 = 1¾" BSP (male)
78 = 2" BSP (male)
79 = 2" NPT (male)
80 = 2½" BSP (male)
81 = 2½" NPT (male)
82 = 3" BSP (male)
83 = 3" NPT (male)
84 = 4" BSP (male)
85 = 4" NPT (male)
86 = Weld.flange 2" Ø 60,5 ID
87 = Flange TW 1 (2" DN 50)
88 = Weld.flange 2" Ø 50 - Ø 70 (flat)
89 = Weld.flange 2" Ø 57 (int. chamfer)
90 = Weld.flange 2" Ø 60 (outer chamfer)
91 = Weld.flange 3" Ø 75 - Ø 90 (flat)
92 = Weld.flange 3" Ø76 (int. chamfer)
93 = Weld.flange 3" Ø 89 (outer chamfer)
94 = Weld.flange 4" Ø 100 - Ø 120 (flat)
95 = Weld.flange 4" Ø 102 (int. chamfer)
96 = Weld.flange 4" Ø 108 (int. chamfer)
97 = Weld.flange 4" Ø 114 (outer chamfer)
98 = Flange TW 1 (2" - DN 50) with drain connection
99 = Flange DN 150 PN 25
100 = Flange 6" ANSI Class 150
101 = Flange 6" ANSI Class 300
102 = Flange DN 200 PN 10
103 = Flange DN 200 PN 16
104 = Flange DN 200 PN 25
105 = Flange 8" ANSI Class 150
106 = Flange 8" ANSI Class 300
107 = Flange Square ISO 45
108 = S60 x 6 (female)
109 = S60 x 6 (male)
110 = 6" BSP (female)
111 = 6" NPT (female)
112 = W2" - 7 (female)
113 = Weld.flange 3" Ø 92 inner
114 = Square flange, 4 holes
115 = 6" BSP (male)
116 = 6" NPT (male)
117 = 8" NPT (female)
118 = 4" Victaulic
119 = Flange DN 50 PN 25/40**
120 = Flange DN 65 PN 25/40**
121 = Flange DN 80 PN 25/40**
122 = Flange DN 100 PN 25/40**
123 = W2" - 7 (male)
124 = 5" NPT (female)
125 = 5" NPT (male)
126 = Flange DN 100 PN 6
127 = Flange DN 80 PN 6
128 = Flange DN 65 PN 6
129 = Flange DN 50 PN 6
130 = Flange 8" ANSI Class 600
131 = W90 x 1/6" (female)
132 = ½" NPT (female)
133 = ½" BSP (female)
134 = Flange Ø 184.2, 6 holes
135 = Flange TW 7 (6" - DN 150)
136 = 4" ASSPT (female)
137 = Triclamp DN 25
138 = M54 x 1,5 (female)
139 = Triclamp DN 50
140 = Weld.flange Ø 73 (outer chamfer)
141 = 3" Victaulic
142 = Flange 5" ANSI Class 150
143 = 3" Ball valve
144 = 2" Victaulic
145 = 3" BSPT (male)
146 = 5" Victaulic
147 = 2" BSPT (female)
148 = 2" BSPT (male)
149 = 1½" Victaulic
150 = 2½" Victaulic
151 = Flange 1" DIN 11864-2
152 = Flange 2" DIN 11864-2
153 = Flange Ø 135, 8 x M6
154 = 4" BSPT (female)
155 = 4" BSPT (male)
156 = Weld flange 2" Ø 61,5 ID
157 = 3" BSPT (female)
158 = Weld end 1½" Ø 48 OD
159 = Thread TR 57 x 4
160 = Flange 2" BS10 Table D
161 = Flange 12" ANSI Class 150
162 = Flange 10" ANSI Class 150
163 = Flange DN 250 PN 16
164 = M130 x 6 (female)
165 = Flange 10" ANSI Class 300
166 = ACME 1¼" (female)
167 = ACME 1¾" (female)
168 = ACME 2¼" (female)
169 = ACME 3¼" (female)
170 = ACME 1¼" (male)
171 = ACME 1¾" (male)
172 = ACME 2¼" (male)
173 = ACME 3¼" (male)
174 = Weld.flange Ø 76 (outer chamfer)
175 = Flange DN 15 PN 10/16
176 = Flange DN 15 PN 25/40
177 = M130 x 6 (male)
178 = Flange 6" T.T.M.A.
179 = Flange DN 80 PN 25/40***
180 = ½" NPT (male)
181 = ½" BSP (male)
182 = 5" BSP (female)
183 = 5" BSP (male)
184 = Weld end 8" Ø 219 OD
185 = Weld end 6" Ø 168 OD
186 = Flange DN 250 PN 25
187 = Flange 2" T.T.M.A.
188 = Flange 3" BS10 Table D
189 = Flange ½" ANSI Class 150
190 = Flange 1" ANSI Class 150 Flat Face
191 = Flange 12" ANSI Class 300
192 = Flange DN 250 PN 10
193 = Weld end Ø 114 Schedule 40
194 = Weld end Ø 114 Schedule 80
195 = 6" Victaulic
196 = 1" Victaulic
197 = DN 125 JIS 5K
198 = DN 100 JIS 5K
199 = DN 80 JIS 5K
200 = DN 50 JIS 5K
201 = DN 40 JIS 5K
202 = Flange 2" DIN 11864-3
203 = 3½" BSP (female)
204 = Flange Ø 110, Ø 86/Ø 5.5 (6x)
205 = Weld end Ø 60 Schedule 80
206 = Weld end Ø 89 Schedule 40
207 = Weld end Ø 89 Schedule 80
208 = Flange DN 25 PN 6
209 = Flange DN 32 PN 6
210 = Flange DN 40 PN 6
211 = DN 125 JIS 10K
212 = DN 100 JIS 10K
213 = DN 80 JIS 10K
214 = DN 50 JIS 10K
215 = DN 40 JIS 10K
216 = Flange DN 80, holes Ø 14 (6x)
217 = Flanges 5" ANSI Class 300

* EN 1092-1:2001 Type E: Spigot
** EN 1092-1:2001 Type F
*** EN 1092-1:2001 Type C

NOTE: When swivels are chosen (first sign: D), the second and the third sign indicate thread side 1 one, the fourth and the fifth sign the second outlet indicate thread side 2.

Design may change without notice.

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Order Number Breakdown – 2/2

4. Version

A = Version No. 1 (machined from bar)
B = Version No. 2 (casted)
C = Version No. 3 (kokill casted)
D = Sep. piston guide
E = Injection moulded
F = 6" Flange Hydrant
G = Drain connection
H = Leaf spring lock
I = Bent (dropped) Tank Unit, short (15°)
J = Bent (dropped) Tank Unit (15°)
K = Short Tank Unit/Swivel
N = Non Return Valve
P = Pressure (Custom)
S = Sight Glass
T = Transparent
U = Stop before disconnected

5. Material Coupling and Body

1 = Aluminium
2 = Brass
3 = Steel
4 = Edelstahl 1.4408 (AISI 316)
5 = Edelstahl 1.4301 (AISI 304)
6 = Titan
7 = Hastelloy
8 = PVDF
9 = PEEK

6. Material in the Innerparts

1 = Aluminium
2 = Brass
3 = Steel
4 = Stainless steel A4 (316)
5 = Stainless steel 1.4301
K = Inconel
6 = Titan
7 = Hastelloy
8 = PVDF
9 = PEEK
K = Inconel

7. O-Rings

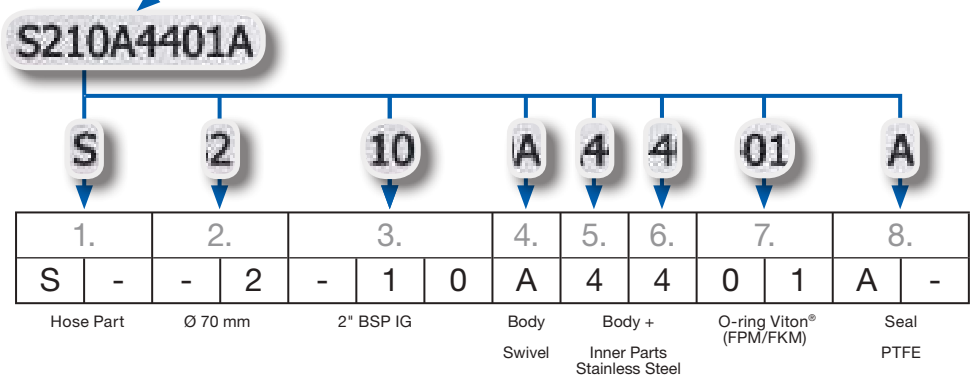
01 = Viton® (FPM/FKM)
02 = Nitrile (NBR)
03 = EPDM
04 = Kalrez® (FFKM) 6375
05 = NBR Low temp
06 = Teflon® (PTFE)
07 = Neoprene® (CR)
08 = Silicone (Q)
09 = Polyurethane (PUR)
10 = Butyl (IIR)
11 = Nitrile (Gasol NBR 70 K-6)
12 = Perfluorelastomer (FFPM/FFKM)
13 = PVC/NBR
14 = Fluorsilicone rubber (MFQ)
15 = FEP encapsulated silicone
16 = Hypalon® (CSM)
17 = Chemraz® 505 (FFKM)
18 = Xyflour® 860 (AFKM)
19 = Zetpol® / Therban® (HNBR)
20 = NBR 90° shore
21 = Viton®-GF (Special Viton quality)
22 = Composite
23 = Viton® GFLT-S
24 = Viton® GLT
25 = Klingerit®
26 = POM
27 = Epiclorhydrin (ECO)
28 = Viton® GFLT-S NMO
31 = Viton® 90 Shore (FPM/FKM)
33 = EPDM 291
34 = Kalrez® 0040
37 = Chemraz® 510 (90 Shore)
40 = FEP or PTFE encapsulated Viton®
50 = Kalrez® (PFPM) 1050LF
51 = Nylon® (PA)
61 = Viton® (FPM), FDA, USP C6 & ADI
62 = Nitrile (NBR), FDA, USP C6 & ADI
63 = EPDM, FDA, USP C6 & ADI
64 = Kalrez® (FFKM) 6230, FDA, USP C6 & ADI
66 = PTFE (Virgin), FDA
71 = FPM/FKM Low Temp
77 = Chemraz® SD517, FDA, USP C6 & ADI
83 = EPDM, BAM certified for oxygene

8. Flat (Thread) Seals / Extras

A = Flat Seal, Teflon® (PTFE)
B = Flat Seal, Vulkollan® (PUR)
C = 2-Way Ball Valve
D = Flat Seal, Viton® (FPM)
DA = Double Acting (Ball Valve)
E = Tank Unit with short piston
F = Flange thickness acc. to standard
G = Flat Seal, Hypalon®
H = Flat Seal, Nitrile (NBR)
I = Emco compatible
J = Flat Seal, EPDM
K = Adapter with locked piston guide
L = Locked thread (maggot screw)
M = Two-Step Cam Curve
N = No Branding
NA = Ball Valve w. square trunnion (no actuator)
P = Pressure Equalizing Valve
Q = Sampling Vent & Drain Unit with reduced bore diameter (Argus, Hydrant)
R = Hose Unit with int. Break Away
S = Sampling Vent & Drain Unit, only with one ball valve
SR = Spring Return (Ball Valve)
T = TW-Flange extended circles
U = Pressure Bleeding Valve 16 bar
U5 = Pressure Bleeding Valve 5 bar
U20 = Pressure Bleeding valve 20 bar
V = Locked Hose Unit
W = Swivel with double ball valve
X = Special surface treatment
Z = Excentric Tank Unit
-RA = Racing
-LC = Locking Cap
-S = FEP O-Ring in Hose Unit swivel
-ST = Steam
-XL = Oversized swivel
-45 = 45 Mesh
-60 = 60 Mesh
-10 = 100 Mesh

ID-Tag on Coupling:

Example:
2" (Ø 70 mm) Hose Unit (Coupler), female thread



Selection of registred trade names from BASF, Bayer AG, B.F. Goodrich, Chemische Werke Hüls, Daikin, Dow Company, DSM, DuPont, DuPont Dow Elastomers, Esso Chemie, Hercules, Hoechst AG, Montedison, Monteflous, Nippon Zeon, Polysar LTD., Rhone Poulenc, 3M Company, Wacker Chemie, Precision Polymer Engineering Ltd.

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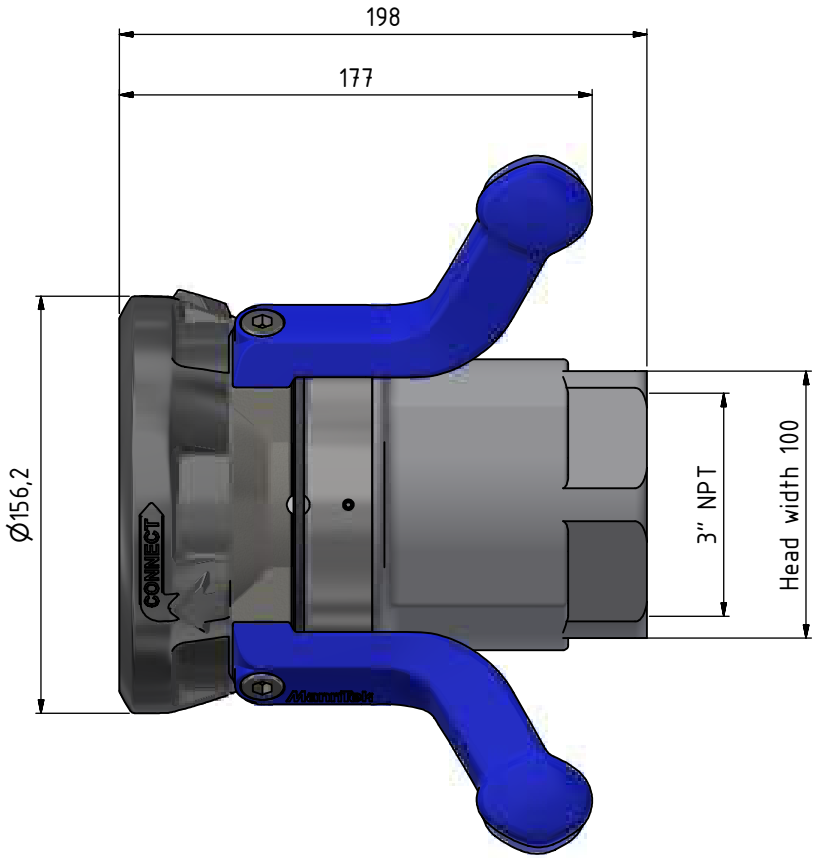
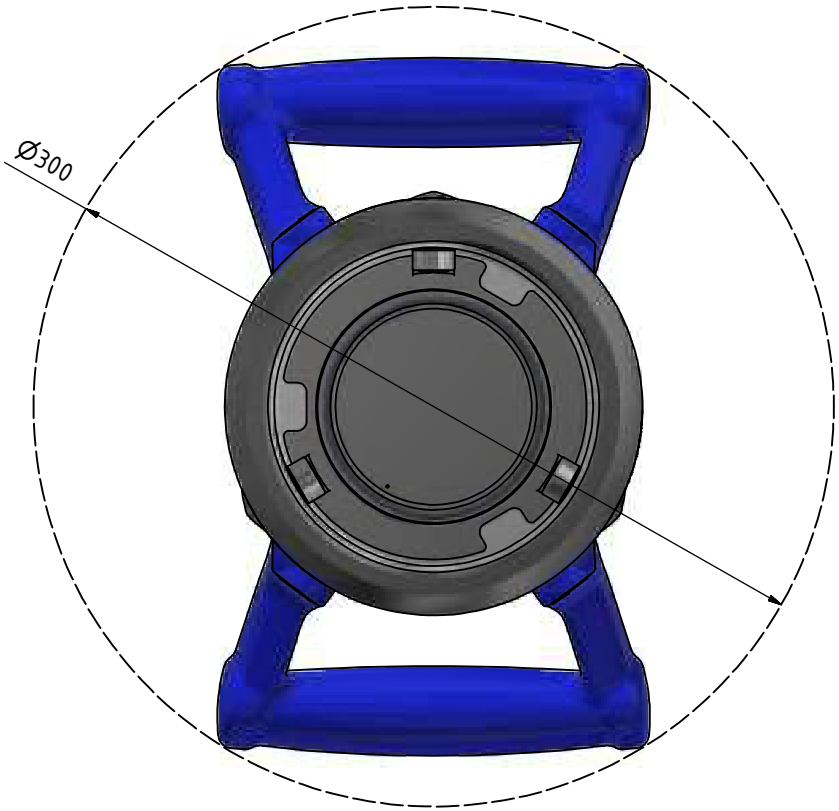
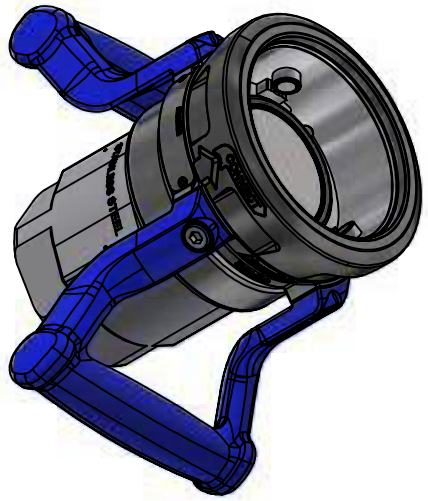






Detail No:	Material	Ämne/Blank	Ytbeh./Treatment	Vikt/Weight

MannTek

Designed/drawn by	Scale:
J. Gustafsson	1 : 2
Checked: G. Kopplin	Date: 2023-09-01
Approved: M. Bäckström	

Dimension with no tolerances acc to

Machined	SS-ISO 2768-1 medium
Weld Forge	SS-ISO 2768-1 coarse
Surface roughness	SS-ISO 1302
Form & position tol	SS-ISO 2768-2-K

Hose unit 3" - DDC

Connection: 3" NPT

Data nr./Comp No. 4S15B4401 | Blad/Sheet 1 / 1

Drawing No. 4S15B4401-1 | Rev. - 221

NO.	Revision comprises	Date	Sign.	Appr by

Sealing Cap for Tank Unit (Adapter) – 10 bar / 150psi

Suitable if a pressure tightness up to 10 bar - but no pressure relief - is required.
Supplied with cable and mounting eyelets.

Inch / DN	Material of Body [1]	Seal [2]	Weight ≈ kg	Weight ≈ lbs	Mann Tek Code No.
¾"-1" Ø 56 mm, DN 20-25	SS	Standard: FPM (Viton®) Other on request.	0,6	1,30	C100P4401
1½"-2" Ø 70 mm, DN 40-50	Al		0,3	0.7	C200P1101
	SS		0,4	0.9	C200P4401
2½" Ø 105 mm, DN 65	SS		1,5	1.1	C300P4401
3" Ø 119 mm, DN 80	Al		0,6	1.3	C400H1101
	SS		1,3	2.9	C400P4401
4" Ø 164 mm, DN 100	SS		2,9	6.4	C500P4401

[1] Material: Al = Aluminium, SS = Stainless Steel
[2] Standard seal FPM/FKM or NBR. Alternative materials, e.g. EPDM, Chemraz® or Kalrez® on request

Dust plug for Hose Unit (Coupler)

To increase the lifespan of couplers, dust plugs should be used. They protect the couplings against the ingress of dirt and water. Materials: polyethylene, aluminum, stainless steel. Supplied with cable attachment and eyelets.

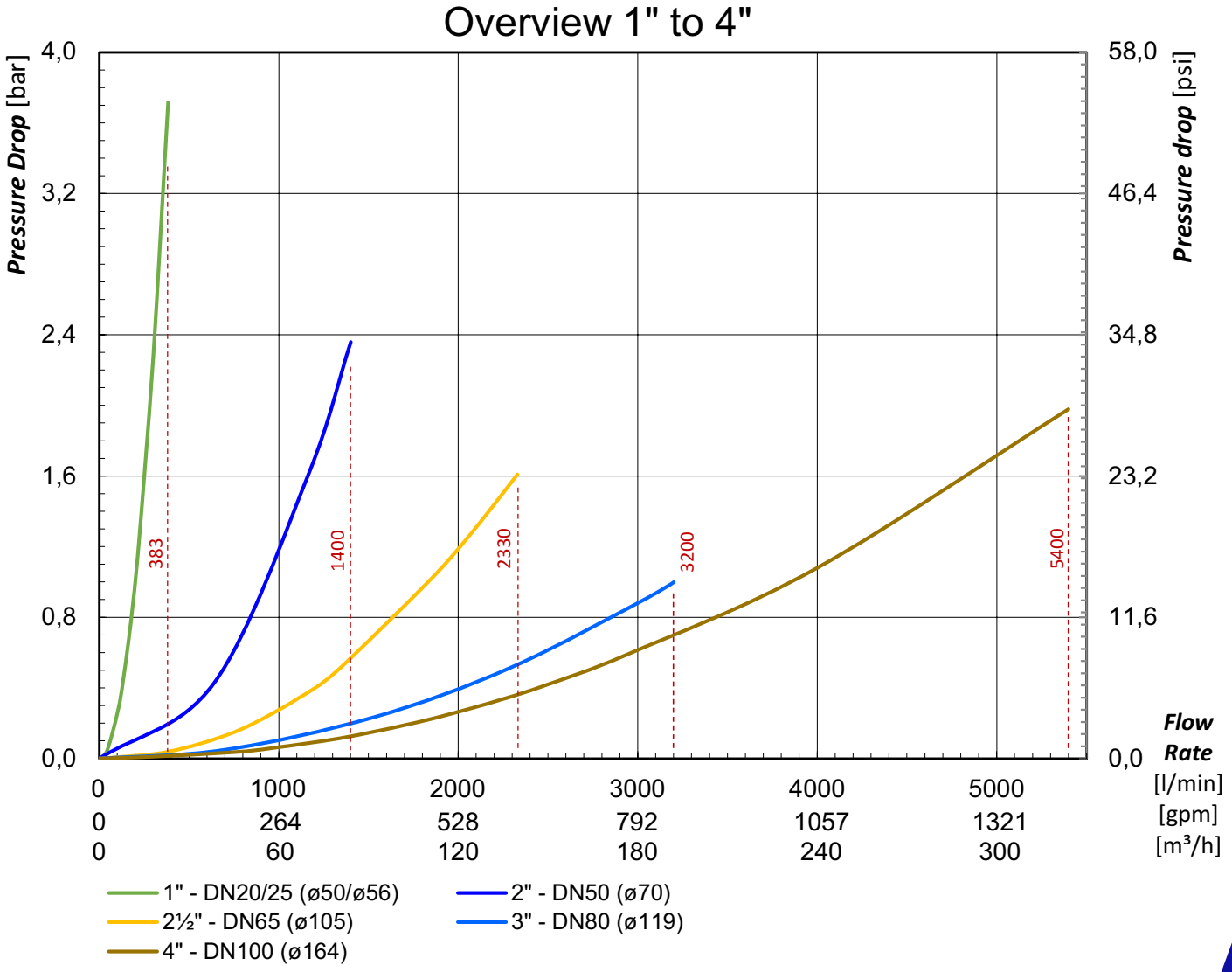
Inch / DN	Material of Body [1]	Seal [2]	Weight ≈ kg	Weight ≈ lbs	Mann Tek Code No.
¾"-1" Ø 56 mm, DN 20-25	PE	SPECIFY TEFLON (PTFE) O-RING	0,1	0.1	P100A2201
	Al		0,1	0.3	P100A1101
	SS		0,3	0.7	P100A4401
1½-2" Ø 70 mm, DN 40-50	PE		0,1	0.2	P200A2201
	Al		0,2	0.5	P200A1101
	SS		0,2	0.6	P200A4401
2½" Ø 105 mm, DN 65	PE		0,2	0.5	P300A2201
	Al		0,6	1.2	P300A1101
	SS				P300A4401
3" Ø 119 mm, DN 80	PE		0,1	0.2	P400A2201
	Al		0,7	1.6	P400A1101
	SS		0,6	1.3	P400A4401 6
4" Ø 164 mm, DN 100	PE		0,2	0.4	P500B2201
	Al		0,3	0.7	P500B1101
	SS				P500A4401
6" Ø 238 mm, DN 150	PE				P600A2201
	Al		1,2	0.4	P600A1101
	SS				P600A4401
8" Ø 272 mm, DN 200	Al		1,2	0.4	P800A1101
	SS				P800A4401

(QTY=2)
DUST PLUGS TO BE ATTACHED TO HOSE ENDS WHEN NOT IN USE.

[1] Material: PE = Polyethylene Composite, Al = Aluminium, SS = Stainless Steel
[2] Standard seal FPM/FKM or NBR. Alternative materials, e.g. EPDM, Chemraz®, Kalrez® or HNBR on request

Viton® is a registred trademark of DuPont, DuPont Elastomers.

Flow Diagram
Dry Disconnect Coupling



Test Fluid: n-paraffin **Temperature:** 20 °C **Density:** 0,75 kg/dm³ **Viscosity:** 1,75 mm²/s
Max. recommended flow velocity: 10m/s

Size	1"	2" (2.0)	2½"	3" (2.0)	4"	Unit
Flow rate at max. rec. flow velocity	383	1400	2330	3200	5400	l/min
Pressure drop at max. rec. Flow velocity	3,7	2,3	1,6	1,0	2,0	bar
Kv	10,3	47,8	95,4	167,2	199,6	[m³/h].[bar]
Cv	11,9	55,7	110,3	194,8	230,8	[gpm].[psi]

Verification method

- *Performance based on combination of physical tests and CFD flow simulations*
- *Test/Simulation method: STANAG 3756, Annex E*
- *Test/Simulation fluid: n-paraffin, Density: 0,75 kg/dm³, Viscosity: 1,75 mm²/s*
- *Design 2.0 performance valid for hose unit and tank unit 2.0 status.*

FLEX HOSE LENGTH NOTES

1. SEE TABLE BELOW FOR MINIMUM HOSE LENGTH REQUIREMENTS. REFER TO *PDI.M003 (FLEX HOSE CONNECTIONS)* FOR DETAILED INSTRUCTIONS.

Pipe Size	Standard Hose Length (Min)
1/4"	18"
3/8"	18"
1/2"	18"
3/4"	18"
1"	24"
1 1/2"	24"
2"	24"
2 1/2"	30"
3"	30"
4"	36"
6"	42"
8"	48"
10"	54"
12"	60"

2. FLEX HOSES THAT DO NOT ALIGN WITH THE STANDARD HOSE LENGTHS LISTED SHALL HAVE LENGTHS IN INCREMENTS OF 6" AS MUCH AS POSSIBLE FOR CONSISTENCY.

3. HOSE LENGTHS SHALL BE LISTED FOR EACH ITEM TAG IN THE SPECIALTY ITEM LIST.

4. FIELD CONTRACTOR TO VERIFY LENGTHS FOR EACH HOSE.

=== PIPING SPECIALTY ITEM LIST ===

CUT SHEET NUMBER:	PCS-FLX-056
DESCRIPTION:	Flexible Hose Assembly
VENDOR:	Swagelok
MODEL:	U Series 316 SS 150# Lap Joint Flange x Female Cam and Groove with Dust Plug

NOTE:

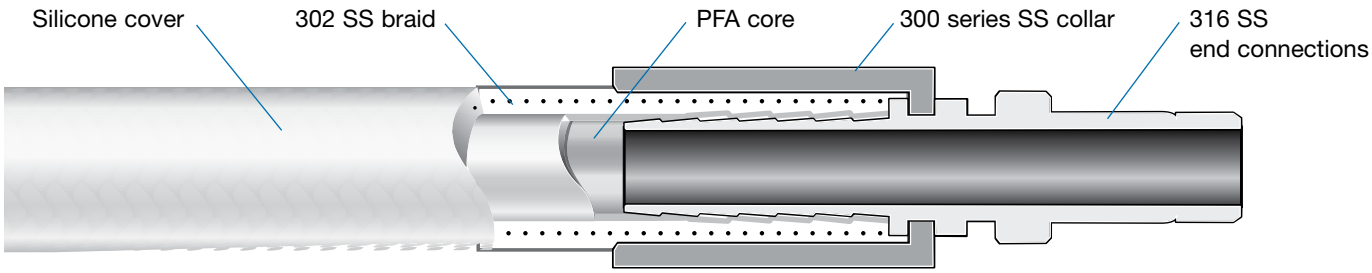
USED TO SUPPLY NITROGEN FOR PURGING AT TRUCK UNLOADING STATIONS.

CUT SHEET BEGINS ON NEXT PAGE

U Series PFA Hose

Features

- Highly flexible PFA hose.
- Smooth-bore PFA core.
- Size range of 1/2 through 2 in. and working pressures up to 300 psig (20.6 bar).
- 302 stainless steel reinforcement ensures hose pressure containment and supports core to resist kinking.
- Silicone cover provides a smooth, noncontaminating, easy-to-clean surface and reduces internal system fluid temperature transfer.
- Hose layers are encapsulated together without adhesives or cements with a patent-pending process, providing high flexibility and outstanding kink resistance.
- PFA material complies with FDA regulation 21CFR Part 177.1550, USP <87, 88> Class VI (121°C), 3-A (for hose sizes 3/4 through 2 in.), and is free of TSE, BSE, and ADI as defined in EMEA/410/01.
- Optional carbon black-filled PFA core is available for applications that require static dissipation.
- Commonly used where high flexibility, chemical compatibility, and a smooth exterior cover are desired.
- Custom hose lengths and end connections available.
- Options include hose covers, hose tags, and additional cleaning. See page 107 for details.
- For electrical properties, see page 5 for details.



Technical Data

Nominal Hose Size in. (mm)	Inside Diameter in. (mm)	Outside Diameter in. (mm)	Minimum Inside Bend Radius in. (cm)		Temperature Range °F (°C)	Vacuum (28.5 in.Hg [96.5 kPa]) Rated to ... °F (°C)	Working Pressure at 70°F (20°C) psig (bar)	Minimum Burst Pressure at 70°F (20°C) psig (bar)	Bulk Hose Weight lb/ft (kg/m)
			Static	Dynamic					
1/2 (12.7)	0.50 (12.7)	0.81 (20.6)	1.50 (3.81)	4.50 (11.4)	–65 to 400 (–53 to 204)	400 (204)	300 (20.6)	1200 (82.6)	0.20 (0.30)
3/4 (19.0)	0.75 (19.0)	1.13 (28.7)	2.50 (6.35)	5.20 (13.2)		400 (204)	300 (20.6)	1200 (82.6)	0.38 (0.57)
1 (25.4)	1.00 (25.4)	1.48 (37.6)	4.00 (10.2)	6.50 (16.5)		250 (121)	250 (17.2)	1000 (68.9)	0.63 (0.94)
1 1/2 (38.1)	1.50 (38.1)	2.00 (50.8)	7.00 (17.8)	9.10 (23.1)		250 (121)	200 (13.7)	800 (55.1)	0.88 (1.3)
2 (50.8)	2.00 (50.8)	2.50 (63.5)	7.00 (17.8)	9.10 (23.1)		150 (65)	150 (10.3)	600 (41.3)	1.3 (1.9)

Pressure-temperature ratings may be limited by the end connections.

Pressure-Temperature Ratings

Nominal Hose Size, in.	1/2	3/4	1	1 1/2	2
Temperature, °F (°C)	Working Pressure, psig (bar)				
–65 (–53)	200 (13.7)	115 (7.9)	250 (17.2)	200 (13.7)	150 (10.3)
0 (–17) to 100 (37)	300 (20.6)	300 (20.6)	250 (17.2)	200 (13.7)	150 (10.3)
200 (93)	280 (19.2)	300 (20.6)	250 (17.2)	200 (13.7)	150 (10.3)
300 (148)	210 (14.4)	270 (18.6)	230 (15.8)	200 (13.7)	150 (10.3)
400 (204)	160 (11.0)	195 (13.4)	175 (12.0)	200 (13.7)	150 (10.3)

U Series PFA Hose

Testing

Every Swagelok U series hose assembly is pressure tested with water at room temperature for 30 seconds to a requirement of no detectable leakage. Testing is performed at 225 psig (15.5 bar).

Cleaning and Packaging

Every Swagelok U series hose component is cleaned in accordance with Swagelok *Standard Cleaning and Packaging (SC-10)* catalog, MS-06-62.

3/4" HOSE - SS - UT 12 GF12 GF12 - 600 - PW
1" HOSE - SS - UT 16 GF16 GF12 - 600 - PW
1" HOSE w/2" FLG - SS - UT 16 GF16 GA32 - 600 - PW

Ordering Information

Custom Hose Assemblies

Build a hose assembly ordering number by combining the designators in the sequence shown below.

Typical Ordering Number

1	2	3	4	4	5	6		5	6	7
SS -	UT	8	TA8	KC16 -	28 -	PB	or	71CM -	PB	
					in.			cm		

1 Material

End Connections

SS = 316 stainless steel
HC = Alloy C-276

2 Hose

UT = U series PFA hose with silicone cover
UC = U series carbon black-filled PFA hose with silicone cover

3 Nominal Hose Size, in.

8 = 1/2
12 = 3/4
16 = 1
24 = 1 1/2
32 = 2

4 End Connections

*See **End Connection Designator** column in tables on pages 73 to 82.*

5 Overall Length

*Inches or centimeters, in whole numbers. Include **CM** as shown for centimeter lengths.*

Typical maximum one-piece hose length:

- *900 in. or 2286 cm for 1/2 in. hose*
- *600 in. or 1524 cm for 3/4 and 1 in. hose*
- *300 in. or 762 cm for 1 1/2 and 2 in. hose.*

*Longer assemblies may be spliced; specify splices under **Options.** See page 4 for more information about splices.*

6 Options

For multiple options, add designators with a dash between each designator.

A = Armor guard
C = ASTM G93 Level C cleaning on hose wetted surfaces
F = Fire jacket
G6 = Spiral guard, black
G7 = Spiral guard, blue
G8 = Spiral guard, yellow
W = Hydrostatic test

Splices

SP1 = 1 splice
SP2 = 2 splices

Mat Tags

MA = Gray
MB = Blue
MC = Brown
MG = Green
MK = Black
MN = Pink
MO = Orange
MP = Purple
MR = Red
MW = White
MY = Yellow

*Add **2** to the end of the Mat Tag designator for two tags.*
*Example: MA**2***

Perma Tags

PA = Gray
PB = Blue
PC = Brown
PG = Green
PK = Black
PN = Pink
PO = Orange
PP = Purple
PR = Red
PW = White
PY = Yellow

*Add **2** to the end of the Perma Tag designator for two tags.*
*Example: PA**2***

Other Tags

T = Lanyard tag
T2 = Two lanyard tags

*Specify text for tags. See **Hose Tag Text** table, page 109.*

See page 107 for detailed descriptions of options.

7 Two Elbow Orientation

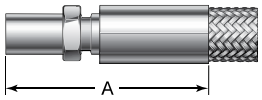
Only include a value in the assembly number when both end connections are elbows. See page 76 for values and further information.

INCLUDE FOLLOWING DETAILS IN HOSE LABEL:
- MANUFACTURER
- HOSE TYPE / MODEL
- MONTH & YEAR OF MANUFACTURE
- P.O. NUMBER
- WETTED PARTS MATERIAL OF CONSTRUCTION (MOC)
- MAXIMUM WORKING PRESSURE AND TEMPERATURE OF ASSEMBLY
- FULL ITEM LOCATION TAG (E.G. LP150FLX1000A)
- APPROVAL / CERTIFICATION MARKS (E.G. FDA, NSF, USP, ETC.)

SEE LAST PAGE FOR HOSE LENGTH DETAILS.

End Connections for B, X, S, C, J, N, W, F and U Series Hose

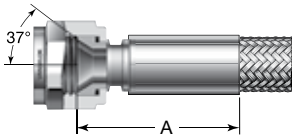
Tube Butt Welds①



① 316 SS material with an I.D. of 15 µin. (0.38 µm) R_a max surface finish applies prior to crimp.

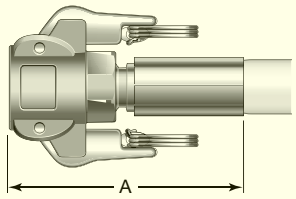
Tube Butt Weld Size in.	Wall Thickness in.	Nominal Hose Size Designator	End Connection Designator	Dimensions, in. (mm)			Pressure Rating psig (bar)
				A Max	Minimum Inside Diameter	Maximum Outside Dimension	
1/2	0.049	8	TB8	2.70 (68.6)	0.34 (8.6)	1.04 (26.4)	Determined by hose
3/4	0.049	12	TB12	3.26 (82.8)	0.54 (13.7)	1.35 (34.3)	
1	0.065	16	TB16	3.26 (82.8)	0.78 (19.8)	1.75 (44.5)	
1 1/2	0.095	24	TB24	4.65 (118)	1.24 (31.4)	2.20 (55.9)	
2	0.109	32	TB32	5.56 (141)	1.68 (42.6)	2.75 (69.9)	

SAE 37° (JIC) Female Swivel



SAE 37° (JIC) Female Swivel Size in.	Nominal Hose Size Designator	End Connection Designator	Dimensions, in. (mm)			Pressure Rating psig (bar)
			A Max	Minimum Inside Diameter	Maximum Outside Dimension	
1/8	2	AS2	1.05 (26.7)	0.070 (1.7)	0.55 (14.0)	Determined by hose
1/4	4	AS4	1.41 (35.8)	0.16 (4.0)	0.66 (16.8)	
3/8	6	AS6	2.05 (52.1)	0.26 (6.6)	0.82 (20.8)	
1/2	8	AS8	2.26 (57.4)	0.34 (8.6)	1.04 (26.4)	
3/4	12	AS12	2.63 (66.8)	0.54 (13.7)	1.35 (34.3)	
1	16	AS16	2.62 (66.5)	0.78 (19.8)	1.75 (44.5)	
1 1/2	24	AS24	3.30 (83.8)	1.24 (31.4)	2.61 (66.3)	
2	32	AS32	4.84 (123)	1.68 (42.6)	3.33 (84.6)	

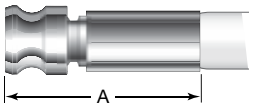
Female Cam and Groove



Female Cam and Groove Size in.	Nominal Hose Size Designator	End Connection Designator	Dimensions, in. (mm)			Pressure Rating psig (bar)
			A Max	Minimum Inside Diameter	Maximum Outside Dimension	
3/4	12	GF12	4.03 (102)	0.54 (13.7)	3.27 (83.1)	250 (17.2)
1	16	GF16	4.53 (115)	0.78 (19.8)	3.50 (88.9)	
1 1/2	24	GF24	5.39 (137)	1.24 (31.4)	4.44 (113)	
2	32	GF32	6.30 (160)	1.68 (42.6)	4.82 (122)	

The A dimension may vary if armor guard hose cover is ordered.

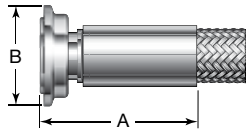
Male Cam and Groove



Male Cam and Groove Size in.	Nominal Hose Size Designator	End Connection Designator	Dimensions, in. (mm)			Pressure Rating psig (bar)
			A Max	Minimum Inside Diameter	Maximum Outside Dimension	
3/4	12	GM12	2.92 (74.2)	0.54 (13.7)	1.27 (32.3)	250 (17.2)
1	16	GM16	3.54 (89.9)	0.78 (19.8)	1.45 (36.8)	
1 1/2	24	GM24	4.40 (112)	1.24 (31.4)	2.11 (53.6)	
2	32	GM32	5.15 (131)	1.68 (42.6)	2.47 (62.7)	

End Connections for B, X, S, C, J, N, W, F and U Series Hose

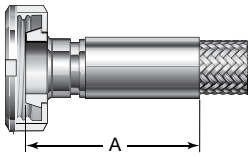
Sanitary DIN 11864-3 Series A, Form A, Clamp Ferrule with Groove①



① 316L SS material with an I.D. of 15 µin. (0.38 µm) R_a max surface finish applies prior to crimp.

Sanitary DIN 11864-3 Series A, Form A, Clamp Ferrule with Groove, Size mm	Nominal Hose Size Designator	End Connection Designator	Dimensions, mm (in.)			Pressure Rating bar (psig)
			A Max	Minimum Inside Diameter	B, Flange Outside Diameter	
10	6	DB10	51.8 (2.04)	6.6 (0.26)	35.6 (1.40)	40.0 (580)
15	8	DB15	56.9 (2.24)	8.6 (0.34)	34.0 (1.34)	
	12	DB15	70.4 (2.77)	13.7 (0.54)	34.0 (1.34)	
20	12	DB20	60.5 (2.38)	13.7 (0.54)	50.3 (1.98)	
25	16	DB25	67.3 (2.65)	19.8 (0.78)	50.3 (1.98)	
40	24	DB40	83.6 (3.29)	31.4 (1.24)	64.0 (2.52)	24.8 (360)
50	32	DB50	101 (3.97)	42.6 (1.68)	77.5 (3.05)	

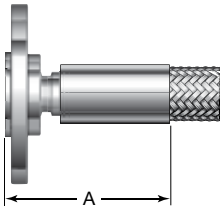
Female DIN 11851 with Nut①



① 316L SS material with an I.D. of 15 µin. (0.38 µm) R_a max surface finish applies prior to crimp.

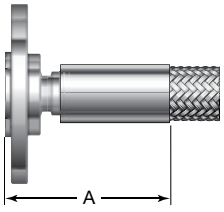
Female DIN 11851 with Nut Size mm	Nominal Hose Size Designator	End Connection Designator	Dimensions, mm (in.)			Pressure Rating bar (psig)
			A Max	Minimum Inside Diameter	Maximum Outside Dimension	
15	8	DF15	57.4 (2.26)	8.6 (0.34)	44.2 (1.74)	40.0 (580)
20	12	DF20	73.7 (2.90)	13.7 (0.54)	54.4 (2.14)	
25	16	DF25	70.6 (2.78)	19.8 (0.78)	63.2 (2.49)	
40	24	DF40	87.6 (3.45)	31.4 (1.24)	78.2 (3.08)	
50	32	DF50	106 (4.19)	42.6 (1.68)	92.2 (3.63)	24.8 (360)

ASME Class 150 Lap Joint Flange



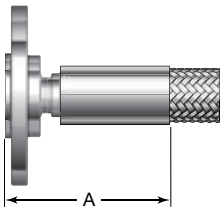
ASME Class 150 Lap Joint Flange Size in.	Nominal Hose Size Designator	End Connection Designator	Dimensions, in. (mm)			Pressure Rating psig (bar)
			A Max	Minimum Inside Diameter	Maximum Outside Dimension	
1/2	8	GA8	2.75 (69.9)	0.34 (8.6)	3.56 (90.4)	275 (18.9)
3/4	12	GA12	3.41 (86.6)	0.54 (13.7)	3.91 (99.3)	
1	16	GA16	3.38 (85.9)	0.78 (19.8)	4.28 (109)	
1 1/2	24	GA24	4.09 (104)	1.24 (31.4)	5.03 (128)	
2	32	GA32	5.06 (129)	1.68 (42.6)	6.03 (153)	

JIS 10K Lap Joint Flange



JIS 10K Lap Joint Flange Size mm	Nominal Hose Size Designator	End Connection Designator	Dimensions, mm (in.)			Pressure Rating bar (psig)
			A Max	Minimum Inside Diameter	Maximum Outside Dimension	
15	8	HA15	69.9 (2.75)	8.6 (0.34)	95.3 (3.75)	9.7 (142)
20	12	HA20	86.6 (3.41)	13.7 (0.54)	100 (3.95)	
25	16	HA25	85.9 (3.38)	19.8 (0.78)	125 (4.93)	
40	24	HA40	104 (4.09)	31.4 (1.24)	140 (5.52)	
50	32	HA50	123 (4.86)	42.6 (1.68)	155 (6.11)	

DIN PN10 Lap Joint Flange



DIN PN10 Lap Joint Flange Size mm	Nominal Hose Size Designator	End Connection Designator	Dimensions, mm (in.)			Pressure Rating bar (psig)
			A Max	Minimum Inside Diameter	Maximum Outside Dimension	
15	8	FA15	69.9 (2.75)	8.6 (0.34)	95.3 (3.75)	10.0 (145)
20	12	FA20	86.6 (3.41)	13.7 (0.54)	105 (4.14)	
25	16	FA25	88.9 (3.50)	19.8 (0.78)	115 (4.54)	
40	24	FA40	121 (4.76)	31.4 (1.24)	150 (5.92)	
50	32	FA50	123 (4.86)	42.6 (1.68)	165 (6.51)	

Type DP Dust Plugs

356T6 aluminum

aluminum hard coat

brass

unplated malleable iron

plated malleable iron

(QTY=1)

316 stainless steel

Feature

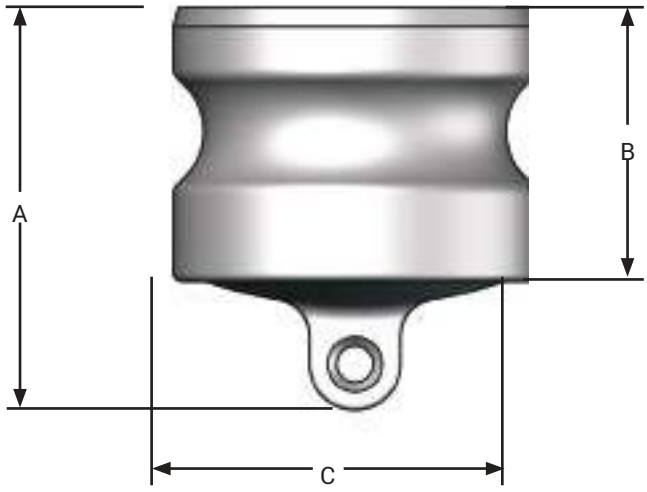
- Dust caps and dust plugs are not to be used in pressure applications for safety and environmental reasons

Size	356T6 Aluminum Part #	Aluminum Hard Coat Part #	Brass Part #
1/2"	---	---	50-DP-BR
3/4"	75-DP-AL	---	75-DP-BR
1"	100-DP-AL	---	100-DP-BR
1-1/4"	125-DP-AL	---	125-DP-BR
1-1/2"	150-DP-AL	150-DP-ALH	150-DP-BR
2"	200-DP-AL	200-DP-ALH	200-DP-BR
2-1/2"	250-DP-AL	---	---
3"	300-DP-AL	300-DP-ALH	300-DP-BR
4"	400-DP-AL	400-DP-ALH	400-DP-BR
5"	500-DP-AL	---	---
6"	600-DP-AL	---	600-DP-BR
8" DIX	800-DP-AL[1]	---	---
8" BL	801-DP-AL[1]	---	---

[1] Dixon® and Boss-Lock cam & groove couplings do not interchange in the 8" size

Size	Unplated Malleable Iron Part #	Plated Malleable Iron Part #	316 Stainless Steel Part #
1/2"	---	---	50-DP-SS[1]
3/4"	---	---	75-DP-SS
1"	---	---	100-DP-SS
1-1/4"	---	---	125-DP-SS
1-1/2"	---	---	150-DP-SS
2"	200-DP-MI	200-DP-PM	200-DP-SS
2-1/2"	---	---	250-DP-SS
3"	300-DP-MI	300-DP-PM	300-DP-SS
4"	400-DP-MI	400-DP-PM	400-DP-SS
6"	---	---	600-DP-SS

Type DP Dust Plugs Dimensions



356T6 Aluminum, Brass, and Iron Dimensions

Size		1/2"	3/4"	1"	1-1/4"	1-1/2"	2"	2-1/2"	3"	4"	5"	6"	8" DIX	8" BL
A	overall length	1-7/16"	1-3/4"	1-29/32"	2-3/8"	2-13/32"	2-13/16"	2-13/16"	2-13/16"	3-1/16"	3-3/8"	3-1/16"	5-3/16"	4-11/16"
B	adapter length	15/16"	1-3/32"	1-5/16"	1-9/16"	1-5/8"	1-7/8"	1-15/16"	2"	2-1/16"	2-11/32"	2-1/4"	3-9/16"	3-1/16"
C	outside O.D.	15/16"	1-1/4"	1-7/16"	1-25/32"	2-3/32"	2-15/32"	2-31/32"	3-19/32"	4-23/32"	5-23/32"	6-15/16"	9-3/32"	8-15/16"

316 Stainless Steel Dimensions

Size		1/2"	3/4"	1"	1-1/4"	1-1/2"	2"	2-1/2"	3"	4"	5"	6"	8" DIX	8" BL
A	overall length	1-7/16"	1-11/16"	1-15/16"	2-1/4"	2-1/2"	2-3/4"	2-7/8"	2-27/32"	3-1/8"	---	3-1/2"	---	---
B	adapter length	15/16"	15/16"	1-15/16"	1-9/16"	1-5/8"	1-7/8"	1-15/16"	2"	2-1/16"	---	2-1/4"	---	---
C	outside O.D.	15/16"	1-1/4"	1-7/16"	1-25/32"	2-3/32"	2-15/32"	2-31/32"	3-19/32"	4-23/32"	---	6-15/16"	---	---

Abrasion Resistant Inserts

Features
- Extends the life of adapters when transferring proppant to storage tanks
- Available for multiple adapters and multiple coupler materials

Description	UHMW Polyethylene Part #
to be used with 400-A-AL, 400-A-BR, 400-A-MI, 400-A-SS	400-A-INSERT
to be used with 400-C-AL	400-C-INSERTAL
to be used with 400-C-MI	400-C-INSERTMI

400-A-INSERT

400-C-INSERTAL

Clip for Boss-Lock Couplings

Feature
- .072 stainless steel wire

Stainless Steel Part #
AC7

Clip Assembly

Features
- For use with Boss-Lock couplings
- Includes pull ring, chain, and AC3 clip

Carbon Steel Part #
BLCA152CS

Jack Chain with S-Hook

Length	Material	Part #	Optional Qty
6"	brass	CH-B-6	10
12"	brass	CH-B-12	10
24"	brass	CH-B-24	---
12"	carbon steel	CH-C-12	10
6"	stainless	CH-SS-6	10
12"	stainless	CH-SS-12	10
24"	stainless	CH-SS-24	10

Locking Cam Arm Velcro® Straps

Application
- Use with cam & groove couplers to prevent spills, secure from damage, and hold in place during transport

Features
- Used on the female end of a cam & groove coupler
- High visibility orange hook and loop with metal buckle provides strength, durability, and long life

Specification
- Temperature range: **-70°F to 200°F (-57°C to 93°C)**

Length	Used On	Part #
12"	fits up to 1-1/2" coupler	CVS12
18"	fits up to 3" coupler	CVS18
24"	fits up to 5" coupler	CVS24
36"	fits up to 8" coupler	CVS36

coupler sold separately

FLEX HOSE LENGTH NOTES

1. SEE TABLE BELOW FOR MINIMUM HOSE LENGTH REQUIREMENTS. REFER TO *PDI.M003 (FLEX HOSE CONNECTIONS)* FOR DETAILED INSTRUCTIONS.

Pipe Size	Standard Hose Length (Min)
1/4"	18"
3/8"	18"
1/2"	18"
3/4"	18"
1"	24"
1 1/2"	24"
2"	24"
2 1/2"	30"
3"	30"
4"	36"
6"	42"
8"	48"
10"	54"
12"	60"

2. FLEX HOSES THAT DO NOT ALIGN WITH THE STANDARD HOSE LENGTHS LISTED SHALL HAVE LENGTHS IN INCREMENTS OF 6" AS MUCH AS POSSIBLE FOR CONSISTENCY.

3. HOSE LENGTHS SHALL BE LISTED FOR EACH ITEM TAG IN THE SPECIALTY ITEM LIST.

4. FIELD CONTRACTOR TO VERIFY LENGTHS FOR EACH HOSE.

=== PIPING SPECIALTY ITEM LIST ===

CUT SHEET NUMBER:	PCS-PC-001
DESCRIPTION:	Pipe Cover
VENDOR:	SL&C
MODEL:	WSL Series

NOTE:

MAXIMUM TEMPERATURE LISTED (280F) REFERS TO CONDITION THAT THE PIPE COVER IS CONSTANTLY EXPOSED TO. RELIEF EVENTS WITH MOMENTARY EXPOSURE TO HIGHER TEMPERATURES WILL NOT AFFECT FUNCTIONALITY AND REUSABILITY OF PIPE COVER.

CUT SHEET BEGINS ON NEXT PAGE

Scientific Linings & Coatings

WCL Series WeatherCap™ Valve Protection

Features:

- Fits 1/2” thru 16” schedule 40 pipe
- High visibility - chrome yellow color
- Installs without tools
- Withstands abrasive weather conditions: rain, ice, snow, and seals out insects, birds, bird nests, sand, dust and salt spray
- Eliminates equipment deterioration caused by external environment
- Provides visual indication of valve activation
- Provides audible warning of leaking valve
- Chemically resistant - will not shrink or discolor
- Non-toxic elastic PVC construction blended with ultraviolet inhibitors
- Flame Retardant
- Minimal interior pressure surge (1/2psi) removes WeatherCap™ from pipe
- Durable and reusable
- Five year unconditional guarantee against environmentally caused deterioration

Industries:

- Petrochemical
- Pharmaceutical
- Gas & Liquid Transportation
- Off-shore Platforms
- Oil and Gas Exploration
- Food Processing
- Ammonia Cold Storage Plants
- Paper Mills

Applications:

- Protection for relief valves
- Manifold piping systems, ductwork
- Common header systems
- Storage tanks

Physical and Chemical Properties & Range of Characteristics

PHYSICAL CHARACTERISTICS	
Hardness	60 Durometer
Tensile Strength	1250 psi Minimum
Elongation	250%
Specific Gravity	1.25 gm/cc
Low Temperature Flex	Less than -25°F.
Maximum Service Temperature	More than 180°F
Outdoor Exposure	Color stability varies. No significant change after ten years of exposure.
Toxicity	None
Abrasion Resistance	E
Impact Resistance	E

PLASTISOL CHEMICAL - RESISTANCE RANGE	
Water	E
Ethyl Alcohol	E
Sodium Chloride	E
Bleach	E
Hydrochloric Acid	F to E*
Sulfuric Acid	F to E*
Nitric Acid	F to E*
Acetic Acid	F
ASTM 1 Oil	F
Gasoline	F
Benzene	P
Acetone	P
Trichloroethylene	P to F

* In general this product has excellent resistance to alkalis. Concentrated oxidizing and organic acids have a deleterious effect on prolonged exposure. Most PVC formulas have fair resistance to aliphatic solvents, fair to poor resistance to aromatic and chlorinated solvents, and poor resistance to ketones and most esters.

CHEMICAL PROPERTIES AND RESISTANCE RANGE	
Exterior Durability	E
Flexibility	E
Salt Spray	E
Solvents-Alcohols	E
Solvents-Gasoline	E
Solvents-Hydrocarbons	G
Solvents-Esters, Ketones	P
Solvents-Chlorinated	P
Salts (Solutions)	E
Ammonia	E
Alkalis	E
Acids, Mineral, Dilute (10%)	E
Acids, Mineral, Concentrated	G
Acids, Oxidizing, Dilute (10%)	E
Acids, Oxidizing, Concentrated	G

DECORATIVE PROPERTIES	
Color Range	E
Color Retention	E
Initial Gloss	E
Gloss Retention	VG

Key: E - Excellent; VG - Very Good; G - Good; F - Fair; P - Poor

WCL Series cannot be used on vent piping that is installed to provide vacuum relief.

WCL Series Pipe Guard should not be used on applications where relieving pressures are less than 0.5psi.

WCL Series is not recommended for use on PVC, Plastic, or Vinyl piping.

The above information is for use as a general guide.

MSDS Sheet is available

To locate a distributor near you, log on to www.weathercap.com or call (800) 348-2585

WeatherCap Installation Instructions:

1) **Make certain you have the correct size WeatherCap.**
1) **Preparation of the discharge stacks, vents, or outlet ports:**
    a) **PRIOR** to installing the WeatherCap you must remove any water or debris that may have accumulated in the opening. Use an approved method to clear the pipe.
    b) The exhaust opening may be cut square or at an angle up to 45°
    c) Make certain the pipe surface which will contact the WeatherCap is clean and smooth.
    d) Paint or other surface coatings must be fully cured.
2) **Installation for sizes ½" thru 4"**
    a) Install the retaining collar first (see figure 1). Slip the collar onto the pipe as shown and slide the collar down the pipe 1 to 2 inches.
    b) Place the WeatherCap on the pipe above the retaining collar. Slide the WeatherCap down onto the pipe about 1 inch. (see figure 2)
    c) Slide the retaining collar up the pipe until it almost touches the WeatherCap.
4) **Installation of 6" thru 24"**
    a) Place the WeatherCap on the pipe. Slide the WeatherCap down onto the pipe about 1 inch. (see figure 2)
    b) Place the vinyl zip tie through the loop at the end of the strap attached to the WeatherCap. Wrap the tapered end of the zip tie around the pipe and slip the end of the tie into the slot at the other end of the zip tie. Draw the zip tie snug; do not cinch tight yet.
    c) Slide the zip tie up until it is next to or touching the WeatherCap.
    d) Cinch the zip tie tight to avoid any further movement and cut off excess zip tie.

If the over pressure device activates, the WeatherCap will "pop" off the pipe and must be reinstalled:

1) Inspect the WeatherCap and replace if damaged.
2) Inspect the stack / pipe for liquid or debris accumulation and remove any fluids and debris using an approved method.

Reinstall the WeatherCap as noted above.

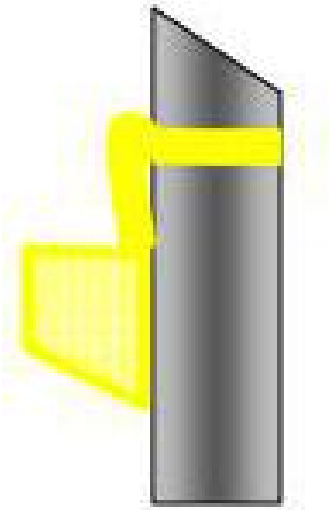


Figure 1

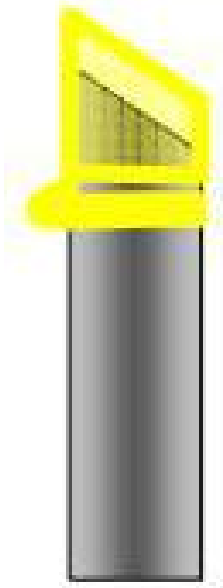


Figure 2

WeatherCaps Fit the OD Dimensions of Schedule 40 Pipe			
Part No.	**Fits Schedule 40 Pipe**	**OD Inches**	**ID Inches**
wcl0.5	1/2"	0.840	0.622
wcl0.75	3/4"	1.050	0.824
wcl1	1"	1.315	1.049
wcl1.25	1 1/4"	1.660	1.380
wcl1.5	1 1/2"	1.900	1.610
wcl2	2"	2.375	2.067
wcl2.5	2 1/2"	2.875	2.469
wcl3	3"	3.500	3.068
wcl4	4"	4.500	4.026
wcl5	5"		
wcl6	6"	6.625	6.065
wcl8	8"	8.625	7.981
wcl10	10"	10.750	10.020
wcl12	12"	12.750	11.938
wcl14	14"	14.000	13.125
wcl16	16"	16.000	15.000
wcl18	18"	18.000	16.874
wcl20	20"	20.000	18.614
wcl24	24"	24.000	22.626

Scientific Linings & Coatings, Inc.

Email: Info@weathercap.com
Website: www.WeatherCap.com
Local: 210-525-8333
Fax: 210-340-5354
Toll Free: 1-800-348-2585

For more inform Page 4 of 6 PCS-PC-001 **ll free, or contact us through our website. International orders are also welcome. A listing of authorized distributors is located on our website.**

B1LG-00-K215-00177-1_LP1Y-250-LST-K215-01.pdf

Benefits:

- Seals out rain, ice, snow, sand, salt spray, insects, birds and their residuals.
- Extends valve life.
- Provides visual evidence of valve activation.
- Minimal backpressure removes the weathercap.
- Reusable.
- Adds dollars to the bottom line.
- 5 Year warranty against any and all weather.
- Most durable, stable and environmentally resistant valve protection available.

Features:

- Exceeds compliance requirements
- Highly visible color (Chrome Yellow)
- Installs without tools
- Withstands abrasive weather conditions
- Airtight type seal
- Chemical resistant
- Non-toxic blend of PVC, plasticizer and ultraviolet inhibitors
- Guaranteed five-year life
- Flame Resistant
- Will not shrink or discolor

Industries:

- Gas & Liquid Transportation
- Hydrocarbon
- Petrochemical
- Off-shore Platforms
- Refineries
- Refrigerated Warehouses
- Paper Mills
- Breweries

- ***WeatherCap™ WCL*** *Series cannot be used on vent piping that is installed to provide vacuum relief.*
- ***WeatherCap™ WCL*** *Series is not recommended for use on PVC, plastic, or vinyl piping.*
- ***WeatherCap™ WCL*** *series valve protectors should not be used on applications where valve relief pressures are less than 1/2psi.*

FOR A DISTRIBUTOR NEAR YOU OR FOR PRODUCT SPECIFICATIONS SEE WWW.WEATHERCAP.COM OR CALL
(800) 348 2585

WeatherCap™

- Patented -
- 1st Choice of Industry Leaders -
- Guaranteed for 5 years -
- Environmentally Compliant -
- Fits Exhaust Nipples 1/2”-16” -
- Reusable -

Applications:

Protection of safety relief valves and rupture disks from rust and corrosion

www.weathercap.com

SL&C WEATHERCAPS™ ARE THE INDUSTRIES FIRST CHOICE FOR VALVE PROTECTION.

IT'S ALL ABOUT SAFETY, COMPLIANCE, ECONOMY & SUCCESS

DID YOU KNOW...? TEXAS COMMISSION ON ENVIRONMENTAL QUALITY'S (TCEQ) MINIMUM NON-COMPLIANCE PENALTIES START AT $10,000.00 PER DAY PER VIOLATION. WEATHERCAPS™ EXCEED EXISTING REQUIREMENTS OF FEDERAL AND STATE AGENCIES. FUGITIVE EMISSIONS CAN RESULT FROM WEATHER AND DEBRIS-INDUCED DETERIORATION OF RELIEF VALVE DISCHARGE MECHANISMS.

PUT WEATHERCAPS™ TO WORK PROTECTING YOUR PRESSURE RELIEF VALVES

Pressure Relief Valves, PRVs, are indispensable to pressure vessel systems, but costly to maintain and costly to replace. Now you can safeguard expensive PRVs with affordable guaranteed protection that isolates the discharge side of your relief valves from harmful corrosive elements, thus protecting against fugitive emission violations. With WEATHERCAPS™ installed on your system you stop expensive corrosion repair or replacement of your Pressure Relief Valves.

AFFORDABLE VALVE PROTECTION ADDS PROFIT TO THE BOTTOM LINE

Use of WEATHERCAP™ protection brings your pressure system into compliance while extending valve life and minimizing valve maintenance. Your Pressure Relief Valves will function correctly for an extended period.

WEATHERCAPS™ take the weather abuse by protecting against corrosion caused by weather, ice, salt spray, sand, birds, insects and their residuals.

WEATHERCAPS™ are the economical way to be in compliance while safe-guarding against penalties resulting from a fugitive emission.

FREQUENTLY ASKED QUESTIONS

1. What about the cost-to-benefit ratio of the WeatherCap™?
   WeatherCaps™ are the most durable, stable, environmentally resistant and cost efficient valve protection on the market today.
2. Is the "Yellow Jacket" available with a flat top surface instead of the 45° angle top surface?
   NO. The slope design acts as a watershed.
3. What are the temperature limits of the "Yellow Jacket" WeatherCap™?
   The WeatherCap™ function remains consistent between temperatures of -100°F to 280° F.
4. Where can I find detailed WeatherCap™ Specifications?
   From your distributer or at www.weathercap.com.

=== PIPING SPECIALTY ITEM LIST ===

CUT SHEET NUMBER:	PCS-QCU-045
DESCRIPTION:	Quick Connect Unit
VENDOR:	MannTek
MODEL:	316 SS Flanged Dry Disconnect Adapter with Dust Cap

NOTE:

USED AT TRUCK UNLOADING AREAS.

USED FOR THE FOLLOWING SERVICES
Acetonitrile
Alcohol, Isopropyl
Dichloromethane
Dimethyl Sulfoxide
Dimethylformamide
General Solvent
Methyl Tert Butyl Ether
Please confirm whether MTBE is applicable to QCU-045 or it needs other type of QCU.
Sodium Hydroxide, Industrial 50%
Sulfuric Acid

CUT SHEET BEGINS ON NEXT PAGE

3" (Ø 119 mm) Technical Information

Tank Unit (Adapter) and Hose Unit (Coupler)

According to NATO STANAG 3756

Material	Maximum Working Pressure	Test Pressure	Minimum Burst Pressure
Aluminium	10 bar / 145 psi	15 bar / 218 psi	50 bar / 726 psi
Brass/Gunmetal	16 bar / 232 psi	24 bar / 348 psi	80 bar / 1160 psi
Stainless Steel	25 bar / 363 psi	37,5 bar / 544 psi	125 bar / 1813 psi
Titan	25 bar / 363 psi	37,5 bar / 544 psi	125 bar / 1813 psi
Hastelloy	25 bar / 363 psi	37,5 bar / 544 psi	125 bar / 1813 psi
PEEK	6 bar / 87 psi	9 bar / 131 psi	30 bar / 435 psi

Connections
3" in BSP, NPT and flanged inlet.

Applications
For road, rail tanker loading/unloading, filling of tanks and containers, refuelling and loading/unloading bunker boats (marine tankers) and small ships.
Also suitable for other applications where high loading rates are required and product contamination and spillage needs to be minimized.

Media
Petroleum products: gasoline, diesel, oil etc.
Chemical products: ethylene oxide, propylene oxide, acrylonitrile, butadiene, ammonia, vinyl chloride, toluene, xylene, sulphuric acid, phenol etc.
Gas: vapor recovery/balance systems
Dry powder.

Material of Body
Aluminium, brass/gunmetal, stainless steel, Hastelloy and PEEK. Other materials on request.

Seals
Standard seals FPM (Viton®), EPDM, FFKM (Chemraz®, Kalrez®), NBR. Other on request.

High Flow Rates / Low Pressure Drop
Allows maximum product transfer with minimal losses.

Recommended Maximum Flow Rates
2000 litres/minute (fuel)

Selectivity
- avoid mixing products
To avoid product contamination caused by connecting a hose unit to the wrong tank unit, selective versions of the hose and tank units are available.
Each unit has a number of selective positions, designated by a coded part number according to the coupling size – specify when placing order.

Interchangeability
Compatible with couplings of other manufacturers.

Hose Unit with Integrated Swivel
All hose units are supplied with an integrated swivel.

3" (Ø 119 mm) Tank Unit (Adapter) – Female Thread

Connection [1] Inch/DN	Body Material [2]	Seal		Weight ≈		Mann Tek Code No.
		O-Ring [3]	Thread Seal	kg	lbs	
F 3" BSP	Al	Standard: FPM/FKM (Viton®) Other on request.	PUR	1,2	2.7	T414D1101B
F 3" BSP				1,3	2.9	T414K1101B*)
F 3" NPT				1,2	2.7	T415D1101
F 3" BSP	GM		PUR	3,2	7.1	T414D2201B
F 3" NPT				3,5	7.7	T415D2201
F 3" BSP	Br		PUR	1,3	2.9	T414K2201B*)
F 3" BSP	SS		PTFE	3,1	6.8	T414B4401A
F 3" NPT				3,4	7.5	T415B4401
F 3" BSP	Hastelloy		PTFE			T414A7701A
F 3" NPT						T415A7701
F 3" BSP	PVDF/Hastelloy		PTFE	3,6	7.9	T414A8701A
F 3" BSP	PEEK		PTFE			T414A9901A
F 3" NPT						T415A9901

According to NATO STANAG 3756

3" (Ø 119 mm) Tank Unit (Adapter)– Flanged Inlet

Flange [1B]	Body Material [2]	Seal [3]	Weight ≈		Mann Tek Code No.
		O-Ring	kg	lbs	
undrilled Ø 210 mm	Al	SPECIFY TEFLON (PTFE) O-RING	2,0	4.4	T420D1101
DN 65 PN 10 / 16 Type A					T433D1101
DN 80 PN 10/16 Type A			2,0	4.4	T436D1101
3" ASA 150			1,8	4.0	T461D1101
DIN 28459 'TW 1' / DN 80			1,4	3.1	T465D1101
DIN 28459 'TW 3' / DN 100			1,6	3.5	T466D1101
3" TTMA			1,8	4.0	T467D1101
4" TTMA			1,5	3.3	T468D1101
undrilled Ø 210 mm	GM		6,2	13.7	T320D2201
DN 65 PN 10/16 Type B			4,9	10.8	T433D2201
DN 65 PN 25/40 Type B					T434D2201
DN 80 PN 10/16 Type B			5,2	11.5	T436D2201
DN 80 PN 25/40 Type B					T437D2201
3" ASA 150			4,9	10.8	T461D2201
3" ASA 300					T462D2201
DIN 28459 'TW 1' / DN 80			4,1	9.0	T465D2201
DIN 28459 'TW 3' / DN 100			4,7	10.4	T466D2201
3" TTMA					T467D2201
4" TTMA			4,6	10.1	T468D2201
undrilled Ø 210 mm	SS				T420B4401
undrilled Ø 210 mm***)					T420B4401F
DN 65 PN 10/16 Type B					T433B4401
DN 65 PN 10/16 Type B***)					T433B4401F
DN 65 PN 25/40 Type B					T434B4401
DN 65 PN 25/40 Type B***)					T434B4401F
DN 80 PN 25/40 Type E**)					T435B4401F
DN 80 PN 10/16 Type B			5,1	11.2	T436B4401
DN 80 PN 10/16 Type B***)					T436B4401F
DN 80 PN 25/40 Type B			5,1	11.2	T437B4401
DN 80 PN 25/40 Type B***)					T437B4401F
3" ASA 150			4,9	10.8	T461B4401
3" ASA 150***)				14.1	T461B4401F
3" ASA 300			4,9	10.8	T462B4401
3" ASA 300***)					T462B4401F
DIN 28459 'TW 1' / DN 80			3,9	8.6	T465B4401
DIN 28459 'TW 3' / DN 100					T466B4401
3" TTMA					T467B4401
4" TTMA					T468B4401
undrilled Ø 210 mm	PEEK				T420A9901
DN 80 PN 10/16 Type B					T436A9901
3" ASA 150					T461A9901
3" TTMA					T467A9901
4" TTMA					T468A9901

According to NATO STANAG 3756

*) shortened version for tight installation situations

**) Type E (with spigot), EN 1092-1, see page 44

***) Flange with standard thickness

[1A] F = Female thread, BSP = EN ISO 228, NPT = ANSI B1.20.1

[1B] PN 10/16/25/40 = EN 1092 (types see page 44), ASA = ANSI B16.5 (150 o. 300 psi), DIN 28459 = old standard TW, 10 bar. TTMA = Truck Trailer Manufacturers Association

[2] Material: Al = Aluminium, GM = Gunmetal, SS = Stainless Steel

[3] Standard seal FPM/FKM. Alternative materials, e.g. EPDM, Chemraz®, Kalrez®, NBR or HNBR on request

Viton® is a registred trademark of DuPont, DuPont Elastomers.

O-Ring Materials

Designation	Trade name	ISO 1629	ASTM 1418	Temp. Range °C		Field of Application
Nitrile Butadiene Rubber / Nitrile Rubber	Buna N® Europrene N® Hycar® Nipol N® Perbunan ®	NBR	NBR	-55	110	Petroleum based products / aliphatic hydrocarbons (gasoline, diesel, oils, petroleum, propane, butane). Animal and vegetal oils and fats. Flame retardant liquids.. Silicone oils and greases. Water up to 80°C.
Ethylene-Propylene Diene Rubber	Dutral Keltan® Vistalon® Buna AP®	EPDM	EPDM	-55	120	Water, hot water, vapour, brake fluids, detergents. Alcohols, ketons, engine coolants, phosphate based flame retardants, organic and inorganic acids and alkalis. Not suitable for petroleum based products.
Fluoroelastomer	Fluorel® Technoflon® Viton®	FPM	FKM	-20	200	Petroleum based products / aliphatic hydrocarbons (gasoline, diesel, oils, petroleum, propane, butane) - also with high additive content. Phosphate based flame retardants. Silicone oils and greases, acids and alkalis.
Fluorosilicone Elastomer	——	MFQ	FVMQ	-60	200	Additional to the properties of silicone elastomers, fluorosilicone elastomers show a consideably higher resistance to oils, fuels and solvents - especially to aromatic and aromatic and chlorinated hydrocarbons and alcohols.
Silicone	Silastic® Silopren®	MVQ	VMQ	-60	200	Very good heat and cold resistance, very good resistance to oxygene, ozone, UV rays and wheather influence. High electrical insulating properties, physiologically safe.
Hydrogenated Nitrile Butadiene Rubber	Therban® Tornac® Zetpol®	HNBR	HNBR	-35	120	Very good resistance to petroleum based products / aliphatic hydrocarbons like NBR, but better resistance to high temperature, ageing and wheather influence. Excellent physical properties also with high temperatures.
Butyl Rubber	Exxon Butyl® Polysar Butyl®	IIR	IIR	-55	100	For O-Rings this material has largely been replaced by EPDM. Butyl is resistant to the same fluid types as EPDM and -except for resistance to gas permeation- it is somewhat inferior to EPDM for O-Ring use. Excellent weather resistance and gas permeation resistance. Not suitable for petroleum based products.
Perfluorinated Elastomer	CHEMRAZ® Kalrez® PERLAST®	FFPM	FFPM	-40	260	Best chemical resistance of all elastomers, including organic acids, acetic acid, benzoic acid, formic acid.
Chloroprene Rubber	Baypren® Neoprene®	CR	CR	-40	120	Resistant to refrigerants, ammonia, carbon dioxide, freon (R12, R13, R21, R22, R113 - R115), silicone oils, water, oxygen (low pressure), alkalis, coustic soda, alcohols, chlorine, ozone, castor oil and vegetable oils. Properties of CR are similar to NBR but slightly lower resistance to petroleum based producs, better resistance to ageing, UV light and ozone.
Polyester / Polyether Urethane Rubber	Polyurethane Adiprene® Urepan® Desmopan®	PU PUR AU EU	AU EU	-40	100	Very good resistance to petroleum based products, also with aromatic content. Good resistance to ozone and ageing. Very good mechanical properties. Not resistant to esters, concentrated acids and lyes, water above +50° C.
Polytetrafluoroethylene	Teflon®	PTFE	PTFE	-200	260	Universal chemical resistance except liquid alkaline metals and some fluorine compounds. Very good anti friction properties, low wear. High temperature resistance. Physiologically safe. Use for statical seals only.
Fluorinated Ethylene Propylene	Teflon FEP®	FEP/MVQ	FEP/VMQ	-60	200	Universal chemical resistance except liquid alkaline metals and some fluorine compounds. Very good anti friction properties, low wear. High temperature resistance. Physiologically safe.
		FEP/FPM	FEP/FKM	-20	200	
Perflouralkoxy	Teflon PFA®	PFA/MVQ	PFA/VMQ	-60	250	Universal chemical resistance except liquid alkaline metals and some fluorine compounds. Very good anti friction properties, low wear. High temperature resistance. Physiologically safe.
		PFA/FPM	PFA/FKM	-20	250	
Tetrafluoroethylene-Propylene Copolymer Elastomer	Aflas®	——	TFE / P	-25	200	Petroleum based products, oils and greases, brake fluids, fuels, alcohols, heat transfer media, oils. amines, acids, alkalis.

***Note**: Specifications in this chart are based on the information given by our suppliers together with published guides. This is not always a guarantee for a proper function. If in doubt, please ask our sales team, statiing medium and operating conditions.*

Selectivity System for 2½" (Ø 105 mm) – Special Version Fulcrum

Coupling diameter:
Ø 105 mm (2½")

Tank Unit

Hose Unit

With 7 Positions

1	2	3	4	5	6	7

Selectivity System:
Fulcrum

Each unit can be provided with selectivity according to two systems – with 7 or 21 positions (see chart).

When ordering, please the chosen selectivity to the Mann+Tek Code No.
Examples: S312A4401/**SEL 1** or S312A4401/**SEL 12** (A)

With 21 Positions

TU	TV	TW	TX	TY	TZ	UV
UW	UX	UY	UZ	VW	VX	VY
WY	WZ	XY	XZ	YZ		

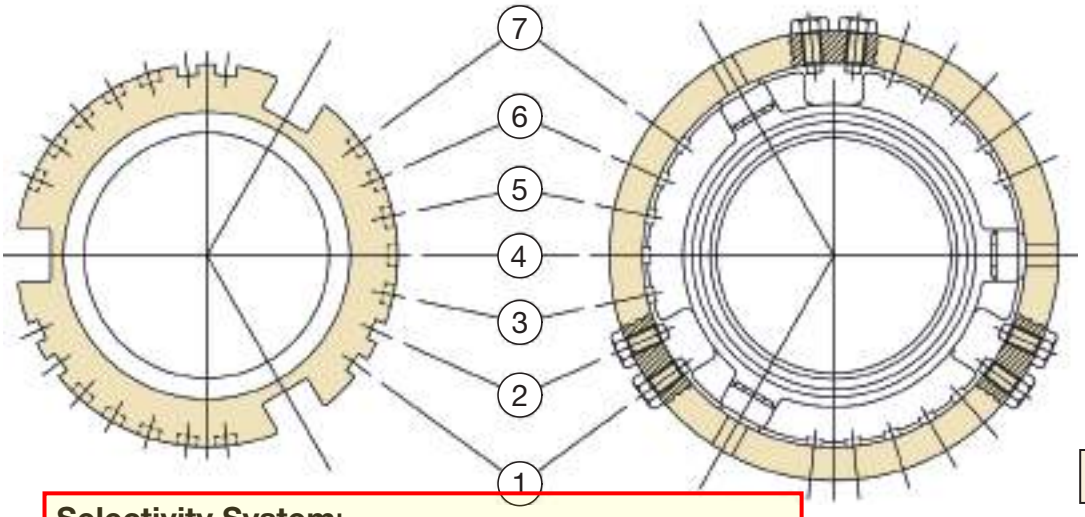
iTEM TAG NO. - LP1Y-QCU-0631A2
SERVICE DESCRIPTION - METHYL TERT BUTYL ETHER
SEL # - 6

Selectivity System for 3" (Ø 119 mm)

Coupling diameter:
Ø 119 mm (3")

Tank Unit

Hose Unit



Media	Position
acrylonitrile	**36** (P)
ethylene oxide	**34** (M)
propylene oxide	**35** (N)

Selectivity positions for acrylonitrile, ethylene oxide and propylene oxide for 3"

With 7 Positions

1	2	3	4	5	6	7

Selectivity System:
NATO STANAG 3756. Interchangable with other manufacturers.

Each unit can be provided with selectivity according to two systems – with 7 or 21 positions (see chart).

When ordering, please add "/SEL" and the chosen selectivity to the Mann Tek Code No.
Examples: S414A4401/**SEL 1** bzw. S414A4401/**SEL 12** (A)

With 21 Positions

12	13	14	15	16	17	23	24	25	26	27
34	35	36	37	45	46	47	56	57	67	

Order Number Breakdown – 1/2

1. Type of Coupling

A = API Adapter
AV = Tank Unit EN 13081 (special version)
B = Ball Valve
C = Dust Cap
CG = Dust Cap LPG
D = Swivel
E = Tank Unit with pressure valves
F = Hose Unit (ISO 45)
G = Tank Unit (ISO 45)
GS = Tank Unit (ISO 45) with selectivity
H = Sampling Vent & Drain Unit
I = Dust Plug ISO 45
K = Dust Cap ISO 45
L = Tank Unit LPG
LC = Tank Unit Cryogenic
M = Hose Unit LPG
MC = Hose Unit Cryogenic
N = Break Away Pin
NC = Break Away Pin Cryogenic
O = Break Away Wire
P = Dust Plug
R = Pressure Cap
RG = Pressure Cap LPG
S = Hose Unit (STANAG 3756)
SN = Hose Unit int. Break Away Pin
SO = Hose Unit int. Break Away Wire
T = Tank Unit (STANAG 3756)
U = Filter / Sight Glass
V = Dust Plug LPG
WA = Hose Fittings

2. Socket Diameter / Nominal Diameter

0 = 50 mm or ¾"
1 = 56 mm or 1", 1¼"
2 = 70 mm or 1½", 2"
3 = 105 mm or 2½"
4 = 119 mm or 3"
5 = 164 mm or 4"
V = 5"
6 = 238 mm or 6"
8 = 272 mm or 8"
10 = 10"
12 = 12"

3. Connection (Thread, Flange etc.)

01 = ¾" BSP (female)
02 = ¾" NPT (female)
03 = 1" BSP (female)
04 = 1" NPT (female)
05 = 1¼" BSP (female)
06 = 1¼" NPT (female)
07 = 1½" BSP (female)
08 = 1½" NPT (female)
09 = 1¾" BSP (female)
10 = 2" BSP (female)
11 = 2" NPT (female)
12 = 2½" BSP (female)
13 = 2½" NPT (female)
14 = 3" BSP (female)
15 = 3" NPT (female)
16 = 4" BSP (female)
17 = 4" NPT (female)
18 = Flange undrilled Ø 156
19 = Flange undrilled Ø 165
20 = Flange undrilled Ø 210
21 = Flange undrilled Ø 230
22 = Flange undrilled Ø 254
23 = Flange DN 25 PN 10/16
24 = Flange DN 25 PN 25/40
25 = Flange DN 32 PN 10/16
26 = Flange DN 32 PN 25/40
27 = Flange DN 40 PN 10/16
28 = Flange DN 40 PN 25/40
29 = Flange DN 50 PN 25/40*
30 = Flange DN 50 PN 10/16
31 = Flange DN 50 PN 25/40
32 = Flange DN 65 PN 25/40*
33 = Flange DN 65 PN 10/16
34 = Flange DN 65 PN 25/40
35 = Flange DN 80 PN 25/40*
36 = Flange DN 80 PN 10/16
37 = Flange DN 80 PN 25/40
38 = Flange DN 100 PN 25/40*
39 = Flange DN 100 PN 10/16
40 = Flange DN 100 PN 25/40
41 = Flange DN 125 PN 6
42 = Flange DN 125 PN 10/16
43 = Flange DN 125 PN 25/40
44 = Flange DN 150 PN 6
45 = Flange DN 150 PN 10/16
46 = Flange DN 150 PN 25/40
47 = Flange DN 20 PN 10/16
48 = Flange DN 20 PN 25/40
49 = Flange ¾" ANSI Class 150
50 = Flange ¾" ANSI Class 300
51 = Flange 1" ANSI Class 150
52 = Flange 1" ANSI Class 300
53 = Flange 1¼" ANSI Class 150
54 = Flange 1¼" ANSI Class 300
55 = Flange 1½" ANSI Class 150
56 = Flange 1½" ANSI Class 300
57 = Flange 2" ANSI Class 150
58 = Flange 2" ANSI Class 300
59 = Flange 2½" ANSI Class 150
60 = Flange 2½" ANSI Class 300
61 = Flange 3" ANSI Class 150
62 = Flange 3" ANSI Class 300
63 = Flange 4" ANSI Class 150
64 = Flange 4" ANSI Class 300
65 = Flange TW 1 (3" - DN 80)
66 = Flange TW 3 (4" - DN 100)
67 = Flange 3" T.T.M.A.
68 = Flange 4" T.T.M.A.
69 = ¾" BSP (male)
70 = ¾" NPT (male)
71 = 1" BSP (male)
72 = 1" NPT (male)
73 = 1¼" BSP (male)
74 = 1¼" NPT (male)
75 = 1½" BSP (male)
76 = 1½" NPT (male)
77 = 1¾" BSP (male)
78 = 2" BSP (male)
79 = 2" NPT (male)
80 = 2½" BSP (male)
81 = 2½" NPT (male)
82 = 3" BSP (male)
83 = 3" NPT (male)
84 = 4" BSP (male)
85 = 4" NPT (male)
86 = Weld.flange 2" Ø 60,5 ID
87 = Flange TW 1 (2" DN 50)
88 = Weld.flange 2" Ø 50 - Ø 70 (flat)
89 = Weld.flange 2" Ø 57 (int. chamfer)
90 = Weld.flange 2" Ø 60 (outer chamfer)
91 = Weld.flange 3" Ø 75 - Ø 90 (flat)
92 = Weld.flange 3" Ø76 (int. chamfer)
93 = Weld.flange 3" Ø 89 (outer chamfer)
94 = Weld.flange 4" Ø 100 - Ø 120 (flat)
95 = Weld.flange 4" Ø 102 (int. chamfer)
96 = Weld.flange 4" Ø 108 (int. chamfer)
97 = Weld.flange 4" Ø 114 (outer chamfer)
98 = Flange TW 1 (2" - DN 50) with drain connection
99 = Flange DN 150 PN 25
100 = Flange 6" ANSI Class 150
101 = Flange 6" ANSI Class 300
102 = Flange DN 200 PN 10
103 = Flange DN 200 PN 16
104 = Flange DN 200 PN 25
105 = Flange 8" ANSI Class 150
106 = Flange 8" ANSI Class 300
107 = Flange Square ISO 45
108 = S60 x 6 (female)
109 = S60 x 6 (male)
110 = 6" BSP (female)
111 = 6" NPT (female)
112 = W2" - 7 (female)
113 = Weld.flange 3" Ø 92 inner
114 = Square flange, 4 holes
115 = 6" BSP (male)
116 = 6" NPT (male)
117 = 8" NPT (female)
118 = 4" Victaulic
119 = Flange DN 50 PN 25/40**
120 = Flange DN 65 PN 25/40**
121 = Flange DN 80 PN 25/40**
122 = Flange DN 100 PN 25/40**
123 = W2" - 7 (male)
124 = 5" NPT (female)
125 = 5" NPT (male)
126 = Flange DN 100 PN 6
127 = Flange DN 80 PN 6
128 = Flange DN 65 PN 6
129 = Flange DN 50 PN 6
130 = Flange 8" ANSI Class 600
131 = W90 x 1/6" (female)
132 = ½" NPT (female)
133 = ½" BSP (female)
134 = Flange Ø 184.2, 6 holes
135 = Flange TW 7 (6" - DN 150)
136 = 4" ASSPT (female)
137 = Triclamp DN 25
138 = M54 x 1,5 (female)
139 = Triclamp DN 50
140 = Weld.flange Ø 73 (outer chamfer)
141 = 3" Victaulic
142 = Flange 5" ANSI Class 150
143 = 3" Ball valve
144 = 2" Victaulic
145 = 3" BSPT (male)
146 = 5" Victaulic
147 = 2" BSPT (female)
148 = 2" BSPT (male)
149 = 1½" Victaulic
150 = 2½" Victaulic
151 = Flange 1" DIN 11864-2
152 = Flange 2" DIN 11864-2
153 = Flange Ø 135, 8 x M6
154 = 4" BSPT (female)
155 = 4" BSPT (male)
156 = Weld flange 2" Ø 61,5 ID
157 = 3" BSPT (female)
158 = Weld end 1½" Ø 48 OD
159 = Thread TR 57 x 4
160 = Flange 2" BS10 Table D
161 = Flange 12" ANSI Class 150
162 = Flange 10" ANSI Class 150
163 = Flange DN 250 PN 16
164 = M130 x 6 (female)
165 = Flange 10" ANSI Class 300
166 = ACME 1¼" (female)
167 = ACME 1¾" (female)
168 = ACME 2¼" (female)
169 = ACME 3¼" (female)
170 = ACME 1¼" (male)
171 = ACME 1¾" (male)
172 = ACME 2¼" (male)
173 = ACME 3¼" (male)
174 = Weld.flange Ø 76 (outer chamfer)
175 = Flange DN 15 PN 10/16
176 = Flange DN 15 PN 25/40
177 = M130 x 6 (male)
178 = Flange 6" T.T.M.A.
179 = Flange DN 80 PN 25/40***
180 = ½" NPT (male)
181 = ½" BSP (male)
182 = 5" BSP (female)
183 = 5" BSP (male)
184 = Weld end 8" Ø 219 OD
185 = Weld end 6" Ø 168 OD
186 = Flange DN 250 PN 25
187 = Flange 2" T.T.M.A.
188 = Flange 3" BS10 Table D
189 = Flange ½" ANSI Class 150
190 = Flange 1" ANSI Class 150 Flat Face
191 = Flange 12" ANSI Class 300
192 = Flange DN 250 PN 10
193 = Weld end Ø 114 Schedule 40
194 = Weld end Ø 114 Schedule 80
195 = 6" Victaulic
196 = 1" Victaulic
197 = DN 125 JIS 5K
198 = DN 100 JIS 5K
199 = DN 80 JIS 5K
200 = DN 50 JIS 5K
201 = DN 40 JIS 5K
202 = Flange 2" DIN 11864-3
203 = 3½" BSP (female)
204 = Flange Ø 110, Ø 86/Ø 5.5 (6x)
205 = Weld end Ø 60 Schedule 80
206 = Weld end Ø 89 Schedule 40
207 = Weld end Ø 89 Schedule 80
208 = Flange DN 25 PN 6
209 = Flange DN 32 PN 6
210 = Flange DN 40 PN 6
211 = DN 125 JIS 10K
212 = DN 100 JIS 10K
213 = DN 80 JIS 10K
214 = DN 50 JIS 10K
215 = DN 40 JIS 10K
216 = Flange DN 80, holes Ø 14 (6x)
217 = Flanges 5" ANSI Class 300

* EN 1092-1:2001 Type E: Spigot
** EN 1092-1:2001 Type F
*** EN 1092-1:2001 Type C

NOTE: When swivels are chosen (first sign: D), the second and the third sign indicate thread side 1 one, the fourth and the fifth sign the second outlet indicate thread side 2.

Design may change without notice.

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Order Number Breakdown – 2/2

4. Version

A = Version No.1 (machined from bar)
B = Version No.2 (casted)
C = Version No.3 (kokill casted)
D = Sep. piston guide
E = Injection moulded
F = 6" Flange Hydrant
G = Drain connection
H = Leaf spring lock
I = Bent (dropped) Tank Unit, short (15°)
J = Bent (dropped) Tank Unit (15°)
K = Short Tank Unit/Swivel
N = Non Return Valve
P = Pressure (Custom)
S = Sight Glass
T = Transparent
U = Stop before disconnected

5. Material Coupling and Body

1 = Aluminium
2 = Brass
3 = Steel
4 = Edelstahl 1.4408 (AISI 316)
5 = Edelstahl 1.4301 (AISI 304)
6 = Titan
7 = Hastelloy
8 = PVDF
9 = PEEK

6. Material in the Innerparts

1 = Aluminium
2 = Brass
3 = Steel
4 = Stainless steel A4 (316)
5 = Stainless steel 1.4301
K = Inconel
6 = Titan
7 = Hastelloy
8 = PVDF
9 = PEEK
K = Inconel

7. O-Rings

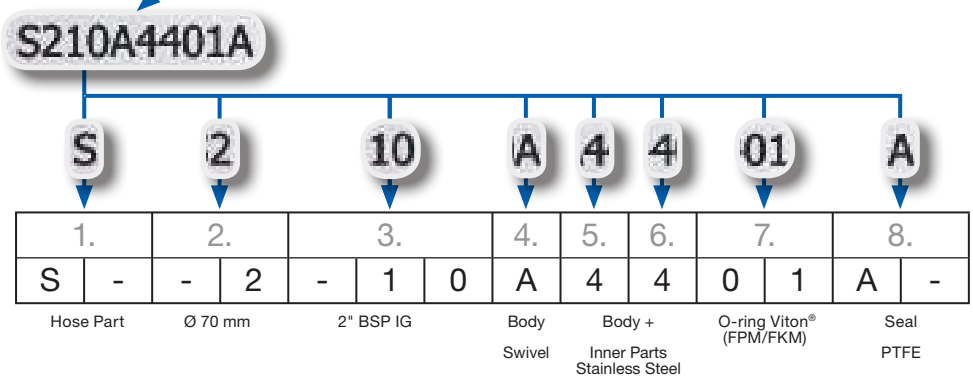
01 = Viton® (FPM/FKM)
02 = Nitrile (NBR)
03 = EPDM
04 = Kalrez® (FFKM) 6375
05 = NBR Low temp
06 = Teflon® (PTFE)
07 = Neoprene® (CR)
08 = Silicone (Q)
09 = Polyurethane (PUR)
10 = Butyl (IIR)
11 = Nitrile (Gasol NBR 70 K-6)
12 = Perfluorelastomer (FFPM/FFKM)
13 = PVC/NBR
14 = Fluorsilicone rubber (MFQ)
15 = FEP encapsulated silicone
16 = Hypalon® (CSM)
17 = Chemraz® 505 (FFKM)
18 = Xyflour® 860 (AFKM)
19 = Zetpol® / Therban® (HNBR)
20 = NBR 90° shore
21 = Viton®-GF (Special Viton quality)
22 = Composite
23 = Viton® GFLT-S
24 = Viton® GLT
25 = Klingerit®
26 = POM
27 = Epiclorhydrin (ECO)
28 = Viton® GFLT-S NMO
31 = Viton® 90 Shore (FPM/FKM)
33 = EPDM 291
34 = Kalrez® 0040
37 = Chemraz® 510 (90 Shore)
40 = FEP or PTFE encapsulated Viton®
50 = Kalrez® (PFPM) 1050LF
51 = Nylon® (PA)
61 = Viton® (FPM), FDA, USP C6 & ADI
62 = Nitrile (NBR), FDA, USP C6 & ADI
63 = EPDM, FDA, USP C6 & ADI
64 = Kalrez® (FFKM) 6230, FDA, USP C6 & ADI
66 = PTFE (Virgin), FDA
71 = FPM/FKM Low Temp
77 = Chemraz® SD517, FDA, USP C6 & ADI
83 = EPDM, BAM certified for oxygene

8. Flat (Thread) Seals / Extras

A = Flat Seal, Teflon® (PTFE)
B = Flat Seal, Vulkollan® (PUR)
C = 2-Way Ball Valve
D = Flat Seal, Viton® (FPM)
DA = Double Acting (Ball Valve)
E = Tank Unit with short piston
F = Flange thickness acc. to standard
G = Flat Seal, Hypalon®
H = Flat Seal, Nitrile (NBR)
I = Emco compatible
J = Flat Seal, EPDM
K = Adapter with locked piston guide
L = Locked thread (maggot screw)
M = Two-Step Cam Curve
N = No Branding
NA = Ball Valve w. square trunnion (no actuator)
P = Pressure Equalizing Valve
Q = Sampling Vent & Drain Unit with reduced bore diameter (Argus, Hydrant)
R = Hose Unit with int. Break Away
S = Sampling Vent & Drain Unit, only with one ball valve
SR = Spring Return (Ball Valve)
T = TW-Flange extended circles
U = Pressure Bleeding Valve 16 bar
U5 = Pressure Bleeding Valve 5 bar
U20 = Pressure Bleeding valve 20 bar
V = Locked Hose Unit
W = Swivel with double ball valve
X = Special surface treatment
Z = Excentric Tank Unit
-RA = Racing
-LC = Locking Cap
-S = FEP O-Ring in Hose Unit swivel
-ST = Steam
-XL = Oversized swivel
-45 = 45 Mesh
-60 = 60 Mesh
-10 = 100 Mesh

ID-Tag on Coupling:

Example:
2" (Ø 70 mm) Hose Unit (Coupler), female thread



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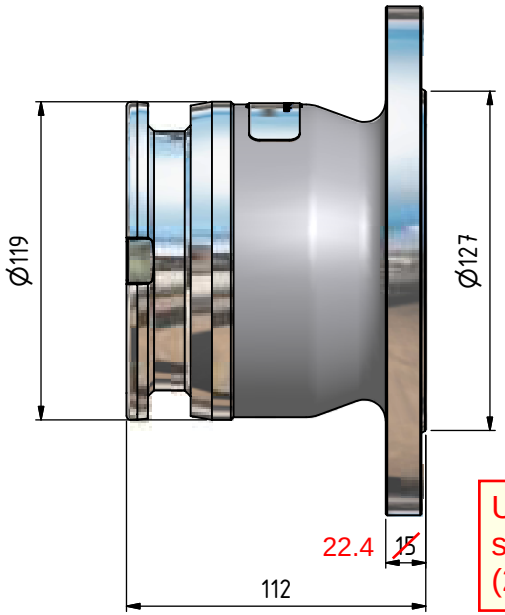
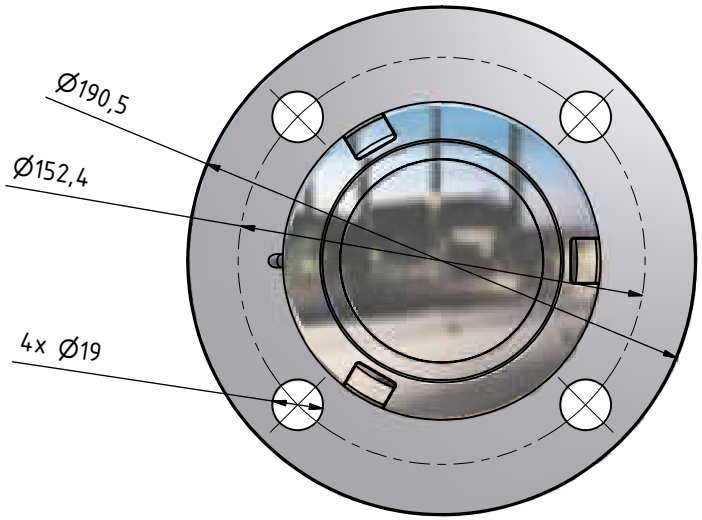
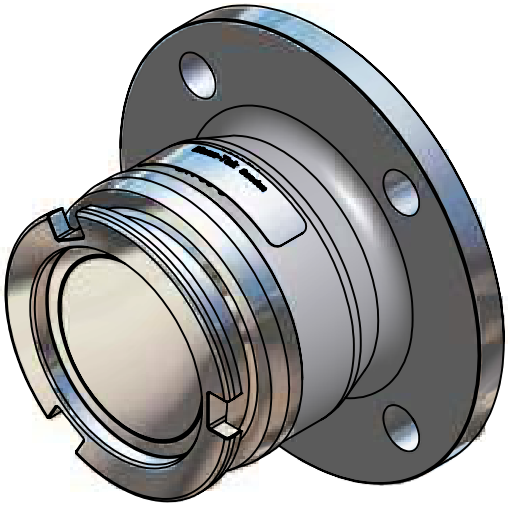
Selection of registred trade names from BASF, Bayer AG, B.F. Goodrich, Chemische Werke Hüls, Daikin, Dow Company, DSM, DuPont, DuPont Dow Elastomers, Esso Chemie, Hercules, Hoechst AG, Montedison, Monteflous, Nippon Zeon, Polysar LTD., Rhone Poulenc, 3M Company, Wacker Chemie, Precision Polymer Engineering Ltd.

Notice. This drawing contains poprietary information and shall not be used or reproduced or its contents disclosed in whole or in part, without the prior written concent of the Mann Teknik AB



Use ASME B16.5
standard flange thickness
(22.4 mm or 0.88 in)





Dimensions are in metric.

Detail No:	Material		Ytbeh./Treatment
Mann Tek	Designed/drawn by A. Olofsson Checked M. Bäckström Approved T. Mann	Scale: 1:2 Date 2005-12-19	Dimension with no tolerances acc to Machined SS-ISO 2768-1 medium Weld Forge SS-ISO 2768-1 coarse Surface roughness ISO 1302 Form & position tol SS ISO 1101

Tank Unit 3"	Data nr./Comp No. T461B4401	Blad/Sheet 1 / 1	
Flange 3" ANSI 150 PSI	Drawing No. T461B4401-1	Rev. 300	

NO.	Revision comprises	Date	Sign.	Appr by

Dust Cap for Tank Unit (Adapter)

To increase the lifespan of DDC tank units, dust caps should be used. They protect the couplings againstthe ingress of dirt and water.

Materials: polyethylene, aluminum, stainless steel or rubber. Supplied with cable attachment and eyelets (rubber design DN 40 - 50 with integrated eyelet).

With the exception of the simple rubber type, the caps feature a safeguard against inadvertent release: before removing the cap, a spring-loaded locking pin is pulled.

Polyethylene composite caps in sizes in sizes DN 40-80 mm include a patented design with lip seal to facilitate the handling.

PE-Composite
Patent No. 000840780-0001

Aluminium

Stainless Steel

Rubber

Inch / DN	Material [1]	Seal [2]	Weight ≈ kg	Weight ≈ lbs	Mann Tek Code No.
¾"-1" Ø 56 mm, DN 20-25	PE	Standard: FPM (Viton®)	0,1	0.29	C100A2201
	Al		0,2	0.49	C100A1101
	SS		0,6	1.30	C100A4401
1½-2" Ø 70 mm, DN 40-50	PE	NBR	0,1	0.2	C200E2202
	Al	Standard: FPM (Viton®)	0,3	0.7	C200A1101
	SS		0,6	1.3	C200C4401
	Rubber		0,1	0.3	C200D1300
2½" Ø 105 mm, DN 65	PE	NBR	0,2	0,4	C300E2202
	Al	Standard: FPM (Viton®)			C300A1101
	SS		1,5	3,3	C300C4401
3" Ø 119 mm, DN 80	PE	NBR	0,2	0,5	C400E2202
	Al	SPECIFY TEFLON (PTFE) SEAL			C400A1101
	SS				C400C4401
	Rubber	—			C400D1300
4" Ø 164 mm, DN 100	PE	Standard: FPM (Viton®)	0,2	0,5	C500E2202
	Al		1,2	2,5	C500B1101
	SS		2,5	5,5	C500C4401
6" Ø 238 mm, DN 150	PE				C600A2201
	Al		1,6	3,4	C600A1101
	SS				C600A4401
8" Ø 272 mm, DN 200	Al				C800B1101
	SS				C800B4401

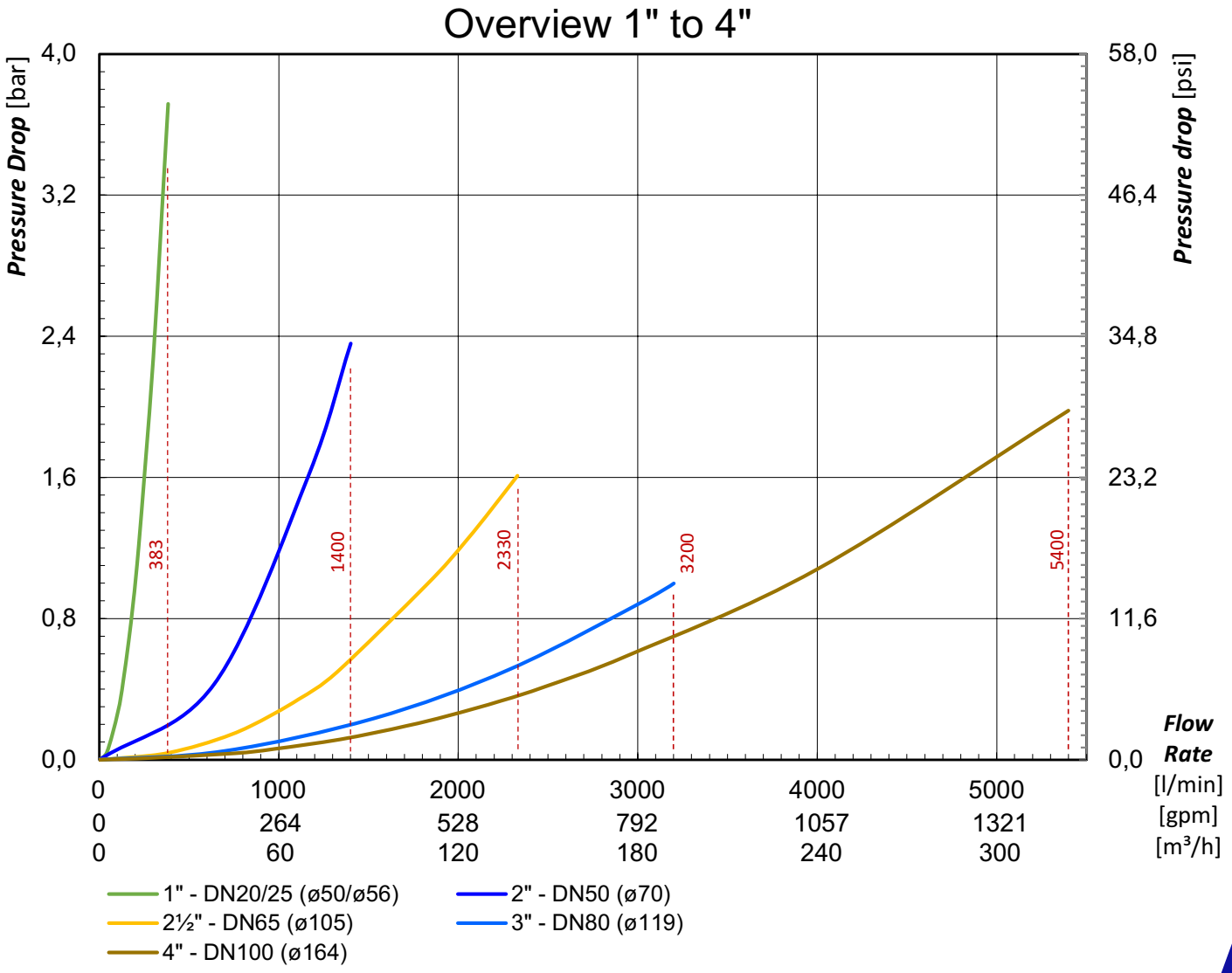
Rubber

[1] Material: PE = Polyethylen-Composite, Al = Aluminium, SS = Stainless Steel, Rubber = NBR/PVC

[2] Standard seal FPM/FKM or NBR. Alternative materials, e.g. EPDM, Chemraz®, Kalrez® or HNBR on request

Viton® is a registred trademark of DuPont, DuPont Elastomers.

Flow Diagram
Dry Disconnect Coupling



Test Fluid: n-paraffin **Temperature:** 20 °C **Density:** 0,75 kg/dm³ **Viscosity:** 1,75 mm²/s
Max. recommended flow velocity: 10m/s

Size	1"	2" (2.0)	2½"	3" (2.0)	4"	Unit
Flow rate at max. rec. flow velocity	383	1400	2330	3200	5400	l/min
Pressure drop at max. rec. Flow velocity	3,7	2,3	1,6	1,0	2,0	bar
Kv	10,3	47,8	95,4	167,2	199,6	[m³/h].[bar]
Cv	11,9	55,7	110,3	194,8	230,8	[gpm].[psi]

Verification method

- *Performance based on combination of physical tests and CFD flow simulations*
- *Test/Simulation method: STANAG 3756, Annex E*
- *Test/Simulation fluid: n-paraffin, Density: 0,75 kg/dm³, Viscosity: 1,75 mm²/s*
- *Design 2.0 performance valid for hose unit and tank unit 2.0 status.*

=== PIPING SPECIALTY ITEM LIST ===

CUT SHEET NUMBER:	PCS-QCU-046
DESCRIPTION:	Quick Connect Unit
VENDOR:	MannTek
MODEL:	316 SS FNPT Dry Disconnect Adapter with Dust Cap

NOTE:

USED AT TRUCK UNLOADING AREAS.

USED FOR THE FOLLOWING SERVICES
Acetonitrile
Alcohol, Isopropyl
Dichloromethane
Dimethyl Sulfoxide
Dimethylformamide
General Solvent
Methyl Tert Butyl Ether Please confirm whether MTBE is applicable to QCU-046 or it needs other type of QCU.
Sodium Hydroxide, Industrial 50%
Sulfuric Acid

CUT SHEET BEGINS ON NEXT PAGE

3" (Ø 119 mm) Technical Information

Tank Unit (Adapter) and Hose Unit (Coupler)

According to NATO STANAG 3756

Material	Maximum Working Pressure	Test Pressure	Minimum Burst Pressure
Aluminium	10 bar / 145 psi	15 bar / 218 psi	50 bar / 726 psi
Brass/Gunmetal	16 bar / 232 psi	24 bar / 348 psi	80 bar / 1160 psi
Stainless Steel	25 bar / 363 psi	37,5 bar / 544 psi	125 bar / 1813 psi
Titan	25 bar / 363 psi	37,5 bar / 544 psi	125 bar / 1813 psi
Hastelloy	25 bar / 363 psi	37,5 bar / 544 psi	125 bar / 1813 psi
PEEK	6 bar / 87 psi	9 bar / 131 psi	30 bar / 435 psi

Connections
3" in BSP, NPT and flanged inlet.

Applications
For road, rail tanker loading/unloading, filling of tanks and containers, refuelling and loading/unloading bunker boats (marine tankers) and small ships.
Also suitable for other applications where high loading rates are required and product contamination and spillage needs to be minimized.

Media
Petroleum products: gasoline, diesel, oil etc.
Chemical products: ethylene oxide, propylene oxide, acrylonitrile, butadiene, ammonia, vinyl chloride, toluene, xylene, sulphuric acid, phenol etc.
Gas: vapor recovery/balance systems
Dry powder.

Material of Body
Aluminium, brass/gunmetal, stainless steel, Hastelloy and PEEK. Other materials on request.

Seals
Standard seals FPM (Viton®), EPDM, FFKM (Chemraz®, Kalrez®), NBR. Other on request.

High Flow Rates / Low Pressure Drop
Allows maximum product transfer with minimal losses.

Recommended Maximum Flow Rates
2000 litres/minute (fuel)

Selectivity
- avoid mixing products
To avoid product contamination caused by connecting a hose unit to the wrong tank unit, selective versions of the hose and tank units are available.
Each unit has a number of selective positions, designated by a coded part number according to the coupling size – specify when placing order.

Interchangeability
Compatible with couplings of other manufacturers.

Hose Unit with Integrated Swivel
All hose units are supplied with an integrated swivel.

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3" (Ø 119 mm) Tank Unit (Adapter) – Female Thread

Connection [1] Inch/DN	Body Material [2]	Seal		Weight ≈		Mann Tek Code No.
		O-Ring [3]	Thread Seal	kg	lbs	
F 3" BSP	Al	SPECIFY TEFLON (PTFE) O-RING	PUR	1,2	2.7	T414D1101B
F 3" BSP				1,3	2.9	T414K1101B*)
F 3" NPT				1,2	2.7	T415D1101
F 3" BSP	GM		PUR	3,2	7.1	T414D2201B
F 3" NPT				3,5	7.7	T415D2201
F 3" BSP	Br		PUR	1,3	2.9	T414K2201B*)
F 3" BSP	SS		PTFE	3,1	6.8	T414B4401A
F 3" NPT				3,4	7.5	T415B4401A
F 3" BSP	Hastelloy		PTFE			T414A7701A
F 3" NPT						T415A7701
F 3" BSP	PVDF/Hastelloy		PTFE	3,6	7.9	T414A8701A
F 3" BSP	PEEK		PTFE			T414A9901A
F 3" NPT						T415A9901

According to NATO STANAG 3756

3" (Ø 119 mm) Tank Unit (Adapter)– Flanged Inlet

Flange [1B]	Body Material [2]	Seal [3]	Weight ≈		Mann Tek Code No.
		O-Ring	kg	lbs	
undrilled Ø 210 mm	Al	Standard: **FPM/FKM** (Viton®) Other on request.	2,0	4.4	T420D1101
DN 65 PN 10 / 16 Type A					T433D1101
DN 80 PN 10/16 Type A			2,0	4.4	T436D1101
3" ASA 150			1,8	4.0	T461D1101
DIN 28459 'TW 1' / DN 80			1,4	3.1	T465D1101
DIN 28459 'TW 3' / DN 100			1,6	3.5	T466D1101
3" TTMA			1,8	4.0	T467D1101
4" TTMA			1,5	3.3	T468D1101
undrilled Ø 210 mm	GM		6,2	13.7	T320D2201
DN 65 PN 10/16 Type B			4,9	10.8	T433D2201
DN 65 PN 25/40 Type B					T434D2201
DN 80 PN 10/16 Type B			5,2	11.5	T436D2201
DN 80 PN 25/40 Type B					T437D2201
3" ASA 150			4,9	10.8	T461D2201
3" ASA 300					T462D2201
DIN 28459 'TW 1' / DN 80			4,1	9.0	T465D2201
DIN 28459 'TW 3' / DN 100			4,7	10.4	T466D2201
3" TTMA					T467D2201
4" TTMA			4,6	10.1	T468D2201
undrilled Ø 210 mm	SS				T420B4401
undrilled Ø 210 mm***)					T420B4401F
DN 65 PN 10/16 Type B					T433B4401
DN 65 PN 10/16 Type B***)					T433B4401F
DN 65 PN 25/40 Type B					T434B4401
DN 65 PN 25/40 Type B***)					T434B4401F
DN 80 PN 25/40 Type E**)					T435B4401F
DN 80 PN 10/16 Type B			5,1	11.2	T436B4401
DN 80 PN 10/16 Type B***)					T436B4401F
DN 80 PN 25/40 Type B			5,1	11.2	T437B4401
DN 80 PN 25/40 Type B***)					T437B4401F
3" ASA 150			4,9	10.8	T461B4401
3" ASA 150***)				14.1	T461B4401F
3" ASA 300			4,9	10.8	T462B4401
3" ASA 300***)					T462B4401F
DIN 28459 'TW 1' / DN 80			3,9	8.6	T465B4401
DIN 28459 'TW 3' / DN 100					T466B4401
3" TTMA					T467B4401
4" TTMA					T468B4401
undrilled Ø 210 mm	PEEK				T420A9901
DN 80 PN 10/16 Type B					T436A9901
3" ASA 150					T461A9901
3" TTMA					T467A9901
4" TTMA					T468A9901

According to NATO STANAG 3756

*) shortened version for tight installation situations

**) Type E (with spigot), EN 1092-1, see page 44

***) Flange with standard thickness

[1A] F = Female thread, BSP = EN ISO 228, NPT = ANSI B1.20.1

[1B] PN 10/16/25/40 = EN 1092 (types see page 44), ASA = ANSI B16.5 (150 o. 300 psi), DIN 28459 = old standard TW, 10 bar. TTMA = Truck Trailer Manufacturers Association

[2] Material: Al = Aluminium, GM = Gunmetal, SS = Stainless Steel

[3] Standard seal FPM/FKM. Alternative materials, e.g. EPDM, Chemraz®, Kalrez®, NBR or HNBR on request

Viton® is a registred trademark of DuPont, DuPont Elastomers.

O-Ring Materials

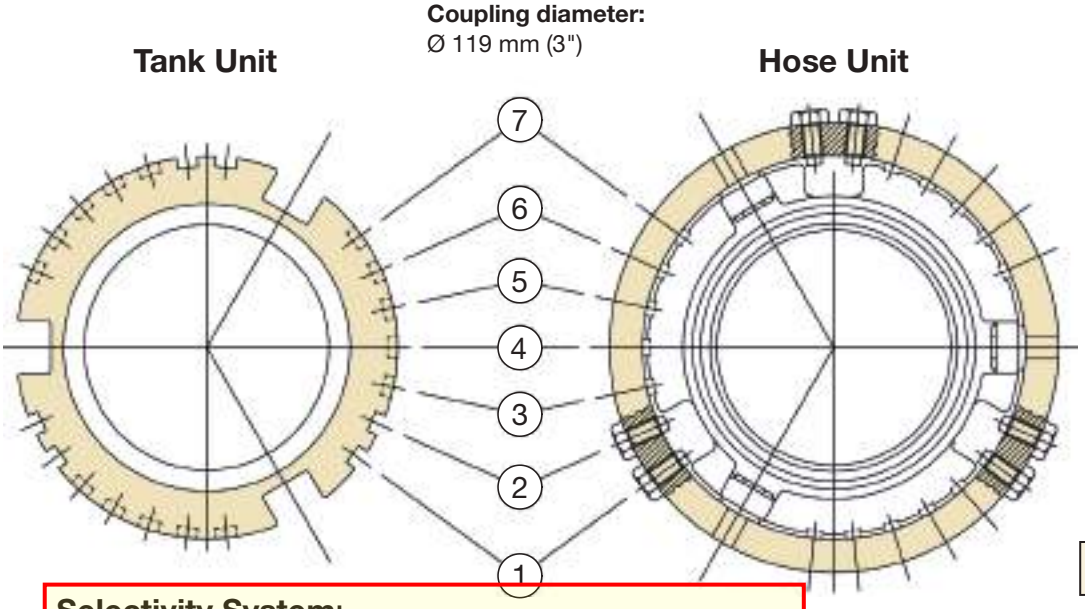
Designation	Trade name	ISO 1629	ASTM 1418	Temp. Range °C		Field of Application
Nitrile Butadiene Rubber / Nitrile Rubber	Buna N® Europrene N® Hycar® Nipol N® Perbunan ®	NBR	NBR	-55	110	Petroleum based products / aliphatic hydrocarbons (gasoline, diesel, oils, petroleum, propane, butane). Animal and vegetal oils and fats. Flame retardant liquids.. Silicone oils and greases. Water up to 80°C.
Ethylene-Propylene Diene Rubber	Dutral Keltan® Vistalon® Buna AP®	EPDM	EPDM	-55	120	Water, hot water, vapour, brake fluids, detergents. Alcohols, ketons, engine coolants, phosphate based flame retardants, organic and inorganic acids and alkalis. Not suitable for petroleum based products.
Fluoroelastomer	Fluorel® Technoflon® Viton®	FPM	FKM	-20	200	Petroleum based products / aliphatic hydrocarbons (gasoline, diesel, oils, petroleum, propane, butane) - also with high additive content. Phosphate based flame retardants. Silicone oils and greases, acids and alkalis.
Fluorosilicone Elastomer	——	MFQ	FVMQ	-60	200	Additional to the properties of silicone elastomers, fluorosilicone elastomers show a consideably higher resistance to oils, fuels and solvents - especially to aromatic and aromatic and chlorinated hydrocarbons and alcohols.
Silicone	Silastic® Silopren®	MVQ	VMQ	-60	200	Very good heat and cold resistance, very good resistance to oxygene, ozone, UV rays and wheather influence. High electrical insulating properties, physiologically safe.
Hydrogenated Nitrile Butadiene Rubber	Therban® Tornac® Zetpol®	HNBR	HNBR	-35	120	Very good resistance to petroleum based products / aliphatic hydrocarbons like NBR, but better resistance to high temperature, ageing and wheather influence. Excellent physical properties also with high temperatures.
Butyl Rubber	Exxon Butyl® Polysar Butyl®	IIR	IIR	-55	100	For O-Rings this material has largely been replaced by EPDM. Butyl is resistant to the same fluid types as EPDM and -except for resistance to gas permeation- it is somewhat inferior to EPDM for O-Ring use. Excellent weather resistance and gas permeation resistance. Not suitable for petroleum based products.
Perfluorinated Elastomer	CHEMRAZ® Kalrez® PERLAST®	FFPM	FFPM	-40	260	Best chemical resistance of all elastomers, including organic acids, acetic acid, benzoic acid, formic acid.
Chloroprene Rubber	Baypren® Neoprene®	CR	CR	-40	120	Resistant to refrigerants, ammonia, carbon dioxide, freon (R12, R13, R21, R22, R113 - R115), silicone oils, water, oxygen (low pressure), alkalis, coustic soda, alcohols, chlorine, ozone, castor oil and vegetable oils. Properties of CR are similar to NBR but slightly lower resistance to petroleum based producs, better resistance to ageing, UV light and ozone.
Polyester / Polyether Urethane Rubber	Polyurethane Adiprene® Urepan® Desmopan®	PU PUR AU EU	AU EU	-40	100	Very good resistance to petroleum based products, also with aromatic content. Good resistance to ozone and ageing. Very good mechanical properties. Not resistant to esters, concentrated acids and lyes, water above +50° C.
Polytetrafluoroethylene	Teflon®	PTFE	PTFE	-200	260	Universal chemical resistance except liquid alkaline metals and some fluorine compounds. Very good anti friction properties, low wear. High temperature resistance. Physiologically safe. Use for statical seals only.
Fluorinated Ethylene Propylene	Teflon FEP®	FEP/MVQ	FEP/VMQ	-60	200	Universal chemical resistance except liquid alkaline metals and some fluorine compounds. Very good anti friction properties, low wear. High temperature resistance. Physiologically safe.
		FEP/FPM	FEP/FKM	-20	200	
Perflouralkoxy	Teflon PFA®	PFA/MVQ	PFA/VMQ	-60	250	Universal chemical resistance except liquid alkaline metals and some fluorine compounds. Very good anti friction properties, low wear. High temperature resistance. Physiologically safe.
		PFA/FPM	PFA/FKM	-20	250	
Tetrafluoroethylene-Propylene Copolymer Elastomer	Aflas®	——	TFE / P	-25	200	Petroleum based products, oils and greases, brake fluids, fuels, alcohols, heat transfer media, oils. amines, acids, alkalis.

***Note**: Specifications in this chart are based on the information given by our suppliers together with published guides. This is not always a guarantee for a proper function. If in doubt, please ask our sales team, statiing medium and operating conditions.*

Selectivity Coding Table

iTEM TAG NO. - LP1Y-QCU-0631A1
SERVICE DESCRIPTION - METHYL TERT BUTYL ETHER
SEL # - 6

Selectivity System for 3" (Ø 119 mm)



Media	Position
acrylonitrile	**36** (P)
ethylene oxide	**34** (M)
propylene oxide	**35** (N)

Selectivity positions for acrylonitrile, ethylene oxide and propylene oxide for 3"

With 7 Positions

1	2	3	4	5	6	7

Selectivity System:
NATO STANAG 3756. Interchangable with other manufacturers.

Each unit can be provided with selectivity according to two systems – with 7 or 21 positions (see chart).

When ordering, please add "/SEL" and the chosen selectivity to the Mann Tek Code No.
Examples: S414A4401**/SEL 1** bzw. S414A4401**/SEL 12** (A)

With 21 Positions

12	13	14	15	16	17	23	24	25	26	27
34	35	36	37	45	46	47	56	57	67	

1. Type of Coupling

A = API Adapter
AV = Tank Unit EN 13081 (special version)
B = Ball Valve
C = Dust Cap
CG = Dust Cap LPG
D = Swivel
E = Tank Unit with pressure valves
F = Hose Unit (ISO 45)
G = Tank Unit (ISO 45)
GS = Tank Unit (ISO 45) with selectivity
H = Sampling Vent & Drain Unit
I = Dust Plug ISO 45
K = Dust Cap ISO 45
L = Tank Unit LPG
LC = Tank Unit Cryogenic
M = Hose Unit LPG
MC = Hose Unit Cryogenic
N = Break Away Pin
NC = Break Away Pin Cryogenic
O = Break Away Wire
P = Dust Plug
R = Pressure Cap
RG = Pressure Cap LPG
S = Hose Unit (STANAG 3756)
SN = Hose Unit int. Break Away Pin
SO = Hose Unit int. Break Away Wire
T = Tank Unit (STANAG 3756)
U = Filter / Sight Glass
V = Dust Plug LPG
WA = Hose Fittings

2. Socket Diameter / Nominal Diameter

0 = 50 mm or ¾"
1 = 56 mm or 1", 1¼"
2 = 70 mm or 1½", 2"
3 = 105 mm or 2½"
4 = 119 mm or 3"
5 = 164 mm or 4"
V = 5"
6 = 238 mm or 6"
8 = 272 mm or 8"
10 = 10"
12 = 12"

3. Connection (Thread, Flange etc.)

01 = ¾" BSP (female)
02 = ¾" NPT (female)
03 = 1" BSP (female)
04 = 1" NPT (female)
05 = 1¼" BSP (female)
06 = 1¼" NPT (female)
07 = 1½" BSP (female)
08 = 1½" NPT (female)
09 = 1¾" BSP (female)
10 = 2" BSP (female)
11 = 2" NPT (female)
12 = 2½" BSP (female)
13 = 2½" NPT (female)
14 = 3" BSP (female)
15 = 3" NPT (female)
16 = 4" BSP (female)
17 = 4" NPT (female)
18 = Flange undrilled Ø 156
19 = Flange undrilled Ø 165
20 = Flange undrilled Ø 210
21 = Flange undrilled Ø 230
22 = Flange undrilled Ø 254
23 = Flange DN 25 PN 10/16
24 = Flange DN 25 PN 25/40
25 = Flange DN 32 PN 10/16
26 = Flange DN 32 PN 25/40
27 = Flange DN 40 PN 10/16
28 = Flange DN 40 PN 25/40
29 = Flange DN 50 PN 25/40*
30 = Flange DN 50 PN 10/16
31 = Flange DN 50 PN 25/40
32 = Flange DN 65 PN 25/40*
33 = Flange DN 65 PN 10/16
34 = Flange DN 65 PN 25/40
35 = Flange DN 80 PN 25/40*
36 = Flange DN 80 PN 10/16
37 = Flange DN 80 PN 25/40
38 = Flange DN 100 PN 25/40*
39 = Flange DN 100 PN 10/16
40 = Flange DN 100 PN 25/40
41 = Flange DN 125 PN 6
42 = Flange DN 125 PN 10/16
43 = Flange DN 125 PN 25/40
44 = Flange DN 150 PN 6
45 = Flange DN 150 PN 10/16
46 = Flange DN 150 PN 25/40
47 = Flange DN 20 PN 10/16
48 = Flange DN 20 PN 25/40
49 = Flange ¾" ANSI Class 150
50 = Flange ¾" ANSI Class 300
51 = Flange 1" ANSI Class 150
52 = Flange 1" ANSI Class 300
53 = Flange 1¼" ANSI Class 150
54 = Flange 1¼" ANSI Class 300
55 = Flange 1½" ANSI Class 150
56 = Flange 1½" ANSI Class 300
57 = Flange 2" ANSI Class 150
58 = Flange 2" ANSI Class 300
59 = Flange 2½" ANSI Class 150
60 = Flange 2½" ANSI Class 300
61 = Flange 3" ANSI Class 150
62 = Flange 3" ANSI Class 300
63 = Flange 4" ANSI Class 150
64 = Flange 4" ANSI Class 300
65 = Flange TW 1 (3" - DN 80)
66 = Flange TW 3 (4" - DN 100)
67 = Flange 3" T.T.M.A.
68 = Flange 4" T.T.M.A.
69 = ¾" BSP (male)
70 = ¾" NPT (male)
71 = 1" BSP (male)
72 = 1" NPT (male)
73 = 1¼" BSP (male)
74 = 1¼" NPT (male)
75 = 1½" BSP (male)
76 = 1½" NPT (male)
77 = 1¾" BSP (male)
78 = 2" BSP (male)
79 = 2" NPT (male)
80 = 2½" BSP (male)
81 = 2½" NPT (male)
82 = 3" BSP (male)
83 = 3" NPT (male)
84 = 4" BSP (male)
85 = 4" NPT (male)
86 = Weld.flange 2" Ø 60,5 ID
87 = Flange TW 1 (2" DN 50)
88 = Weld.flange 2" Ø 50 - Ø 70 (flat)
89 = Weld.flange 2" Ø 57 (int. chamfer)
90 = Weld.flange 2" Ø 60 (outer chamfer)
91 = Weld.flange 3" Ø 75 - Ø 90 (flat)
92 = Weld.flange 3" Ø76 (int. chamfer)
93 = Weld.flange 3" Ø 89 (outer chamfer)
94 = Weld.flange 4" Ø 100 - Ø 120 (flat)
95 = Weld.flange 4" Ø 102 (int. chamfer)
96 = Weld.flange 4" Ø 108 (int. chamfer)
97 = Weld.flange 4" Ø 114 (outer chamfer)
98 = Flange TW 1 (2" - DN 50) with drain connection
99 = Flange DN 150 PN 25
100 = Flange 6" ANSI Class 150
101 = Flange 6" ANSI Class 300
102 = Flange DN 200 PN 10
103 = Flange DN 200 PN 16
104 = Flange DN 200 PN 25
105 = Flange 8" ANSI Class 150
106 = Flange 8" ANSI Class 300
107 = Flange Square ISO 45
108 = S60 x 6 (female)
109 = S60 x 6 (male)
110 = 6" BSP (female)
111 = 6" NPT (female)
112 = W2" - 7 (female)
113 = Weld.flange 3" Ø 92 inner
114 = Square flange, 4 holes
115 = 6" BSP (male)
116 = 6" NPT (male)
117 = 8" NPT (female)
118 = 4" Victaulic
119 = Flange DN 50 PN 25/40**
120 = Flange DN 65 PN 25/40**
121 = Flange DN 80 PN 25/40**
122 = Flange DN 100 PN 25/40**
123 = W2" - 7 (male)
124 = 5" NPT (female)
125 = 5" NPT (male)
126 = Flange DN 100 PN 6
127 = Flange DN 80 PN 6
128 = Flange DN 65 PN 6
129 = Flange DN 50 PN 6
130 = Flange 8" ANSI Class 600
131 = W90 x 1/6" (female)
132 = ½" NPT (female)
133 = ½" BSP (female)
134 = Flange Ø 184.2, 6 holes
135 = Flange TW 7 (6" - DN 150)
136 = 4" ASSPT (female)
137 = Triclamp DN 25
138 = M54 x 1,5 (female)
139 = Triclamp DN 50
140 = Weld.flange Ø 73 (outer chamfer)
141 = 3" Victaulic
142 = Flange 5" ANSI Class 150
143 = 3" Ball valve
144 = 2" Victaulic
145 = 3" BSPT (male)
146 = 5" Victaulic
147 = 2" BSPT (female)
148 = 2" BSPT (male)
149 = 1½" Victaulic
150 = 2½" Victaulic
151 = Flange 1" DIN 11864-2
152 = Flange 2" DIN 11864-2
153 = Flange Ø 135, 8 x M6
154 = 4" BSPT (female)
155 = 4" BSPT (male)
156 = Weld flange 2" Ø 61,5 ID
157 = 3" BSPT (female)
158 = Weld end 1½" Ø 48 OD
159 = Thread TR 57 x 4
160 = Flange 2" BS10 Table D
161 = Flange 12" ANSI Class 150
162 = Flange 10" ANSI Class 150
163 = Flange DN 250 PN 16
164 = M130 x 6 (female)
165 = Flange 10" ANSI Class 300
166 = ACME 1¼" (female)
167 = ACME 1¾" (female)
168 = ACME 2¼" (female)
169 = ACME 3¼" (female)
170 = ACME 1¼" (male)
171 = ACME 1¾" (male)
172 = ACME 2¼" (male)
173 = ACME 3¼" (male)
174 = Weld.flange Ø 76 (outer chamfer)
175 = Flange DN 15 PN 10/16
176 = Flange DN 15 PN 25/40
177 = M130 x 6 (male)
178 = Flange 6" T.T.M.A.
179 = Flange DN 80 PN 25/40***
180 = ½" NPT (male)
181 = ½" BSP (male)
182 = 5" BSP (female)
183 = 5" BSP (male)
184 = Weld end 8" Ø 219 OD
185 = Weld end 6" Ø 168 OD
186 = Flange DN 250 PN 25
187 = Flange 2" T.T.M.A.
188 = Flange 3" BS10 Table D
189 = Flange ½" ANSI Class 150
190 = Flange 1" ANSI Class 150 Flat Face
191 = Flange 12" ANSI Class 300
192 = Flange DN 250 PN 10
193 = Weld end Ø 114 Schedule 40
194 = Weld end Ø 114 Schedule 80
195 = 6" Victaulic
196 = 1" Victaulic
197 = DN 125 JIS 5K
198 = DN 100 JIS 5K
199 = DN 80 JIS 5K
200 = DN 50 JIS 5K
201 = DN 40 JIS 5K
202 = Flange 2" DIN 11864-3
203 = 3½" BSP (female)
204 = Flange Ø 110, Ø 86/Ø 5.5 (6x)
205 = Weld end Ø 60 Schedule 80
206 = Weld end Ø 89 Schedule 40
207 = Weld end Ø 89 Schedule 80
208 = Flange DN 25 PN 6
209 = Flange DN 32 PN 6
210 = Flange DN 40 PN 6
211 = DN 125 JIS 10K
212 = DN 100 JIS 10K
213 = DN 80 JIS 10K
214 = DN 50 JIS 10K
215 = DN 40 JIS 10K
216 = Flange DN 80, holes Ø 14 (6x)
217 = Flanges 5" ANSI Class 300

* EN 1092-1:2001 Type E: Spigot
** EN 1092-1:2001 Type F
*** EN 1092-1:2001 Type C

NOTE: When swivels are chosen (first sign: D), the second and the third sign indicate thread side 1 one, the fourth and the fifth sign the second outlet indicate thread side 2.

Design may change without notice.

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Order Number Breakdown – 2/2

4. Version

A = Version No. 1 (machined from bar)
B = Version No. 2 (casted)
C = Version No. 3 (kokill casted)
D = Sep. piston guide
E = Injection moulded
F = 6" Flange Hydrant
G = Drain connection
H = Leaf spring lock
I = Bent (dropped) Tank Unit, short (15°)
J = Bent (dropped) Tank Unit (15°)
K = Short Tank Unit/Swivel
N = Non Return Valve
P = Pressure (Custom)
S = Sight Glass
T = Transparent
U = Stop before disconnected

5. Material Coupling and Body

1 = Aluminium
2 = Brass
3 = Steel
4 = Edelstahl 1.4408 (AISI 316)
5 = Edelstahl 1.4301 (AISI 304)
6 = Titan
7 = Hastelloy
8 = PVDF
9 = PEEK

6. Material in the Innerparts

1 = Aluminium
2 = Brass
3 = Steel
4 = Stainless steel A4 (316)
5 = Stainless steel 1.4301
K = Inconel
6 = Titan
7 = Hastelloy
8 = PVDF
9 = PEEK
K = Inconel

7. O-Rings

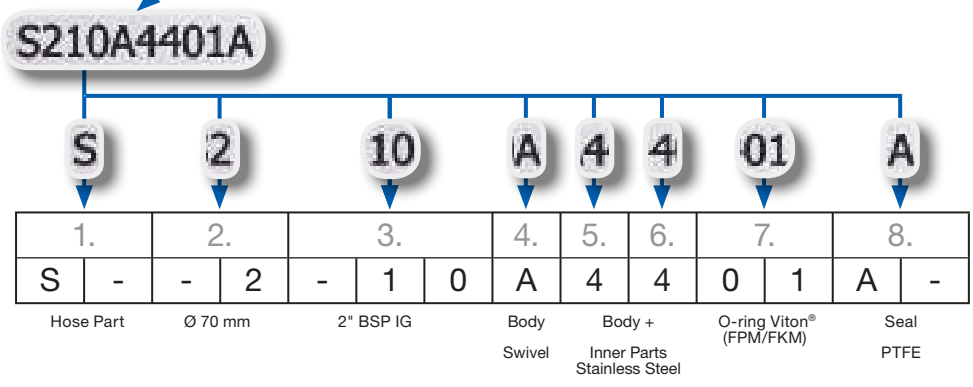
01 = Viton® (FPM/FKM)
02 = Nitrile (NBR)
03 = EPDM
04 = Kalrez® (FFKM) 6375
05 = NBR Low temp
06 = Teflon® (PTFE)
07 = Neoprene® (CR)
08 = Silicone (Q)
09 = Polyurethane (PUR)
10 = Butyl (IIR)
11 = Nitrile (Gasol NBR 70 K-6)
12 = Perfluorelastomer (FFPM/FFKM)
13 = PVC/NBR
14 = Fluorsilicone rubber (MFQ)
15 = FEP encapsulated silicone
16 = Hypalon® (CSM)
17 = Chemraz® 505 (FFKM)
18 = Xyflour® 860 (AFKM)
19 = Zetpol® / Therban® (HNBR)
20 = NBR 90° shore
21 = Viton®-GF (Special Viton quality)
22 = Composite
23 = Viton® GFLT-S
24 = Viton® GLT
25 = Klingerit®
26 = POM
27 = Epiclorhydrin (ECO)
28 = Viton® GFLT-S NMO
31 = Viton® 90 Shore (FPM/FKM)
33 = EPDM 291
34 = Kalrez® 0040
37 = Chemraz® 510 (90 Shore)
40 = FEP or PTFE encapsulated Viton®
50 = Kalrez® (PFPM) 1050LF
51 = Nylon® (PA)
61 = Viton® (FPM), FDA, USP C6 & ADI
62 = Nitrile (NBR), FDA, USP C6 & ADI
63 = EPDM, FDA, USP C6 & ADI
64 = Kalrez® (FFKM) 6230, FDA, USP C6 & ADI
66 = PTFE (Virgin), FDA
71 = FPM/FKM Low Temp
77 = Chemraz® SD517, FDA, USP C6 & ADI
83 = EPDM, BAM certified for oxygene

8. Flat (Thread) Seals / Extras

A = Flat Seal, Teflon® (PTFE)
B = Flat Seal, Vulkollan® (PUR)
C = 2-Way Ball Valve
D = Flat Seal, Viton® (FPM)
DA = Double Acting (Ball Valve)
E = Tank Unit with short piston
F = Flange thickness acc. to standard
G = Flat Seal, Hypalon®
H = Flat Seal, Nitrile (NBR)
I = Emco compatible
J = Flat Seal, EPDM
K = Adapter with locked piston guide
L = Locked thread (maggot screw)
M = Two-Step Cam Curve
N = No Branding
NA = Ball Valve w. square trunnion (no actuator)
P = Pressure Equalizing Valve
Q = Sampling Vent & Drain Unit with reduced bore diameter (Argus, Hydrant)
R = Hose Unit with int. Break Away
S = Sampling Vent & Drain Unit, only with one ball valve
SR = Spring Return (Ball Valve)
T = TW-Flange extended circles
U = Pressure Bleeding Valve 16 bar
U5 = Pressure Bleeding Valve 5 bar
U20 = Pressure Bleeding valve 20 bar
V = Locked Hose Unit
W = Swivel with double ball valve
X = Special surface treatment
Z = Excentric Tank Unit
-RA = Racing
-LC = Locking Cap
-S = FEP O-Ring in Hose Unit swivel
-ST = Steam
-XL = Oversized swivel
-45 = 45 Mesh
-60 = 60 Mesh
-10 = 100 Mesh

ID-Tag on Coupling:

Example:
2" (Ø 70 mm) Hose Unit (Coupler), female thread



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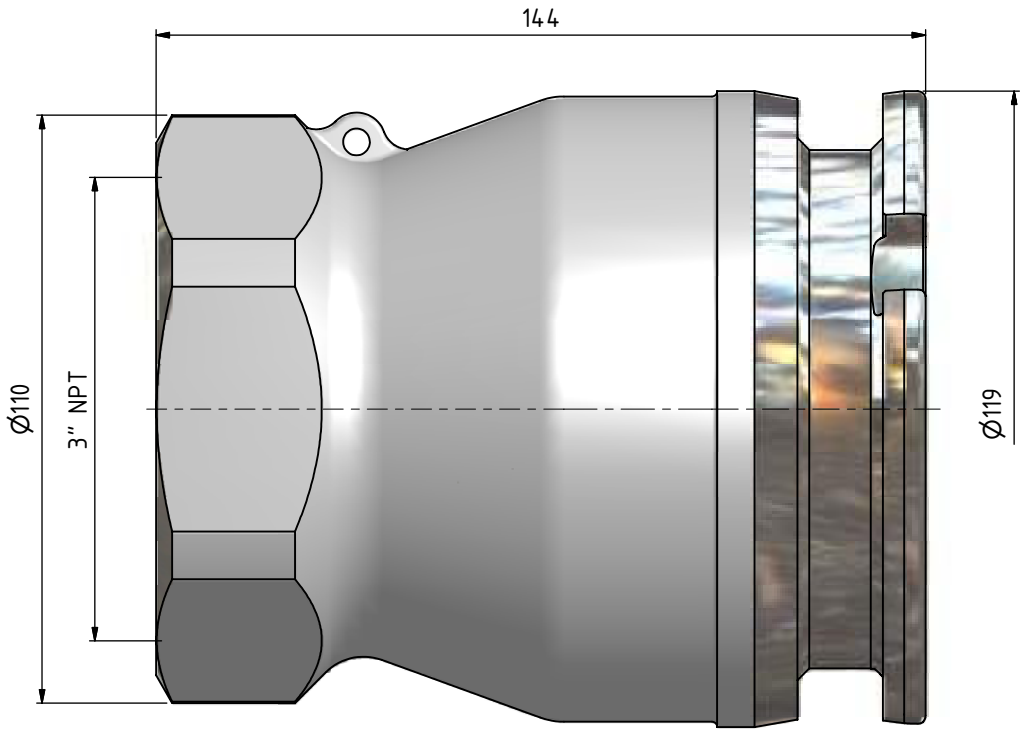
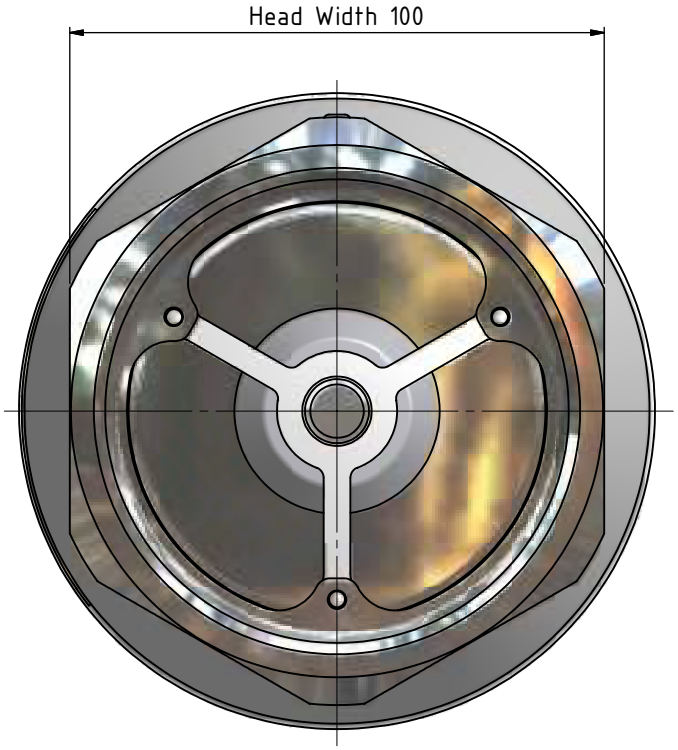
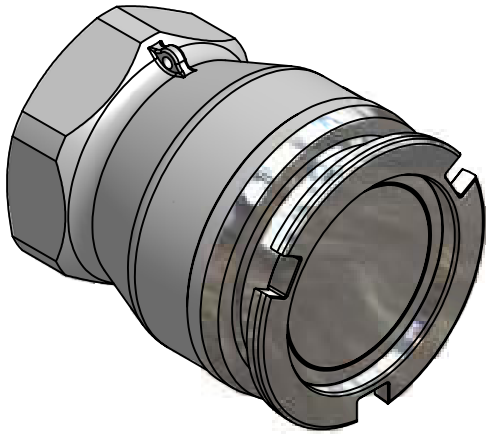






Dimensions are in metric.

Detail No:	Material			Ytbeh./Treatment
Mann Tek	Designed/drawn by P. Nyberg	Scale: 1:1	Dimension with no tolerances acc to	
	Checked G. Kopplin	Date 2007-05-15	Machined	SS-ISO 2768-1 medium
	Approved T. Mann		Weld Forge	SS-ISO 2768-1 coarse
			Surface roughness	ISO 1302
			Form & position tol	SS ISO 1101

Tank Unit 3"

Connection: 3" NPT

Data nr./Comp No.	Blad/Sheet
T415B4401	1 / 1

Drawing No.	Rev.
T415B4401-1	310

NO.	Revision comprises	Date	Sign.	Appr by

Dust Cap for Tank Unit (Adapter)

To increase the lifespan of DDC tank units, dust caps should be used. They protect the couplings againstthe ingress of dirt and water.

Materials: polyethylene, aluminum, stainless steel or rubber. Supplied with cable attachment and eyelets (rubber design DN 40 - 50 with integrated eyelet).

With the exception of the simple rubber type, the caps feature a safeguard against inadvertent release: before removing the cap, a spring-loaded locking pin is pulled.

Polyethylene composite caps in sizes in sizes DN 40-80 mm include a patented design with lip seal to facilitate the handling.

PE-Composite
Patent No. 000840780-0001

Stainless Steel

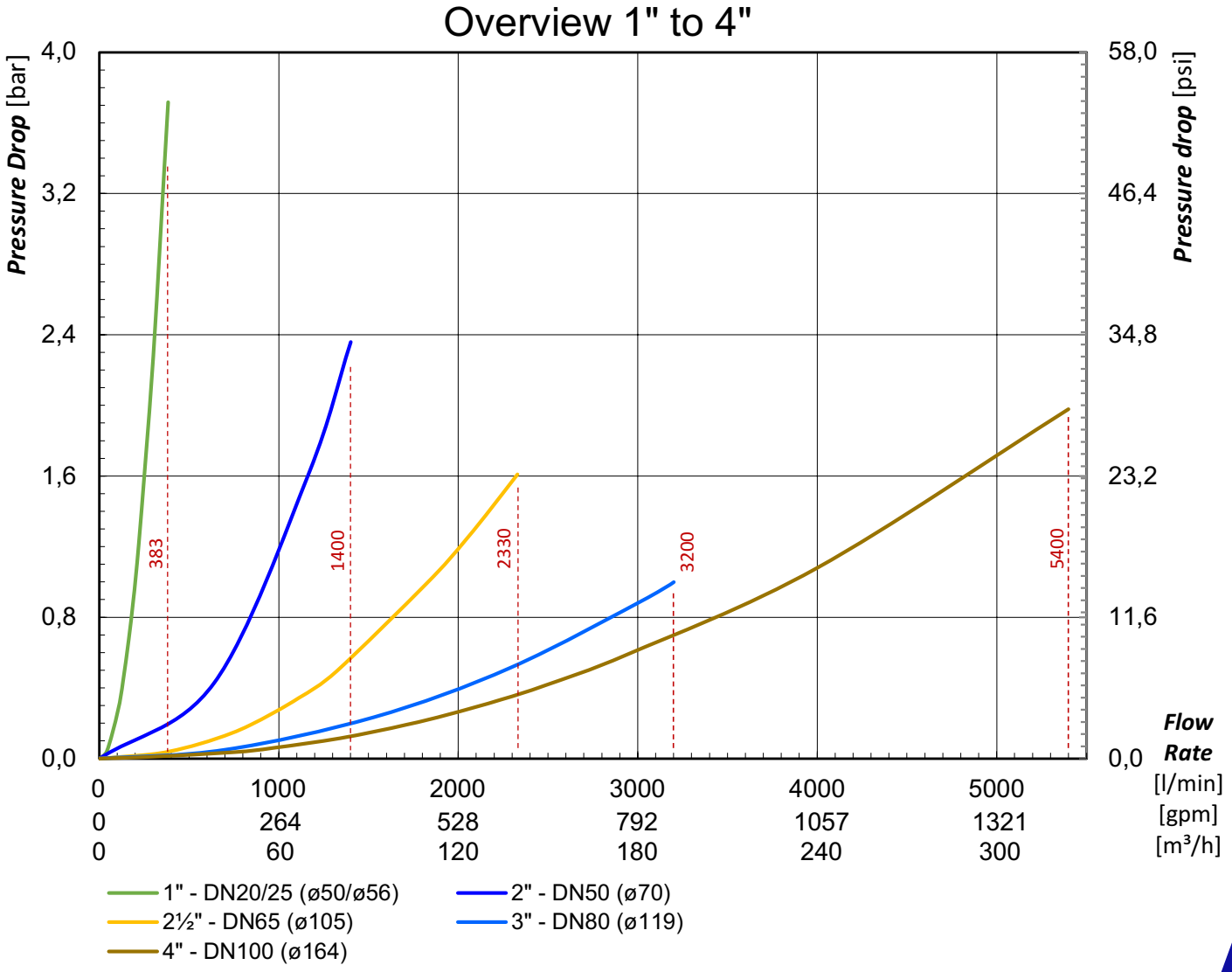
Rubber

Inch / DN	Material [1]	Seal [2]	Weight ≈ kg	Weight ≈ lbs	Mann Tek Code No.
¾"-1" Ø 56 mm, DN 20-25	PE	Standard: FPM (Viton®)	0,1	0.29	C100A2201
	Al		0,2	0.49	C100A1101
	SS		0,6	1.30	C100A4401
1½-2" Ø 70 mm, DN 40-50	PE	NBR	0,1	0.2	C200E2202
	Al	Standard: FPM (Viton®)	0,3	0.7	C200A1101
	SS		0,6	1.3	C200C4401
	Rubber		0,1	0.3	C200D1300
2½" Ø 105 mm, DN 65	PE	NBR	0,2	0,4	C300E2202
	Al	Standard: FPM (Viton®)			C300A1101
	SS		1,5	3,3	C300C4401
3" Ø 119 mm, DN 80	PE	NBR	0,2	0,5	C400E2202
	Al	SPECIFY TEFLON (PTFE) SEAL			C400A1101
	SS				C400C4401
	Rubber	—			C400D1300
4" Ø 164 mm, DN 100	PE	Standard: FPM (Viton®)	0,2	0,5	C500E2202
	Al		1,2	2,5	C500B1101
	SS		2,5	5,5	C500C4401
6" Ø 238 mm, DN 150	PE				C600A2201
	Al		1,6	3,4	C600A1101
	SS				C600A4401
8" Ø 272 mm, DN 200	Al				C800B1101
	SS				C800B4401

[1] Material: PE = Polyethylen-Composite, Al = Aluminium, SS = Stainless Steel, Rubber = NBR/PVC
[2] Standard seal FPM/FKM or NBR. Alternative materials, e.g. EPDM, Chemraz®, Kalrez® or HNBR on request

Viton® is a registred trademark of DuPont, DuPont Elastomers.

Flow Diagram
Dry Disconnect Coupling



Test Fluid: n-paraffin **Temperature:** 20 °C **Density:** 0,75 kg/dm³ **Viscosity:** 1,75 mm²/s
Max. recommended flow velocity: 10m/s

Size	1"	2" (2.0)	2½"	3" (2.0)	4"	Unit
Flow rate at max. rec. flow velocity	383	1400	2330	3200	5400	l/min
Pressure drop at max. rec. Flow velocity	3,7	2,3	1,6	1,0	2,0	bar
Kv	10,3	47,8	95,4	167,2	199,6	[m³/h].[bar]
Cv	11,9	55,7	110,3	194,8	230,8	[gpm].[psi]

Verification method

- *Performance based on combination of physical tests and CFD flow simulations*
- *Test/Simulation method: STANAG 3756, Annex E*
- *Test/Simulation fluid: n-paraffin, Density: 0,75 kg/dm³, Viscosity: 1,75 mm²/s*
- *Design 2.0 performance valid for hose unit and tank unit 2.0 status.*

=== PIPING SPECIALTY ITEM LIST ===

CUT SHEET NUMBER:	PCS-RCA-001
DESCRIPTION:	Rail Car Adapter
VENDOR:	Dixon
MODEL:	RTCA30 (316 SS)

CUT SHEET BEGINS ON NEXT PAGE

C

Railroad Tank Car Connections

Features
- All swivel nuts have standard 5" Railroad Tank Car thread
- Supplied with a nitrile rubber gasket in the threaded end, other gaskets available on page 129

Style A

Male Adapter	Swivel Nut	Elbow	316 Stainless Steel Part #	Aluminum Part #
3"	5"	120°	RTCA30	---
4"	5"	110°	RTCA40[1]	ATCA40

[1] Part is produced as a welded fabrication

NOTE: Stainless steel railroad tank car connections are not compatible with aluminum and brass.

Style F

Features
- 4" TTMA flange can be bolted to sight glass; API adapter or other dry disconnect
- Supplied with PTFE nitrile rubber envelope gasket

110° elbow

TTMA Flange	Swivel Nut	Elbow	Aluminum Part #
4"	5"	110°	ATCF40[1]
	5"	90°	ATCF40-90

[1] Part is produced as a welded fabrication

NOTE: For flange dimensions, diagrams, and additional information, please reference dixonvalve.com.

Style GF

Feature
- 4" TTMA flange can be bolted to sight glass; API adapter or other dry disconnect

110° elbow

TTMA Flange	Boss-lock Coupler	Elbow	Aluminum Part #
4"	4"	110°	ATCGF40
	4"	90°	ATCGF40-90

NOTE: For flange dimensions, diagrams, and additional information, please reference dixonvalve.com.

Screen

Features
- Stainless steel screen encapsulated in a nitrile rubber gasket
- For use in Style GF elbows

Size	Hole Size	316L Stainless Steel Part #
4"	1"	400-G-BUSS

Railroad Tank Car Gaskets

Old Style

Feature
- For use with Dixon® fittings

Size	O.D.	I.D.	Thick	Leather Part #
5"	5-1/4"	3-1/2"	3/16"	G9

Standard

Feature
- For use with Dixon fittings

Size	O.D.	I.D.	Thick	Material	Part #
5"	5-1/4"	3-7/8"	1/8"	nitrile rubber	500GTCABU
	5-1/4"	3-7/8"	1/8"	FKM-A	500GTCAVI
	5-1/4"	3-7/8"	1/8"	fuel grade nitrile rubber	500GTCABF
	5-5/16"	3-7/8"	3/16"	FKM-A	500GTCAVITHK

PTFE Envelope

Features
- For use with Dixon fittings
- May be too thick for insertion in some older Dixon fittings

Size	O.D.	I.D.	Thick	Material	Part #
5"	5-1/4"	3-13/16"	3/16"	PTFE with nitrile rubber filler	500GTCATF
	5-1/4"	3-13/16"	3/16"	PTFE with FKM filler	500GTCATV

Pin Lug Wrench

Feature
- Used on aluminum and iron tank car nuts

Iron Part #
G10

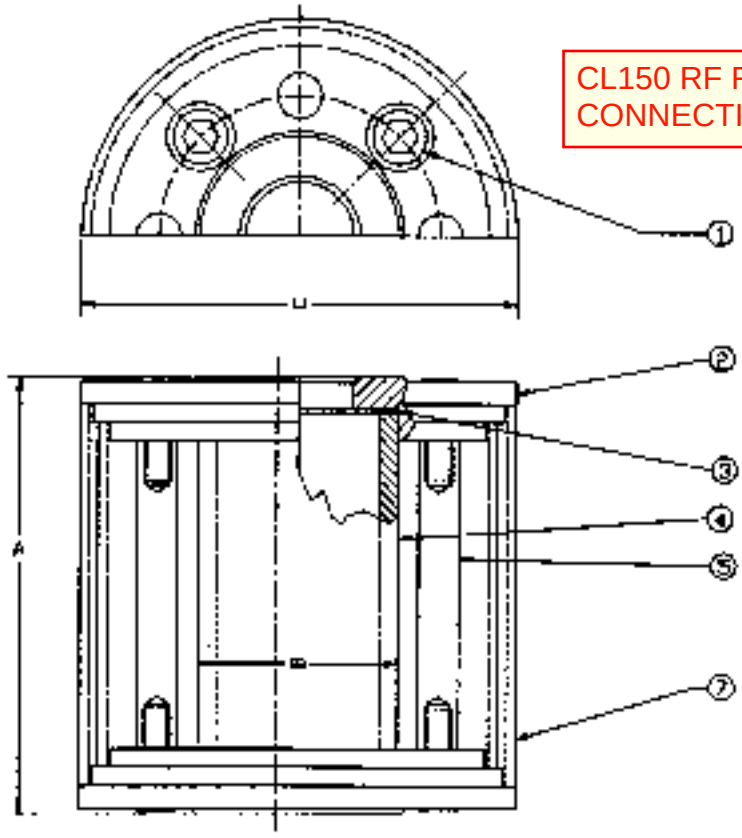
=== PIPING SPECIALTY ITEM LIST ===

CUT SHEET NUMBER:	PCS-SG-016
DESCRIPTION:	Sight Glass
VENDOR:	Jacoby Tarbox
MODEL:	Style F-830HP-PFA (316 SS)

NOTE: USED FOR LINES DESIGNED FOR FULL VACUUM CONDITIONS.

CUT SHEET BEGINS ON NEXT PAGE

MATERIAL: SEE B/M		JACOBY-TARBOX A Division Of The Dark-Reliance Corporation	SIGHT FLOW INDICATOR STYLE F-830HP, FD-830HP
DRAWN: MBV	CHECKED: DEP		Section: T400 Bulletin: T400.36 Date: 11/10/98 Supersedes: 06/01/97
SCALE: NONE	DATE: 5/31/00	VARIOUS PRESSURES	FILE NO.T400-36HP.DWG

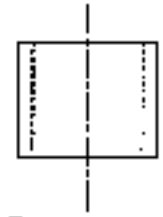


CL150 RF FLANGE CONNECTIONS

PARTS LIST

PIECE NO.	NAME OF PART	QUAN.	MATERIAL	COMMENTS
1	CAP SCREWS	8	STEEL	
2	HEAD	2	SEE BELOW	
3	SEAL GASKET	2		
4	CYLINDER	1	A-BORO	
5	SPACERS	4	STEEL	
6	INDICATOR	1		
7	SHIELD	2	ACRYLIC	

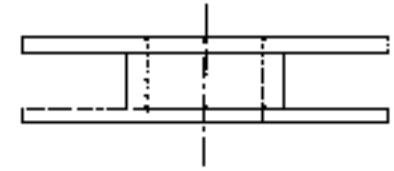
STANDARD OPTIONS:



⑥ DRIP TUBE

STYLE FD-830HP (STD)

STYLE FD-830HP-TFE (TFE LINED DRIP)
ADD 3/8 (IN) TO "A" DIM



PTFE LINING

STYLE F-830HP-TFE (PLAIN)
ADD 1/4 (IN) TO "A" DIM
(NOT SUITABLE FOR VACUUM SVC)
(USE "-PFA" FOR VACUUM SVC)

STANDARD DESIGN MATERIALS:

HEAD: ASTM A-216 WCB/A515 OR 70 (CARBON STEEL)
ASTM A-351 CF8M/A240/A276 (316 SS)

FASTENERS: STEEL
DRIP TUBE: 316 SS/PTFE

OPTIONAL ALLOYS FURNISHED UPON REQUEST

OPTIONAL LININGS: (MODELS F-830HP-PFA, FD-830HP-PFA)
PFA TEFLON 30-40 MILS FOR FULL VACUUM SERVICE

OPTIONAL FEATURES:

SPECIAL LENGTHS UP TO 48 INCHES AVAILABLE ("A" DIM)
MODEL: FL-830HP, FL-830HP-TFE
SPECIAL CALIBRATED TUBES (VOLUME/LENGTH) PER APPLICATION
304/316 SS TRIM AVAILABLE

PSI	SIZE		DIMENSIONS A		B		C		WEIGHT	
	IN	DN	IN	mm	IN	mm	IN	mm	LBS	KGS
200	1/2	15	6.000	152.4	1.250	31.8	4.500	114	53	2.4
175	3/4	20	6.000	152.4	1.500	38.1	4.750	121	6.0	2.7
175	1	25	6.000	152.4	1.750	40.0	5.250	133	7.5	3.5
150	1-1/2	40	6.000	152.4	2.500	55.0	6.000	152	9.0	4.0
150	2	50	6.000	152.4	3.000	69.0	7.000	178	17.25	7.8
150	2-1/2	65	6.000	152.4	3.500	90.0	8.000	203	22.5	10.2
150	3	80	6.000	152.4	4.000	95.63	8.500	216	28.5	12.9
150	4	100	6.000	152.4	5.000	133.75	10.000	254	33	15.0
150	6	150	6.000	152.4	7.000	177.80	12.00	305	84	38.1
70	8	200	6.000	152.4	8.875	225	14.500	368	102	46.2
60	10	250	12.000	304.8	11.000	270	17.000	432	87	39.4
60	12	300	12.000	304.8	13.000	325	20.000	508	131	59.4

=== PIPING SPECIALTY ITEM LIST ===

CUT SHEET NUMBER:	PCS-STR-005
DESCRIPTION:	Strainer
VENDOR:	Keckley
MODEL:	Style SB7

NOTES:
- USE IN CARBON STEEL PIPING SYSTEMS FOR STEAM AND STEAM CONDENSATE
- SIZES 1/2" - 1 1/2" TYPICAL.
- 3/64" PERFORATIONS FOR STEAM SERVICE.

CUT SHEET BEGINS ON NEXT PAGE

KECKLEY

Style SB7

Y-Strainer
Carbon Steel (ASTM A 216, Grade WCB)
Class 600 NPT

Style SB7

Y-Strainer
Carbon Steel (ASTM A 216, Grade WCB)
Class 600 Socket Weld

Carbon Steel Y-Strainer

APPLICATIONS

Where protection from foreign matter in a pipeline is required.

CONSTRUCTION

The Keckley Style SB7 strainers are constructed from rugged carbon steel castings that are machined to exacting specifications.

Socket Weld bore is in compliance with ASME B16.11 unless otherwise specified.

FEATURES

The Keckley Style SB7 strainer features a machined groove in the body and cap for proper alignment and to ensure accurate reseating when servicing is required. This strainer has a straight threaded cap and is furnished standard with a NPT blow-off connection. The gasket is 304 stainless steel spiral wound and is compressed between the body and cap (for maximum strength and durability) and designed for both high pressure and high temperature service. Keckley Style SB7 strainers can be supplied with a carbon steel blow-off plug upon request.

SCREENS

Standard perforated 304 stainless steel screens are spot welded along the seam for maximum strength. Different size perforations and meshes are available in stainless steel, monel, and brass to meet specific media requirements. If media is not indicated, screens for *liquid* will be supplied.

SELF CLEANING

Self cleaning is accomplished by opening the valve or drain plug connected to the blow-off port. **Warning:** See Maintenance Instructions on page S6 of the Strainer Information Section for additional precautions and detailed information on servicing the strainer.

WORKING PRESSURES - NON SHOCK

NOM. RATING	1/4" to 3"	8 mm to 80 mm
CLASS 600	1480 PSI @ 100ºF	10208 KPa @ 38ºC
	600 PSI @ 838ºF	4138 KPa @ 448ºC

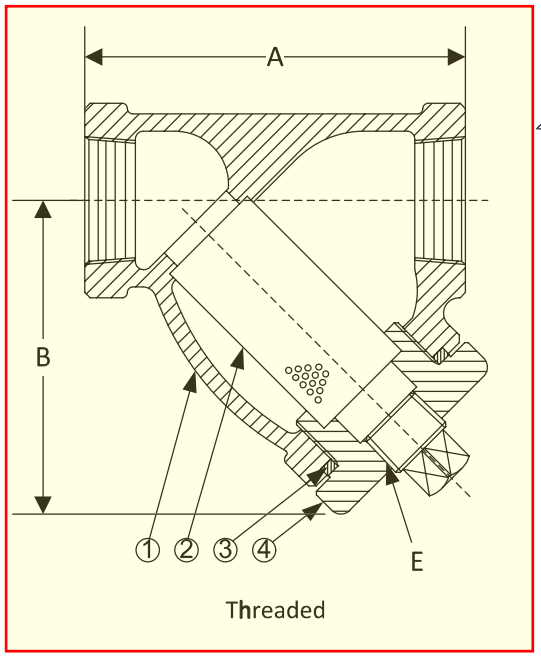
Values listed represent typical market and service applications. Due to numerous variables (concentrations, temperatures, and flow) present in any application, no representation or guarantee, expressed or implied, is given.

KECKLEY

TECHNICAL DATA
DIMENSIONS AND WEIGHTS

Style SB7

Y-Strainer, Class 600 NPT & Socket Weld Carbon Steel (ASTM A 216, Grade WCB)



PARTS LIST		
ITEM	DESCRIPTION	MATERIAL
1	Body	Carbon Steel (ASTM A 216, Grade WCB)
2	Screen	Stainless Steel (304)
3	Gasket	Spiral Wound Stainless Steel (304)
4	Cap	Carbon Steel (ASTM A 216, Grade WCB)

Optional: Blow-off Plug, Carbon Steel (ASTM A 105).
*Optional Body Materials Available in LCB, WC6, and WC9.

STANDARD SCREENS SUPPLIED

SIZE		SCREEN PERFORATION					
		FOR LIQUID		OPEN AREA	FOR STEAM		OPEN AREA
in	mm	in	mm		in	mm	
1/4 to 3	8 to 80	1/16"	1.6	30%	3/64	1.2	33%

Standard screens supplied are for **liquid service**, unless otherwise specified.
Options: Other perforations, meshes, and screen materials are available.

SIZE		DIMENSIONS										WEIGHTS	
		A		B		C		D		E			
in	mm	in	mm	in	mm	in	mm	in	mm	in	mm	lbs	kgs
1/4	8	2-15/16	75	2-7/16	62	0.555	14	3/8	10	1/4	8	2	0.76
3/8	10	2-15/16	75	2-7/16	62	0.690	18	3/8	10	1/4	8	2	0.76
1/2	15	2-15/16	75	2-7/16	62	0.855	22	3/8	10	1/4	8	2	0.76
3/4	20	3-11/16	94	3	76	1.065	27	1/2	13	3/8	10	3	1.21
1	22	4-9/16	116	4-5/16	110	1.330	34	1/2	13	3/8	10	6	2.33
1-1/4	32	4-15/16	125	4-3/16	106	1.675	43	1/2	13	3/4	20	7	3.02
1-1/2	40	5-9/16	141	4-11/16	119	1.915	49	1/2	13	3/4	20	9	3.98
2	50	6-15/16	176	6-1/4	159	2.406	61	5/8	16	1	25	15	6.80
2-1/2	65	12	305	9-3/8	238	2.906	74	5/8	16	1-1/4	32	34	15.03
3	80	12	305	9-3/8	238	3.535	90	5/8	16	1-1/4	32	36	15.97

†This table reflects only the nearest metric equivalents.
Dimensions and weights are for reference only. When required, request certified drawings.
Face to face values for threaded strainers have a tolerance in compliance with ASME B16.34 and socket weld strainers have a tolerance in compliance with ASME B16.11.

FLOW COEFFICIENTS

Size	C_v	Size	C_v	Size	C_v
1/4"	9.5	1"	30	2-1/2"	129.7
3/8"	9.5	1-1/4"	44.9	3"	161.3
1/2"	9.5	1-1/2"	61		
3/4"	18.7	2"	98		

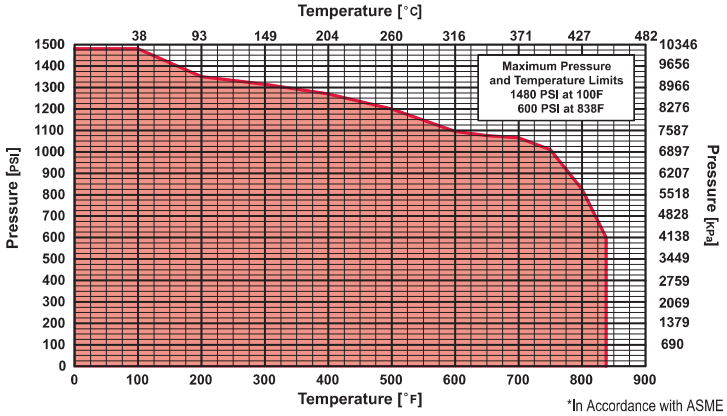
TOTAL SCREEN AREA

Size	(in^2)	Size	(in^2)	Size	(in^2)
1/4"	2.75	1"	10.08	2-1/2"	78.14
3/8"	2.75	1-1/4"	12.79	3"	78.14
1/2"	2.75	1-1/2"	16.33		
3/4"	4.71	2"	27.04		

*See DETERMINING RATIOS on page S5 of the Strainer Information Section for calculating NET FREE AREA of the screen to inside pipe area.

PRESSURE vs. TEMPERATURE CHART

Class 600 NPT & Socket Weld Carbon Steel (ASTM A 216, Grade WCB)



*In Accordance with ASME B16.34

KECKLEY

PRESSURE DROP CHART

"Y" Pattern Strainers

(Styles B, BDI, E150, F150, E300, F300, E7, F7, SB7, SB7BC, SBF, SSB7, SSB7BC and SSBF)

This pressure drop chart is based on the flow of clean water through the Keckley "Y" strainers listed above with screen perforations ranging from 3/64" through 1/8" and is additionally for use with those units equipped with a 20 mesh screen as standard.

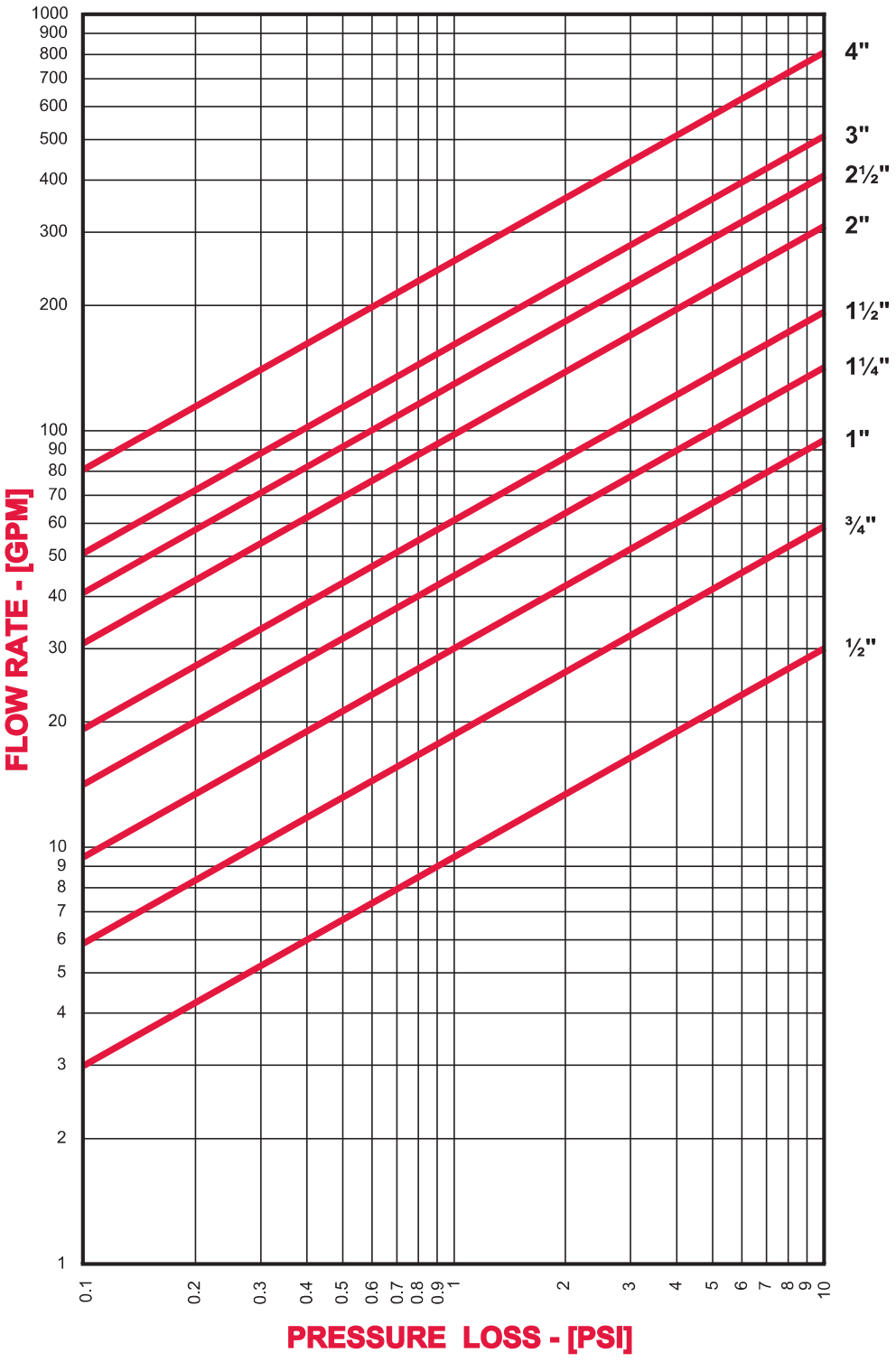
TO USE CHARTS:

Find your desired rate of flow (GPM) on the left hand side of the chart. Follow its corresponding horizontal line to the point where it intersects the diagonal line indicating the strainer pipe size. From this point of intersection, follow the vertical line down to the bottom of the chart to determine the approximate pressure drop.

CORRECTION FACTORS:

For finer mesh screens that are backed with a perforated sheet, multiply the pressure drops shown at right by the following:

40 mesh	x 1.2
60 mesh	x 1.4
80 mesh	x 1.6
100 mesh	x 1.7
150 mesh	x 1.8
200 mesh	x 2.0



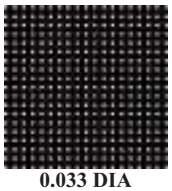
KECKLEY

Strainer Information

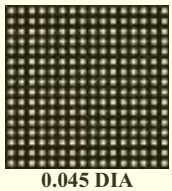
SCREEN OPTIONS

The screen or basket is the heart of the Keckley strainer. The media flows into the open end of the screen or basket and is strained as it passes through the screen towards the outlet. All particles larger than the screen opening are trapped inside. Screens are provided in perforated metal or wire mesh, depending on strainer size and/or material being strained. Only the best materials of the proper gauge to suit the service are used. All seams are spot welded for maximum strength. Double or reinforced screens are spot welded on the end peripheries as well as the seams. Reinforced screens consist of a perforated sheet lined with wire mesh. Keckley engineers have designed the screens to provide maximum total screen area.

Perforated Sheet Metal Sizes

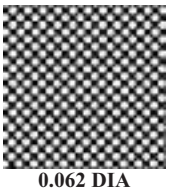


0.033 DIA
1/32" Approximately
331 Holes Per Sq. In.
29% Open Area

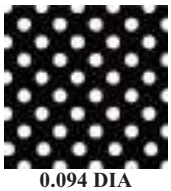


0.045 DIA
3/64" Approximately
225 Holes Per Sq. In.
33% Open Area

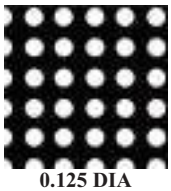
FOR STEAM AND CONDENSATE SERVICE



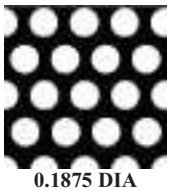
0.062 DIA
1/16" Approximately
98 Holes Per Sq. In.
30% Open Area



0.094 DIA
3/32" Approximately
51 Holes Per Sq. In.
36% Open Area

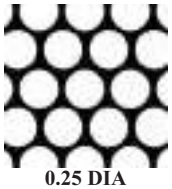


0.125 DIA
1/8" Approximately
29 Holes Per Sq. In.
43% Open Area

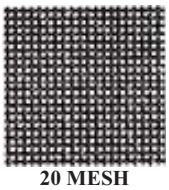


0.1875 DIA
3/16" Approximately
18 Holes Per Sq. In.
51% Open Area

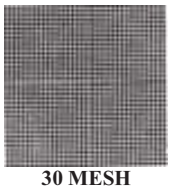
Mesh Sizes



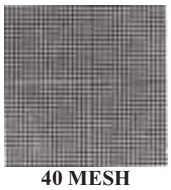
0.25 DIA
1/4" Approximately
12 Holes Per Sq. In.
58% Open Area



20 MESH
Wire Dia. 0.015
Opening 0.034
49% Open Area



30 MESH
Wire Dia. 0.011
Opening 0.021
45% Open Area

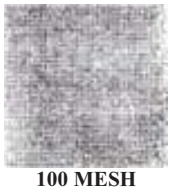


40 MESH
Wire Dia. 0.009
Opening 0.016
41% Open Area

50 MESH
Wire Dia. 0.0085
Opening 0.011
33% Open Area

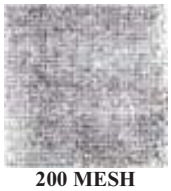
60 MESH
Wire Dia. 0.0065
Opening 0.010
38% Open Area

80 MESH
Wire Dia. 0.0055
Opening 0.0070
31% Open Area



100 MESH
Wire Dia. 0.0045
Opening 0.0055
30% Open Area

150 MESH
Wire Dia. 0.0026
Opening 0.0041
37% Open Area



200 MESH
Wire Dia. 0.0021
Opening 0.0029
34% Open Area

***300 MESH**
Wire Dia. 0.0012
Opening 0.002
41% Open Area

***300 Mesh available in Duplex Strainers only.**

Stainless steel screens are standard in all strainers except for Style F-300, E-300 and flanged bronze strainers; these strainers are supplied with brass screens. Other screen materials are available upon request (i.e. 316 SS, Monel, Hastelloy C276, Alloy 20, Duplex Stainless Steel, Titanium). In stainless steel, the smallest perforation obtainable is generallly twice the thickness of the metal itself. Therefore, perforations from 0.033" through 0.250", dependent on metal thickness, are readily available. When extra fine straining is required of the larger strainers, reinforced screens consisting of a perforated sheet lined with wire mesh are recommended. This a ıe particles with added durability.

KECKLEY

Strainer Information

MAGNETS

Magnets can be provided as an option which, when placed inside the strainer screen, will remove very fine iron or steel particles present in fluid.

Magnets provide protection for equipment against abrasive damage.

Strainer Size	Magnets required
2½" – 4"	1 magnets
5" – 6"	2 magnets
8" – 10"	3 magnets
12" – 14"	4 magnets
16" – 18"	5 magnets

*Sizes 2" and smaller strainers can be furnished with magnetic plugs.

REINFORCING BANDS

Reinforcing bands can be used to add additional strength and durability to the screens or baskets when straining conditions have higher than normal pressure drops.

DETERMINING NET FREE AREA RATIOS

To calculate the ratio, use the following formula:

Formula:

1. Choose the size perforation or mesh needed to remove particles from the media passing through the strainer.
2. Multiply the *TOTAL SCREEN AREA* by the *PERCENT OF OPEN AREA of the screen*. The result equals the *OPEN AREA of the screen.*
3. Divide the result (*OPEN AREA of the screen*) by *the INSIDE AREA of the pipe* to give *the ratio of net free area of the screen to the pipe.*

Example: (2" Style B screwed "Y" strainer with a 20 mesh 304 stainless steel screen)

36.23 (*total screen area in²*)
x .49 (*20 mesh = 49% open area*)
17.753 (*total open area of screen*)

17.753" / 3.356" (*inside area of 2" pipe*) = 5.29:1
(RATIO OF NET FREE AREA OF THE SCREEN TO PIPE AREA)

INSIDE AREA OF THE PIPE (in²)							
Size	(in²)	Size	(in²)	Size	(in²)	Size	(in²)
1/4"	0.104	1-1/4"	1.496	4"	12.732	12"	111.946
3/8"	0.191	1-1/2"	2.036	5"	20.008	14"	135.294
1/2"	0.304	2"	3.356	6"	28.894	16"	176.738
3/4"	0.534	2-1/2"	4.788	8"	48.914	18"	223.71
1"	0.864	3"	7.394	10"	78.865	20"	[illegible]

Screen Opening Equivalents

Fractional Inches	Decimal Inches	Millimeters	Microns	Mesh
--	0.001	--	25	--
--	0.0015	--	37	400
--	0.002	--	50	300
--	0.003	--	75	**200**
--	0.004	1/10	100	**150**
--	0.005	1/8	125	115
--	0.006	--	149	**100**
--	0.007	--	177	**80**
--	0.010	1/4	250	**60**
--	0.011	--	280	**50**
--	0.016	--	406	**40**
--	0.020	1/2	500	--
--	0.021	--	533	**30**
--	0.030	3/4	750	--
1/32	0.033	--	838	--
--	0.034	--	840	**20**
--	0.039	1	1000	16
3/64	0.045	--	1143	--
--	0.046	--	1190	14
--	0.055	--	1410	12
--	0.059	1-1/2	1500	--
1/16	0.062	--	1575	--
--	0.065	--	1680	10
--	0.079	2	2000	9
--	0.093	--	2380	8
3/32	0.094	--	2388	--
--	0.110	--	2790	7
--	0.118	3	3000	--
1/8	0.125	--	3175	--
--	0.131	--	3330	6
--	0.156	4	4000	5
--	0.185	--	4700	4
3/16	0.1875	--	4763	--
--	0.197	5	5000	--
--	0.236	6	6000	--
1/4	0.250	--	6350	--
	0.263	--	6700	3

old red are available from stock at Keckley Company.
y for the availability of other sizes including those not listed.

=== PIPING SPECIALTY ITEM LIST ===

CUT SHEET NUMBER:	PCS-STT-003
DESCRIPTION:	Steam Trap
VENDOR:	Armstrong International
MODEL:	2000 Series Inverted Bucket

Note: To be used along outdoor steam headers

CUT SHEET BEGINS ON NEXT PAGE

Armstrong®

2000 aoktob Il vokfo8 B6akof afo%5 Tk%i

ANWafo%il Mdbb afooMWtfh St0r 2Kl loafKk/la-2/TVa-4000

For Pressures to 650 psig (45 bar)...Capacities to 1,300 lb/hr (590 kg/hr)

0Gal /ra1121- a1S
0Gal /raC21- EEN2Il 016

nobakti ftKl

y .nd ndp uggg ApSpv, z/ gOxD.RpSverfoDDpfmSatox feD.Dvmnr.DRpSpd hxfMpm pIl.f.pDft eDdmoDi vpSR.fp rlp .DeDt z.z.Di foDLixSemoDs .nd rmp oSDo Spz.z.Di 0. ox ipmndp Spreh.rmtol ndp .DRpSpd hxfMpmozpSemDi zSDf.zrpazrxv emndp hpDpl.mtol em3vm.Dpvv vnppr foDvrfkfmoDD

A1 vpenpdam3 zpSzSsol zefMei p
A1 fo3 zefmri dms p.i dmrfiez
A2dp eh.rmtm s .ndvmDdlSpp"p3xzv s .ndoxmde3 ei p
AHPfpzmoDerfoSSov.oDSpv.vmDfp
A1 ndSpp3tpeSixeSeDpp ei e.Dvmdplpfm Rp 3 empServadplpfm Rp s oSMi eDvd.z0

uggg ApSpv vrpe3 rfiezv fo3 h.Dp veR.Di v .DndSpp .3 zoSmDmeSpevDpDpSi ta .DvmnemoDeDd Spzrefp3 pDft2dp z/ gOxD.RpSverfoDDpfmSzSoRdpv qx.fMi pevt .D3rDp Spzrefp3 pDmeroDi s .nd emndp zSoRpDedReDmi pv ol .DRpSpd hxfMpmozpSemoD0

1rvo eRe.rehp s .nd ozmoDerIA3u .Dpi Servrfie.DpSfoDDpfmS0

9 %4t5 65 5i ok%ftl 6 2Kl 8tftKl b

0 eP.3 x3 emos ehp zSpvvxSp 9Rpvvprd pv.i DID

0 odprug8gaug88D	7gg zv.i H WggO7 9uWheSH 7u*OF1
0 odpruguuD	/-g zv.i H /ggO7 97- heSH z8-OF1
	/u* zv.i H *ggO7 97z heSH z*8OF1
	/gg zv.i H WggO7 978 heSH 7u*OF1

0 eP.3 x3 ozpSemDi zSpvvxSpD

0 odprug8gD	ugg zv.i 987 heSl
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benRp eDd vpenD	neSdpDpd fdSo3 p vnppnv 8*37; n oS 2.mDx3
FoDDpfmShodt 9vmd u IA3uID	Am.Dpvv vnppnv zg7
7reDi p	1A20 18g- p.Df zrempd

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AlIDvx3 eMi .Dvxremo DloS0 odprv ug8gyug88
AAm.Dpvv vnppr zoz dSe.DloS0 odprv ug88yuguu
A; Sohp foDDpfmoDloS0 odprv ug88yuguu
AArfie.DpShnos dos DRenRp loSIA3u foDDpfmS
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Azpf.ltD
A0 odprDx3 hpS
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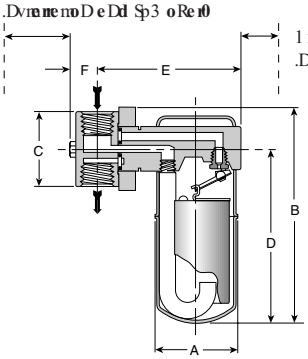
#38 R60

If oa fully detailed cedified dcawing, cefeotf :
2000 Sedes with Standaod Cf nnectf o CD #1003
2000 Sedes with IS-2 Cf nncectf o CD #1113

1nos u38yu" fipeSeDfp loShorm
.DvmnemoDeDd Sp3 oRen0



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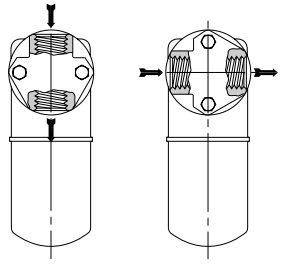
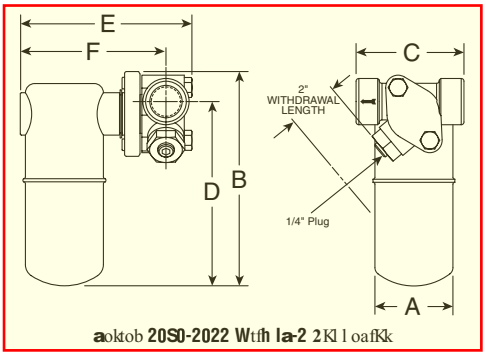


9 K8oM20SS Tk%i Wtfh af%i 8%k8 2Kl loafKk



aoktob 20S0-2022 Wtfh la-2 2Kl loafKk

2000 aoktob Tk%i b Wtfh af%i 8%k8 2Kl loafKk

9 K8oMNK.	20S0		20SS		2022	
Pipe Connections	tl	55	tl	55	tl	55
	1/2, 3/4	15, 20	1/2, 3/4	15, 20	1/2, 3/4	15, 20
"A" (Diameter)	2-11/16	68	2-11/16	68	3-7/8	98
"B" (Height)	6	152	6-15/16	176	8-11/16	221
"C" (Face to Face)	2-3/8	60	2-3/8	60	2-3/8	60
"D" (Bottom to ℄)	4-19/32	117	5-9/16	141	7-3/8	187
"E" (℄ to Outside)	4-9/16	115	4-9/16	115	5-3/4	146
"F" (℄ to Bolt)	1	25	1	25	1	25
Weight lb (kg)	4-1/4 (1.9)		4-1/2 (2.0)		7 (3)	

2000 aoktob Tk%i b Wtfh la-2 Il fo6k%Mafk%il ok 2Kl loafKk

9 K8oMNK.	20S0				20SS				2022			
Pipe Connections	tl	55	tl	55	tl	55	tl	55	tl	55	tl	55
	1/2, 3/4	15, 20	1	25	1/2, 3/4	15, 20	1	25	1/2, 3/4	15, 20	1	25
"A" (Diameter)	2-11/16	68	2-11/16	68	2-11/16	68	2-11/16	68	3-7/8	98	3-7/8	98
"B" (Height)*	6	152	6	152	6-15/16	176	6-15/16	176	8-11/16	221	8-11/16	221
"C" (Face to Face)	3-1/2	89	4	102	3-1/2	89	4	102	3-1/2	89	4	102
"D" (Bottom to ℄)*	5	127	5	127	6	152	6	152	7-3/4	197	7-3/4	197
"E" (Outside to Bolt)	5-1/2	140	5-11/16	144	5-1/2	140	5-11/16	144	6-11/16	170	6-7/8	175
"F" (℄ to Outside)	4-5/8	117	4-13/16	122	4-5/8	117	4-13/16	122	5-13/16	148	6	152
Weight lb (kg)	4-3/4 (2.2)		5-1/4 (2.4)		5 (2.3)		5-1/2 (2.5)		7 (3)			

*Add 1/2" (15 mm) to "B" and "D" dimensions when optional probe

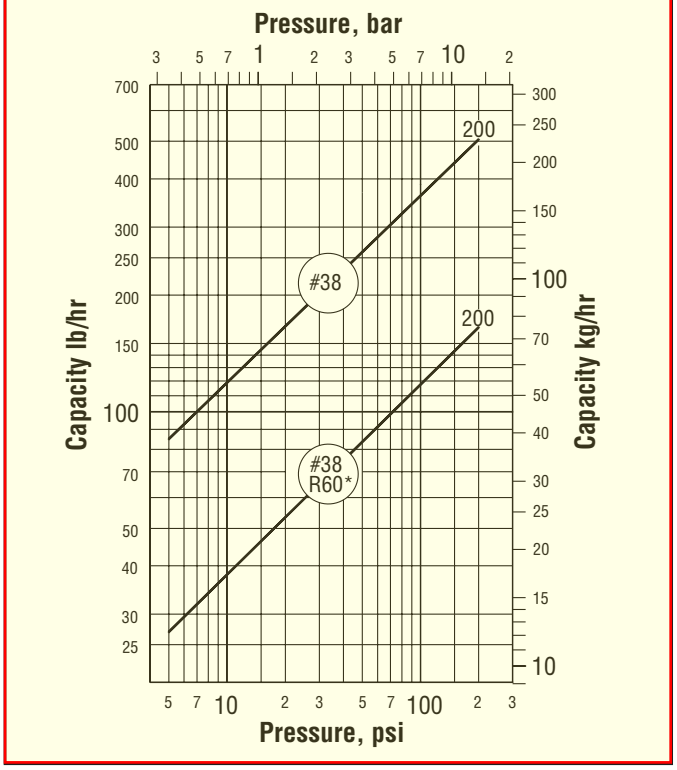
D0s2-1sOl a6r2aIsOT02-16s a1S 10r17rl a1C0 ra621-s ar0 a11r7h2 a6 Page 2 of 3 PCS-STT-003 5176X0s c2s26arl s671-216r1a6271aIsC7l 17r N1r67rSa60 2117rl a6271s

2000 Series Inverted Bucket Steam Trap

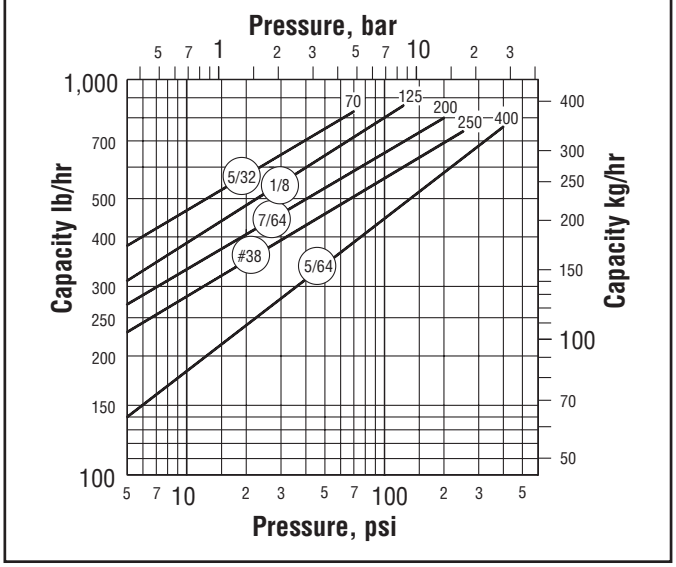
All Stainless Steel With 360° Connector/IS-2/TVS-4000

For Pressures to 650 psig (45 bar)...Capacities to 1,300 lb/hr (590 kg/hr)

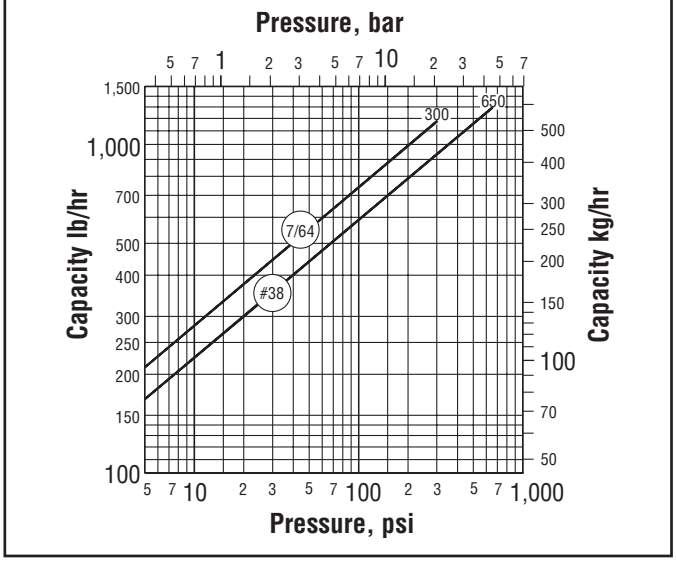
Model 2010 Capacity



Model 2011 Capacity



Model 2022 Capacity



*NOTE: Because the orifice is located at the top, inverted bucket steam traps handle dirt and scale better than other types of traps. However, in applications where extremely dirty conditions exist, care should be exercised in the use of all types of restricted-orifice, reduced-capacity traps.

Connectors

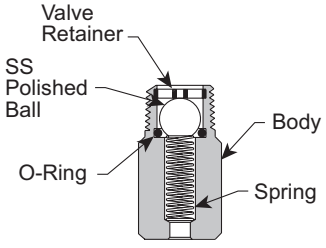
Besides the inverted bucket traps, the standard connector, IS-2 connector, and TVS-4000 connector with integral strainer can also be used on thermostatic, thermostatic wafer and disc traps.

CD-3300 Disc Trap
With IS-2 Integral Strainer Connector

Options

Pop Drain for Freeze Protection
In general, a properly selected and installed Armstrong trap will not freeze as long as steam is coming to the trap. If the steam supply is shut off, a pop drain should be used to automatically drain the trap. Stainless steel pop drain available for Models 2011 and 2022.

Maximum Operating Conditions
Pressure: 600 psig (41 bar)
Temperature: 350°F (177°C)



Insu-Pak
Now you can insulate the in-line traps in your plant without complicating regular trap maintenance. Insu-Pak, a simple reusable insulation package, cuts the time and cost of in-field installation because it goes on in a snap. And it comes off just as easily. The Insu-Pak can prevent trap freeze-up when used with a properly designed condensate manifold. Designed for use with Model 2010 and Model 2011 traps.

Probe connections are available for trap monitoring for Models 2011

Designs, materials, weights and performance ratings are approxima otice. Visit armstronginternational.com for up-to-date information.

=== PIPING SPECIALTY ITEM LIST ===

CUT SHEET NUMBER:	PCS-WH-003
DESCRIPTION:	Water Heater
VENDOR:	Eemax
MODEL:	SafeAdvantage

CUT SHEET BEGINS ON NEXT PAGE

SafeAdvantage™ (NEMA 4 Incl.)

For Safety Applications – Eye/Face Wash and Drench Showers

Specifications
Tankless Electric Water Heater

Applications

- Eye/face wash
- Emergency drench showers
- Where tepid water is needed

Features

- 90°F max. output temperature
- Fully Modulating - Predictive control algorithm and diverse safety features ensures conformity to ANSI Z358.1 tepid water without additional mixing valves (check local codes)
- Fast response rate eliminates the requirement for any additional hot water purge system (no drainage required)
- T&P not required per UL499 - Check local codes
- Thermo-Optical Sensor for infrared element monitoring
- Field programmable, updatable firmware, and adjustable turn-on
- LCD display and control with built in diagnostics
- Standard model includes NEMA 4 cabinet; alternative cabinets are available for upgrade

Optional Features

- N4X (304SS) N4X6 (316SS) enclosures
- Free standing legs
- Freeze protection for harsh climate, up to -30°F
- Non-fused or fused electrical disconnect
- GFCI
- Explosion proof – C1D2 Compliant, local certification required. Class Z purge and pressurization system provided with pressure switch for alarm controls. For classification other than C1D2 please contact the factory to discuss options.
- Siren and Beacon – audible and visual alarm (C1D2 rated when paired with Explosion proof package)

Product Specifications

Min. Operating Pressure:	35 PSI
Max. Operating Pressure:	150 PSI
Optimum Operating Pressure:	60-90 PSI

U.S. Patent #'s: US20140023354 and US20140178057

Installation Requirements

- Properly sized water hammer arrestor
- Minimum 35 PSI dynamic pressure at inlet
- Sediment filter installed on inlet

Special Design Service

- Inquiries for units for unique applications are welcome. Call our Technical Service department at **1 800 543 6163**.
- Custom orders non-refundable

LISTED

Intertek

NO LEAD*
*The wetted surface of this product contacted by water contains less than 0.25% lead and meets NSF/ANSI 372

Assembled In USA

bimobject Find our products on bimobject.com BIM

Suggested Specification

Tankless water heater shall be an Eemax SafeAdvantage model number AP_____.

Factory installation in a _____(N4/N4X/N4X6) enclosure.

Enclosure to be fitted with the following features:

_____ **FP**	Freeze protection (-30°F)
_____ **EDS**	Non-fused disconnect
_____ **FDS**	Fused disconnect
_____ **EP**	Explosion proof (C1D2 compliant)
_____ **GFCI**	True RMS GFCI with digital display and reset
_____ **SK**	24" legs for free standing applications
_____ **RD**	Remote display
_____ **SB**	Siren and Beacon
_____ **DC**	Dry contact
_____ **ES**	Emergency stop - push button

Tankless water heater must have water connections on the bottom, and be constructed with NSF61 listed materials. Direct heating element to be non-ferrous, cartridge style, designed for field replacement. Tankless water heater to utilize a dual PID algorithm, actively managing power application to real-time system demand. Integrated flow meter capable of volumes in excess of 30 GPM drives predictive control algorithm. Water heater must be protected by redundant safeties. Redundant safeties to include thermo mechanical safety switches, infrared element monitoring via thermo optical sensors, and dual temperature monitoring via master control board. Tankless water heater user interface must have the following capabilities:

- Selectable display including Celsius/Fahrenheit, inlet temperature, outlet temperature, flow rate, and setpoint temperature.
- Capable of displaying flow rate in gallons per minute or liters per minute.
- Diagnostic features to include error and fault code display.
- Control board must maintain error/fault history of 9 events.
- Capable of factory coded temperature setting (max. and min.)
- Capable of firmware upgrades via USB port
- Capable of BMS integration - contact factory
- Conforms to ANSI Z358.1 tepid water without additional mixing or purge features (inlet temperatures must not exceed 100ºF).

Inform ontained in this document are subject to change without notice. REV 2022-05

SafeAdvantage (NEMA 4 Incl.)

For Safety Applications – Eye/Face Wash and Drench Showers

Specifications
Tankless Electric Water Heater

Suffix Definitions

EE Emergency Eyewash. Max. outlet temperature 90°F. Conforms to ANSI Z3581.1 tepid water without additional mixing valve. Shipped with display "Locked."

EFD Emergency Eye, Face & Drench. Max. outlet temperature 90°F. Conforms to ANSI Z3581.1 tepid water without additional mixing valve. Shipped with display "Locked."

Note: Models with an EE or EFD suffix are not to be fed to a thermostatic mixing valve. If a thermostatic mixing valve is required, heater of non EE or EFD must be specified.

Note: Models with an EE or EFD suffix have a 90°F maximum temperature. Temperature rise data is provided for reference, but temperature is electronically limited to factory preset not to exceed temperature.

	MODEL NUMBER	kW	BTU/H	AMPS PER PHASE	TURN ON (GPM)	RECOMMENDED WIRE SIZE (75° C/CU)	TEMPERATURE RISE °F 3.0 GPM	4.0 GPM	5.0 GPM	6.0 GPM	20.0 GPM	23.0 GPM	26.0 GPM	30.0 GPM
	VOLTS 208 Three Phase Delta													
	AP032208 EE N4	32	109,189	89	1.0	3AWG	73°	55°	44°	36°	11°	10°	8°	7°
	AP032208 EE N4X	32	109,189	89	1.0	3 AWG	73°	55°	44°	36°	11°	10°	8°	7°
	AP036208 EE N4	36	122,832	100	1.0	3 AWG	82°	61°	49°	41°	12°	11°	9°	8°
	AP036208 EE N4X	36	122,832	100	1.0	3 AWG	82°	61°	49°	41°	12°	11°	9°	8°
	AP041208 EFD N4	41	139,189	113	1.0	2 AWG	†	70°	56°	47°	14°	12°	11°	9°
	AP041208 EFD N4X	41	139,189	113	1.0	2 AWG	†	70°	56°	47°	14°	12°	11°	9°
	AP054208 EFD N4	54	184,256	150	1.5	1/0 AWG	†	†	74°	61°	18°	16°	14°	12°
	AP054208 EFD N4X	54	184,256	150	1.5	1/0 AWG	†	†	74°	61°	18°	16°	14°	12°
	AP064208 EFD N4	64	218,377	178	2.5	3/0 AWG	†	†	87°	73°	22°	19°	17°	15°
	AP064208 EFD N4X	64	218,377	178	2.5	3/0 AWG	†	†	87°	73°	22°	19°	17°	15°
	VOLTS 480 Three Phase Delta													
	AP036480 EE N4	36	122,837	43	1.0	8 AWG	82°	61°	49°	41°	12°	11°	9°	8°
	AP036480 EE N4X	36	122,837	43	1.0	8 AWG	82°	61°	49°	41°	12°	11°	9°	8°
	AP039480 EE N4	39	133,074	47	1.0	8 AWG	89°	67°	53°	44°	13°	12°	10°	9°
	AP039480 EE N4X	39	133,074	47	1.0	8 AWG	89°	67°	53°	44°	13°	12°	10°	9°
	AP048480 EFD N4	48	163,783	58	1.0	6 AWG	†	82°	66°	55°	16°	14°	13°	11°
	AP048480 EFD N4X	48	163,783	58	1.0	6 AWG	†	82°	66°	55°	16°	14°	13°	11°
	AP054480 EFD N4	54	184,256	65	1.5	6 AWG	†	†	74°	61°	18°	16°	14°	12°
	AP054480 EFD N4X	54	184,256	65	1.5	6 AWG	†	†	74°	61°	18°	16°	14°	12°
	AP063480 EFD N4	63	214,965	76	2.5	4 AWG	†	†	86°	72°	22°	19°	17°	14°
	AP063480 EFD N4X	63	214,965	76	2.5	4 AWG	†	†	86°	72°	22°	19°	17°	14°
	AP072480 EFD N4	72	245,674	87	2.5	3 AWG	†	†	†	82°	25°	21°	19°	16°
	AP072480 EFD N4X	72	245,674	87	2.5	3 AWG	†	†	†	82°	25°	21°	19°	16°
	AP096480 EFD N4	96	327,552	116	2.5	1 AWG	†	†	†	†	33°	29°	25°	22°
	AP096480 EFD N4X	96	327,552	116	2.5	1 AWG	†	†	†	†	33°	29°	25°	22°
	AP108480 EFD N4	108	368,511	130	2.5	1 AWG	†	†	†	†	37°	32°	28°	25°
	AP108480 EFD N4X	108	368,511	130	2.5	1 AWG	†	†	†	†	37°	32°	28°	25°
	AP126480 EFD N4	126	429,930	151	2.5	2/0 AWG	†	†	†	†	43°	37°	33°	29°
	AP126480 EFD N4X	126	429,930	151	2.5	2/0 AWG	†	†	†	†	43°	37°	33°	29°
C	**AP144480 EFD N4**	144	491,348	173	2.5	2/0 AWG	†	†	†	†	49°	43°	38°	33°
C	**AP144480 EFD N4X**	144	491,348	173	2.5	2/0 AWG	†	†	†	†	49°	43°	38°	33°
	VOLTS 600 Three Phase Delta													
C	**AP061600 EFD N4**	61	208,141	59	2.5	6 AWG	†	†	83°	69°	21°	18°	16°	14°
C	**AP061600 EFD N4X**	61	208,141	59	2.5	6 AWG	†	†	83°	69°	21°	18°	16°	14°
C	**AP071600 EFD N4**	71	242,262	68	2.5	4 AWG	†	†	†	81°	24°	21°	19°	16°
C	**AP071600 EFD N4X**	71	242,262	68	2.5	4 AWG	†	†	†	81°	24°	21°	19°	16°
C	**AP102600 EFD N4**	102	348,038	98	2.5	3 AWG	†	†	†	†	35°	30°	27°	23°
C	**AP102600 EFD N4X**	102	348,038	98	2.5	3 AWG	†	†	†	†	35°	30°	27°	23°
C	**AP130600 EFD N4**	130	443,578	125	2.5	1 AWG	†	†	†	†	44°	39°	34°	34°
C	**AP130600 EFD N4X**	130	443,578	125	2.5	1 AWG	†	†	†	†	44°	39°	34°	34°
C	**AP150600 EFD N4**	150	511,821	144	2.5	1/0 AWG	†	†	†	†	51°	45°	39°	34°
C	**AP150600 EFD N4X**	150	511,821	144	2.5	1/0 AWG	†	†	†	†	51°	45°	39°	34°

† Temperature electronically limited to factory preset temperature.

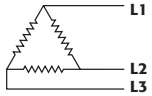
"C" indicates evaluation and compliance to either Underwriters Laboratories (UL) or Intertek (ETL) under CAN/CSA-C22.2 No. 64/No. 88.

Electrical configuration and requirements

All Eemax three phase units are custom made to order and as such, are non-returnable and non-refundable. Check your electrical supply, making sure all criteria for operating your Eemax water heater are met.

Eemax 600V, 480V and 208V Three Phase Units Delta Configuration

Requires: 3 Lives and 1 Ground (earth)



Information and product specifications contained in this document are subject t :2-05

SafeAdvantage (NEMA 4 Incl.)

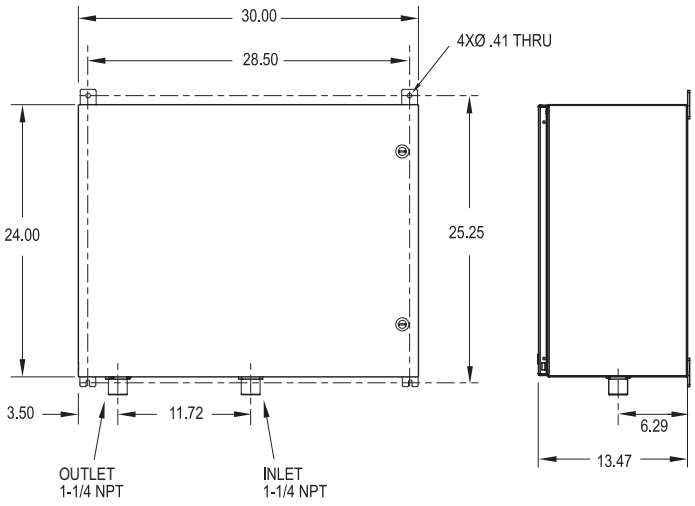
For Safety Applications – Eye/Face Wash and Drench Showers

NEMA Cabinet Options*

Dimensions: 24″H x 30″W x 13.5″D (Total weight est 130 lb)

N4	Powder coated steel
N4X	Corrosion-resistant 304 stainless steel
N4X6	Corrosion-resistant 316 stainless steel

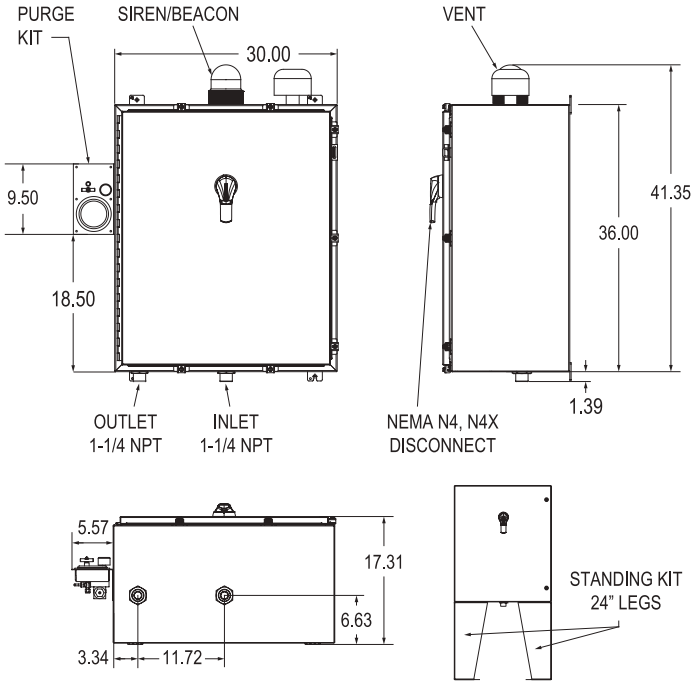
Designed for wall mounted installation. Free standing legs and other options available.



NEMA Cabinet Options Accessories

Dimensions: 36″H x 30″W x 17.3″D

FP	Freeze protection (-30°F)
EDS	Non-fused disconnect
FDS	Fused disconnect
EP	Explosion proof (C1D2 compliant)
GFCI	True RMS GFCI with digital display and reset
SK	24″ legs for free standing applications
RD	Remote display
SB	Siren and Beacon
DC	Dry contact
ES	Emergency stop - push button



Est. total weight 225 lb, varies based on options.
Designed for wall mounted installation. Free standing legs and other options available.

*Refer to page 47 of this specification guide for more details regarding selection.

Informa ontained in this document are subject to change without notice. REV 2022-05

SafeAdvantage (NEMA 4 Incl.)

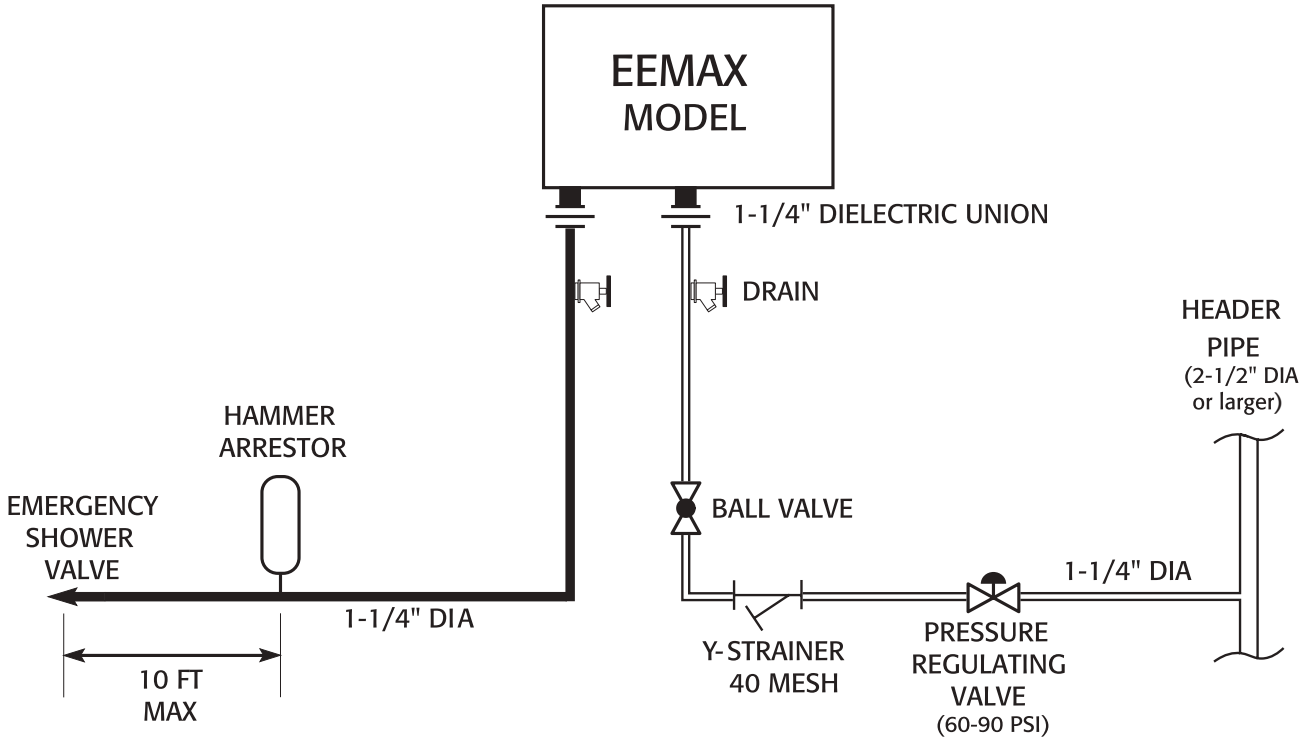
For Safety Applications – Eye/Face Wash and Drench Showers

Typical Safety Shower - Piping Installation Schematic

Below represents an installation of our SafeAvantage heater for safety shower application. Please note the required use of water hammer arrestors, Y-strainer, ball valves and drains.

System Criteria

Max Flow Rate:	30 GPM
Pipe Diameter:	1-1/4" or larger
Optimum Operating Pressure:	60-90 PSI



Water Hammer Arrestor Sizing Chart

Pipe Length (header to shower)	**Arrestor Size**
15'	B
15' - 30'	C
30' - 50'	D
50' - 75'	E
75' - 90'	F
90' +	Contact Arrestor Mfgr

Information and product specifications contained in this document are subject

=== PIPING SPECIALTY ITEM LIST ===

CUT SHEET NUMBER:	PCS-WHA-002
DESCRIPTION:	Water Hammer Arrester
VENDOR:	Jay R Smith
MODEL:	5000 Series

CUT SHEET BEGINS ON NEXT PAGE

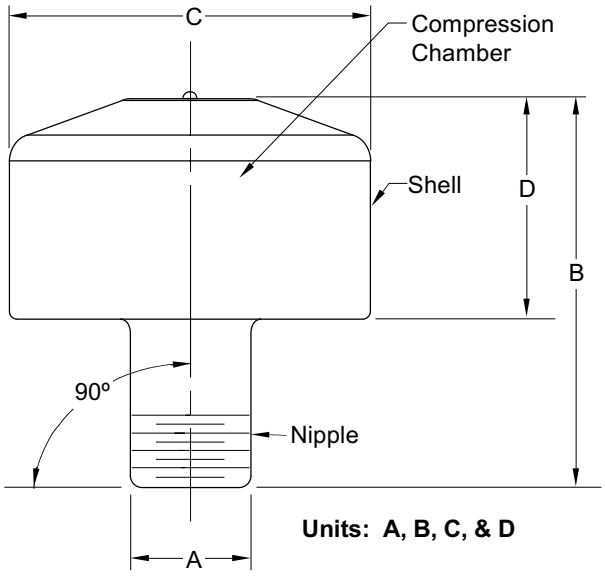
SMITH® JAY R. SMITH MFG. CO.®
CUSTOMER DRIVEN
MEMBER OF ACORN'S FAMILY OF COMPANIES
POST OFFICE BOX 3237
MONTGOMERY, ALABAMA 36109-0237 (USA)
TEL: 334-277-8520 FAX: 334-272-7396 www.jrsmith.com
MEMBER OF:

LOCATION

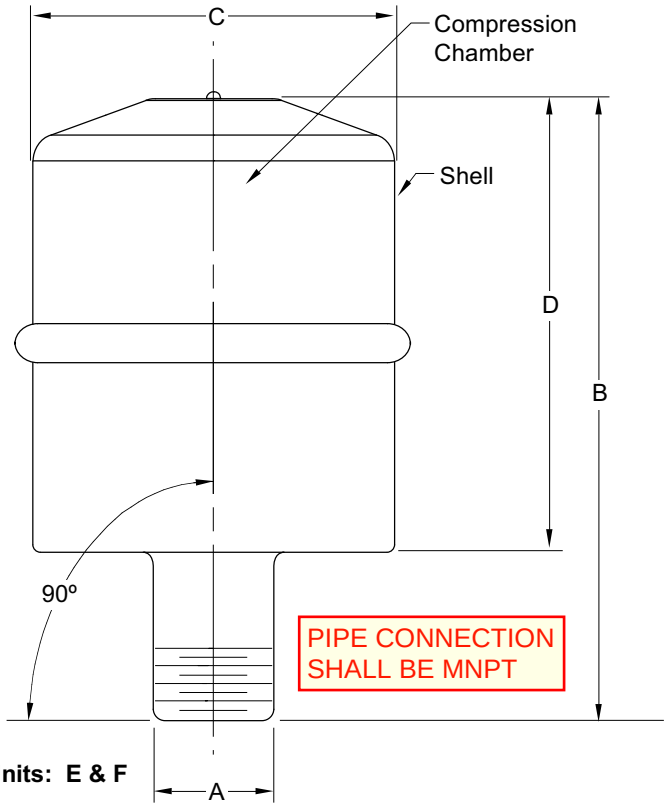
HYDROTROLS

ENGINEERED WATER HAMMER ARRESTERS

FUNCTION: Quick closing electrical, pnuematic, spring loaded valves or devices, and the quick hand closure of fixture trim can cause destructive "water hammer". Engineered water hammer arresters ("Hydrotrols") employ a permanently sealed cushion of air or gas which absorbs the energy of water hammer and reduces pressure rise in the piping system to a safe level. Hydrotrol units correctly sized and placed at specific locations in the water piping system, will control the destructive shock of water hammer.



Units: A, B, C, & D



Units: E & F

Hydrotrols are pre-charged and permanently sealed at the factory. All hydrotrols are constructed entirely of stainless steel.

PCN/ Fig. No.	P.D.I. Symbol	Fixture Unit Rating	A SIZE	B	C	D
5005	A	1-11	3/4 (19)	3.18 (81)	3.25 (83)	1.80 (46)
5010	B	12-32	1 (25)	3.93 (100)	3.25 (83)	2.13 (54)
5020	C	33-60	1 (25)	4.30 (109)	3.25 (83)	2.42 (61)
5030	D	61-113	1 (25)	5.25 (133)	3.25 (83)	3.38 (86)
5040	E	114-154	1 (25)	5.57 (141)	3.25 (83)	3.57 (91)
5050	F	155-330	1 (25)	6.53 (166)	3.25 (83)	4.50 (114)

Hydrotrols Fig. 5005 to 5050 inclusive have been tested and certified in accordance with the Plumbing and Drainage Institute "Standard P.D.I. WH-201"

RECOMMENDED SPECIFICATION FOR HYDROSTATIC SHOCK CONTROL
Smith series 5000 "Hydrotrol" all stainless steel shock absorbers shall be installed at all solenoid, remote operated or quick closing valves and at each plumbing fixture or battery of plumbing fixtures. Install on both hot and cold water branch lines in an upright position as close as possible to the valve or valves being served. Sizes and locations as indicated on drawings.

NOTE: Sizing information on reverse side.

NOTE: Stainless steel units are lead-free.

NOTE: Dimensions shown in parentheses are in millimeters.

REV.	DATE	DESCRIPTION		
M	11-2-11	Rev. "B" & "D" Dim., Drawing	TBW	JM
L	01/20/11	Updated Logo	JJ	TW
K	02/16/10	Revised "B" & "D" Dimensions		

WEIGHT POUNDS | VOLUME CUBIC FEET | FIGURE NUMBER 5005 To 5050

FIGURE NUMBER 5005 To 5050 | DRAWN BY: CM | CHECKED BY: KG | APPROVED BY: TD | DATE: 2-12-86 | SCALE: NONE | SIZE A | DRAWING NUMBER S5005 | M

DIMENSIONS ARE SUBJECT TO MANUFACTURERS TOLERANCE AND CHANGE WITHOUT NOTICE
WE CAN ASSUME NO RESPONSIBILITY FOR USE OF SUPERSEDED OR VOID DATA

C | DRAWING NUMBER S5005BS | SIZE A | SCALE: NONE | DATE: 2-12-86 | APPROVED BY: TD | CHECKED BY: KG | DRAWN BY: CM | 5005BS | FIGURE NUMBER

WE CAN ASSUME NO RESPONSIBILITY FOR USE OF SUPERSEDED OR VOID DATA

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MEMBER OF:

LOCATION

HYDROTROLS

SIMPLIFIED SIZING METHOD

The selection and specification for the correct size Hydrotrol for each piping run may be easily accomplished by the following procedure which incorporates length of piping run, water pressure and other pertinent data into the Sizing Chart.

The Hydrotrol Sizing Chart shows the number of fixture units that may be accommodated by each size Hydrotrol. The U.S. Bureau of Standards "Fixture Unit Rating Table" designates the fixture unit rating for each plumbing fixture when installed in public or private buildings.

Referring to the "Fixture Unit Rating Table", determine the fixture unit rating of each plumbing fixture on the line. Add the total number of fixture units on the line, and use this figure to select and specify the size Hydrotrol required as indicated on the Hydrotrol Sizing Chart.

PLACEMENT OF HYDROTROLS

EXAMPLE:

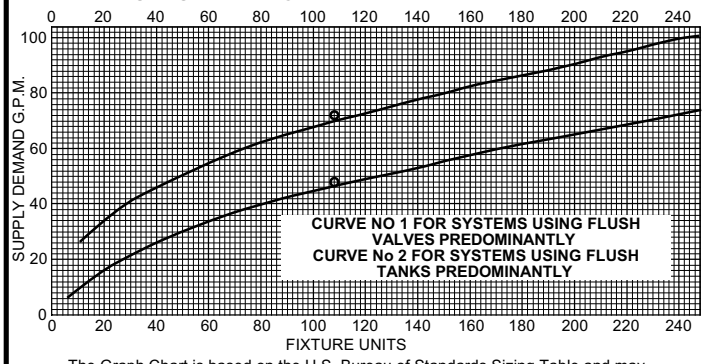
The drawing shows plumbing fixtures installed on a typical piping layout. The figures to the right of each fixture run are the total fixture units for the run based on the fixture unit rating table. The total number of fixture units for each plumbing fixture on the line has been added and is shown for both hot and cold water lines. The total number of fixture units for each run determines the size of Hydrotrol required. If the fixture unit total ends up in a 1/2 number, round up to the next larger whole number.

The additional Hydrotrols shown on batteries of over 20'(6096) in length are the same size as those specified in the chart ... shown for each floor.

The following general rules are applicable to proper placement of "Hydrotrols". The locations shown in the installation layout are recommended to give the maximum in water hammer arresting effectiveness:

1. Single fixtures or appliances --Locate Hydrotrol just before fixture of valve.
2. Battery of fixtures -- Length of battery 20'(6096) or less. Locate Hydrotrol between the last and next to last fixtures. (See 5th Floor Examples.)
3. For a battery of fixtures having a branch line over 20 (6096) ft. in length, two Hydrotrols should be installed. The two Hydrotrols selected should have combined fixture unit rating equal to or greater than the total fixture unit demand of the plumbing fixtures on the entire branch line. Locate one Hydrotrol betweeen the last and next-to-the-last fixture on the branch line. The other Hydrotrol to be located at a point equal to half the length of the branch line. (See first floor example).
4. On hot water branch lines over 20 (6096) ft. in length, where plumbing fixtures are installed on a portion of the line, the second Hydrotrol should be located at a point half the distance of the plumbing fixtures installed. (See second and third floor examples).
5. For special or unique battery arrangements or piping layouts, please consult the Smith Engineering Dept. for the proper specification on sizing and placement.

FOR APPLICATIONS NOT COVERED BY THE FIXTURE UNIT RATING TABLE -



The Graph Chart is based on the U.S. Bureau of Standards Sizing Table and may be used to determine the fixture unit rating based on GPM supplied to the fixture in question.

HYDROTROL SIZING CHART

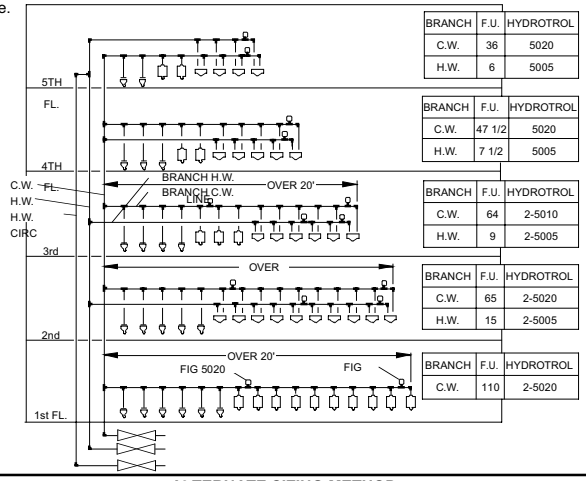
P.D.I. SYMBOLS	A	B	C	D	E	F
HYDROTROL	5005	5010	5020	5030	5040	5050
Fixture Unit Rating	1-11	12-32	33-60	61-113	114-154	155-330

NOTE: When Water Pressure in line exceeds 65 psi, specify the next larger Hydrotrol.

FIXTURE UNIT RATING TABLE GENERAL BUILDING AREAS

FIXTURE	TYPE OF SUPPLY CONTROL	WEIGHT IN FIXTURE - UNITS PUBLIC C.W.	PUBLIC H.W.	PRIVATE C.W.	PRIVATE H.W.
Water Closet	Flush Valve	10	-	6	-
Water Closet	Flush Tank	5	-	3	-
Pedestal Urinal	Flush Valve	10	-	-	-
Stall or Wall Urinal	Flush Valve	5	-	-	-
Stall or Wall Urinal	Flush Tank	3	-	-	-
Lavatory	Faucet	1 1/2	1 1/2	1	1
Bathtub	Faucet	2	3	1 1/2	1 1/2
Shower head	Mixing Valve	2	3	1	2
Bathroom Group	Flush Valve Closet	-	-	8	3
Bathroom Group	Flush Tank Closet	-	-	6	3
Separate Shower	Mixing Valve	-	-	1	2
Service Sink	Faucet	3	3	-	-
Laundry Tubs (1-3)	Faucet	-	-	3	3
Combination Fixture	Faucet	-	-	3	3

NOTE: This Fixture Unit Rating Table is established in accordance with the Plumbing and Drainage Institute "Standard - WH-201" and is based upon information offered in the National Plumbing Code. It takes into consideration all design factors, including simultaneous usage, pipe size, lenth, flow pressure and velocity. This table, therefore, provides an easy, accurate method for sizing Hydrotrols on each multiple fixture branch line.



ALTERNATE SIZING METHOD

The simplified sizing method covers 90% of most applications encountered. Where fixture unit ratings are not available or where water is piped to remote items of equipment, the following method of Hydrotrol selection should be used:

INFORMATION REQUIRED TO USE CHART:
1. LENGTH OF PIPE RUN.
2. PIPE SIZE.
3. WATER PRESSURE.

Length of pipe run should be determined from the point of valve closure to a point of relief, such as a large pipe riser twice the size of the branch line, main line or water tank.

HYDROTROL SELECTION CHART

LENGTH OF PIPE	NOMINAL PIPE SIZE 1/2"(13)	3/4"(19)	1"(25)	1 1/4"(32)	1 1/2"(38)	2"(51)
25	5005	5005	5010	5020	5030	5040
50	5005	5010	5020	5030	5040	5050
75	5010	5020	5030	1-5005 1-5040	5050	1-5040 1-5050
100	5020	5030	5040	5050	1-5020 1-5050	2-5050
125	5020	5030	5050	1-5005 1-5050	1-5040 1-5050	1-5040 2-5050
150	5030	5040	5050	1-5030 1-5050	2-5050	3-5050

NOTE: Data in table is based on a maximum operating water pressure of 65 psi, and an average velocity between 5 to 10 feet per second. If operating pressures are over 65 psi, use next larger Hydrotrol unit.

REV.	DATE	DESCRIPTION		
C	01/20/11	Updated Logo	JJ	TW
B	4-16-97	Added Millimeters		

WEIGHT POUNDS | VOLUME CUBIC FEET | FIGURE NUMBER 5005BS

Document Title:	Loading Arms Data Sheet
Document Number:	LP1Y-255-DST-K215-01
Security Class:	Restricted

Loading Arms Data Sheet

Lilly - Lebanon Plant 1 (B1LG)

K215 - Lilly Rail Offloading Rack

Hanson Project Number: 25i0007

A1 - DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT
Authorization to proceed does not relieve Contractor/Supplier of its responsibility or liability under the Contract and or Purchase Order.
By: Ricardo DaCosta dac54131 March 18, 2026

Revision	Issue Status	Originator	Reviewer	Approver	Date
E	IFP	SS	QR	MW	3/4/2026
D	IFP	SS	QR	MW	2/6/2026
C	IFP	SS	QR	MW	1/14/2026
B	IFQ	SS	QR	MW	11/6/2025
A	IFR	SS	QR	MW	10/9/2025

Document Number: LP1Y-255-DST-K215-01
Revision:E
Revision Date: 2026-03-04

HANSON

Rail Unloading Station

Tag No.:	as per Description Chart
P&ID No.:	as per Description Chart
RFQ No.:	LP1Y-255-PRO-K215-01

Rail Unloading Station:	Rail Stations 1, 2, 3, & 4
Station Use:	as per Description Chart
P&IDs:	as per Description Chart
Environmental Conditions	Refer to project document LP100-225-DBD-001
Min/Max Temperature:	-27.5°C to 45°C
Wind Load (mph/sqf):	120 mph
Seismic Design:	Refer to project document LP100-215-EMS-05002
Hazardous Area Classification:	Class 1, Division 2, Group B, T3

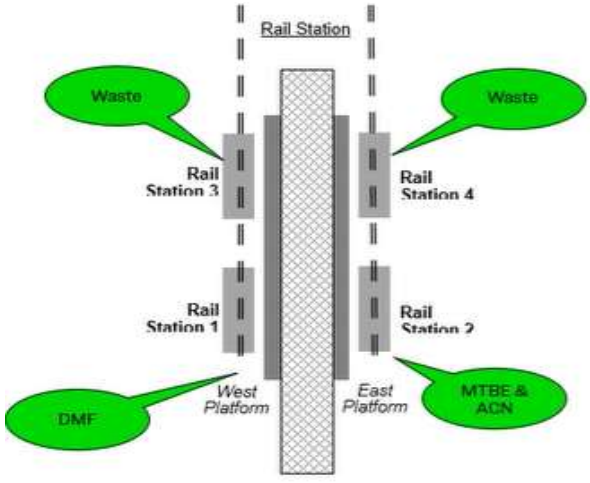
Design Specifications

Mechanical Specifications:	N/A
Structural Specifications:	N/A
Pipe Schedules:	LP100-250-EMS-15100, LP100-250-EMS-15110, LP100-250-EMS-15112, LP100-250-EMS-15100PS315, LP100-250-EMS-15100PS344

Description Chart

Rev	#	Equipment Description	Tag	QTY	Size	Material	Station	Description	P&ID
	1	Unloading Arm - DMF	LA-0622 A	1	4 x 3	SS 316 L	1	Top Unloading with piggyback N2 pressoff	LP1Y-DI-00000062
	2	Unloading Arm - ACN	LA-0632 A	1	4 x 3	SS 316 L	2	Top Unloading with piggyback N2 pressoff	LP1Y-DI-00000063
	3	Unloading Arm - MTBE	LA-0631 A	1	4 x 3	SS 316 L	2	Vapor Return to Top from Pumped Bottom Unloading	LP1Y-DI-00000063
E	4	Loading Arm - Waste	LA-0640 A	1	4 x 3	ALLOY 2205	3	Top Loading with 2" piggyback vent hose	LP1Y-DI-00000064
E	5	Loading Arm - Waste	LA-0650 A	1	4 x 3	ALLOY 2205	4	Top Loading with 2" piggyback vent hose	LP1Y-DI-00000065

Station Overview



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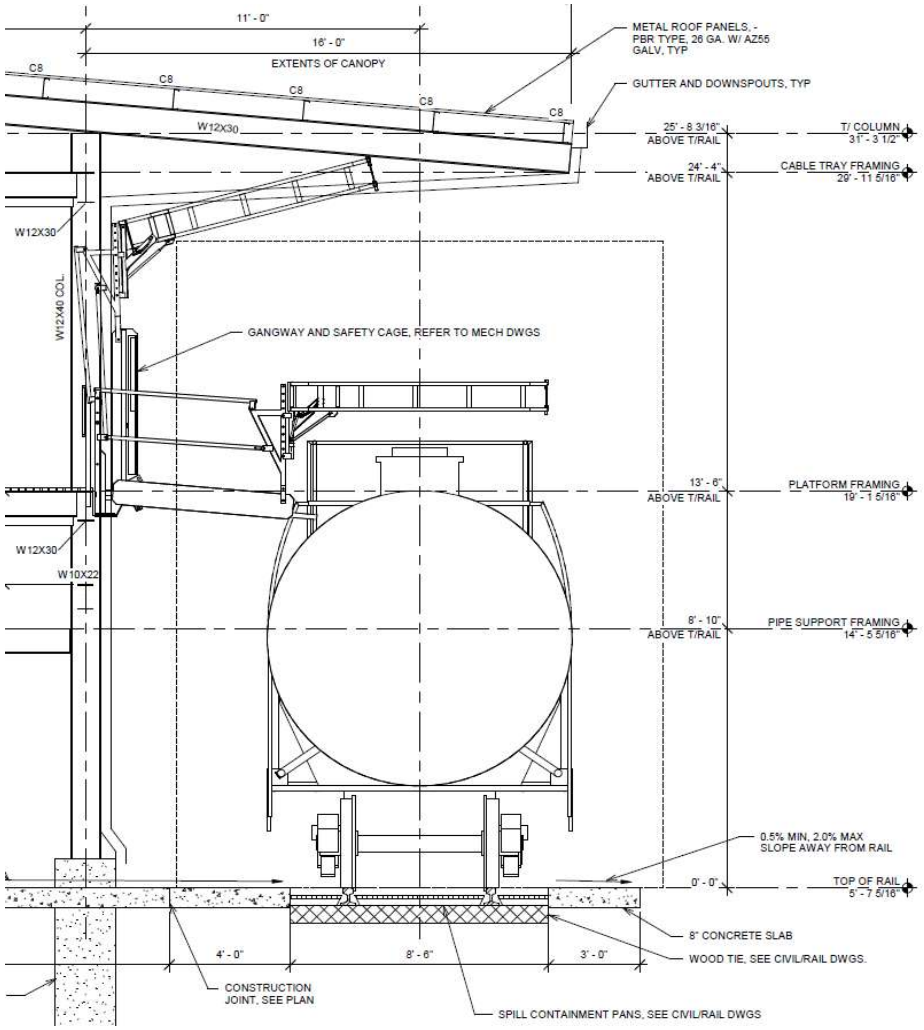
Document Number: LP1Y-255-DST-K215-01
Revision:E
Revision Date: 2026-03-04

Rail Unloading Station

Tag No.:	as per Description Chart
P&ID No.:	as per Description Chart
RFQ No.:	LP1Y-255-PRO-K215-01

Rev	Platform Layout (Note 1)					
	Railcar center to Platform center - A:	17' - 6"	ft	Top of railcar from TOR - High	12' - 6"	ft
	Railcar Center to Platform Edge - B:	11'		Top of railcar from TOR - Low	11' - 6"	ft
	Hatch Height - C:	TBD	ft	Top of Platform from TOR	13' - 6"	ft
						ft
				Gangway upright - TOR to Botton of safety hoop (min)	22' - 4"	ft
				Gangway upright - TOR to Top of safety hoop (max)	23' - 6"	ft
				Railcar Type	DOT 111 and DOT117 (~30,600 Gal)	

TOR = Top of Rail



Document Number: LP1Y-255-DST-K215-01
Revision:E
Revision Date: 2026-03-04

HANSON

Rail Unloading Station

Tag No.: as per Description Chart
P&ID No.: as per Description Chart
RFQ No.: LP1Y-255-PRO-K215-01

DMF Top Unloading Arm

Loading Arm Overview:	4" x 3" Supported Boom Unloading Arm with Nitrogen Piggyback Hose					
Unloading Arm:						
Tag:	LA-0622A		Quantity Required:	1		
Application:	Top Arm		Product:	DMF - Dimethylformamide		
Arm Operation:	Manually Spring Assisted.		Viscosity:	0.87	cP	
Arm Type	Supported Boom Style		Specific Gravity:	0.94	g/mL	
Pipe/hose Material:	316L SS / 316L SS		Temperature (Operating/Design):	15 / 65	°C	
Seal Material:	Chemraz SD517		Pressure (Operating/Design):	40 / 150	psig	
Coating/Paint:	Not Required		Flowrate (Operating/Design):	150 / 180	gpm	
Tie-In Flange Size / Rating:	4" 150# Flange, 3" Pipe (See Note 1)		Pressure Drop (Operating/Design):	Minimal (VTA)	psig	
Rail Connection Type:	316SS Autolock Coupler		Arm Pressoff Product:	Nitrogen		
Rail Connection Size:	3"		Grounding Cable Required:	No (See Note 2)		
Local Shut Off Valve Required:			Heat Tracing Required (Voltage/Ph/Hz)	N/A		
Valve Type:	3" SS Ball Valve (See Note 1)		Insulation Required:	N/A		
Valve Used for Line Shutdo	Yes		Lubrication Required:	MFG Standard		
Product Retention	No		Drain Connection:	At Local Low Point		
Working Envelope:			Breakaway coupling:	3" 316 SS NTS-PU		
Minimum Working Radius:	78	in	Minimum Working Height:		in	
Maximum Working Radius:	156	in	Maximum Working Height:		in	
Piggyback Nitrogen Hose						
Tag:	LA - 0622A		Quantity Required:	1		
Application:	Nitrogen Pressoff unloading		Product:	Nitrogen		
Hose Material Tube	PFA Core		Viscosity:	0.018	cP	
Hose Material Cover	Silicone Cover with 302 SS Braid		Specific Gravity:	0.00419	g/mL	
Tie-in Flange Size/Rating	2" 150#		Temperature (Operating/Design):	15 / 65	oC	
Rail Connection Type:	316 SS Cam Lock w/ Chemraz SD517		Pressure (Operating/Design):	40 / 150	psig	
Rail connection Size:	2"		Flowrate (Operating/Design):	150 / 180 (See Note 3)	gpm	
Local Shut Off Valve Required:	Ball Valve		Pressure Drop (Operating/Design):	Minimal - VTA	psig	
Valve Type:	N/A		Arm Purging Product:	N/A		
Valve Used for Line Shut Down:	N/A		Grounding Cable Required:	Yes		
Product Retention	No		Heat Tracing (Voltage/Ph/Hz):	No		
Hose Size	1" ID (See Note 3)		Insulation Required:	No		

Note 1: Arm and Hose Size limited by Railcar Connection of 3" with 3" Diptube
At Design 180 GPM Flow Rate, 3" S/10S Flow Velocity 6.6 ft/s, and ~ 2.9 psi pressure loss per 100 feet of Pipe

Note 2: Grounding of Railcars outside of K215 Scope

Note 3: 1" Sized by vendor quote.
At Design 180 GPM flow rate, Nitrogen flow at 40 psig of regulated pressure is 24 cfm, or 89.5 scfm,
~ 3.77 psi pressure loss per 100 ft of 1" hose

HANSON

Document Number: LP1Y-255-DST-K215-01
Revision:E
Revision Date: 2026-03-04

Rail Unloading Station

Tag No.: as per Description Chart
P&ID No.: as per Description Chart
RFQ No.: LP1Y-255-PRO-K215-01

ACN Unloading Arm

Loading Arm Overview:	4" x 3" Supported Boom Unloading Arm with Nitrogen Piggyback Hose				
Unloading Arm:					
Tag:	LA-0632A	Quantity Required:	1		
Application:	Top Arm	Product:	ACN - Acetonitrile		
Arm Operation:	Manually Spring Assisted.	Viscosity:	0.00419	cP	
Arm Type	Supported Boom Style	Specific Gravity:	0.94	g/mL	
Pipe/hose Material:	316L SS / 316L SS	Temperature (Operating/Design):	15 / 65	°C	
Seal Material:	Chemraz SD517	Pressure (Operating/Design):	40 / 150	psig	
Coating/Paint:	Not Required	Flowrate (Operating/Design):	150 / 180	gpm	
Tie-In Flange Size / Rating:	4" 150# Flange, 3" Pipe (See Note 1)	Pressure Drop (Operating/Design):	Minimal (VTA)	psig	
Rail Connection Type:	316SS Autolock Coupler	Arm Pressoff Product:	Nitrogen		
Rail Connection Size:	3"	Grounding Cable Required:	No (See Note 2)		
Local Shut Off Valve Required:		Heat Tracing Required (Voltage/Ph/Hz)	N/A		
Valve Type:	3" SS Ball Valve (See Note 1)	Insulation Required:	N/A		
Valve Used for Line Shutdo	Yes	Lubrication Required:	MFG Standard		
Product Retention	No	Drain Connection:	At Local Low Point		
Working Envelope:		Breakaway coupling:	3" 316 SS NTS-PU		
Minimum Working Radius:	78 in	Minimum Working Height:	in		
Maximum Working Radius:	156 in	Maximum Working Height:	in		
Piggyback Nitrogen Hose					
Tag:	LA - 0632A	Quantity Required:	1		
Application:	Nitrogen Pessoff unloading	Product:	Nitrogen		
Hose Material Tube	PFA Core	Viscosity:	0.018	cP	
Hose Material Cover	Silicone Cover with 302 SS Braid	Specific Gravity:	0.00419	g/mL	
Tie-in Flange Size/Rating	2" 150#	Temperature (Operating/Design):	15 / 65	oC	
Rail Connection Type:	316 SS Cam Lock w/ Chemraz SD517	Pressure (Operating/Design):	40 / 150	psig	
Rail connection Size:	2"	Flowrate (Operating/Design):	150 / 180 (See Note 3)	gpm	
Local Shut Off Valve Required:	Ball Valve	Pressure Drop (Operating/Design):	Minimal - VTA	psig	
Valve Type:	N/A	Arm Purging Product:	N/A		
Valve Used for Line Shut Down:	N/A	Grounding Cable Required:	Yes		
Product Retention	No	Heat Tracing (Voltage/Ph/Hz):	No		
Hose Size	1" ID (See Note 3)	Insulation Required:	No		

Note 1: Arm and Hose Size limited by Railcar Connection of 3" with 3" Diptube
At Design 180 GPM Flow Rate, 3" S/10S Flow Velocity 6.6 ft/s, and ~ 2.9 psi pressure loss per 100 feet of Pipe

Note 2: Grounding of Railcars outside of K215 Scope

Note 3: 1" Sized by vendor quote.
At Design 180 GPM flow rate, Nitrogen flow at 40 psig of regulated pressure is 24 cfm, or 89.5 scfm,
~ 3.77 psi pressure loss per 100 ft of 1" hose

HANSON

Document Number: LP1Y-255-DST-K215-01
Revision:E
Revision Date: 2026-03-04

Rail Unloading Station

Tag No.:	as per Description Chart
P&ID No.:	as per Description Chart
RFQ No.:	LP1Y-255-PRO-K215-01

MTBE Unloading Arm

Loading Arm Overview:	4" x 3" Supported Boom Vapor Balance for MTBE Bottom Unloading				
Vent Arm:					
Tag:	LA-0631A		Quantity Required:	1	
Application:	Top Arm		Product:	Vapor Return	
Arm Operation:	Manually Spring Assisted.		Viscosity:	0.018	cP
Arm Type	Supported Boom Style		Specific Gravity:	0.0014	g/mL
Pipe/hose Material:	316L SS / 316L SS		Temperature (Operating/Design):	15 / 65	°C
Seal Material:	Chemraz SD517		Pressure (Operating/Design):	2.5 / 150	psig
Coating/Paint:	Not Required		Flowrate (Operating/Design):	150 / 180 (See Note 1)	gpm
Tie-In Flange Size / Rating:	4" 150# Flange, 3" Pipe (See Note 1)		Pressure Drop (Operating/Design):	Minimal (VTA)	psig
Rail Connection Type:	316SS Autolock Coupler		Arm Purging Product:	Nitrogen	
Rail Connection Size:	2" for Vent		Grounding Cable Required:	No (See Note 2)	
Local Shut Off Valve Required:			Heat Tracing Required (Voltage/Ph/Hz)	N/A	
Valve Type:	3" SS Ball Valve (See Note 1)		Insulation Required:	N/A	
Valve Used for Line Shutdow	Yes		Lubrication Required:	MFG Standard	
Product Retention	No		Drain Connection:	At Local Low Point	
Working Envelope:			Breakaway coupling:	3" 316 SS NTS-PU	
Minimum Working Radius:	78 in		Minimum Working Height:	in	
Maximum Working Radius:	156 in		Maximum Working Height:	in	

Vapor Return Hose: N/A

Note 1: Arm and Hose Size limited by Railcar Connection of 2" for Vent Connection.
At Design 180 GPM Flow Rate, vapor return flow is 24 cfm at 2.5 psig of pressure, or 28.15 scfm.

Note 2: Grounding of Railcars outside of K215 Scope

HANSON

Document Number: LP1Y-255-DST-K215-01
Revision:E
Revision Date: 2026-03-04

Rail Unloading Station

Tag No.:	as per Description Chart
P&ID No.:	as per Description Chart
RFQ No.:	LP1Y-255-PRO-K215-01

Waste Loading and Piggybacked Vapor Return Arm

Loading Arm Overview.	4" x 3" Supported Boom Loading Arm with 2" Piggyback Vapor Hose					
Loading Arm:						
Tag:	LA-0640A & LA-0650A		Quantity Required:	2		
Application:	Top Loading		Product:	Waste - Primary & Secndary Waste		
Arm Operation:	Manually Spring Assisted		Viscosity:	1.12	cP	
Arm Type	Supported Boom Style		Specific Gravity:	1	g/mL	
Pipe/hose Material:	DUPLEX 2205 Pipe / Hastelloy Hose		Temperature (Operating/Design):	15 / 53	°C	
Seal Material:	Kalrez 0040		Pressure (Operating/Design):	15 / 150	psig	
Coating/Paint:	Not Required		Flowrate (Operating/Design):	350 / 394	gpm	
Tie-In Flange Size / Rating:	4" 150# Flange, 3" Pipe (See Note 1)		Pressure Drop (Operating/Design):	11.80 / 14.95 (preliminary)	psi	
Rail Connection Type:	3" Hastelloy female Camlok Coupler		Arm Purging Product:	Nitrogen		
Rail Connection Size:	3" (See note 1)		Grounding Cable Required:	No (See Note 2)		
Local Shut Off Valve Required:			Heat Tracing Required (Voltage/Ph/Hz)	277V/1 Ph/ 60 Hz (By Others)		
Valve Type:	3" Duplex 2205 Ball Valve (See Note 1)		Insulation Required:	Yes (By Others)		
Valve Used for Line Shutdown	Yes		Lubrication Required:	MFG Standard		
Product Retention	No		Drain Connection:	At Local Low Point		
Working Envelope:			Breakaway coupling:	3" -Hastelloy		
Minimum Working Radius:	78	in	Minimum Working Height:		in	
Maximum Working Radius:	156	in	Maximum Working Height:		in	
Vapor Piggyback Return Hose:						
Tag:	LA-0640A & LA-0650A		Quantity Required:	2		
Application:	Top Vapor Return		Product:	Waste Vapor		
Hose Material:	Hastelloy		Viscosity:	0.018	cP	
Seal Material:	Kalrez 0040		Specific Gravity:	0.0014	g/mL	
Coating/Paint:	Not Required		Temperature (Operating/Design):	15 / 53	°C	
Tie-In Flange Size / Rating:	3" - ANSI 150# Flange		Pressure (Operating/Design):	2.5 / 150	psig	
Rail Connection Type:	2" Hastelloy Female Camlock Coupler		Flowrate (Operating/Design):	350 / 394 (See Note 3)	gpm	
Rail Connection Size:	2"		Pressure Drop (Operating/Design):	Minimal (VTA)	psig	
Local Shut Off Valve Required:			Arm Purging Product:	Nitrogen		
Valve Type:	2" - Duplex Ball Valve		Heat Tracing Required (Voltage/Ph/Hz)	No		
Valve Used for Line Shutdown:	Yes		Insulation Required:	No		
Product Retention	No		Lubrication Required:	MFG Standard		
			Drain Connection:	At Local Low Point.		
			Breakaway coupling:	2" -Hastelloy		

Note 1: 4" Flange & Boom, 3" Primary & Hose Size will connect to 3" railcar connection and 3" diptube via 3" hastelloy camlock coupler
3" Duplex Alloy 2205 Adapter
At Design 394 GPM Flow Rate, 3" S/40S Flow Velocity ~17.3 ft/s, and ~14.7 psi pressure loss per 100 feet of 3" Pipe

Note 2: Grounding of Railcars outside of K215 Scope

Note 3: 2" Sized by vendor quote, along with limit of railcar 2" connection for venting
At Design 394 GPM of design flow, vapor flow is 53 cfm @ 2.5 psig, or 62 scfm
~ 0.03 psi pressure loss per 100 ft of 2" hose

Document Number: LP1Y-255-DST-K215-01
Revision:E
Revision Date: 2026-03-04

HANSON

Rail Unloading Station

Tag No.:	as per Description Chart
P&ID No.:	as per Description Chart
RFQ No.:	LP1Y-255-PRO-K215-01

1	Rev	General Notes
2		1) Loading and vapor arm piping shall include a manual drain valve and in-line manual valve to stop flow and isolate the arms.
3		2) - Note Deleted
4		3) The loading arms shall have a breakaway coupling to safely detach loading arm and halt flow in the event of a accidental
5		disconnection from the railcar while unloading or loading chemicals.
6		4) Dimensions of the working envelope, to be confirmed by the supplier
7		5) Seller to specify the weight of each loading arm.
8		6) Swivel joints shall be capable of being lubricated without dismantling. Lubrication points shall be readily accessible
9		in an area not subject to ice build-up.
10		7) Swivel joints shall be split flange type to allow removal of the balls when replacing seals.
11		8) The basic material of swivel joints shall be resistance to corrosive and abrasive action or temperature condition
12		of the product being handled.
13		9) Seller to recommend proper configuration of the arms (left or right handed) to avoid any future interference specially
14		for spots with more than one arm
15		10) Seller to consider and recommend the draining sequence of the offloading arms after offloading is completed.
16		11) At client request, Seller to disregard project spec regarding duplex hoses for duplex wast loading arms. Hastelloy hoses shall be utilized
17		in lieu due to availability and Client preference.
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A1 - DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT
Authorization to proceed does not relieve Contractor/Supplier of its responsibility or liability under the Contract and or Purchase Order.
By: Ricardo DaCosta dac54131 December 3, 2025

HANSON

Document Title:	Gangways Data Sheet
Document Number:	LP1Y-255-DST-K215-05
Security Class:	Restricted

REVIEWED
No comments

Approved

REVIEWED
By:Brydon Coombe,coo11790,November 26, 2025

Gangways Data Sheet

Lilly - Lebanon Plant 1 (B1LG)

K215 - Lilly Rail Offloading Rack

Hanson Project Number: 25I0007

Revision	Issue Status	Originator	Reviewer	Approver	Date
B	IFQ	SS	QR	MW	11/3/2025
A	IFR	SS	QR	MW	10/9/2025

No Lilly comment

HANSON

Document Number: LP1Y-255-DST-K215-05
Revision: B
Revision Date: 2025-11-03

Rail Unloading Station

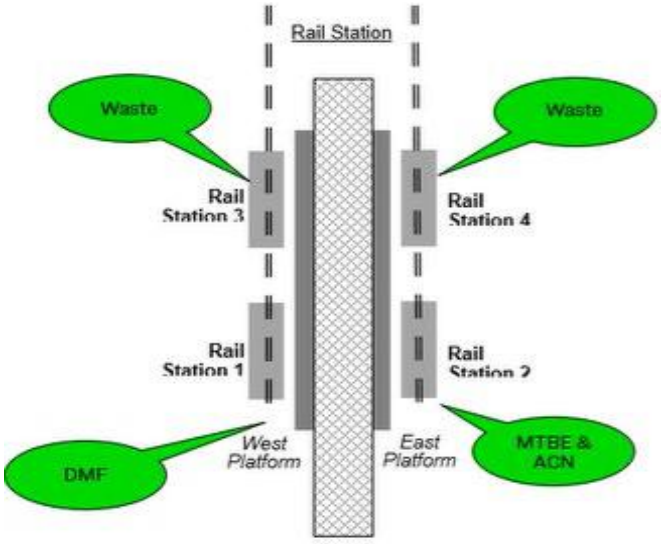
Tag No.:	N/A
P&ID No.:	N/A
RFQ No.:	LP1Y-255-PRO-K215-03

1	Rev	General Information	
2		Rail Unloading Station:	Rail Stations 1, 2, 3 & 4
3		Station Use:	as per Description Chart
4		P&IDs:	as per Description Chart
5		Environmental Conditions	Refer to project document LP100-225-DBD-001
6		Min/Max Temperature:	-27.5°C to 45°C
7		Wind Load (mph/sqf):	120 mph
8		Seismic Design:	Refer to project document LP100-215-EMS-05002
9		Hazardous Area Classification:	Class 1, Division 2
10		Design Specifications	
11		Mechanical Specifications:	N/A
12		Structural Specifications:	N/A
13		Pipe Specifications:	N/A
14			

Description Chart

#	Equipment Description	Tag	QTY	Note
1	Railcar Gangway - Spot 1	N/A	1	
2	Railcar Gangway - Spot 2	N/A	1	
3	Railcar Gangway - Spot 3	N/A	1	
4	Railcar Gangway - Spot 4	N/A	1	

Station Overview



HANSON

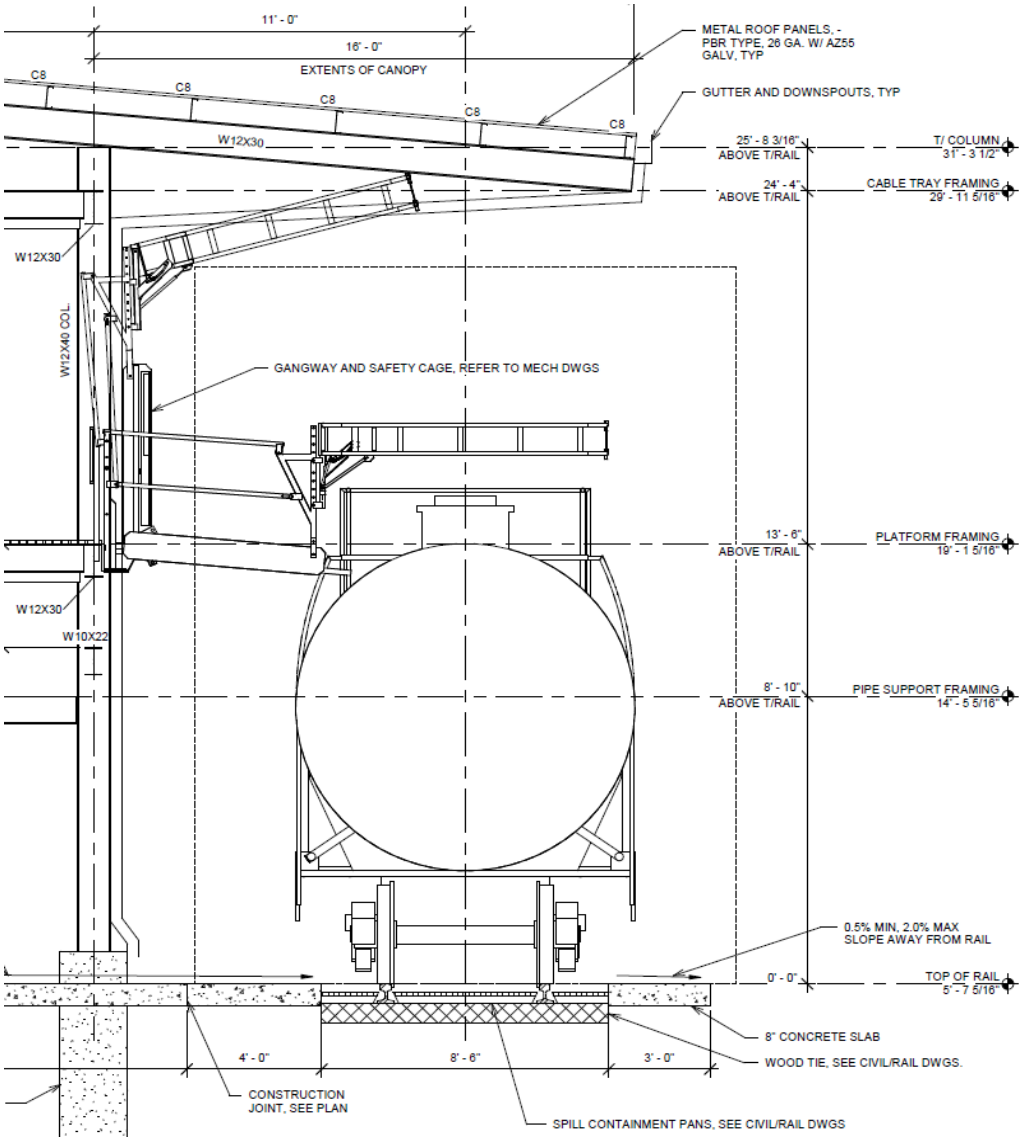
Document Number: LP1Y-255-DST-K215-05
Revision: B
Revision Date: 2025-11-03

Rail Unloading Station

Tag No.:	N/A
P&ID No.:	N/A
RFQ No.:	LP1Y-255-PRO-K215-03

Platform Layout

Maximum Required Rail Clearance - A:	17' - 6"	ft		Top of railcar from TOR - High	12' - 6"	ft
Railcar Center to Platform Edge - B:	11'			Top of railcar from TOR - Low	11' - 6"	ft
Hatch Height - C:	TBD	ft		Top of Platform from TOR	13' - 6"	ft
Hatch Diameter - D:	TBD					ft
				Gangway upright - Botton of gangway to TOR (min)	22' - 4"	ft
General Gangway Width:		ft		Gangway upright - Top of gangway to TOR (max)	23' - 6"	ft
General Gangway Length:		ft		Railcar Type	DOT 111 and DOT117 (~30,600 Gal)	
				TOR = Top of Rail		



HANSON

Document Number: LP1Y-255-DST-K215-05
Revision: B
Revision Date: 2025-11-03

Rail Unloading Station

Tag No.:	N/A
P&ID No.:	N/A
RFQ No.:	LP1Y-255-PRO-K215-03

	Rev	Gangway Design Notes
2		Access Gangway Type: Spring balanced self-adjusting (aluminum)
3		Platform Scope and Features: Design and supply of rail foldable access gangway with swing gate and fall protection cage,
4		to provide safe access to the top hatch of the railcars
5		Structural Steel Design and Materials of Construction: Aluminum
6		Guardrail Material of Construction: See Specification Below
7		Grounding Connections: Two ground lugs to be provided at opposing corners at the bottom of the station steel
8		Foundation Tie-In Location: N/A
9		
10		
11		
12		1. Gangway shall be inherently safe and provide fall protection while in the fully retracted position.
13		2. The entrance to the gangway shall have a swing gate that must be opened inward by the operator to access the top of the railcar.
14		3. The gangway shall be self-balanced and operatable without any need for Nitrogen or pneumatic sources.
15		4. Cages, shall be bolted to the offloading structure/platform
16		5. Gangway shall provide safe access to all different types of rail cars,
17	B	6. Folding stairs or flat ramp and Cage subframes to be galvanized steel. Aluminum cage and handrail structure to be considered
18		7. Seller to suggest a proper, safe design for the top access gangways, so that operator can access the top of the rail car in
19		a safe manner
20	B	8. Access gangway shall be equipped with a self-adjusting stair or flat ramp gangway. If the railcars come in taller or shorter
21	B	than the platform, these stairs or ramp provide a level surface to transition back and forth
22		
23		Specifications
24		Total Load Capacity - 500 lbs, Handrail 200 lbs., Midrail - 100 lbs.
25		Available Working Range - +/- 15 Deg to Horizontal
26		2.9" Diameter Handrail with 1" Diameter Grab Bar
27		Base Tread Hot Dipped Galvanized per ASTM A123
28		Self Leveling stairs
29		Gangway Body constructed of Aluminum
30		Self Supporting by use of Adjustable Chain Stops
31		Spring Bearing Pivot Points Contain Oil Impregnated Bronze Bearings
32		AWS Welding Spec D 1.1 - D 1.2
33		All critical Hardware is Stainless Steel
34		Self Leveling Safety Hoop with 200 lb top rail rating
35		
36		
37		
38		

A1 - DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT
Authorization to proceed does not relieve Contractor/Supplier of its responsibility or liability under the Contract and or Purchase Order.
By: Ricardo DaCosta dac54131 December 3, 2025

HANSON

Document Title:	Safety Shower & Eye Wash Data Sheet
Document Number:	LP1Y-255-DST-K215-06
Security Class:	Restricted

REVIEWED
By:Gale Yuan,yux15445,November 20, 2025

SSEW model was dedicated by Lilly and Model already sent to K215. just follow the specs to purchase.

No Comments

Approved with comments incorporated.

NO COMMENTS. SUBMITTAL ALIGNS WITH PROJECT STANDARDS.

Safety Shower & Eye Wash Data Sheet

Lilly - Lebanon Plant 1 (B1LG)

K215 - Lilly Rail Offloading Rack

Hanson Project Number: 25i0007

Revision	Issue Status	Originator	Reviewer	Approver	Date
B	IFQ	SS	QR	MW	11/10/2025
A	IFR	SS	QR	MW	10/9/2025

No Lilly comment

HANSON

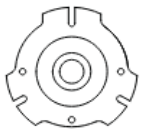
Document Number: LP1Y-255-DST-K215-06
Revision: B
Revision Date: 2025-11-10

Rail Loading Station

Tag No.: SSEW-15-0621-01, SSEW-15-0621-02
P&ID No.: LP1Y-DI-00000061
Doc. No.: LP1Y-255-DST-K215-06

Rev	Safety Shower & Eye Wash Design Notes
	Type: Safety station with Eyewash, Freeze and Scald Protection Valves — Quantity: **2**
	Requirement: Deliver water at 20 GPM for a minimum 15 minutes
	Valve Type: Stay-open ball valves
	Flow Rate and Pressure: per ISEA Z358.1
	Standard: ANSI Z358.1-2014, OSHA 1910.151
	LOCATION:
	○ INDOOR ○ HEATED ● OUTDOOR ○ UNHEATED
	ELECTRICAL AREA CLASSIFICATION **N/A**
	● CL **1** DIV **2** GROUP **C & D, T4** ○
	● WINTERIZATION REQ'D ● TROPICALIZATION REQ'D. **(See Note)**
	SITE DATA **Lebanon, Indiana**
	ALTITUDE (ft) BAROMETER (psia)
	○ AMBIENT TEMP RANGE: MIN/MAX. / (°F)
	○ RELATIVE HUMIDITY: MIN / MAX / (%)
	CAPACITY: POWER RATING: **N/A**
	OPERATING CONDITION:
	WEIGHT: DIMENSION:
	MATERIAL OF TANK: **N/A**
	POWER SUPPLY Volts Hz PHASE:
	OTHER
	Note: Freeze resistant, scald protection valve, freestanding, no bowl, with self-limiting heat trace cable, thermostat, stainless steel showerhead, alarm beacon, pull rod on stay-open ball valve, insulation with weatherproof cover, ANSI-compliant identification sign, lead free, with factory mounted Form C dry contact type proximity switches for both eyewash and shower valves.
	Acceptable Manufacturers: Guardian Equipment GFR3500-SFP-PS-SSH-TL for Class 1 Division 2 hazardous environments.
B	**Note 2**: Top inlet 1-1/4" NPT

UNIVERSAL FLOOR FLANGE



Ø7/16" MOUNTING SLOTS
ON 7-5/8" BOLT CIRCLE

Figure 1 - Floor Mounting Flange

A1 - DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT
Authorization to proceed does not relieve Contractor/Supplier of its responsibility or liability under the Contract and or Purchase Order.
By: Ricardo DaCosta dac54131 December 3, 2025

HANSON

Document Title:	Nitrogen Distribution Filter Data Sheet
Document Number:	LP1Y-255-DST-K215-07
Security Class:	Restricted

REVIEWED
By:Gale Yuan,yux15445,November 20, 2025
No comments

REVIEWED
By:Brydon Coombe,coo11790,November 26, 2025
No Comments

Reviewed

Nitrogen Distribution Filter Data Sheet

Lilly - Lebanon Plant 1 (B1LG)

K215 - Lilly Rail Offloading Rack

Hanson Project Number: 25i0007

A	IFR	SS	QR	MW	11/10/2025
Revision	**Issue Status**	**Originator**	**Reviewer**	**Approver**	**Date**

No Lilly comment

HANSON

Document Number: LP1Y-255-DST-K215-07
Revision: A
Revision Date: 2025-11-10

Rail Unloading Station

Tag No.:	FLT-0610
P&ID No.:	LP1Y-DI-00000061
RFQ No.:	N/A

General Information

Environmental Conditions	Refer to project document LP100-225-DBD-001
Min/Max Temperature:	-27.5°C to 45°C
Wind Load (mph/sqf):	120 mph
Seismic Design:	Refer to project document LP100-215-EMS-05002
Hazardous Area Classification:	Class 1, Division 2

PROCESS DATA / REQUIREMENTS

Process Information

Fluid:	Nitrogen	
Specific Gravity (max):	0.58 lb/ft^3	
Viscosity (max):	0.02	cP
Operating Pressure:	100.0	psig
Design Pressure:	150 / FV	psig
Operating Temperature:	25	°C
Design Temperature:	75.0	°C
Filtration Flow Rate:	400.0	scfm
Clean Pressure Drop:	*	psi
Dirty Pressure Drop:	* (Note T.8)	psi
Element Type:	Hydrophobic	
Flow Direction:	Uni-Directional	
Rated Pore Size:	0.25 micron	
Rated Retention:	*	

Vessel Information (If applicable)

Vessel Mounted:	No	
Volume:	N/A	L
Vessel ID:	N/A	in
Vessel T/T:	N/A	in
Vessel Fill Rate:	N/A	L/min
Vessel Drawdown Rate:	N/A	L/min
Vessel Pressure Rating:	N/A	psig

CLEAN IN PLACE (CIP) / STEAM IN PLACE (SIP)

Clean In Place (CIP)

Element In Place/Removed:	N/A	
CIP Flow Rate:	N/A	L/min
CIP Pressure:	N/A	psi
CIP Temperature:	N/A	ºC
CIP Duration:	N/A	min
CIP Fluid Description:	N/A	

Steam In Place (SIP)

Element In Place/Removed:	N/A	
SIP Flow Rate:	N/A	0
SIP Pressure:	N/A	psi
SIP Temperature:	N/A	ºC
SIP Duration:	N/A	min

HANSON

Document Number: LP1Y-255-DST-K215-07
Revision: A
Revision Date: 2025-11-10

Rail Unloading Station

Tag No.:	FLT-0610
P&ID No.:	LP1Y-DI-00000061
RFQ No.:	N/A

MECHANICAL DATA/REQUIREMENTS

		Rev		
Manufacturer:	*		***Housing Information***	
Quantity of Housings:	1		Filter Housing Design:	Non-Sanitary
Housing Model:	*		Base Type:	T Style
Element Model:	*		Housing Length:	* in
Element By:	Others		Flow Pattern:	Uni-Directional
Design Code:	ASME		Housing Orientation:	Vertical
Code Stamp:	TBD**		Housing Supports:	**
			Closure Type	*
Connections			Closure Size:	* in
Inlet Size:	2 in		Jacketed	no
Inlet Type:	RF		***Element Information***	
Outlet Size:	2 in		Elements per Housing:	*
Outlet Type:	RF		Element Length:	* in
Vent Size:	0.75 in		Element Diameter:	* in
Vent Type:	RF		Stacking Arrangement:	xR xy" H*
Drain Size:	N/A in		Filter Area per Element:	* m2
Drain Type:	N/A		Total Filter Area:	* m2
			Top End Configuration:	Bayonet w/Spear
Note that Flanges should be Class 150			Configuration Code:	Code 7

Mechanical Notes

F.A. The filter housing must be fully drainable through the inlet/outlet nozzles

F.B. Filter elements to be supplied by Owner.

**Required if multi-round

MATERIALS OF CONSTRUCTION / SURFACE FINISH

		Rev		
Material			***Surface Finish***	
Housing:	316L SS		Product Contact Finish:	Vendor Standard Finish
Housing O-ring:	EPDM		External Surface Finish:	Vendor Standard Finish
Housing Gasket:	EPDM		Product Contact Weld:	Vendor Standard Finish
Bolts/Nuts:	*		External Weld Finish:	Vendor Standard Finish
Element Media:	*		Legs / Supports Finish:	Vendor Standard Finish
Element Supports:	*		Electropolish Internals:	N/A
			Passivate Internals:	Yes(*)

SITE DATA

		Rev	
Classification			***Seismic Data***
Room Classification:	NC (outdoor)		N/A

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Rail Unloading Station

Tag No.:	FLT-0610
P&ID No.:	LP1Y-DI-00000061
RFQ No.:	N/A

WEIGHTS AND DIMENSIONS

Weights			Rev	***Dimensions***		
Empty Weight:	*	lb		Overall Height:	*	in
Flooded Weight:	*	lb		Outside Diameter:	*	in

GENERAL NOTES

T.1. Seller to complete this data sheet & submit this with their proposal including those blanks marked by an asterisk.

T.2. Vendor shall stamp the following information on each 300 series steel continuously welded nameplate:

1.0 Manufacturer's serial number, name, and model number.
2.0 Year Built.
3.0 Pressure and temperature ratingns.
4.0 Material of construction.
5.0 Ownere's equipment number and purchase order number.
6.0 Manufacturer's drawing number.
7.0 Project number.
8.0 National Board Number (If applica le for pressure vessels).
9.0 For ASME Code stamped housings, nameplate shall be in accordance with ASME BPVC Section VIII Division 1 and include all additional information as required above.

T.3. N/A

T.4. Exterior surfaces shall be easily cleanable without cracks. crevices, exposed insulation.

T.5. Gasket/Elastomer Material must be USP Class VI or 21 CFR Part 177 Compliant.

T.6. Nuts thread shall be made of dissimilar metals insert made of Nitronic 60 or equivalent and not subject to galling when used with stainless steel threaded bolts and studs.

T.7. The use of rouge or polishing compounds is prohibited. The use of rouge or polishing compounds ON PRODUCT CONTACT SURFACES shall be cause for automatic rejection.
The finish of weld shall be flush, smooth, polish and blended.

T.8 Max pressure drop less than 5 psi at dirty condition

T.9. Welds inside the housing and on the external surfaces shall be ground smooth and flush to match adjacent surface with the requested Ra value of surface finish.

T.10. If multi-round, housing should be provided with appropriate supports (legs, lifting lugs, etc.)

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Document Title:	Mechanical Equipment List
Document Number:	LP1Y-255-LST-K215-01
Security Class:	Restricted

Mechanical Equipment List

Lilly - Lebanon Plant 1 (B1LG)

K215 - Lilly Rail Offloading Rack

Hanson Project Number: 25i0007

Revision	Issue Status	Originator	Reviewer	Approver	Date
C	IFR	MW	QR	SS	4/14/2025
B	IFR	QR	SS	MW	11/17/2025
A	IFR	QR	SS	MW	10/14/2025

Pumps

TAG No.	PARENT TAG No.	DESCRIPTION	EQUIP. COUNT	TYPE	NPSH (Available/Required) ft	FLOW RATE GPM	DESIGN DATA TEMP. º C	DESIGN DATA PRESS. PSIG	Viscosity cP	Inlet/Outlet Size inch	MATERIAL	RPM	POWER KW	DRIVER RATING SELECTED HP	DRIVER RATING SELECTED KW	DRIVER TYPE	WEIGHT LBS	DIMENSIONS LxWxH inch	P&ID No.	P&ID Rev	LLI	Lead Time	Vendor	COST	AREA	REV
P-0631A	N/A	Unloading Pump - MTBE	1	Centriugal	13.6 / 11.5	150	-27 TO 37	60	0.37	1.5" / 1.0"	316L SS	3600	20	14.9	20	Motor (ABB)	609	48 x 18 x 20	LP1Y-DI-00000063	2	No	Procured by Fluor	Flowserve	N/A	Rail Station 2 - Ground	C

Filter

TAG No.	PARENT TAG No.	DESCRIPTION	EQUIP. COUNT	TYPE	CAPACITY SCFM	CONNECTION	DESIGN DATA TEMP. º C	DESIGN DATA PRESS. PSIG	MATERIAL	SERVICE	OTHER INFORMATION	WEIGHT LBS	DIMENSIONS	P&ID No.	P&ID Rev	LLI	Lead Time	Vendor	COST	AREA	REV
FLT-0610	N/A	Nitrogen Filter	1	Cartridge	400	2" 150#	-29 TO 177	150 / FV	316L SS	NITROGEN	Sanitary Modified T-Style Filter Housing - 20" High, 6" Diameter Housing with 2" Flg. I/O connections and 3/4" Vent - Type 226 Code 7 Cartridge, 3.367 psi pressure drop.	39	OD: 6" High: 20"	LP1Y-DI-00000061	2	TBD	18 wk ARAD	Allegheny Bradford Corp.	TBD	Rail Station 2 - Ground	C

Loading Arms

TAG No.	PARENT TAG No.	DESCRIPTION	EQUIP. COUNT	TYPE	CAPACITY GPM	CONNECTION	DESIGN DATA TEMP. º C	DESIGN DATA PRESS. PSIG	MATERIAL	SERVICE	OTHER INFORMATION	WEIGHT	DIMENSIONS	P&ID No.	P&ID Rev	LLI	LEAD TIME	VENDOR	COST	AREA	REV
LA-0622A	N/A	Unloading Arm - DMF	1	Manual	180	4" 150#	65	150	316L SS	DMF	Top Unloading 3", Manual, with 3" Hose End, Ball Valve and Break Away Coupling, 1" N2 Piggyback Pressoff Hose Sizing confirmed by vendor quote.	TBD	TBD	LP1Y-DI-00000062	2	TBD	TBD	TBD	TBD	Rail Station 1 - Platform	C
LA-0631A	N/A	Unloading Arm - MTBE	1	Manual	180	4" 150#	65	150	316L SS	MTBE	Vapor Return Top 3", Manual, with 3" Hose End, Ball Valve and Break Away Coupling Sizing confirmed by vendor quote	TBD	TBD	LP1Y-DI-00000063	2	TBD	TBD	TBD	TBD	Rail Station 2 - Platform	C
LA-0632A	N/A	Unloading Arm - ACN	1	Manual	180	4" 150#	65	150	316L SS	ACN	Top Unloading 3", Manual, with 3" Hose End, Ball Valve and Break Away Coupling, 1" N2 Piggyback Pressoff Hose Sizing confirmed by vendor quote.	TBD	TBD	LP1Y-DI-00000063	2	TBD	TBD	TBD	TBD	Rail Station 2 - Platform	C
LA-0640A	N/A	Loading Arm - Waste	1	Manual	394	4" 150#	53	150	DUPLEX 2205 PIPING/ Hastelloy Hoses	WASTE	Top Unloading 3", Manual, with 3" Hose End, Ball Valve and Break Away Coupling Sizing confirmed by vendor quote.	TBD	TBD	LP1Y-DI-00000064	2	TBD	TBD	TBD	TBD	Rail Station 3 - Platform	C
N/A	LA-0640A	Vapor Return Hose	1	Manual	394	3" 150#	53	150	Hastelloy	WASTE VAPOR	Top Connection - Piggy backed flexible 2" Hose, Ball Valve and Break Away Coupling Sizing confirmed by vendor quote.	TBD	TBD	LP1Y-DI-00000064	2	TBD	TBD	TBD	TBD	Rail Station 3 - Platform	C
LA-0650A	N/A	Loading Arm - Waste	1	Manual	394	4" 150#	53	150	DUPLEX 2205 PIPING/ Hastelloy Hoses	WASTE	Top Unloading 3", Manual, with 3" Hose End, Ball Valve and Break Away Coupling Sizing confirmed by vendor quote.	TBD	TBD	LP1Y-DI-00000065	2	TBD	TBD	TBD	TBD	Rail Station 4 - Platform	C
N/A	LA-0650A	Vapor Return Hose	1	Manual	394	3" 150#	53	150	Hastelloy	WASTE VAPOR	Top Connection - Piggy backed flexible 2" Hose, Ball Valve and Break Away Coupling Sizing confirmed by vendor quote.	TBD	TBD	LP1Y-DI-00000065	2	TBD	TBD	TBD	TBD	Rail Station 4 - Platform	C

Access Gangway

TAG No.	PARENT TAG No.	DESCRIPTION	EQUIP. COUNT	TYPE	TYPE	DIMENSION	DESIGN DATA TEMP. º C	DESIGN DATA PRESS. PSIG	MATERIAL	SERVICE	OTHER INFORMATION	WEIGHT	DIMENSIONS	Data Sheet No.	Data Sheet Rev	LLI	LEAD TIME	VENDOR	COST	AREA	REV
N/A	N/A	Railcar Access Gangway	4	self balanced	Spring or counterweight	N/A	N/A	N/A	Galvanized and/or Aluminum Construction	Personal Access to Railcars	To provide safe access and fit DOT-111 and DOT-117 Railcar top cage Includes safety hoop/cage.	TBD	TBD	LP1Y-255-DST-K215-05	B	TBD	TBD	TBD	TBD	Rail Station - Platform	C

Eyewash & Safety Shower

TAG No.	PARENT TAG No.	DESCRIPTION	EQUIP. COUNT	TYPE	SOURCE	CAPACITY Gallon	DESIGN DATA TEMP. º C	DESIGN DATA PRESS. PSIG	MATERIAL	SERVICE	OTHER INFORMATION	WEIGHT	DIMENSIONS	P&ID No.	P&ID Rev	LLI	LEAD TIME	VENDOR	COST	AREA	REV
SSEW-15-0621-01, & 02	N/A	Safety Shower and Eyewash	2	EHT	WPC	20 GPM (for 15 min)	35	195	316L SS	WPC	Outdoor, Heat Traced, Freeze Resistantc Combination Eyewash and Shower Safety Station, Alarm Horn, Freeze and Scald Protection Valves, Remote Sensing Proximity Switches, & Heat Trace Indicator Light 120 VAC Single Phase for Heat Trace and Alarm Light	TBD	TBD	LP1Y-DI-00000061	2	TBD	TBD	Guardian Equipment	TBD	Rail Station - Platform and Ground	C

Note:
1) Data on this equipment list are preliminary and shall be confirmed in the next stage of the project
2) TBD = To Be Defined, TBC = To Be Confirmed, VTA - Vendor To Advise

Document Title:	Request For Purchase (RFP) for Loading Arms
Document Number:	LP1Y-255-PRO-K215-01
Security Classification:	Restricted

Request For Purchase (RFP) for Loading Arms

Eli Lilly Lebanon Plant 1 (B1LG)

K215 - Rail Unloading Rack

Hanson Project Number:25I0007.00

Revision	Issue Status	Originator	Reviewer	Approver	Date
0	Issued for Purchase	SS/QR	MW	JR	2026-03-26
B	Issued for Quotation	SS	QR	MW	2025-11-10
A	Issued for Review	SS	QR	MW	2025-10-29

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1. Project Summary

The equipment listed under this RFP forms part of the LP1 Project for Eli Lilly at their plant located in Lebanon, Indiana, USA.

Fluor serves as the overall EPC Contractor for the complete project scope.

Hanson Professional Services Inc. (hereafter referred to as Hanson) is the engineering partner responsible for the detailed design of the rail station area, while Standard Constructors Inc. (hereafter referred to as SCI) is responsible for procurement and construction of the rail station area and will be in charge for this purchase order.

Purchase Order Contact point:

- To: Jamie Reingold – SCI's Director and Project Mgr
- email: reingold@standardconstructors.com

2. Scope of Supply

The Seller shall design, manufacture, test, and supply complete top loading/unloading arms suitable for installation on the K215 Rail Unloading Rack in accordance with the following scope:

#	Item Tag no.	Item Description	Qty
1	All	Seller is responsible for the complete design, engineering, design authentication (Professional Engineering Stamp, where required), supply of materials, fabrication, shop inspection, and testing, surface preparation, painting certification (where required), documentation, and preparation for shipment in accordance with this material requisition for the following equipment: Quantities, dimensions, and lengths where provided are used for planning & estimating purposes only. Final design quantities are to be determined by the seller.	
1.1	LA-0622 A	DMF Unloading Arm – Top Unloading	1 set
1.2	LA-0632 A	ACN Unloading Arm – Top Unloading	1 set
1.3	LA-0631 A	MTBE Unloading Arm – Top Unloading Vapor Return	1 set
1.4	LA-0640 A	Waste Loading Arm – Top Loading with Vent	1 set
1.5	LA-0650 A	Waste Loading Arm – Top Loading with Vent	1 set
2	All	Seller scope also include the followings for items 1.1 to 1.5 - NDE as project specifications and also required by code. - Surface preparation (if & where required) - Design calculations and drawings (Indiana PE stamp, only if required) - Grounding lugs (2 nos.)	

		- Lifting lugs - Shipping drawings and calculations. - Drilled matched anchor bolt template. - Emergency Release Coupling - Hydrotest - Teflon blind flanges for the inlet and outlet of the arms - Shipment packing	
3	All	Start-up and Commissioning Spares for each arm - 100% of all packing and gaskets, 10% of studs, nuts and bolts for all flanges and swivel joints - Grease for the Swivel Joints, (minimum for 2 time greasing all swivel Joint) - Seller's recommended spare parts.	1 set per arm
4	All	Recommended Spares for Two Years operation for each arm - 200% of all packing and gaskets, 10% of studs, nuts and bolts (min two studs or bolts and four nuts) for all flanges and swivel joints - Grease for the Swivel Joints, Proper amount for two years greasing of swivel Joint) - Seller's recommended spare parts.	1 set per arm
5	All	Capital Spares - Seller's recommended spare parts	1 set per arm

Note: All arms are manual operation and rack-mounted. Vapor return applies to MTBE only; Waste arms include 3" vent take-offs.

2.1. Included in Seller's Scope

- Complete mechanical assembly for each arm including:
    - Drop-tube/nozzle (where required), multi-plane swivel joints, and counterbalance mechanism (spring or weight).
    - Vapor return and vent branches as defined above.
    - Parking posts, and stow locks.
    - Bonding/grounding lugs and provisions per NFPA 77 / API RP 2003.
- Factory fabrication, assembly, and testing.
- Materials, welding, and finishing per applicable codes.
- Anchor bolt loads and connection details for rack mounting.

- Documentation, testing, inspection, and packaging for shipment.
- Recommended spare parts (start-up, 2-year, capital).
- Optional field services for installation supervision and training.

2.2. Excluded (by Buyer/others)

- Rack structure, gangways, fall protection, and foundations.
- Piping, vapor, and vent lines beyond Seller's terminal flanges.
- Electrical power or controls (arms are manual).
- Heat tracing, insulation, or painting beyond Seller's assembly.

3. List of Documents

#	Document Number	Document Description
1	LP1Y-255-DST-K215-01 _ Rev E	Mechanical Data Sheets – Loading Arms
3	LP1Y-D1-00000062_R2	P&ID
4	LP1Y-D1-00000063_R2	P&ID
5	LP1Y-D1-00000064_R2	P&ID
6	LP1Y-D1-00000065_R2	P&ID

Any discrepancy between the attached documents, the "Mechanical Data Sheets – Loading Arms" will take precedent.

4. Technical Requirements

4.1. The Seller shall have sole responsibility for the design, fabrication, assembly, testing, and documentation of the supplied equipment.

4.2. Each arm shall include a permanently affixed stainless-steel nameplate, stamped with at least the following (in U.S. customary units):

- Manufacturer name and year of manufacture
- Tag number and service
- Maximum allowable working pressure (MAWP)
- Design temperature

- Empty and operating weight
- Purchase order number

4.3. Water used for hydrostatic testing or cleaning shall be potable, containing less than 100 ppm chloride ions. After successful hydrotesting, all equipment shall be thoroughly drained, cleaned, and dried prior to shipment.

4.4. Loose parts shall be suitably packed, identified, and protected from mechanical damage and corrosion during storage and shipment.

4.5. The Seller shall provide adequate packaging and preparation for shipment; all preparation shall be subject to inspection and approval by the Buyer.

4.6. All assemblies and parts shall be protected against mechanical damage and atmospheric corrosion for a minimum of twelve (12) months outdoor storage at the job site prior to installation.

4.7. Each package shall be clearly marked inside and outside with item number, service, tag number, purchase order number, and gross weight.

4.8. Units of measurement shall follow U.S. Customary Units.

4.9. The Seller's quotation and any subsequently agreed technical clarifications shall form part of this purchase requisition.

4.10. The loading arms shall be designed in accordance with the applicable provisions of ASME Section VIII Div. 1 and Section IX (for welding), and general industry practice for process loading systems.

4.11. Connection flanges at the battery limit shall be ASME B16.5 Class 300 raised-face with a surface finish between 125–250 AARH.

4.12. The Seller shall define and specify:

- Method of counterbalance (spring or counterweight),
- Swivel joint type, manufacturer, and seal material,
- Emergency release coupling (ERC) type (if applicable),
- Any required instrumentation, and
- Associated piping interfaces for the loading arms.
- Quick Connect Coupling & Full Port Ball Valves

4.13. Seller shall advise the minimum allowable spacing between adjacent loading arms to suit the existing rack configuration and any future expansion provisions.

4.14. Seller shall provide reaction loads and moments to enable Buyer to design supporting structures.

4.15. Connection between the loading arm and fixed piping shall conform to ASME B16.5 and the attached data sheets.

4.16. Loading arms shall be protected against overpressure via bypass and/or safety control valves as applicable.

4.17. Swivel Joints

- Swivels shall be grease-lubricated without disassembly; all lubrication points shall be readily accessible.
- Pressure-relief ports shall prevent over-pressurization during greasing.
- Swivels shall use ball or roller bearings, and split-flange construction to allow seal replacement without removing bearing elements.
- Seal faces shall be stainless steel for corrosion protection.
- Swivel body materials shall be compatible with the handled product's corrosivity, abrasiveness, and temperature.

4.18. Emergency Release & Safety Devices

- Arms handling flammable or toxic liquids, or incorporating vapor-return systems, shall include Breakaway Couplings as applicable.

4.19. Arm Locking, Draining & Venting

- Each arm shall have a mechanical tie-down to secure the arm in stored position under high-wind conditions.
- Arms shall be lockable against horizontal rotation in stored and maintenance positions.
- All lubricated parts shall be serviceable without dismantling, and grease fittings and lines shall be sized for adequate flow in expected ambient temperatures.
- A portable or integrated grease-pumping device shall be supplied for lubrication of all loading arms.

4.20 System Configuration & Alternatives

- The loading system shall consist of a mounting column with locking device.

- Each arm shall include a balancing device to offset arm weight and ensure ease of operation.
- Vapor return or press off lines shall be integrated, or piggy backed, with the main loading arm to eliminate separate vapor arm.

5. Supplier Document & Data Control (SDDC)

The Seller shall provide all engineering, fabrication, testing, and quality documentation necessary to demonstrate compliance with this Technical RFQ, the applicable codes and standards, and the approved data sheets.

All submittals shall be issued electronically in PDF format for review and approval. Native editable files (e.g., AutoCAD, Excel, Word) shall be provided for all drawings, data sheets, and lists at the time of final/as-built issue.

Each document shall include, at a minimum, the Project Name, Equipment Tag Number(s), Purchase Order Number, Revision Number, and Date of Issue.

All submittals shall be transmitted through the Buyer's document control system or by email as instructed at time of award.

All dates are referenced After Receipt of Order (ARO) unless otherwise stated.

All drawings and 3D models shall show interface points, swing envelopes, and clearances relevant to installation and maintenance.

Documentation must demonstrate compliance with the applicable design standards (ASME B16.5, B16.47, Section VIII, API RP 2003, NFPA 30).

The final Record Book shall include WPS/PQRs, MTRs, Inspection Reports, Test Certificates, Photographic Records (where available), and As-Built Drawings.

Seller shall maintain all quality and inspection documentation for a minimum of five (5) years from shipment.

Item	Document Description	With Proposal	After Order	For Review / Approval	Final / As-Built	Include with Shipment / Record Book / Due Date
A – Commercial Data						
A-1	Sellers List of Document	X				2 wks ARO
A-2	Procurement / Engineering / Production Schedule			X		4 wks ARO

A-3	Monthly Progress Report					Monthly
A-4	Unpriced Sub-Vendor PO Confirmations					As Completed
A-5	Packing List			X		4 wks ARO
B – Drawings & Engineering Data						
B-1	Dimensional Outline Drawing & BOM, weight	X		X	X	2 wks ARO
B-2	General Arrangement / Assembly Drawing			X	X	4 wks ARO
B-3	Loads & Moment Details			X	X	2 wks ARO
C – Procedures & Calculations						
C-1	WPS / PQR / Welder Qualifications			X	X	Before Fabrication
C-2	Hydrostatic Test Procedure			X	X	6 wks ARO
D – Quality & Inspection Documents						
D-1	QA/QC Program Registration Certificate			X		6 wks ARO
D-2	Inspection & Test Plan (ITP)			X	X	6 wks ARO
D-3	Mill Test Reports / Material Traceability					At Completion
D-4	Non-Conformance Reports & Resolutions					As Completed
E – Test & Certification Reports						
E-1	Hydrostatic Test Report / Chart					At Completion
E-2	Dimensional Check Report					At Completion
F – Compliance Certificates						
F-1	Code / Standard Compliance Certificate					Prior to Shipment
F-2	Certificate of Compliance with Specifications					Prior to Shipment

G – Manuals & Record Books						
G-1	Record Book Index					8 wks ARO
G-2	Record Book (Full PDF + Native Files)					2 wks After Shipment
G-3	Installation / Operation / Maintenance Manual			X	X	8 wks ARO

6. Supplier Inspection and Test Plan Requirements

The Seller shall prepare and submit an Inspection and Test Plan (ITP) detailing all fabrication, inspection, and testing activities required to demonstrate compliance with this specification and the approved drawings.

The ITP shall identify the sequence of inspection and test operations, the responsible party (Seller or Buyer), and the required inspection status (Witness, Hold, Review, or Surveillance).

Buyer's representative shall be notified of a minimum of five (5) working days before any Hold or Witness inspection point.

All inspection records and test reports shall be compiled into the final Record Book / Manufacturer's Data Book submitted with the shipment.

- H – Hold Point: Work shall not proceed without Buyer's written release.
- W – Witness Point: Activity may proceed after inspection if Buyer is present or waives attendance.
- R – Review: Documents submitted for desk review after completion of the activity.
- S – Surveillance: Buyer may observe at their discretion without affecting schedule.

Item	Component / Activity	Seller's Inspection Points	Buyer's Inspection Points	Remarks / Applicable Documents
A – Materials & Preparation				
A-1	Material Receiving and Identification	In-Process Check	Review of MTRs	Verify traceability and specification compliance
A-2	Mechanical / Chemical Testing of Base Materials	Witness	Random Surveillance	Test certificates to ASTM/ASME requirements
A-3	Welding Procedure and Welder Qualification	Witness	Review	Submit WPS/PQR and Qualification Records
B – Fabrication Activities				
B-1	Fit-Up and Welding Inspection	In-Process	Random Surveillance	Per approved WPS and drawings
B-2	Non-Destructive Examination (NDE) – Visual / MT / PT	Witness	Review of Reports	Performed per ASME Sec. V

B-3	Dimensional Check of Assembly	Witness	Random Surveillance	Confirm critical dimensions and mounting interfaces
B-4	Surface Preparation and Coating	In-Process Check	Final Review	Per approved coating procedure
C – Testing and Functional Verification				
C-1	Hydrostatic Test of Pressure Components	Witness	Hold Point	As per approved test procedure
C-2	Leak Test / Pressure Retention Check	Witness	Review	After Hydrotest – no leak permitted
C-3	Function Test of Counter-Balance Mechanism	Witness	Random Surveillance	Confirm smooth operation and seal integrity
C-4	Emergency Release Coupling (ERC) Functional Test	Witness	Random Surveillance	Demonstrate safe separation and re-engagement
D – Final Inspection and Documentation				
D-1	Final Visual Inspection and Marking Check	Witness	Review	Confirm tag numbers, orientation, and finish
D-2	Packing and Preservation Inspection	Witness	Review	Verify protection for 12-month outdoor storage
D-3	Record Book / Data Book Compilation	Review	Review	All test certificates and inspection records included
D-4	Final Release for Shipment	Hold Point	Hold Point	Joint sign-off required prior to dispatch

7. Recommended Spare Parts

The Seller shall provide a list of recommended spare parts for start-up, commissioning, and two years of operation. All parts shall be identical in design and quality to those used in the supplied equipment and shall be clearly identified with part number, description, and equipment tag.

Spare parts shall be properly packaged, labeled, and protected for long-term storage and safe transport.

Labels shall include equipment tag number, PO number, and manufacturer part reference.

A spare-part list and corresponding data sheets shall be submitted for Buyer's review and approval before shipment.

8. Exception

The Seller is required to provide list of exceptions, clarifications, and recommendations on a separate section.

Document Title:	Loading Arms Bid Tabulation Sheet - Carbis
Document Number:	LP1Y-255-PRO-K215-02
Security Class:	Restricted

Loading Arms Bid Tabulation Sheet - Carbis

Lilly - Lebanon Plant 1 (B1LG)

K215 - Lilly Rail Offloading Rack .

Hanson Project Number: 25i0007

A1 - DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

Authorization to proceed does not relieve Contractor/Supplier of its responsibility or liability under the Contract and or Purchase Order.

By: Ricardo DaCosta dac54131 April 13, 2026

A	IFR	MW	QR	SS	3/25/2026
Revision	**Issue Status**	**Originator**	**Reviewer**	**Approver**	**Date** .

HANSON

Document Number: LP1Y-255-PRO-K215-02
Revision: A
Revision Date: 2026-03-25

Tag Nos	Description	Vendor		Hanson
		Confirm (Yes/No)	Notes	Acceptable (Yes/No)
LA-0622A & LA-0632A	Type: 4" x 3" Supported Boom Top Unloading Arm - Downfeed (x 2)	Yes		Yes
	Wetted Path -316L SS	Yes		Yes
	Swivel Modules Non Wetted - 316L SS	Yes		Yes
	Dimensions (List dimensions quoted in **Notes** column)			
	Boom : 78" Minimum 4" 40S Pipe, 316L SS	Yes	84"	Yes
	Primary: 78" Minimum 3" 40S Pipe, 316L Ss	Yes	96"	Yes
	Drop: 60" Minimum 3" 316L SS Metal Hose with single Braid	Yes	60"	Yes
	Inlet Flange to Pillow Block: 84" Maximum	Yes	60" (Assumption)	Yes
	Spring Balance - Manual Operation	Yes		Yes
	Metal Hose End Connection to Rail Car 3" Dip Tube Valve			
	3" 316L SS Ball Valve at End of Metal Hose	Yes		Yes
	3" 316L SS Breakaway Coupling	Yes		Yes
	3" 316L SS Coupler with Dust Plug	Yes		Yes
	Seal Material: Chemraz SD517	Yes		Yes
	1" Piggy Back Nitrogen Hose	Yes		Yes
	Hose Material Tube PFA Core, Silicone Cover with 302 SS Braid	Yes		Yes
	Hose End Connection to Rail Car 2" Vent Valve			
	1" 316L SS Ball Valve	Yes	**1"** ball valve	Yes
	1" 316L SS Breakaway Coupling	Yes		Yes
	1" 316L SS Coupler with Dust Plug	No	2" Coupler	Yes
	Seal Material: Chemraz SD517	Yes		Yes
LA-0622A DMF Unloading	Viscosity: 0.87 cP S.G.: 0.94 Temperature (Oper./Design): 15 / 65 C Pressure (Oper./Design): 40/150 psig Flowrate (Oper./Design): 150/180 gpm			
	Unloading Arm Pressure Drop Off Estimate by Vendor based on above	Yes		Yes
LA-0632A ACN Unloading	Viscosity: 0.00419 cP S.G.: 0.94 Temperature (Oper./Design): 15 / 65 C Pressure (Oper./Design): 40/150 psig Flowrate (Oper./Design): 150/180 gpm			
	Unloading Arm Pressure Drop Off Estimate by Vendor based on above	Yes		Yes

HANSON

Document Number: LP1Y-255-PRO-K215-02
Revision: A
Revision Date: 2026-03-25

Tag Nos	Description	Vendor		Hanson
		Confirm (Yes/No)	Notes	Acceptable (Yes/No)
LA-0631A	Type: 4" x 3" Supported Boom Top Vapor Return Arm -Downfeed	Yes		Yes
	Wetted Path -316L SS	Yes		Yes
	Swivel Modules Non Wetted - 316L SS	Yes		Yes
	Dimensions (List dimensions quoted in **Notes** column)			
	Boom : 78" Minimum 4" 40S Pipe, 316L SS	Yes	84"	Yes
	Primary: 78" Minimum 3" 40S Pipe, 316L Ss	Yes	96"	Yes
	Drop: 60" Minimum 3" 316L SS Metal Hose with single Braid	Yes	60"	Yes
	Inlet Flange to Pillow Block: 84" Maximum	Yes	60" (Assumption)	Yes
	Spring Balance - Manual Operation	Yes		Yes
	Metal Hose End Connection to Rail Car 2" Vent			
	2" 316L SS Ball Valve at End of Metal Hose	No	3" ball valve prior to 2" reducer	Yes
	2" 316L SS Breakaway Coupling	No	3" breakaway coupling prior to 2" reducer	Yes
	2" 316L SS Coupler with Dust Plug	Yes		Yes
	Seal Material: Chemraz SD517	Yes		Yes

HANSON

Document Number: LP1Y-255-PRO-K215-02
Revision: A
Revision Date: 2026-03-25

Tag Nos	Description	Vendor		Hanson
		Confirm (Yes/No)	Notes	Acceptable (Yes/No)
LA-0640A & LA-0650A	Type: 4" x 3" Supported Boom Top Loading Arm - Downfeed (x 2)	Yes		Yes
	Wetted Path -Duplex Steel S32205-W	Yes		Yes
	Swivel Modules Non Wetted - 316L SS	Yes		Yes
	Dimensions (List dimensions quoted in **Notes** column)			
	Boom : 78" Minimum 4" 40S Pipe, Duplex Stl S32205-W	Yes	84"	Yes
	Primary: 78" Minimum 3" 40S Pipe, Duplex Stl S32205-W	Yes	96"	Yes
	Drop: 60" Minimum 3" Hastelloy Braided Hose	Yes	60"	Yes
	Inlet Flange to Pillow Block: 84" Maximum	Yes	60" (Assumption)	Yes
	Spring Balance - Manual Operation	Yes		Yes
	Metal Hose End Connection to Rail Car 3" Dip Tube Valve			
	3" Duplex S32205-W Ball Valve at End of Metal Hose	Yes		Yes
	3" Hastelloy Breakaway Coupling	Yes		Yes
	3" Hastelloy Coupler with Dust Plug	Yes	3" **Hastelloy** cam and groove coupler	Yes
	Seal Material: Kalrez 0040	Yes		Yes
	Heat Tracing and Insulation Required - By Others?	Yes	By Others	Yes
	Heat Tracing and Insulation Weight Accounted for in Spring Balance?	Yes	We will need to know lbs. per ft. and total weight being added to the arm to properly calculate the load being added for spring set up and stress analysis	Yes
	2" Piggy Back Vent Metal Hose	Yes		Yes
	Hose Material Hastelloy	Yes		Yes
	Hose End Connection to Rail Car 2" Vent Valve	Yes		Yes
	2" Duplex S32205-W Ball Valve at End of Metal Hose	Yes		Yes
	2" Hastelloy Breakaway Coupling	Yes		Yes
	2" Hastelloy Coupler with Dust Plug	Yes	2" **Hastelloy** cam and groove coupler	Yes
	Seal Material: Kalrez 0040	Yes		Yes
LA-0640A & LA-0650A Waste Loading	Viscosity: 1.12 cP S.G.: 1 Temperature (Oper./Design): 15 / 53 C Pressure (Oper./Design): 15/150 psig Flowrate (Oper./Design): 350/394 gpm			
	Unloading Arm Pressure Drop Off Estimate by Vendor based on above	Yes		Yes

HANSON

Document Number: LP1Y-255-PRO-K215-02
Revision: A
Revision Date: 2026-03-25

Tag Nos	Description	Vendor		Hanson
		Confirm (Yes/No)	Notes	Acceptable (Yes/No)
NDE For All Arms	Inspection and Test Plan	**Yes**		Yes
	WPS, PQR, WPQ and Weld Map Submittal	Yes		Yes
	PMI Report	Yes		Yes
	Hydro Report with Certification	Yes		Yes
	5% X-Ray and 100% Visual with Report	Yes		Yes
	Full Certified Drawing Set with Weights and Support Requirements	Yes		Yes
	Calculated pressure drops per operation and design flows indicated	Yes		Yes

Document Title:	Request For Quotation (RFQ) for Gangway
Document Number:	LP1Y-255-PRO-K215-03
Security Classification:	Restricted

Request For Quotation (RFQ) for Gangway

Eli Lilly Lebanon Plant 1 (B1LG)

K215 - Rail Unloading Rack

Hanson Project Number:25I0007.00

A1 - DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

Authorization to proceed does not relieve Contractor/Supplier of its responsibility or liability under the Contract and or Purchase Order.

By: Ricardo DaCosta dac54131 December 1, 2025

Revision	Issue Status	Originator	Reviewer	Approver	Date
B	Issued for Quotation	SS	QR	MW	2025-11-10
A	Issued for Review	SS	QR	MW	2025-10-19

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Document Title:	Request For Quotation (RFQ) for Gangway		
Document Number:	LP1Y-255-PRO-K215-03		
Revision:	B	Revision Date:	11/10/2025

1. Project Summary

The equipment listed under this RFQ forms part of the LP1 Project for Eli Lilly at their plant located in Lebanon, Indiana, USA.

Fluor serves as the overall EPC Contractor for the complete project scope.

Hanson Professional Services Inc. (hereafter referred to as Hanson) is the engineering partner responsible for the detailed design of the rail station area, while Standard Constructors Inc. (hereafter referred to as SCI) is responsible for procurement and construction of the rail station area.

Technical Contact point:

- To: Saeed Sabah – Hanson's Project Manager – email: ssabah@hanson-inc.com
- cc: Jamie Reingold – SCI's Project Mgr – email: reingold@standardconstructors.com

Commercial Contact point:

- To: Jamie Reingold – SCI's Project Mgr – email: reingold@standardconstructors.com

2. Scope of Supply

The Seller shall design, manufacture, test, and supply complete spring-balanced aluminum gangway assemblies with safety cages and mounting hardware suitable for installation on the K215 Rail Unloading Rack in accordance with the following scope:

#	Item Tag no.	Item Description	Qty
1	All	Seller is responsible for the complete design, engineering, design authentication (Professional Engineering Stamp, where required), supply of materials, fabrication, shop inspection, testing, documentation, and preparation for shipment in accordance with this material requisition for the following equipment: Quantities, dimensions, and lengths where provided are used for planning & estimating purposes only. Final design quantities are to be determined by the seller.	
1.1	N/A	Railcar Access Gangway Assembly (including safety cage, swing gate, and mounting base)	4 set
2	All	Seller scope also includes the following for item 1.1 - Aluminum gangway body with self-leveling stairs and handrails - Hot-dip galvanized subframe and safety cage - Spring-balanced counterweight system (no pneumatic assist) - Bolted base-plate connection with drilling template - Adjustable chain stops or equivalent restraint system - Two grounding lugs per station - Factory assembly, inspection, and function test prior to shipment	

		- Shop finishing, packaging, and corrosion protection for 12-month outdoor storage - Assembly and installation instructions, recommended spare parts, and maintenance manual	
3	All	Start-up and Commissioning Spares for each gangway - Seller's recommended spare parts.	1 set per gangway
4	All	Recommended Spares for Two Years of operation for each gangway - Seller's recommended spare parts.	1 set per gangway
5	All	Capital Spares - Seller's recommended spare parts	1 set per gangway

2.1. Excluded (by Buyer/others)

- Supporting rack structure, foundations
- Loading arms, piping, and vapor/vent systems
- Electrical, pneumatic, or control systems
- Field installation, unless specifically requested
- Painting or coating beyond factory-supplied aluminum and galvanized finishes.

3. List of Documents

#	Document Number	Document Description
1	LP1Y-255-DST-K215-05 _ Rev A	Mechanical Data Sheets – Gangways
2	LP1Y-SF-00000250_Rev A	Structural Drawing – General Arrangement Plan
3	LP1Y-SF-00000253_Rev A	Structural Drawing – Platform Framing Plan
4	LP1Y-SF-00000901_Rev A	Structural Drawing – Unloading Rack Sections
5	LP100-215-EMS-05002	Specification – Structural Data for Mechanical Equipment

4. Technical Requirements

4.1. The Seller shall have full responsibility for the design, fabrication, assembly, testing, and documentation of the supplied gangway system, including all mechanical, structural, and safety components.

4.2. The gangway and safety cage shall provide safe, ergonomic, and reliable access for operators to reach the top of railcars for loading or unloading operations. All supplied equipment shall be factory-assembled and tested prior to shipment.

4.3. Gangways shall comply with applicable OSHA 29 CFR 1910.23 and API RP 2003 for access and fall-protection systems.

4.4. Structural materials and welds shall conform to AWS D1.1/D1.2 and ASTM B221 (for aluminum) or ASTM A36 / A123 (for galvanized steel) as applicable.

4.5. Gangway body and handrails shall be constructed of aluminum for corrosion resistance and ease of operation.

4.6. Folding stair and cage subframes shall be hot-dip galvanized steel.

4.7. Access to the top of railcars shall be via a minimum 2.5 ft wide adjustable gangway connected to the rack platform.

4.8. The gangway shall be spring-balanced, self-adjusting, and operable without external pneumatic or assist.

4.9. The gangway shall be self-leveling, maintaining a horizontal walking surface for railcars of varying heights, including DOT-117

4.10. The length shall be sufficient to bridge between the platform and the railcar’s gap without gaps; if required, the design shall allow adjustable reach to accommodate different railcar configurations.

4.11. The gangway shall provide inherent fall protection and remain stable when fully retracted or deployed.

4.12. The entrance shall include an inward-opening swing gate for safe operator entry.

4.13. Each gangway shall be equipped with a fall-protection safety cage, bolted to the supporting ramp.

4.14. The cage shall be designed and sized to minimize open gaps between the railcar and cage perimeter.

4.15. The safety cage shall be self-leveling, adjustable, and rated for a 200 lb. top-rail concentrated load per OSHA requirements.

4.16. Cage and gangway design shall allow quick replacement or adjustment for future variations in railcar configuration.

4.17. The gangway, safety cage, and loading arm shall be designed in coordination to ensure safe, ergonomic operation and sufficient clearance through the defined working envelope.

4.18. Supplier shall confirm the reach and clearance geometry and provide dimensional drawings for Buyer review.

4. 19. Each gangway shall have a permanently affixed stainless-steel nameplate indicating:

- Manufacturer name and year of manufacture
- Model or serial number
- Maximum working load
- Purchase order number

4.20. Seller shall provide:

- Dimensional and general-arrangement drawings
- Assembly and installation instructions
- Recommended spare parts list
- Maintenance and lubrication manual

4.21. All assemblies and subcomponents shall be protected against mechanical damage and corrosion for at least 12 months of outdoor storage.

4.22. Loose parts shall be tagged with service, tag number, and purchase order reference.

4.23. Aluminum components shall be packaged to prevent contact with carbon-steel parts during transport and storage.

4.24. Two grounding lugs shall be provided at opposing corners at the bottom of the station steel.

4.25. Seller shall provide all interface and loading information required for Buyer to design the supporting structure.

4.26. Seller shall submit detailed base-plate and anchor drawings showing bolt pattern, spacing, hole diameter, plate thickness, and elevation.

4.27. Reaction loads shall be specified, including vertical, horizontal, and moment loads for both operating and stored positions.

4.28. Seller shall state the total installed weight and center of gravity of the complete gangway and cage assembly.

4.29. Any uplift or side-load forces resulting from gangway motion shall be clearly identified.

4.30. Seller shall furnish a working-envelope diagram showing the full range of motion of the gangway and cage relative to the platform edge and railcar.

4.31. A 3D CAD or Revit model of the gangway assembly, including base interface and motion limits, shall be provided for integration into the overall rack model.

5. Supplier Document & Data Control (SDDC)

The Seller shall provide all engineering, fabrication, testing, and quality documentation necessary to demonstrate compliance with this Technical RFQ, the applicable codes and standards, and the approved data sheets.

All submittals shall be issued electronically in PDF format for review and approval. Native editable files (e.g., AutoCAD, Excel, Word) shall be provided for all drawings, data sheets, and lists at the time of final/as-built issue.

Each document shall include, at a minimum, the Project Name, Equipment Tag Number(s), Purchase Order Number, Revision Number, and Date of Issue.

All submittals shall be transmitted through the Buyer's document control system or by email as instructed at time of award.

All dates are referenced After Receipt of Order (ARO) unless otherwise stated.

All drawings and 3D models shall show interface points, foundation bolt layouts, swing envelopes, and clearances relevant to installation and maintenance.

Documentation must demonstrate compliance with the applicable design standards.

The final Record Book shall include WPS/PQRs, MTRs, Inspection Reports, Test Certificates, Photographic Records (where available), and As-Built Drawings.

Seller shall maintain all quality and inspection documentation for a minimum of five (5) years from shipment.

Item	Document Description	With Proposal	After Order	For Review / Approval	Final / As-Built	Include with Shipment / Record Book / Due Date
A – Commercial Data						
A-1	Seller's List of Documents	X				2 wks ARO
A-2	Procurement / Engineering / Production Schedule			X		4 wks ARO
A-3	Monthly Progress Report					Monthly

A-4	Unpriced Sub-Vendor PO Confirmations					As Completed
A-5	Packing List			X		4 wks ARO
B – Drawings & Engineering Data						
B-1	Dimensional Outline Drawing & BOM, weight	X		X	X	2 wks ARO
B-2	General Arrangement / Assembly Drawing			X	X	4 wks ARO
C – Procedures & Calculations						
C-1	WPS / PQR / Welder Qualifications			X	X	Before Fabrication
D – Quality & Inspection Documents						
D-1	QA/QC Program Registration Certificate			X		6 wks ARO
D-2	Inspection & Test Plan (ITP)			X	X	6 wks ARO
D-3	Mill Test Reports / Material Traceability					At Completion
D-4	Non-Conformance Reports & Resolutions					As Completed
E – Test & Certification Reports						
E-1	Dimensional Check Report					At Completion
F – Compliance Certificates						
F-1	Code / Standard Compliance Certificate					Prior to Shipment
F-2	Certificate of Compliance with Specifications					Prior to Shipment
G – Manuals & Record Books						
G-1	Record Book Index					8 wks ARO
G-2	Record Book (Full PDF + Native Files)					2 wks After Shipment
G-3	Installation / Operation / Maintenance Manual			X	X	8 wks ARO

6. Supplier Inspection and Test Plan Requirements

The Seller shall prepare and submit an Inspection and Test Plan (ITP) detailing all fabrication, inspection, and testing activities required to demonstrate compliance with this specification and the approved drawings.

The ITP shall identify the sequence of inspection and test operations, the responsible party (Seller or Buyer), and the required inspection status (Witness, Hold, Review, or Surveillance).

Buyer's representative shall be notified a minimum of five (5) working days before any Hold or Witness inspection point.

All inspection records and test reports shall be compiled into the final Record Book / Manufacturer's Data Book submitted with the shipment.

- H – Hold Point: Work shall not proceed without Buyer's written release.
- W – Witness Point: Activity may proceed after inspection if Buyer is present or waives attendance.
- R – Review: Documents submitted for desk review after completion of the activity.
- S – Surveillance: Buyer may observe at their discretion without affecting schedule.

Item	Component / Activity	Seller's Inspection Points	Buyer's Inspection Points	Remarks / Applicable Documents
A – Materials & Preparation				
A-1	Material Receiving and Identification	In-Process Check	Review of MTRs	Verify traceability and specification compliance
A-2	Mechanical / Chemical Testing of Base Materials	Witness	Random Surveillance	Test certificates to ASTM/ASME requirements
A-3	Welding Procedure and Welder Qualification	Witness	Review	Submit WPS/PQR and Qualification Records
B – Fabrication Activities				
B-1	Fit-Up and Welding Inspection	In-Process	Random Surveillance	Per approved WPS and drawings
B-2	Non-Destructive Examination (NDE) – Visual / MT / PT	Witness	Review of Reports	
B-3	Dimensional Check of Assembly	Witness	Random Surveillance	Confirm critical dimensions and mounting interfaces
B-4	Surface Preparation and Coating	In-Process Check	Final Review	Per approved coating procedure
C – Testing and Functional Verification				
C-1	Function Test of Counter-Balance Mechanism	Witness	Random Surveillance	Confirm smooth operation through full range of motion

D – Final Inspection and Documentation				
D-1	Final Visual Inspection and Marking Check	Witness	Review	Confirm tag numbers, orientation, and finish
D-2	Packing and Preservation Inspection	Witness	Review	Verify protection for 12-month outdoor storage
D-3	Record Book / Data Book Compilation	Review	Review	All test certificates and inspection records included
D-4	Final Release for Shipment	Hold Point	Hold Point	Joint sign-off required prior to dispatch

7. Recommended Spare Parts

The Seller shall provide a list of recommended spare parts for start-up, commissioning, and two years of operation. Parts shall include replacement springs, pins, bushings, bearings, locking mechanisms, safety gate latches, and hardware kits.

Spare parts shall be properly packaged, labeled, and protected for long-term storage and safe transport.

Labels shall include equipment tag number, PO number, and manufacturer part reference.

A spare-part list and corresponding data sheets shall be submitted for Buyer's review and approval before shipment.

8. Exception

The Seller is required to provide list of exceptions, clarifications, and recommendations on a separate section.

Document Title:	Technical Bid Evaluation - Gangways
Document Number:	LP1Y-255-PRO-K215-04
Security Class:	Restricted

Technical Bid Evaluation - Gangways

Lilly - Lebanon Plant 1 (B1LG)

K215 - Lilly Rail Offloading Rack

Hanson Project Number: 25i0007

0	IFI	QR	SS	MW	12/4/2025
Revision	**Issue Status**	**Originator**	**Reviewer**	**Approver**	**Date**

TECHNICAL BID EVALUATION

	Equipment: Railcar Access Gangway Assembly				**Project No.: 25i0007**					**Document No.: LP1Y-255-PRO-K215-04**		
	RFQ No.: LP1Y-255-PRO-K215-03				**Project: K215 - Lilly Rail Offloading Rack**					**Revision No.: 0**		
	Tag No.: N/A				**Client: Lilly - Lebanon Plant 1 (B1LG)**					**Date:** 12/4/2025		
Technical Description (List Tendered Requirements)	**Specification**	**Bidder: Bullard** **Bid No: PQ 10-32-25** **Bid Date: 10/27/25**	Designation	Cost Adder	**Bidder: S&S Technical** **Bid No: 774-25** **Bid Date: 11/24/25**	Designation	Cost Adder	**Bidder: SafeRack** **Bid No: SAS-195396R3** **Bid Date: 11/20/25**	Designation	Cost Adder		Designation	Cost Adder
Legend: Y=Complies with Specification N=Not Acceptable A=Acceptable Alternative INA=Information Not Available N/A=Not Applicable VTS=Vendor to Specify													
1. GENERAL													
Vendor Proposal Number		PQ 10-32-25			774-25			SAS-195396R3					
Vendor Proposal Current Revision Number / Date		10/27/2025			11/24/2025			11/20/2025					
Fabrication Facility Location		7080 Schnipke Drive Ottawa Lake, Michigan 49267			1900 Grassland Pakway Alpharatta, GA 30004			219 Safety Ave Andres, SC 29510					
2. DESIGN													
Item 1: Railcar Access Gangway Assembly													
General								G4 Series FRT					
Quantity	4	4	Y		4	Y		4	Y				
Min / Max Temperature Deg C	-27.5 to 45		INA			INA			INA				
Wind Load mph	120		INA			INA			INA				
Hazardous Area Classification	Class 1, Division 2		INA			INA			INA				
Max Required Rail Clearance ft	17' - 6"												
General Gangway Width ft	2.5 (min)	3	Y		3	Y		3					
General Gangway Length ft	VTA	6'11 3/8" when flush mounted and positioned horizontal; 8'6 3/8" fully extended.	Y		22" TELESCOPING EXTENTION	INA		TELESCOPING EXTENTION	INA				
Gangway Upright - Bottom of Gangway to TOR (min) ft	22' - 4"	23'10 1/4" (slightly encroaches on CSX envelope)	A		Unclear without GA drawing	INA		Unclear without GA drawing	INA				
Gangway Upright - Bottom of Gangway to TOR (max) ft	23' - 6"	23'10 1/4" (slightly encroaches on CSX envelope)	A		Unclear without GA drawing	INA		Unclear without GA drawing	INA				
Gangway Type	Spring Balanced Self-Adjusting	Spring Balanced Self-Adjusting	Y		Spring Balanced Self-Adjusting	Y		Spring Balanced Self-Adjusting	Y				
Structural Steel Design & Materials of Construction	Aluminum	Aluminum	Y		Aluminum	Y		Aluminum	Y				
Guardrail Material of Construction	Aluminum	Aluminum	Y		Aluminum	Y		Stainless Steel	Y				
Grounding Connections	Two Ground Lugs at opposing corners at bottom of the station steel	NOT INCLUDED	N			INA			INA				
Base Tread	Hot Dipped Galvanized per ASTM A123	Stainless Steel	A		Hot Dipped Galvanized	Y		Included	Y				
Stair Type	Self Leveling Stairs	Self Leveling OR Flat Ramp (both options provided)	Y		Flat Ramp	A		Flat Ramp	A				
Maneuverability	Self-balanced and operable without need for N2 or pneumatic sources.	Spring counterbalanced system	Y		Spring counterbalanced system	Y		Spring counterbalanced system	Y				
Total Load Capacity lbs	Total: 500, Handrail: 200, Midrail: 100	Working load limit: 500lb Handrails: 200lb	Y		500 lb load capacity	Y		Working load limit: 500lb Handrails: 200lb	Y				
Safety Cage													
Safety Cage	Self Leveling Safety Hoop with 200 lb top rail rating	Included	Y		Included	INA		Included	INA				
Safety Cage Dimensions	VTA	7'0" long x 8'0" wide x 1'0" high	Y		8'0" long x 10'0" wide x 1'0" high (2 Rail High)	Y		8'0" long x 8'0" wide x 1'0" high (2 Rail High)	Y				
Safety Cage Material	Aluminum	Aluminum	Y		Aluminum			Aluminum					
Available Working Range	+/- 15 degree to horizontal	Included	Y			INA			INA				
Handrail Dimensions, Size in	2.9" Diameter Handrail w/ 1" Diameter Grab Bar	1.5" tubing meets OSHA diameter specifications	A			INA		1.5" tubing meets OSHA diameter specifications	A				
Swing Gate													
Entrance Requirement	Swing Gate - Opens Inward	Swing gate included, open direction unclear	INA		Swing gate included, open direction unclear	INA		Included	Y				
Other	VTA	Sliding Track & Swing Gates; Spring loaded gates at 3' increments; Carbon steel construction	A		N/A			N/A					
Mounting Base													
Mounting Base Connection	Bolted base-plate connection with drilling template	Galvanized Steel base with Grade 5 bolts with nylon lock nuts	Y		Hot Dipped Galvanized	Y		Included	Y				
Support Method	Self Supporting by use of Adjustable Chain Stops	Included	Y		Included	Y			INA				
Accessories	- Spring Bearing Pivot Points Contain Oil Impregnated Bronze Bearings												
Operating Load	VTA	INITIAL PULL TO RISE <15-20 LBS	Y										
Other	VTA	Locking foot latch			Locking foot latch				INA				

TECHNICAL BID EVALUATION

	Equipment: Railcar Access Gangway Assembly				**Project No.: 25i0007**					**Document No.: LP1Y-255-PRO-K215-04**							
	RFQ No.: LP1Y-255-PRO-K215-03				**Project: K215 - Lilly Rail Offloading Rack**					**Revision No.: 0**							
	Tag No.: N/A				**Client: Lilly - Lebanon Plant 1 (B1LG)**					**Date:** 12/4/2025							
Technical Description (List Tendered Requirements)	**Specification**	**Bidder: Bullard** **Bid No: PQ 10-32-25** **Bid Date: 10/27/25**	Designation	Cost Adder	**Bidder: S&S Technical** **Bid No: 774-25** **Bid Date: 11/24/25**	Designation	Cost Adder	**Bidder: SafeRack** **Bid No: SAS-195396R3** **Bid Date: 11/20/25**	Designation	Cost Adder		Designation	Cost Adder
4. NON-DESTRUCTIVE TESTING & INSPECTION REQUIREMENTS														
Vibration Testing														
PMI														
NDE		Included	Y			INA			INA					
Running test														
Function test	Included	NOT INCLUDED	N		NOT INCLUDED	N			INA					
Hydrotest														
5. SHIPPING:														
Shipping Weight (lbs)	By Vendor	Included	Y		Included	Y			INA					
Shipping Outside Dimensions (ft)	By Vendor													
Shipping Drawings	Yes													
Shipment Packing	Yes	Included	Y						INA					
Other					Buyer responsible for arranging transportation to required location.									
6. ADDITIONAL REQUIREMENTS														
Assemblies & Subcomponents Requirements & Warranty	- Factory assembly, inspection, and function test prior to shipment - Shop finishing, packaging, and corrosion protection for 12-month outdoor storage - Assembly and installation instructions, recommended spare parts, and maintenance manual		INA			INA			INA					
Reaction Loads														
Vertical, Horizontal, and Moment Loads - Operating Position	VTA		INA			INA			INA					
Vertical, Horizontal, and Moment Loads - Stored Position	VTA		INA			INA			INA					
Total Installed Weight of Gangway Assembly lbs	Included	1129 (OR 563 less the slide track assembly)	Y		375				INA					
Center of Gravity of Gangway Assembly	Included		INA			INA			INA					
Working Envelope Diagram - Full Range of Motion	Included	Included	Y			INA			INA					
CAD or Revit Model of Gangway Assembly	Included; base interface and motion limits	Included	Y		To be provided	INA			INA					
Start-up and Commissioning Spares for each gangway	Seller's Recommended Spare Parts		INA		To be quoted after approved design									
Recommended Spares for Two Years Operation for each gangway	Seller's Recommended Spare Parts		INA		To be quoted after approved design									
Capital Spares	Seller's Recommended Spare Parts		INA		To be quoted after approved design									
Equipment Documentation	Match SDDC or provide mark up of exclusions.		INA		Per specification; Drawings consist of P&ID and GA CAD drawings. Seller will provide an initial set of P&ID and GA drawings, submitted for customer approval									
7. BONUS / MALUS														
Delivery Weeks : (week)		Bid includes schedule for construction of structural steel rack.			6 weeks (After receipt of approved drawings)									
Delivery Terms:		Progressive payments; refer to proposal; Generally Net 30 Days			Net 30 Days			Net 30 Days						
Bid Validity		Material Cost: 15 days Bid validity 30 days			30 days			41 days						

TECHNICAL BID EVALUATION

	Equipment: Railcar Access Gangway Assembly				**Project No.: 25i0007**					**Document No.: LP1Y-255-PRO-K215-04**			
	RFQ No.: LP1Y-255-PRO-K215-03				**Project: K215 - Lilly Rail Offloading Rack**					**Revision No.: 0**			
	Tag No.: N/A				**Client: Lilly - Lebanon Plant 1 (B1LG)**					**Date:** 12/4/2025			
Technical Description (List Tendered Requirements)	**Specification**	**Bidder: Bullard** **Bid No: PQ 10-32-25** **Bid Date: 10/27/25**	Designation	Cost Adder	**Bidder: S&S Technical** **Bid No: 774-25** **Bid Date: 11/24/25**	Designation	Cost Adder	**Bidder: SafeRack** **Bid No: SAS-195396R3** **Bid Date: 11/20/25**	Designation	Cost Adder		Designation	Cost Adder
9. SUMMARY													
Exception													
Technically Acceptable	Yes or No	Yes											
Comments	As needed	**Remarks/Cost Adders:** **- Bullard's quote mostly meets the requested intent. Clarifications required: Do the safety gates open inward, will these be factory assembled and delivered to site, what are the design conditions? Can the self-leveling stair option be raised any higher to remain outside the CSX envelope for the hoop?** **- Grounding lugs are not included.** **- Some items are unclear due to the nature of inclusion with the structural steel erection portion of the proposal.** **- If Bullard is selected, these can be seamlessly integrated into the procurement of the structural steel for construction.** **- Spare parts were not included in this proposal, nor a list of major documentation to be provided with purchase.** **- Not signed off by a licensed engineer**			**Remarks/Cost Adders:** **- As there is not a GA drawing provided, it is difficult to confirm if the gangway will meet the desired intent. However, S&S did provide a photo layout view portraying the correct / adequate placement of the gangway.** **- There is no information on warranties, but that likely won't be provided until the vendor is selected and confirmed on the commerical and technical requirements prior to purchase.** **- Additional information required in some areas for clarification / confirmation if purchase is pursued.**			**Remarks/Cost Adders:** **- SafeRack did not provide much information. About half of the above was found by searching applicable catalogue's on their website.** **- A GA drawing was provided showing the stowed position, but the dimensions were not as clear as required. However, this generally shows it should meet the clearnace requirements with a potential for permit modification with CSX clearance envelope.** **- Additional information required in some areas for clarification / confirmation if purchase is pursued.**			**Remarks/Cost Adders:** -		
PREPARED BY: (HANSON) Quade Robertson	**DATE:** 12/2/2025												
REVIEWED BY: (HANSON) Saeed Sabah	**DATE:** 12/4/2025												
APPROVED BY: (HANSON) Mike Walther	**DATE:** 12/4/2025												
PROJ. APPROVAL:	**DATE:**	**RECOMMENDATION:** Based on the information provided by all three bidders, Bullard provided the most clarity, but the other two are also technically acceptable based on desired function. For ease of constructability, Bullard and S&S Technical are quoting on providing the structure as well. As such, it is recommended to go with either of those two on that benefit.											
CLIENT APPROVAL:	**DATE:**												

Key

Y	Meets Specification
N	Does Not Meet Specification
INA:	Information Not Available
N/A:	Not Applicable
VTS:	Vendor to Specify
A:	Acceptable Alternative
	Potential Cost Adder
Text	Critical Item / Cost Impact

Document Title:	Request For Quotation (RFQ) for Safety Shower and Eyewash Assembly
Document Number:	LP1Y-255-PRO-K215-05
Security Classification:	Restricted

A1 - DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT
Authorization to proceed does not relieve Contractor/Supplier of its responsibility or liability under the Contract and or Purchase Order.
By: Ricardo DaCosta dac54131 December 1, 2025

Request For Quotation (RFQ) for Safety Shower and Eyewash Assembly

Eli Lilly Lebanon Plant 1 (B1LG)

K215 - Rail Unloading Rack

Hanson Project Number:25I0007.00

Revision	Issue Status	Originator	Reviewer	Approver	Date
B	Issued for Quotation	SS	QR	MW	2025-11-10
A	Issued for Review	SS	QR	MW	2025-10-29

Document Title:	Request For Quotation (RFQ) for Safety Shower and Eyewash Assembly		
Document Number:	LP1Y-255-PRO-K215-05		
Revision:	B	Revision Date:	11/10/2025

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Document Title:	Request For Quotation (RFQ) for Safety Shower and Eyewash Assembly		
Document Number:	LP1Y-255-PRO-K215-05		
Revision:	B	Revision Date:	11/10/2025

1. Project Summary

The equipment listed under this RFQ forms part of the LP1 Project for Eli Lilly at their plant located in Lebanon, Indiana, USA.

Fluor serves as the overall EPC Contractor for the complete project scope.

Hanson Professional Services Inc. (hereafter referred to as Hanson) is the engineering partner responsible for the detailed design of the rail station area, while Standard Constructors Inc. (hereafter referred to as SCI) is responsible for procurement and construction of the rail station area.

Technical Contact point:

- To: Saeed Sabah – Hanson's Project Manager – email: ssabah@hanson-inc.com
- cc: Jamie Reingold – SCI's Project Mgr – email: reingold@standardconstructors.com

Commercial Contact point:

- To: Jamie Reingold – SCI's Project Mgr – email: reingold@standardconstructors.com

2. Scope of Supply

The Seller shall furnish a complete, factory-assembled, freeze-protected, and heat-traced combination safety shower and eyewash assembly suitable for outdoor installation in a Class I, Div 2, Groups C & D, T4 hazardous location.

The assembly shall be based on Guardian Model GFR3500 or approved equivalent, meeting ANSI Z358.1.

#	Item Tag no.	Item Description	Qty
1	All	The Seller shall supply a complete, factory-assembled, freeze-protected, and heat-traced combination safety shower and eyewash assembly. The scope includes all materials, fabrication, shop inspection, testing, documentation, and preparation for shipment in accordance with this material requisition. Quantities, dimensions, and lengths where provided are used for planning & estimating purposes only. Final design quantities are to be determined by the seller.	
1.1	SSEW-15-0621-01	Combination Safety Shower and Eye Wash Assembly (freeze-resistant, heat-traced)	1 set
1.2	SSEW-15-0621-02	Combination Safety Shower and Eye Wash Assembly (freeze-resistant, heat-traced)	1 set
2	All	The Seller shall provide a fully factory-assembled and tested unit, complete with all valves, insulation, heat tracing, thermostatic control, and electrical alarm accessories as per the manufacturer's brochure - Two grounding lugs per station - Factory assembly, inspection, and function test prior to shipment	

		- Shop finishing, packaging, and corrosion protection for 12-month outdoor storage - Assembly and installation instructions, recommended spare parts, and maintenance manual	
3	All	Start-up and Commissioning Spares for each unit - Seller's recommended spare parts.	1 set per unit
4	All	Recommended Spares for Two Years operation for each unit - Seller's recommended spare parts.	1 set per unit
5	All	Capital Spares - Seller's recommended spare parts	1 set per unit

2.1. Included in the Seller's Scope

- Guardian GFR3500 or approved equivalent, including:
  - Stainless-steel frame and piping, self-draining design
  - Freeze-protected shower with self-closing valve
  - Heated eyewash station, integral drain, and flow regulator
  - Electrical heat trace, insulation, and weatherproof cladding
  - Explosion-proof alarm light and horn with flow-actuated switch
  - Thermostatic heater control (factory-wired junction box)
  - Identification signage and operation tag
  - Mounting base and anchor plate (drilling template included)
- Factory pressure test, function test, and electrical verification
- Shipment in weather-protected packaging suitable for 12-month outdoor storage
- Installation, operation, and maintenance (IOM) manual

2.2. Excluded (by Buyer/others)

- Civil foundations and embedded anchor bolts
- Field electrical wiring and conduit to power source

- Commissioning or field start-up services (optional only if requested)
- Supply of water piping beyond 1-inch NPT inlet connection.

3. List of Documents

#	Document Number	Document Description
1	LP1Y-255-DST-K215-06 _ Rev A	Mechanical Data Sheets – Safety Shower & Eye Wash
2	LP100-653-PHL-003	SAFETY SHOWER AND EYEWASH PHILOSOPHY
3	PCS-SSEW-002	Cut Sheet – Safety Shower / Eyewash
4	LP100-240-EMS-15401_R2	SAFETY SHOWER AND EYEWASH

4. Technical Requirements

4.1. The Seller shall have full responsibility for the design, fabrication, assembly, testing, and documentation of the supplied system.

4.2. Equipment shall conform to ANSI Z358.1 – 2014, OSHA 1910.151, and LP100-653-PHL-003.

4.3. Heat tracing – self-limiting electric type, complete with insulation and vapor-sealed jacket.

4.4. Enclosure and components shall be rated for Class I Div 2 Groups C & D T4.

4.5. Seller shall provide anchor bolt template, installation drawings, and total unit weight. The Seller shall also specify bolt size, pattern, and maximum installation loads to assist Buyer in foundation design.

4.6. Unit shall be labeled with manufacturer name, model number, serial number, electrical data, and compliance markings.

4.7. Unit shall be completely assembled and function-tested at the factory.

4.8. Drain ports shall be sealed for transport and clearly marked.

4.9. Seller shall ensure compatibility with outdoor environment: ambient −25 °F to +110 °F.

4.10. Proximity switches shall be terminated inside a junction box and ready for field connection, no pigtails allowed.

5. Supplier Document & Data Control (SDDC)

The Seller shall provide all engineering, fabrication, testing, and quality documentation necessary to demonstrate compliance with this Technical RFQ, the applicable codes and standards, and the approved data sheets.

All submittals shall be issued electronically in PDF format for review and approval. Native editable files (e.g., AutoCAD, Excel, Word) shall be provided for all drawings, data sheets, and lists at the time of final/as-built issue.

Each document shall include, at a minimum, the Project Name, Equipment Tag Number(s), Purchase Order Number, Revision Number, and Date of Issue.

All submittals shall be transmitted through the Buyer's document control system or by email as instructed at time of award.

All dates are referenced After Receipt of Order (ARO) unless otherwise stated.

All drawings and 3D models shall show interface points, foundation bolt layouts, swing envelopes, and clearances relevant to installation and maintenance.

Documentation must demonstrate compliance with the applicable design standards.

The final Record Book shall include WPS/PQRs, MTRs, Inspection Reports, Test Certificates, Photographic Records (where available), and As-Built Drawings.

Seller shall maintain all quality and inspection documentation for a minimum of five (5) years from shipment.

Item	Document Description	With Proposal	After Order	For Review / Approval	Final / As-Built	Include with Shipment / Record Book / Due Date
A-1	Product data sheet and catalog	X				
A-2	Dimensional outline and baseplate drawing		X	X	X	
A-3	Electrical wiring diagram		X	X	X	
A-4	Electrical Heat Tracing Load List		X	X	X	
A-5	Material list and weights		X	X	X	

A-6	Factory test certificate					
A-7	Installation, Operation & Maintenance manual			X	X	

6. Supplier Inspection and Test Plan Requirements

- All assemblies shall be factory pressure-tested to 1.5 × working pressure.
- Electrical circuits (heat trace and alarm) shall be continuity-tested and megger-tested before shipment.
- Final inspection shall confirm labeling, marking, and finish.
- Buyer reserves right to witness functional test upon request.

7. Recommended Spare Parts

Seller shall list recommended spare parts for two years' service, including:

- Shower and eyewash valves
- Flow actuators and O-rings
- Heater thermostat and heat-trace terminations
- Alarm lamp and horn assemblies.

8. Exception

The Seller is required to provide list of exceptions, clarifications, and recommendations on a separate section.

A1 - DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT
Authorization to proceed does not relieve Contractor/Supplier of its responsibility or liability under the Contract and or Purchase Order.
By: Ricardo DaCosta dac54131 January 5, 2026

No RE comment

HANSON

Document Title:	TBE - Safety Shower & Eye Wash
Document Number:	LP1Y-255-PRO-K215-06
Security Class:	Restricted

No HSE comment

TBE - Safety Shower & Eye Wash

Lilly - Lebanon Plant 1 (B1LG)

K215 - Lilly Rail Offloading Rack

Hanson Project Number: 25i0007

A	IFR	QR	SS	MW	12/18/2025
Revision	**Issue Status**	**Originator**	**Reviewer**	**Approver**	**Date**

HANSON

TECHNICAL BID EVALUATION

Equipment: Safety Shower & Eye Wash Station | **Project No.: 25i0007** | **Document No.: LP1Y-255-PRO-K215-06**

RFQ No.: LP1Y-255-PRO-K215-05 | **Project: K215 - Lilly Rail Offloading Rack** | **Revision No.: 0**

Tag No.: N/A | **Client: Lilly - Lebanon Plant 1 (B1LG)** | **Date:**

Technical Description (List Tendered Requirements)	Specification	Bidder: Moore Supply Bid No: S178050895 Bid Date: 12/11/2025	Designation	Cost Adder	Bidder: Barry Company Bid No: Q037601 Bid Date: 12/16/2025	Designation	Cost Adder	Bidder: Bid No: Bid Date:	Designation	Cost Adder		Designation	Cost Adder
Legend: Y=Complies with Specification N=Not Acceptable A=Acceptable Alternative INA=Information Not Available N/A=Not Applicable VTS=Vendor to Specify													
1. GENERAL													
Vendor Proposal Number		S178050895			Q037601								
Vendor Proposal Current Revision Number / Date		Rev 0, 12/11/2025			Rev 0, 12/16/2025								
Fabrication Facility Location		1140 N North Branch St Chicago, IL 60642			1145 East Maryland Street Indianapolis, Indiana 46202								
2. DESIGN													
Item 1: Safety Shower & Eye Wash Station													
General													
Quantity	2	2	Y		1 (Double the quote and it is acceptable)	A							
Type	Freeze resistant, scald protection valve, freestanding, no bowl, with self-limiting heat trace cable, thermostat, stainless steel showerhead, alarm beacon, pull rod on stay-open ball valve, insulation with weatherproof cover, ANSI-compliant identification sign, lead free, with factory mounted Form C dry contact type proximity switches for both eyewash and shower valves.	Included	Y		Included	Y							
Flow Requirement	Deliver water at 20 GPM for a minimum of 15 minutes	Included	Y		Included	Y							
Application	Electric heat-traced stations prevent freezing even in extreme conditions. Thermostat regulated units are heated by self-limiting heat-tracing cable wrapped around internal pipe. Units are intended for cold weather use down to -40F. See 'Thermostatic Mixing Valves' section to ensure tepid water compliance with ANSI Z358.1-2014	Included	Y		Included	Y							
Heating Cable	Self-regulated with external termination guards. Thermostat activates when internal piping temperature falls below 75F. FM, UL, and CSA approved for Class 1, Division 2, Groups B, C, and D	Included	Y		Included	Y							
Electrical System	Unit is fully wired at factory. Junction box and components rated for Class 1, Division 2, Groups B, C, and D. Requires 120 VAC single phase power supply.	Included	Y		Included	Y							
Jacket / Insulation	UV-resistant orange ABS plastic with polyethylene foam insulation. Jacket sections secured with removable stainless steel screws for easy maintenance.	Included	Y		Included	Y							
Shower Head	10" Diam. Stainless steel with 20 GPM flow control	Included	Y		Included	Y							
Shower Valve	1" IPS chrome-plated brass stay-open ball valve. US-made with brass ball and PTFE seals. Includes stainless steel actuating arm and 29" stainless stell pull rod.	Included	Y		Included	Y							
Spray Head Assembly	Freeze-resistant with float-off dust covers and internal flow control	Included	Y		Included	Y							
Eyewash Valve	1/2" IPS chrome plated brass three-way self-draining stay-open ball valve. US-made with brass ball and PTFE seals. Includes 1/2" NPT port to rdain water from spray heads after use.	Included	Y		Included	Y							
Pipe and Fittings	Sch 40 Type 304 Stainless Steel	Included	Y		Included	Y							
Supply	1-1/4" NPT female bottom inlet	Included	Y		Included	Y							
Sign	ANSI-compliant identification sign.	Included	Y		Included	Y							
Quality Assurance	Factory tested prior to shipment. Ships fully assembled and crated.	Included	Y		Included	Y							
					Included	Y							
Additional Options					Included	Y							
Scald Protection Valve	Include	Included	Y		Included	Y							
Freeze Protection Valve	Include	Included	Y		Included	Y							
Proximity Switch	Magnetically Activated (MAP) Switch	Included	Y		Included	Y							
Alarm Horn and Remote Sensing Capability	Include	Included	Y		Included	Y							
					Included	Y							
					Included	Y							
Electrical Hazardous Area Classification	Class 1, Division 2; Group: C&D, T4 Winterization Required; Tropicalization Required	Included	Y		Included	Y							
Standard	ANSI Z358.1-204, OSHA 1910.151	Included	Y		Included	Y							
Location	Outdoor	Included	Y		Included	Y							
Top Inlet Size in	1-1/4" NPT	Included	Y		Included	Y							
Mounting Flange	7/16" diam. mounting slots on 7-5/8" bolt circle	Included	Y		Included	Y							
Accessories		Included	Y		Included	Y							
Operating Load	VTA	Included	Y		Included	Y							
Other	Guardian Equipment GFR3500-SFP-PS-SSH-TL	GFR3503-SFP-PS-TL	Y		GFR3503-SFP-PS-TL	Y							
Notes	VTA	Freeze resistant sprayhead assembly with internal 3.2 GPM flow control and float off dust covers. Designed for use in a Class 1, Div. 2, Groups B, C, & D Environment. Peak Heat trace power consumption is 75 watts. In order to meet class 1, div 2 requirements, approved conduit and fittings must be used. All threaded electrical connectison must have hazardous location rated sealant applied to threads.	Y		Same as left (based off model number from Guardian)	Y							
4. NON-DESTRUCTIVE TESTING & INSPECTION REQUIREMENTS													
Vibration Testing													
PMI													
NDE													
Running test													
Function test	Included	Included	Y			INA							
Hydrotest													
5. SHIPPING:													
Shipping Weight (lbs)	By Vendor	VTS			VTS								
Shipping Outside Dimensions (ft)	By Vendor	VTS			VTS								
Shipping Drawings	Yes	VTS			VTS								
Shipment Packing	Yes	VTS			VTS								
Other													
6. ADDITIONAL REQUIREMENTS													
Start-up and Commissioning Spares for each gangway	Seller's Recommended Spare Parts	VTS			VTS								
Recommended Spares for Two Years Operation for each gangway	Seller's Recommended Spare Parts	VTS			VTS								
Capital Spares	Seller's Recommended Spare Parts	VTS			VTS								
Equipment Documentation	Match SDDC or provide mark up of exclusions.	VTS			VTS								
7. BONUS / MALUS													
Delivery Weeks : (week)		5-6 Weeks											
Delivery Terms:		Freight estimate and taxes not included			Freight not included, tax included								
Bid Validity		45 Days			30 Days								
9. SUMMARY													
Exception													
Technically Acceptable	Yes or No	Yes											
Comments	As needed	**Remarks/Cost Adders:** **- Supplier very responsive. Worked at length to provide requested information and clarifications.** **- Supplier based out of Houston, TX** **- Provided GA drawing and electrical wiring diagrams.**			**Remarks/Cost Adders:** **- Local to Indianapolis** **- Provided quote, but no additional information. Original contact provided quote and then is on leave for ~3 months. Reached out to backups, but no responses have been provided for additional documentation / information.**			**Remarks/Cost Adders:** -			**Remarks/Cost Adders:** -		

PREPARED BY: (HANSON) Quade Robertson **DATE:** 12/18/2025

REVIEWED BY: (HANSON) Saeed Sabah **DATE:**

APPROVED BY: (HANSON) Mike Walther **DATE:**

PROJ. APPROVAL: **DATE:**

CLIENT APPROVAL: **DATE:**

RECOMMENDATION:

Based on the information provided, Moore Supply has been more responsive and willing to coordinate with Guardian. Technically, both bids are acceptable based on the Model number provided.

Key

Y	Meets Specification
N	Does Not Meet Specification
INA:	Information Not Available
N/A:	Not Applicable
VTS:	Vendor to Specify
A:	Acceptable Alternative
	Potential Cost Adder
Text	Critical Item / Cost Impact

Document Title:	Loading Arms Bid Tabulation Sheet - SafeRack
Document Number:	LP1Y-255-PRO-K215-07
Security Class:	Restricted

Loading Arms Bid Tabulation Sheet - SafeRack

Lilly - Lebanon Plant 1 (B1LG)

K215 - Lilly Rail Offloading Rack

Hanson Project Number: 25i0007

B1 - DATA COMPLETE - WITH COMMENTS - PROCEED - RESUBMIT

Authorization to proceed does not relieve Contractor/Supplier of its responsibility or liability under the Contract and or Purchase Order.

By: Ricardo DaCosta dac54131 April 13, 2026

REVIEWED

By:Brydon Coombe,coo11790,April 10, 2026

A	IFR	MW	QR	SS	3/25/2026
Revision	**Issue Status**	**Originator**	**Reviewer**	**Approver**	**Date**

HANSON

Document No: LP1Y-255-PRO-K215-07
Revision: A
Revision Date: 2026-03-25

Tag Nos	Description	Vendor		Hanson
		Confirm (Yes/No)	Notes	Acceptable (Yes/No)
LA-0622A & LA-0632A	Type: 4" x 3" Supported Boom Top Unloading Arm - Downfeed (x 2)	Yes		Yes
	Wetted Path -316L SS	Yes		Yes
	Swivel Modules Non Wetted - 316L SS	Yes		Yes
	Dimensions (List dimensions quoted in **Notes** column)			
	Boom : 78" Minimum 4" 40S Pipe, 316L SS	Yes	96" max	Yes
	Primary: 78" Minimum 3" 40S Pipe, 316L Ss	Yes	96" max	Yes
	Drop: 60" Minimum 3" 316L SS Metal Hose with single Braid	Yes	72" max	Yes
	Inlet Flange to Pillow Block: 84" Maximum	Yes	84" max	Yes
	Spring Balance - Manual Operation	Yes		Yes
	Metal Hose End Connection to Rail Car 3" Dip Tube Valve			
	3" 316L SS Ball Valve at End of Metal Hose	Yes		Yes
	3" 316L SS Breakaway Coupling	Yes		Yes
	3" 316L SS Coupler with Dust Plug	Yes		Yes
	Seal Material: Chemraz SD517	Yes		Yes
	1" Piggy Back Nitrogen Hose	Yes		Yes
	Hose Material Tube PFA Core, Silicone Cover with 302 SS Braid	No	Quoted 1" x 25' EPDM N2 hose with Chicago fittings on each end	Yes
	Hose End Connection to Rail Car 2" Vent Valve	No	Stinger needed for car connection not included	No - Stinger not req'd
	1" 316L SS Ball Valve	Yes	Viton Seals	Yes
	1" 316L SS Breakaway Coupling	Yes	Viton Seals	Yes
	1" 316L SS Coupler with Dust Plug	Yes	Buna Seal	Yes
	Seal Material: Chemraz SD517	No		Yes
LA-0622A DMF Unloading	Viscosity: 0.87 cP			
	S.G.: 0.94			
	Temperature (Oper./Design): 15 / 65 C			
	Pressure (Oper./Design): 40/150 psig			
	Flowrate (Oper./Design): 150/180 gpm			
	Unloading Arm Pressure Drop Off Estimate by Vendor based on above	See note	Verified during final engineering	Yes
LA-0632A ACN Unloading	Viscosity: 0.00419 cP			
	S.G.: 0.94			
	Temperature (Oper./Design): 15 / 65 C			
	Pressure (Oper./Design): 40/150 psig			
	Flowrate (Oper./Design): 150/180 gpm			
	Unloading Arm Pressure Drop Off Estimate by Vendor based on above		Verified during final engineering	Yes

These seal materials need to be confirmed for acceptance with Lilly. If the N2 Line is considered "impact", only Chemraz SD517 is acceptable. FDA/USP certs would be required.

RD: N2 system, including hose, is impact system. the seal material to follow line specification. Change to Chemraz SD517

HANSON

Document Number: LP1Y-255-PRO-K215-07
Revision: A
Revision Date: 2026-03-25

Tag Nos	Description	Vendor		Hanson
		Confirm (Yes/No)	Notes	Acceptable (Yes/No)
LA-0631A	Type: 4" x 3" Supported Boom Top Vapor Return Arm -Downfeed	Yes		Yes
	Wetted Path -316L SS	Yes		Yes
MTBE Vapor	Swivel Modules Non Wetted - 316L SS	Yes		Yes
	Dimensions (List dimensions quoted in **Notes** column)			
	Boom : 78" Minimum 4" 40S Pipe, 316L SS	Yes	96" max	Yes
	Primary: 78" Minimum 3" 40S Pipe, 316L Ss	Yes	96" max	Yes
	Drop: 60" Minimum 3" 316L SS Metal Hose with single Braid	Yes	72" max	Yes
	Inlet Flange to Pillow Block: 84" Maximum	Yes	84" max	Yes
	Spring Balance - Manual Operation	Yes		Yes
	Metal Hose End Connection to Rail Car 2" Vent	No	Quoted 3" - can change to 2" if needed	
	2" 316L SS Ball Valve at End of Metal Hose	No	Quoted 3" - can change to 2" if needed	Yes
	2" 316L SS Breakaway Coupling	No	Quoted 3" - can change to 2" if needed	Yes
	2" 316L SS Coupler with Dust Plug	No	Quoted 3" - can change to 2" if needed	Yes
	Seal Material: Chemraz SD517	Yes		Yes

HANSON

Document Number: LP1Y-255-PRO-K215-07
Revision: A
Revision Date: 2026-03-25

Tag Nos	Description	Vendor		Hanson
		Confirm (Yes/No)	Notes	Acceptable (Yes/No)
LA-0640A & LA-0650A	Type: 4" x 3" Supported Boom Top Loading Arm - Downfeed (x 2)	Yes		Yes
	Wetted Path -Duplex Steel S32205-W	Yes	Duplex 2205	Yes
	Swivel Modules Non Wetted - 316L SS	No	Duplex 2205	Yes
	Dimensions (List dimensions quoted in **Notes** column)			
	Boom : 78" Minimum 4" 40S Pipe, Duplex Stl S32205-W	Yes	104" max	Yes
	Primary: 78" Minimum 3" 40S Pipe, Duplex Stl S32205-W	Yes	88" max	Yes
	Drop: 60" Minimum 3" Hastelloy Braided Hose	Yes	72" max	Yes
	Inlet Flange to Pillow Block: 84" Maximum	Yes	84" max	Yes
	Spring Balance - Manual Operation	Yes		Yes
	Metal Hose End Connection to Rail Car 3" Dip Tube Valve			
	3" Duplex S32205-W Ball Valve at End of Metal Hose	Yes		Yes
	3" Hastelloy Breakaway Coupling	No	Duplex 2205 wetted	Yes
	3" Hastelloy Coupler with Dust Plug	Yes	Hastelloy coupler with FFKM156 seals - SST Dust plug included	Yes
	Seal Material: Kalrez 0040	Yes		Yes
	Heat Tracing and Insulation Required - By Others?	By others	3.5 lbs per ft	Yes
	Heat Tracing and Insulation Weight Accounted for in Spring Balance?	Yes		Yes
	2" Piggy Back Vent Metal Hose	Yes		Yes
	Hose Material Hastelloy	Yes		Yes
	Hose End Connection to Rail Car 2" Vent Valve	Yes		Yes
	2" Duplex S32205-W Ball Valve at End of Metal Hose	Yes		Yes
	2" Hastelloy Breakaway Coupling	No	Duplex 2205 wetted	Yes
	2" Hastelloy Coupler with Dust Plug	Yes	Hastelloy coupler with FFKM156 seals - SST Dust plug included	Yes
	Seal Material: Kalrez 0040	Yes		Yes
LA-0640A & LA-0650A	Viscosity: 1.12 cP			
Waste Loading	S.G.: 1			
	Temperature (Oper./Design): 15 / 53 C			
	Pressure (Oper./Design): 15/150 psig			
	Flowrate (Oper./Design): 350/394 gpm			
	Unloading Arm Pressure Drop Off Estimate by Vendor based on above	See Note	Verified during final engineering	Yes

HANSON

Document Number: LP1Y-255-PRO-K215-07
Revision: A
Revsion Date: 2026-03-25

Tag Nos	Description	Vendor		Hanson
		Confirm (Yes/No)	Notes	Acceptable (Yes/No)
NDE For All Arms	Inspection and Test Plan	**Yes**		Yes
	WPS, PQR, WPQ and Weld Map Submittal	Yes		Yes
	PMI Report	No	Can be quoted as adder / not included on truck arms	Yes
	Hydro Report with Certification	Yes		Yes
	5% X-Ray and 100% Visual with Report	Yes		Yes
	Full Certified Drawing Set with Weights and Support Requirements	Yes	Final drawings have weights, dims and moment loads	Yes
	Calculated pressure drops per operation and design flows indicated	See note	To be confirmed in final engineering.	Yes

Document Title:	Electrical Bill of Material
Document Number:	LP1Y-265-MTO-K215-01
Security Class:	Restricted

B1 - DATA COMPLETE - WITH COMMENTS - PROCEED - RESUBMIT
Authorization to proceed does not relieve Contractor/Supplier of its responsibility or liability under the Contract and or Purchase Order.
By: Ricardo DaCosta dac54131 April 10, 2026

Electrical Bill of Material

Lilly Rail Containment

Hanson Project Number: 25I0007

REVIEWED
By:LDowns,dow81230,April 9, 2026

General Comments
- Please use the Electrical BOQ Input File provided Feb.27.2026
- Include updated BOQ in Electrical IFR Submittal for reference and review against design.

K215 to cross reference BOQ with Rev1 IFR package for accuracy. Additionally, please provide revised BOQ in latest request format .

A	IFR	AT	BM	RN	3/26/2026
Revision	**Issue Status**	**Originator**	**Reviewer**	**Approver**	**Date**

RD: No additional comments

HANSON

ITEM NO	Description	Unit	LP1 Elec K215 RAIL CAR Total Quantity	Remarks
6.1	**SID**			
6.2	**Electrical Equipment**			
6.2.1	Low voltage Adjustable Speed Drive (ASD) (EA)	EA	1	Integral to MCC Bucket for Pump
6.2.2	Medium voltage Adjustable Speed Drive (ASD)	EA	0	
6.2.3	LV MCCs	EA	0	Utilizing E-house 48 MCCs
6.2.4	LV Switchgears (unit Substation) includes 2 transformer and1 SWG	EA	0	
6.2.5	MV Switchgears	EA	0	
6.2.6	MV Transformer	EA	0	
6.2.7	UPS and DC System	EA	0	Utilizing E-house 48 UPS & Panels
	UPS	EA	0	Utilizing E-house 48 UPS & Panels
	UPS Battery	EA	0	Utilizing E-house 48 UPS & Panels
6.2.8	Dry Type Transformer (<225kVA)	EA	0	
6.2.9	Low voltage Lighting/Receptacle Panelboards/ UPS Panel board			Utilizing E-house 48 Panels
	LP	EA	0	Utilizing a Fluor Panel
	RP	EA	0	Utilizing E-house 48 Panels
	UDP	EA	0	Utilizing E-house 48 Panels
	UPP	EA	0	Utilizing E-house 48 Panels
	PDP	EA	0	Utilizing E-house 48 Panels
6.2.10	Electrical JB (Lighting)	EA	60	(1) JB per fixture + (5) additional
6.2.11	Disconnect Switch			
	120V Power Field Disconnect Switches	EA	10	Probes, Safety Showers, and 120V panels
	Motor Space Heater Disconnect Switch	EA	1	
	Motor Disconnect Switch	EA	1	
	Water Heater Disconnect Switch	EA	1	Integral to Heater Package
6.2.12	Local Control Station (LCS)			
	Probe Local Control Station	EA	2	LSC0640WP & LSC0650WP
	Unloading Pump JOA LCS	EA	1	
6.2.13	EHT Equipment			
	Terminator End-of-Circuit Light	EA	15	Provided by Thermon
	Line Sensing RTD	EA	4	Provided by Thermon
	SS Banding (fits up to 10" pipe)	EA	4	Provided by Thermon
	Fiberglass Fixing Tape	LF	4320	Provided by Thermon
	Caution Labels	EA	150	Provided by Thermon
	EHT Circuit ID Tags	EA	34	Provided by Thermon
	EHT Design and Deliverable Package	LS	1	Provided by Thermon
	EHT JB (Power to Self Regulating Connections)	EA	15	
6.3	**Receptacle and Lighting**			
6.3.1	Receptacle (EA)			Receptacle quantities to be updated in IFR April 2026 submittal
6.3.2	Lighting Fixtures			
	H1 - Class 1, Div 2, 23W LED, Type 3 Optics, Surface Mount	EA	10	Includes Mounting Hardware
	H2 - Class 1, Div 2, 39W LED, Type 3 Optics, Surface Mount	EA	3	Includes Mounting Hardware

HANSON

Document Number: LP1Y-265-MTO-K215-01
Document Revision: A
Revision Date: 2026-03-26

ITEM NO	Description	Unit	LP1 Elec K215 RAIL CAR Total Quantity	Remarks
	H3 - Class 1, Div 2, 54W LED, Type 5 Optics, Surface Mount	EA	2	Includes Mounting Hardware
	P33 - Class 1, Div 2, 39W LED, Type 1 Optics, Stanchion Mount	EA	6	Includes Mounting Hardware and Pole
	W2 - Class 1, Div 2, 39W LED, Type 3 Optics, Wall Mount	EA	12	Includes Mounting Hardware
	W6 - Class 1, Div 2, 23W LED, Type 3 Optics, Wall Mount	EA	22	Includes Mounting Hardware
6.4	**Grounding System**			
6.4.1	Grounding Cable			
	AG Grounding Cable -#4/0 AWG Copper Conductor	LF	1000	Refer to Fluor Standard Details for Bare or Insulation Recommendations
	UG Grounding Cable - #4/0 AWG Bare Copper Conductor	LF	1000	
6.4.2	Ground Rod	EA	6	
6.4.3	Ground Bar	EA	2	(1) Ground Bar at Ground Level & (1) at Platform Level
6.4.4	Ground Well	EA	4	Test Well with Rod
6.4.5	Grounding Connections	EA	200	
6.5	**Wires and Cable**			
6.5.1	UG Cable			
	MV Cable	LF	0	Not Used on K215 Project
	LV Power and Control Cable	LF	0	Not Used on K215 Project
6.5.2	AG Cable			
	MV Cable	LF	0	Not Used on K215 Project
	LV Power and Control Cable			
	FO CABLE , 6 FIBER, 300V, ITC	LF	600	Instrumentation Cable
	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC	LF	3200	Instrumentation Cable
	1-PR, #18 AWG, TWISTED W/ SHLD, ARMORED, 300V, ITC	LF	600	Instrumentation Cable
	1-TRIAD, #16 AWG, ARMORED, NO SHLD, 300V, ITC	LF	900	Instrumentation Cable
	Water Heater Circuit - 3/C-#2, 1-GND, TC-ER, ARMORED, 600V	LF	355	1 Circuit Total, 480V Power Cable
	Unloading Pump Motor Power Circuit - 3/C-#8, 1-GND, TC-ER, ARMORED, 600V	LF	315	1 Circuit Total, 480V Power Cable
	Unloading Pump Motor JOA Control Circuit - 2/C-#16, 1-GND, TC-ER, ARMORED, 600V	LF	300	1 Circuit Total, 480V Power Cable
	Unloading Pump Motor Space Heater Power Circuit - 2/C-#10, 1-GND, TC-ER, ARMORED, 600V	LF	315	1 Circuit Total, 480V Power Cable
	UPS, Normal, and Utility Panelboards Branch Circuits - 2/C-#10, 1-GND, TC-ER, ARMORED, 600V	LF	3450	11 Circuits Total, 120V Power Cable
	277V Lighting Power Cable - 2/C-#10, 1-GND, TC-ER, ARMORED, 600V	LF	2500	2 Circuits Total, 277V Power Cable (1 emergency, 1 non-emergency)
	EHT 277V Power Cable - 2/C-#4, 1-GND, TC-ER, ARMORED, 600V	LF	600	2 Circuits Total, 277V Power Cable
	EHT 277V Power Cable - 2/C-#10, 1-GND, TC-ER, ARMORED, 600V	LF	2400	6 Circuits Total, 277V Power Cable
	EHT RTD control cable (LF)	LF	1200	Provided by Thermon
	EHT self-regulating cable, 208V Low Temperature	LF	1109	Provided by Thermon
	EHT self-regulating cable, 208V Medium Temperature	LF	925	Provided by Thermon
6.5.4	Cable Termination			
	MV Cable	EA	0	Not Used in K215
	Low Voltage Control and Power Terminations	EA	100	
	EHT Terminator Power Connection	EA	15	Provided by Thermon
	EHT Power Connection Kits	EA	15	Provided by Thermon
	End-of-Circuit Light Connection Kit	EA	15	Provided by Thermon

HANSON

Document Number: LP1Y-265-MTO-K215-01
Document Revision: A
Revision Date: 2026-03-26

Lilly Spec. Outdoor cable trays shall be 6" deep, 20 ft spans, heavy-duty aluminum, ladder type with 9" rung spacing, minimum, NEMA Class 20C, (B-line H46A or equivalent), unless specified otherwise

ITEM NO	Description	Unit	LP1 Elec K215 RAIL CAR Total Quantity	Remarks
	End-of-Circuit Light Connection Kit	EA	15	Provided by Thermon
6.6	**Cable Tray/ Rack and Coduit**			
6.6.1	Cable tray			
	6"W x 4"D Ladder Type SS Cable Tray - Emergency Power	LF	205	
	6"W x 4"D Ladder Type SS Cable Tray - Emergency Power	LF	120	Tray outside of Unloading Rack Area
	6"W x 4"D Ladder Type SS Cable Tray - Power	LF	130	Bottom Level
	12"W x 4"D Ladder Type SS Cable Tray - Auxiliary	LF	195	
	12"W x 4"D Ladder Type SS Cable Tray - Instrumentation	LF	130	
	12"W x 4"D Ladder Type SS Cable Tray - Instrumentation	LF	130	Bottom Level
	15"W x 4"D Ladder Type SS Cable Tray - Power	LF	130	
	18"W x 4"D Ladder Type SS Cable Tray - Instrumentation	LF	115	
	20"W x 4"D Ladder Type SS Cable Tray - Power	LF	260	
	24"W x 4"D Ladder Type SS Cable Tray - Instrumentation	LF	50	
	30"W x 4"D Ladder Type SS Cable Tray - Instrumentation	LF	130	
	Cable tray Splice/Expansion Fittings (aprox 3ft per fitting)	EA	540	
	6"W x 4"D tray, 24-inch radius, 90-degree horizontal bend	EA	1	
	12"W x 4"D tray, 24-inch radius, 90-degree horizontal bend	EA	1	
	18"W x 4"D tray, 24-inch radius, 90-degree horizontal bend	EA	1	
	20"W x 4"D tray, 24-inch radius, 90-degree horizontal bend	EA	2	
	6"W x 4"D tray, 22-inch radius, 90-degree vertical bend	EA	2	
	12"W x 4"D tray, 22-inch radius, 90-degree vertical bend	EA	1	
	18"W x 4"D tray, 22-inch radius, 90-degree vertical bend	EA	1	
	24"W x 4"D tray, 22-inch radius, 90-degree vertical bend	EA	1	
	20"W to 15"W Cable Tray Reducer (4"D)	EA	1	
	24"W to 20"W Cable Tray Reducer (4"D)	EA	2	
	18"W to 12"W Cable Tray Reducer (4"D)	EA	2	
6.6.2	Conduit			
	Control - Field Routed Conduit (below 2" diameter)	LF	800	
	Lighting Power - Field Routed Conduit (below 2" diameter)	LF	500	
	Power - Field Routed Conduit (below 2" diameter)	LF	200	

Document Title:	Sensory Assy, Portable Cane
Document Number:	LP1Y-270-DIA-K215-01
Security Classification:	Restricted

Sensory Assy, Portable Cane

LSH-0640WP

LSH-0650WP

Lilly Rail Containment

Hanson Project Number:25I0007.00

A1 - DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

Authorization to proceed does not relieve Contractor/Supplier of its responsibility or liability under the Contract and or Purchase Order.

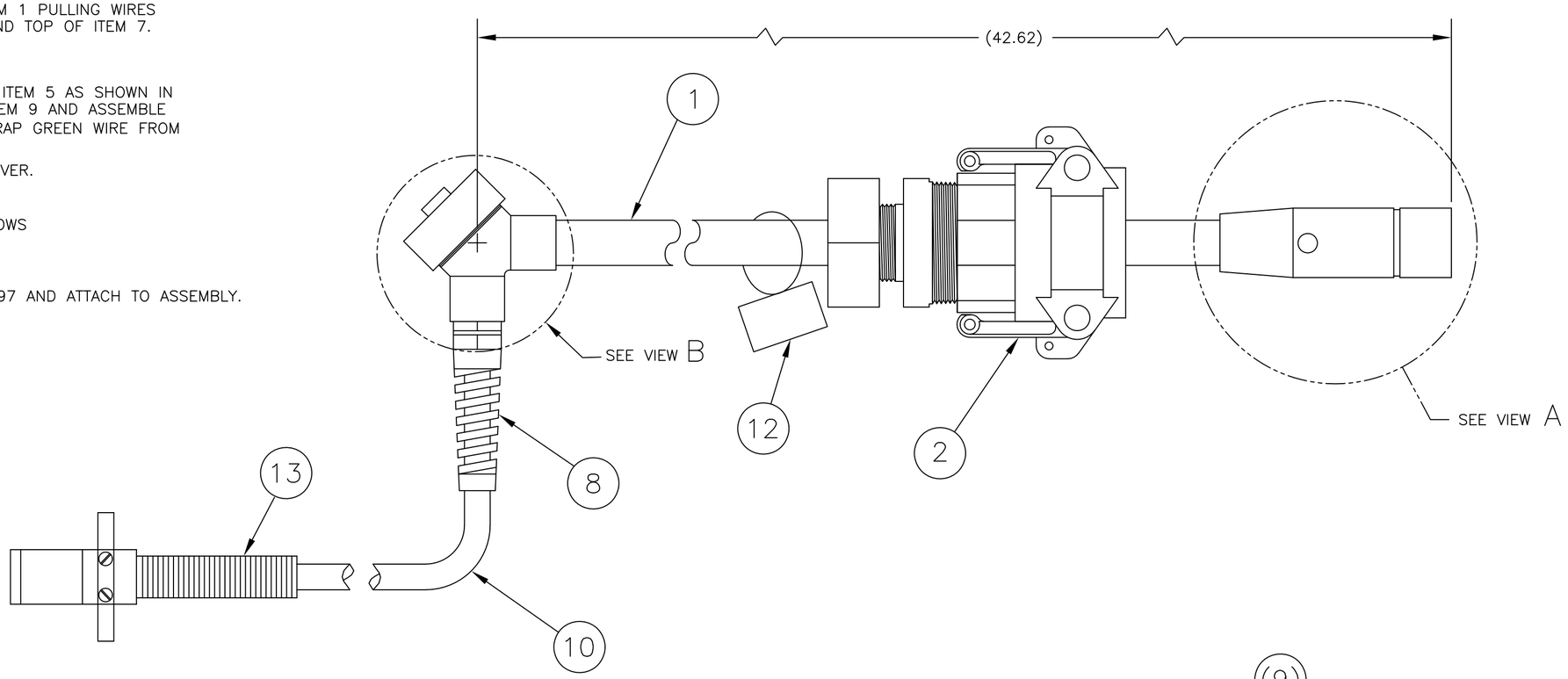
By: Ricardo DaCosta dac54131 April 7, 2026

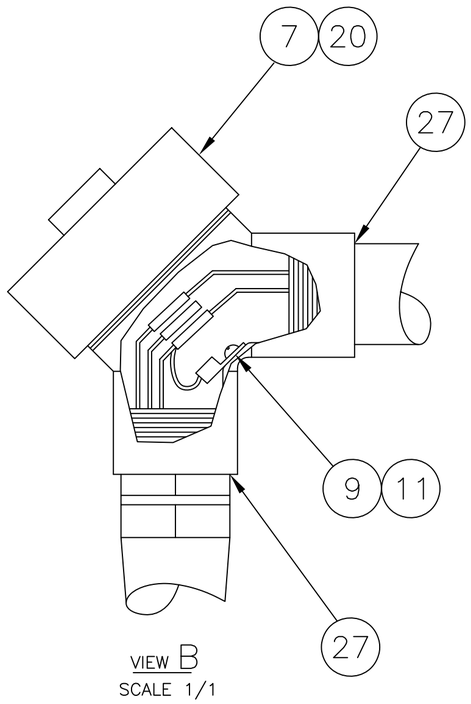
Revision	Issue Status	Originator	Reviewer	Approver	Date
0	IFQ	JN	SS	MW	3/9/2026
A	IFR	JN	SS	MW	2/4/2026

REVISIONS						
REV	ECO	DESCRIPTION	DWN	DATE	CHKR	APVD
A	3857	CREATED PER ECO	JRM	1/29/26		

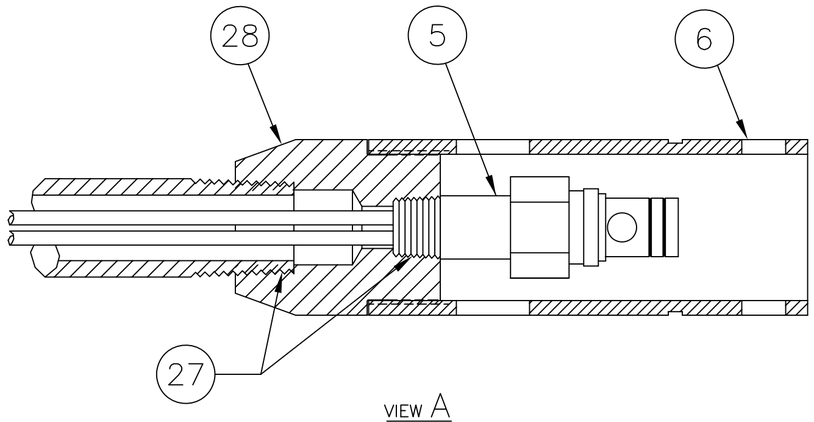
NOTES: UNLESS OTHERWISE SPECIFIED:

1. ASSEMBLE AS FOLLOWS
   - A) APPLY ITEM 27 TO PIPE THREADS PRIOR TO ASSEMBLY
   - B) ASSEMBLE ITEM 5 TO ITEM 28 AND ITEM 6 TO ITEM 28
   - C) ASSEMBLE ITEM 1 TO ITEM 7
   - D) SLIDE ITEM 2 ONTO ITEM 1
   - E) ASSEMBLE ITEM 28 ASSEMBLY ON TO ITEM 1 PULLING WIRES FROM ITEM 5 ASSEMBLY THRU ITEM 1 AND TOP OF ITEM 7.
   - F) ASSEMBLE ITEM 10 THRU ITEM 8, PULLING ITEM 10 THRU TOP OF ITEM 7
   - G) USING ITEM 15 BUTT SPLICE ITEM 10 TO ITEM 5 AS SHOWN IN WIRING DIAGRAM, CONNECT ITEM 29 TO ITEM 9 AND ASSEMBLE TO ITEM 7 AS SHOWN USING ITEM 11. WRAP GREEN WIRE FROM ITEM 10 WITH TAPE.
   - H) DRESS WIRES BACK INTO ITEM 7 AND COVER.
   - J) ASSEMBLE ITEM 8 TO ITEM 7
   - K) ASSEMBLE ITEM 13 TO ITEM 10 AS FOLLOWS
     GREEN WIRE TO PIN G
     BLACK WIRE TO PIN BK
     WHITE WIRE TO PIN W
   - L) PLACE ITEM 98 AND ITEM 99 INTO ITEM 97 AND ATTACH TO ASSEMBLY.
2. TEST PER SPECIFICATION NUMBER 43011.

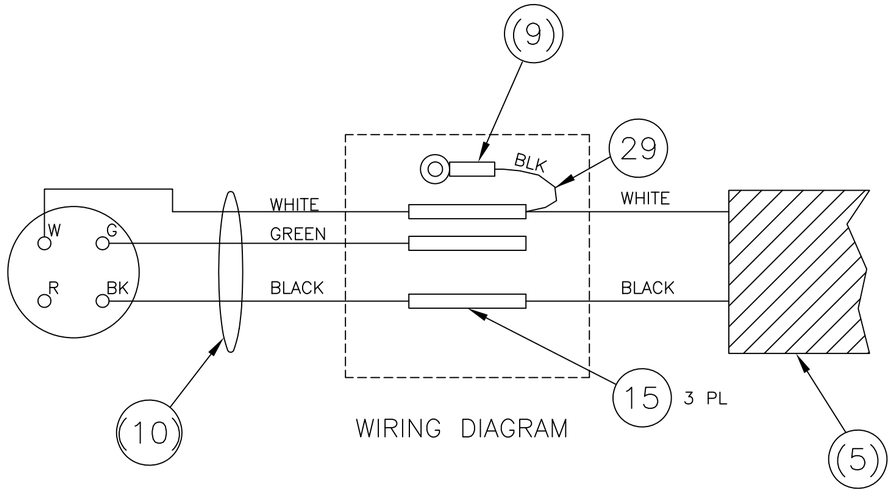




VIEW B
SCALE 1/1



VIEW A
SCALE NONE



WIRING DIAGRAM

UNLESS OTHERWISE SPECIFIED:	DRAWN	DATE
1. ALL DIMENSIONS ARE IN INCHES 2. DIMENSIONS ARE AFTER MACHINED AND/OR APPLIED FINISHES 3. DO NOT SCALE PRINT 4. REMOVE BURRS. BREAK SHARP EDGES AND CORNERS .005–.020 5. TOLERANCES: .XX=±.01 ANGLES ±0° 30' .XXX=±.005 6. MIN. SURFACE FINISH √63 7. THDS PER ANSI B1 SERIES 8. GEOMETRIC TOLERANCES PER ANSI Y14.5–1982 9. ELECTRONIC SYMBOLS PER ANSI 32.2	J. MANSEAU	1/29/26
	CHECKER	
	ENGINEER	
	APPROVED	

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scully SIGNAL COMPANY
WILMINGTON, MA. U.S.A.

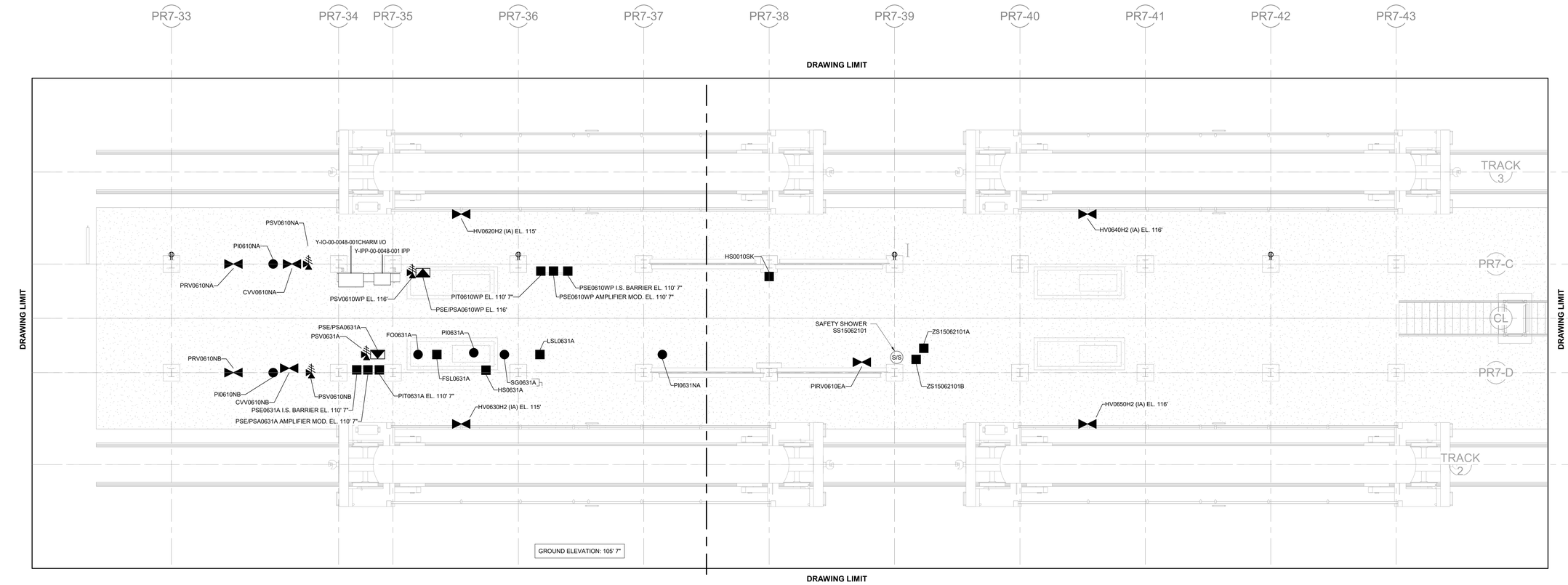
TITLE:
SENSOR ASSY,PORTABLE CANE
SP–PYU, (DUPLEX 2205 SS)

MATERIAL: SEE BOM	SIZE	DWG NO.	REV
FINISH:	C	091000063	A
	SCALE 1/2	SHEET 1 OF 1	

Single Level Current BOM Report

2/26/2026 1:28:18 PM

Bubble#	Item/ Components	Description	Quantity Per	U/M	Per	Rev	Drawing#	Ref Designator Ref Des
	091000063	F SP-PYU(2205 SS)/4P CANE,36",20' STR CA				000	D091000063	
1	717000006	P PIPE,SST,SCHED 40,TYPE 2205,36",1/2IPS	1.00000	EA	Unit	000	D717000006	
2	101000034	PH CPLG ASSY, GATX CANES, HASTELLOY	1.00000	EA	Unit	000	D32377	
5	101000033	S SNSR ASSY,SP-PYU(SS 2205),PORT CANE	1.00000	EA	Unit	000	D31852	
6	724000095	M DRAFT SHIELD,2205 SS,THD,BLANK	1.00000	EA	Unit	000	D31509	
7	22416	M CONN,MACH,"Y"	1.00000	EA	Unit	A	D22416	
8	27593	P STR RLF,METAL,.24-.47 CA DIA,1/2NPT	1.00000	EA	Unit	B	D27593	
9	27914	P CONN,RT,RED,22-18AWG,#6	1.00000	EA	Unit	C	D27914	
10	28713	P CA, 3 CNDCT,BLK,STR	20.00000	FT	Unit	C	D28713	
11	50545	P SCR,PNH,SLTD,SST,6-32 X .25	1.00000	EA	Unit	D	D50584	
12	60278	P TAG,MANILA,CAUTION,PORT THMS SNSRS,RED	1.00000	EA	Unit	E	D60278	
13	27402	P PL,ZINC,4 POLE	1.00000	EA	Unit	A	D27402	
15	27916	P CONN,BUTT,BLU,14-16 AWG	3.00000	EA	Unit	F	D27916	
16	53124	P SHOE,TOGGLE SCREW,CLAMP	1.00000	EA	Unit	B	D53124	
20	31853	P GSKT,RUBBER,1.5 DIA X .09 THK	1.00000	EA	Unit	A	D31853	
27	40005	P SEALANT,PIPE W/TEFLON 592	0.00010	TB	Unit	B	D40005	
28	724000096	M CPLG,DRAFT SHIELD TOP,2205SS	1.00000	EA	Unit	000	D31510	
29	27228	P WIRE,STRD,PVC,UL1007,BLK,#18	0.19000	FT	Unit	K	D27227	
69	43011	REF TSPEC,PORTABLE CANE,SP-PY TYPE	0.00000	REF	Unit	A	D43011	
97	60479	P BAG,POLY,9 X 12	1.00000	EA	Unit	A	D60479	
99	60972	P CARD,LIMITED WARRANTY	1.00000	EA	Unit	J	D60972	



1 INSTRUMENT PLAN - GROUND LEVEL - ENLARGED

FROM ELEVATION 105'-7" TO 116'-0"

LEGEND & SYMBOLS

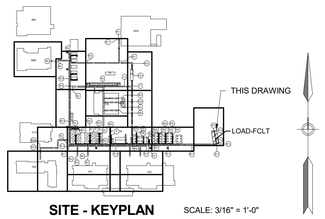
- ELECTRONIC INSTRUMENT
- LOCAL INSTRUMENT (PI, TI, FO, ETC.)
- PRESSURE SAFETY VALVE
- VALVE: INSTRUMENT (PNEU.) VALVE, IA ROOT VALVE, REGULATOR, OR CONSERVATION VENT VALVE.
- RUPTURE DISK ASSEMBLY
- XX LOCAL PANEL (LP) OR JUNCTION BOX (JB)

NOTES:

1. UNLESS SPECIFIED OTHERWISE ALL INSTRUMENTS WILL BE AT ELEVATION 108' 7" (APPROX.)
2. ALL INSTRUMENTS ARE PREFIXED WITH "LP1Y"

REFERENCE DRAWINGS:

1. LP1Y-270-LST-K215-01 - RAIL STATION INSTRUMENT INDEX.
2. LP1Y-270-LST-K215-02 - RAIL STATION I/O LIST.
3. LP1Y-270-LST-K215-03 - RAIL STATION INSTRUMENT CABLE SCHEDULE.
4. LP1Y-270-DIA-K215-01 THRU 31 - RAIL STATION INSTRUMENT WIRING DIAGRAMS.
5. LP1Y-DI-00000061 THRU 00000065 - RAIL STATION P&ID.



SITE - KEYPLAN SCALE: 3/16" = 1'-0"

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
0	ALL	ISSUED FOR CONSTRUCTION	NA	10-APR-26
A	ALL	ISSUED FOR REVIEW	NA	16-JAN-26

REVISIONS

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Houston, TX 77043
(346) 297-1720

REV	DATE	DES	ENG	CHK	APP	PM APP
1	10-APR-26	MGL	JN	BM	AJT	SS
A	16-JAN-26	MGL	JN	BM	AJT	SS

Lilly

INSTRUMENT LOCATION PLAN
RAIL CAR LOADING AND UNLOADING AREA
GROUND LEVEL

SYSTEM TYPE: NA			EDIFICE/BUILDING: Y	LEVEL/FLOOR: 0
SUB DISCIPLINE:	LOCATION CODE: LP1	LOCATION/GRID: COL C-D / 33-43	CCI CODE: YELLOW	PROJECT NAME: LP1-306-1181
CRITICAL QUALITY		ENTERPRISE GROUP: API	DRAWING INTENTION	DRAWING SCALE: As indicated
CRITICAL BUSINESS		EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA	MERIDIAN REV: B
CRITICAL HSE		VENDOR: NA	DOCUMENT ID	
NON-CRITICAL		ENGINEERING AGENCY DOC ID: NA	LP1Y-270-LAY-K215-01	

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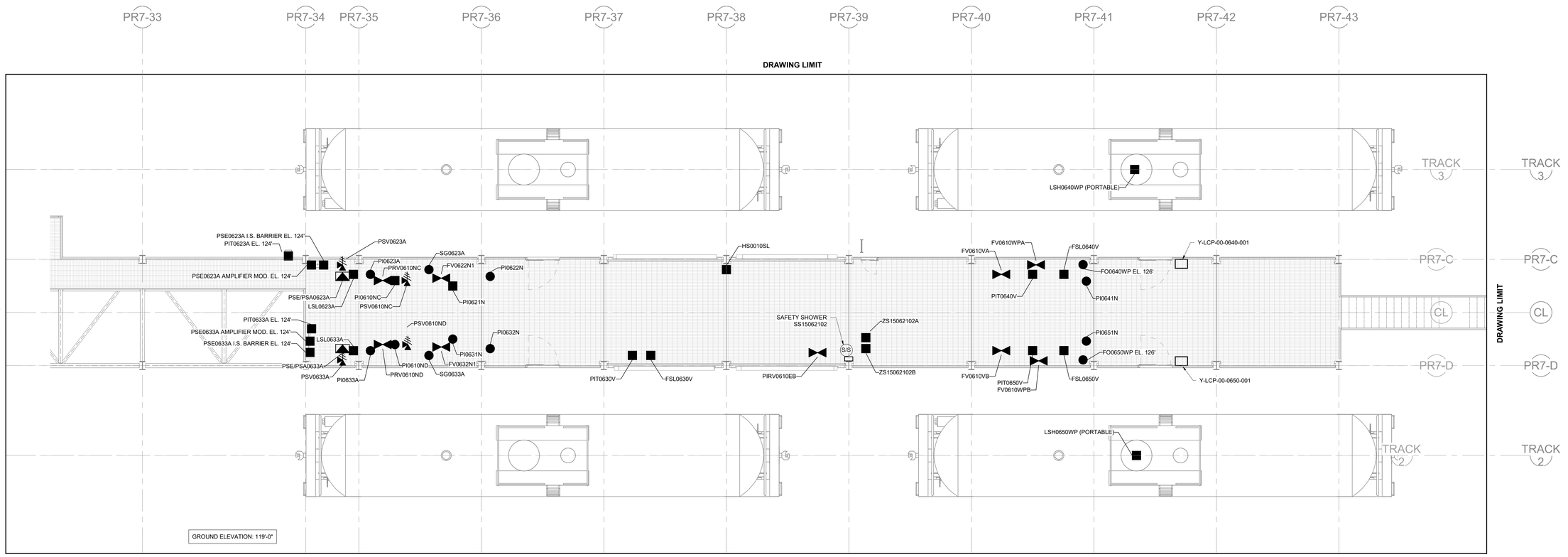


1 INSTRUMENT PLAN - OPERATING PLATFORM - ENLARGED VIEW

FROM ELEVATION 119'-0" TO 129'-0"

LEGEND & SYMBOLS

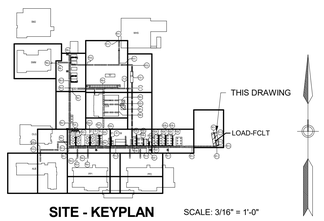
- ELECTRONIC INSTRUMENT
- LOCAL INSTRUMENT (PI, TI, FO, ETC.)
- PRESSURE SAFETY VALVE
- VALVE: INSTRUMENT (PNEU.) VALVE, IA ROOT VALVE, REGULATOR, OR CONSERVATION VENT VALVE.
- RUPTURE DISK ASSEMBLY
- XX LOCAL PANEL (LP) OR JUNCTION BOX (JB)

NOTES:

1. UNLESS SPECIFIED OTHERWISE ALL INSTRUMENTS WILL BE AT ELEVATION 122' (APPROX.)
2. ALL INSTRUMENTS ARE PREFIXED WITH "LP1Y"

REFERENCE DRAWINGS:

1. LP1Y-270-LST-K215-01 - RAIL STATION INSTRUMENT INDEX.
2. LP1Y-270-LST-K215-02 - RAIL STATION I/O LIST.
3. LP1Y-270-LST-K215-03 - RAIL STATION INSTRUMENT CABLE SCHEDULE.
4. LP1Y-270-DIA-K215-01 THRU 31 - RAIL STATION INSTRUMENT WIRING DIAGRAMS.
5. LP1Y-DI-00000061 THRU 00000065 - RAIL STATION P&ID.



SITE - KEYPLAN SCALE: 3/16" = 1'-0"

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
△	ALL	ISSUED FOR CONSTRUCTION	NA	10-APR-26
△	ALL	ISSUED FOR REVIEW	NA	16-JAN-26

REVISIONS

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REV	DATE	DES	ENG	CHK	APP	PM APP
△						
△						
△						
△	10-APR-26	MGL	JN	BM	AJT	SS
△	16-JAN-26	MGL	JN	BM	AJT	SS

INSTRUMENT LOCATION PLAN
RAIL CAR LOADING AND UNLOADING AREA
PLATFORM LEVEL

SYSTEM TYPE: NA			EDIFICE/BUILDING: Y	LEVEL/FLOOR: 1
SUB DISCIPLINE	LOCATION CODE: LP1	LOCATION/GRID: COL C-D / 33-43	CCI CODE: YELLOW	PROJECT NAME: LP1-306-1181
CRITICAL QUALITY		ENTERPRISE GROUP: API	DRAWING INTENTION	DRAWING SCALE: As indicated
CRITICAL BUSINESS		EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA	MERIDIAN REV
CRITICAL HSE		VENDOR: NA	DOCUMENT ID	
NON-CRITICAL		ENGINEERING AGENCY DOC ID: NA	LP1Y-270-LAY-K215-02	

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Document Title:	25I0007- Rail Station Instrument Index
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25I0007- Rail Station Instrument Index

Lilly Rail Containment

Hanson Project Number: 25I0007

Revision	Issue Status	Originator	Reviewer	Approver	Date
0	ISSUED FOR CONST.	JN	AT	SS	10-Apr-2026
A	ISSUED FOR REVIEW	JN	AT	SS	12-Jan-2026

Rev.	Tag No.	Loop No.	Service Desc.	P&ID	Line No.	Instrument Desc.	Location	Status	Manufacturer	Model	P.O.	Signal Type	System	Charm I/O Panel	BasePlate	Channel	24VDC Power Panel	Design Scope	Provided by	Wiring Diag.	Typical Wiring Template	Location Plan	Process/Pneu. Detail	Elect Detail	Mounting Detail	Notes
0	LP1YHS0010SK	H-0010SK	RAIL STATION WALKWAY HAND SWITCH	LP1Y-DI-00000001	N/A	Emergency Hand Switch	Field	New	ALLEN-BRADLEY	800H-FRXT6AF	TBD	DI-DCS	DCS	LP1Y-IO-00-0048-001	CBP103	01	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-01	LP100-DQ-00000906C	Y-270-LAY-K215-01	N/A	IE900B-ZC19B	IY905C	
0	LP1YHS0010SL	H-0010SL	RAIL STATION GROUND HAND SWITCH	LP1Y-DI-00000001	N/A	Emergency Hand Switch	Field	New	ALLEN-BRADLEY	800H-FRXT6AF	TBD	DI-DCS	DCS	LP1Y-IO-00-0048-001	CBP105	01	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-02	LP100-DQ-00000906C	Y-270-LAY-K215-02	N/A	IE900B-ZC19B	IY905C	
0	LP1YZSC0610VA	F-0610VA	RAIL STATION #3 VAPOR BALANCE	LP1Y-DI-00000061	N/A	Limit Switch	Field	New	Westlock	9479NRG4B2M0600	B1LG-7-0003	DI-DCS	DCS	LP1Y-IO-00-0048-001	CBP102	01	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-03	LP100-DQ-0000908D	Y-270-LAY-K215-02	N/A	N/A	On-Valve	Conn. to Limit Switch JB
0	LP1YZSO0610VA	F-0610VA	RAIL STATION #3 VAPOR BALANCE	LP1Y-DI-00000061	N/A	Limit Switch	Field	New	Westlock	9479NRG4B2M0600	B1LG-7-0003	DI-DCS	DCS	LP1Y-IO-00-0048-001	CBP102	02	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-03	LP100-DQ-0000908D	Y-270-LAY-K215-02	N/A	N/A	On-Valve	Conn. to Limit Switch JB
0	LP1YSV0610VA	F-0610VA	RAIL STATION #3 VAPOR BALANCE	LP1Y-DI-00000061	N/A	Solenoid Valve	Field	New	ASCO	F8316H38124VDC	B1LG-7-0003	DO-DCS	DCS	LP1Y-IO-00-0048-001	CBP102	03	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-03	LP100-DQ-0000908D	Y-270-LAY-K215-02	N/A	N/A	On-Valve	Conn. to Limit Switch JB
0	LP1YFV0610VA	F-0610VA	RAIL STATION #3 VAPOR BALANCE	LP1Y-DI-00000061	3" DUP WV (344) (108) 064001	Isolation Valve	Field	New	ITT Richter	30-KNA1100-Series	B1LG-7-0003	N/A	N/A	N/A	N/A	N/A	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-03	LP100-DQ-0000908D	Y-270-LAY-K215-02	IA901A-ZA61	IE103	In-Line	
0	LP1YZSC0610VB	F-0610VB	RAIL STATION #4 VAPOR BALANCE	LP1Y-DI-00000061	N/A	Limit Switch	Field	New	Westlock	9479NRG4B2M0600	B1LG-7-0003	DI-DCS	DCS	LP1Y-IO-00-0048-001	CBP102	04	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-04	LP100-DQ-0000908D	Y-270-LAY-K215-02	N/A	N/A	On-Valve	Conn. to Limit Switch JB
0	LP1YZSO0610VB	F-0610VB	RAIL STATION #4 VAPOR BALANCE	LP1Y-DI-00000061	N/A	Limit Switch	Field	New	Westlock	9479NRG4B2M0600	B1LG-7-0003	DI-DCS	DCS	LP1Y-IO-00-0048-001	CBP102	05	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-04	LP100-DQ-0000908D	Y-270-LAY-K215-02	N/A	N/A	On-Valve	Conn. to Limit Switch JB
0	LP1YSV0610VB	F-0610VB	RAIL STATION #4 VAPOR BALANCE	LP1Y-DI-00000061	N/A	Solenoid Valve	Field	New	ASCO	F8316H38124VDC	B1LG-7-0003	DO-DCS	DCS	LP1Y-IO-00-0048-001	CBP102	06	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-04	LP100-DQ-0000908D	Y-270-LAY-K215-02	N/A	N/A	On-Valve	Conn. to Limit Switch JB
0	LP1YFV0610VB	F-0610VB	RAIL STATION #4 VAPOR BALANCE	LP1Y-DI-00000061	3" DUP WV (344) (108) 065001	Isolation Valve	Field	New	ITT Richter	30-KNA1100-Series	B1LG-7-0003	N/A	N/A	N/A	N/A	N/A	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-04	LP100-DQ-0000908D	Y-270-LAY-K215-02	IA901A-ZA61	IE103	In-Line	
0	LP1YZSC0610WPA	F-0610WPA	1st/2nd WASTE TK TO RAIL STATION #3	LP1Y-DI-00000061	N/A	Limit Switch	Field	New	Westlock	9479NRG4B2M0600	B1LG-7-0003	DI-DCS	DCS	LP1Y-IO-00-0048-001	CBP102	07	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-05	LP100-DQ-0000908D	Y-270-LAY-K215-02	N/A	N/A	On-Valve	Conn. to Limit Switch JB
0	LP1YZSO0610WPA	F-0610WPA	1st/2nd WASTE TK TO RAIL STATION #3	LP1Y-DI-00000061	N/A	Limit Switch	Field	New	Westlock	9479NRG4B2M0600	B1LG-7-0003	DI-DCS	DCS	LP1Y-IO-00-0048-001	CBP102	08	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-05	LP100-DQ-0000908D	Y-270-LAY-K215-02	N/A	N/A	On-Valve	Conn. to Limit Switch JB
0	LP1YSV0610WPA	F-0610WPA	1st/2nd WASTE TK TO RAIL STATION #3	LP1Y-DI-00000061	N/A	Solenoid Valve	Field	New	ASCO	F8316H38124VDC	B1LG-7-0003	DO-DCS	DCS	LP1Y-IO-00-0048-001	CBP102	09	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-05	LP100-DQ-0000908D	Y-270-LAY-K215-02	N/A	N/A	On-Valve	Conn. to Limit Switch JB
0	LP1YFV0610WPA	F-0610WPA	1st/2nd WASTE TK TO RAIL STATION #3	LP1Y-DI-00000061	6" DUP PWP (344) (108) 061031	Isolation Valve	Field	New	ITT Richter	30-KNA1100-Series	B1LG-7-0003	N/A	N/A	N/A	N/A	N/A	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-05	LP100-DQ-0000908D	Y-270-LAY-K215-02	IA902A-ZA60	IE103	In-Line	
0	LP1YZSC0610WPB	F-0610WPB	1st/2nd WASTE TK TO RAIL STATION #4	LP1Y-DI-00000061	N/A	Limit Switch	Field	New	Westlock	9479NRG4B2M0600	B1LG-7-0003	DI-DCS	DCS	LP1Y-IO-00-0048-001	CBP102	10	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-06	LP100-DQ-0000908D	Y-270-LAY-K215-02	N/A	N/A	On-Valve	Conn. to Limit Switch JB
0	LP1YZSO0610WPB	F-0610WPB	1st/2nd WASTE TK TO RAIL STATION #4	LP1Y-DI-00000061	N/A	Limit Switch	Field	New	Westlock	9479NRG4B2M0600	B1LG-7-0003	DI-DCS	DCS	LP1Y-IO-00-0048-001	CBP102	11	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-06	LP100-DQ-0000908D	Y-270-LAY-K215-02	N/A	N/A	On-Valve	Conn. to Limit Switch JB
0	LP1YSV0610WPB	F-0610WPB	1st/2nd WASTE TK TO RAIL STATION #4	LP1Y-DI-00000061	N/A	Solenoid Valve	Field	New	ASCO	F8316H38124VDC	B1LG-7-0003	DO-DCS	DCS	LP1Y-IO-00-0048-001	CBP102	12	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-06	LP100-DQ-0000908D	Y-270-LAY-K215-02	N/A	N/A	On-Valve	Conn. to Limit Switch JB
0	LP1YFV0610WPB	F-0610WPB	1st/2nd WASTE TK TO RAIL STATION #4	LP1Y-DI-00000061	6" DUP PWP (344) (108) 061025	Isolation Valve	Field	New	ITT Richter	30-KNA1100-Series	B1LG-7-0003	N/A	N/A	N/A	N/A	N/A	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-06	LP100-DQ-0000908D	Y-270-LAY-K215-02	IA902A-ZA60	IE103	In-Line	
0	LP1YZSO0622N1	F-622N1	N2 TO RAIL STATION #1	LP1Y-DI-00000062	N/A	Limit Switch	Field	New	Westlock	9479NRG4B2M0600	B1LG-7-0003	DI-DCS	DCS	LP1Y-IO-00-0048-001	CBP105	07	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-07	LP100-DQ-0000908D	Y-270-LAY-K215-02	N/A	N/A	On-Valve	Conn. to Limit Switch JB
0	LP1YZSC0622N1	F-622N1	N2 TO RAIL STATION #1	LP1Y-DI-00000062	N/A	Limit Switch	Field	New	Westlock	9479NRG4B2M0600	B1LG-7-0003	DI-DCS	DCS	LP1Y-IO-00-0048-001	CBP105	08	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-07	LP100-DQ-0000908D	Y-270-LAY-K215-02	N/A	N/A	On-Valve	Conn. to Limit Switch JB
0	LP1YSV0622N1	F-622N1	N2 TO RAIL STATION #1	LP1Y-DI-00000062	N/A	Solenoid Valve	Field	New	ASCO	F8316H38124VDC	B1LG-7-0003	DO-DCS	DCS	LP1Y-IO-00-0048-001	CBP105	09	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-07	LP100-DQ-0000908D	Y-270-LAY-K215-02	N/A	N/A	On-Valve	Conn. to Limit Switch JB
0	LP1YFV0622N1	F-622N1	N2 TO RAIL STATION #1	LP1Y-DI-00000062	2" SS N2 (315) E 061053	Isolation Valve	Field	New	Worcester	20E596666TTBW1N4433 Series	B1LG-7-0003	N/A	N/A	N/A	N/A	N/A	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-07	LP100-DQ-0000908D	Y-270-LAY-K215-02	IA901A-ZA61	IE103	In-Line	
0	LP1YZSO0632N1	F-0632N1	N2 TO RAIL STATION #2	LP1Y-DI-00000063	N/A	Limit Switch	Field	New	Westlock	9479NRG4B2M0600	B1LG-7-0003	DI-DCS	DCS	LP1Y-IO-00-0048-001	CBP105	10	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-08	LP100-DQ-0000908D	Y-270-LAY-K215-02	N/A	N/A	On-Valve	Conn. to Limit Switch JB
0	LP1YZSC0632N1	F-0632N1	N2 TO RAIL STATION #2	LP1Y-DI-00000063	N/A	Limit Switch	Field	New	Westlock	9479NRG4B2M0600	B1LG-7-0003	DI-DCS	DCS	LP1Y-IO-00-0048-001	CBP105	11	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-08	LP100-DQ-0000908D	Y-270-LAY-K215-02	N/A	N/A	On-Valve	Conn. to Limit Switch JB
0	LP1YSV0632N1	F-0632N1	N2 TO RAIL STATION #2	LP1Y-DI-00000063	N/A	Solenoid Valve	Field	New	ASCO	F8316H38124VDC	B1LG-7-0003	DO-DCS	DCS	LP1Y-IO-00-0048-001	CBP105	12	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-08	LP100-DQ-0000908D	Y-270-LAY-K215-02	N/A	N/A	On-Valve	Conn. to Limit Switch JB
0	LP1YFV0632N1	F-0632N1	N2 TO RAIL STATION #2	LP1Y-DI-00000063	2" SS N2 (315) E 061055	Isolation Valve	Field	New	Worcester	20E596666TTBW1N4433 Series	B1LG-7-0003	N/A	N/A	N/A	N/A	N/A	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-08	LP100-DQ-0000908D	Y-270-LAY-K215-02	IA901A-ZA61	IE103	In-Line	
0	LP1YLSL0623A	L-0623A	DMF VAP FR RAIL STATION #1 TO DMF TK	LP1Y-DI-00000062	3" SS PVP (315) P 065006	Level Switch-Tuning Folk	Field	New	Endress+Hauser	FTL51B-CCA4AMIAA2DJAAAGJ+JAZ1	B1LG-7-0111	DI-DCS	DCS	LP1Y-IO-00-0048-001	CBP101	11	LP1Y-IPP-00-0048-001/X6-FS07	Hanson	Fluor	LP1Y-275-DIA-K215-09	LP100-DQ-0000907B	Y-270-LAY-K215-02	IF905A	IE903A-ZC19B	Direct Mounted	2" 150#RF Conn., U-Length = 7.8"
0	LP1YFSL0630V	F-0630V	MTBE FR TK-1121/1122 TO RAIL STATION #2	LP1Y-DI-00000063	3" SS PVP (315) P 063005	Flow Switch	Field	New	FCI Fluid Comp.	FLT93S-1B1A4F4H4A003000	B1LG-7-0112	DI-DCS	DCS	LP1Y-IO-00-0048-001	CBP103	03	LP1Y-IPP-00-0048-001/X6-FS03	Hanson	Fluor	LP1Y-275-DIA-K215-10	LP100-DQ-0000907A	Y-270-LAY-K215-02	IF905A	IE903A-ZC19B	Direct Mounted	2" 150#RF Conn., U-Length=6"
0	LP1YFSL0631A	F-0631A	MTBE FR PMP P-0631A TO TK-1121/1122	LP1Y-DI-00000063	3" SS MTBE (315) P 063106	Flow Switch	Field	New	FCI Fluid Comp.	FLT93S-1B1A4F4H4A003000	B1LG-7-0112	DI-DCS	DCS	LP1Y-IO-00-0048-001	CBP103	05	LP1Y-IPP-00-0048-001/X6-FS04	Hanson	Fluor	LP1Y-275-DIA-K215-11	LP100-DQ-0000907A	Y-270-LAY-K215-01	IF905A	IE903A-ZC19B	Direct Mounted	2" 150#RF Conn., U-Length=6"
0	LP1YLSL0633A	L-0633A	ACN FR RAIL STATION #2 TO TK T-1112	LP1Y-DI-00000063	3" SS ACN (315) P 063302	Level Switch-Tuning Folk	Field	New	Endress+Hauser	FTL51B-CCA4AMIAA2DJAAAGJ+JAZ1	B1LG-7-0111	DI-DCS	DCS	LP1Y-IO-00-0048-001	CBP103	02	LP1Y-IPP-00-0048-001/X6-FS14	Hanson	Fluor	LP1Y-275-DIA-K215-12	LP100-DQ-0000907B	Y-270-LAY-K215-02	IF905A	IE903A-ZC19B	Direct Mounted	2" 150#RF Conn., U-Length = 7.8"
0	LP1YFSL0640V	F-0640V	WASTE VAP FR RAIL STATION #3 TO WASTE TK	LP1Y-DI-00000064	3" DUP WV (344) (108) 064001	Flow Switch	Field	New	FCI Fluid Comp.	FLT93S-1B1W4F4H4A003000	B1LG-7-0112	DI-DCS	DCS	LP1Y-IO-00-0048-001	CBP103	09	LP1Y-IPP-00-0048-001/X6-FS05	Hanson	Fluor	LP1Y-275-DIA-K215-13	LP100-DQ-0000907A	Y-270-LAY-K215-02	IF905A	IE903A-ZC19B	Direct Mounted	2" 150#RF Conn., U-Length=6"
0	LP1YFSL0650V	F-0650V	WASTE VAP FR RAIL STATION #4 TO WASTE TK	LP1Y-DI-00000065	3" DUP WV (344) (108) 065001	Flow Switch	Field	New	FCI Fluid Comp.	FLT93S-1B1W4F4H4A003000	B1LG-7-0112	DI-DCS	DCS	LP1Y-IO-00-0048-001	CBP104	06	LP1Y-IPP-00-0048-001/X6-FS08	Hanson	Fluor	LP1Y-275-DIA-K215-14	LP100-DQ-0000907A	Y-270-LAY-K215-02	IF905A	IE903A-ZC19B	Direct Mounted	2" 150#RF Conn., U-Length=6"
0	LP1YLSL0631A	L-0631A	MTBE FR RAIL STATION #2 TO P-0631A	LP1Y-DI-00000063	4" SS MTBE (315) P 063101	Level Switch-Tuning Folk	Field	New	Endress+Hauser	FTL51B-CCA4AMIAA2DJAAAGJ+JAZ1	B1LG-7-0111	DI-DCS	DCS	LP1Y-IO-00-0048-001	CBP101	12	LP1Y-IPP-00-0048-001/X6-FS13	Hanson	Fluor	LP1Y-275-DIA-K215-15	LP100-DQ-0000907B	Y-270-LAY-K215-01	IF905A	IE903A-ZC19B	Direct Mounted	2" 150#RF Conn., U-Length = 7.8"
0	LP1YLSH0640WP	L-0640WP	1st/2nd WASTE RAIL CAR STATION #3	LP1Y-DI-00000064	N/A	Level Switch	Field	New	Scully	SP-PYU(2205 SS)/4P & ST-15-115-ELK	TBD	DI-DCS	DCS	LP1Y-IO-00-0048-001	CBP101	07	N/A	Hanson	SCI	LP1Y-275-DIA-K215-16	LP100-DQ-0000907B	Y-270-LAY-K215-02	N/A	IE903A-ZC19B	IS907, IS900-ZS19	Include 120VAC Level Controller
0	LP1YLSH0650WP	L-0650WP	WASTE RAIL CAR STATION #4	LP1Y-DI-00000065	N/A	Level Switch	Field	New	Scully	SP-PYU(2205 SS)/4P & ST-15-115-ELK	TBD	DI-DCS	DCS	LP1Y-IO-00-0048-001	CBP101	10	N/A	Hanson	SCI	LP1Y-275-DIA-K215-17	LP100-DQ-0000907B	Y-270-LAY-K215-02	N/A	IE903A-ZC19B	IS907, IS900-ZS19	Include 120VAC Level Controller
0	LP1YPIT0610WP	P-0610WP	RUPTURE DISK PSE-0610WP	LP1Y-DI-00000061	2" SS RV (315) (108) 061050	Pressure Transmitter	Field	New	Rosemount	3051TG3A2B21AS5K5D4M5/w-R306ET22	B1LG-7-0103	AI-DCS	DCS	LP1Y-IO-00-0048-001	CBP101	08	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-18	LP100-DQ-00000900A	Y-270-LAY-K215-01	IP906B-Z23	IE900B-ZC19B	N/A	Direct Mounted w/ Elect Trace
0	LP1YPIT0623A	P-0623A	RUPTURE DISK PSE-0623A	LP1Y-DI-00000062	2" SS DMF (315) 062301	Pressure Transmitter	Field	New	Rosemount	3051TG3A2B21ASXXXX	B1LG-7-0103	AI-DCS	DCS	LP1Y-IO-00-0048-001	CBP104	09	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-19	LP100-DQ-00000900A	Y-270-LAY-K215-02	IP906B-Z23	IE900B-ZC19B	IM902-ZS19A, IS900-ZS19	w/ PSA-0623A
0	LP1YPIT0630V	P-0630V	MTBE FR TK-1121/1122 TO RAIL STATION #2	LP1Y-DI-00000063	3" SS PVP (315) P 063005	Pressure Transmitter	Field	New	Rosemount	3051TG3A2B21AS1B4K5Q8D4M5/w-1199	B1LG-7-0103	AI-DCS	DCS	LP1Y-IO-00-0048-001	CBP103	04	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-20	LP100-DQ-00000900A	Y-270-LAY-K215-02	IP904A	IE900B-ZC19B	IS900-ZS19	Remote Mounted w/ Diaph Seal & Capillary
0	LP1YPIT0631A	P-0631A	RUPTURE DISK PSE-0631A	LP1Y-DI-00000063	3" SS MTBE (315) P 063106	Pressure Transmitter	Field	New	Rosemount	3051TG3A2B21AS5K5D4M5Q8/w-R306ET	B1LG-7-0103	AI-DCS	DCS	LP1Y-IO-00-0048-001	CBP103	06	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-21	LP100-DQ-00000900A	Y-270-LAY-K215-01	IP906B-Z23	IE900B-ZC19B	N/A	w/ PSA-0631A
0	LP1YPIT0633A	P-0633A	RUPTURE DISK PSE-0633A	LP1Y-DI-00000063	2" SS ACN (315) P 063302	Pressure Transmitter	Field	New	Rosemount	3051TG3A2B21ASXXXX	B1LG-7-0103	AI-DCS	DCS	LP1Y-IO-00-0048-001	CBP104	03	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-22	LP100-DQ-00000900A	Y-270-LAY-K215-02	IP906B-Z23	IE900B-ZC19B	N/A	w/ PSA-0633A
0	LP1YPIT0640V	P-0640V	WASTE VAP FR RAIL STATION #3 TO WASTE TK	LP1Y-DI-00000064	3" DUP WV (344) (108) 064001	Pressure Transmitter	Field	New	Rosemount	3051TG3A2B21AS1B4K5D4M5/w-1199WP	B1LG-7-0103	AI-DCS	DCS	LP1Y-IO-00-0048-001	CBP103	10	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-23	LP100-DQ-00000900A	Y-270-LAY-K215-02	IP904A	IE900B-ZC19B	IS900-ZS19	Remote Mounted w/ Diaph Seal & Capillary
0	LP1YPIT0650V	P-0650V	WASTE VAP FR RAIL STATION #4 TO WASTE TK	LP1Y-DI-00000065	3" DUP WV (344) (108) 065001	Pressure Transmitter	Field	New	Rosemount	3051TG3A2B21AS1B4K5D4M5/w-1199WP	B1LG-7-0103	AI-DCS	DCS	LP1Y-IO-00-0048-001	CBP104	07	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-24	LP100-DQ-00000900A	Y-270-LAY-K215-02	IP904A	IE900B-ZC19B	IS900-ZS19	Remote Mounted w/ Diaph Seal & Capillary
0	LP1YPSA0610WP	P-0610WP	PSV-0610WP RUPTURE DISK	LP1Y-DI-00000061	2" SS RV (315) (108) 061050	Burst Disc Sensor	Field	New	Continental Disc	2" BDI-FLX	B1LG-7-0008	DI-DCS	DCS	LP1Y-IO-00-0048-001	CBP101	09	LP1Y-IPP-00-0048-001/X6-FS09	Hanson	Fluor	LP1Y-275-DIA-K215-26	LP100-DQ-00000906A	Y-270-LAY-K215-01	IP901K	IJB901A	IM900-ZS23A, IS902-ZS19	
0	LP1YPSE0610WP	P-0610WP	PSV-0610WP RUPTURE DISK	LP1Y-DI-00000061	2" SS RV (315) (108) 061050	Rupture Disk	Field	New	Continental Disc	2" HPX-90 Series	B1LG-7-0008	N/A	N/A	N/A	N/A	N/A	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-26	LP100-DQ-00000906A	Y-270-LAY-K215-01	IP901H	N/A	In-Line	w/ HPX Insert Holder
0	LP1YPSA0623A	P-0623A	PSV-0623A RUPTURE DISK	LP1Y-DI-00000062	2" SS DMF (315) 062301	Burst Disc Sensor	Field	New	Continental Disc	2" BDI-FLX	B1LG-7-0008	DI-DCS	DCS	LP1Y-IO-00-0048-001	CBP104	10	LP1Y-IPP-00-0048-001/X6-FS12	Hanson	Fluor	LP1Y-275-DIA-K215-27	LP100-DQ-00000906A	Y-270-LAY-K215-02	IP901K	IJB901A	IM001-ZS23A, IS901-ZS19	
0	LP1YPSE0623A	P-0623A	PSV-0623A RUPTURE DISK	LP1Y-DI-00000062	2" SS DMF (315) 062301	Rupture Disk	Field	New	Continental Disc	2" HPX-90 Series	B1LG-7-0008	N/A	N/A	N/A	N/A	N/A	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-27	LP100-DQ-00000906A	Y-270-LAY-K215-02	IP901H	N/A	In-Line	w/ HPX Insert Holder
0	LP1YPSA0631A	P-0631A	PSV-0631A RUPTURE DISK	LP1Y-DI-00000063	2" SS MTBE (315) P 063106	Burst Disc Sensor	Field	New	Continental Disc	2" BDI-FLX	B1LG-7-0008	DI-DCS	DCS	LP1Y-IO-00-0048-001	CBP103	07	LP1Y-IPP-00-0048-001/X6-FS10	Hanson	Fluor	LP1Y-275-DIA-K215-28	LP100-DQ-00000906A	Y-270-LAY-K215-01	IP901K	IJB901A	IM900-ZS23A, IS902-ZS19	
0	LP1YPSE0631A	P-0631A	PSV-0631A RUPTURE DISK	LP1Y-DI-00000063	2" SS MTBE (315) P 063106	Rupture Disk	Field	New	Continental Disc	2" HPX-90 Series	B1LG-7-0008	N/A	N/A	N/A	N/A	N/A	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-28	LP100-DQ-00000906A	Y-270-LAY-K215-01	IP901H	N/A	In-Line	w/ HPX Insert Holder
0	LP1YPSA0633A	P-0633A	PSV-0633A RUPTURE DISK	LP1Y-DI-00000063	2" SS ACN (315) P 063302	Burst Disc Sensor	Field	New	Continental Disc	2" BDI-FLX	B1LG-7-0008	DI-DCS	DCS	LP1Y-IO-00-0048-001	CBP104	04	LP1Y-IPP-00-0048-001/X6-FS11	Hanson	Fluor	LP1Y-275-DIA-K215-29	LP100-DQ-00000906A	Y-270-LAY-K215-02	IP901K	IJB901A	IM001-ZS23A, IS902-ZS19	
0	LP1YPSE0633A	P-0633A	PSV-0633A RUPTURE DISK	LP1Y-DI-00000063	2" SS ACN (315) P 063302	Rupture Disk	Field	New	Continental Disc	2" HPX-90 Series	B1LG-7-0008	N/A	N/A	N/A	N/A	N/A	N/A	Hanson	Fluor	LP1Y-275-DIA-K215-29	LP100-DQ-00000906A	Y-270-LAY-K215-02	IP901H	N/A	In-Line	w/ HPX Insert Holder
0	LP1YCVV0610NA	--	N2 HDR TO RAIL STATIONS	LP1Y-DI-00000061	2" SS N2 (100) (315) E 061029	Conservation Vent Valve	Field	New	Protego	DZ/E-50	B1LG-7-0016	N/A	N/A	N/A	N/A	N/A	N/A	Hanson	TBD	N/A	N/A	Y-270-LAY-K215-01	IY900	N/A	In-Line	
0	LP1YCVV0610NB	--	N2 HDR TO RAIL STATIONS	LP1Y-DI-00000061	1.5" SS N2 (315) (315) E 061005	Conservation Vent Valve	Field	New	Protego	DZ/E-50	B1LG-7-0016	N/A	N/A	N/A	N/A	N/A	N/A	Hanson	Fluor	N/A	N/A	Y-270-LAY-K215-01	IY900	N/A	In-Line	
0	LP1YFAL0623A	F-0623A	DMF FR RAIL STATION #1 TO TK-1021/1022	LP1Y-DI-00000062	N/A	DCS Alarm	DCS	New	Emerson	N/A		Soft	DCS	N/A	N/A	N/A	N/A	Emerson	Emerson	N/A	N/A	N/A	N/A	N/A	N/A	
0	LP1YFAL0630V	F-0630V	MTBE FR TK-1121/1122 TO RAIL STATION #2	LP1Y-DI-00000063	N/A	DCS Alarm	DCS	New	Emerson	N/A		Soft	DCS	N/A	N/A	N/A	N/A	Emerson	Emerson	N/A	N/A	N/A	N/A	N/A	N/A	
0	LP1YFAL0631A	F-0631A	MTBE FR PMP P-0631A TO TK-1121/1122	LP1Y-DI-00000063	N/A	DCS Alarm	DCS	New	Emerson	N/A		Soft	DCS	N/A	N/A	N/A	N/A	Emerson	Emerson	N/A	N/A	N/A	N/A	N/A	N/A	
0	LP1YFAL0633A	F-0633A	ACN FR RAIL STATION #2 TO TK T-1112	LP1Y-DI-00000063	N/A	DCS Alarm	DCS	New	Emerson	N/A		Soft	DCS	N/A	N/A	N/A	N/A	Emerson	Emerson	N/A	N/A	N/A	N/A	N/A	N/A	
0	LP1YFAL0640V	F-0640V	WASTE VAP FR RAIL STATION #3 TO WASTE TK	LP1Y-DI-00000064	N/A	DCS Alarm	DCS	New	Emerson	N/A		Soft	DCS	N/A	N/A	N/A	N/A	Emerson	Emerson	N/A	N/A	N/A	N/A	N/A	N/A	
0	LP1YFAL0650V	F-0650V	WASTE VAP FR RAIL STATION #4 TO WASTE TK	LP1Y-DI-00000065	N/A	DCS Alarm	DCS	New	Emerson	N/A		Soft	DCS	N/A	N/A	N/A	N/A	Emerson	Emerson	N/A	N/A	N/A	N/A	N/A	N/A	
0	LP1YFO0631A	--	MTBE FR PMP P-0631A TO TK-1121/1122	LP1Y-DI-00000063	3" SS MTBE (315) P 063106	Flow Orifice	Field	New	Rosemount	1495PG030A1SA01XXXX	B1LG-7-0014	N/A	N/A	N/A	N/A	N/A	N/A	Hanson	Fluor	N/A	N/A	Y-270-LAY-K215-01	N/A	N/A	In-Line	
0	LP1YFO0640WP	--	1st/2nd WASTE TK TO RAIL STATION #3	LP1Y-DI-00000064	6" DUP PWP (344) (108) 061031	Flow Orifice	Field	New	Rosemount	1495PG030A1SA01XXXX	B1LG-7-0014	N/A	N/A	N/A	N/A	N/A	N/A	Hanson	Fluor	N/A	N/A	Y-270-LAY-K215-02	N/A	N/A	In-Line	
0	LP1YFO0650WP	--	1st/2nd WASTE TK TO RAIL STATION #4	LP1Y-DI-00000065	6" DUP PWP (344) (108) 061025	Flow Orifice	Field	New	Rosemount	1495PG030A1SA01XXXX	B1LG-7-0014	N/A	N/A	N/A	N/A	N/A	N/A	Hanson	Fluor	N/A	N/A	Y-270-LAY-K215-02	N/A	N/A	In-Line	
0	LP1YFZIC0610VA	F-0610VA	RAIL STATION #3 VAPOR BALANCE	LP1Y-DI-00000061	N/A	DCS Indicator	DCS	New	Emerson	N/A		Soft	DCS	N/A	N/A	N/A	N/A	Emerson	Emerson	N/A	N/A	N/A	N/A	N/A	N/A	
0	LP1YFZIC0610VB	F-0610VB	RAIL STATION #4 VAPOR BALANCE	LP1Y-DI-00000061	N/A	DCS Indicator	DCS	New	Emerson	N/A		Soft	DCS	N/A	N/A	N/A	N/A	Emerson	Emerson	N/A	N/A	N/A	N/A	N/A	N/A	
0	LP1YFZIC0610WPA	F-0610WPA	1st/2nd WASTE TK TO RAIL STATION #3	LP1Y-DI-00000061	N/A	DCS Indicator	DCS	New	Emerson	N/A		Soft	DCS	N/A	N/A	N/A	N/A	Emerson	Emerson	N/A	N/A	N/A	N/A	N/A	N/A	
0	LP1YFZIC0610WPB	F-0610WPB	1st/2nd WASTE TK TO RAIL STATION #4	LP1Y-DI-00000061	N/A	DCS Indicator	DCS	New	Emerson	N/A		Soft	DCS	N/A	N/A	N/A	N/A	Emerson	Emerson	N/A	N/A	N/A	N/A	N/A	N/A	
0	LP1YFZIC0622N1	F-622N1	N2 TO RAIL STATION #1	LP1Y-DI-00000062	N/A	DCS Indicator	DCS	New	Emerson	N/A		Soft	DCS	N/A	N/A	N/A	N/A	Emerson	Emerson	N/A	N/A	N/A	N/A	N/A	N/A	
0	LP1YFZIC0632N1	F-0632N1	N2 TO RAIL STATION #2	LP1Y-DI-00000063	N/A	DCS Indicator	DCS	New	Emerson	N/A		Soft	DCS	N/A	N/A	N/A	N/A	Emerson	Emerson	N/A	N/A	N/A	N/A	N/A	N/A	
0	LP1YFZIO0610VA	F-0610VA	RAIL STATION #3 VAPOR BALANCE	LP1Y-DI-00000061	N/A	DCS Indicator	DCS	New	Emerson	N/A		Soft	DCS	N/A	N/A	N/A	N/A	Emerson	Emerson	N/A	N/A	N/A	N/A	N/A	N/A	
0	LP1YFZIO0610VB	F-0610VB	RAIL STATION #4 VAPOR BALANCE	LP1Y-DI-00000061	N/A	DCS Indicator	DCS	New	Emerson	N/A		Soft	DCS	N/A	N/A	N/A	N/A	Emerson	Emerson	N/A	N/A	N/A	N/A	N/A	N/A	
0	LP1YFZIO0610WPA	F-0610WPA	1st/2nd WASTE TK TO RAIL STATION #3	LP1Y-DI-00000061	N/A	DCS Indicator	DCS	New	Emerson	N/A		Soft	DCS	N/A	N/A	N/A	N/A	Emerson	Emerson	N/A	N/A	N/A	N/A	N/A	N/A	
0	LP1YFZIO0610WPB	F-0610

Rev.	Tag No.	Loop No.	Service Desc.	P&ID	Line No.	Instrument Desc.	Location	Status	Manufacturer	Model	P.O.	Signal Type	System	Charm I/O Panel	BasePlate	Channel	24VDC Power Panel	Design Scope	Provided by	Wiring Diag.	Typical Wiring Template	Location Plan	Process/Pneu. Detail	Elect Detail	Mounting Detail	Notes
0	LP1YPI0641N	--	N2 LINE PURGING TO RAIL STATION #3	LP1Y-DI-00000064	2" SS N2 (315) E 061003	Pressure Gauge	Field	New	Ashcroft	451279SS Series	B1LG-7-0101	N/A	N/A	N/A	N/A	N/A	N/A	Hanson	Fluor	N/A	N/A	Y-270-LAY-K215-02	IPG900A	N/A	Direct Mounted	w/10202SS04TXCK 150RF Diaph. Seal
0	LP1YPI0650V	P-0650V	WASTE VAP FR RAIL STATION #4 TO WASTE TK	LP1Y-DI-00000065	N/A	DCS Indicator	DCS	New	Emerson	N/A		Soft	DCS	N/A	N/A	N/A	N/A	Emerson	Emerson	N/A	N/A	N/A	N/A	N/A	N/A	
0	LP1YPI0651N	--	N2 TO RAIL STATION #4	LP1Y-DI-00000065	2" SS N2 (315) E 061004	Pressure Gauge	Field	New	Ashcroft	451279SS Series	B1LG-7-0101	N/A	N/A	N/A	N/A	N/A	N/A	Hanson	Fluor	N/A	N/A	Y-270-LAY-K215-02	IPG900A	N/A	IS900-ZS19	Remote Mount w/10202SS 150RF Diaph. Seal
0	LP1YPIRV0610EA	--	PORTABLE H2O TO SS EW-15-0621-01	LP1Y-DI-00000061	2" SS WPC (36) (108) E 061027	Pressure Regulator	Field	New	Fisher	M95H-3063-Series	B1LG-7-0005	N/A	N/A	N/A	N/A	N/A	N/A	Hanson	Fluor	N/A	N/A	Y-270-LAY-K215-01	IP900A	N/A	In-Line	W/ Press Gauge on Outlet
0	LP1YPIRV0610EB	--	PORTABLE H2O TO SS EW-15-0621-02	LP1Y-DI-00000061	2" SS WPC (36) (108) E 061028	Pressure Regulator	Field	New	Fisher	M95H-3063-Series	B1LG-7-0005	N/A	N/A	N/A	N/A	N/A	N/A	Hanson	Fluor	N/A	N/A	Y-270-LAY-K215-02	IP900A	N/A	In-Line	W/ Press Gauge on Outlet
0	LP1YPRV0610NA	--	N2 HDR TO RAIL STATIONS	LP1Y-DI-00000061	2" SS N2 (100) (315) E 061029	Pressure Regulator	Field	New	Fisher	M95H-3063-Series	B1LG-7-0005	N/A	N/A	N/A	N/A	N/A	N/A	Hanson	Fluor	N/A	N/A	Y-270-LAY-K215-01	IP900A	N/A	In-Line	
0	LP1YPRV0610NB	--	N2 HDR TO RAIL STATIONS	LP1Y-DI-00000061	2" SS N2 (315) (315) E 061005	Pressure Regulator	Field	New	Fisher	M95L-3063-Series	B1LG-7-0005	N/A	N/A	N/A	N/A	N/A	N/A	Hanson	Fluor	N/A	N/A	Y-270-LAY-K215-01	IP900A	N/A	In-Line	
0	LP1YPRV0610NC	--	N2 HDR TO RAIL STATION #1	LP1Y-DI-00000061	2" SS N2 (100) (315) E 061053	Pressure Regulator	Field	New	Fisher	M95H-3063-Series	B1LG-7-0005	N/A	N/A	N/A	N/A	N/A	N/A	Hanson	Fluor	N/A	N/A	Y-270-LAY-K215-02	IP900A	N/A	In-Line	
0	LP1YPRV0610ND	--	N2 HDR TO RAIL STATION #2	LP1Y-DI-00000061	2" SS N2 (100) (315) E 061055	Pressure Regulator	Field	New	Fisher	M95H-3063-Series	B1LG-7-0005	N/A	N/A	N/A	N/A	N/A	N/A	Hanson	Fluor	N/A	N/A	Y-270-LAY-K215-02	IP900A	N/A	In-Line	
0	LP1YPSAA0610WP	P-0610WP	PSV-0610WP RUPTURE DISK	LP1Y-DI-00000061	N/A	DCS Alarm	DCS	New	Emerson	N/A		Soft	DCS	N/A	N/A	N/A	N/A	Emerson	Emerson	N/A	N/A	N/A	N/A	N/A	N/A	
0	LP1YPSAA0623A	P-0623A	PSV-0623A RUPTURE DISK	LP1Y-DI-00000062	N/A	DCS Alarm	DCS	New	Emerson	N/A		Soft	DCS	N/A	N/A	N/A	N/A	Emerson	Emerson	N/A	N/A	N/A	N/A	N/A	N/A	
0	LP1YPSAA0631A	P-0631A	PSV-0631A RUPTURE DISK	LP1Y-DI-00000063	N/A	DCS Alarm	DCS	New	Emerson	N/A		Soft	DCS	N/A	N/A	N/A	N/A	Emerson	Emerson	N/A	N/A	N/A	N/A	N/A	N/A	
0	LP1YPSAA0633A	P-0633A	PSV-0633A RUPTURE DISK	LP1Y-DI-00000063	N/A	DCS Alarm	DCS	New	Emerson	N/A		Soft	DCS	N/A	N/A	N/A	N/A	Emerson	Emerson	N/A	N/A	N/A	N/A	N/A	N/A	
0	LP1YPSV0610NA	--	N2 HDR TO RAIL STATIONS	LP1Y-DI-00000061	1.5" SS N2 (100) (315) E 061029	Relief Valve	Field	New	Farris	26GA10A-120/S4	B1LG-7-0033	N/A	N/A	N/A	N/A	N/A	N/A	Hanson	Fluor	N/A	N/A	Y-270-LAY-K215-01	IP905	N/A	In-Line	
0	LP1YPSV0610NB	--	N2 HDR TO RAIL STATIONS	LP1Y-DI-00000061	1.5" SS N2 (315) (315) E 061005	Relief Valve	Field	New	Farris	26GA10A-120/S4	B1LG-7-0033	N/A	N/A	N/A	N/A	N/A	N/A	Hanson	Fluor	N/A	N/A	Y-270-LAY-K215-01	IP905	N/A	In-Line	
0	LP1YPSV0610NC	--	N2 HDR TO RAIL STATION #1	LP1Y-DI-00000061	1.5" SS N2 (100) (315) E 061053	Relief Valve	Field	New	Farris	26GA10A-120/S4	B1LG-7-0033	N/A	N/A	N/A	N/A	N/A	N/A	Hanson	Fluor	N/A	N/A	Y-270-LAY-K215-02	IP905	N/A	In-Line	
0	LP1YPSV0610ND	--	N2 HDR TO RAIL STATION #2	LP1Y-DI-00000061	1.5" SS N2 (100) (315) E 061055	Relief Valve	Field	New	Farris	26GA10A-120/S4	B1LG-7-0033	N/A	N/A	N/A	N/A	N/A	N/A	Hanson	Fluor	N/A	N/A	Y-270-LAY-K215-02	IP905	N/A	In-Line	
0	LP1YPSV0610WP	--	1st/2nd WASTE TK TO RAIL STATION #3& 4	LP1Y-DI-00000061	2" SS RV (315) (108) 061050	Relief Valve	Field	New	Farris	27DAAL-211-311-20S4	B1LG-7-0033	N/A	N/A	N/A	N/A	N/A	N/A	Hanson	Fluor	N/A	N/A	Y-270-LAY-K215-01	IP905	N/A	In-Line	
0	LP1YPSV0623A	--	DMF FR RAIL STATION #1 TO TK-1021/1022	LP1Y-DI-00000062	0.75" SS DMF (315) 062301	Relief Valve	Field	New	Farris	27DAAL-211-311-20S4	B1LG-7-0033	N/A	N/A	N/A	N/A	N/A	N/A	Hanson	Fluor	N/A	N/A	Y-270-LAY-K215-02	IP905	N/A	In-Line	
0	LP1YPSV0631A	--	MTBE FR PMP P-0631A TO TK-1121/1122	LP1Y-DI-00000063	0.75" SS MTBE (315) P 063106	Relief Valve	Field	New	Farris	27DAAL-211-311-20S4	B1LG-7-0033	N/A	N/A	N/A	N/A	N/A	N/A	Hanson	Fluor	N/A	N/A	Y-270-LAY-K215-01	IP905	N/A	In-Line	
0	LP1YPSV0633A	--	ACN FR RAIL STATION #3 TO TK T-1112	LP1Y-DI-00000063	2" SS ACN (315) P 063302	Relief Valve	Field	New	Farris	27DAAL-211-311-20S4	B1LG-7-0033	N/A	N/A	N/A	N/A	N/A	N/A	Hanson	Fluor	N/A	N/A	Y-270-LAY-K215-02	IP905	N/A	In-Line	
0	LP1YSG0623A	--	DMF FR RAIL STATION #1 TO TK-1021/1022	LP1Y-DI-00000062	3" SS DMF (315) 062301	Sight Glass	Field	New	TBD	TBD	TBD	N/A	N/A	N/A	N/A	N/A	N/A	Hanson	Fluor	N/A	N/A	Y-270-LAY-K215-02	N/A	N/A	In-Line	
0	LP1YSG0631A	--	MTBE FR RAIL STATION #3 TO TK T-1121/1122	LP1Y-DI-00000063	4" SS MTBE (315) P 063101	Sight Glass	Field	New	TBD	TBD	TBD	N/A	N/A	N/A	N/A	N/A	N/A	Hanson	Fluor	N/A	N/A	Y-270-LAY-K215-01	N/A	N/A	In-Line	
0	LP1YSG0633A	--	ACN FR RAIL STATION #3 TO TK T-1112	LP1Y-DI-00000063	3" SS ACN (315) P 063101	Sight Glass	Field	New	TBD	TBD	TBD	N/A	N/A	N/A	N/A	N/A	N/A	Hanson	Fluor	N/A	N/A	Y-270-LAY-K215-02	N/A	N/A	In-Line	
0	LP1YZA15062101A	Z-15062101A	SAFETY SHOWER EW-15-0621-01	LP1Y-DI-00000061	N/A	DCS Alarm	DCS	New	Emerson	N/A		Soft	DCS	N/A	N/A	N/A	N/A	Emerson	Emerson	N/A	N/A	N/A	N/A	N/A	N/A	
0	LP1YZA15062101B	Z-15062101B	SAFETY SHOWER EW-15-0621-01	LP1Y-DI-00000061	N/A	DCS Alarm	DCS	New	Emerson	N/A		Soft	DCS	N/A	N/A	N/A	N/A	Emerson	Emerson	N/A	N/A	N/A	N/A	N/A	N/A	
0	LP1YZA15062102A	Z-15062102A	SAFETY SHOWER EW-15-0621-02	LP1Y-DI-00000061	N/A	DCS Alarm	DCS	New	Emerson	N/A		Soft	DCS	N/A	N/A	N/A	N/A	Emerson	Emerson	N/A	N/A	N/A	N/A	N/A	N/A	
0	LP1YZA15062102B	Z-15062102B	SAFETY SHOWER EW-15-0621-02	LP1Y-DI-00000061	N/A	DCS Alarm	DCS	New	Emerson	N/A		Soft	DCS	N/A	N/A	N/A	N/A	Emerson	Emerson	N/A	N/A	N/A	N/A	N/A	N/A	
0	LP1YZS15062101A	Z-15062101A	SAFETY SHOWER EW-15-0621-01	LP1Y-DI-00000061	N/A	Safety Shower Pos. Switch	Field	New	Guardian	TBD	TBD	DI-FGS	FAS	N/A	TBD	TBD	N/A	Hanson	Hanson	TBD	N/A	Y-270-LAY-K215-01	N/A	IE900B-ZC19B	On Safety-Shower	Include w/ Safety Shower
0	LP1YZS15062101B	Z-15062101B	SAFETY SHOWER EW-15-0621-01	LP1Y-DI-00000061	N/A	Safety Shower Pos. Switch	Field	New	Guardian	TBD	TBD	DI-FGS	FAS	N/A	TBD	TBD	N/A	Hanson	Hanson	TBD	N/A	Y-270-LAY-K215-01	N/A	IE900B-ZC19B	On Safety-Shower	Include w/ Safety Shower
0	LP1YZS15062102A	Z-15062102A	SAFETY SHOWER EW-15-0621-02	LP1Y-DI-00000061	N/A	Safety Shower Pos. Switch	Field	New	Guardian	TBD	TBD	DI-FGS	FAS	N/A	TBD	TBD	N/A	Hanson	Hanson	TBD	N/A	Y-270-LAY-K215-02	N/A	IE900B-ZC19B	On Safety-Shower	Include w/ Safety Shower
0	LP1YZS15062102B	Z-15062102B	SAFETY SHOWER EW-15-0621-02	LP1Y-DI-00000061	N/A	Safety Shower Pos. Switch	Field	New	Guardian	TBD	TBD	DI-FGS	FAS	N/A	TBD	TBD	N/A	Hanson	Hanson	TBD	N/A	Y-270-LAY-K215-02	N/A	IE900B-ZC19B	On Safety-Shower	Include w/ Safety Shower

Document Title:	25I0007- Rail Station Instrument I/O List
Document Number:	LP1Y-270-LST-K215-02
Security Class:	Confidential

25I0007- Rail Station Instrument I/O List

Lilly Rail Containment

Hanson Project Number: 25I0007

0	ISSUED FOR CONST.	JN	AT	SS	10-Apr-2026
A	ISSUED FOR REVIEW	JN	AT	SS	12-Jan-2026
Revision	**Issue Status**	**Originator**	**Reviewer**	**Approver**	**Date**

											POWER SUPPLY INFO					
Rev.	Bldg	System Type	Tag Name	Tag Prefix (TP)	Tag Suffix (TS)	Parent Tag	DeltaV DST	Tag Service Description	P&ID	I/O	Status	Power Supply	Power Panel Tag	Circuit Location	Controller Cabinet Tag	Controller (DN)
0	Y	DeltaV 5	LP1YXS00004801A	XS	00004801A	LP1YXS00004801A	XS-00004801A	CHARM I/O Cab.Pwr Sup. Fault Alarm	LP1Y-DI-00000003	DI	NEW	---	---	---	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YTE00004801A	TE	00004801A	LP1YTE00004801A	TE-00004801A	CHARM I/O Cabinet Temp	LP1Y-DI-00000003	T/C	NEW	---	---	---	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YXS00004801B	XS	00004801B	LP1YXS00004801B	XS-00004801B	IPP Cabinet Pwr Sup. Fault Alarm	LP1Y-DI-00000003	DI	NEW	---	---	---	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YTS00004801L	TS	00004801L	LP1YTS00004801L	TS-00004801L	IPP Cabinet Temp Switch Low	LP1Y-DI-00000003	DI	NEW	---	---	---	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YTS00004801H	TS	00004801H	LP1YTS00004801H	TS-00004801H	IPP Cabinet Temp Switch High	LP1Y-DI-00000003	DI	NEW	---	---	---	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YTS10004801D	TS	10004801D	LP1YTS10004801D	TS-10004801D	CHARM IO Cab. Temp Switch High	LP1Y-DI-00000003	DI	NEW	---	---	---	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YLSH0640WP	LSH	0640WP	LP1YLSH0640WP	LSH-0640WP	Rail Car #3 Level Switch	LP1Y-DI-00000064	DI	CONTRACTOR	120 VAC	48-UDP/00-00-01	29	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YPIT0610WP	PIT	0610WP	LP1YPIT0610WP	PIT-0610WP	RV Pressure	LP1Y-DI-00000061	AI	NEW	---	---	---	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YPSA0610WP	PSA	0610WP	LP1YPSE0610WP	PSA-0610WP	PWP Pressure Safety Element	LP1Y-DI-00000061	DI	w/ NEW	24VDC	Y-IPP-00-0048-001	X6/F9	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YLSH0650WP	LSH	0650WP	LP1YLSH0650WP	LSH-0650WP	Rail Car #4 Level Switch	LP1Y-DI-00000065	DI	CONTRACTOR	120 VAC	48-UDP/00-00-02	29	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YLSL0623A	LSL	0623A	LP1YLSL0623A	LSL-0623A	DMF Level Switch	LP1Y-DI-00000062	DI	NEW	24VDC	Y-IPP-00-0048-001	X6/F7	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YLSL0631A	LSL	0631A	LP1YLSL0631A	LSL-0631A	MTBE Level Switch	LP1Y-DI-00000063	DI	NEW	24VDC	Y-IPP-00-0048-001	X6/F13	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YZSC0610VA	ZSC	0610VA	LP1YFV0610VA	ZSC-0610VA	WV Flow Valve	LP1Y-DI-00000061	DI	w/ NEW	---	---	---	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YZSO0610VA	ZSO	0610VA	LP1YFV0610VA	ZSO-0610VA	WV Flow Valve	LP1Y-DI-00000061	DI	w/ NEW	---	---	---	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YSV0610VA	SV	0610VA	LP1YFV0610VA	SV-0610VA	WV Flow Valve	LP1Y-DI-00000061	DO	w/ NEW	---	---	---	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YZSC0610VB	ZSC	0610VB	LP1YFV0610VB	ZSC-0610VB	WV Flow Valve	LP1Y-DI-00000061	DI	w/ NEW	---	---	---	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YZSO0610VB	ZSO	0610VB	LP1YFV0610VB	ZSO-0610VB	WV Flow Valve	LP1Y-DI-00000061	DI	w/ NEW	---	---	---	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YSV0610VB	SV	0610VB	LP1YFV0610VB	SV-0610VB	WV Flow Valve	LP1Y-DI-00000061	DO	w/ NEW	---	---	---	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YZSC0610WPA	ZSC	0610WPA	LP1YFV0610WPA	ZSC-0610WPA	PWP Flow Valve	LP1Y-DI-00000061	DI	w/ NEW	---	---	---	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YZSO0610WPA	ZSO	0610WPA	LP1YFV0610WPA	ZSO-0610WPA	PWP Flow Valve	LP1Y-DI-00000061	DI	w/ NEW	---	---	---	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YSV0610WPA	SV	0610WPA	LP1YFV0610WPA	SV-0610WPA	PWP Flow Valve	LP1Y-DI-00000061	DO	w/ NEW	---	---	---	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YZSC0610WPB	ZSC	0610WPB	LP1YFV0610WPB	ZSC-0610WPB	PWP Flow Valve	LP1Y-DI-00000061	DI	w/ NEW	---	---	---	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YZSO0610WPB	ZSO	0610WPB	LP1YFV0610WPB	ZSO-0610WPB	PWP Flow Valve	LP1Y-DI-00000061	DI	w/ NEW	---	---	---	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YZSC0610WPB	ZSC	0610WPB	LP1YFV0610WPB	ZSC-0610WPB	PWP Flow Valve	LP1Y-DI-00000061	DI	w/ NEW	---	---	---	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YZSO0610WPB	ZSO	0610WPB	LP1YFV0610WPB	ZSO-0610WPB	PWP Flow Valve	LP1Y-DI-00000061	DI	w/ NEW	---	---	---	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YSV0610WPB	SV	0610WPB	LP1YFV0610WPB	SV-0610WPB	PWP Flow Valve	LP1Y-DI-00000061	DO	w/ NEW	---	---	---	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YHS0010SK	HS	0010SK	LP1YHS0010SK	HS-0010SK	Rail Station Walkway Hand Switch	LP1Y-DI-00000001	DI	CONTRACTOR	---	---	---	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YLSL0633A	LSL	0633A	LP1YLSL0633A	LSL-0633A	ACN Level Switch	LP1Y-DI-00000063	DI	NEW	24VDC	Y-IPP-00-0048-001	X6/F14	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YFSL0630V	FSL	0630V	LP1YFSL0630V	FSL-0630V	PVP Flow Switch	LP1Y-DI-00000063	DI	NEW	24VDC	Y-IPP-00-0048-001	X6/F3	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YPIT0630V	PIT	0630V	LP1YPIT0630V	PIT-0630V	PVP Pressure	LP1Y-DI-00000063	AI	NEW	---	---	---	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YFSL0631A	FSL	0631A	LP1YFSL0631A	FSL-0631A	MTBE Flow Switch	LP1Y-DI-00000063	DI	NEW	24VDC	Y-IPP-00-0048-001	X6/F4	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YPIT0631A	PIT	0631A	LP1YPIT0631A	PIT-0631A	MTBE Pressure	LP1Y-DI-00000063	AI	NEW	---	---	---	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YPSA0631A	PSA	0631A	LP1YPSE0631A	PSA-0631A	MTBE Pressure Safety Element	LP1Y-DI-00000063	DI	w/ NEW	24VDC	Y-IPP-00-0048-001	X6/F10	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YFSL0640V	FSL	0640V	LP1YFSL0640V	FSL-0640V	WV Flow Switch	LP1Y-DI-00000064	DI	NEW	24VDC	Y-IPP-00-0048-001	X6/F5	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YPIT0640V	PIT	0640V	LP1YPIT0640V	PIT-0640V	WV Pressure	LP1Y-DI-00000064	AI	NEW	---	---	---	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YPIT0633A	PIT	0633A	LP1YPIT0633A	PIT-0633A	ACN Pressure	LP1Y-DI-00000063	AI	NEW	---	---	---	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YPSA0633A	PSA	0633A	LP1YPSE0633A	PSA-0633A	ACN Pressure Safety Element	LP1Y-DI-00000063	DI	w/ NEW	24VDC	Y-IPP-00-0048-001	X6/F11	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YFSL0650V	FSL	0650V	LP1YFSL0650V	FSL-0650V	WV Flow Switch	LP1Y-DI-00000065	DI	NEW	24VDC	Y-IPP-00-0048-001	X6/F8	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YPIT0650V	PIT	0650V	LP1YPIT0650V	PIT-0650V	WV Pressure	LP1Y-DI-00000065	AI	NEW	---	---	---	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YPIT0623A	PIT	0623A	LP1YPIT0623A	PIT-0623A	DMF Pressure	LP1Y-DI-00000062	AI	NEW	---	---	---	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YPSA0623A	PSA	0623A	LP1YPSE0623A	PSA-0623A	DMF Pressure Safety Element	LP1Y-DI-00000062	DI	w/ NEW	24VDC	Y-IPP-00-0048-001	X6/F12	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YHS0010SL	HS	0010SL	LP1YHS0010SL	HS-0010SL	Rail Station Ground Hand Switch	LP1Y-DI-00000001	DI	CONTRACTOR	---	---	---	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YZSO0622N1	ZSO	0622N1	LP1YFV0622N1	ZSO-0622N1	N2 Flow Valve	LP1Y-DI-00000062	DI	w/ NEW	---	---	---	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YZSC0622N1	ZSC	0622N1	LP1YFV0622N1	ZSC-0622N1	N2 Flow Valve	LP1Y-DI-00000062	DI	w/ NEW	---	---	---	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YSV0622N1	SV	0622N1	LP1YFV0622N1	SV-0622N1	N2 Flow Valve	LP1Y-DI-00000062	DO	w/ NEW	---	---	---	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YZSO0632N1	ZSO	0632N1	LP1YFV0632N1	ZSO-0632N1	N2 Flow Valve	LP1Y-DI-00000063	DI	w/ NEW	---	---	---	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YZSC0632N1	ZSC	0632N1	LP1YFV0632N1	ZSC-0632N1	N2 Flow Valve	LP1Y-DI-00000063	DI	w/ NEW	---	---	---	Y-CTL-10-0043-001	LP1Y5C02
0	Y	DeltaV 5	LP1YSV0632N1	SV	0632N1	LP1YFV0632N1	SV-0632N1	N2 Flow Valve	LP1Y-DI-00000063	DO	w/ NEW	---	---	---	Y-CTL-10-0043-001	LP1Y5C02

				DELTA V					CONTROLLER CABINET								
Rev.	**Bldg**	**System Type**	**Tag Name**	**CIOC No. (FN)**	**Baseplate (SN)**	**Channel (CN)**	**Charm TB Type (CTT)**	**Charm Type (CT)**	**Slot No.**	**IO Module Type**	**Channel No.**	**Terminal Strip**	**Terminal Number (+)**	**Terminal Number (-)**	**Panel Tag 1 (PN1) *(IO Panel)***	**Panel Tag 2 (PN2)**	**Panel Tag 3 (PN3)**
0	Y	DeltaV 5	LP1YXS00004801A	LY500004800101	CBP101	01	3-Wire DI Fused Injected Power Term Blk	DI 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YTE00004801A	LY500004800101	CBP101	02	Thermocouple/mV Term Blk	Thermocouple/mV	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YXS00004801B	LY500004800101	CBP101	03	3-Wire DI Fused Injected Power Term Blk	DI 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	Y-IPP-00-0048-001	---
0	Y	DeltaV 5	LP1YTS00004801L	LY500004800101	CBP101	04	3-Wire DI Fused Injected Power Term Blk	DI 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	Y-IPP-00-0048-001	---
0	Y	DeltaV 5	LP1YTS00004801H	LY500004800101	CBP101	05	3-Wire DI Fused Injected Power Term Blk	DI 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	Y-IPP-00-0048-001	---
0	Y	DeltaV 5	LP1YTS10004801D	LY500004800101	CBP101	06	3-Wire DI Fused Injected Power Term Blk	DI 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	Y-NER-10-0048-001	---
0	Y	DeltaV 5	LP1YLSH0640WP	LY500004800101	CBP101	07	3-Wire DI Fused Injected Power Term Blk	DI 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	Y-LCP-00-0640-001	---
0	Y	DeltaV 5	LP1YPIT0610WP	LY500004800101	CBP101	08	3-Wire AI Fused Injected Power Term Blk	AI 4-20 mA HART	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YPSA0610WP	LY500004800101	CBP101	09	3-Wire DI Fused Injected Power Term Blk	DI 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	Y-FJB-00-0610-001	---
0	Y	DeltaV 5	LP1YLSH0650WP	LY500004800101	CBP101	10	3-Wire DI Fused Injected Power Term Blk	DI 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	Y-LCP-00-0650-001	---
0	Y	DeltaV 5	LP1YLSL0623A	LY500004800101	CBP101	11	3-Wire DI Fused Injected Power Term Blk	DI 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YLSL0631A	LY500004800101	CBP101	12	3-Wire DI Fused Injected Power Term Blk	DI 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YZSC0610VA	LY500004800101	CBP102	01	3-Wire DI Fused Injected Power Term Blk	DI 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YZSO0610VA	LY500004800101	CBP102	02	3-Wire DI Fused Injected Power Term Blk	DI 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YSV0610VA	LY500004800101	CBP102	03	Fused Injected Power Term Blk	DO 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YZSC0610VB	LY500004800101	CBP102	04	3-Wire DI Fused Injected Power Term Blk	DI 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YZSO0610VB	LY500004800101	CBP102	05	3-Wire DI Fused Injected Power Term Blk	DI 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YSV0610VB	LY500004800101	CBP102	06	Fused Injected Power Term Blk	DO 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YZSC0610WPA	LY500004800101	CBP102	07	3-Wire DI Fused Injected Power Term Blk	DI 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YZSO0610WPA	LY500004800101	CBP102	08	3-Wire DI Fused Injected Power Term Blk	DI 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YSV0610WPA	LY500004800101	CBP102	09	Fused Injected Power Term Blk	DO 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YZSC0610WPB	LY500004800101	CBP102	10	3-Wire DI Fused Injected Power Term Blk	DI 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YZSO0610WPB	LY500004800101	CBP102	11	3-Wire DI Fused Injected Power Term Blk	DI 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YSV0610WPB	LY500004800101	CBP102	12	Fused Injected Power Term Blk	DO 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YHS0010SK	LY500004800101	CBP103	01	3-Wire DI Fused Injected Power Term Blk	DI 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YLSL0633A	LY500004800101	CBP103	02	3-Wire DI Fused Injected Power Term Blk	DI 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YFSL0630V	LY500004800101	CBP103	03	3-Wire DI Fused Injected Power Term Blk	DI 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YPIT0630V	LY500004800101	CBP103	04	3-Wire AI Fused Injected Power Term Blk	AI 4-20 mA HART	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YFSL0631A	LY500004800101	CBP103	05	3-Wire DI Fused Injected Power Term Blk	DI 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YPIT0631A	LY500004800101	CBP103	06	3-Wire AI Fused Injected Power Term Blk	AI 4-20 mA HART	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YPSA0631A	LY500004800101	CBP103	07	3-Wire DI Fused Injected Power Term Blk	DI 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	Y-FJB-00-0631-001	---
0	Y	DeltaV 5	LP1YFSL0640V	LY500004800101	CBP103	09	3-Wire DI Fused Injected Power Term Blk	DI 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YPIT0640V	LY500004800101	CBP103	10	3-Wire AI Fused Injected Power Term Blk	AI 4-20 mA HART	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YPIT0633A	LY500004800101	CBP104	03	3-Wire AI Fused Injected Power Term Blk	AI 4-20 mA HART	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YPSA0633A	LY500004800101	CBP104	04	3-Wire DI Fused Injected Power Term Blk	DI 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	Y-FJB-00-0633-001	---
0	Y	DeltaV 5	LP1YFSL0650V	LY500004800101	CBP104	06	3-Wire DI Fused Injected Power Term Blk	DI 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YPIT0650V	LY500004800101	CBP104	07	3-Wire AI Fused Injected Power Term Blk	AI 4-20 mA HART	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YPIT0623A	LY500004800101	CBP104	09	3-Wire AI Fused Injected Power Term Blk	AI 4-20 mA HART	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YPSA0623A	LY500004800101	CBP104	10	3-Wire DI Fused Injected Power Term Blk	DI 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	Y-FJB-00-0623-001	---
0	Y	DeltaV 5	LP1YHS0010SL	LY500004800101	CBP105	01	3-Wire DI Fused Injected Power Term Blk	DI 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YZSO0622N1	LY500004800101	CBP105	07	3-Wire DI Fused Injected Power Term Blk	DI 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YZSC0622N1	LY500004800101	CBP105	08	3-Wire DI Fused Injected Power Term Blk	DI 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YSV0622N1	LY500004800101	CBP105	09	Fused Injected Power Term Blk	DO 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YZSO0632N1	LY500004800101	CBP105	10	3-Wire DI Fused Injected Power Term Blk	DI 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YZSC0632N1	LY500004800101	CBP105	11	3-Wire DI Fused Injected Power Term Blk	DI 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	---	---
0	Y	DeltaV 5	LP1YSV0632N1	LY500004800101	CBP105	12	Fused Injected Power Term Blk	DO 24V DC Isolated	---	---	---	---	---	---	Y-IO-00-0048-001	---	---

Rev.	Bldg	System Type	Tag Name	Cable Number 1 (CN1)	Cable Number 2 (CN2)	Cable Number 3 (CN3)	Cable Number 4 (CN4)	Cable Number 5 (CN5)	Cable Number 6 (CN6)	Wiring Diagram
0	Y	DeltaV 5	LP1YXS00004801A	---	---	---	---	---	---	LP1Y-275-DIA-K215-34
0	Y	DeltaV 5	LP1YTE00004801A	---	---	---	---	---	---	NA
0	Y	DeltaV 5	LP1YXS00004801B	XS00004801B_Y-IPP-00-0048-001/Y-IO-00-0048-001	---	---	---	---	---	NA
0	Y	DeltaV 5	LP1YTS00004801L	TS00004801L_Y-IPP-00-0048-001/Y-IO-00-0048-001	---	---	---	---	---	LP1Y-275-DIA-K215-33
0	Y	DeltaV 5	LP1YTS00004801H	TS00004801H_Y-IPP-00-0048-001/Y-IO-00-0048-001	---	---	---	---	---	LP1Y-275-DIA-K215-32
0	Y	DeltaV 5	LP1YTS10004801D	TS10004801D_Y-NER-10-0048-001/Y-IO-00-0048-001	---	---	---	---	---	NA
0	Y	DeltaV 5	LP1YLSH0640WP	LSH0640WP_Y-LCP-00-0640-001/Y-IO-00-0048-001	LSH0640WP/Y-LCP-00-0640-001	YLCP000640001DS-P/CKT #29	YLCP000640001-P	---	---	LP1Y-275-DIA-K215-16
0	Y	DeltaV 5	LP1YPIT0610WP	PIT0610WP/Y-IO-00-0048-001	---	---	---	---	---	LP1Y-275-DIA-K215-18
0	Y	DeltaV 5	LP1YPSA0610WP	PSA0610WP_Y-FJB-00-0610-001/Y-IO-00-0048-001	PSE0610WP/Y-FJB-00-0610-001	Y-FJB-00-0610-001/Y-IPP-00-0048-001/X6/F9	PSE0610WP/PSA0610WP_1	PSE0610WP/PSA0610WP_2	---	LP1Y-275-DIA-K215-26
0	Y	DeltaV 5	LP1YLSH0650WP	LSH0650WP_Y-LCP-00-0650-001/Y-IO-00-0048-001	LSH0650WP/Y-LCP-00-0650-001	YLCP000650001DS-P/CKT #29	YLCP000650001-P	---	---	LP1Y-275-DIA-K215-17
0	Y	DeltaV 5	LP1YLSL0623A	LSL0623A/Y-IO-00-0048-001	LSL0623A/Y-IPP-00-0048-001/X6/F7	---	---	---	---	LP1Y-275-DIA-K215-09
0	Y	DeltaV 5	LP1YLSL0631A	LSL0631A/Y-IO-00-0048-001	LSL0631A/Y-IPP-00-0048-001/X6/F13	---	---	---	---	LP1Y-275-DIA-K215-15
0	Y	DeltaV 5	LP1YZSC0610VA	---	FV0610VA_ZSC/Y-IO-00-0048-001	---	---	---	---	LP1Y-275-DIA-K215-03
0	Y	DeltaV 5	LP1YZSO0610VA	FV0610VA_ZSO/Y-IO-00-0048-001	---	---	---	---	---	LP1Y-275-DIA-K215-03
0	Y	DeltaV 5	LP1YSV0610VA	---	---	FV0610VA_SV/Y-IO-00-0048-001	---	---	---	LP1Y-275-DIA-K215-03
0	Y	DeltaV 5	LP1YZSC0610VB	---	FV0610VB_ZSC/Y-IO-00-0048-001	---	---	---	---	LP1Y-275-DIA-K215-04
0	Y	DeltaV 5	LP1YZSO0610VB	FV0610VB_ZSO/Y-IO-00-0048-001	---	---	---	---	---	LP1Y-275-DIA-K215-04
0	Y	DeltaV 5	LP1YSV0610VB	---	---	FV0610VB_SV/Y-IO-00-0048-001	---	---	---	LP1Y-275-DIA-K215-04
0	Y	DeltaV 5	LP1YZSC0610WPA	---	FV0610WPA_ZSC/Y-IO-00-0048-001	---	---	---	---	LP1Y-275-DIA-K215-05
0	Y	DeltaV 5	LP1YZSO0610WPA	FV0610WPA_ZSO/Y-IO-00-0048-001	---	---	---	---	---	LP1Y-275-DIA-K215-05
0	Y	DeltaV 5	LP1YSV0610WPA	---	---	FV0610WPA_SV/Y-IO-00-0048-001	---	---	---	LP1Y-275-DIA-K215-05
0	Y	DeltaV 5	LP1YZSC0610WPB	---	FV0610WPB_ZSC/Y-IO-00-0048-001	---	---	---	---	LP1Y-275-DIA-K215-06
0	Y	DeltaV 5	LP1YZSO0610WPB	FV0610WPB_ZSO/Y-IO-00-0048-001	---	---	---	---	---	LP1Y-275-DIA-K215-06
0	Y	DeltaV 5	LP1YSV0610WPB	---	---	FV0610WPB_SV/Y-IO-00-0048-001	---	---	---	LP1Y-275-DIA-K215-06
0	Y	DeltaV 5	LP1YHS0010SK	HS0010SK/Y-IO-00-0048-001	---	---	---	---	---	LP1Y-275-DIA-K215-01
0	Y	DeltaV 5	LP1YLSL0633A	LSL0633A/Y-IO-00-0048-001	LSL0633A/Y-IPP-00-0048-001/X6/F14	---	---	---	---	LP1Y-275-DIA-K215-12
0	Y	DeltaV 5	LP1YFSL0630V	FSL0630V/Y-IO-00-0048-001	FSL0630V/Y-IPP-00-0048-001/X6/F3	---	---	---	---	LP1Y-275-DIA-K215-10
0	Y	DeltaV 5	LP1YPIT0630V	PIT0630V/Y-IO-00-0048-001	---	---	---	---	---	LP1Y-275-DIA-K215-20
0	Y	DeltaV 5	LP1YFSL0631A	FSL0631A/Y-IO-00-0048-001	FSL0631A/Y-IPP-00-0048-001/X6/F4	---	---	---	---	LP1Y-275-DIA-K215-11
0	Y	DeltaV 5	LP1YPIT0631A	PIT0631A/Y-IO-00-0048-001	---	---	---	---	---	LP1Y-275-DIA-K215-21
0	Y	DeltaV 5	LP1YPSA0631A	PSA0631A_Y-FJB-00-0631-001/Y-IO-00-0048-001	PSE0631A/Y-FJB-00-0631-001	Y-FJB-00-0631-001/Y-IPP-00-0048-001/X6/F10	PSE0631A/PSA0631A_1	PSE0631A/PSA0631A_2	---	LP1Y-275-DIA-K215-28
0	Y	DeltaV 5	LP1YFSL0640V	FSL0640V/Y-IO-00-0048-001	FSL0640V/Y-IPP-00-0048-001/X6/F5	---	---	---	---	LP1Y-275-DIA-K215-13
0	Y	DeltaV 5	LP1YPIT0640V	PIT0640V/Y-IO-00-0048-001	---	---	---	---	---	LP1Y-275-DIA-K215-23
0	Y	DeltaV 5	LP1YPIT0633A	PIT0633A/Y-IO-00-0048-001	---	---	---	---	---	LP1Y-275-DIA-K215-22
0	Y	DeltaV 5	LP1YPSA0633A	PSA0633A_Y-FJB-00-0633-001/Y-IO-00-0048-001	PSE0633A/Y-FJB-00-0633-001	Y-FJB-00-0633-001/Y-IPP-00-0048-001/X6/F11	PSE0633A/PSA0633A_1	PSE0633A/PSA0633A_2	---	LP1Y-275-DIA-K215-29
0	Y	DeltaV 5	LP1YFSL0650V	FSL0650V/Y-IO-00-0048-001	FSL0650V/Y-IPP-00-0048-001/X6/F8	---	---	---	---	LP1Y-275-DIA-K215-14
0	Y	DeltaV 5	LP1YPIT0650V	PIT0650V/Y-IO-00-0048-001	---	---	---	---	---	LP1Y-275-DIA-K215-24
0	Y	DeltaV 5	LP1YPIT0623A	PIT0623A/Y-IO-00-0048-001	---	---	---	---	---	LP1Y-275-DIA-K215-19
0	Y	DeltaV 5	LP1YPSA0623A	PSA0623A_Y-FJB-00-0623-001/Y-IO-00-0048-001	PSE0623A/Y-FJB-00-0623-001	Y-FJB-00-0623-001/Y-IPP-00-0048-001/X6/F12	PSE0623A/PSA0623A_1	PSE0623A/PSA0623A_2	---	LP1Y-275-DIA-K215-27
0	Y	DeltaV 5	LP1YHS0010SL	HS0010SL/Y-IO-00-0048-001	---	---	---	---	---	LP1Y-275-DIA-K215-02
0	Y	DeltaV 5	LP1YZSO0622N1	FV0622N1_ZSO/Y-IO-00-0048-001	---	---	---	---	---	LP1Y-275-DIA-K215-07
0	Y	DeltaV 5	LP1YZSC0622N1	---	FV0622N1_ZSC/Y-IO-00-0048-001	---	---	---	---	LP1Y-275-DIA-K215-07
0	Y	DeltaV 5	LP1YSV0622N1	---	---	FV0622N1_SV/Y-IO-00-0048-001	---	---	---	LP1Y-275-DIA-K215-07
0	Y	DeltaV 5	LP1YZSO0632N1	FV0632N1_ZSO/Y-IO-00-0048-001	---	---	---	---	---	LP1Y-275-DIA-K215-08
0	Y	DeltaV 5	LP1YZSC0632N1	---	FV0632N1_ZSC/Y-IO-00-0048-001	---	---	---	---	LP1Y-275-DIA-K215-08
0	Y	DeltaV 5	LP1YSV0632N1	---	---	FV0632N1_SV/Y-IO-00-0048-001	---	---	---	LP1Y-275-DIA-K215-08

							SOFTWARE PORT (DELTAV)				HARDWARE PORT (DELTAV)		REMOTE SLOT	CONTROL MODULE	
Rev.	**Bldg**	**System Type**	**Tag Name**	**Calibrated Range From**	**Calibrated Range To**	**Calibrated Range UOM**	**CHARMS**	**P01**	**PDT**	**LDT**	**CIOC**	**IP Adress [Ethernet/IP Gateway]**	**Card Address [ASD, DeltaV]**	**Control Module Name**	**Comments**
0	Y	DeltaV 5	LP1YXS00004801A	---	---	---									
0	Y	DeltaV 5	LP1YTE00004801A	---	---	---									
0	Y	DeltaV 5	LP1YXS00004801B	---	---	---									For additional wiring details, see B1LG-7-0303-00-00008 drawing
0	Y	DeltaV 5	LP1YTS00004801L	---	---	---									For additional wiring details, see B1LG-7-0303-00-00009 drawing
0	Y	DeltaV 5	LP1YTS00004801H	---	---	---									For additional wiring details, see B1LG-7-0303-00-00009 drawing
0	Y	DeltaV 5	LP1YTS10004801D	---	---	---									For additional wiring details, see B1LG-7-0303-00-00007 drawing
0	Y	DeltaV 5	LP1YLSH0640WP	---	---	---									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YPIT0610WP	0	200	psig									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YPSA0610WP	---	---	---									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YLSH0650WP	---	---	---									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YLSL0623A	---	---	---									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YLSL0631A	---	---	---									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YZSC0610VA	---	---	---									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YZSO0610VA	---	---	---									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YSV0610VA	---	---	---									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YZSC0610VB	---	---	---									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YZSO0610VB	---	---	---									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YSV0610VB	---	---	---									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YZSC0610WPA	---	---	---									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YZSO0610WPA	---	---	---									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YSV0610WPA	---	---	---									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YZSC0610WPB	---	---	---									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YZSO0610WPB	---	---	---									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YSV0610WPB	---	---	---									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YHS0010SK	---	---	---									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YLSL0633A	---	---	---									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YFSL0630V	---	---	---									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YPIT0630V	-14.7	100	psig									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YFSL0631A	---	---	---									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YPIT0631A	0	200	psig									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YPSA0631A	---	---	---									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YFSL0640V	---	---	---									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YPIT0640V	-14.7	100	psig									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YPIT0633A	0	200	psig									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YPSA0633A	---	---	---									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YFSL0650V	---	---	---									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YPIT0650V	-14.7	100	psig									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YPIT0623A	0	200	psig									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YPSA0623A	---	---	---									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YHS0010SL	---	---	---									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YZSO0622N1	---	---	---									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YZSC0622N1	---	---	---									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YSV0622N1	---	---	---									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YZSO0632N1	---	---	---									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YZSC0632N1	---	---	---									Wiring and Installation is under K215 scope.
0	Y	DeltaV 5	LP1YSV0632N1	---	---	---									Wiring and Installation is under K215 scope.

Document Title:	25I0007- Rail Station Instrument Cable Schedule
Document Number:	LP1Y-270-LST-K215-03
Security Class:	Confidential

25I0007- Rail Station Instrument Cable Schedule

Lilly Rail Containment

Hanson Project Number: 25I0007

0	ISSUED FOR CONST.	JN	AT	SS	10-Apr-2026
A	ISSUED FOR REVIEW	JN	AT	SS	12-Jan-2026
Revision	**Issue Status**	**Originator**	**Reviewer**	**Approver**	**Date**

INSTRUMENT CABLE SCHEDULE

Rev.	No.	Cable Tag	From	To	Description	Cable Type	Voltage	Notes
0	01	FSL0630V/Y-IO-00-0048-001	FSL0630V	LP1Y-IO-00-0048-001	Discrete Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	02	FSL0631A/Y-IO-00-0048-001	FSL0631A	LP1Y-IO-00-0048-001	Discrete Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	03	FSL0640V/Y-IO-00-0048-001	FSL0640V	LP1Y-IO-00-0048-001	Discrete Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	04	FSL0650V/Y-IO-00-0048-001	FSL0650V	LP1Y-IO-00-0048-001	Discrete Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	05	FV0610VA_SV/Y-IO-00-0048-001	FV0610VA_SV	LP1Y-IO-00-0048-001	Discrete Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	06	FV0610VA_ZSC/Y-IO-00-0048-001	FV0610VA_ZSC	LP1Y-IO-00-0048-001	Discrete Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	07	FV0610VA_ZSO/Y-IO-00-0048-001	FV0610VA_ZSO	LP1Y-IO-00-0048-001	Discrete Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	08	FV0610VB_SV/Y-IO-00-0048-001	FV0610VB_SV	LP1Y-IO-00-0048-001	Discrete Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	09	FV0610VB_ZSC/Y-IO-00-0048-001	FV0610VB_ZSC	LP1Y-IO-00-0048-001	Discrete Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	10	FV0610VB_ZSO/Y-IO-00-0048-001	FV0610VB_ZSO	LP1Y-IO-00-0048-001	Discrete Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	11	FV0610WPA_SV/Y-IO-00-0048-001	FV0610WPA_SV	LP1Y-IO-00-0048-001	Discrete Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	12	FV0610WPA_ZSC/Y-IO-00-0048-001	FV0610WPA_ZSC	LP1Y-IO-00-0048-001	Discrete Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	13	FV0610WPA_ZSO/Y-IO-00-0048-001	FV0610WPA_ZSO	LP1Y-IO-00-0048-001	Discrete Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	14	FV0610WPB_SV/Y-IO-00-0048-001	FV0610WPB_SV	LP1Y-IO-00-0048-001	Discrete Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	15	FV0610WPB_ZSC/Y-IO-00-0048-001	FV0610WPB_ZSC	LP1Y-IO-00-0048-001	Discrete Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	16	FV0610WPB_ZSO/Y-IO-00-0048-001	FV0610WPB_ZSO	LP1Y-IO-00-0048-001	Discrete Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	17	FV0622N1_SV/Y-IO-00-0048-001	FV0622N1_SV	LP1Y-IO-00-0048-001	Discrete Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	18	FV0622N1_ZSC/Y-IO-00-0048-001	FV0622N1_ZSC	LP1Y-IO-00-0048-001	Discrete Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	19	FV0622N1_ZSO/Y-IO-00-0048-001	FV0622N1_ZSO	LP1Y-IO-00-0048-001	Discrete Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	20	FV0632N1_SV/Y-IO-00-0048-001	FV0632N1_SV	LP1Y-IO-00-0048-001	Discrete Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	21	FV0632N1_ZSC/Y-IO-00-0048-001	FV0632N1_ZSC	LP1Y-IO-00-0048-001	Discrete Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	22	FV0632N1_ZSO/Y-IO-00-0048-001	FV0632N1_ZSO	LP1Y-IO-00-0048-001	Discrete Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	23	HS0010SK/Y-IO-00-0048-001	HS0010SK	LP1Y-IO-00-0048-001	Discrete Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	24	HS0010SL/Y-IO-00-0048-001	HS0010SL	LP1Y-IO-00-0048-001	Discrete Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	25	LSH0640WP/Y-LCP-00-0640-001	LSH0640WP	Y-LCP-00-0640-001 Level Controller	Signal from Portable LSH Probe to Level Controller	Vendor Cable with Plugs	300V	Roll on hanger
0	26	LSH0640WP_Y-LCP-00-0640-001/Y-IO-00-0048-001	Y-LCP-00-0640-001	LP1Y-IO-00-0048-001	Discrete Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	27	LSH0650WP/Y-LCP-00-0650-001	LSH0650WP	Y-LCP-00-0650-001 Level Controller	Signal from Portable LSH Probe to Level Controller	Vendor Cable with Plugs	300V	Roll on hanger
0	28	LSH0650WP_Y-LCP-00-0650-001/Y-IO-00-0048-001	Y-LCP-00-0650-001	LP1Y-IO-00-0048-001	Discrete Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	29	LSL0623A/Y-IO-00-0048-001	LSL0623A	LP1Y-IO-00-0048-001	Discrete Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	30	LSL0631A/Y-IO-00-0048-001	LSL0631A	LP1Y-IO-00-0048-001	Discrete Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	31	LSL0633A/Y-IO-00-0048-001	LSL0633A	LP1Y-IO-00-0048-001	Discrete Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	32	PSE0610WP/Y-FJB-00-0610-001	PSE0610WP AMPLIFIER MODULE	LP1Y-FJB-00-0610-001	Rupture Disk Amplifier Module to I.S. Barrier JB	1-PR, TWISTED, #18, SHLD, 300V, ITC (JACKET COLOR: BLU)	300V	Field routed
0	33	PSE0623A/Y-FJB-00-0623-001	PSE0610WP AMPLIFIER MODULE	LP1Y-FJB-00-0623-001	Rupture Disk Amplifier Module to I.S. Barrier JB	1-PR, TWISTED, #18, SHLD, 300V, ITC (JACKET COLOR: BLU)	300V	Field routed
0	34	PSE0631A/Y-FJB-00-0631-001	PSE0610WP AMPLIFIER MODULE	LP1Y-FJB-00-0631-001	Rupture Disk Amplifier Module to I.S. Barrier JB	1-PR, TWISTED, #18, SHLD, 300V, ITC (JACKET COLOR: BLU)	300V	Field routed
0	35	PSE0633A/Y-FJB-00-0633-001	PSE0610WP AMPLIFIER MODULE	LP1Y-FJB-00-0633-001	Rupture Disk Amplifier Module to I.S. Barrier JB	1-PR, TWISTED, #18, SHLD, 300V, ITC (JACKET COLOR: BLU)	300V	Field routed
0	36	PSA0610WP_Y-FJB-00-0610-001/Y-IO-00-0048-001	Y-FJB-00-0610-001	LP1Y-IO-00-0048-001	Rupture Disk I.S. Safety Barrier to CHARM I/O	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	37	PSA0623A_Y-FJB-00-0623-001/Y-IO-00-0048-001	Y-FJB-00-0623-001	LP1Y-IO-00-0048-001	Rupture Disk I.S. Safety Barrier to CHARM I/O	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	38	PSA0631A_Y-FJB-00-0631-001/Y-IO-00-0048-001	Y-FJB-00-0631-001	LP1Y-IO-00-0048-001	Rupture Disk I.S. Safety Barrier to CHARM I/O	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	39	PSA0633A_Y-FJB-00-0633-001/Y-IO-00-0048-001	Y-FJB-00-0633-001	LP1Y-IO-00-0048-001	Rupture Disk I.S. Safety Barrier to CHARM I/O	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	40	PSE0610WP/PSA0610WP_1	PSE0610WP RUTURE DISK	PSA0610WP INTERFACE MODULE	Rupture Sensor to Interface Module	Rupture Sensor Pre-Amp Cable with Plug (Vendor Cable)	300V	Field routed
0	41	PSE0623A/PSA0623A_1	PSE0610WP RUTURE DISK	PSE0610WP INTERFACE MODULE	Rupture Sensor to Interface Module	Rupture Sensor Pre-Amp Cable with Plug (Vendor Cable)	300V	Field routed
0	42	PSE0631A/PSA0631A_1	PSE0610WP RUTURE DISK	PSE0610WP INTERFACE MODULE	Rupture Sensor to Interface Module	Rupture Sensor Pre-Amp Cable with Plug (Vendor Cable)	300V	Field routed
0	43	PSE0633A/PSA0633A_1	PSE0610WP RUTURE DISK	PSE0610WP INTERFACE MODULE	Rupture Sensor to Interface Module	Rupture Sensor Pre-Amp Cable with Plug (Vendor Cable)	300V	Field routed
0	44	PSE0610WP/PSA0610WP_2	PSA0610WP INTERFACE MODULE	PSA0610WP CONDUIT BOX	Rupture Interface Module to Vendor Conduit Box	Rupture Sensor Output Cable with Plug (Vendor Cable)	300V	Field routed
0	45	PSE0623A/PSA0623A_2	PSA0610WP INTERFACE MODULE	PSA0610WP CONDUIT BOX	Rupture Interface Module to Vendor Conduit Box	Rupture Sensor Output Cable with Plug (Vendor Cable)	300V	Field routed
0	46	PSE0631A/PSA0631A_2	PSA0610WP INTERFACE MODULE	PSA0610WP CONDUIT BOX	Rupture Interface Module to Vendor Conduit Box	Rupture Sensor Output Cable with Plug (Vendor Cable)	300V	Field routed
0	47	PSE0633A/PSA0633A_2	PSA0610WP INTERFACE MODULE	PSA0610WP CONDUIT BOX	Rupture Interface Module to Vendor Conduit Box	Rupture Sensor Output Cable with Plug (Vendor Cable)	300V	Field routed
0	48	PIT0610WP/Y-IO-00-0048-001	PIT0610WP	LP1Y-IO-00-0048-001	Analog Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, TWISTED W/ SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	49	PIT0623A/Y-IO-00-0048-001	PIT0623A	LP1Y-IO-00-0048-001	Analog Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, TWISTED W/ SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	50	PIT0630V/Y-IO-00-0048-001	PIT0630V	LP1Y-IO-00-0048-001	Analog Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, TWISTED W/ SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	51	PIT0631A/Y-IO-00-0048-001	PIT0631A	LP1Y-IO-00-0048-001	Analog Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, TWISTED W/ SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	52	PIT0633A/Y-IO-00-0048-001	PIT0633A	LP1Y-IO-00-0048-001	Analog Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, TWISTED W/ SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed

INSTRUMENT CABLE SCHEDULE

Rev.	No.	Cable Tag	From	To	Description	Cable Type	Voltage	Notes
0	53	PIT0640V/Y-IO-00-0048-001	PIT0640V	LP1Y-IO-00-0048-001	Analog Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, TWISTED W/ SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	54	PIT0650V/Y-IO-00-0048-001	PIT0650V	LP1Y-IO-00-0048-001	Analog Signal from Field Instrument to CHARM I/O Panel	1-PR, #18 AWG, TWISTED W/ SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	55	FSL0630V/Y-IPP-00-0048-001/TSX6/FS03	FSL0630V	LP1Y-IPP-00-0048-001/X6/F3	24VDC from IPP to Field Instrument	1-TRIAD, #16 AWG, ARMORED, NO SHLD, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	56	FSL0631A/Y-IPP-00-0048-001/TSX6/FS04	FSL0631A	LP1Y-IPP-00-0048-001/X6/F4	24VDC from IPP to Field Instrument	1-TRIAD, #16 AWG, ARMORED, NO SHLD, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	57	FSL0640V/Y-IPP-00-0048-001/TSX6/FS05	FSL0640V	LP1Y-IPP-00-0048-001/X6/F5	24VDC from IPP to Field Instrument	1-TRIAD, #16 AWG, ARMORED, NO SHLD, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	58	FSL0650V/Y-IPP-00-0048-001/TSX6/FS08	FSL0650V	LP1Y-IPP-00-0048-001/X6/F8	24VDC from IPP to Field Instrument	1-TRIAD, #16 AWG, ARMORED, NO SHLD, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	59	LSL0623A/Y-IPP-00-0048-001/TSX6/FS07	LSL0623A	LP1Y-IPP-00-0048-001/X6/F7	24VDC from IPP to Field Instrument	1-TRIAD, #16 AWG, ARMORED, NO SHLD, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	60	LSL0631A/Y-IPP-00-0048-001/TSX6/FS13	LSL0631A	LP1Y-IPP-00-0048-001/X6/F13	24VDC from IPP to Field Instrument	1-TRIAD, #16 AWG, ARMORED, NO SHLD, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	61	LSL0633A/Y-IPP-00-0048-001/TSX6/FS14	LSL0633A	LP1Y-IPP-00-0048-001/X6/F14	24VDC from IPP to Field Instrument	1-TRIAD, #16 AWG, ARMORED, NO SHLD, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	62	Y-FJB-00-0610-001/Y-IPP-00-0048-001/TSX6/FS09	Y-FJB-00-0610-001	LP1Y-IPP-00-0048-001/X6/F9	24VDC from IPP to Field Instrument	1-TRIAD, #16 AWG, ARMORED, NO SHLD, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	63	Y-FJB-00-0623-001/Y-IPP-00-0048-001/TSX6/FS12	Y-FJB-00-0623-001	LP1Y-IPP-00-0048-001/X6/F12	24VDC from IPP to Field Instrument	1-TRIAD, #16 AWG, ARMORED, NO SHLD, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	64	Y-FJB-00-0631-001/Y-IPP-00-0048-001/TSX6/FS10	Y-FJB-00-0631-001	LP1Y-IPP-00-0048-001/X6/F10	24VDC from IPP to Field Instrument	1-TRIAD, #16 AWG, ARMORED, NO SHLD, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	65	Y-FJB-00-0633-001/Y-IPP-00-0048-001/TSX6/FS11	Y-FJB-00-0633-001	LP1Y-IPP-00-0048-001/X6/F11	24VDC from IPP to Field Instrument	1-TRIAD, #16 AWG, ARMORED, NO SHLD, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	66	LP1Y-IO-00-0048-001/LP1Y-CTL-10-0043-001-001	LP1Y-IO-00-0048-001	LP1Y-CTL-10-0043-001	Home-run from CHARM I/O Panel to DCS Controller	FO CABLE , 6 FIBER, 300V, ITC (JACKET COLOR: BLK)	300V	Instrument tray, Aux Scope
0	67	LP1Y-IO-00-0048-001/LP1Y-CTL-10-0043-001-002	LP1Y-IO-00-0048-001	LP1Y-CTL-10-0043-001	Home-run from CHARM I/O Panel to DCS Controller	FO CABLE , 6 FIBER, 300V, ITC (JACKET COLOR: BLK)	300V	Instrument tray, Aux Scope
0	68	Y-IPP-00-0048-001/Y-IO-00-0048-001_1	LP1Y-IPP-00-0048-001	LP1Y-IO-00-0048-001	Panel LP1Y-IPP-00-0048-001 Power Supply Fault	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	69	Y-IPP-00-0048-001/Y-IO-00-0048-001_2	LP1Y-IPP-00-0048-001	LP1Y-IO-00-0048-001	Panel LP1Y-IPP-00-0048-001 Temp Low Fault	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed
0	70	Y-IPP-00-0048-001/Y-IO-00-0048-001_3	LP1Y-IPP-00-0048-001	LP1Y-IO-00-0048-001	Panel LP1Y-IPP-00-0048-001 Temp High Fault	1-PR, #18 AWG, NO SHLD, ARMORED, 300V, ITC (JACKET COLOR: BLK)	300V	Field routed

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25I0007- Rail Station Instrument Material Take-off

Lilly Rail Containment

Hanson Project Number: 25I0007

0	ISSUED FOR CONST.	JN	AT	SS	10-Apr-2026
A	ISSUED FOR REVIEW	JN	AT	SS	12-Jan-2026
Revision	**Issue Status**	**Originator**	**Reviewer**	**Approver**	**Date**

Instrument MTO

Rev.	Item #	Item Code	Size	Desc.	Qty	25% Contingency
0	01	7122205	1/2 IN	TUBING, SMLS, .065 WALL, 316L SS	100 FT	125 FT
0	02	612031900	1/2 IN	CONDUIT, FLEXIBLE, LIQUID-TIGHT, TYPE "UA", W/GROUND	54 FT	68 FT
0	03	612320310	1/2 IN x 18 IN	COUPLING, FLEXIBLE BRAIDED, MXM ENDS, BRONZE, GROUPS A, B, C, D	14	18
0	04	612351900	3/4 IN x 1/2 IN	REDUCER, CONDUIT, IRON ALLOY, GROUPS A, B, C, D	64	80
0	05	612371900	1/2 IN	UNION, CONDUIT, MXF, STEEL, GROUPS A, B, C, D	14	18
0	06	612451900	1/2 IN	CONNECTOR, LIQUID-TIGHT, STRAIGHT, MI	36	45
0	07	615000671	A/R	GROUND LUG	4	5
0	08	618312010	260 - .550 x 1/2	GLAND, TC/TC-ER CABLE, CLASS 1, DIV 2, 316SS, CMP	4	5
0	09	619001911	1 5/8 IN x 1 5/8 IN	STRUT, SLOTTED, SINGLE SIDE, CS-HDG	48 FT	60 FT
0	10	619502002	2 IN x 3/8 IN	U-BOLT, ROUND, W/NUTS, 316SS	4	5
0	11	619532002	3/8 IN	NUT, HEX, NC 316SS	20	25
0	12	619542012	3/8 IN	WASHER, LOCK SPRING, 316SS	28	35
0	13	619542014	1/4 IN	A/R WASHER, LOCK TOOTH/SERRATED, 316SS	32	40
0	14	619562011	3/8 IN x 5 IN	ANCHOR STUD, WEDGE TYPE, W/NUT & WASHER, 316SS	12	15
0	15	619572001	3/8 IN x 3 IN	HOOK BOLT, SQUARE BEND, 316SS	8	10
0	16	619582002	1/2 IN	ROD, ALL-THREAD, NC, 316SS	8 FT	10 FT
0	17	619602001	3/8 IN	SPRING NUT FOR 1-5/8" x 1-5/8" STRUT 316SS	24	30
0	18	619711901	C3 x 4.1	CHANNEL, "C", CS-HDG	32 FT	40 FT
0	19	5713971L	1/4 IN	PRESTO POLYAMIDE TUBING, WHITE	32 FT	40 FT
0	20	5713971L1	3/8 IN	TUBING .062 WALL POLYETHYLENE	16 FT	20 FT
0	21	612170000L	1/2 IN x 1/2 IN	FITTING, CONDUIT, OUTLET BOX, "GUAC14", ALUMINUM, 2"	4	5
0	22	619511901-2	3/8 IN x 2 IN	BOLT, HEX HEAD, NC, CS-HDG	4	5
0	23	619511901-3	1/2 IN x 2 IN	BOLT, HEX HEAD, NC, CS-HDG	4	5
0	24	619512002-1	1/4 IN x 3/4 IN	BOLT, HEX HEAD, NC, 316SS	32	40
0	25	619512002-2	3/8 IN x 3 IN	BOLT, HEX HEAD, NC, 316SS	24	30
0	26	619531901-2	3/8 IN	NUT, HEX, NC, CS-HDG	4	5
0	27	619531901-3	1/2 IN	NUT, HEX, NC,	4	5
0	28	619532002-1	1/4 IN	NUT, HEX, NC 316SS	32	40
0	29	619532002-3	1/2 IN	NUT, HEX, NC 316SS	32	40
0	30	619541901-1	1/4 IN	WASHER, FLAT ROUND, CS-HDG	24	30
0	31	619541901-2	3/8 IN	WASHER, FLAT ROUND, CS-HDG	8	10
0	32	619541901-3	1/2 IN	WASHER, FLAT ROUND, CS-HDG	8	10
0	33	619541902-2	3/8 IN	WASHER, LOCK SPRING, CS-HDG	4	5
0	34	619541902-3	1/2 IN	WASHER, LOCK SPRING, CS-HDG	4	5
0	35	619542011-1	1/4 IN	WASHER, FLAT ROUND, 316SS	8	10
0	36	619542011-2	3/8 IN	WASHER, FLAT ROUND, 316SS	76	95

Instrument MTO

Rev.	Item #	Item Code	Size	Desc.	Qty	25% Contingency
0	37	619542011-3	1/2 IN	WASHER, FLAT ROUND, 316SS	32	40
0	38	7112380L-1	1/4 IN	PLUG VALVE, LOCKING HANDLE, TUBE END, 316SS	4	5
0	39	7112380L-2	3/8 IN	PLUG VALVE, LOCKING HANDLE, TUBE END, 316SS	2	3
0	40	7122203-1	1/4 IN	TUBING, SMLS, .035 WALL, 316L SS	32 FT	40 FT
0	41	7122203-2	3/8 IN	TUBING, SMLS, .035 WALL, 316L SS	16 FT	20 FT
0	42	7142310L-1	1/4 IN x 1/4 IN	TUBE ADAPTER, STRAIGHT, MALE (TxMNPT), 316SS	4	5
0	43	7142310L-2	3/8 IN x 3/8 IN	TUBE ADAPTER, STRAIGHT, MALE (TxMNPT), 316SS	2	3
0	44	7142310L-3	1/2 IN x 1/2 IN	TUBE ADAPTER, STRAIGHT, MALE (TxMNPT), 316SS	8	10
0	45	7142323-1	1/4 IN x 1/4 IN	TUBE FITTING, UNION (TxT), 316SS	4	5
0	46	7142323-2	3/8 IN x 3/8 IN	TUBE FITTING, UNION (TxT), 316SS	2	3
0	47	7142390L	6 IN x 1/2 IN	TUBE FITTING, LONG HEX NIPPLE, MALE (MNPTxMNPT), 316SS	4	5
0	48	7152392L	1/2 IN x 1/2 IN	FLANGE ADAPTER, TOE x 150# ANSI, 316SS	4	5
0	49	7181901L	10 IN x 10 IN	BACKPLANE, SHEET METAL, 11GA, CS-HDG	4 FT	5 FT
0	50	7181903L	2"	INSTRUMENT STAND, SINGLE, 2 NPS SCHEDULE 40, PREFAB, CS-HDG	7	9
0	51	7181906L	2"	NSTRUMENT STAND, DUAL, 2 NPS SCHEDULE 40, PREFAB, CS-HDG	1	1
0	52	7181909L	2"	INSTRUMENT STAND, TRIPLE, 2 NPS SCHEDULE 40, PREFAB, CS-HDG	3	4
0	53	7181928L	20 IN x 10 IN x 7 IN	OUTSIDE PLATFORM MOUNTING BRACKET, PREFAB, CS-HDG	1	1
0	54	7190091L	No size	EMERGENCY STOP IEC RING W/ LEGEND, ALLEN-BRADLEY, 800T-X646EM	2	3
0	55	7192300L	3.5 x 6.42	PUSH BUTTON ENCLOSURE, ALLEN-BRADLEY, 800R-1HZ4	2	3
0	56	7192301L	3.5 x 6.42	PUSH BUTTON ENCLOSURE, ALLEN-BRADLEY, 800H-1HZ4	2	3
0	57	7192319L	10.24"x10.24"x20.5"	ZONEX ENCLOSURE, ATEX, IECEX, TYPE 4X, 316SS,HOFFMAN ATEX262616SS61 OR EQUAL	4	5
0	58	7192340L	1.5" x 7.35"	DIN RAIL, 316SS	4	5
0	59	7192343L	1.5"	DIN RAIL END BARRIER	8	10
0	60	7192344L	0.374"	DIN RAIL TERMINAL MARKER	4	5
0	61	7192360L	8.44" x 8.44"	ENCLOSURE BACK PANEL, CONDUCTIVE, HOFFMAN ATEX26P26G OR EQUAL	4	5
0	62	7192390L	No size	PUSH BUTTON PROTECTIVE RING, ALLEN-BRADLEY, 800T-N310	2	3
0	63	7195940L	4" x 1.25" x 6.75"	WIRING DUCT WITH COVER, BLUE, PANDUCT F1X4IB6 /C1BIB6 OR EQUAL	4	5
0	64	7195941L	4" x 1.25" x 6"	WIRING DUCT WITH COVER, GRAY, PANDUCT F1X4LG6 / C1LG6 OR EQUAL	4	5
0	65	7197460L	0.5" x 4.7" x 4.5"	SWITCH AMPLIFIER, 1 CH. ISOLATED BARRIER PEPPERL+FUCHS KCD2-SR-Ex1 OR EQUAL	4	5
0	66	7197493L	No size	PUSH BUTTON, ALLEN-BRADLEY, 800H-FRXT6AF	2	3
0	67	7197494L	No size	PUSH BUTTON, ALLEN-BRADLEY, 800T-FX6D4	2	3

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Scully ST-15 Technical Manual

LSH-0640WP

LSH-0650WP

Lilly Rail Containment

Hanson Project Number:25I0007.00

A1 - DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

Authorization to proceed does not relieve Contractor/Supplier of its responsibility or liability under the Contract and or Purchase Order.

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Scully ST-15

Technical Manual

Single Point Overfill Prevention Control Monitor

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ST-15 - Technical Manual

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General

The Scully ST-15 is an optic or thermistor secondary overfill prevention control unit that connects to vehicles equipped with the Scully model SP-TO two-wire optic sensor or SP-BL (Thermistor sensor), BLU (Uniprobe) and BLH series thermistor sensors. The ST-15 outputs Dynacheck® signals for automatic and continuous monitoring of system status. In the unlikely event of a fault, Scully's Faylsafe™ system will not permit loading.

The ST-15 controller is available in an explosion proof and weatherproof housing which is denoted by an "E" in the model description.

Available configurations in 115V and corresponding 240V models covered by this manual are listed below:

In Explosion Proof Housings

ST-15-115-EL	ST-15-240-EL
ST-15-115-ELK	ST-15-240-ELK

1.1 The ST-15-115H & ST-15-240H Module

Model ST-15 is the basic single channel modular monitor circuit for use with Scully Thermistor and Two-Wire Optic Liquid Level Sensors. This unit is powered by line voltage and provides DPDT relay contacts for auxiliary load control.

The ST-15-115H is the building block on which all Scully ST-15-115 Series Control Units are based. The ST-15-240H is the building block on which all Scully ST-15-240 Series Control Units are based.

The ST-15C-115H is designed to be powered by a 105 - 125VAC 50-60 Hz supply.

Models for operation from 210 - 250VAC 50-60 Hz main power supplies are designated ST-15-240H.

The designations "115" or "240" indicating main power supply voltage are carried throughout the product line model numbering system. ST-15 will be used as the default reference throughout this manual.

NOTE: The Latest Module has both FM & ATEX approvals and is labeled ST-15C-115 or ST-15C-240. Diagrams in this manual may not indicate a "C" moniker, but still applies. Also, the "H" moniker is left in this manual to indicate the module not system.

1.2 Technical Specifications

Temperature Range	-40°C to +60°C
Power Requirements	120VAC Models: 105 – 125 VAC, 50/60 Hz, 20VA Max.* 240VAC Models: 210 – 250 VAC, 50/60 Hz, 20VA Max.* *20VA does not include circuits controlled by the output relay contact
Sensors:	One of the following sensors: SP-BLUK Series Thermistor Sensor (-40˚C to +48˚C) SP-BLHK Series Thermistor Sensor (-28˚C to +93˚C) SP-TO, SP-IO OR SP-IR Series 2-Wire Optic Sensor (-40˚C to +60˚C)
Output Relay:	One normally open volt-free contact rated 250VAC, 5A resistive max. The output contact closes when all sensors are dry and functional.
Response Time:	0.5 second maximum
Indicator Lights:	Output status indicator lights: Red: Non-permissive Green: Permissive
Output Control Fuse:	Internal 5 Ampere
Connections:	Provided by internal terminal strips.
Enclosure:	Explosion-proof, weatherproof IP65.
Dimensions:	See Appendix 7.5 for outline drawing.
Weight:	12.7 Kg (28 Lbs.)
Approvals:	FM Explosion Proof for Class 1, Division 1, Group C & D, T6 (-40°C ≤ Ta ≤ +60°C)

Degree of protection of enclosure IP65. The rating of the enclosure is negated if the holes in the enclosure are unplugged/unscrewed or any holes are drilled into the enclosure. Cover screws to minimum grade A2-70 stainless steel.

The ST-15 control unit has been designed to be impact-resistant. Do not subject to excessive mechanical or thermal stresses. Do not allow the unit to come in contact with aggressive substances.

The ST-15 may only be repaired by replacing the internal control module. The internal control module is non-repairable.

Do not open when an explosive gas atmosphere is present.

2.1 Thermistor Sensors

Scully Liquid Level Thermistor sensors are a two-lead device which connects via a pair of wires to the input terminals of the ST-15. Thermistor sensors are available in a wide range of constructions to suit different applications and environments. Descriptions and specifications are available in Scully Technical Data Catalog Sheets.

The general application areas for thermistor sensors are liquid level sensing of petroleum products ranging from gasoline through No. 4 oil at temperatures of -20° F to +130° F (-28° C to +55° C). They may also be used to sense a variety of other liquid products compatible with the construction materials and approval listings of these sensors.

Due to variations in manufacturing processes, thermistor sensors from an individual manufacturing runs may be overly sensitive to airflow. This applies to all thermistor sensors as listed in sections 2.2 and 2.3 as well. If the ST-15 module with a thermistor sensor seems sensitive to air flow, refer to section 4.8.1 of this manual or contact Scully Service.

2.2 Uniprobes

Model SP-BLU Uniprobes are Scully thermistor sensors with extended low temperature performance thermistors permitting their application in temperature environments of -40° F to +120° F (-40° C to +48° C). Uniprobes are otherwise identical to corresponding models of thermistor sensors. A suffixed “U” in a Scully thermistor sensor model number denotes the Uniprobe thermistor.

2.3 High Temperature Sensors

Model SP-BLH Sensors are Scully Uniprobes with extended high temperature performance thermistors permitting their application for heated products up to +200° F (-18° C to +93° C). The high temperature sensor differs from all other thermistor Sensors in that it uses three wires. A suffixed “H” in a Scully thermistor sensor model number denotes the high temperature thermistor.

2.4 Scully Instant-On™ Two Wire Optic Sensors

Model’s SP-IO, SP-TO, and SP-IR Instant-On Sensors are two lead devices using an infrared optic principle to detect liquids. Descriptions and specifications are available in Scully Technical Data Catalog Sheets.

The general application areas for two-wire Sensors are liquid level sensing of petroleum products ranging from gasoline through No. 4 oil at temperatures of -40° F to +140° F (-40° C to +60° C). They may also be used to sense a variety of other liquid products compatible with the construction materials and approval listings of these sensors.

3.1 Housings

ST-15 modules are available in NEMA-7 explosion-proof housings.

3.2 Model Numbering System

All ST-15 modules in the ST-15 series have model numbers beginning with ST-15-115 or ST-15-240, according to the main power supply. The suffixed letters and numbers describe the optional controls and indicators. The letters and numbers used and their interpretations follow:

Model Configurations and Suffix Definitions

“E”	Indicates the ST-15 module has an explosion-proof housing.
“L”	Indicates the ST-15 module is fitted with indicator lamps.
“K”	Indicates that the ST-15 unit is fitted with a lockable “Bypass Switch” permitting key controller over-ride of automatic overfill shutdown systems.

ST-15 Module

4.1 Description

The model ST-15 is a ST-15 control module for use with Scully Dynaprobe Thermistor, Uniprobe and Instant-On Optic Liquid Level Sensors. The module uses the Scully Dynacheck®` system for dynamically self-checking the sensor, lead wires and all control circuit components up to 30 times per second. The circuit will fail safe (indicating high level) under all expected component or power failures.

The sensor circuitry is intrinsically safe and has been designed so that the control unit can not deliver sufficient energy to the sensor circuit to ignite Class I, Group D, Division 1 hazardous materials. The ST-15 module itself is not rated for use in hazardous locations without being enclosed in an explosion-proof housing.

4.2 Power Requirements

The ST-15-115H is designed to operate from 105 -125VAC 50-60 Hz and the ST-15-240H from 210 - 250VAC 50-60 Hz. Current draw is less than 1/4 ampere for the 115 volt models and less than 1/8 ampere for the 240 volt models.

4.3 Control Circuit Ratings

The ST-15 has a DPDT output relay for controlling external alarms and other loads. Contact ratings are 5 amperes resistive at 240 VAC. Power relays or motor starters must be used if the controlled load exceeds this rating.

4.4 Dynacheck Monitor Lamp

This flashing lamp, located on the ST-15 module, provides a visual indication of the Dynamic Self-Checking, taking place when the ST-15 unit is connected to a “dry” sensor.

The lamp should flash every few seconds and will stop flashing when the sensor is “wet” or if a malfunction in the sensor wiring or ST-15 module circuit exists. Burnout of this indicator lamp bulb will not impair the operation of the ST-15 module or the Dynamic-Self-Checking function.

Use only a 12ESB lamp (Scully Part No. 27002) for replacement purposes. The lamp pulls straight out of its socket. Do not replace this lamp unless main power has been removed from the ST-15.

4.5 Mechanical Installation Of the ST-15 Module

The ST-15 module is designed for mounting inside an enclosure suitable for protection against damage and environmental conditions to which it will be exposed. Hole locations are shown on Diagram 20269, Section 7.2.

4.6 Electrical Connections

When making electrical connections to the ST-15 module, keep the intrinsically safe wiring to the Sensors separate from the 115V power and control wiring. Separate entrance holes and conduits must be used for the sensor circuits and other wiring.

CAUTION: THE IMPORTANCE OF MAINTAINING ABSOLUTE SEPARATION OF INTRINSICALLY SAFE AND UNSAFE WIRING CAN NOT BE OVEREMPHASIZED!

Connect line power to terminals 2 and 3 of the power terminal strip on the module. This source may be switched externally if desired, but consideration should be given to someone inadvertently switching off power to the ST-15 module causing an alarm or shutdown to take place. Terminal 1 on this strip should be wired to ground or a grounded point on the enclosure, assuming ground is provided through conduit.

Connect the controlled circuits to the appropriate relay contacts on the power terminal strip. The diagram on the module shows the position of the contacts with the control relay released. This corresponds to the ST-15 module powered up with a “wet” sensor connected. When the sensor is “dry” the control relay will be energized. A failure anywhere in the sensor wiring or the sensor will result in the control relay being released.

4.6.1 Sensor Connections

4.6.1.1 Standard Dynaprobe Thermistor, Uniprobe, and Two-wire Optic Sensors

Connect the two leads from the Thermistor sensor to terminal strip TB2 pins 4 and 3, black to pin 4 and white to pin 3. Leave factory installed jumper connected to TB2, terminals 1 and 2. When using Uniprobes, be sure the BLACK lead connects to TB2-4 terminal and the WHITE lead connects to TB2-3 terminal.

4.6.1.2 “H” (High Product Temperature) Thermistors

Remove the factory installed jumper connected to TB2 terminals 1 and 2. Connect the GREEN, WHITE, and BLACK Thermistor sensor leads to TB2 terminals 2, 3, and 4 respectively. Be very careful to preserve the continuity of the three leads.

4.6.1.3 Sensor Lead Extension

The sensor leads may be extended with normal electrical wire. The resistance of each extension conductor should be kept below 2Ω. Do not use a test lamp or other high current device to check the sensor circuit. If a low current ohmmeter is available, the sensor circuit may be checked and should display the resistance given below shown in DWG 61420 (see section 7.1). When measuring Uniprobes, the positive lead of the ohmmeter must connect to the BLACK lead of the sensor.

Wire Size (mm2)	Wire Size (AWG)	Maximum Run Length (Feet)	Maximum Run Length (Meters)
1.5	#16	400	120
2.5	#14	700	210
4.0	#12	1000	300
6.0	#10	1600	480
10.0	#8	2500	760

CAUTION: If extending Sensor leads beyond 2,500 feet (760 m), please consult the factory for specific wiring recommendations to prevent jeopardizing the intrinsically safe nature of the system.

The sensor lead wires should be run in conduit or other protection and should never be intermingled with line voltage wiring. It is not necessary to use shielded cable to extend the sensor leads.

Do not use a test lamp or other high current device to check the sensor circuit. If a low current ohmmeter is available, the sensor circuit may be checked and should display the resistance shown in DWG 61420 (see section 7.1). When measuring Uniprobes, the positive lead of the ohmmeter must connect to the BLACK lead of the sensor.

4.7 Operation

4.7.1 Warm-Up Time

A short time delay will occur between the application of power to the ST-15 module unit and control relay pull in with a “dry” sensor. This delay is characteristic of the thermal warm-up of a cold sensor and will take approximately 30 seconds. A similar delay will also occur when sensing the change from “wet” to “dry” condition. There is no delay in responding to a change from “dry” to “wet” sensor condition (high level sensing).

The warm-up time should be taken into consideration when designing alarm installations. Otherwise, the alarm may sound for approximately 30 seconds at initial turn-on of the alarm system. This feature may be desirable in some applications to provide a full system check if the alarm system is infrequently used.

NOTE: Model SP-IO and SP-TO Instant-On Sensors require no warm-up time.

4.7.2 Functional Testing

The ST-15 should be tested per Table 1 as part of the installation and commissioning process before being approved for service, and any time the function of the system may come into question. It is important to note that the ST-15 meets the requirements for IEC 61508 SIL 3 (Certificate # SSC 080841 C001). Systems installed as part of a Safety Instrumented System must undergo periodic proof tests, of which the Functional Test is a required process. Please refer to The Scully ST-15 Safety Manual PN 61626 if the process controls require Functional Safety compliance.

4.8 Maintenance

The model ST-15-115H control unit is designed for continuous operation. The only component which may require field replacement is the Dynacheck indicator lamp.

All other components are mounted onto the printed circuit board and are not field replaceable. The equipment should be serviced by field replacement of the ST-15 module and the defective unit returned for shop service.

Please remember that improper field wiring, component substitutions or “jumping” connections while testing can defeat the intrinsically safe design of this equipment and result in the application of dangerously high voltages to the sensor lead wires.

4.8.1 Thermistor Calibration

The ST-15 module control circuit has a trip point adjustment which is preset at manufacture to work properly with all IEC EN 13922 two-wire optic and thermistor sensors. Due to variations in manufacturing processes, thermistor sensors from individual manufacturing runs may be overly sensitive to airflow. If the ST-15 module with a thermistor sensor seems sensitive to air flow, the adjustment may be reset in the field as follows:

Test Equipment Required

1 - Digital or Analog Multimeter with frequency measurement capability.
1 - Screwdriver (Approx. 1/8-inch head).
1 - Set of test leads with alligator clips at the other end.

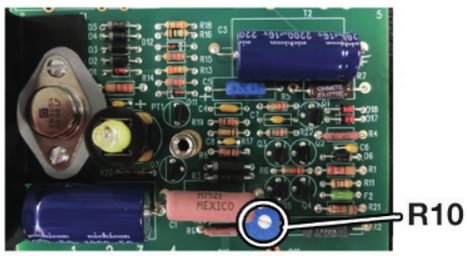
4.8.1.1 Adjustment Procedure

NOTE – This procedure matches a specific ST-15 module with a specific thermistor sensor as a set. If the ST-15 module or thermistor sensor is replaced, it will be necessary to recalibrate the module to the thermistor.

DO NOT PERFORM THIS PROCEDURE WHEN USING AN OPTIC SENSOR.

1. Turn off the power to the ST-15 module to be calibrated.
2. Remove the module metal circuit cover by removing the screw in the center of the cover. Locate the adjustment potentiometer R10 on the circuit card. (See Figure 1)

Figure 1



3. Connect the thermistor sensor to be used. If the factory installed jumper is not already in place across TB2 terminals 1 and 2, one should be installed. For older models, the factory installed 680 ohm resistor should be in place across the 'W' and 'G' terminals.
4. Ensure the thermistor is in an environment with no air flow. If the ambient temperature is below 0°C or it is not possible to eliminate airflow, it is acceptable to cover the tip of the thermistor with a bag or piece of cloth to isolate the sensor.
5. Set the Multi-Meter to measure frequency. Connect the + test lead from the multimeter to TB2-4. Connect the – lead from the multimeter to TB2-3. For older models, connect the + lead to the 'B' terminal and the – lead to the 'W' terminal.
6. Power the system up. Allow the ST-15 to reach a permit condition. This may take as long as 2 minutes.

CAUTION – LIVE AC MAINS VOLTAGE WILL BE PRESENT ON TB1.

7. Slowly adjust R10 clockwise until the frequency measurement stops increasing. Do not turn R10 past this point.
8. Turn R10 slowly counterclockwise until the frequency begins to decrease, then slowly turn back clockwise until the frequency stops increasing.

NOTE: Due to manufacturing variations of the thermistor, the final frequency will vary somewhat from unit to unit. The ST-15 and Scully Thermistors are designed to IEC EN 13922 standards, which require the operating frequency to be between 20 Hz and 125 Hz. If the final frequency is beyond these limits, contact Scully Service.

9. The ST-15 should be in a permit state at this time. If not, repeat steps 7 and 8. If the unit is still not permitting, contact Scully service.
10. Disconnect Mains power. Remove the multi-meter test leads from TB2, and reattach the module metal circuit cover. Close the ST-15 enclosure, then reapply mains power. The unit is now ready for service.

ST-15 Systems Using The ST-15 Module

Features, connections, and operation of specific ST-15 control systems which are built around the ST-15 module are described in this section.

5.1 ST-15 "L" and "K" Options

The ST-15-115 and 240-ELK monitor contains a set of red and green indicator lamps and a lockable power control and bypass switch. They are often used in conjunction with rail car loading or single compartment tank truck loading systems to provide overfill protection via automatic high level shutoff or loading valves or pumps.

5.1.1 Output Control

Output control provision on these models is provided by SPDT contact set in the ST-15modules output control relay. These contacts are rated for 5A - 240 VAC resistive loads. Power relays or motor starters should be used to buffer these relay contacts if high power loads must be controlled. See appropriate field wiring diagrams for the location of these terminals.

5.1.2 Indicator Lamp Functions (“L” option)

The GREEN lamp lit indicates a normally functioning unit, connected to a dry sensor.

The RED lamp lit indicates a wet sensor, or a system fault which has caused a fail safe “wet” condition.

5.1.3 Control Switch Functions (“K” option)

Switch Position	Function
OFF	Switch in far left position. Removes mains power from the ST-15 module and the indicator lamps. The output control contacts will be in their “wet” state.
ON	Switch in center position. Powers the ST-15-115H module and the indicator lamp circuits for normal operation.
BYPASS	Switch in far right position. Removes power from the ST-15 module and indicator lamp circuits and creates a switch closure across the “dry” contact position of the systems output circuit. This removes all automatic control by the ST-15 MODULE system. **USE THE BYPASS SWITCH WITH CARE!**

Caution: Placing the switch on BYPASS removes all fail-safe operation and secondary overfill prevention.

Tank Vehicle Overfill Protection

A common application for single channel ST-15 monitors is providing of primary or secondary overfill protection during the loading of single compartment tank vehicles or railcars.

Secondary overfill protection implies that the filling operation is expected to be stopped automatically by a product flow meter pre-set to deliver a specific amount of product before stopping the flow. The ST-15 module overfill detection system acts as a backup system to stop product flow in instances where the flow meter is pre-set incorrectly or suffers an unsafe failure.

6.1 Applicable ST-15 Module Monitors

Any single channel ST-15 monitor in the ST-15 series may be used for these applications, but the most commonly used model is the ST-15-115 or 240 -ELK. This model includes an explosion-proof housing for mounting in hazardous locations.

6.2 Sensor Location

Single channel tank vehicle overfill control systems can use a Scully sensor in the following applications:

6.2.1 Tank Vehicle Mounted Sensors

In this application, a sensor is permanently mounted in the tank vehicle with the sensing level set at the desired overfill sensing level.

6.2.1.1 Two-Wire Sensor to ST-15 Connection

The sensor in the tank vehicle is connected to the ST-15 monitor at the loading point via a plug and socket connection. Scully manufactures tank vehicle sockets and loading point plugs and cables for these applications. The sensor is connected to the ST-15 via a vehicle socket to loading plug interface (Sculcon).

6.2.2 Temporarily Installed Sensors

In applications where the sensor cannot be permanently installed in the tank vehicles, Scully manufacturers sensors suitable for temporary installation during loading and removal when loading is completed.

Typical applications for these types of sensors are rail car and chemical tanker loading, where for various reasons the sensors cannot be permanently installed in the vehicles. In these applications the sensors are installed in a “cane” or portable style holder which is placed in the loading hatch prior to loading. These units are vertically adjustable to permit setting the overfill detection level to suit the vehicle being loaded. This style of sensor installation is generally connected by a flexible cord to the loading rack equipment, permitting a range of movement to suit the application.

6.2.3 Top Loading - Vapor Recovery Applications

A Scully sensor may be mounted inside the vapor collection passages of top loading vapor recovery heads providing automatic overfill protection. The electrical leads from the sensor are run along the loading arm to a connection in the ST-15 monitor.

Some vapor recovery heads are fitted with electrical switches which operate when the head is secured to the vehicle. This type of switch may be wired in series with the Black colored sensor lead to prevent loading when the vapor recovery head is not secured.

6.2.4 Loading Arm Mounted Sensors

In some applications a sensor may be mounted on the outside of a conventional loading arm where the normal use of the arm will locate the sensor at an overfill cut off level inside the vehicle. These applications require that the loading arm insertion into the tank vehicle be predictably constant to insure that the overfill detection level is properly established.

6.3 Tank Vehicle Sensor And Socket Installation

6.3.1 Sensor Types and Their Typical Uses

For unheated light products use Thermistor sensor Dynaprobe Model SP-BL, Uniprobe Model SP-BLU, and Instant-On Optic Models: SP-TO, SP-IO, mounted in an appropriate Scully sensor holder. Model SP-IR is a one piece unit that includes a sensor holder.

For heated products, use Thermistor sensor Model SP-BLH in an appropriate Scully sensor holder.

For corrosive products use stainless steel Thermistor sensor Dynaprobe Model SP-BL(SS), Uniprobe Model SP-BLU(SS), and Optic Model SP-TO(SS) in an appropriate Scully sensor holder.

6.3.2 Tanker Socket Types

Two models of Scully sockets are commonly used for single compartment overfill detection/sensor connection. These are:

SJ-6S	A Scully pattern socket featuring rugged construction with bayonet action connection. This socket mates with a Scully SC-6A Plug and Cable Assembly.
SJ-4D	A four (4) pole “trailer” style socket. This socket mates with any of the Scully Plug and Cable systems shown on Drawing 61425 Tanker and Loading Rack Wiring, located in the Appendix.

6.3.3 Vehicle Wiring

6.3.3.1 Regular Thermistor Sensors

Wire according to Drawing 61424 (SJ-6S socket) or Drawing 61423 (SJ-4R and SJ-4D sockets). Drawings located in Appendix.

6.3.3.2 Heated Product Thermistor Sensors, SP-BLH

Wire according to Drawing 61423 & 61424 (SJ-6S Socket). Drawing located in Appendix.

6.4 Loading Rack Equipment Installation

The single channel ST-15 monitor used is wired to the plug and cable assembly according to the documentation listed in this manual.

6.4.1 SC-6A Plug and Cable Assemblies

Use wiring shown on DWG. 61424 (SC-6A) in Appendix. This wiring is typical of any single channel ST-15 series module. The factory installed jumper connected between TB2 terminals 1 and 2 on the sensor terminal strip is left in place. A three-wire connection to the plug and cable assembly is required for the SC -6A. The jumper wire between TB2 terminals 1 and 2 on the ST-15 module's terminal strip is left in place. However, if the tankers are equipped with an SP-BLH sensor, the factory jumper wire must be removed.

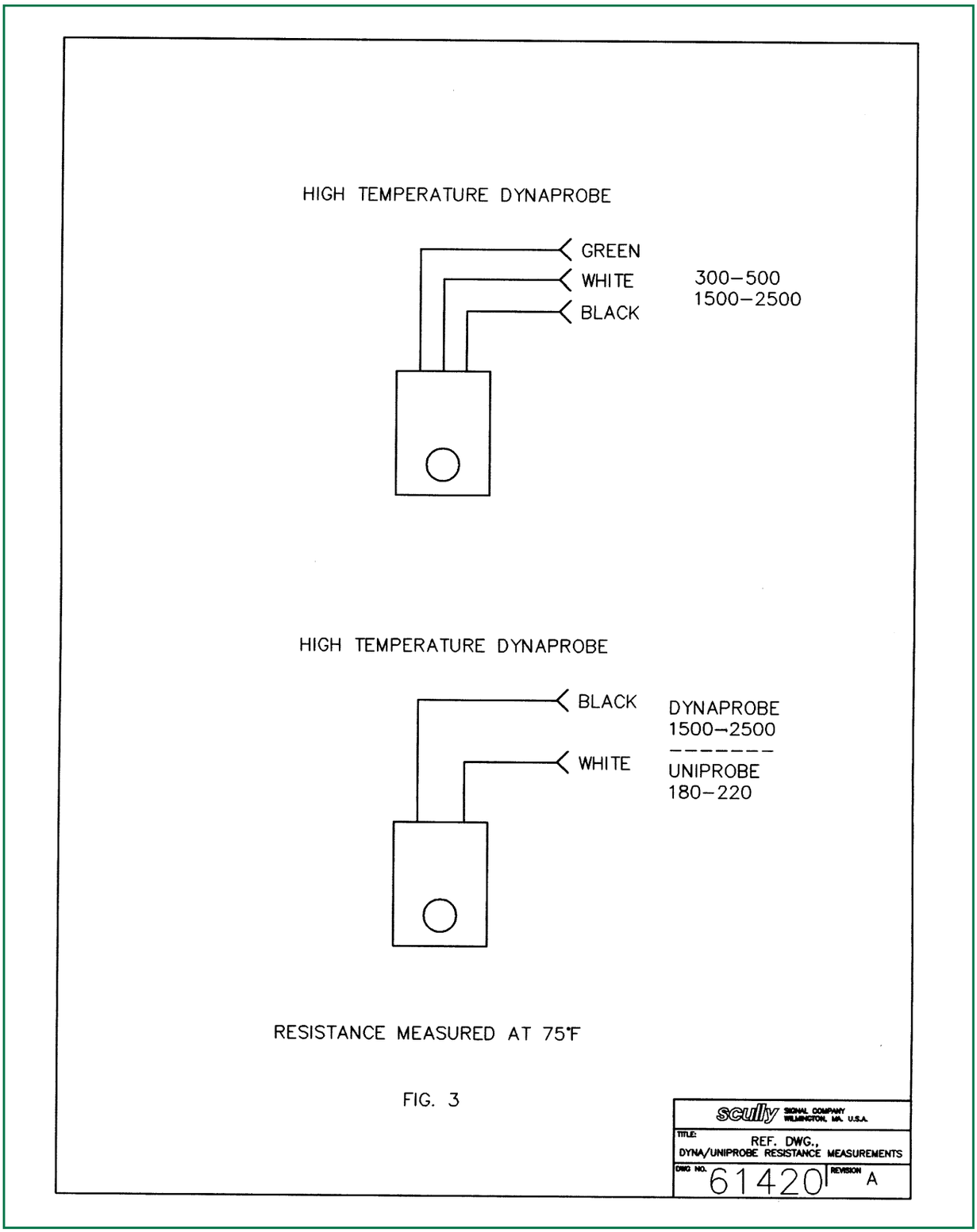
6.4.2 SC-1 Series Plug and Cable Assemblies

This series of four pole plug and cable assemblies, shown in DWG. 61425 in the Appendix, are wired to the sensor connection terminals on the ST-15 module in the ST-15 monitor. A three-wire connection to the plug and cable assembly is required for the SC-1. The factory installed jumper wire between TB2 terminals 1 and 2 on the ST-15 module's terminal strip is left in place. If the tanker or portable sensor is equipped with a SP-BLH sensor, the jumper wire must be removed.

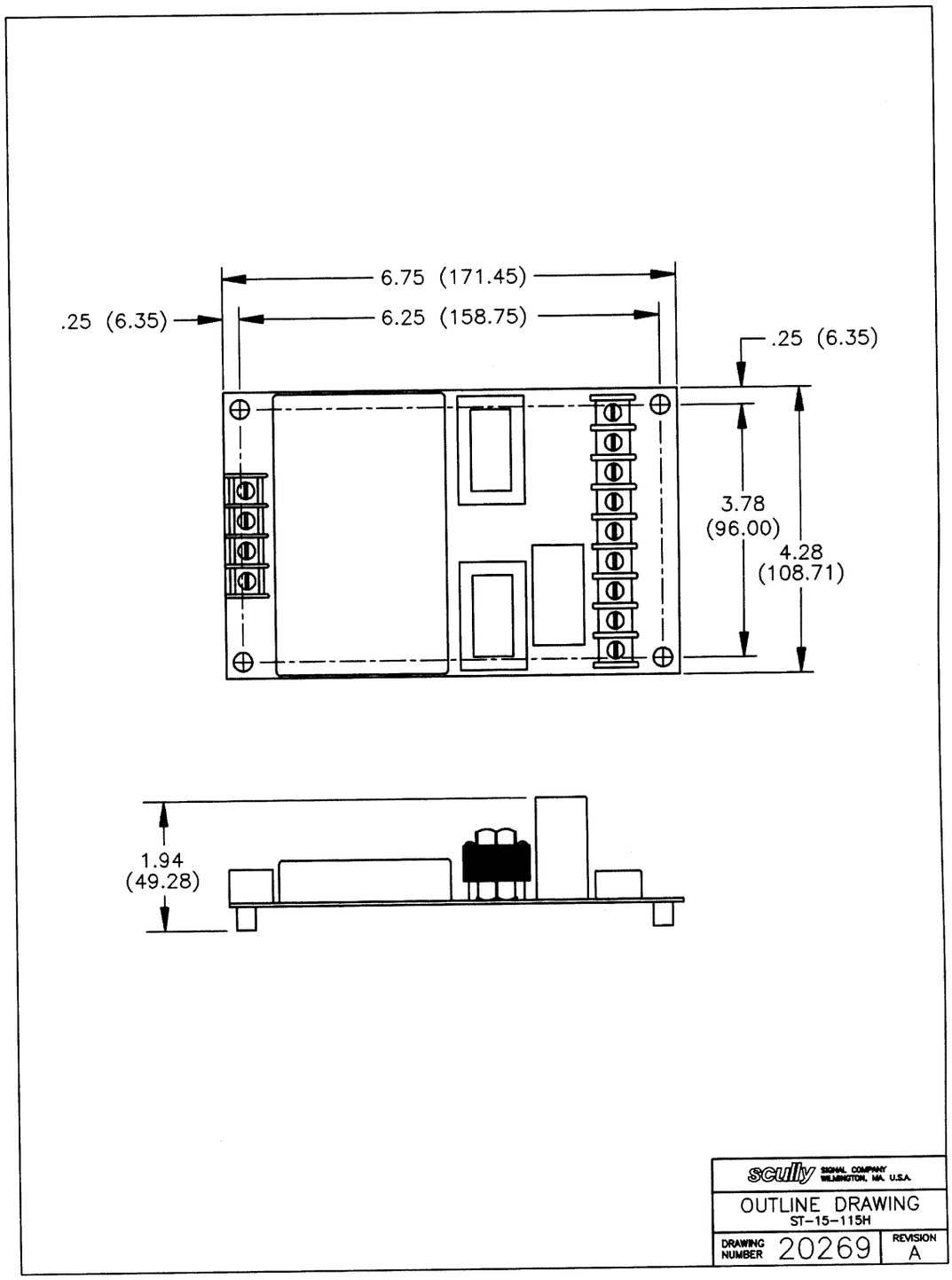
6.4.3 SJ-4R and SJ-4D Sockets

A three-wire connection to the socket is required. The factory installed jumper wire between TB2 terminals 1 and 2 on the ST-15 module's terminal strip is left in place. If the tanker or portable sensor is equipped with an SP-BLH sensor, the jumper wire must be removed.

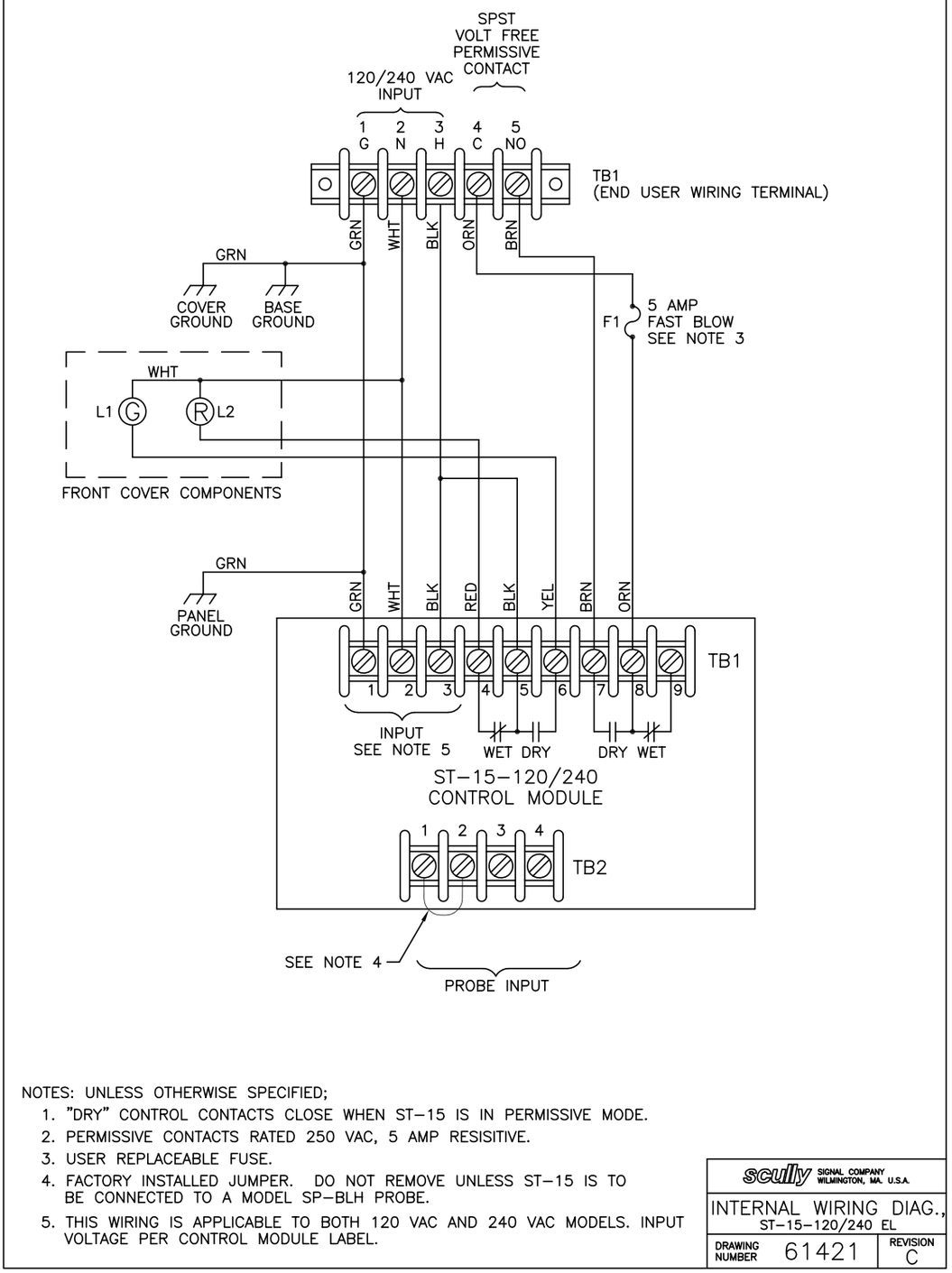
7.1 DWG 61420 - Thermistor Sensor And Uniprobe Resistance Measurements



7.2 DWG. 20269 - Outline Drawing ST-15-115H Module



7.3 DWG. 61421 - Internal Wiring Diagram ST-15-120/240 EL

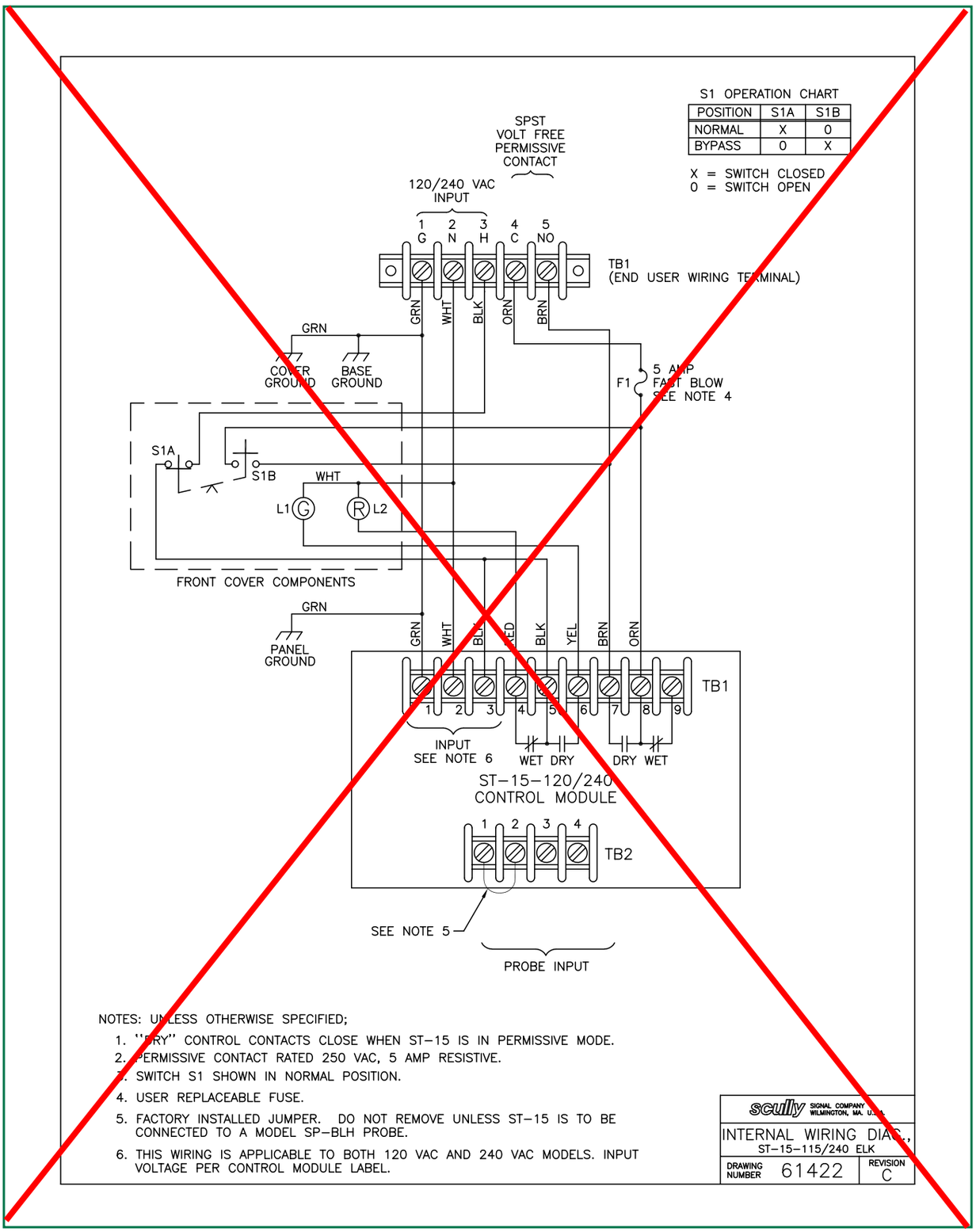


NOTES: UNLESS OTHERWISE SPECIFIED;

1. "DRY" CONTROL CONTACTS CLOSE WHEN ST–15 IS IN PERMISSIVE MODE.
2. PERMISSIVE CONTACTS RATED 250 VAC, 5 AMP RESISITIVE.
3. USER REPLACEABLE FUSE.
4. FACTORY INSTALLED JUMPER. DO NOT REMOVE UNLESS ST–15 IS TO BE CONNECTED TO A MODEL SP–BLH PROBE.
5. THIS WIRING IS APPLICABLE TO BOTH 120 VAC AND 240 VAC MODELS. INPUT VOLTAGE PER CONTROL MODULE LABEL.

scully SIGNAL COMPANY WILMINGTON, MA. U.S.A.	
INTERNAL WIRING DIAG., ST–15–120/240 EL	
DRAWING NUMBER 61421	REVISION C

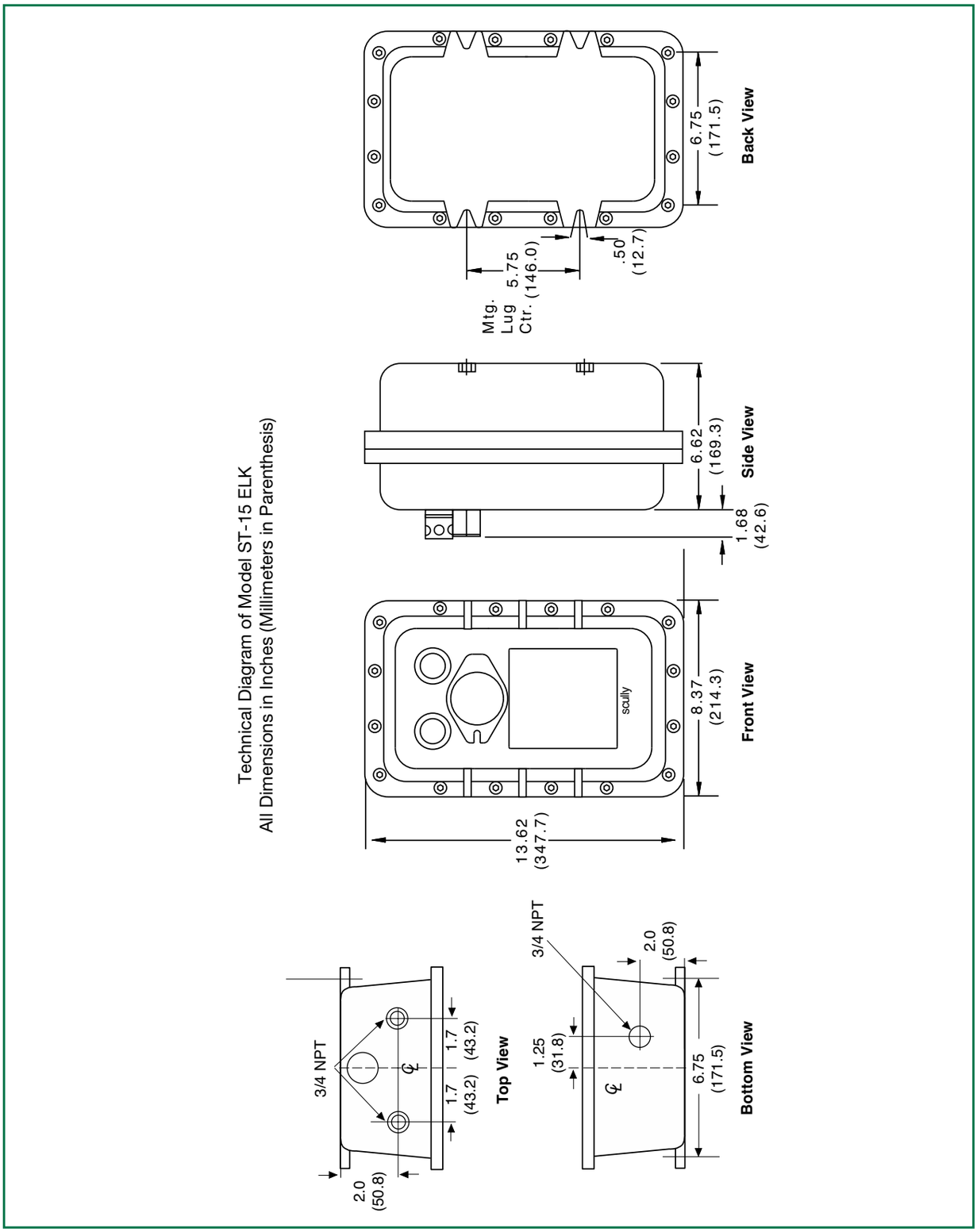
7.4 DWG. 61422 - Internal Wiring Diagram, ST-15-115/240 ELK



ST-15 - Technical Manual

Diagram Appendix

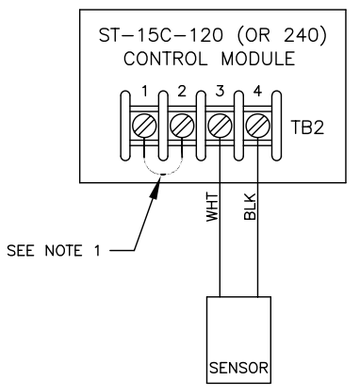
7.5 ST-15 Outline Drawing



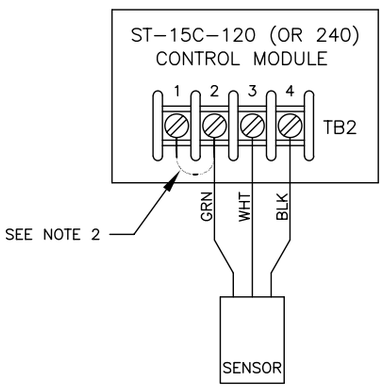
7.6 DWG 61423 - Field Wiring Diagram, ST-15 Direct Sensor Wiring

NOTES:

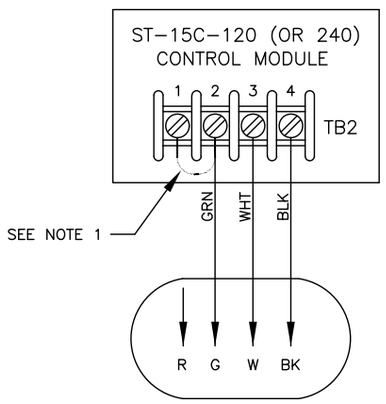
1. DO NOT REMOVE FACTORY INSTALLED JUMPER WHEN CONNECTING TO A TWO-WIRE SENSOR.
2. REMOVE FACTORY INSTALLED JUMPER BETWEEN TERMINALS 1 AND 2 WHEN CONECTING TO A THREE-WIRE SENSOR, MODEL SP-BLHK.



TYPICAL TWO-WIRE SENSOR WIRING



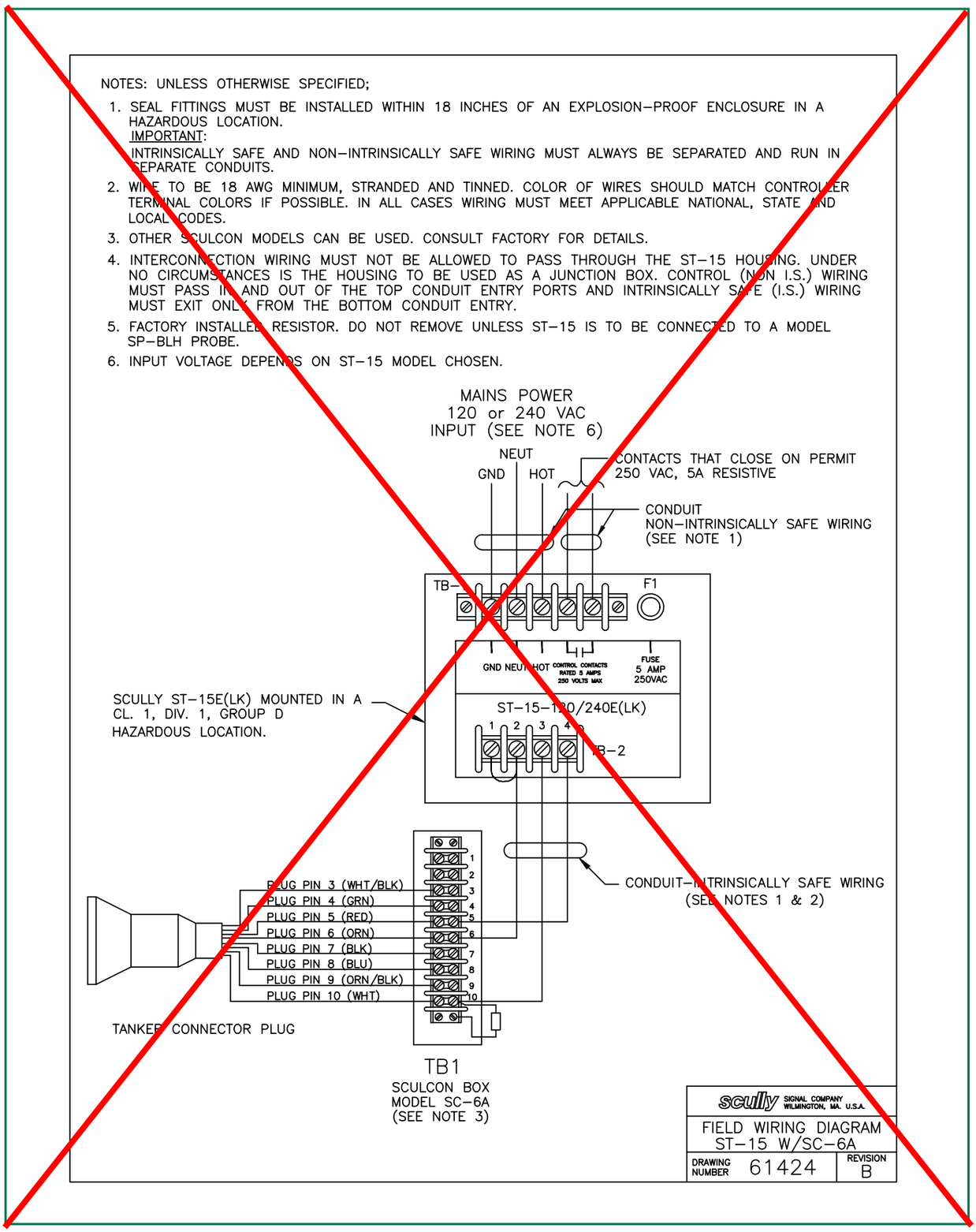
TYPICAL THREE-WIRE SENSOR WIRING



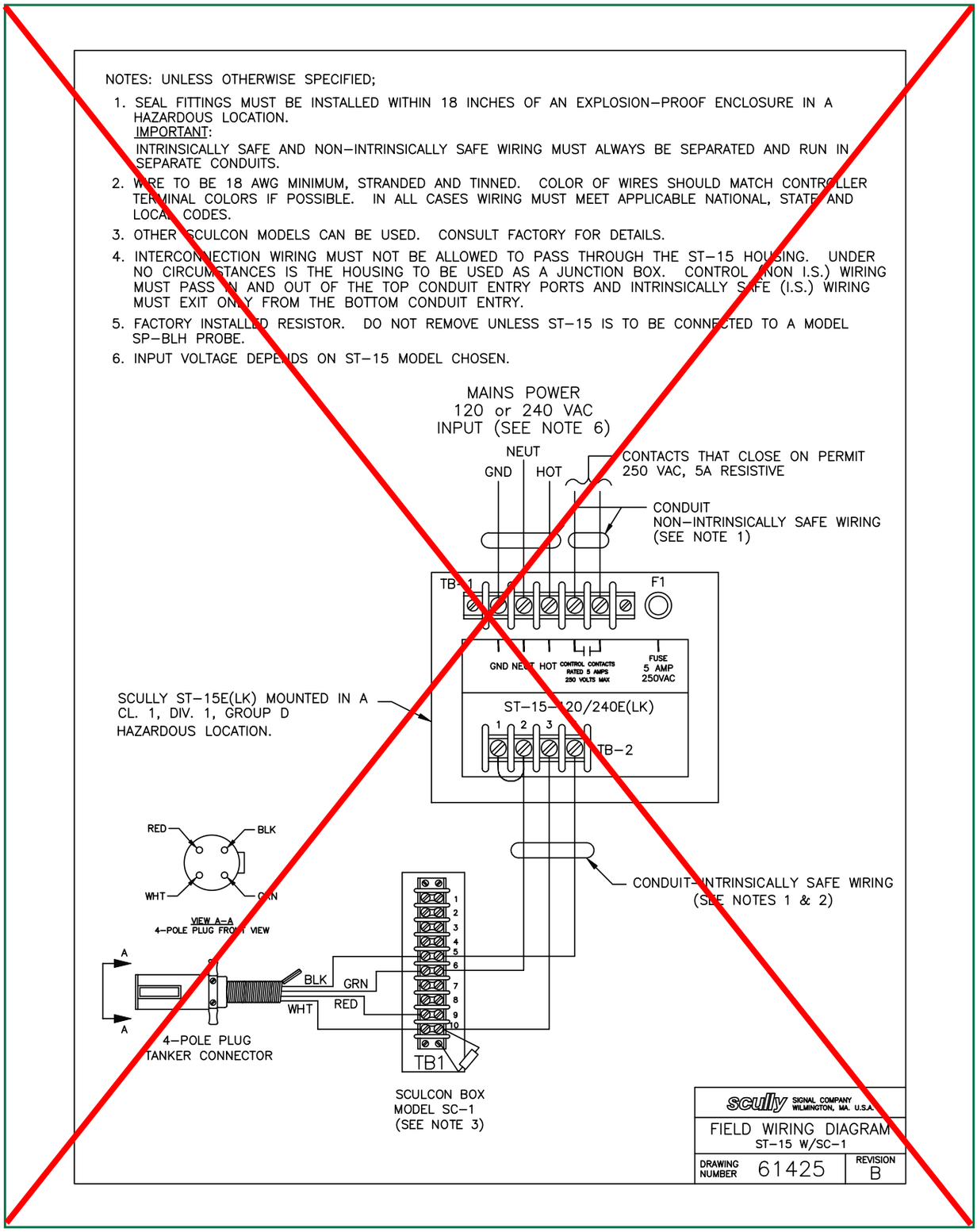
WIRING TO SCULLY SOCKETS
MODELS SJ-4D OR SJ-4R

scully SIGNAL COMPANY WILMINGTON, MA. U.S.A.	
FIELD WIRING DIAGRAM ST-15 DIRECT SENSOR WIRING	
DRAWING NUMBER 61423	REVISION B

7.7 DWG. 61424 - Field Wiring Diagram, ST-15 with SC-6A



7.8 DWG 61425 - Field Wiring Diagram, ST-15 with SC-1



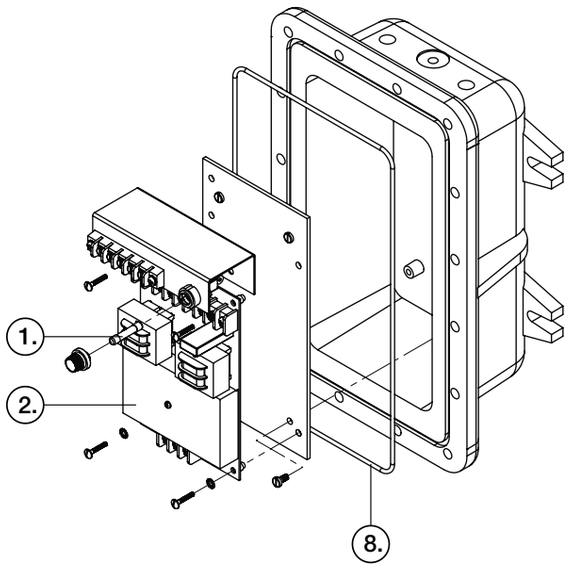
7.9 DWG. 61201 - Replacement Parts Diagram, ST-15 115VAC

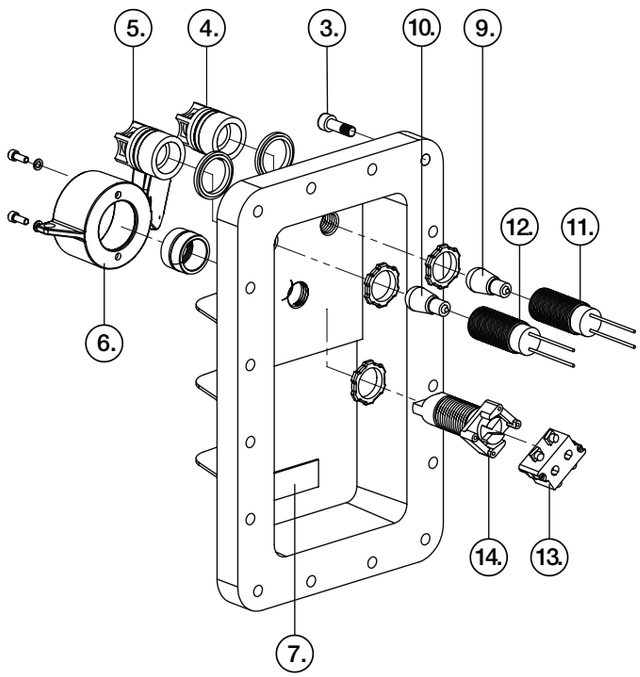
ST-15 115VAC

Single Point Thermister Controller

Part No.	Model
07528	ST-15-115 EL
07677	ST-15-115 ELK

Note: Controller Models Suffix:
Explosion-proof housing (E), Indicator Lights (L),
Key (K) Lockable Bypass Switch.





Replacement Parts

Item	Part No.	Description	Qty.
1	26311	Mains Fuse, 5 Ampere	1
2	08756	ST-15C-115H Circut Module	1
3	50041	Screw, Hex Head	16
4	09124	Red Replacement Head and Guard Assembly	1
5	09125	Green Replacemnt Head and Guard Assembly	1
6	08360	Bypass Switch Lock Box (includes 2 mounting screws)	1
7	21733	Corrosion Capsule	1
8	31340	O-ring cover seal	1
9	09384	LED- Red	1
10	09385	LED- Green	1
11	08343	Pilot Light Replacement Kit Red (includes 5,9)	1
12	08342	Pilot Light Replacement Kit Green (includes 5,9)	1
13	26022	Bypass Switch Contact Block	1
14	26055	Bypass Switch Operator 3-Position	1

Notes:
Item 3: Hex head screw used on models built after 10/92.
Item 8: O-ring used on models built after 10/92

61201 Rev C
October 2014

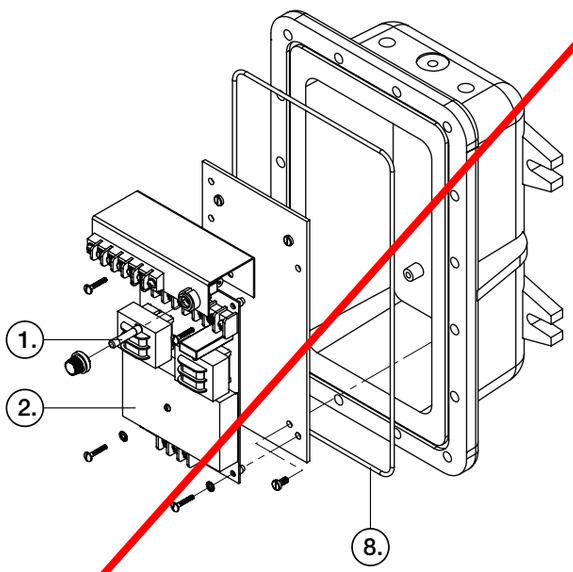
7.10 DWG. 61561 - Replacement Parts Diagram, ST-15 240VAC

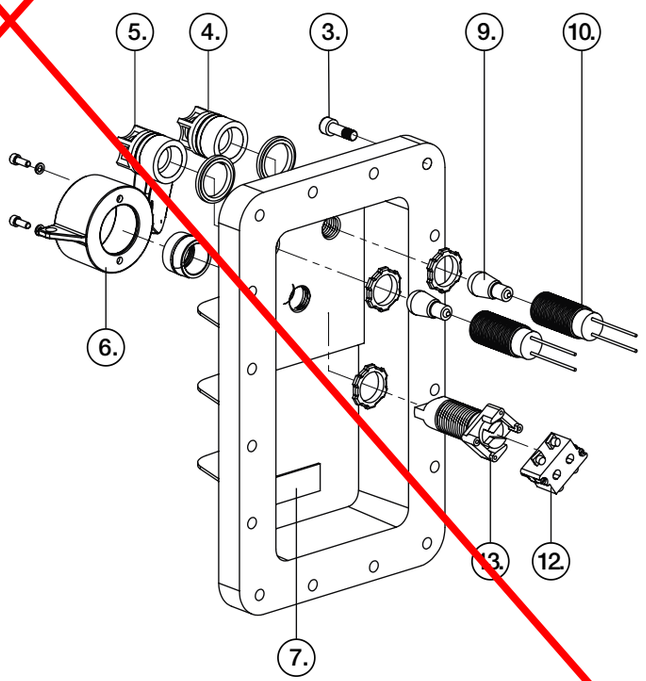
ST-15 240VAC

Single Point Thermister Controller

Part No.	Model
07640	ST-15-240 EL
07547	ST-15-240 ELK

Note: Controller Models Suffix:
Explosion-proof housing (E), Indicator Lights (L),
Key (K) Lockable Bypass Switch.





Replacement Parts

Item	Part No.	Description	Qty.
1	26311	Mains Fuse, 5 Ampere	1
2	08656	ST-15C-240H Circut Module	1
3	50041	Screw, Hex Head	6
4	09122	Red Replacement Head and Guard Assembly	1
5	09123	Green Replacemnt Head and Guard Assembly	1
6	08360	Bypass Switch Lock Box (includes 2 mounting screws)	1
7	21733	Corrosion Capsule	1
8	31340	O-ring cover seal	1
9	27006	Bulb, 250V, 10W	2
10	08346	Pilot Light Replacement Kit Red (includes 4,9,10)	1
11	08345	Pilot Light Replacement Kit Green (includes 5,9,11)	1
12	26022	Bypass Switch Contact Block	1
13	26055	Bypass Switch Operator 3-Position	1

Notes:
Item 3: Hex head screw used on models built after 10/92.
Item 8: O-ring used on models built after 10/92

61561 Rev A
October 2014

ST-15 OVERFILL PREVENTION SYSTEM FUNCTIONAL TEST CHECKLIST													
This test is intended to be used as part of commissioning a completed Scully ST-15 WX or ST-15Exx system, but can be used as a pre-installation checklist or for any periodic functional verification. For systems requiring IEC 61508 or IEC 61511 Functional Safety Proof Testing, please refer to to The Scully ST-15 Safety Manual PN 61626.													
System Model Name:													
System Model Number:													
System Serial Number:													
Customer Name and Site Address:													
Step	**Action**	**Chan 1**	**Chan 2**	**Chan 3**	**Chan 4**	**Chan 5**	**Chan 6**	**Chan 7**	**Chan 8**	**Chan 9**	**Chan 10**	**Chan 11**	**Chan 12**
1	Confirm that the control module is indicating a dry condition (output relay closed).												
2[a]	Bypass the safety function and take appropriate action to avoid a false trip.												
3[b]	On WX systems, remove power from the WX relay of the channel under test. Confirm that the relay opens within one (1) second. Reconnect power.												
4	Disconnect one of the wires to the sensor. Confirm that the relay opens within one (1) second. Reconnect sensor.												
5	Short the sensor inputs at the ST-15 module together. Confirm that the relay opens within one (1) second. Remove short.												
6[c]	Wet-test the sensor by submersing the tip in a dark liquid (such as coffee). Confirm that the relay opens within one (1) second. Remove sensor from liquid and confirm that the relay closes.												
7	Remove the bypass and otherwise restore normal operation												

[a]Step 2 implies that the output relay may be bypassed prior to testing. This can only be done if site policies allow for the safety function to be bypassed.

[b]On single point systems in explosion-proof housings, disconnect power to the module and confirm that the output relay is open. Reconnect power before proceding.

[c]If it is impractical to immerse the sensor associated with the channel under test due to installation limitations, it is acceptable to use a representative sensor to test the individual channels in the cabinet. HOWEVER: the sensors must still be tested to confirm proper operation independently. Contact Scully for guidance.

B1LG-00-K215-00092-2_LP1Y-270-PRC-K215-01.pdf

Scully - Setting Standards in Safety and Dependability since 1936.

For over seventy-five years Scully has been engineering and building products to the highest safety and reliability standards. We design and manufacture all of our systems under one roof to ensure complete quality control over our manufacturing and testing operations. Scully is ISO certified and all of our products are 100% made in the U.S.A. In addition, we back up our products with the best service in the industry. We have direct sales and service personnel in the U.S.A., The United Kingdom, and Europe and are represented in over 50 countries.

For more information and 24 hour technical assistance, call Scully Signal Company at 1-800-2SCULLY (1-800-272-8559).

Scully Headquarters in Wilmington, MA U.S.A.

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60905 Rev S
September 2020

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37 Road One Winsford Industrial Estate,
Winsford Cheshire CW7 3QG, UK
+44 (0) 1606 553805 • sales@scullyuk.com

Dynamic Self-Testing® Overfill Prevention Systems

Document Title:	Estimate Response Form Report
Document Number:	LP1Y-270-RPT-K215-01
Security Classification:	Restricted

Estimate Response Form Report

LSH-0640WP

LSH-0650WP

Lilly Rail Containment

Hanson Project Number:25I0007.00

A1 - DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

Authorization to proceed does not relieve Contractor/Supplier of its responsibility or liability under the Contract and or Purchase Order.

By: M. Despa des63440 March 31, 2026

Revision	Issue Status	Originator	Reviewer	Approver	Date
0	IFQ	JN	SS	MW	3/9/2026
A	IFR	JN	SS	MW	2/4/2026

Estimate Response Form Report

3/9/2026 10:21:05 AM

Customer:

Prospect: 2214

Saeed Sabah
Hanson Professional Services, Inc.
11931 Wickchester Ln., #370
Houston TX 77043
USA

Phone:

Fax:

Estimate	Terms	Quote Date	Expiration Date	Salesperson	Customer Currency
89949	To Be Determined	12/15/2025	2/13/2026	MWB	USD US Dollar

UM	Quantity	Item	Net Price(Disc Incl)	Extended Price
EA	2.000	07528 F ST-15-115-EL SGL PT THMS CONT		
EA	2.000	091000063 F SP-PYU(2205 SS)/4P CANE,36",20' STR CA		
EA	2.000	07707 F SJ-4D POLE PATTERN SKT, 4 PIN		

Lead time: Ship 3 weeks after receiving order.
Please reference this estimate when placing order..

** Can probe specifications to be determined **

Document Title:	SJ-4D 4-Pole Socket Assembly
Document Number:	LP1Y-270-SPC-K215-01
Security Classification:	Restricted

SJ-4D 4-Pole Socket Assembly

LSH-0640WP

LSH-0650WP

Lilly Rail Containment

Hanson Project Number:25I0007.00

A1 - DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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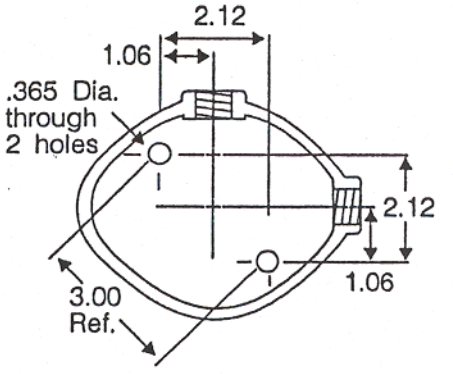
By: M. Despa des63440 March 31, 2026

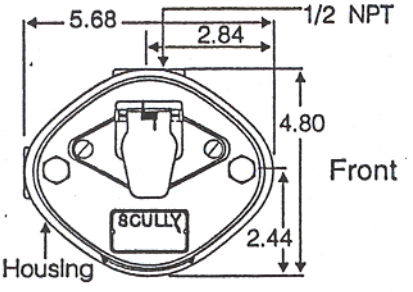
0	IFQ	JN	SS	MW	3/9/2026
A	IFR	JN	SS	MW	2/4/2026
Revision	**Issue Status**	**Originator**	**Reviewer**	**Approver**	**Date**

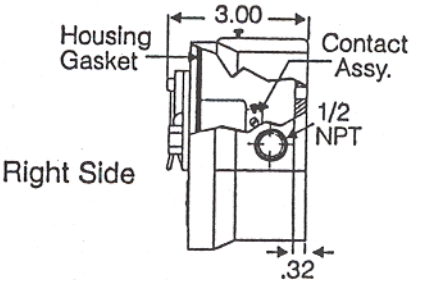
SJ-4D
4-POLE SOCKET ASSEMBLY

These sockets have many uses. They may be rack mounted for connection to the Scully "cane" style sensor. As a truck mounted socket, they may be used for connection to a rack mounted overfill system via the Scully SC-1 or SC-1D Sculcon® Junction Box with Plug and Cable. The socket's four electrical contacts allow its use with may styles of Scully liquid level sensing probes. When the socket is not in use, a spring-loaded cover protects the contact pins. Two 1/2" NPT conduit entry holes allow easy wiring connection using readily available watertight strain relief fittings. The four electrical contact pins are color coded for easy identification.



Mounting Dimensions





Technical Specifications	
Housing:	Die cast zinc alloy
Contact Pins:	Plated brass
Screw Terminals:	Accepts up to 12 Gauge Wire

Ordering Information and Spare Parts	
Description	**Part Number**
SJ-4D Pole Pattern Socket, 4 Pin	07707
Socket Housing	21810
Housing Gasket	20628

Biclops, Dynacheck, Dynamic Self-checking, Dynamic Self-Test, Faylsafe, Load Anywhere, Maxsafety, Multe-Cept, Scultrol, Scully are registered trade marks of the Scully Signal company.

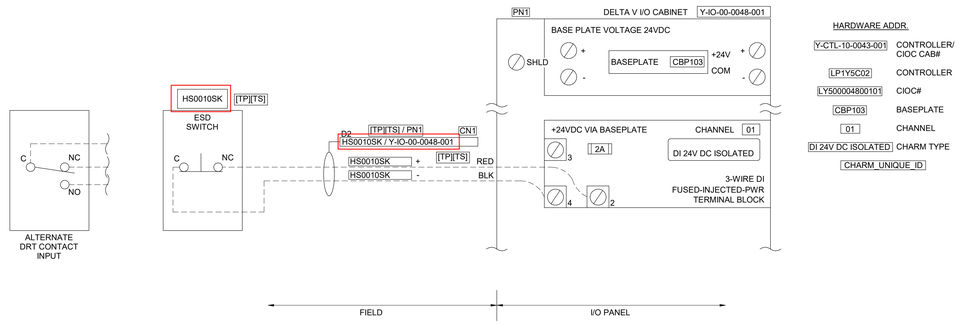
Specifications are subject to change without notice.

Scully Signal Company

Worldwide Headquarters · 70 Industrial Way · Wilmington, MA 01887 USA
phone: 1-617-692-8600 · toll free: 1-800-272-8559 · fax: 1-617-692-8620
www.scully.com · sales@scully.com

LABELING DESCRIPTION AND EXAMPLE

TP = TAG PREFIX = ESD SWITCH = HS
TS = TAG SUFFIX = LOOP NUMBER = 0010SK
PN1 = IO PANEL = <BLDG>-<TYPE>-<FLOOR>-<ROOM # OR SKID#>-<INDEX> = Y-IO-00-0048-001
CN1 = SIGNAL CABLE TAG = TP TS / PN1 = HS0010SK / Y-IO-00-0048-001
CHARM_UNIQUE_ID = PN1 - SN1 - CN1
D2 = CABLE CODE: 1 PAIR, #18, NO SHLD, 300V, ITC (JACKET COLOR: BLK)



HARDWARE ADDR.

Y-CTL-10-0043-001	CONTROLLER/ CIOC CAB#
LP1Y5C02	CONTROLLER
LY50004800101	CIOC#
CBP103	BASEPLATE
01	CHANNEL
DI 24V DC ISOLATED	CHARM TYPE
CHARM_UNIQUE_ID	

NOTES:

1. ALL CABLES FROM FIELD INSTRUMENT TO CHARM I/O PANEL OR IPP TO BE FIELD ROUTED USING SNAP TRACK CHANNEL TRAY OR EQUIVALENT.
2. ALL TAGS WILL HAVE PREFIX "LP1Y".

1 INSTRUMENT WIRING DIAGRAM - LP1YHS0010SK - DISCRETE INPUT 2-WIRE HAND SWITCH

NTS

REFERENCE DRAWINGS:

1. LP1Y-270-LST-K215-02: INSTRUMENT I/O LIST.
2. LP1Y-270-LST-K215-03: INSTRUMENT CABLE SCHEDULE.
3. LP1Y-270-LAY-L215-01 & 02: INSTRUMENT LOCATION PLAN.
4. LP1Y-275-DIA-K215-30: CHARM I/O PANEL WIRING.

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
0	ALL	ISSUED FOR CONSTRUCTION	NA	10-APR-26
A	ALL	ISSUED FOR REVIEW	NA	16-JAN-26

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REV	DATE	DES	ENG	CHK	APP	PM APP
0	10-APR-26	MGL	JN	BM	AJT	SS
A	16-JAN-26	MGL	JN	BM	AJT	SS

INSTRUMENT WIRING DIAGRAM - LP1YHS0010SK - DISCRETE INPUT 2-WIRE HAND SWITCH

RAIL UNLOADING RACK

SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: 0
SUB DISCIPLINE	LOCATION CODE: LP1	LOCATION/GRID: COL C-D / 33-43	CCI CODE: YELLOW / PROJECT NAME: LP1-306-1181
CRITICAL QUALITY		ENTERPRISE GROUP: API	DRAWING INTENTION / DRAWING SCALE: NONE
CRITICAL BUSINESS		EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA / MERIDIAN REV: B
CRITICAL HSE		VENDOR: NA	DOCUMENT ID: LP1Y-275-DIA-K215-01
NON-CRITICAL		ENGINEERING AGENCY DOC ID: NA	

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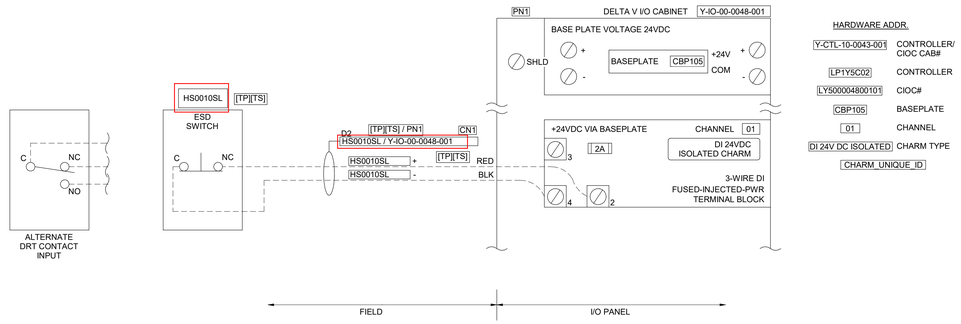
LABELING DESCRIPTION AND EXAMPLE		
TP	=	TAG PREFIX = ESD SWITCH = HS
TS	=	TAG SUFFIX = LOOP NUMBER = 0010SL
PN1	=	IO PANEL = <BLDG>-<TYPE>-<FLOOR>-<ROOM # OR SKID#>-<INDEX> = Y-IO-00-0048-001
CNT	=	SIGNAL CABLE TAG = TP TS / PN1 = HS0010SL / Y-IO-00-0048-001
CHARM_UNIQUE_ID	=	FN - SN - CN
D2	=	CABLE CODE: 1 PAIR, #18, NO SHLD, 300V, ITC (JACKET COLOR: BLK)



HARDWARE ADDR.

Y-CTL-10-0043-001	CONTROLLER/ CIOC CAB#
LP1Y5C02	CONTROLLER
LY500004800101	CIOC#
CBP105	BASEPLATE
01	CHANNEL
DI 24V DC ISOLATED	CHARM TYPE
CHARM_UNIQUE_ID	

NOTES:

1. ALL CABLES FROM FIELD INSTRUMENT TO CHARM I/O PANEL OR IPP TO BE FIELD ROUTED USING SNAP TRACK CHANNEL TRAY OR EQUIVALENT.
2. ALL TAGS WILL HAVE PREFIX "LP1Y".

1 INSTRUMENT WIRING DIAGRAM - LP1YHS0010SL - DISCRETE INPUT 2-WIRE HAND SWITCH

NTS

REFERENCE DRAWINGS:

1. LP1Y-270-LST-K215-02: INSTRUMENT I/O LIST.
2. LP1Y-270-LST-K215-03: INSTRUMENT CABLE SCHEDULE.
3. LP1Y-270-LAY-L215-01 & 02: INSTRUMENT LOCATION PLAN.
4. LP1Y-275-DIA-K215-30: CHARM I/O PANEL WIRING.

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
0	ALL	ISSUED FOR CONSTRUCTION	NA	10-APR-26
A	ALL	ISSUED FOR REVIEW	NA	16-JAN-26

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REV	DATE	DES	ENG	CHK	APP	PM APP
0	10-APR-26	MGL	JN	BM	AJT	SS
A	16-JAN-26	MGL	JN	BM	AJT	SS

INSTRUMENT WIRING DIAGRAM - LP1YHS0010SL - DISCRETE INPUT 2-WIRE HAND SWITCH

RAIL UNLOADING RACK

SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: 1
SUB DISCIPLINE	LOCATION CODE: LP1	LOCATION/GRID: COL C-D / 33-43	CCI CODE: YELLOW	PROJECT NAME: LP1-306-1181
CRITICAL QUALITY		ENTERPRISE GROUP: API	DRAWING INTENTION	DRAWING SCALE: NONE
CRITICAL BUSINESS		EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA	MERIDIAN REV: B
CRITICAL HSE		VENDOR: NA	DOCUMENT ID: LP1Y-275-DIA-K215-02	
NON-CRITICAL		ENGINEERING AGENCY DOC ID: NA		

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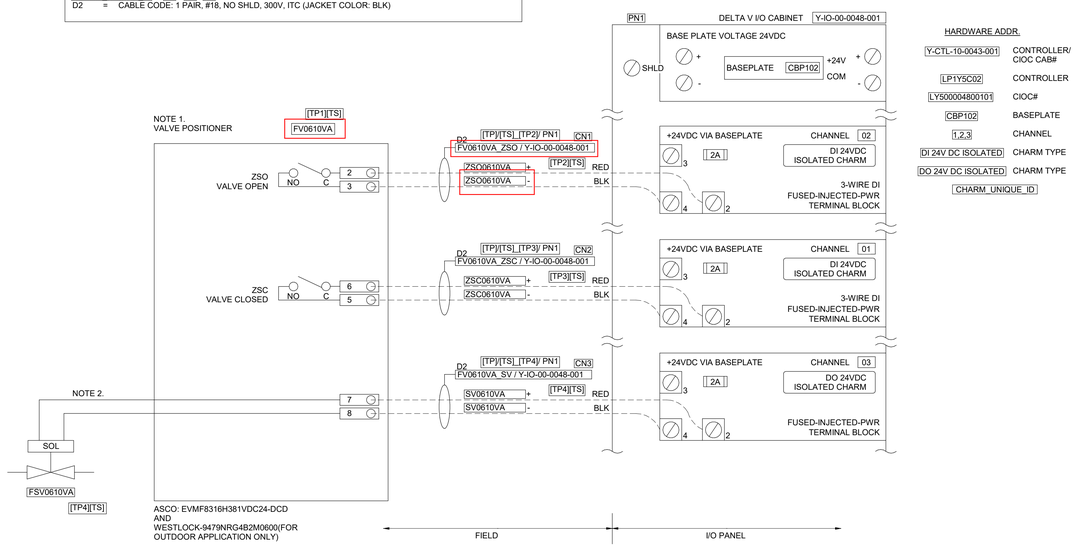
LABELING DESCRIPTION AND EXAMPLE		
TP1	=	TAG PREFIX = ON/OFF VALVE = FV
TP2	=	TAG PREFIX = POSITION SWITCH, OPEN = ZSO
TP3	=	TAG PREFIX = POSITION SWITCH, CLOSED = ZSC
TP4	=	TAG PREFIX = SOLENOID VALVE = SV
TS	=	TAG SUFFIX = LOOP NUMBER = 0610VA
PN1	=	IO PANEL = <BLDG>-<TYPE>-<FLOOR>-<ROOM # OR SKID#>-<INDEX> = Y-IO-00-0048-001
CN1	=	SIGNAL CABLE TAG = TP1 TS TP2 / PN1 = FV0610VA_ZSO / Y-IO-00-0048-001
CN2	=	SIGNAL CABLE TAG = TP1 TS TP3 / PN1 = FV0610VA_ZSC / Y-IO-00-0048-001
CN3	=	SIGNAL CABLE TAG = TP1 TS TP4 / PN1 = FV0610VA_SV / Y-IO-00-0048-001
CHARM_UNIQUE_ID	=	FN - SN - CN
D2	=	CABLE CODE: 1 PAIR, #18, NO SHLD, 300V, ITC (JACKET COLOR: BLK)



NOTES:
1. USE WESTLOCK LIMIT SWITCHES (9479NRG4B2M0600) FOR OUTDOOR USE ONLY.
2. ASCO SOLENOID VALVE IS PRE-WIRED TO VALVE POSITIONER.
3. ALL CABLES FROM FIELD INSTRUMENT TO CHARM I/O PANEL OR IPP TO BE FIELD ROUTED USING SNAP TRACK CHANNEL TRAY OR EQUIVALENT.
4. ALL TAGS WILL HAVE PREFIX "LP1Y".

1 INSTRUMENT WIRING DIAGRAM - LP1YFV0610VA - DISCRETE INPUT AND OUTPUT DELTA V CHARMS IO

NTS

REFERENCE DRAWINGS:

1. LP1Y-270-LST-K215-02: INSTRUMENT I/O LIST.
2. LP1Y-270-LST-K215-03: INSTRUMENT CABLE SCHEDULE.
3. LP1Y-270-LAY-L215-01 & 02: INSTRUMENT LOCATION PLAN.
4. LP1Y-275-DIA-K215-30: CHARM I/O PANEL WIRING.

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REV	DATE	DES	ENG	CHK	APP	PM APP
1	10-APR-26	MGL	JN	BM	AJT	SS
0	16-JAN-26	MGL	JN	BM	AJT	SS

INSTRUMENT WIRING DIAGRAM - LP1YFV0610VA - DISCRETE INPUT AND OUTPUT DELTA-V CHARMS IO

RAIL UNLOADING RACK

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
1	ALL	ISSUED FOR CONSTRUCTION	NA	10-APR-26
0	ALL	ISSUED FOR REVIEW	NA	16-JAN-26

REVISIONS

SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: 1
SUB DISCIPLINE	LOCATION CODE: LP1 / LOCATION/GRID: COL C-D / 33-43	CCI CODE: YELLOW	PROJECT NAME: LP1-306-1181
CRITICAL QUALITY	ENTERPRISE GROUP: API	DRAWING INTENTION	DRAWING SCALE: NONE
CRITICAL BUSINESS	EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA	MERIDIAN REV: B
CRITICAL HSE	VENDOR: NA	DOCUMENT ID: LP1Y-275-DIA-K215-03	
NON-CRITICAL	ENGINEERING AGENCY DOC ID: NA		

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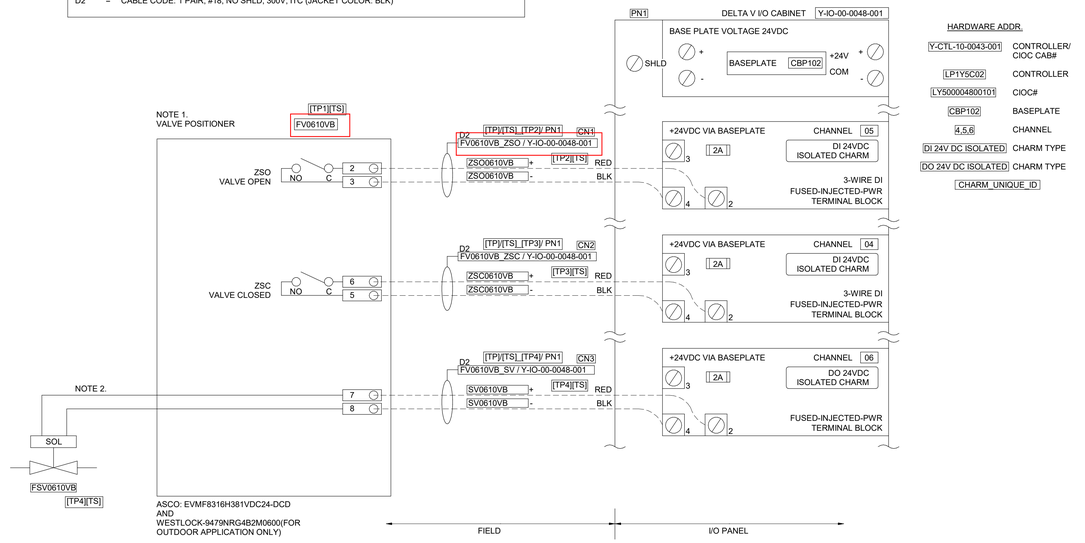
LABELING DESCRIPTION AND EXAMPLE		
TP1	=	TAG PREFIX = ON/OFF VALVE = FV
TP2	=	TAG PREFIX = POSITION SWITCH, OPEN = ZSO
TP3	=	TAG PREFIX = POSITION SWITCH, CLOSED = ZSC
TP4	=	TAG PREFIX = SOLENOID VALVE = SV
TS	=	TAG SUFFIX = LOOP NUMBER = 0610VB
PN1	=	IO PANEL = <BLDG>-<TYPE>-<FLOOR>-<ROOM # OR SKID#>-<INDEX> = Y-IO-00-0048-001
CN1	=	SIGNAL CABLE TAG = TP1 TS TP2 / PN1 = FV0610VB_ZSO / Y-IO-00-0048-001
CN2	=	SIGNAL CABLE TAG = TP1 TS TP3 / PN1 = FV0610VB_ZSC / Y-IO-00-0048-001
CN3	=	SIGNAL CABLE TAG = TP1 TS TP4 / PN1 = FV0610VB_SV / Y-IO-00-0048-001
CHARM_UNIQUE_ID	=	FN - SN - CN
D2	=	CABLE CODE: 1 PAIR, #18, NO SHLD, 300V, ITC (JACKET COLOR: BLK)



NOTES:
1. USE WESTLOCK LIMIT SWITCHES (9479NRG4B2M0600) FOR OUTDOOR USE ONLY.
2. ASCO SOLENOID VALVE IS PRE-WIRED TO VALVE POSITIONER.
3. ALL CABLES FROM FIELD INSTRUMENT TO CHARM I/O PANEL OR IPP TO BE FIELD ROUTED USING SNAP TRACK CHANNEL TRAY OR EQUIVALENT.
4. ALL TAGS WILL HAVE PREFIX "LP1Y".

1 INSTRUMENT WIRING DIAGRAM - LP1YFV0610VB - DISCRETE INPUT AND OUTPUT DELTA V CHARMS IO

NTS

REFERENCE DRAWINGS:

1. LP1Y-270-LST-K215-02: INSTRUMENT I/O LIST.
2. LP1Y-270-LST-K215-03: INSTRUMENT CABLE SCHEDULE.
3. LP1Y-270-LAY-L215-01 & 02: INSTRUMENT LOCATION PLAN.
4. LP1Y-275-DIA-K215-30: CHARM I/O PANEL WIRING.

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
2	ALL	ISSUED FOR CONSTRUCTION	NA	10-APR-26
1	ALL	ISSUED FOR REVIEW	NA	16-JAN-26

REVISIONS

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REV	DATE	DES	ENG	CHK	APP	PM APP
2	10-APR-26	MGL	JN	BM	AJT	SS
1	16-JAN-26	MGL	JN	BM	AJT	SS

INSTRUMENT WIRING DIAGRAM - LP1YFV0610VB - DISCRETE INPUT AND OUTPUT DELTA-V CHARMS IO

RAIL UNLOADING RACK

SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: 1
SUB DISCIPLINE:	LOCATION CODE: LP1	LOCATION/GRID: COL C-D / 33-43	CCI CODE: YELLOW
			PROJECT NAME: LP1-306-1181
CRITICAL QUALITY		ENTERPRISE GROUP: API	DRAWING INTENTION
			DRAWING SCALE: NONE
CRITICAL BUSINESS		EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA
			MERIDIAN REV: B
CRITICAL HSE		VENDOR: NA	DOCUMENT ID
NON-CRITICAL		ENGINEERING AGENCY DOC ID: NA	LP1Y-275-DIA-K215-04

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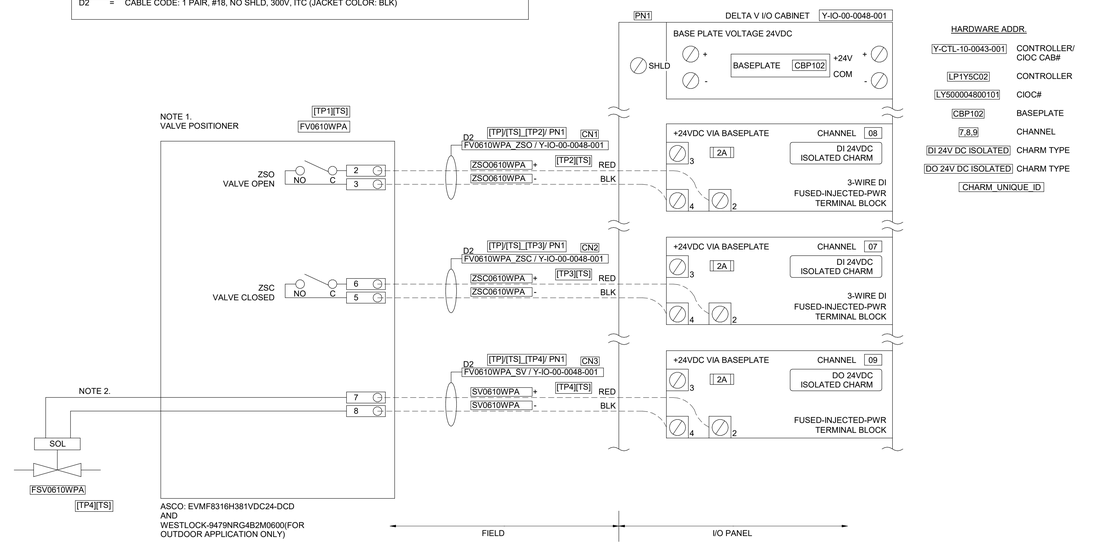
LABELING DESCRIPTION AND EXAMPLE		
TP1	=	TAG PREFIX = ON/OFF VALVE = FV
TP2	=	TAG PREFIX = POSITION SWITCH, OPEN = ZSO
TP3	=	TAG PREFIX = POSITION SWITCH, CLOSED = ZSC
TP4	=	TAG PREFIX = SOLENOID VALVE = SV
TS	=	TAG SUFFIX = LOOP NUMBER = 0610WPA
PN1	=	IO PANEL = <BLDG>-<TYPE>-<FLOOR>-<ROOM # OR SKID#>-<INDEX> = Y-IO-00-0048-001
CN1	=	SIGNAL CABLE TAG = TP1 TS TP2 / PN1 = FV0610WPA_ZSO / Y-IO-00-0048-001
CN2	=	SIGNAL CABLE TAG = TP1 TS TP3 / PN1 = FV0610WPA_ZSC / Y-IO-00-0048-001
CN3	=	SIGNAL CABLE TAG = TP1 TS TP4 / PN1 = FV0610WPA_SV / Y-IO-00-0048-001
CHARM_UNIQUE_ID	=	FN - SN - CN
D2	=	CABLE CODE: 1 PAIR, #18, NO SHLD, 300V, ITC (JACKET COLOR: BLK)



NOTES:
1. USE WESTLOCK LIMIT SWITCHES (9479NRG4B2M0600) FOR OUTDOOR USE ONLY.
2. ASCO SOLENOID VALVE IS PRE-WIRED TO VALVE POSITIONER.
3. ALL CABLES FROM FIELD INSTRUMENT TO CHARM I/O PANEL OR IPP TO BE FIELD ROUTED USING SNAP TRACK CHANNEL TRAY OR EQUIVALENT.
4. ALL TAGS WILL HAVE PREFIX "LP1Y".

1 INSTRUMENT WIRING DIAGRAM - LP1YFV0610WPA - DISCRETE INPUT AND OUTPUT DELTA V CHARMS IO

NTS

REFERENCE DRAWINGS:

1. LP1Y-270-LST-K215-02: INSTRUMENT I/O LIST.
2. LP1Y-270-LST-K215-03: INSTRUMENT CABLE SCHEDULE.
3. LP1Y-270-LAY-L215-01 & 02: INSTRUMENT LOCATION PLAN.
4. LP1Y-275-DIA-K215-30: CHARM I/O PANEL WIRING.

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1	10-APR-26	MGL	JN	BM	AJT	SS
0	16-JAN-26	MGL	JN	BM	AJT	SS

INSTRUMENT WIRING DIAGRAM - LP1YFV0610WPA - DISCRETE INPUT AND OUTPUT DELTA-V CHARMS IO

RAIL UNLOADING RACK

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
0	ALL	ISSUED FOR CONSTRUCTION	NA	10-APR-26
A	ALL	ISSUED FOR REVIEW	NA	16-JAN-26

REVISIONS

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SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: 1
SUB DISCIPLINE	LOCATION CODE: LP1	LOCATION/GRID: COL C-D / 33-43	CCI CODE: YELLOW
			PROJECT NAME: LP1-306-1181
CRITICAL QUALITY		ENTERPRISE GROUP: API	DRAWING INTENTION
			DRAWING SCALE: NONE
CRITICAL BUSINESS		EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA
			MERIDIAN REV: B
CRITICAL HSE		VENDOR: NA	DOCUMENT ID: LP1Y-275-DIA-K215-05
NON-CRITICAL		ENGINEERING AGENCY DOC ID: NA	

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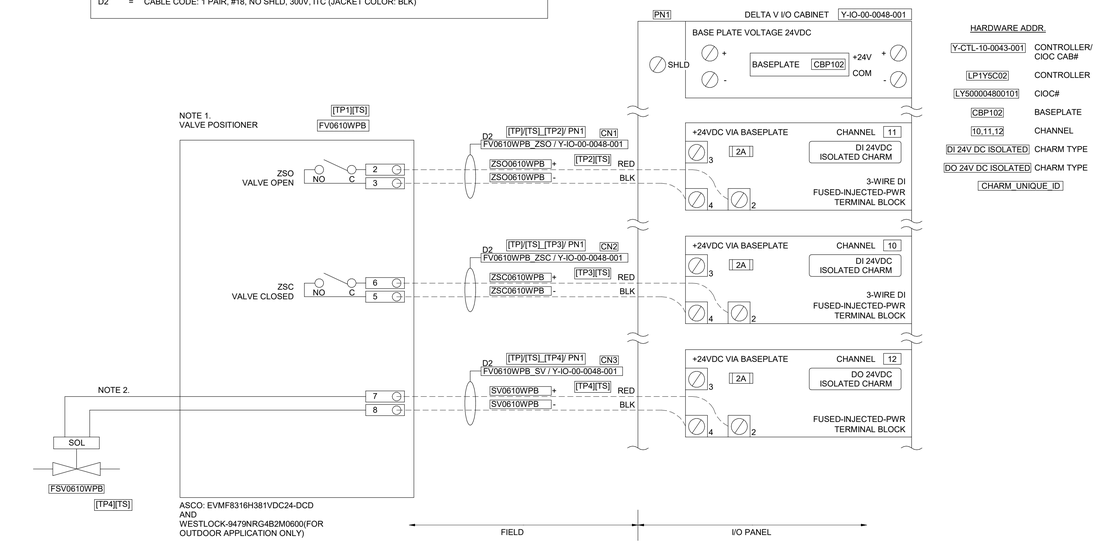
LABELING DESCRIPTION AND EXAMPLE		
[TP1]	=	TAG PREFIX = ON/OFF VALVE = FV
[TP2]	=	TAG PREFIX = POSITION SWITCH, OPEN = ZSO
[TP3]	=	TAG PREFIX = POSITION SWITCH, CLOSED = ZSC
[TP4]	=	TAG PREFIX = SOLENOID VALVE = SV
[TS]	=	TAG SUFFIX = LOOP NUMBER = 0610WPB
[PN1]	=	IO PANEL = <BLDG>-<TYPE>-<FLOOR>-<ROOM # OR SKID#>-<INDEX> = Y-IO-00-0048-001
[CN1]	=	SIGNAL CABLE TAG = [TP1][TS]_[TP2] / [PN1] = FV0610WPB_ZSO / Y-IO-00-0048-001
[CN2]	=	SIGNAL CABLE TAG = [TP1][TS]_[TP3] / [PN1] = FV0610WPB_ZSC / Y-IO-00-0048-001
[CN3]	=	SIGNAL CABLE TAG = [TP1][TS]_[TP4] / [PN1] = FV0610WPB_SV / Y-IO-00-0048-001
[CHARM_UNIQUE_ID] = [FN]-[SN]-[CN]		
D2	=	CABLE CODE: 1 PAIR, #18, NO SHLD, 300V, ITC (JACKET COLOR: BLK)



NOTES:
1. USE WESTLOCK LIMIT SWITCHES (9479NRG4B2M0600) FOR OUTDOOR USE ONLY.
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4. ALL TAGS WILL HAVE PREFIX "LP1Y".

1 INSTRUMENT WIRING DIAGRAM - LP1YFV0610WPB - DISCRETE INPUT AND OUTPUT DELTA V CHARMS IO

NTS

REFERENCE DRAWINGS:

1. LP1Y-270-LST-K215-02: INSTRUMENT I/O LIST.
2. LP1Y-270-LST-K215-03: INSTRUMENT CABLE SCHEDULE.
3. LP1Y-270-LAY-L215-01 & 02: INSTRUMENT LOCATION PLAN.
4. LP1Y-275-DIA-K215-30: CHARM I/O PANEL WIRING.

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REV	DATE	DES	ENG	CHK	APP	PM APP
1	10-APR-26	MGL	JN	BM	AJT	SS
A	16-JAN-26	MGL	JN	BM	AJT	SS

INSTRUMENT WIRING DIAGRAM - LP1YFV0610WPB - DISCRETE INPUT AND OUTPUT DELTA-V CHARMS IO

RAIL UNLOADING RACK

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
1	ALL	ISSUED FOR CONSTRUCTION	NA	10-APR-26
A	ALL	ISSUED FOR REVIEW	NA	16-JAN-26

REVISIONS

SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: 1
SUB DISCIPLINE	LOCATION CODE: LP1	LOCATION/GRID: COL C-D / 33-43	CCI CODE: YELLOW
CRITICAL QUALITY		ENTERPRISE GROUP: API	DRAWING INTENTION: PROJECT NAME: LP1-306-1181
CRITICAL BUSINESS		EQUIPMENT CLASSIFICATION: NA	DRAWING SCALE: NONE
CRITICAL HSE		VENDOR: NA	ALTERNATE DOCUMENT ID: NA / MERIDIAN REV: B
NON-CRITICAL		ENGINEERING AGENCY DOC ID: NA	DOCUMENT ID: LP1Y-275-DIA-K215-06

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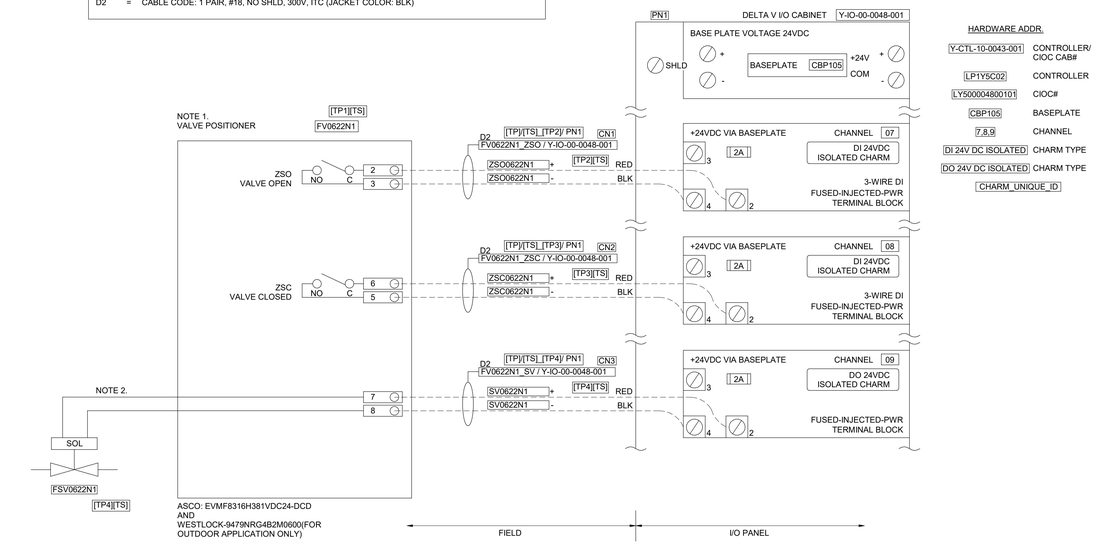
LABELING DESCRIPTION AND EXAMPLE		
TP1	=	TAG PREFIX = ON/OFF VALVE = FV
TP2	=	TAG PREFIX = POSITION SWITCH, OPEN = ZSO
TP3	=	TAG PREFIX = POSITION SWITCH, CLOSED = ZSC
TP4	=	TAG PREFIX = SOLENOID VALVE = SV
TS	=	TAG SUFFIX = LOOP NUMBER = 0622N1
PN1	=	IO PANEL = <BLDG>-<TYPE>-<FLOOR>-<ROOM # OR SKID#>-<INDEX> = Y-IO-00-0048-001
CN1	=	SIGNAL CABLE TAG = TP1 TS _TP2 / PN1 = FV0622N1_ZSO / Y-IO-00-0048-001
CN2	=	SIGNAL CABLE TAG = TP1 TS _TP3 / PN1 = FV0622N1_ZSC / Y-IO-00-0048-001
CN3	=	SIGNAL CABLE TAG = TP1 TS _TP4 / PN1 = FV0622N1_SV / Y-IO-00-0048-001
CHARM_UNIQUE_ID	=	FN _ SN - CN
D2	=	CABLE CODE: 1 PAIR, #18, NO SHLD, 300V, ITC (JACKET COLOR: BLK)



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4. ALL TAGS WILL HAVE PREFIX "LP1Y".

1 INSTRUMENT WIRING DIAGRAM - LP1YFV0622N1 - DISCRETE INPUT AND OUTPUT DELTA V CHARMS IO

NTS

REFERENCE DRAWINGS:

1. LP1Y-270-LST-K215-02: INSTRUMENT I/O LIST.
2. LP1Y-270-LST-K215-03: INSTRUMENT CABLE SCHEDULE.
3. LP1Y-270-LAY-L215-01 & 02: INSTRUMENT LOCATION PLAN.
4. LP1Y-275-DIA-K215-30: CHARM I/O PANEL WIRING.

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0	10-APR-26					
A	16-JAN-26					

INSTRUMENT WIRING DIAGRAM - LP1YFV0622N1 - DISCRETE INPUT AND OUTPUT DELTA-V CHARMS IO

RAIL UNLOADING RACK

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
0	ALL	ISSUED FOR CONSTRUCTION	NA	10-APR-26
A	ALL	ISSUED FOR REVIEW	NA	16-JAN-26

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SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: 1
SUB DISCIPLINE	LOCATION CODE: LP1	LOCATION/GRID: COL C-D / 33-43	CCI CODE: YELLOW / PROJECT NAME: LP1-306-1181
CRITICAL QUALITY	ENTERPRISE GROUP: API	DRAWING INTENTION	DRAWING SCALE: NONE
CRITICAL BUSINESS	EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA	MERIDIAN REV: B
CRITICAL HSE	VENDOR: NA	DOCUMENT ID: LP1Y-275-DIA-K215-07	
NON-CRITICAL	ENGINEERING AGENCY DOC ID: NA		

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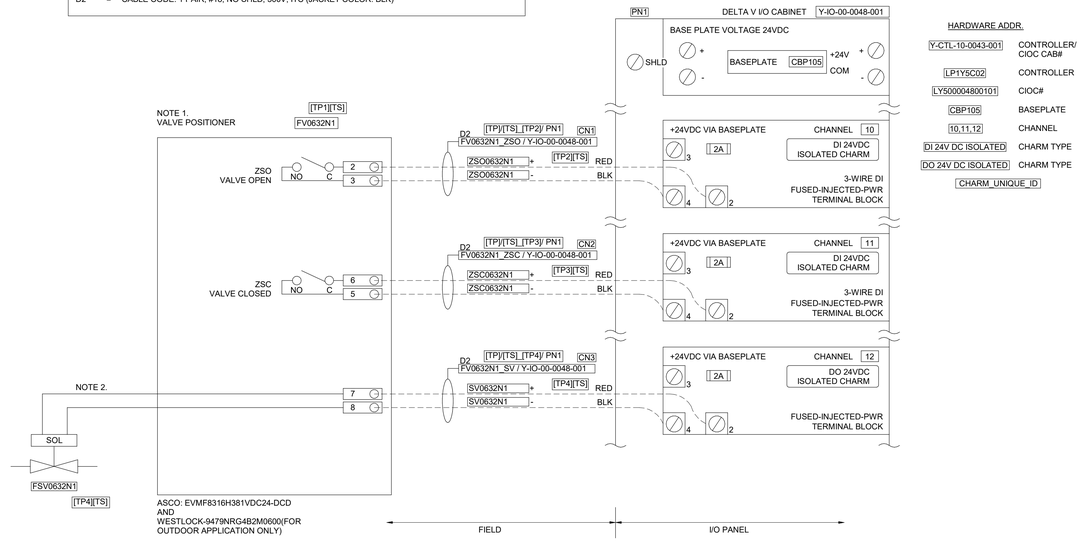
LABELING DESCRIPTION AND EXAMPLE		
[TP1]	=	TAG PREFIX = ON/OFF VALVE = FV
[TP2]	=	TAG PREFIX = POSITION SWITCH, OPEN = ZSO
[TP3]	=	TAG PREFIX = POSITION SWITCH, CLOSED = ZSC
[TP4]	=	TAG PREFIX = SOLENOID VALVE = SV
[TS]	=	TAG SUFFIX = LOOP NUMBER = 0632N1
[PN1]	=	IO PANEL = <BLDG>-<TYPE>-<FLOOR>-<ROOM # OR SKID#>-<INDEX> = Y-IO-00-0048-001
[CN1]	=	SIGNAL CABLE TAG = [TP1][TS]_TP2 / [PN1] = FV0632N1_ZSO / Y-IO-00-0048-001
[CN2]	=	SIGNAL CABLE TAG = [TP1][TS]_TP3 / [PN1] = FV0632N1_ZSC / Y-IO-00-0048-001
[CN3]	=	SIGNAL CABLE TAG = [TP1][TS]_TP4 / [PN1] = FV0632N1_SV / Y-IO-00-0048-001
[CHARM_UNIQUE_ID] = [FN]-[SN]-[CN]		
D2	=	CABLE CODE: 1 PAIR, #18, NO SHLD, 300V, ITC (JACKET COLOR: BLK)



HARDWARE ADDR.

Y-CTL-10-0043-001	CONTROLLER/ CIOC CAB#
LP1Y5C02	CONTROLLER
LY500004800101	CIOC#
CBP105	BASEPLATE
10,11,12	CHANNEL
DI 24V DC ISOLATED	CHARM TYPE
DO 24V DC ISOLATED	CHARM TYPE
CHARM_UNIQUE_ID	

NOTES:
1. USE WESTLOCK LIMIT SWITCHES (9479NRG4B2M0600) FOR OUTDOOR USE ONLY.
2. ASCO SOLENOID VALVE IS PRE-WIRED TO VALVE POSITIONER.
3. ALL CABLES FROM FIELD INSTRUMENT TO CHARM I/O PANEL OR IPP TO BE FIELD ROUTED USING SNAP TRACK CHANNEL TRAY OR EQUIVALENT.
4. ALL TAGS WILL HAVE PREFIX "LP1Y".

1 INSTRUMENT WIRING DIAGRAM - LP1YFV0632N1 - DISCRETE INPUT AND OUTPUT DELTA V CHARMS IO

NTS

REFERENCE DRAWINGS:

1. LP1Y-270-LST-K215-02: INSTRUMENT I/O LIST.
2. LP1Y-270-LST-K215-03: INSTRUMENT CABLE SCHEDULE.
3. LP1Y-270-LAY-L215-01 & 02: INSTRUMENT LOCATION PLAN.
4. LP1Y-275-DIA-K215-30: CHARM I/O PANEL WIRING.

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
0	ALL	ISSUED FOR CONSTRUCTION	NA	10-APR-26
A	ALL	ISSUED FOR REVIEW	NA	16-JAN-26

REVISIONS

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1	10-APR-26	MGL	JN	BM	AJT	SS
0	16-JAN-26	MGL	JN	BM	AJT	SS

INSTRUMENT WIRING DIAGRAM - LP1YFV0632N1 - DISCRETE INPUT AND OUTPUT DELTA-V CHARMS IO

RAIL UNLOADING RACK

SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: 1
SUB DISCIPLINE	LOCATION CODE: LP1	LOCATION/GRID: COL C-D / 33-43	CCI CODE: YELLOW
			PROJECT NAME: LP1-306-1181
CRITICAL QUALITY		ENTERPRISE GROUP: API	DRAWING INTENTION:
			DRAWING SCALE: NONE
CRITICAL BUSINESS		EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA
			MERIDIAN REV: B
CRITICAL HSE		VENDOR: NA	DOCUMENT ID: LP1Y-275-DIA-K215-08
NON-CRITICAL		ENGINEERING AGENCY DOC ID: NA	

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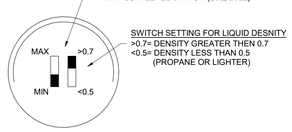
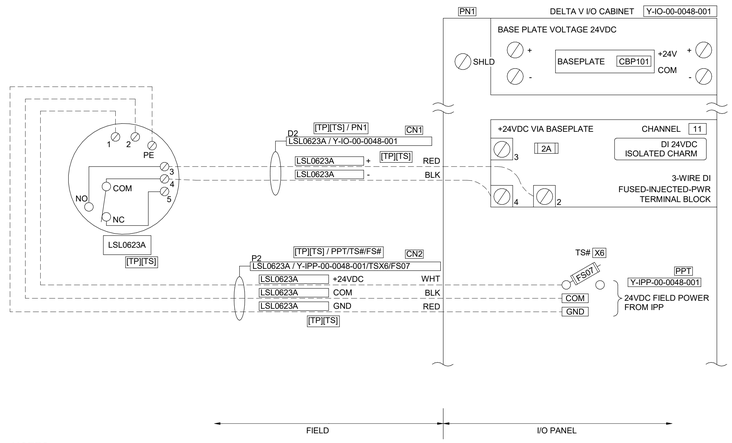
LABELING DESCRIPTION AND EXAMPLE		
TP	=	TAG PREFIX = LEVEL SWITCH = LSL
TS	=	TAG SUFFIX = LOOP NUMBER = 0623A
PN1	=	IO PANEL = <BLDG>-<TYPE>-<FLOOR>-<ROOM # OR SKID#>-<INDEX> = Y-IO-00-0048-001
PPT	=	POWER PANEL TAG = <BLDG>-<TYPE>-<FLOOR>-<ROOM# OR SKID#>-<INDEX> = Y-IPP-00-0048-001
CN1	=	SIGNAL CABLE TAG = TP TS / PN1 = LSL0623A / Y-IO-00-0048-001
CN2	=	POWER CABLE TAG = TP TS / PPT / TS# / FS# = LSL0623A / Y-IPP-00-0048-001 / TSX6 / FS9
CHARM_UNIQUE_ID = PN - SN - CN		
D2	=	CABLE CODE: 1 PAIR, #18, NO SHLD, 300V, ITC (JACKET COLOR: BLK)
P2	=	CABLE CODE: 1 TRAID, #16, NO SHLD, 300V, ITC (JACKET COLOR: BLK)

SWITCH SETTING FOR LIQUID LEVEL APPLICATIONS LEVEL SWITCH

MAX= HIGH LEVEL SWITCH (LSH, LSHH)
MIN= LOW LEVEL SWITCH (LSL, LSLL)

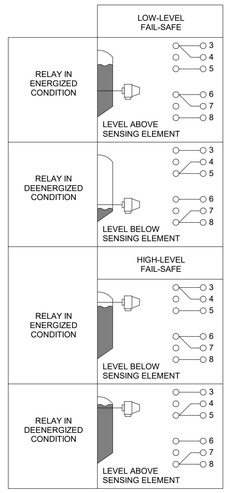


SWITCH SETTING FOR LIQUID DESNITY
>0.7= DENSITY GREATER THEN 0.7
<0.5= DENSITY LESS THAN 0.5
(PROPANE OR LIGHTER)



HARDWARE ADDR.

Y-CTL-10-0043-001	CONTROLLER/ CIOC CAB#
LP1Y5C02	CONTROLLER
LY500004800101	CIOC#
CBP101	BASEPLATE
11	CHANNEL
DI 24V DC ISOLATED	CHARM TYPE
CHARM_UNIQUE_ID	



NOTES:

1. ALL CABLES FROM FIELD INSTRUMENT TO CHARM I/O PANEL OR IPP TO BE FIELD ROUTED USING SNAP TRACK CHANNEL TRAY OR EQUIVALENT.
2. ALL TAGS WILL HAVE PREFIX "LP1Y".

1 INSTRUMENT WIRING DIAGRAM - LP1YLSL0623A - DISCRETE INPUT 4-WIRE, 24VDC THERMAL MASS FLOW SWITCH

NTS

REFERENCE DRAWINGS:

1. LP1Y-270-LST-K215-02: INSTRUMENT I/O LIST.
2. LP1Y-270-LST-K215-03: INSTRUMENT CABLE SCHEDULE.
3. LP1Y-270-LAY-L215-01 & 02: INSTRUMENT LOCATION PLAN.
4. LP1Y-275-DIA-K215-30: CHARM I/O PANEL WIRING.
5. LP1Y-275-DIA-K215-31: 24VDC IPP PANEL WIRING.

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0	16-JAN-26	MGL	JN	BM	AJT	SS

INSTRUMENT WIRING DIAGRAM - LP1YLSL0623A - DISCRETE INPUT 4-WIRE, 24VDC THERMAL MASS FLOW SWITCH - RAIL UNLOADING RACK

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
1	ALL	ISSUED FOR CONSTRUCTION	NA	10-APR-26
0	ALL	ISSUED FOR REVIEW	NA	16-JAN-26

REVISIONS

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SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: 0
SUB DISCIPLINE	LOCATION CODE: LP1	LOCATION/GRID: COL C-D / 33-43	CCI CODE: YELLOW
			PROJECT NAME: LP1-306-1181
CRITICAL QUALITY		ENTERPRISE GROUP: API	DRAWING INTENTION
			DRAWING SCALE: NONE
CRITICAL BUSINESS		EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA
			MERIDIAN REV: B
CRITICAL HSE		VENDOR: NA	DOCUMENT ID
NON-CRITICAL		ENGINEERING AGENCY DOC ID: NA	LP1Y-275-DIA-K215-09

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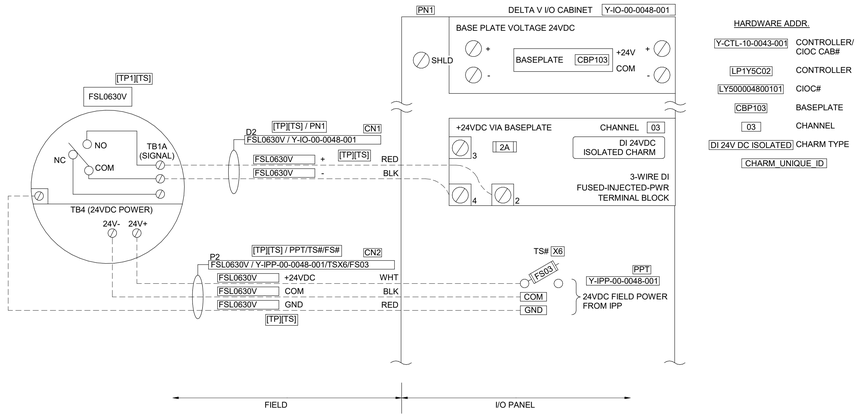
LABELING DESCRIPTION AND EXAMPLE		
TP	=	TAG PREFIX = FLOW SWITCH = FSL, FSLL, FSH, FSHH
TS	=	TAG SUFFIX = LOOP NUMBER = 0630V
PN1	=	IO PANEL = <BLDG>-<TYPE>-<FLOOR>-<ROOM # OR SKID#>-<INDEX> = Y-IO-00-0048-001
PPT	=	POWER PANEL TAG = <BLDG>-<TYPE>-<FLOOR>-<ROOM # OR SKID#>-<INDEX> = Y-IPP-00-0048-001
CN1	=	SIGNAL CABLE TAG = TP TS / PN1 = FSL0630V / Y-IO-00-0048-001
CN2	=	POWER CABLE TAG = TP TS / PPT / TS# / FS# = FSL0630V / Y-IPP-00-0048-001 / TSX6 / FS9
CHARM_UNIQUE_ID = PN - SN - CN		
D2	=	CABLE CODE: 1 PAIR, #18, NO SHLD, 300V, ITC (JACKET COLOR: BLK)
P1	=	CABLE CODE: 1 TRIAD, #16, NO SHLD, 300V,ITC (JACKET COLOR: BLK)



NOTE:
1. HEAT SHRINK RED CONDUCTOR GREEN THREE INCHES IN LENGTH TO IDENTIFY AS GROUND.
2. ALL CABLES FROM FIELD INSTRUMENT TO CHARM I/O PANEL OR IPP TO BE FIELD ROUTED USING SNAP TRACK CHANNEL TRAY OR EQUIVALENT.
3. ALL TAGS WILL HAVE PREFIX "LP1Y".

1 INSTRUMENT WIRING DIAGRAM - LP1YFSL0630V - DISCRETE INPUT 4-WIRE, 24VDC THERMAL MASS FLOW SWITCH

NTS

REFERENCE DRAWINGS:

1. LP1Y-270-LST-K215-02: INSTRUMENT I/O LIST.
2. LP1Y-270-LST-K215-03: INSTRUMENT CABLE SCHEDULE.
3. LP1Y-270-LAY-L215-01 & 02: INSTRUMENT LOCATION PLAN.
4. LP1Y-275-DIA-K215-30: CHARM I/O PANEL WIRING.
5. LP1Y-275-DIA-K215-31: 24VDC IPP PANEL WIRING.

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A	16-JAN-26	MGL	JN	BM	AJT	SS

INSTRUMENT WIRING DIAGRAM - LP1YFSL0630V - DISCRETE INPUT 4-WIRE, 24VDC THERMAL MASS FLOW SWITCH - RAIL UNLOADING RACK

REVISIONS			
REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
0	ALL	ISSUED FOR CONSTRUCTION	NA	10-APR-26
A	ALL	ISSUED FOR REVIEW	NA	16-JAN-26

SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: 1
SUB DISCIPLINE: LP1	LOCATION CODE: COL C-D / 33-43	CCI CODE: YELLOW	PROJECT NAME: LP1-306-1181
CRITICAL QUALITY: API	ENTERPRISE GROUP	DRAWING INTENTION	DRAWING SCALE: NONE
CRITICAL BUSINESS: NA	EQUIPMENT CLASSIFICATION	ALTERNATE DOCUMENT ID: NA	MERIDIAN REV: B
CRITICAL HSE: NA	VENDOR	DOCUMENT ID: LP1Y-275-DIA-K215-10	
NON-CRITICAL: NA	ENGINEERING AGENCY DOC ID		

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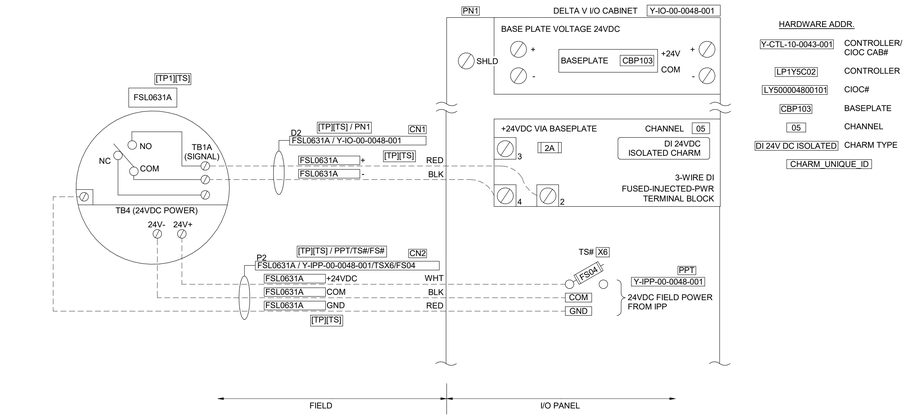
LABELING DESCRIPTION AND EXAMPLE		
TP	=	TAG PREFIX = FLOW SWITCH = FSL, FSLL, FSH, FSHH
TS	=	TAG SUFFIX = LOOP NUMBER = 0631A
PN1	=	IO PANEL = <BLDG>-<TYPE>-<FLOOR>-<ROOM # OR SKID#>-<INDEX> = Y-IO-00-0048-001
PPT	=	POWER PANEL TAG = <BLDG>-<TYPE>-<FLOOR>-<ROOM # OR SKID#>-<INDEX> = Y-IPP-00-0048-001
CN1	=	SIGNAL CABLE TAG = TP TS / PN1 = FSL0631A / Y-IO-00-0048-001
CN2	=	POWER CABLE TAG = TP TS / PPT / TS# / FS# = FSL0631A / Y-IPP-00-0048-001 / TSX6 / FS9
CHARM_UNIQUE_ID	=	FN - SN - CN
D2	=	CABLE CODE: 1 PAIR, #18, NO SHLD, 300V, ITC (JACKET COLOR: BLK)
P1	=	CABLE CODE: 1 TRIAD, #16, NO SHLD, 300V,ITC (JACKET COLOR: BLK)



NOTE:
1. HEAT SHRINK RED CONDUCTOR GREEN THREE INCHES IN LENGTH TO IDENTIFY AS GROUND.
2. ALL CABLES FROM FIELD INSTRUMENT TO CHARM I/O PANEL OR IPP TO BE FIELD ROUTED USING SNAP TRACK CHANNEL TRAY OR EQUIVALENT.
3. ALL TAGS WILL HAVE PREFIX "LP1Y".

1 INSTRUMENT WIRING DIAGRAM - LP1YFSL0631A - DISCRETE INPUT 4-WIRE, 24VDC THERMAL MASS FLOW SWITCH

NTS

REFERENCE DRAWINGS:

1. LP1Y-270-LST-K215-02: INSTRUMENT I/O LIST.
2. LP1Y-270-LST-K215-03: INSTRUMENT CABLE SCHEDULE.
3. LP1Y-270-LAY-L215-01 & 02: INSTRUMENT LOCATION PLAN.
4. LP1Y-275-DIA-K215-30: CHARM I/O PANEL WIRING.
5. LP1Y-275-DIA-K215-31: 24VDC IPP PANEL WIRING.

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INSTRUMENT WIRING DIAGRAM - LP1YFSL0631A - DISCRETE INPUT 4-WIRE, 24VDC THERMAL MASS FLOW SWITCH - RAIL UNLOADING RACK

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
1	ALL	ISSUED FOR CONSTRUCTION	NA	10-APR-26
0	ALL	ISSUED FOR REVIEW	NA	16-JAN-26

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SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: 0
SUB DISCIPLINE:	LOCATION CODE: LP1	LOCATION/GRID: COL C-D / 33-43	CCI CODE: YELLOW
CRITICAL QUALITY:	ENTERPRISE GROUP: API	DRAWING INTENTION:	PROJECT NAME: LP1-306-1181
CRITICAL BUSINESS:	EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA	DRAWING SCALE: NONE
CRITICAL HSE:	VENDOR: NA	DOCUMENT ID: LP1Y-275-DIA-K215-11	MERIDIAN REV: B
NON-CRITICAL:	ENGINEERING AGENCY DOC ID: NA		

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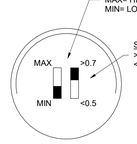
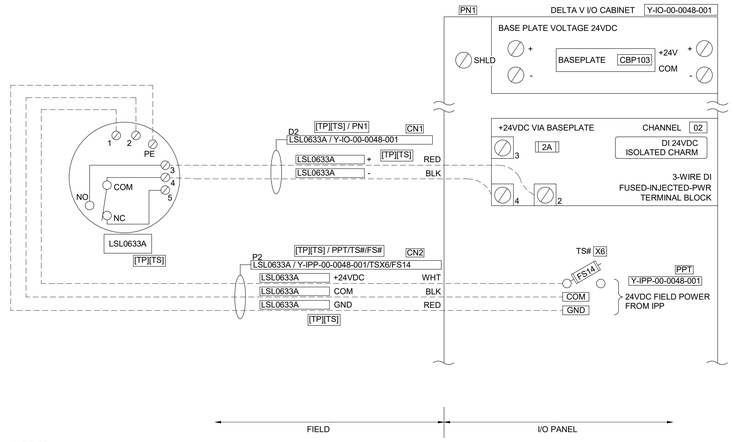
LABELING DESCRIPTION AND EXAMPLE		
TP	=	TAG PREFIX = LEVEL SWITCH = LSL
TS	=	TAG SUFFIX = LOOP NUMBER = 0633A
PN1	=	IO PANEL = <BLDG>-<TYPE>-<FLOOR>-<ROOM # OR SKID#>-<INDEX> = Y-IO-00-0048-001
PPT	=	POWER PANEL TAG = <BLDG>-<TYPE>-<FLOOR>-<ROOM# OR SKID#>-<INDEX> = Y-IPP-00-0048-001
CN1	=	SIGNAL CABLE TAG = TP TS / PN1 = LSL0633A / Y-IO-00-0048-001
CN2	=	POWER CABLE TAG = TP TS / PPT / TS# / FS# = LSL0633A / Y-IPP-00-0048-001 / TSX6 / FS9
CHARM_UNIQUE_ID	=	FN - SN - CN
D2	=	CABLE CODE: 1 PAIR, #18, NO SHLD, 300V, ITC (JACKET COLOR: BLK)
P2	=	CABLE CODE: 1 TRAID, #16, NO SHLD, 300V, ITC (JACKET COLOR: BLK)

SWITCH SETTING FOR LIQUID LEVEL APPLICATIONS LEVEL SWITCH

MAX= HIGH LEVEL SWITCH (LSH, LSHH)
MIN= LOW LEVEL SWITCH (LSL, LSLL)

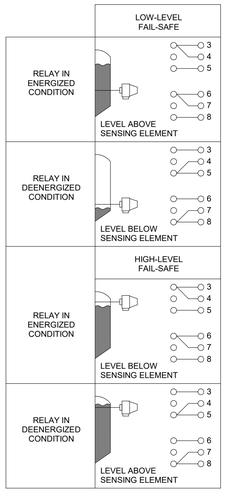
SWITCH SETTING FOR LIQUID DESNITY
>0.7= DENSITY GREATER THEN 0.7
<0.5= DENSITY LESS THAN 0.5
(PROPANE OR LIGHTER)





HARDWARE ADDR.

Y-CTL-10-0043-001	CONTROLLER/ CIOC CAB#
LP1YSC02	CONTROLLER
LY500004800101	CIOC#
CBP103	BASEPLATE
02	CHANNEL
DI 24V DC ISOLATED	CHARM TYPE
CHARM_UNIQUE_ID	



NOTES:

1. ALL CABLES FROM FIELD INSTRUMENT TO CHARM I/O PANEL OR IPP TO BE FIELD ROUTED USING SNAP TRACK CHANNEL TRAY OR EQUIVALENT.
2. ALL TAGS WILL HAVE PREFIX "LP1Y".

1 INSTRUMENT WIRING DIAGRAM - LP1YLSL0633A - DISCRETE INPUT 4-WIRE, 24VDC THERMAL MASS FLOW SWITCH

NTS

REFERENCE DRAWINGS:

1. LP1Y-270-LST-K215-02: INSTRUMENT I/O LIST.
2. LP1Y-270-LST-K215-03: INSTRUMENT CABLE SCHEDULE.
3. LP1Y-270-LAY-L215-01 & 02: INSTRUMENT LOCATION PLAN.
4. LP1Y-275-DIA-K215-30: CHARM I/O PANEL WIRING.
5. LP1Y-275-DIA-K215-31: 24VDC IPP PANEL WIRING.

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0	16-JAN-26	MGL	JN	BM	AJT	SS

INSTRUMENT WIRING DIAGRAM - LP1YLSL0633A - DISCRETE INPUT 4-WIRE, 24VDC THERMAL MASS FLOW SWITCH - RAIL UNLOADING RACK

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
1	ALL	ISSUED FOR CONSTRUCTION	NA	10-APR-26
0	ALL	ISSUED FOR REVIEW	NA	16-JAN-26

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SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: 1
SUB DISCIPLINE	LOCATION CODE: LP1	LOCATION/GRID: COL C-D / 33-43	CCI CODE: YELLOW
			PROJECT NAME: LP1-306-1181
CRITICAL QUALITY	ENTERPRISE GROUP: API	DRAWING INTENTION	DRAWING SCALE: NONE
CRITICAL BUSINESS	EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA	MERIDIAN REV: B
CRITICAL HSE	VENDOR: NA	DOCUMENT ID: LP1Y-275-DIA-K215-12	
NON-CRITICAL	ENGINEERING AGENCY DOC ID: NA		

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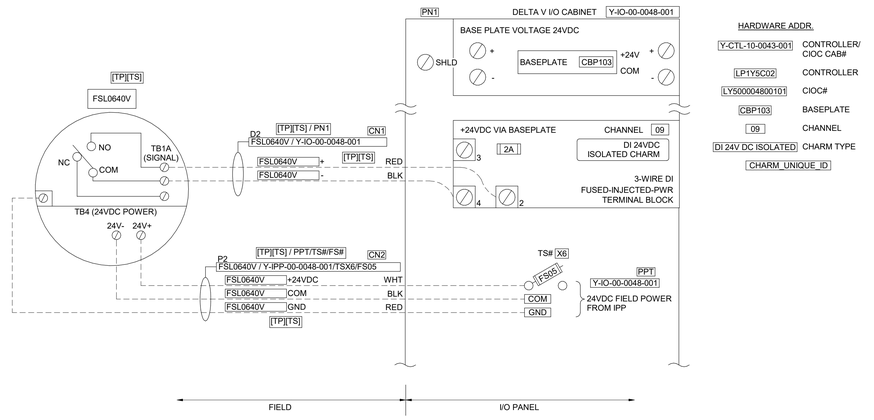
LABELING DESCRIPTION AND EXAMPLE		
TP	=	TAG PREFIX = FLOW SWITCH = FSL, FSLL, FSH, FSHH
TS	=	TAG SUFFIX = LOOP NUMBER = 0640V
PN1	=	IO PANEL = <BLDG>-<TYPE>-<FLOOR>-<ROOM # OR SKID#>-<INDEX> = Y-IO-00-0048-001
PPT	=	POWER PANEL TAG = <BLDG>-<TYPE>-<FLOOR>-<ROOM # OR SKID#>-<INDEX> = Y-IPP-00-0048-001
CN1	=	SIGNAL CABLE TAG = [TP][TS] / [PN1] = FSL0640V / Y-IO-00-0048-001
CN2	=	POWER CABLE TAG = [TP][TS] / [PPT] / TS# / FS# = FSL0640V / Y-IPP-00-0048-001 / TSX6 / FS9
CHARM_UNIQUE_ID	=	FN - SN - CN
D2	=	CABLE CODE: 1 PAIR, #18, NO SHLD, 300V, ITC (JACKET COLOR: BLK)
P1	=	CABLE CODE: 1 TRIAD, #16, NO SHLD, 300V,ITC (JACKET COLOR: BLK)



HARDWARE ADDR.	
Y-CTL-10-0043-001	CONTROLLER/ CIOC CAB#
LP1Y5C02	CONTROLLER
LY500004800101	CIOC#
CBP103	BASEPLATE
09	CHANNEL
DI 24V DC ISOLATED	CHARM TYPE
CHARM_UNIQUE_ID	

NOTE:
1. HEAT SHRINK RED CONDUCTOR GREEN THREE INCHES IN LENGTH TO IDENTIFY AS GROUND.
2. ALL CABLES FROM FIELD INSTRUMENT TO CHARM I/O PANEL OR IPP TO BE FIELD ROUTED USING SNAP TRACK CHANNEL TRAY OR EQUIVALENT.
3. ALL TAGS WILL HAVE PREFIX "LP1Y".

1 INSTRUMENT WIRING DIAGRAM - LP1YFSL0640V - DISCRETE INPUT 4-WIRE, 24VDC THERMAL MASS FLOW SWITCH

NTS

REFERENCE DRAWINGS:

1. LP1Y-270-LST-K215-02: INSTRUMENT I/O LIST.
2. LP1Y-270-LST-K215-03: INSTRUMENT CABLE SCHEDULE.
3. LP1Y-270-LAY-L215-01 & 02: INSTRUMENT LOCATION PLAN.
4. LP1Y-275-DIA-K215-30: CHARM I/O PANEL WIRING.
5. LP1Y-275-DIA-K215-31: 24VDC IPP PANEL WIRING.

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
0	ALL	ISSUED FOR CONSTRUCTION	NA	10-APR-26
A	ALL	ISSUED FOR REVIEW	NA	16-JAN-26

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0	10-APR-26	MGL	JN	BM	AJT	SS
A	16-JAN-26	MGL	JN	BM	AJT	SS

INSTRUMENT WIRING DIAGRAM - LP1YFSL0640V - DISCRETE INPUT 4-WIRE, 24VDC THERMAL MASS FLOW SWITCH - RAIL UNLOADING RACK

SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: 1
SUB DISCIPLINE	LOCATION CODE: LP1	LOCATION/GRID: COL C-D / 33-43	CCI CODE: YELLOW
			PROJECT NAME: LP1-306-1181
CRITICAL QUALITY		ENTERPRISE GROUP: API	DRAWING INTENTION: NONE
			DRAWING SCALE: NONE
CRITICAL BUSINESS		EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA
			MERIDIAN REV: B
CRITICAL HSE		VENDOR: NA	DOCUMENT ID: LP1Y-275-DIA-K215-13
NON-CRITICAL		ENGINEERING AGENCY DOC ID: NA	

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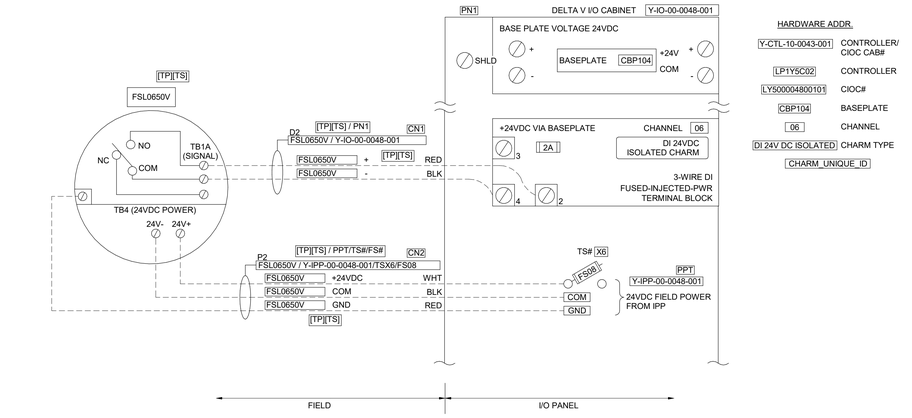
LABELING DESCRIPTION AND EXAMPLE		
TP	=	TAG PREFIX = FLOW SWITCH = FSL, FSLL, FSH, FSHH
TS	=	TAG SUFFIX = LOOP NUMBER = 0650V
PN1	=	IO PANEL = <BLDG>-<TYPE>-<FLOOR>-<ROOM # OR SKID#>-<INDEX> = Y-IO-00-0048-001
PPT	=	POWER PANEL TAG = <BLDG>-<TYPE>-<FLOOR>-<ROOM # OR SKID#>-<INDEX> = Y-IPP-00-0048-001
CN1	=	SIGNAL CABLE TAG = TP TS / PN1 = FSL0650V / Y-IO-00-0048-001
CN2	=	POWER CABLE TAG = TP TS / PPT / TS# / FS# = FSL0650V / Y-IPP-00-0048-001 / TSX6 / FS9
CHARM_UNIQUE_ID = FN - SN - CN		
D2	=	CABLE CODE: 1 PAIR, #18, NO SHLD, 300V, ITC (JACKET COLOR: BLK)
P1	=	CABLE CODE: 1 TRIAD, #16, NO SHLD, 300V,ITC (JACKET COLOR: BLK)



HARDWARE ADDR.	
Y-CTL-10-0043-001	CONTROLLER/ CIOC CAB#
LP1YSC02	CONTROLLER
LY50000480010 1	CIOC#
CBP104	BASEPLATE
06	CHANNEL
DI 24V DC ISOLATED	CHARM TYPE
CHARM_UNIQUE_ID	

NOTE:
1. HEAT SHRINK RED CONDUCTOR GREEN THREE INCHES IN LENGTH TO IDENTIFY AS GROUND.
2. ALL CABLES FROM FIELD INSTRUMENT TO CHARM I/O PANEL OR IPP TO BE FIELD ROUTED USING SNAP TRACK CHANNEL TRAY OR EQUIVALENT.
3. ALL TAGS WILL HAVE PREFIX "LP1Y".

1 INSTRUMENT WIRING DIAGRAM - LP1YFSL0650V - DISCRETE INPUT 4-WIRE, 24VDC THERMAL MASS FLOW SWITCH

NTS

REFERENCE DRAWINGS:

1. LP1Y-270-LST-K215-02: INSTRUMENT I/O LIST.
2. LP1Y-270-LST-K215-03: INSTRUMENT CABLE SCHEDULE.
3. LP1Y-270-LAY-L215-01 & 02: INSTRUMENT LOCATION PLAN.
4. LP1Y-275-DIA-K215-30: CHARM I/O PANEL WIRING.
5. LP1Y-275-DIA-K215-31: 24VDC IPP PANEL WIRING.

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0	16-JAN-26	MGL	JN	BM	AJT	SS

Lilly

INSTRUMENT WIRING DIAGRAM - LP1YFSL0650V - DISCRETE INPUT 4-WIRE, 24VDC THERMAL MASS FLOW SWITCH - RAIL UNLOADING RACK

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
1	ALL	ISSUED FOR CONSTRUCTION	NA	10-APR-26
0	ALL	ISSUED FOR REVIEW	NA	16-JAN-26

REVISIONS

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SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: 1
SUB DISCIPLINE	LOCATION CODE: LP1	LOCATION/GRID: COL C-D / 33-43	CCI CODE: YELLOW
			PROJECT NAME: LP1-306-1181
CRITICAL QUALITY		ENTERPRISE GROUP: API	DRAWING INTENTION: NONE
			DRAWING SCALE:
CRITICAL BUSINESS		EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA
			MERIDIAN REV: B
CRITICAL HSE		VENDOR: NA	DOCUMENT ID: LP1Y-275-DIA-K215-14
NON-CRITICAL		ENGINEERING AGENCY DOC ID: NA	

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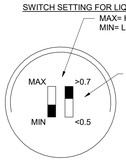
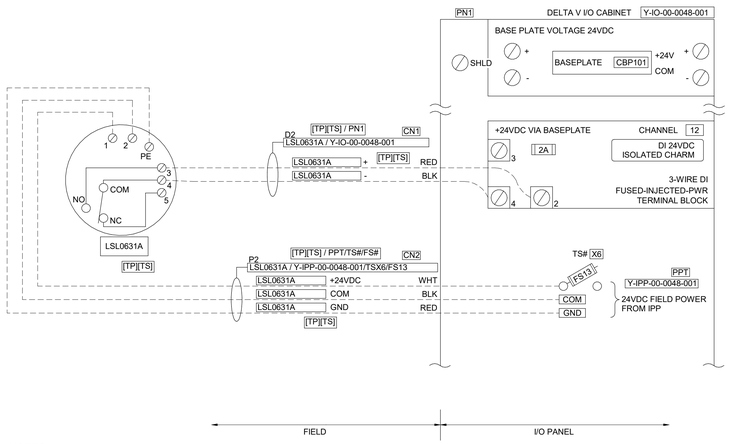
LABELING DESCRIPTION AND EXAMPLE		
TP	=	TAG PREFIX = LEVEL SWITCH = LSL
TS	=	TAG SUFFIX = LOOP NUMBER = 0631A
PN1	=	IO PANEL = <BLDG>-<TYPE>-<FLOOR>-<ROOM # OR SKID#>-<INDEX> = Y-IO-00-0048-001
PPT	=	POWER PANEL TAG = <BLDG>-<TYPE>-<FLOOR>-<ROOM# OR SKID#>-<INDEX> = Y-IPP-00-0048-001
CN1	=	SIGNAL CABLE TAG = TP TS / PN1 = LSL0631A / Y-IO-00-0048-001
CN2	=	POWER CABLE TAG = TP TS / PPT / TS# / FS# = LSL0631A / Y-IPP-00-0048-001 / TSX6 / FS9
CHARM_UNIQUE_ID	=	FN - SN - CN
D2	=	CABLE CODE: 1 PAIR, #18, NO SHLD, 300V, ITC (JACKET COLOR: BLK)
P2	=	CABLE CODE: 1 TRAID, #16, NO SHLD, 300V, ITC (JACKET COLOR: BLK)

SWITCH SETTING FOR LIQUID LEVEL APPLICATIONS LEVEL SWITCH

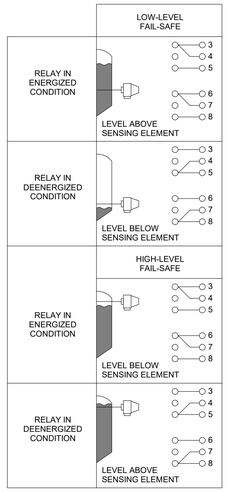
MAX= HIGH LEVEL SWITCH (LSH, LSHH)
MIN= LOW LEVEL SWITCH (LSL, LSLL)

SWITCH SETTING FOR LIQUID DESNITY
>0.7= DENSITY GREATER THEN 0.7
<0.5= DENSITY LESS THAN 0.5
(PROPANE OR LIGHTER)





HARDWARE ADDR.	
Y-CTL-10-0043-001	CONTROLLER/ CIOC CAB#
LP1YSC02	CONTROLLER
LY500004800101	CIOC#
CBP101	BASEPLATE
12	CHANNEL
DI 24V DC ISOLATED	CHARM TYPE
CHARM_UNIQUE_ID	



NOTES:

1. ALL CABLES FROM FIELD INSTRUMENT TO CHARM I/O PANEL OR IPP TO BE FIELD ROUTED USING SNAP TRACK CHANNEL TRAY OR EQUIVALENT.
2. ALL TAGS WILL HAVE PREFIX "LP1Y".

1 INSTRUMENT WIRING DIAGRAM - LP1YLSL0631A - DISCRETE INPUT 4-WIRE, 24VDC THERMAL MASS FLOW SWITCH

NTS

REFERENCE DRAWINGS:

1. LP1Y-270-LST-K215-02: INSTRUMENT I/O LIST.
2. LP1Y-270-LST-K215-03: INSTRUMENT CABLE SCHEDULE.
3. LP1Y-270-LAY-L215-01 & 02: INSTRUMENT LOCATION PLAN.
4. LP1Y-275-DIA-K215-30: CHARM I/O PANEL WIRING.
5. LP1Y-275-DIA-K215-31: 24VDC IPP PANEL WIRING.

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
0	ALL	ISSUED FOR CONSTRUCTION	NA	10-APR-26
A	ALL	ISSUED FOR REVIEW	NA	16-JAN-26

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1	10-APR-26	MGL	JN	BM	AJT	SS
0	16-JAN-26	MGL	JN	BM	AJT	SS

Lilly

INSTRUMENT WIRING DIAGRAM - LP1YLSL0631A - DISCRETE INPUT 4-WIRE, 24VDC THERMAL MASS FLOW SWITCH - RAIL UNLOADING RACK

SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: 0
SUB DISCIPLINE:	LOCATION CODE: LP1	LOCATION/GRID: COL C-D / 33-43	CCI CODE: YELLOW
			PROJECT NAME: LP1-306-1181
CRITICAL QUALITY	ENTERPRISE GROUP: API	DRAWING INTENTION	DRAWING SCALE: NONE
CRITICAL BUSINESS	EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA	MERIDIAN REV: B
CRITICAL HSE	VENDOR: NA	DOCUMENT ID: LP1Y-275-DIA-K215-15	
NON-CRITICAL	ENGINEERING AGENCY DOC ID: NA		

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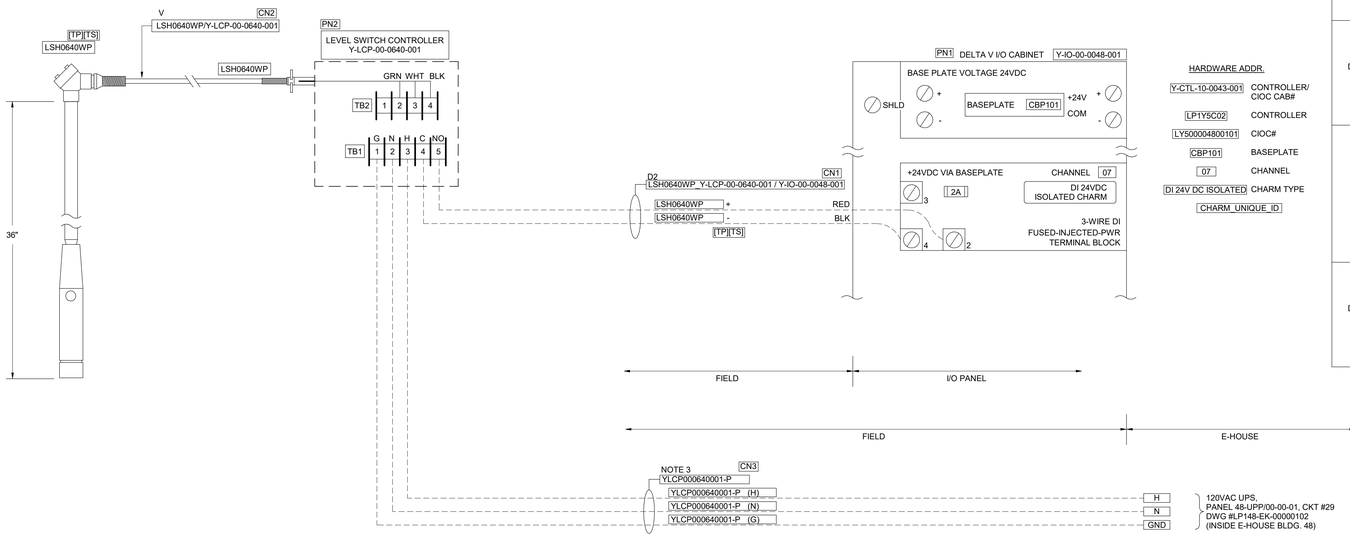
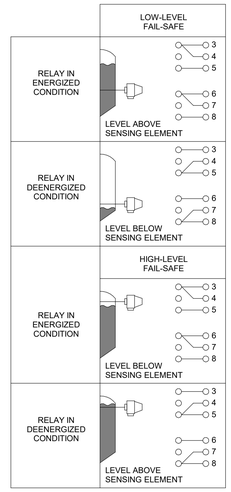
LABELING DESCRIPTION AND EXAMPLE		
TP	=	TAG PREFIX = LEVEL SWITCH = LSH
TS	=	TAG SUFFIX = LOOP NUMBER = 0640WP
PN1	=	IO PANEL = <BLDG>-<TYPE>-<FLOOR>-<ROOM # OR SKID#>-<INDEX> = Y-IO-00-0048-001
PN2	=	LEVEL SWITCH CONTROLLER = <BLDG>-<TYPE>-<FLOOR>-<SKID#>-<INDEX> = Y-LCP-00-0640-001
CN1	=	SIGNAL CABLE TAG = TP TS / PN1 = LSH0640WP_Y-LCP-00-0640-001/Y-IO-00-0048-001
CN2	=	SENSOR CABLE TAG = TP TS / PN2 = LSH0640WP_Y-LCP-00-0640-001
CN3	=	POWER CABLE TAG = SUPPLIED BY ELECTRICAL
CHARM_UNIQUE_ID	=	PN - SN - CN
D2	=	CABLE CODE: 1 PAIR, #18, NO SHLD, 300V, ITC (JACKET COLOR: BLK)
V	=	VENDOR PROVIDED CABLE WITH PLUG-IN CONNECTION



HARDWARE ADDR.

Y-CTL-10-0043-001	CONTROLLER/ CIOC CAB#
LP1Y5C02	CONTROLLER
LY50000480010 1	CIOC#
CBP101	BASEPLATE
07	CHANNEL
DI 24V DC ISOLATED	CHARM TYPE
CHARM_UNIQUE_ID	



NOTES:

(*) VENDOR PROVIDED CABLE WITH PLUG-IN CONNECTION.
1. ALL CABLES FROM FIELD INSTRUMENT TO CHARM I/O PANEL OR IPP TO BE FIELD ROUTED USING SNAP TRACK CHANNEL TRAY OR EQUIVALENT.
2. ALL TAGS WILL HAVE PREFIX "LP1Y".
3. CABLE TYPE AND TAG NUMBER TO BE PROVIDED BY K-215 AS PER DRAWING LP148-265-LST-002 "ELECTRICAL CABLE LIST - E HOUSE #48".

1 INSTRUMENT WIRING DIAGRAM - LP1YLSH0640WP - DISCRETE INPUT 2-WIRE LEVEL SWITCH

NTS

REFERENCE DRAWINGS:
1) LP1Y-270-LST-K215-02: INSTRUMENT I/O LIST.
2) LP1Y-270-LST-K215-03: INSTRUMENT CABLE SCHEDULE.
3) LP1Y-270-LAY-K215-01 & 02: INSTRUMENT LOCATION PLAN.
4) LP1Y-275-DIA-K215-30: CHARM I/O PANEL WIRING.
5) LP148-EK-00000102: UPS PANEL 48-UDP/00-00-01.
6) LP148-265-LST-002: ELECTRICAL CABLE LIST - EHOUSE #48.

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
1	ALL	ISSUED FOR REVIEW	NA	10-APR-26
2	ALL	ISSUED FOR REVIEW	NA	12-FEB-26
3	ALL	ISSUED FOR REVIEW	NA	16-JAN-26

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1	10-APR-26	MGL	JN	BM	AJT	SS
2	12-FEB-26	MGL	JN	BM	AJT	SS
3	16-JAN-26	MGL	JN	BM	AJT	SS

Lilly

INSTRUMENT WIRING DIAGRAM - LP1YLSH0640WP - DISCRETE INPUT 2-WIRE LEVEL SWITCH

RAIL UNLOADING RACK

SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: 1
SUB DISCIPLINE:	LOCATION CODE: LP1	LOCATION/GRID: COL C-D / 33-43	CCI CODE: YELLOW
			PROJECT NAME: LP1-306-1181
CRITICAL QUALITY	ENTERPRISE GROUP: API	DRAWING INTENTION	DRAWING SCALE: NONE
CRITICAL BUSINESS	EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA	MERIDIAN REV: B
CRITICAL HSE	VENDOR: NA	DOCUMENT ID: LP1Y-275-DIA-K215-16	
NON-CRITICAL	ENGINEERING AGENCY DOC ID: NA		

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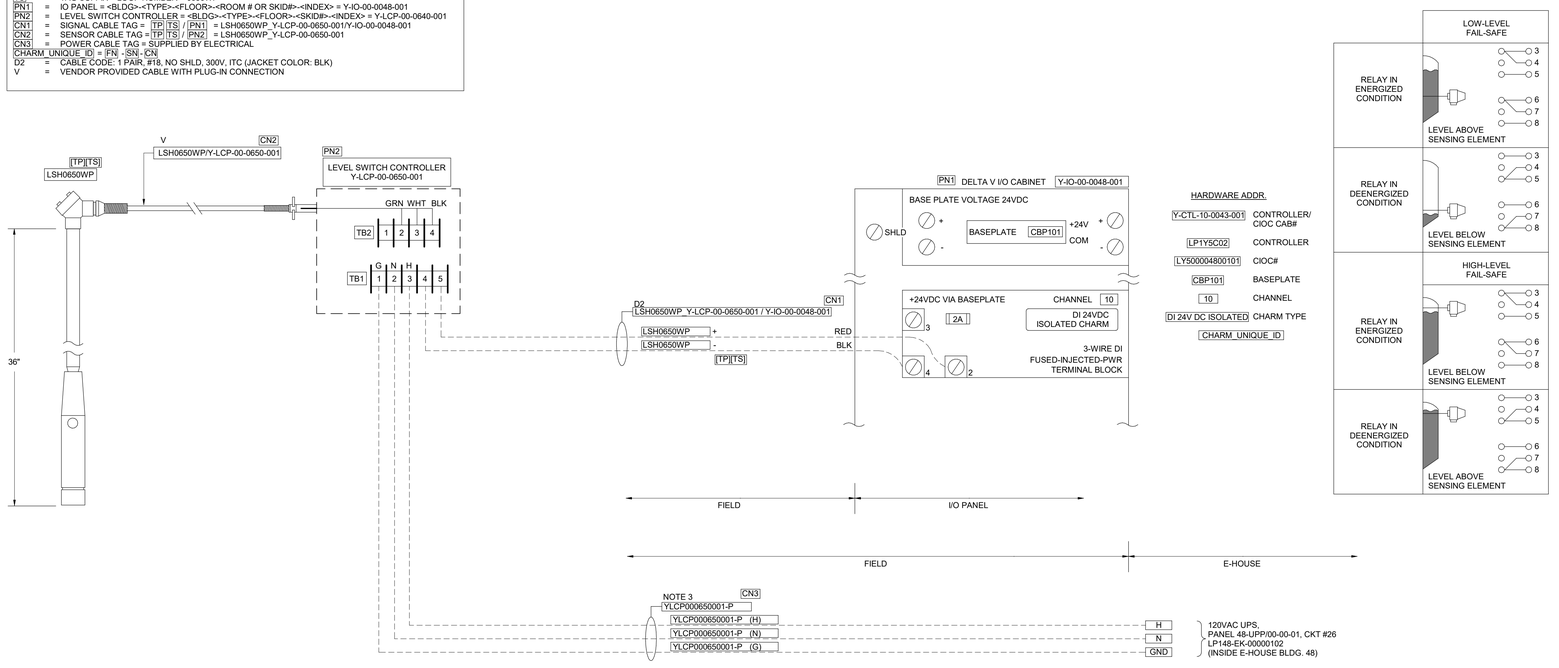
LABELING DESCRIPTION AND EXAMPLE		
TP	=	TAG PREFIX = LEVEL SWITCH = LSH
TS	=	TAG SUFFIX = LOOP NUMBER = 0650WP
PN1	=	IO PANEL = <BLDG>-<TYPE>-<FLOOR>-<ROOM # OR SKID#>-<INDEX> = Y-IO-00-0048-001
PN2	=	LEVEL SWITCH CONTROLLER = <BLDG>-<TYPE>-<FLOOR>-<SKID#>-<INDEX> = Y-LCP-00-0640-001
CN1	=	SIGNAL CABLE TAG = TP TS / PN1 = LSH0650WP_Y-LCP-00-0650-001/Y-IO-00-0048-001
CN2	=	SENSOR CABLE TAG = TP TS / PN2 = LSH0650WP_Y-LCP-00-0650-001
CN3	=	POWER CABLE TAG = SUPPLIED BY ELECTRICAL
CHARM_UNIQUE_ID = FN -SN- CN		
D2	=	CABLE CODE: 1 PAIR, #18, NO SHLD, 300V, ITC (JACKET COLOR: BLK)
V	=	VENDOR PROVIDED CABLE WITH PLUG-IN CONNECTION



NOTES:

(*) VENDOR PROVIDED CABLE WITH PLUG-IN CONNECTION.
1. ALL CABLES FROM FIELD INSTRUMENT TO CHARM I/O PANEL OR IPP TO BE FIELD ROUTED USING SNAP TRACK CHANNEL TRAY OR EQUIVALENT.
2. ALL TAGS WILL HAVE PREFIX "LP1Y".
3. CABLE TYPE AND TAG NUMBER TO BE PROVIDED BY K-215 AS PER DRAWING LP148-265-LST-002 "ELECTRICAL CABLE LIST - EHOUSE #48".

1 INSTRUMENT WIRING DIAGRAM - LP1YLSH0650WP - DISCRETE INPUT 2-WIRE LEVEL SWITCH

NTS

REFERENCE DRAWINGS:
1) LP1Y-270-LST-K215-02: INSTRUMENT I/O LIST.
2) LP1Y-270-LST-K215-03: INSTRUMENT CABLE SCHEDULE.
3) LP1Y-270-LAY-K215-01 & 02: INSTRUMENT LOCATION PLAN.
4) LP1Y-275-DIA-K215-30: CHARM I/O PANEL WIRING.
5) LP148-EK-00000102: UPS PANEL 48-UDP-00-00-01.
6) LP148-265-LST-002: ELECTRICAL CABLE LIST - EHOUSE #48.

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
0	ALL	ISSUED FOR REVIEW	NA	10-APR-26
B	ALL	ISSUED FOR REVIEW	NA	12-FEB-26
A	ALL	ISSUED FOR REVIEW	NA	16-JAN-26

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B	12-FEB-26	MGL	JN	BM	AJT	SS
A	16-JAN-26	MGL	JN	BM	AJT	SS

INSTRUMENT WIRING DIAGRAM - LP1YLSH0650WP - DISCRETE INPUT 2-WIRE LEVEL SWITCH

RAIL UNLOADING RACK

SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: 1
SUB DISCIPLINE: LP1	LOCATION CODE	LOCATION/GRID: COL C-D / 33-43	CCI CODE: YELLOW / PROJECT NAME: LP1-306-1181
CRITICAL QUALITY		ENTERPRISE GROUP: API	DRAWING INTENTION: DRAWING SCALE: NONE
CRITICAL BUSINESS		EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA / MERIDIAN REV: B
CRITICAL HSE		VENDOR: NA	DOCUMENT ID: LP1Y-275-DIA-K215-17
NON-CRITICAL		ENGINEERING AGENCY DOC ID: NA	

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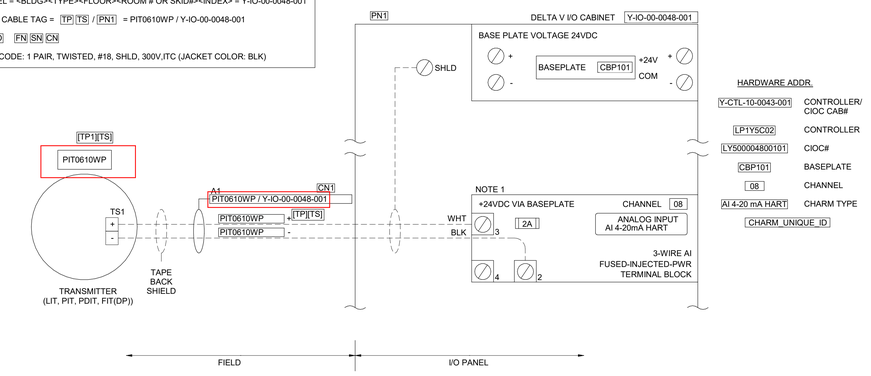
LABELING DESCRIPTION AND EXAMPLE		
TP	=	TAG PREFIX = 2-WIRE ANALOG INSTRUMENT - PIT
TS	=	TAG SUFFIX = LOOP NUMBER = 0610WP
PN1	=	IO PANEL = <BLDG><TYPE><FLOOR><ROOM # OR SKID#><INDEX> = Y-IO-00-0048-001
CN1	=	SIGNAL CABLE TAG = TP TS / PN1 = PIT0610WP / Y-IO-00-0048-001
CHARM_UNIQUE_ID		FN SN CN
A1	=	CABLE CODE: 1 PAIR, TWISTED, #18, SHLD, 300V,ITC (JACKET COLOR: BLK)



NOTE:
1. 3-WIRE FUSE INJECTED POWER TERMINAL BLOCKS ARE USED WITH 2-WIRE HART DEVICES TO ENSURE 15VDC MINIMUM AT HIGHER mA SIGNALS.
2. ALL CABLES FROM FIELD INSTRUMENT TO CHARM I/O PANEL OR IPP TO BE FIELD ROUTED USING SNAP TRACK CHANNEL TRAY OR EQUIVALENT.
3. ALL TAGS WILL HAVE PREFIX "LP1Y".

1 INSTRUMENT WIRING DIAGRAM - LP1YPIT0610WP - ANALOG INPUT DELTA-V CHARMS

NTS

REFERENCE DRAWINGS:

1. LP1Y-270-LST-K215-02: INSTRUMENT I/O LIST.
2. LP1Y-270-LST-K215-03: INSTRUMENT CABLE SCHEDULE.
3. LP1Y-270-LAY-L215-01 & 02: INSTRUMENT LOCATION PLAN.
4. LP1Y-275-DIA-K215-30: CHARM I/O PANEL WIRING.

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REV	DATE	DES	ENG	CHK	APP	PM APP
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0	16-JAN-26	MGL	JN	BM	AJT	SS

INSTRUMENT WIRING DIAGRAM - LP1YPIT0610WP - ANALOG INPUT DELTA-V CHARMS

RAIL UNLOADING RACK

REVISIONS			
REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
1	ALL	ISSUED FOR CONSTRUCTION	NA	10-APR-26
0	ALL	ISSUED FOR REVIEW	NA	16-JAN-26

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SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: 0
SUB DISCIPLINE	LOCATION CODE: LP1	LOCATION/GRID: COL C-D / 33-43	CCI CODE: YELLOW	PROJECT NAME: LP1-306-1181
CRITICAL QUALITY		ENTERPRISE GROUP: API	DRAWING INTENTION	DRAWING SCALE: NONE
CRITICAL BUSINESS		EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA	MERIDIAN REV: B
CRITICAL HSE		VENDOR: NA	DOCUMENT ID: LP1Y-275-DIA-K215-18	
NON-CRITICAL		ENGINEERING AGENCY DOC ID: NA		

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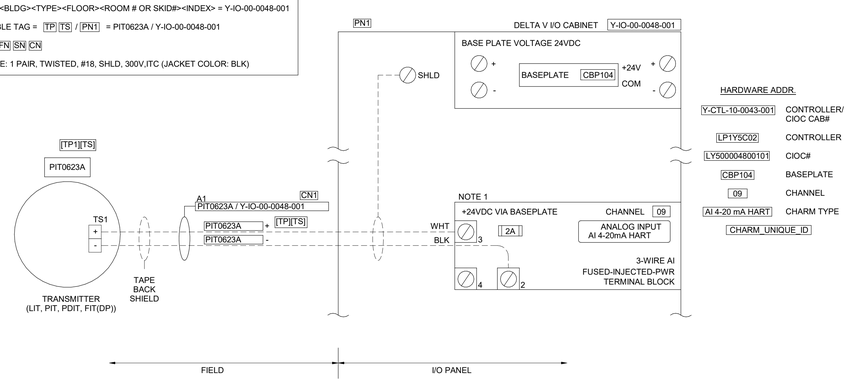
LABELING DESCRIPTION AND EXAMPLE		
[TP]	=	TAG PREFIX = 2-WIRE ANALOG INSTRUMENT - PIT
[TS]	=	TAG SUFFIX = LOOP NUMBER = 0623A
[PN1]	=	IO PANEL = <BLDG><TYPE><FLOOR><ROOM # OR SKID#><INDEX> = Y-IO-00-0048-001
[CN1]	=	SIGNAL CABLE TAG = [TP] [TS] / [PN1] = PIT0623A / Y-IO-00-0048-001
[CHARM_UNIQUE_ID]		[FN] [SN] [CN]
A1	=	CABLE CODE: 1 PAIR, TWISTED, #18, SHLD, 300V,ITC (JACKET COLOR: BLK)



NOTE:
1. 3-WIRE FUSE INJECTED POWER TERMINAL BLOCKS ARE USED WITH 2-WIRE HART DEVICES TO ENSURE 15VDC MINIMUM AT HIGHER mA SIGNALS.
2. ALL CABLES FROM FIELD INSTRUMENT TO CHARM I/O PANEL OR IPP TO BE FIELD ROUTED USING SNAP TRACK CHANNEL TRAY OR EQUIVALENT.
3. ALL TAGS WILL HAVE PREFIX "LP1Y".

1 INSTRUMENT WIRING DIAGRAM - LP1YPIT0623A - ANALOG INPUT DELTA-V CHARMS

NTS

REFERENCE DRAWINGS:

1. LP1Y-270-LST-K215-02: INSTRUMENT I/O LIST.
2. LP1Y-270-LST-K215-03: INSTRUMENT CABLE SCHEDULE.
3. LP1Y-270-LAY-L215-01 & 02: INSTRUMENT LOCATION PLAN.
4. LP1Y-275-DIA-K215-30: CHARM I/O PANEL WIRING.

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1	10-APR-26	MGL	JN	BM	AJT	SS
A	16-JAN-26	MGL	JN	BM	AJT	SS

INSTRUMENT WIRING DIAGRAM - LP1YPIT0623A - ANALOG INPUT DELTA-V CHARMS

RAIL UNLOADING RACK

REVISIONS			
REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
1	ALL	ISSUED FOR CONSTRUCTION	NA	10-APR-26
A	ALL	ISSUED FOR REVIEW	NA	16-JAN-26

SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: 1
SUB DISCIPLINE	LOCATION CODE: LP1	LOCATION/GRID: COL C-D / 33-43	CCI CODE: YELLOW	PROJECT NAME: LP1-306-1181
CRITICAL QUALITY		ENTERPRISE GROUP: API	DRAWING INTENTION	DRAWING SCALE: NONE
CRITICAL BUSINESS		EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA	MERIDIAN REV: B
CRITICAL HSE		VENDOR: NA	DOCUMENT ID: LP1Y-275-DIA-K215-19	
NON-CRITICAL		ENGINEERING AGENCY DOC ID: NA		

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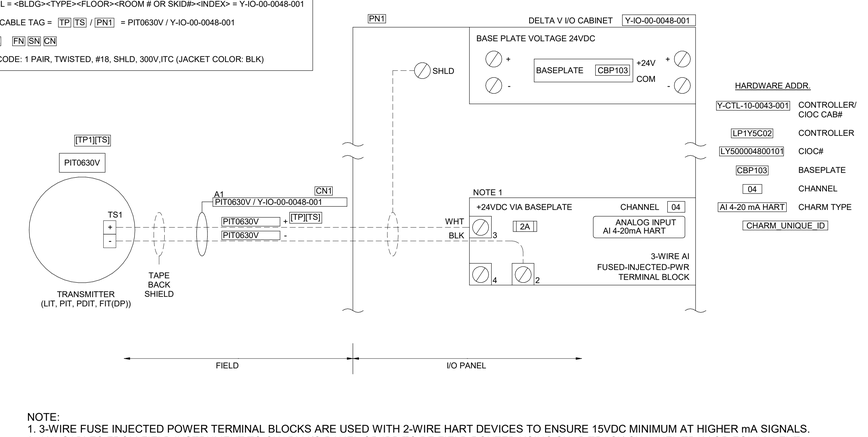
LABELING DESCRIPTION AND EXAMPLE		
TP	=	TAG PREFIX = 2-WIRE ANALOG INSTRUMENT - PIT
TS	=	TAG SUFFIX = LOOP NUMBER = 0630V
PN1	=	IO PANEL = <BLDG><TYPE><FLOOR><ROOM # OR SKID#><INDEX> = Y-IO-00-0048-001
CN1	=	SIGNAL CABLE TAG = TP TS / PN1 = PIT0630V / Y-IO-00-0048-001
CHARM_UNIQUE_ID		FN SN CN
A1	=	CABLE CODE: 1 PAIR, TWISTED, #18, SHLD, 300V,ITC (JACKET COLOR: BLK)



NOTE:
1. 3-WIRE FUSE INJECTED POWER TERMINAL BLOCKS ARE USED WITH 2-WIRE HART DEVICES TO ENSURE 15VDC MINIMUM AT HIGHER mA SIGNALS.
2. ALL CABLES FROM FIELD INSTRUMENT TO CHARM I/O PANEL OR IPP TO BE FIELD ROUTED USING SNAP TRACK CHANNEL TRAY OR EQUIVALENT.
3. ALL TAGS WILL HAVE PREFIX "LP1Y".

1 INSTRUMENT WIRING DIAGRAM - LP1YPIT0630V - ANALOG INPUT DELTA-V CHARMS
NTS

REFERENCE DRAWINGS:

1. LP1Y-270-LST-K215-02: INSTRUMENT I/O LIST.
2. LP1Y-270-LST-K215-03: INSTRUMENT CABLE SCHEDULE.
3. LP1Y-270-LAY-L215-01 & 02: INSTRUMENT LOCATION PLAN.
4. LP1Y-275-DIA-K215-30: CHARM I/O PANEL WIRING.

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
0	ALL	ISSUED FOR CONSTRUCTION	NA	10-APR-26
A	ALL	ISSUED FOR REVIEW	NA	16-JAN-26

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0	10-APR-26	MGL	JN	BM	AJT	SS
A	16-JAN-26	MGL	JN	BM	AJT	SS

INSTRUMENT WIRING DIAGRAM - LP1YPIT0630V - ANALOG INPUT DELTA-V CHARMS

RAIL UNLOADING RACK

SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: 1
SUB DISCIPLINE	LOCATION CODE: LP1	LOCATION/GRID: COL C-D / 33-43	CCI CODE: YELLOW / PROJECT NAME: LP1-306-1181
CRITICAL QUALITY		ENTERPRISE GROUP: API	DRAWING INTENTION / DRAWING SCALE: NONE
CRITICAL BUSINESS		EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID / MERIDIAN REV: B
CRITICAL HSE		VENDOR: NA	DOCUMENT ID: LP1Y-275-DIA-K215-20
NON-CRITICAL		ENGINEERING AGENCY DOC ID: NA	

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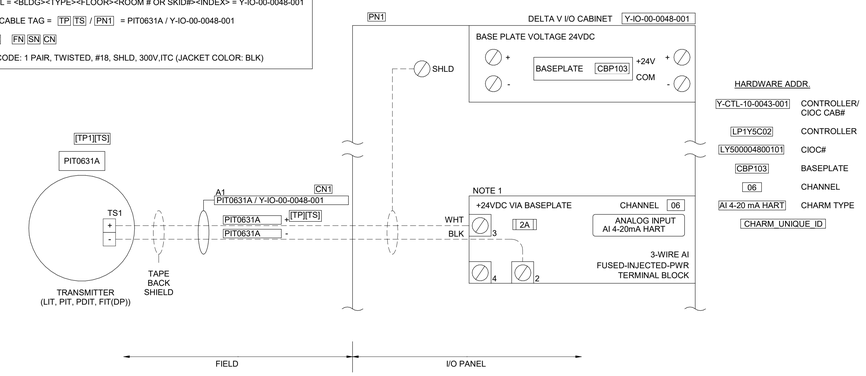
LABELING DESCRIPTION AND EXAMPLE		
TP	=	TAG PREFIX = 2-WIRE ANALOG INSTRUMENT - PIT
TS	=	TAG SUFFIX = LOOP NUMBER = 0631A
PN1	=	IO PANEL = <BLDG><TYPE><FLOOR><ROOM # OR SKID#><INDEX> = Y-IO-00-0048-001
CN1	=	SIGNAL CABLE TAG = TP TS / PN1 = PIT0631A / Y-IO-00-0048-001
CHARM_UNIQUE_ID		FN SN CN
A1	=	CABLE CODE: 1 PAIR, TWISTED, #18, SHLD, 300V,ITC (JACKET COLOR: BLK)



NOTE:
1. 3-WIRE FUSE INJECTED POWER TERMINAL BLOCKS ARE USED WITH 2-WIRE HART DEVICES TO ENSURE 15VDC MINIMUM AT HIGHER mA SIGNALS.
2. ALL CABLES FROM FIELD INSTRUMENT TO CHARM I/O PANEL OR IPP TO BE FIELD ROUTED USING SNAP TRACK CHANNEL TRAY OR EQUIVALENT.
3. ALL TAGS WILL HAVE PREFIX "LP1Y".

1 INSTRUMENT WIRING DIAGRAM - LP1YPIT0631A - ANALOG INPUT DELTA-V CHARMS

NTS

REFERENCE DRAWINGS:

1. LP1Y-270-LST-K215-02: INSTRUMENT I/O LIST.
2. LP1Y-270-LST-K215-03: INSTRUMENT CABLE SCHEDULE.
3. LP1Y-270-LAY-L215-01 & 02: INSTRUMENT LOCATION PLAN.
4. LP1Y-275-DIA-K215-30: CHARM I/O PANEL WIRING.

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0	16-JAN-26	MGL	JN	BM	AJT	SS

INSTRUMENT WIRING DIAGRAM - LP1YPIT0631A - ANALOG INPUT DELTA-V CHARMS

RAIL UNLOADING RACK

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
1	ALL	ISSUED FOR CONSTRUCTION	NA	10-APR-26
0	ALL	ISSUED FOR REVIEW	NA	16-JAN-26

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SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: 0
SUB DISCIPLINE	LOCATION CODE: LP1	LOCATION/GRID: COL C-D / 33-43	CCI CODE: YELLOW
			PROJECT NAME: LP1-306-1181
CRITICAL QUALITY		ENTERPRISE GROUP: API	DRAWING INTENTION
			DRAWING SCALE: NONE
CRITICAL BUSINESS		EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA
			MERIDIAN REV: B
CRITICAL HSE		VENDOR: NA	DOCUMENT ID: LP1Y-275-DIA-K215-21
NON-CRITICAL		ENGINEERING AGENCY DOC ID: NA	

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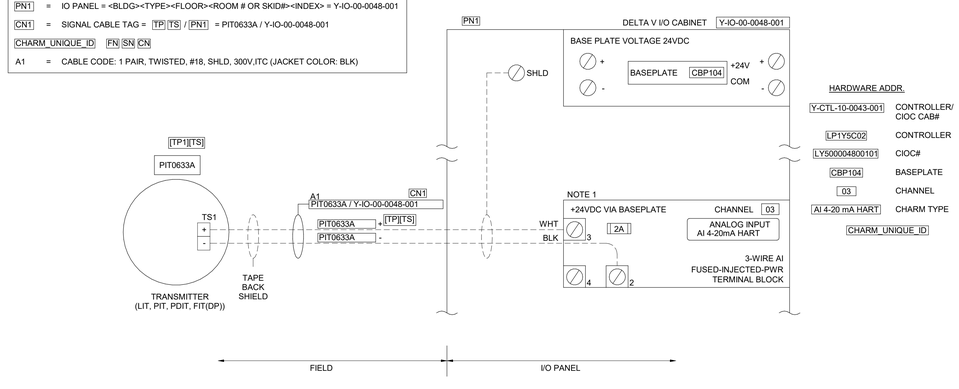
LABELING DESCRIPTION AND EXAMPLE		
TP	=	TAG PREFIX = 2-WIRE ANALOG INSTRUMENT - PIT
TS	=	TAG SUFFIX = LOOP NUMBER = 0633A
PN1	=	IO PANEL = <BLDG><TYPE><FLOOR><ROOM # OR SKID#><INDEX> = Y-IO-00-0048-001
CN1	=	SIGNAL CABLE TAG = TP TS / PN1 = PIT0633A / Y-IO-00-0048-001
CHARM_UNIQUE_ID		FN SN CN
A1	=	CABLE CODE: 1 PAIR, TWISTED, #18, SHLD, 300V,ITC (JACKET COLOR: BLK)



NOTE:
1. 3-WIRE FUSE INJECTED POWER TERMINAL BLOCKS ARE USED WITH 2-WIRE HART DEVICES TO ENSURE 15VDC MINIMUM AT HIGHER mA SIGNALS.
2. ALL CABLES FROM FIELD INSTRUMENT TO CHARM I/O PANEL OR IPP TO BE FIELD ROUTED USING SNAP TRACK CHANNEL TRAY OR EQUIVALENT.
3. ALL TAGS WILL HAVE PREFIX "LP1Y".

1 INSTRUMENT WIRING DIAGRAM - LP1YPIT0633A - ANALOG INPUT DELTA-V CHARMS

NTS

REFERENCE DRAWINGS:

1. LP1Y-270-LST-K215-02: INSTRUMENT I/O LIST.
2. LP1Y-270-LST-K215-03: INSTRUMENT CABLE SCHEDULE.
3. LP1Y-270-LAY-L215-01 & 02: INSTRUMENT LOCATION PLAN.
4. LP1Y-275-DIA-K215-30: CHARM I/O PANEL WIRING.

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
1	ALL	ISSUED FOR CONSTRUCTION	NA	10-APR-26
A	ALL	ISSUED FOR REVIEW	NA	16-JAN-26

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1	10-APR-26	MGL	JN	BM	AJT	SS
A	16-JAN-26	MGL	JN	BM	AJT	SS

INSTRUMENT WIRING DIAGRAM - LP1YPIT0633A - ANALOG INPUT DELTA-V CHARMS
RAIL UNLOADING RACK

SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: 1
SUB DISCIPLINE	LOCATION CODE: LP1	LOCATION/GRID: COL C-D / 33-43	CCI CODE: YELLOW
			PROJECT NAME: LP1-306-1181
CRITICAL QUALITY		ENTERPRISE GROUP: API	DRAWING INTENTION
			DRAWING SCALE: NONE
CRITICAL BUSINESS		EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA
			MERIDIAN REV: B
CRITICAL HSE		VENDOR: NA	DOCUMENT ID
NON-CRITICAL		ENGINEERING AGENCY DOC ID: NA	LP1Y-275-DIA-K215-22

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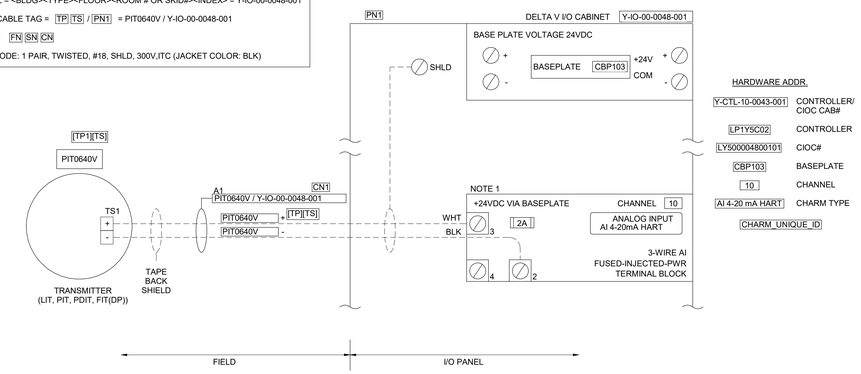
LABELING DESCRIPTION AND EXAMPLE		
TP	=	TAG PREFIX = 2-WIRE ANALOG INSTRUMENT - PIT
TS	=	TAG SUFFIX = LOOP NUMBER = 0640V
PN1	=	IO PANEL = <BLDG><TYPE><FLOOR><ROOM # OR SKID#><INDEX> = Y-IO-00-0048-001
CN1	=	SIGNAL CABLE TAG = TP TS / PN1 = PIT0640V / Y-IO-00-0048-001
CHARM_UNIQUE_ID		FN SN CN
A1	=	CABLE CODE: 1 PAIR, TWISTED, #18, SHLD, 300V,ITC (JACKET COLOR: BLK)



NOTE:
1. 3-WIRE FUSE INJECTED POWER TERMINAL BLOCKS ARE USED WITH 2-WIRE HART DEVICES TO ENSURE 15VDC MINIMUM AT HIGHER mA SIGNALS.
2. ALL CABLES FROM FIELD INSTRUMENT TO CHARM I/O PANEL OR IPP TO BE FIELD ROUTED USING SNAP TRACK CHANNEL TRAY OR EQUIVALENT.
3. ALL TAGS WILL HAVE PREFIX "LP1Y".

1 INSTRUMENT WIRING DIAGRAM - LP1YPIT0640V - ANALOG INPUT DELTA-V CHARMS

NTS

REFERENCE DRAWINGS:

1. LP1Y-270-LST-K215-02: INSTRUMENT I/O LIST.
2. LP1Y-270-LST-K215-03: INSTRUMENT CABLE SCHEDULE.
3. LP1Y-270-LAY-L215-01 & 02: INSTRUMENT LOCATION PLAN.
4. LP1Y-275-DIA-K215-30: CHARM I/O PANEL WIRING.

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
0	ALL	ISSUED FOR CONSTRUCTION	NA	10-APR-26
A	ALL	ISSUED FOR REVIEW	NA	16-JAN-26

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A	16-JAN-26	MGL	JN	BM	AJT	SS

INSTRUMENT WIRING DIAGRAM - LP1YPIT0640V - ANALOG INPUT DELTA-V CHARMS

RAIL UNLOADING RACK

SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: 1
SUB DISCIPLINE	LOCATION CODE: LP1	LOCATION/GRID: COL C-D / 33-43	CCI CODE: YELLOW
CRITICAL QUALITY		ENTERPRISE GROUP: API	DRAWING INTENTION: PROJECT NAME: LP1-306-1181
CRITICAL BUSINESS		EQUIPMENT CLASSIFICATION: NA	DRAWING SCALE: NONE
CRITICAL HSE		VENDOR: NA	ALTERNATE DOCUMENT ID: NA / MERIDIAN REV: B
NON-CRITICAL		ENGINEERING AGENCY DOC ID: NA	DOCUMENT ID: LP1Y-275-DIA-K215-23

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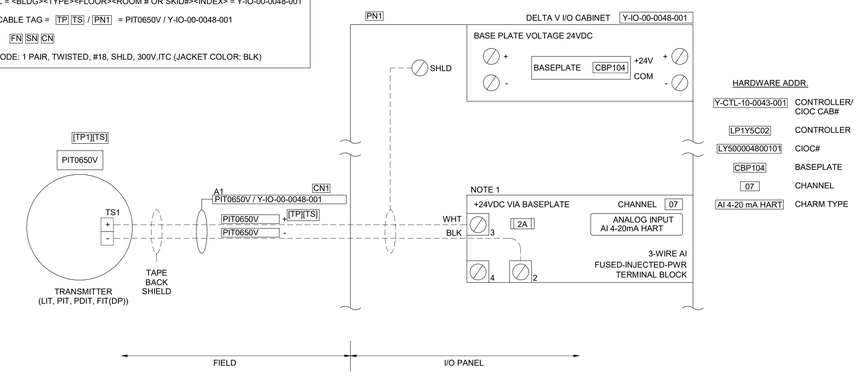
LABELING DESCRIPTION AND EXAMPLE		
TP	=	TAG PREFIX = 2-WIRE ANALOG INSTRUMENT - PIT
TS	=	TAG SUFFIX = LOOP NUMBER = 0650V
PN1	=	IO PANEL = <BLDG><TYPE><FLOOR><ROOM # OR SKID#><INDEX> = Y-IO-00-0048-001
CN1	=	SIGNAL CABLE TAG = TP TS / PN1 = PIT0650V / Y-IO-00-0048-001
CHARM_UNIQUE_ID		FN SN CN
A1	=	CABLE CODE: 1 PAIR, TWISTED, #18, SHLD, 300V,ITC (JACKET COLOR: BLK)



NOTE:
1. 3-WIRE FUSE INJECTED POWER TERMINAL BLOCKS ARE USED WITH 2-WIRE HART DEVICES TO ENSURE 15VDC MINIMUM AT HIGHER mA SIGNALS.
2. ALL CABLES FROM FIELD INSTRUMENT TO CHARM I/O PANEL OR IPP TO BE FIELD ROUTED USING SNAP TRACK CHANNEL TRAY OR EQUIVALENT.
3. ALL TAGS WILL HAVE PREFIX "LP1Y".

1 INSTRUMENT WIRING DIAGRAM - LP1YPIT0650V - ANALOG INPUT DELTA-V CHARMS

NTS

REFERENCE DRAWINGS:

1. LP1Y-270-LST-K215-02: INSTRUMENT I/O LIST.
2. LP1Y-270-LST-K215-03: INSTRUMENT CABLE SCHEDULE.
3. LP1Y-270-LAY-L215-01 & 02: INSTRUMENT LOCATION PLAN.
4. LP1Y-275-DIA-K215-30: CHARM I/O PANEL WIRING.

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REV	DATE	DES	ENG	CHK	APP	PM APP
2						
1						
0	10-APR-26	MGL	JN	BM	AJT	SS
A	16-JAN-26	MGL	JN	BM	AJT	SS

INSTRUMENT WIRING DIAGRAM - LP1YPIT0650V - ANALOG INPUT DELTA-V CHARMS

RAIL UNLOADING RACK

REVISIONS			
REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
0	ALL	ISSUED FOR CONSTRUCTION	NA	10-APR-26
A	ALL	ISSUED FOR REVIEW	NA	16-JAN-26

SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: 1
SUB DISCIPLINE	LOCATION CODE: LP1	LOCATION/GRID: COL C-D / 33-43	CCI CODE: YELLOW	PROJECT NAME: LP1-306-1181
CRITICAL QUALITY		ENTERPRISE GROUP: API	DRAWING INTENTION	DRAWING SCALE: NONE
CRITICAL BUSINESS		EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA	MERIDIAN REV: B
CRITICAL HSE		VENDOR: NA	DOCUMENT ID: LP1Y-275-DIA-K215-24	
NON-CRITICAL		ENGINEERING AGENCY DOC ID: NA		

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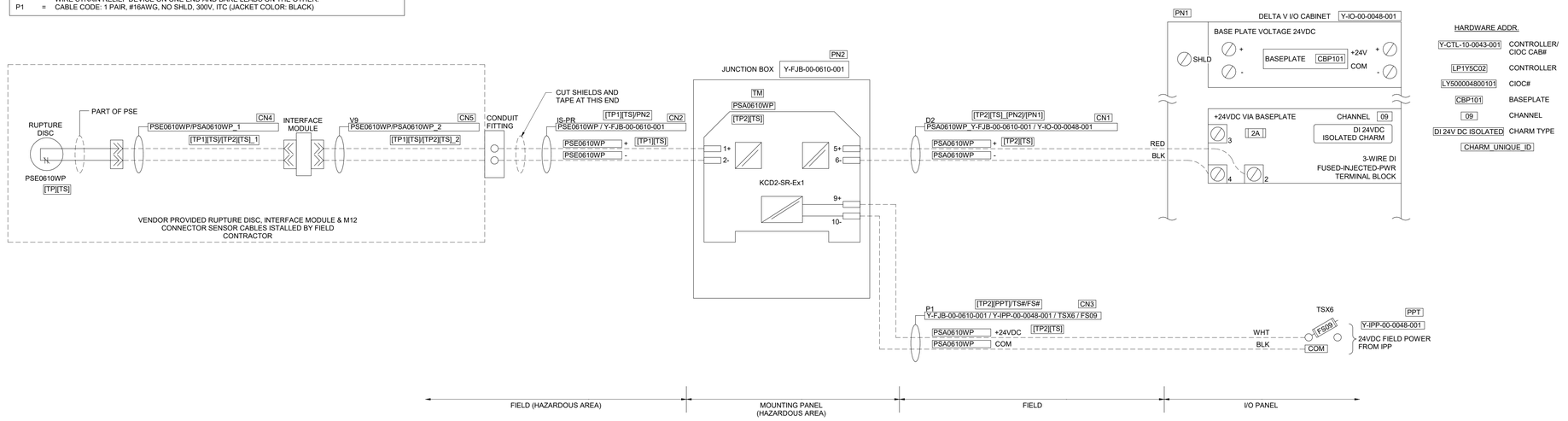
LABELING DESCRIPTION AND EXAMPLE		
TP1	=	TAG PREFIX = RUPTURE DISK ELEMENT = PSE
TP2	=	TAG PREFIX = RUPTURE DISK I.S. BARRIER = PSA
TS	=	TAG SUFFIX = LOOP NUMBER = 0610WP
PN1	=	IO PANEL = <BLDG>-<TYPE>-<FLOOR>-<ROOM # OR SKID#>-<INDEX> = Y-IO-00-0048-001
PN2	=	FIELD JUNCTION BOX = <BLDG>-<TYPE>-<FLOOR>-<ROOM# OR SKID#>-<INDEX> = Y-FJB-00-0610-001
PPT	=	POWER PANELTAG = <BLDG>-<TYPE>-<FLOOR>-<ROOM# OR SKID#>-<INDEX> = Y-IPP-00-0048-001
CN1	=	SIGNAL CABLE TAG = TP2[TS]_PN2/[PN1] = PSA0610WP_Y-FJB-00-0610-001 / Y-IO-00-0048-001
CN2	=	SIGNAL CABLE TAG = TP1[TS] / PN2 = PSE0610WP / Y-FJB-00-0610-001
CN3	=	POWER CABLE TAG = PN2 / PPT/ TS# / FS# = Y-FJB-00-0610-001 / Y-IPP-00-0048-001 / TSX6 / FS9
CN4	=	SIGNAL CABLE TAG = TP1[TS] / TP2[TS]_1 = PSE0610WP/PSA0610WP_1
CN5	=	SIGNAL CABLE TAG = TP1[TS] / TP2[TS]_2 = PSE0610WP/PSA0610WP_2
TM	=	TERMINAL MARKER STRIP = TP2 TS = PSA0610WP
CHARM_UNIQUE_ID = FN|-SN|-CN		
IS-PR	=	CABLE CODE: 1 PAIR, TWISTED, #18 SHLD, 300V, ITC (JACKET COLOR: BLU)
V9	=	CABLE CODE: B.D.I. LEAD WIRE ASSEMBLY WITH A STANDARD PLUG B.D.I. CONNECTION AND A LEAD WIRE STRAIN RELIEF DEVICE ON ONE END AND BARE LEADS ON THE OTHER.
P1	=	CABLE CODE: 1 PAIR, #16AWG, NO SHLD, 300V, ITC (JACKET COLOR: BLACK)



NOTES:

1. ALL CABLES FROM FIELD INSTRUMENT TO CHARM I/O PANEL OR IPP TO BE FIELD ROUTED USING SNAP TRACK CHANNEL TRAY OR EQUIVALENT.
2. ALL TAGS WILL HAVE PREFIX "LP1Y".

1 INSTRUMENT WIRING DIAGRAM - LP1YPSA0610WP - DISCRETE INPUT4-WIRE, RUPTURE DISC WITH I.S. BARRIER FOR HAZADOUS AREA

NTS

REFERENCE DRAWINGS:

1. LP1Y-270-LST-K215-02: INSTRUMENT I/O LIST.
2. LP1Y-270-LST-K215-03: INSTRUMENT CABLE SCHEDULE.
3. LP1Y-270-LAY-L215-01 & 02: INSTRUMENT LOCATION PLAN.
4. LP1Y-275-DIA-K215-30: CHARM I/O PANEL WIRING.
5. LP1Y-275-DIA-K215-31: 24VDC IPP PANEL WIRING.

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1	10-APR-26	MGL	JN	BM	AJT	SS
1	16-JAN-26	MGL	JN	BM	AJT	SS

INSTRUMENT WIRING DIAGRAM - LP1YPSA0610WP - DISCRETE INPUT 4-WIRE RUPTURE DISC WITH I.S. BARRIER FOR HAZARDOUS AREA RAIL UNLOADING RACK

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
1	ALL	ISSUED FOR CONSTRUCTION	NA	10-APR-26
1	ALL	ISSUED FOR REVIEW	NA	16-JAN-26

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SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: 0
SUB DISCIPLINE	LOCATION CODE: LP1	LOCATIONGRID: COL C-D / 33-43	CCI CODE: YELLOW
			PROJECT NAME: LP1-306-1181
CRITICAL QUALITY		ENTERPRISE GROUP: API	DRAWING INTENTION
			DRAWING SCALE: NONE
CRITICAL BUSINESS		EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA
			MERIDIAN REV: B
CRITICAL HSE		VENDOR: NA	DOCUMENT ID: LP1Y-275-DIA-K215-26
NON-CRITICAL		ENGINEERING AGENCY DOC ID: NA	

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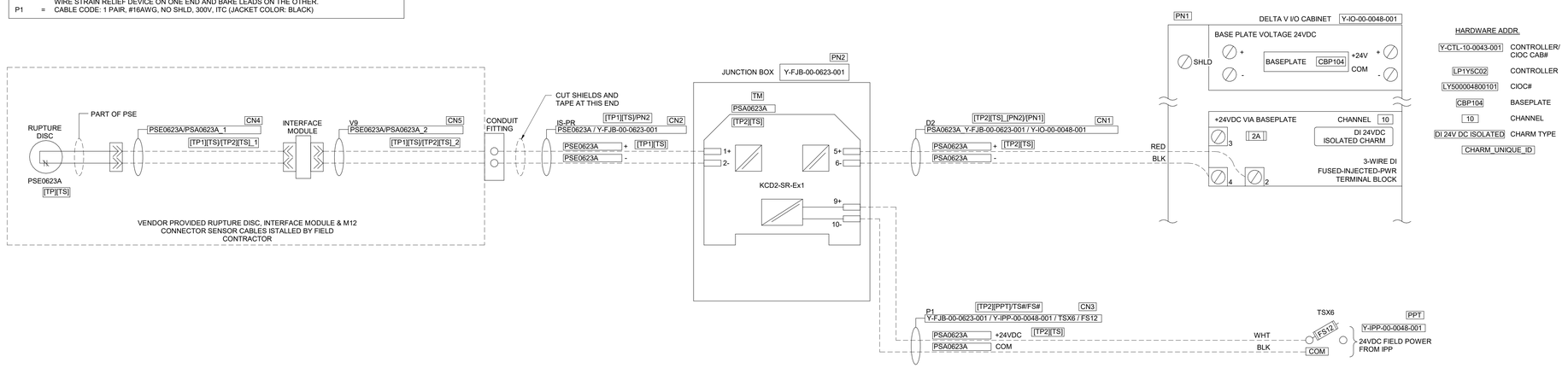
LABELING DESCRIPTION AND EXAMPLE		
TP1	=	TAG PREFIX = RUPTURE DISK ELEMENT = PSE
TP2	=	TAG PREFIX = RUPTURE DISK I.S. BARRIER = PSA
TS	=	TAG SUFFIX = LOOP NUMBER = 0623A
PN1	=	IO PANEL = <BLDG>-<TYPE>-<FLOOR>-<ROOM # OR SKID#>-<INDEX> = Y-IO-00-0048-001
PN2	=	FIELD JUNCTION BOX = <BLDG>-<TYPE>-<FLOOR>-<ROOM# OR SKID#>-<INDEX> = Y-FJB-00-0623-001
PPT	=	POWER PANELTAG = <BLDG>-<TYPE>-<FLOOR>-<ROOM# OR SKID#>-<INDEX> = Y-IPP-00-0048-001
CN1	=	SIGNAL CABLE TAG = TP2[TS]_PN2/ PN1 = PSA0623A_Y-FJB-00-0623-001 / Y-IO-00-0048-001
CN2	=	SIGNAL CABLE TAG = TP1[TS]/ PN2 = PSE0623A / Y-FJB-00-0623-001
CN3	=	POWER CABLE TAG = PN2 / PPT/ TS# / FS# = Y-FJB-00-0623-001 / Y-IPP-00-0048-001 / TSX6 / FS9
CN4	=	SIGNAL CABLE TAG = TP1[TS] / TP2[TS] _1 = PSE0623A/PSA0623A_1
CN5	=	SIGNAL CABLE TAG = TP1[TS] / TP2[TS] _2 = PSE0623A/PSA0623A_2
TM	=	TERMINAL MARKER STRIP = TP2 TS = PSA0623A
CHARM_UNIQUE_ID = FN) - SN) - CN		
IS-PR	=	CABLE CODE: 1 PAIR, TWISTED #18, SHLD, 300V, ITC (JACKET COLOR: BLU)
V9	=	CABLE CODE: B.D.I. LEAD WIRE ASSEMBLY WITH A STANDARD PLUG B.D.I. CONNECTION AND A LEAD WIRE STRAIN RELIEF DEVICE ON ONE END AND BARE LEADS ON THE OTHER.
P1	=	CABLE CODE: 1 PAIR, #16AWG, NO SHLD, 300V, ITC (JACKET COLOR: BLACK)



NOTES:

1. ALL CABLES FROM FIELD INSTRUMENT TO CHARM I/O PANEL OR IPP TO BE FIELD ROUTED USING SNAP TRACK CHANNEL TRAY OR EQUIVALENT.
2. ALL TAGS WILL HAVE PREFIX "LP1Y".

1 INSTRUMENT WIRING DIAGRAM - LP1YPSA0623A - DISCRETE INPUT4-WIRE, RUPTURE DISC WITH I.S. BARRIER FOR HAZADOUS AREA

NTS

REFERENCE DRAWINGS:

1. LP1Y-270-LST-K215-02: INSTRUMENT I/O LIST.
2. LP1Y-270-LST-K215-03: INSTRUMENT CABLE SCHEDULE.
3. LP1Y-270-LAY-L215-01 & 02: INSTRUMENT LOCATION PLAN.
4. LP1Y-275-DIA-K215-30: CHARM I/O PANEL WIRING.
5. LP1Y-275-DIA-K215-31: 24VDC IPP PANEL WIRING.

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1	10-APR-26	MGL	JN	BM	AJT	SS
A	16-JAN-26	MGL	JN	BM	AJT	SS

INSTRUMENT WIRING DIAGRAM -_LP1YPSA0623A - DISCRETE INPUT 4-WIRE RUPTURE DISC WITH I.S. BARRIER FOR HAZARDOUS AREA RAIL UNLOADING RACK

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
0	ALL	ISSUED FOR CONSTRUCTION	NA	10-APR-26
A	ALL	ISSUED FOR REVIEW	NA	16-JAN-26

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SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: 1
SUB DISCIPLINE: LP1	LOCATION CODE / LOCATION/GRID: COL C-D / 33-43	CCI CODE: YELLOW	PROJECT NAME: LP1-306-1181
CRITICAL QUALITY	ENTERPRISE GROUP: API	DRAWING INTENTION	DRAWING SCALE: NONE
CRITICAL BUSINESS	EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA	MERIDIAN REV: B
CRITICAL HSE	VENDOR: NA	DOCUMENT ID: LP1Y-275-DIA-K215-27	
NON-CRITICAL	ENGINEERING AGENCY DOC ID: NA		

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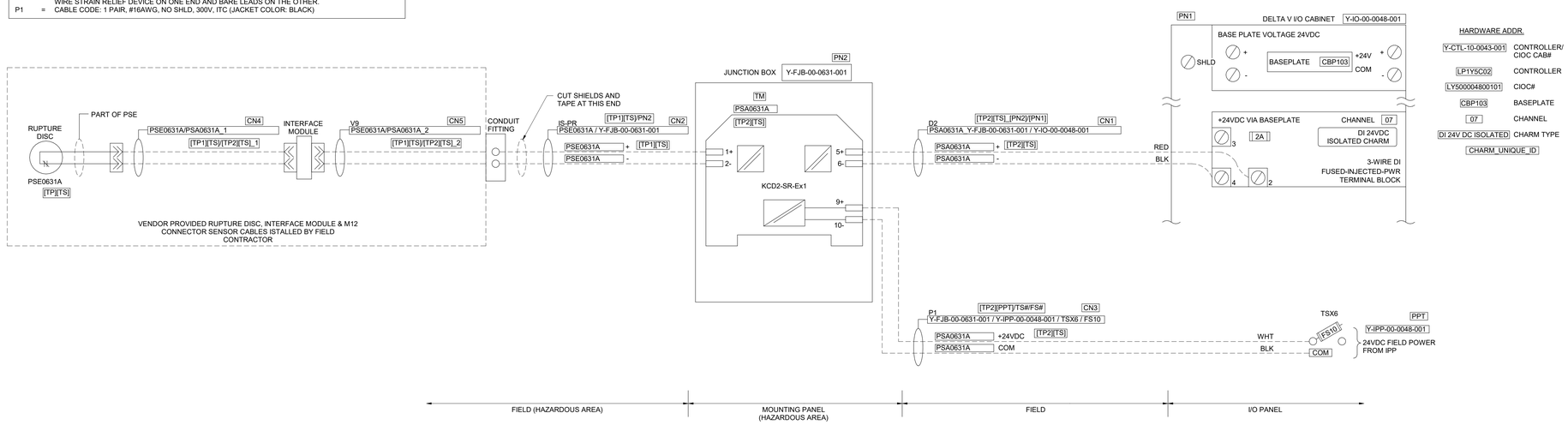
LABELING DESCRIPTION AND EXAMPLE		
TP1	=	TAG PREFIX = RUPTURE DISK ELEMENT = PSE
TP2	=	TAG PREFIX = RUPTURE DISK I.S. BARRIER = PSA
TS	=	TAG SUFFIX = LOOP NUMBER = 0631A
PN1	=	IO PANEL = <BLDG>-<TYPE>-<FLOOR>-<ROOM # OR SKID#>-<INDEX> = Y-IO-00-0048-001
PN2	=	FIELD JUNCTION BOX = <BLDG>-<TYPE>-<FLOOR>-<ROOM# OR SKID#>-<INDEX> = Y-FJB-00-0631-001
PPT	=	POWER PANELTAG = <BLDG>-<TYPE>-<FLOOR>-<ROOM# OR SKID#>-<INDEX> = Y-IPP-00-0048-001
CN1	=	SIGNAL CABLE TAG = TP2[TS]_PN2/ PN1 = PSA0631A_Y-FJB-00-0631-001 / Y-IO-00-0048-001
CN2	=	SIGNAL CABLE TAG = TP1[TS]/ PN2 = PSE0631A / Y-FJB-00-0631-001
CN3	=	POWER CABLE TAG = PN2 / PPT/ TS# / FS# = Y-FJB-00-0631-001 / Y-IPP-00-0048-001 / TSX6 / FS9
CN4	=	SIGNAL CABLE TAG = TP1[TS] / TP2[TS] _1 = PSE0631A/PSA0631A_1
CN5	=	SIGNAL CABLE TAG = TP1[TS] / TP2[TS] _2 = PSE0631A/PSA0631A_2
TM	=	TERMINAL MARKER STRIP = TP2 TS = PSA0631A
CHARM_UNIQUE_ID = FN - SN - CN		
IS-PR	=	CABLE CODE: 1 PAIR, TWISTED #18, SHLD, 300V, ITC (JACKET COLOR: BLU)
V9	=	CABLE CODE: B.D.I. LEAD WIRE ASSEMBLY WITH A STANDARD PLUG B.D.I. CONNECTION AND A LEAD WIRE STRAIN RELIEF DEVICE ON ONE END AND BARE LEADS ON THE OTHER.
P1	=	CABLE CODE: 1 PAIR, #16AWG, NO SHLD, 300V, ITC (JACKET COLOR: BLACK)



HARDWARE ADDR.	
Y-CTL-10-0043-001	CONTROLLER/ CIOC CAB#
LP1YSC02	CONTROLLER
LY500004800101	CIOC#
CBP103	BASEPLATE
07	CHANNEL
DI 24V DC ISOLATED	CHARM TYPE
CHARM_UNIQUE_ID	

NOTES:

1. ALL CABLES FROM FIELD INSTRUMENT TO CHARM I/O PANEL OR IPP TO BE FIELD ROUTED USING SNAP TRACK CHANNEL TRAY OR EQUIVALENT.
2. ALL TAGS WILL HAVE PREFIX "LP1Y".

1 INSTRUMENT WIRING DIAGRAM - LP1YPSA0631A - DISCRETE INPUT4-WIRE, RUPTURE DISC WITH I.S. BARRIER FOR HAZADOUS AREA

NTS

REFERENCE DRAWINGS:

1. LP1Y-270-LST-K215-02: INSTRUMENT I/O LIST.
2. LP1Y-270-LST-K215-03: INSTRUMENT CABLE SCHEDULE.
3. LP1Y-270-LAY-L215-01 & 02: INSTRUMENT LOCATION PLAN.
4. LP1Y-275-DIA-K215-30: CHARM I/O PANEL WIRING.
5. LP1Y-275-DIA-K215-31: 24VDC IPP PANEL WIRING.

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A	16-JAN-26	MGL	JN	BM	AJT	SS

INSTRUMENT WIRING DIAGRAM -_LP1YPSA0631A - DISCRETE INPUT 4-WIRE RUPTURE DISC WITH I.S. BARRIER FOR HAZARDOUS AREA RAIL UNLOADING RACK

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
0	ALL	ISSUED FOR CONSTRUCTION	NA	10-APR-26
A	ALL	ISSUED FOR REVIEW	NA	16-JAN-26

REVISIONS

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SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: 0
SUB DISCIPLINE: LP1	LOCATION CODE / LOCATION/GRID: COL C-D / 33-43	CCI CODE: YELLOW	PROJECT NAME: LP1-306-1181
CRITICAL QUALITY	ENTERPRISE GROUP: API	DRAWING INTENTION	DRAWING SCALE: NONE
CRITICAL BUSINESS	EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA	MERIDIAN REV: B
CRITICAL HSE	VENDOR: NA	DOCUMENT ID: LP1Y-275-DIA-K215-28	
NON-CRITICAL	ENGINEERING AGENCY DOC ID: NA		

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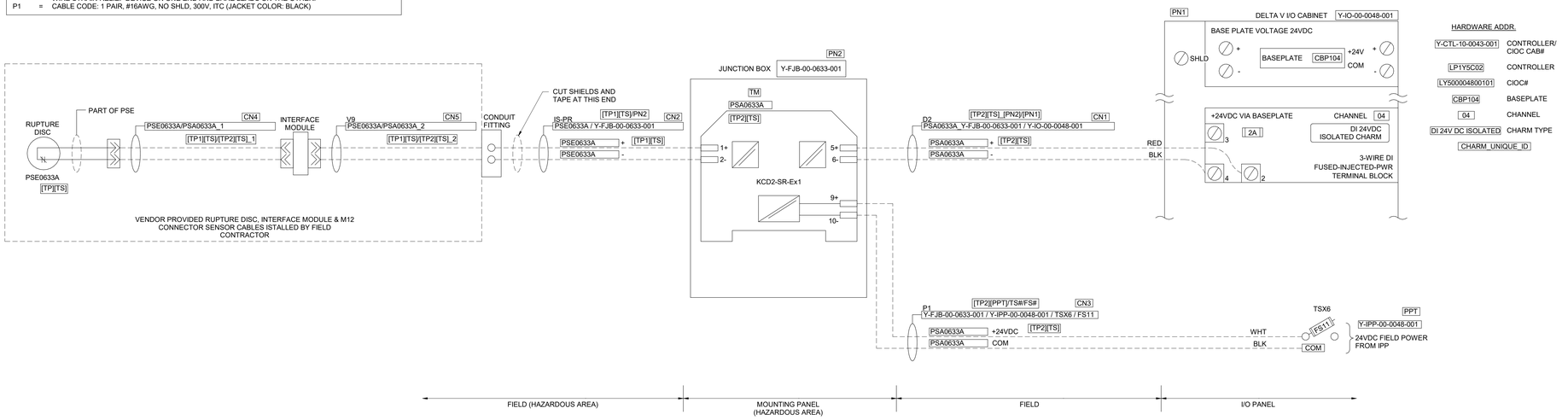
LABELING DESCRIPTION AND EXAMPLE		
TP1	=	TAG PREFIX = RUPTURE DISK ELEMENT = PSE
TP2	=	TAG PREFIX = RUPTURE DISK I.S. BARRIER = PSA
TS	=	TAG SUFFIX = LOOP NUMBER = 0633A
PN1	=	IO PANEL = <BLDG>-<TYPE>-<FLOOR>-<ROOM # OR SKID#>-<INDEX> = Y-IO-00-0048-001
PN2	=	FIELD JUNCTION BOX = <BLDG>-<TYPE>-<FLOOR>-<ROOM# OR SKID#>-<INDEX> = Y-FJB-00-0633-001
PPT	=	POWER PANELTAG = <BLDG>-<TYPE>-<FLOOR>-<ROOM# OR SKID#>-<INDEX> = Y-IPP-00-0048-001
CN1	=	SIGNAL CABLE TAG = TP2[TS]_PN2/ PN1 = PSA0633A_Y-FJB-00-0633-001 / Y-IO-00-0048-001
CN2	=	SIGNAL CABLE TAG = TP1[TS] / PN2 = PSE0633A / Y-FJB-00-0633-001
CN3	=	POWER CABLE TAG = PN2 / PPT / TS# / FS# = Y-FJB-00-0633-001 / Y-IPP-00-0048-001 / TSX6 / FS9
CN4	=	SIGNAL CABLE TAG = TP1[TS] / TP2[TS]_1 = PSE0633A/PSA0633A_1
CN5	=	SIGNAL CABLE TAG = TP1[TS] / TP2[TS]_2 = PSE0633A/PSA0633A_2
TM	=	TERMINAL MARKER STRIP = TP2 TS = PSA0633A
CHARM_UNIQUE_ID = [FN] - [SN] - [CN]		
IS-PR	=	CABLE CODE: 1 PAIR, TWISTED #18, SHLD, 300V, ITC (JACKET COLOR: BLU)
V9	=	CABLE CODE: B.D.I. LEAD WIRE ASSEMBLY WITH A STANDARD PLUG B.D.I. CONNECTION AND A LEAD WIRE STRAIN RELIEF DEVICE ON ONE END AND BARE LEADS ON THE OTHER.
P1	=	CABLE CODE: 1 PAIR, #16AWG, NO SHLD, 300V, ITC (JACKET COLOR: BLACK)



NOTES:

1. ALL CABLES FROM FIELD INSTRUMENT TO CHARM I/O PANEL OR IPP TO BE FIELD ROUTED USING SNAP TRACK CHANNEL TRAY OR EQUIVALENT.
2. ALL TAGS WILL HAVE PREFIX "LP1Y".

1 INSTRUMENT WIRING DIAGRAM - LP1YPSA0633A - DISCRETE INPUT4-WIRE, RUPTURE DISC WITH I.S. BARRIER FOR HAZADOUS AREA

NTS

REFERENCE DRAWINGS:

1. LP1Y-270-LST-K215-02: INSTRUMENT I/O LIST.
2. LP1Y-270-LST-K215-03: INSTRUMENT CABLE SCHEDULE.
3. LP1Y-270-LAY-L215-01 & 02: INSTRUMENT LOCATION PLAN.
4. LP1Y-275-DIA-K215-30: CHARM I/O PANEL WIRING.
5. LP1Y-275-DIA-K215-31: 24VDC IPP PANEL WIRING.

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
0	ALL	ISSUED FOR CONSTRUCTION	NA	10-APR-26
A	ALL	ISSUED FOR REVIEW	NA	16-JAN-26

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1	10-APR-26	MGL	JN	BM	AJT	SS
0	16-JAN-26	MGL	JN	BM	AJT	SS

INSTRUMENT WIRING DIAGRAM -.LP1YPSA0633A - DISCRETE INPUT 4-WIRE RUPTURE DISC WITH I.S. BARRIER FOR HAZARDOUS AREA RAIL UNLOADING RACK

SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: 1
SUB DISCIPLINE:	LOCATION CODE: LP1	LOCATION/GRID: COL C-D / 33-43	CCI CODE: YELLOW
			PROJECT NAME: LP1-306-1181
CRITICAL QUALITY:		ENTERPRISE GROUP: API	DRAWING INTENTION:
			DRAWING SCALE: NONE
CRITICAL BUSINESS:		EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA
			MERIDIAN REV: B
CRITICAL HSE:		VENDOR: NA	DOCUMENT ID: LP1Y-275-DIA-K215-29
NON-CRITICAL:		ENGINEERING AGENCY DOC ID: NA	

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LP1Y-IO-00-0048-001

CBP101

1.	XS00004801A CHARM I/O POWER SUP FAULT
2.	TE00004801A CHARM I/O CAB T/C
3.	XS00004801B IPP PWR SUP FAULT
4.	TS00004801L IPP TEMP SW LOW
5.	TS00004801H IPP TEMP SW HIGH
6.	TS10004801D NER CAB TEMP HIGH
7.	LSH0640WP
8.	PIT0610WP
9.	PSA0610WP
10.	LSH0650WP
11.	LSL0623A
12.	LSL0631A

CBP102

1.	ZSC0610VA
2.	ZSO0610VA
3.	SV0610VA
4.	ZSC0610VB
5.	ZSO0610VB
6.	SV0610VB
7.	ZSC0610WPA
8.	ZSO0610WPA
9.	SV0610WPA
10.	ZSC0610WPB
11.	ZSO0610WPB
12.	SV0610WPB

CBP103

1.	HS0010SK
2.	LSL0633A
3.	FSL0630V
4.	PIT0630V
5.	FSL0631A
6.	PIT0631A
7.	PSA0631A
8.	SPARE
9.	FSL0640V
10.	PIT0640V
11.	SPARE
12.	SPARE

UPS POWER (1ST & 2ND)

CBP104

1.	SPARE
2.	SPARE
3.	PIT0633A
4.	PSA0633A
5.	SPARE
6.	FSL0650V
7.	PIT0650V
8.	SPARE
9.	PIT0623A
10.	PSA0623A
11.	SPARE
12.	SPARE

CBP105

1.	HS0010SL
2.	SPARE
3.	SPARE
4.	SPARE
5.	SPARE
6.	SPARE
7.	ZSO0622N1
8.	ZSC0622N1
9.	SV0622N1
10.	ZSO0632N1
11.	ZSC0632N1
12.	SV0632N1

CBP106

1.	SPARE
2.	SPARE
3.	SPARE
4.	SPARE
5.	SPARE
6.	SPARE
7.	SPARE
8.	SPARE
9.	SPARE
10.	SPARE
11.	SPARE
12.	SPARE

CBP107

1.	SPARE
2.	SPARE
3.	SPARE
4.	SPARE
5.	SPARE
6.	SPARE
7.	SPARE
8.	SPARE
9.	SPARE
10.	SPARE
11.	SPARE
12.	SPARE

CBP108

1.	SPARE
2.	SPARE
3.	SPARE
4.	SPARE
5.	SPARE
6.	SPARE
7.	SPARE
8.	SPARE
9.	SPARE
10.	SPARE
11.	SPARE
12.	SPARE

NOTES:

1. ALL TAGS WILL HAVE PREFIX "LP1Y".

1 DELTA-V 96 I/O CHARM LAYOUT PANEL

NTS

REFERENCE DRAWINGS:

1. B1LG-7-0303-00-00009 WIRING DIAGRAM.
2. LP1Y-275-DIA-K215-01 THRU LP1Y-275-DIA-K215-34 INSTRUMENT WIRING DIAGRAMS.
3. LP1Y-270-LAY-K215-01 & 02: INSTRUMENT LOCATION PLAN.
4. E07-42: UPS PANEL 48-UPP/00-00-01 & E07-43: UPS PANEL 48-UPP/00-00-02.
5. LP1Y-270-LST-K215-01: INSTRUMENT INDEX.
6. LP1Y-270-LST-K215-02: INSTRUMENT I/O LIST.
7. LP1Y-275-DIA-K215-31: 24VDC IPP PANEL WIRING.

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
0	ALL	ISSUED FOR CONSTRUCTION	NA	10-APR-26
A	ALL	ISSUED FOR REVIEW	NA	16-JAN-26

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A	16-JAN-26	MGL	JN	BM	AJIT	SS

PANEL LP1Y-I/O-00-0048-001 DELTA-V 96 I/O CHARM LAYOUT

RAIL UNLOADING RACK

SYSTEM TYPE: NA				EDIFICE/BUILDING: Y	LEVEL/FLOOR: 0
SUB DISCIPLINE:	LOCATION CODE: LP1	LOCATION/GRID: COL C-D / 33-43		CCI CODE: YELLOW	PROJECT NAME: LP1-306-1181
CRITICAL QUALITY		ENTERPRISE GROUP: API		DRAWING INTENTION	DRAWING SCALE: NONE
CRITICAL BUSINESS		EQUIPMENT CLASSIFICATION: NA		ALTERNATE DOCUMENT ID: NA	MERIDIAN REV: B
CRITICAL HSE		VENDOR: NA		DOCUMENT ID	LP1Y-275-DIA-K215-30
NON-CRITICAL		ENGINEERING AGENCY DOC ID: NA			

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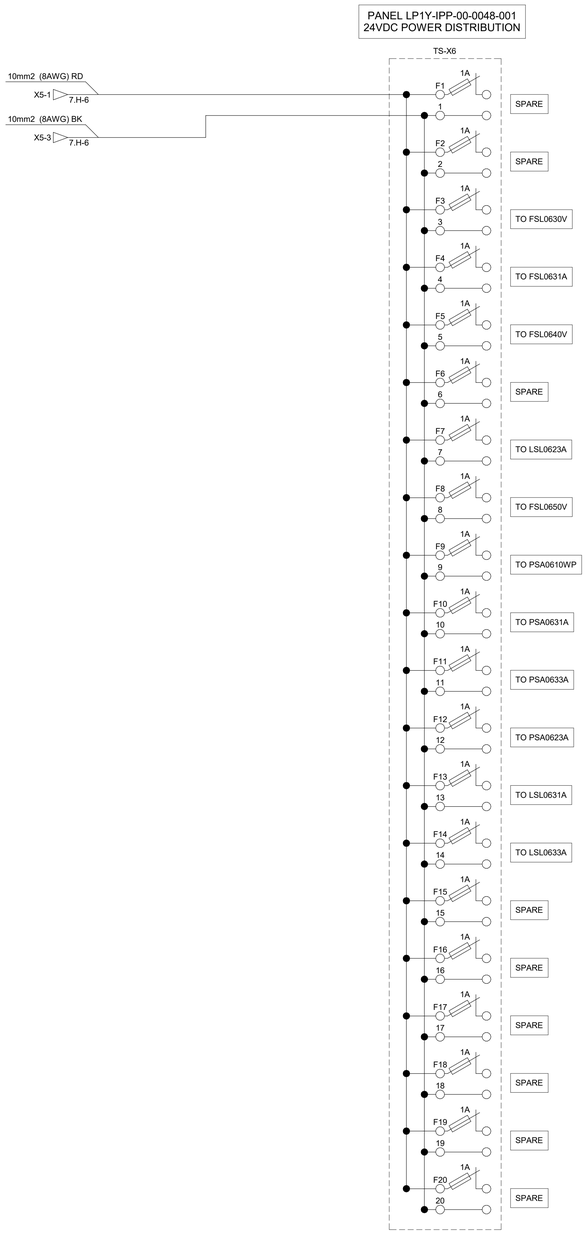

NOTE:

1. ALL TAGS HAVE PREFIX "LP1Y".

1 INSTRUMENT WIRING DIAGRAM - 24VDC POWER DISTRIBUTION

NTS

REFERENCE DRAWINGS:

1. B1LG-7-0303-00-00009 WIRING DIAGRAM.
2. LP1Y-275-DIA-K215-01 THRU LP1Y-275-DIA-K215-34 INSTRUMENT WIRING DIAGRAMS.
3. LP1Y-270-LAY-K215-01 & 02: INSTRUMENT LOCATION PLAN.
4. E07-42: UPS PANEL 48-UPP/00-00-01 & E07-43: UPS PANEL 48-UPP/00-00-02.
5. LP1Y-270-LST-K215-01: INSTRUMENT INDEX.
6. LP1Y-270-LST-K215-02: INSTRUMENT I/O LIST.
7. LP1Y-275-DIA-K215-30: CHARM I/O PANEL WIRING.

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REV	DATE	DES	ENG	CHK	APP	PM APP
1	16-JAN-26	MGL	JN	BM	AJT	SS
0	10-APR-26	MGL	JN	BM	AJT	SS

PANEL LP1Y-IPP-00-0048-001 24VDC FIELD DISTRIBUTION INSTRUMENT POWER PANEL

RAIL UNLOADING RACK

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
0	ALL	ISSUED FOR CONSTRUCTION	NA	10-APR-26
1	ALL	ISSUED FOR REVIEW	NA	16-JAN-26

REVISIONS

SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: 0
SUB DISCIPLINE:	LOCATION CODE: LP1	LOCATION/GRID: COL C-D / 33-43	CCI CODE: YELLOW
CRITICAL QUALITY:		ENTERPRISE GROUP: API	DRAWING INTENTION:
CRITICAL BUSINESS:		EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA
CRITICAL HSE:		VENDOR: NA	DOCUMENT ID: LP1Y-275-DIA-K215-31
NON-CRITICAL:		ENGINEERING AGENCY DOC ID: NA	

PROJECT NAME: LP1-306-1181
DRAWING SCALE: NONE
MERIDIAN REV: B

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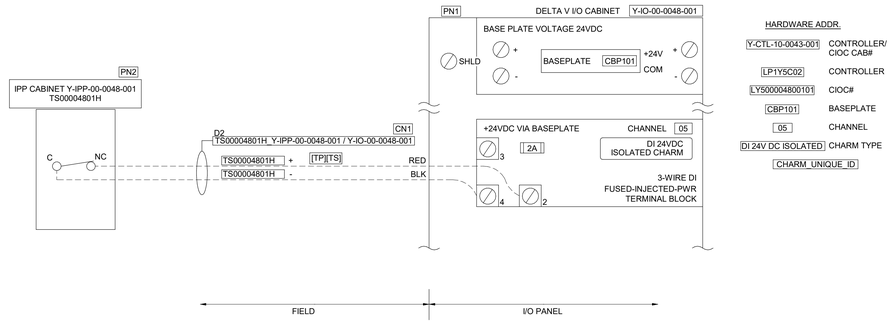
LABELING DESCRIPTION AND EXAMPLE		
TP	=	TAG PREFIX = CABINET TEMP SWITCH = TS
TS	=	TAG SUFFIX = LOOP NUMBER = 00004801H
PN1	=	IO PANEL = <BLDG>-<TYPE>-<FLOOR>-<ROOM # OR SKID#>-<INDEX> = Y-IO-00-0048-001
PN2	=	INSTRUMENT POWER PANEL = <BLDG>-<TYPE>-<FLOOR>-<SKID#>-<INDEX> = Y-IPP-00-0048-001
CN1	=	SIGNAL CABLE TAG = TP|TS , PN2 / PN1 = TS00004801H_Y-IPP-00-0048-001 / Y-IO-00-0048-001
CHARM_UNIQUE_ID	=	FN - SN - CN
D2	=	CABLE CODE: 1 PAIR, #18, NO SHLD, 300V, ITC (JACKET COLOR: BLK)



HARDWARE ADDR.

Y-CTL-10-0043-001	CONTROLLER/ CIOC CAB#
LP1Y5C02	CONTROLLER
LY500004800101	CIOC#
CBP101	BASEPLATE
05	CHANNEL
DI 24V DC ISOLATED	CHARM TYPE
CHARM_UNIQUE_ID	

NOTES:

1. ALL CABLES FROM FIELD INSTRUMENT TO CHARM I/O PANEL OR IPP TO BE FIELD ROUTED USING SNAP TRACK CHANNEL TRAY OR EQUIVALENT.
2. ALL TAGS WILL HAVE PREFIX "LP1Y".

1 INSTRUMENT WIRING DIAGRAM - LP1YTS00004801H - DISCRETE INPUT 2-WIRE IPP TEMPERATURE SWITCH

NTS

REFERENCE DRAWINGS:
1) LP1Y-270-LST-K215-02: INSTRUMENT I/O LIST.
2) LP1Y-270-LST-K215-03: INSTRUMENT CABLE SCHEDULE.
3 LP1Y-270-LAY-K215-01 & 02: INSTRUMENT LOCATION PLAN.
4) LP1Y-275-DIA-K215-30: CHARM I/O PANEL WIRING.
5) LP1Y-275-DIA-K215-31: 24VDC IPP PANEL WIRING.

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
0	ALL	ISSUED FOR REVIEW	NA	10-APR-26
A	ALL	ISSUED FOR REVIEW	NA	12-FEB-26

REVISIONS

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0	10-APR-26	MGL	JN	BM	AJT	SS
A	12-FEB-26	MGL	JN	BM	AJT	SS

INSTRUMENT WIRING DIAGRAM - LP1YTS00004801H
DISCRETE INPUT 2-WIRE IPP CABINET TEMP SWITCH
RAIL UNLOADING RACK

SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: 0
SUB DISCIPLINE	LOCATION CODE: LP1	LOCATION/GRID: COL C-D / 33-43	CCI CODE: YELLOW
			PROJECT NAME: LP1-306-1181
CRITICAL QUALITY		ENTERPRISE GROUP: API	DRAWING INTENTION
			DRAWING SCALE: NONE
CRITICAL BUSINESS		EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA
			MERIDIAN REV: A
CRITICAL HSE		VENDOR: NA	DOCUMENT ID: LP1Y-275-DIA-K215-32
NON-CRITICAL		ENGINEERING AGENCY DOC ID: NA	

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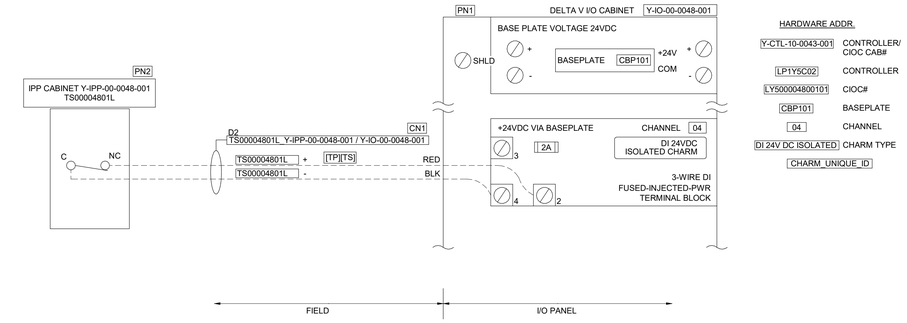
LABELING DESCRIPTION AND EXAMPLE		
TP	=	TAG PREFIX = CABINET TEMP SWITCH = TS
TS	=	TAG SUFFIX = LOOP NUMBER = 00004801L
PN1	=	IO PANEL = <BLDG>-<TYPE>-<FLOOR>-<ROOM # OR SKID#>-<INDEX> = Y-IO-00-0048-001
PN2	=	INSTRUMENT POWER PANEL = <BLDG>-<TYPE>-<FLOOR>-<SKID#>-<INDEX> = Y-IPP-00-0048-001
CN1	=	SIGNAL CABLE TAG = TP TS PN2// PN1 = TS00004801L_Y-IPP-00-0048-001 / Y-IO-00-0048-001
CHARM_UNIQUE_ID	=	FN - SN - CN
D2	=	CABLE CODE: 1 PAIR, #18, NO SHLD, 300V, ITC (JACKET COLOR: BLK)



HARDWARE ADDR.

Y-CTL-10-0043-001	CONTROLLER/ CIOC CAB#
LP1Y5C02	CONTROLLER
LY500004800101	CIOC#
CBP101	BASEPLATE
04	CHANNEL
DI 24V DC ISOLATED	CHARM TYPE
CHARM_UNIQUE_ID	

NOTES:

1. ALL CABLES FROM FIELD INSTRUMENT TO CHARM I/O PANEL OR IPP TO BE FIELD ROUTED USING SNAP TRACK CHANNEL TRAY OR EQUIVALENT.
2. ALL TAGS WILL HAVE PREFIX "LP1Y".

1 INSTRUMENT WIRING DIAGRAM - LP1YTS00004801L - DISCRETE INPUT 2-WIRE IPP CABINET TEMPERATURE SWITCH

NTS

REFERENCE DRAWINGS:
1) LP1Y-270-LST-K215-02: INSTRUMENT I/O LIST.
2) LP1Y-270-LST-K215-03: INSTRUMENT CABLE SCHEDULE.
3 LP1Y-270-LAY-K215-01 & 02: INSTRUMENT LOCATION PLAN.
4) LP1Y-275-DIA-K215-30: CHARM I/O PANEL WIRING.
5) LP1Y-275-DIA-K215-31: 24VDC IPP PANEL WIRING.

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
0	ALL	ISSUED FOR REVIEW	NA	10-APR-26
A	ALL	ISSUED FOR REVIEW	NA	12-FEB-26

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REV	DATE	DES	ENG	CHK	APP	PM APP
0	10-APR-26	MGL	JN	BM	AJT	SS
A	12-FEB-26	MGL	JN	BM	AJT	SS

INSTRUMENT WIRING DIAGRAM - LP1YTS00004801L
DISCRETE INPUT 2-WIRE IPP CABINET TEMP SWITCH
RAIL UNLOADING RACK

SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: 0
SUB DISCIPLINE	LOCATION CODE: LP1	LOCATION/GRID: COL C-D / 33-43	CCI CODE: YELLOW
CRITICAL QUALITY		ENTERPRISE GROUP: API	DRAWING INTENTION:	PROJECT NAME: LP1-306-1181
CRITICAL BUSINESS		EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA	MERIDIAN REV: A
CRITICAL HSE		VENDOR: NA	DRAWING SCALE: NONE	
NON-CRITICAL		ENGINEERING AGENCY DOC ID: NA	DOCUMENT ID: LP1Y-275-DIA-K215-33	

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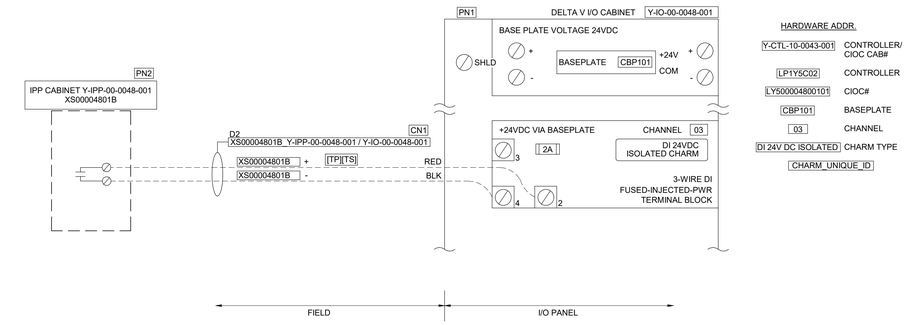
LABELING DESCRIPTION AND EXAMPLE

TP = TAG PREFIX = CABINET TEMP SWITCH = TS
TS = TAG SUFFIX = LOOP NUMBER = 00004801B
PN1 = IO PANEL = <BLDG>-<TYPE>-<FLOOR>-<ROOM # OR SKID#>-<INDEX> = Y-IO-00-0048-001
PN2 = INSTRUMENT POWER PANEL = <BLDG>-<TYPE>-<FLOOR>-<SKID#>-<INDEX> = Y-IPP-00-0048-001
CN1 = SIGNAL CABLE TAG = TP TS_PN2/PN1 = TS00004801B_Y-IPP-00-0048-001 / Y-IO-00-0048-001
CHARM_UNIQUE_ID = PN1 -SN1 -CN1
D2 = CABLE CODE: 1 PAIR, #18, NO SHLD, 300V, ITC (JACKET COLOR: BLK)



NOTES:

1. ALL CABLES FROM FIELD INSTRUMENT TO CHARM I/O PANEL OR IPP TO BE FIELD ROUTED USING SNAP TRACK CHANNEL TRAY OR EQUIVALENT.
2. ALL TAGS WILL HAVE PREFIX "LP1Y".

1 INSTRUMENT WIRING DIAGRAM - LP1YXS00004801B - DISCRETE INPUT 2-WIRE IPP CABINET POWER SUPPLY FAULT

NTS

REFERENCE DRAWINGS:
1) LP1Y-270-LST-K215-02: INSTRUMENT I/O LIST.
2) LP1Y-270-LST-K215-03: INSTRUMENT CABLE SCHEDULE.
3 LP1Y-270-LAY-K215-01 & 02: INSTRUMENT LOCATION PLAN.
4) LP1Y-275-DIA-K215-30: CHARM I/O PANEL WIRING.
5) LP1Y-275-DIA-K215-31: 24VDC IPP PANEL WIRING.

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REV	DATE	DES	ENG	CHK	APP	PM APP
0	10-APR-26	MGL	JN	BM	AJT	SS
A	12-FEB-26	MGL	JN	BM	AJT	SS

INSTRUMENT WIRING DIAGRAM - LP1YXS00004801B
DISCRETE INPUT 2-WIRE IPP CABINET
POWER SUPPLY FAULT - RAIL UNLOADING RACK

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
0	ALL	ISSUED FOR REVIEW	NA	10-APR-26
A	ALL	ISSUED FOR REVIEW	NA	12-FEB-26

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SYSTEM TYPE: NA		EDIFICE/BUILDING: Y	LEVEL/FLOOR: 0
SUB DISCIPLINE	LOCATION CODE: LP1	LOCATION/GRID: COL C-D / 33-43	CCI CODE: YELLOW
CRITICAL QUALITY		ENTERPRISE GROUP: API	DRAWING INTENTION
CRITICAL BUSINESS		EQUIPMENT CLASSIFICATION: NA	ALTERNATE DOCUMENT ID: NA
CRITICAL HSE		VENDOR: NA	DOCUMENT ID: LP1Y-275-DIA-K215-34
NON-CRITICAL		ENGINEERING AGENCY DOC ID: NA	

PROJECT NAME: LP1-306-1181
DRAWING SCALE: NONE
MERIDIAN REV: A

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Lilly	DOCUMENT COVER SHEET PROPRIETARY	FLUOR® EFDEE ENGINEERING PC

Client: Lilly
Lilly Project: LEBANON PLANT 1
Fluor Project No.: B1LG
WBS No.: TANK FARM AND SOLVENT RECOVERY (TFS)

Document Number: LP148-265-LST-002

Document Title: ELECTRICAL CABLE LIST - EHOUSE #48

Rev.	Date	Originated By:	Checked By:	Approvals		
				Disc Lead	PM	Client*
A	23-May-2025	RA	FS	TL	EM	
0	15-Aug-2025	FS	JN	TL	EM	
1	17-Feb-2026	JA	EA	TL	RD	
1A	17-Apr-2026	AT	RN	RN	SS	

Issued For:

☐ Information	☐ Review	☐ Approval	☐ Design
☐ Bid	☐ Purchase	☒ Construction	☐ Other

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Record of Revisions

Revision No.	Date	Description
A	23-May-2025	Issued for Squadcheck
0	15-Aug-2025	Issued for Construction
1	17-Feb-2026	Issued for Construction
1A	17-Apr-2026	Issued for Review

FLUOR
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NOTES AND REFERENCES

NOTES:

1) INSTALLATION CONTRACTOR SHALL PURCHASE CABLES AS SPECIFIED IN THE CABLE SCHEDULE. INSTALLATION CONTRACTOR SHALL SUPPLY AND INSTALL CABLE GLAND AS REQUIRED. FOR ANY DEVIATIONS, CONTRACTOR SHALL OBTAIN APPROVAL BEFORE PURCHASE.
2) CONTRACTOR SHALL VERIFY CABLE LENGTH AND ROUTE BEFORE CUTTING AND INSTALLING THE CABLE. CONTRACTOR SHALL BE ALSO RESPONSIBLE FOR ANY CABLE SIZE CHANGE DUE TO ROUTING DEVIATION RESULTING TO CABLE LENGTH LONGER THAN THE SIZE MENTIONED IN THE CABLE LIST.
3) CONTRACTOR SHALL INSTALL CONDUITS AND FITTINGS AS REQUIRED TO INSTALL THE CABLES.
4) CONDUITS AND FITTINGS BELOW 2 INCH ARE NOT SPECIFIED OR TAGGED IN THE CABLE SCHEDULE. INSTALLATION CONTRACTOR SHALL SIZE, TAG AND FIELD ROUTE AS REQUIRED.
5) E-HOUSE VENDOR INSTALLED CABLES ARE EXCLUDED FROM THIS CABLE LIST. FOR E-HOUSE VENDOR LIST OF CABLES, REFER TO E-HOUSE CABLE SCHEDULE.
6) SPARE CONDUCTOR SHALL BE COILED AND TAPED AND MARKED SPARE (SP) AT BOTH END.
7) CABLE SIZES WERE BASED ON THE ISSUED FOR CONSTRUCTION LOAD LIST REVISION 0.
8) CONTROL SYSTEM CHARMS (RIO) AND IPP CABINETS COMES WITH PHOENIX TERMINAL BLOCKS. CONTRACTOR SHALL REPLACE INCOMER TERMINAL BLOCKS AS NEEDED TO ACCOMMODATE 120/208V POWER SUPPLY INCOMER CABLE.
9) CONTRACTOR SHALL UTILIZE REDUCER PIN OR INLINE SPLICE KIT INSIDE ELECTRICAL PANEL BOARDS TO ACCOMMODATE CABLES IN PROVIDED CIRCUIT BREAKERS.
10) THE CONTRACTOR SHALL SUPPLY AND INSTALL 3M COLD SHRINK CABLE SPLICE KIT, QS-III OR AN APPROVED EQUIVALENT FOR CONNECTION OF THE MV CABLES. WHEN SPLICING CABLES WITHIN THE SAME CABLE TRAY, THE SPLICES SHOULD BE STAGGERED. THE SPLICING LOCATION IN THE TRAY MUST HAVE A CABLE TRAY COVER TO GUARD AGAINST ENVIRONMENTAL ELEMENTS LIKE RAIN, SNOW OR DEBRIS. THE FINAL POSITIONING OF THE SPLICE KITS IS THE RESPONSIBILITY OF THE CONTRACTOR. THE CONTRACTOR SHALL REDLINE THE ELECTRICAL CABLE TRAY LAYOUT DRAWINGS WITH THE EXACT LOCATION OF THE SPLICES.
11) MEDIUM VOLTAGE (MV) CABLE SHIELDS AND ARMOR MUST BE PROPERLY GROUNDED TO ENSURE SAFETY. THE CABLE SHIELD SHALL BE SINGLE-POINT BONDED AT THE SOURCE, WHILE THE CABLE ARMOR SHALL BE BONDED AT BOTH ENDS.
12) CABLES FROM E HOUSE TO RAIL CAR LOADING AREA SHALL BE DESIGNED, ROUTED, SUPPLIED AND INSTALLED BY K215 CONTRACTOR AND ARE EXCLUDED FROM THIS LIST.
13) TWO POWER SUPPLY FEEDER CABLES ARE CONSIDERED FOR OPERATION SHELTER (480V TO DISCONNECT SWITCH AND 208/120V FROM UPS TO JUNCTION BOX) FROM E-HOUSE 48. INTERNAL POWER DISTRIBUTION FOR OPERATION SHELTER TO BE DONE BY VENDOR.

HOLD:

1) FIELD AUXILIARY SYSTEM EQUIPMENT ARE NOT INCLUDED IN THE CABLE SCHEDULE DUE TO ABSENCE OF ANY DESIGN AND LOCATION DATA. CABLE SCHEDULE TO BE UPDATED UPON RECEIPT OF VENDOR DATA.
2) CABLE QUANTITY, LENGTH AND SIZE BETWEEN E-HOUSE AND OPERATION SHELTER SHOWN IN THE CABLE SCHEDULE ARE PRELIMINARY AND SHALL BE UPDATED BASED ON RECEIPT OF THE VENDOR DATA. EQUIPMENT TAGS FOR JBS FOR OPERATION SHELTER IS ALSO ON HOLD.

REFERENCES:

1) REFER TO DRAWING LP1-EJ-00000001 FOR GENERAL NOTES.
2) REFER TO DOCUMENT LP1Y-265-LST-001 FOR ELECTRICAL LOAD LIST - TFS.
3) REFER TO DRAWING LP148-EK-00000602 FOR UPS SINGLE LINE DIAGRAM.
4) REFER TO DRAWING LP1Y-EL-00000301 FOR POWER PLAN DRAWING INDEX.
5) REFER TO LP100-EJ-00000007 ELECTRICAL AND INSTRUMENTATION INSTALLATION PRACTICE - CIRCUIT SEPARATION BUILDING 00 SITE/GENERAL.
6) REFER TO LP1Y-270-LST-003 FOR INSTRUMENTATION CABLE SCHEDULE.
7) REFER TO DOCUMENT LP100-265-LST-001 FOR CABLE LIST.

Client: Lilly
Project: LEBANON PLANT 1
Project No.: B1LG
Plant / Area / Unit: TANK FARM AND SOLVENT RECOVERY (TFS)

LP148-265-LST-002
ELECTRICAL CABLE LIST - EHOUSE # 48

FLUOR®

REV. NO.	ITEM NO.	CABLE DATA			FROM EQUIPMENT				TO EQUIPMENT				ROUTE (FROM SOURCE TO DESTINATION)	CABLE LENGTH (FT) (NOTE 2)	CONDUIT OR SLEEVE SIZE (NOTE 4)	REMARKS
		CIRCUIT GROUP OR MULTICONDUCTOR CABLE TAG NUMBER	SINGLE CONDUCTOR TAG NUMBER	CABLE TYPE	EQUIPMENT TAG	DESCRIPTION	UNIT LOCATION / ROOM NUMBER	EQUIP. DWG. NO. (REF.)	EQUIPMENT TAG	DESCRIPTION	UNIT LOCATION / ROOM NUMBER	EQUIP. DWG. NO. (REF.)				
1	1	**48MCC/000001-TBD-P**	**N/A**	**3X4_1X6 GND_CU_XHHW-2_TC-ER**	48-MCC/00-00-01	RAIL CAR LOADING - 480V MCC	TFS BLDG #48	**LP148-ES-00000501**	TBD	OPERATION SHELTER DISCONNECT SWITCH	OPERATION SHELTER	**TBD**	51E01005M15, YE00051M15, YE00055M15	160	**N/A**	HOLD-2 , NOTE 13
1	2	**48UPP/000001-TDB-P**	**N/A**	**3X4_1X6 GND_CU_XHHW-2_TC-ER**	48-UPP/00-00-01	208/120V 3PH 4W UPS DIST PANEL	TFS BLDG #48	**LP148-ES-00000501**	**TBD**	OPERATION SHELTER JUNCTION BOX	OPERATION SHELTER	**TBD**	51E01007M15, YE00053M15, YE00056M15	160	**N/A**	HOLD-2 , NOTE 13
1	3	**48UPP/000001-YNER100048001-P**	**N/A**	**3X12_1X12 GND_CU_XHHW-2_TC-ER**	48-UPP/00-00-01	208/120V 3PH 4W UPS DIST PANEL	TFS BLDG #48	**LP148-ES-00000501**	Y-NER-10-0048-001	DELTAV NETWORK SWITCH CABINET **(CKT 1) (UPS)**	TFS BLDG #48	**LP148-ES-00000501**	46E00003P, YE00330P, YE00376P, FIELD ROUTED	23	**N/A**	K217 SCOPE
1	4	**48UPP/000001-YNER100048001-P**	**N/A**	**3X12_1X12 GND_CU_XHHW-2_TC-ER**	48-UPP/00-00-01	208/120V 3PH 4W UPS DIST PANEL	TFS BLDG #48	**LP148-ES-00000501**	Y-NER-10-0048-001	DELTAV NETWORK SWITCH CABINET **(CKT 2) (UPS)**	TFS BLDG #48	**LP148-ES-00000501**	46E00003P, YE00330P, YE00376P, FIELD ROUTED	23	**N/A**	K217 SCOPE
1	5	**48RP/000002-YNER100048001-P**	**N/A**	**3X12_1X12 GND_CU_XHHW-2_TC-ER**	48-RP/00-00-02	208/120V 3PH 4W GEN. SERVICE POWER PANEL	TFS BLDG #48	**LP148-ES-00000501**	Y-NER-10-0048-001	DELTAV NETWORK SWITCH CABINET **(CKT 1) (NORMAL)**	TFS BLDG #48	**LP148-ES-00000501**	FIELD ROUTED	23	**N/A**	K217 SCOPE
1	6	**48RP/000002-YNER100048001-P**	**N/A**	**3X12_1X12 GND_CU_XHHW-2_TC-ER**	48-RP/00-00-02	208/120V 3PH 4W GEN. SERVICE POWER PANEL	TFS BLDG #48	**LP148-ES-00000501**	Y-NER-10-0048-001	DELTAV NETWORK SWITCH CABINET **(CKT 2) (NORMAL)**	TFS BLDG #48	**LP148-ES-00000501**	FIELD ROUTED	23	**N/A**	K217 SCOPE
1	~~7~~	~~**00TX/000015-48MCC/000001-P1A**~~	~~**N/A**~~	~~**1X600_CU_XHHW-2**~~	~~**00-TX/00-00-15**~~	~~**12.47kV-480Y/277V 750kVA,3Ø,4W PADMOUNT TRANSFORMER**~~	~~**TFS BLDG #48**~~	~~**LP148-ES-00000501**~~	~~**48-MCC/00-00-01**~~	~~**RAIL CAR LOADING - 480V MCC**~~	~~**TFS BLDG #48**~~	~~**LP148-ES-00000501**~~	~~**UG CONDUIT ROUTED**~~	~~**80**~~	~~**N/A**~~	~~**K217 SCOPE**~~
1	~~8~~	~~**00TX/000015-48MCC/000001-P1B**~~	~~**N/A**~~	~~**1X600_CU_XHHW-2**~~	~~**00-TX/00-00-15**~~	~~**12.47kV-480Y/277V 750kVA,3Ø,4W PADMOUNT TRANSFORMER**~~	~~**TFS BLDG #48**~~	~~**LP148-ES-00000501**~~	~~**48-MCC/00-00-01**~~	~~**RAIL CAR LOADING - 480V MCC**~~	~~**TFS BLDG #48**~~	~~**LP148-ES-00000501**~~	~~**UG CONDUIT ROUTED**~~	~~**80**~~	~~**N/A**~~	~~**K217 SCOPE**~~
1	~~9~~	~~**00TX/000015-48MCC/000001-P1C**~~	~~**N/A**~~	~~**1X600_CU_XHHW-2**~~	~~**00-TX/00-00-15**~~	~~**12.47kV-480Y/277V 750kVA,3Ø,4W PADMOUNT TRANSFORMER**~~	~~**TFS BLDG #48**~~	~~**LP148-ES-00000501**~~	~~**48-MCC/00-00-01**~~	~~**RAIL CAR LOADING - 480V MCC**~~	~~**TFS BLDG #48**~~	~~**LP148-ES-00000501**~~	~~**UG CONDUIT ROUTED**~~	~~**80**~~	~~**N/A**~~	~~**K217 SCOPE**~~
1	~~10~~	~~**00TX/000015-48MCC/000001-P2A**~~	~~**N/A**~~	~~**1X600_CU_XHHW-2**~~	~~**00-TX/00-00-15**~~	~~**12.47kV-480Y/277V 750kVA,3Ø,4W PADMOUNT TRANSFORMER**~~	~~**TFS BLDG #48**~~	~~**LP148-ES-00000501**~~	~~**48-MCC/00-00-01**~~	~~**RAIL CAR LOADING - 480V MCC**~~	~~**TFS BLDG #48**~~	~~**LP148-ES-00000501**~~	~~**UG CONDUIT ROUTED**~~	~~**80**~~	~~**N/A**~~	~~**K217 SCOPE**~~
1	~~11~~	~~**00TX/000015-48MCC/000001-P2B**~~	~~**N/A**~~	~~**1X600_CU_XHHW-2**~~	~~**00-TX/00-00-15**~~	~~**12.47kV-480Y/277V 750kVA,3Ø,4W PADMOUNT TRANSFORMER**~~	~~**TFS BLDG #48**~~	~~**LP148-ES-00000501**~~	~~**48-MCC/00-00-01**~~	~~**RAIL CAR LOADING - 480V MCC**~~	~~**TFS BLDG #48**~~	~~**LP148-ES-00000501**~~	~~**UG CONDUIT ROUTED**~~	~~**80**~~	~~**N/A**~~	~~**K217 SCOPE**~~
1	~~12~~	~~**00TX/000015-48MCC/000001-P2C**~~	~~**N/A**~~	~~**1X600_CU_XHHW-2**~~	~~**00-TX/00-00-15**~~	~~**12.47kV-480Y/277V 750kVA,3Ø,4W PADMOUNT TRANSFORMER**~~	~~**TFS BLDG #48**~~	~~**LP148-ES-00000501**~~	~~**48-MCC/00-00-01**~~	~~**RAIL CAR LOADING - 480V MCC**~~	~~**TFS BLDG #48**~~	~~**LP148-ES-00000501**~~	~~**UG CONDUIT ROUTED**~~	~~**80**~~	~~**N/A**~~	~~**K217 SCOPE**~~
1	~~13~~	~~**00TX/000015-48MCC/000001-P3A**~~	~~**N/A**~~	~~**1X600_CU_XHHW-2**~~	~~**00-TX/00-00-15**~~	~~**12.47kV-480Y/277V 750kVA,3Ø,4W PADMOUNT TRANSFORMER**~~	~~**TFS BLDG #48**~~	~~**LP148-ES-00000501**~~	~~**48-MCC/00-00-01**~~	~~**RAIL CAR LOADING - 480V MCC**~~	~~**TFS BLDG #48**~~	~~**LP148-ES-00000501**~~	~~**UG CONDUIT ROUTED**~~	~~**80**~~	~~**N/A**~~	~~**K217 SCOPE**~~
1	~~14~~	~~**00TX/000015-48MCC/000001-P3B**~~	~~**N/A**~~	~~**1X600_CU_XHHW-2**~~	~~**00-TX/00-00-15**~~	~~**12.47kV-480Y/277V 750kVA,3Ø,4W PADMOUNT TRANSFORMER**~~	~~**TFS BLDG #48**~~	~~**LP148-ES-00000501**~~	~~**48-MCC/00-00-01**~~	~~**RAIL CAR LOADING - 480V MCC**~~	~~**TFS BLDG #48**~~	~~**LP148-ES-00000501**~~	~~**UG CONDUIT ROUTED**~~	~~**80**~~	~~**N/A**~~	~~**K217 SCOPE**~~
1	~~15~~	~~**00TX/000015-48MCC/000001-P3C**~~	~~**N/A**~~	~~**1X600_CU_XHHW-2**~~	~~**00-TX/00-00-15**~~	~~**12.47kV-480Y/277V 750kVA,3Ø,4W PADMOUNT TRANSFORMER**~~	~~**TFS BLDG #48**~~	~~**LP148-ES-00000501**~~	~~**48-MCC/00-00-01**~~	~~**RAIL CAR LOADING - 480V MCC**~~	~~**TFS BLDG #48**~~	~~**LP148-ES-00000501**~~	~~**UG CONDUIT ROUTED**~~	~~**80**~~	~~**N/A**~~	~~**K217 SCOPE**~~
1	~~16~~	~~**00TX/000015-48MCC/000001-P4A**~~	~~**N/A**~~	~~**1X600_CU_XHHW-2**~~	~~**00-TX/00-00-15**~~	~~**12.47kV-480Y/277V 750kVA,3Ø,4W PADMOUNT TRANSFORMER**~~	~~**TFS BLDG #48**~~	~~**LP148-ES-00000501**~~	~~**48-MCC/00-00-01**~~	~~**RAIL CAR LOADING - 480V MCC**~~	~~**TFS BLDG #48**~~	~~**LP148-ES-00000501**~~	~~**UG CONDUIT ROUTED**~~	~~**80**~~	~~**N/A**~~	~~**K217 SCOPE**~~
1	~~17~~	~~**00TX/000015-48MCC/000001-P4B**~~	~~**N/A**~~	~~**1X600_CU_XHHW-2**~~	~~**00-TX/00-00-15**~~	~~**12.47kV-480Y/277V 750kVA,3Ø,4W PADMOUNT TRANSFORMER**~~	~~**TFS BLDG #48**~~	~~**LP148-ES-00000501**~~	~~**48-MCC/00-00-01**~~	~~**RAIL CAR LOADING - 480V MCC**~~	~~**TFS BLDG #48**~~	~~**LP148-ES-00000501**~~	~~**UG CONDUIT ROUTED**~~	~~**80**~~	~~**N/A**~~	~~**K217 SCOPE**~~
1	~~18~~	~~**00TX/000015-48MCC/000001-P4C**~~	~~**N/A**~~	~~**1X600_CU_XHHW-2**~~	~~**00-TX/00-00-15**~~	~~**12.47kV-480Y/277V 750kVA,3Ø,4W PADMOUNT TRANSFORMER**~~	~~**TFS BLDG #48**~~	~~**LP148-ES-00000501**~~	~~**48-MCC/00-00-01**~~	~~**RAIL CAR LOADING - 480V MCC**~~	~~**TFS BLDG #48**~~	~~**LP148-ES-00000501**~~	~~**UG CONDUIT ROUTED**~~	~~**80**~~	~~**N/A**~~	~~**K217 SCOPE**~~
1	19	**00TX/000015-48NGR/000001-N**	**N/A**	**1X4/0_CU_XHHW-2**	00-TX/00-00-15	12.47kV-480Y/277V 750kVA,3Ø,4W PADMOUNT TRANSFORMER	TFS BLDG #48 OUTDOOR	**TBD**	48-NGR/00-00-01	NGR FOR 00-TX/00-00-15	TFS BLDG #48	**LP148-ES-00000501**	UG CONDUIT ROUTED	95	**N/A**	K217 SCOPE
1	20	**48BAT/000001-48UPS/000001-C1**	**N/A**	**1PRX16_600V_SHLD_CU_PVC_TC**	**48-BAT/00-00-01**	**BATTERY CABINET**	TFS BLDG #48	**LP148-ES-00000501**	**48-UPS/00-00-01**	**UPS PANEL**	TFS BLDG #48	**LP148-ES-00000501**	FIELD ROUTED	**10**	**N/A**	**FIELD WIRING NOT TO EXCEED 100M**
1	21	**48BAT/000001-48UPS/000001-C2**	**N/A**	**1PRX16_600V_SHLD_CU_PVC_TC**	**48-BAT/00-00-01**	**BATTERY CABINET**	TFS BLDG #48	**LP148-ES-00000501**	**48-UPS/00-00-01**	**UPS PANEL**	TFS BLDG #48	**LP148-ES-00000501**	FIELD ROUTED	**10**	**N/A**	**FIELD WIRING NOT TO EXCEED 100M**
1	22	**48BAT/000001-48UPS/000001-C3**	**N/A**	**1PRX16_600V_SHLD_CU_PVC_TC**	**48-BAT/00-00-01**	**BATTERY CABINET**	TFS BLDG #48	**LP148-ES-00000501**	**48-UPS/00-00-01**	**UPS PANEL**	TFS BLDG #48	**LP148-ES-00000501**	FIELD ROUTED	**10**	**N/A**	**FIELD WIRING NOT TO EXCEED 100M**
1	23	**48UPP/000001-TRLP14801A1-P**	**N/A**	**3X8_1X10 GND_CU_XHHW-2_TC-ER**	48-UPP/00-00-01	208/120V 3PH 4W UPS DIST PANEL	TFS BLDG #48	**LP148-ES-00000501**	TR-LP1-48-01-A-1	CABLING PANEL **(UPS)**	TFS BLDG #48	**LP148-ES-00000501**	FIELD ROUTED	**55**	**N/A**	
1	24	**48UPP/000001-TRLP14801A2-P**	**N/A**	**3X8_1X10 GND_CU_XHHW-2_TC-ER**	48-UPP/00-00-01	208/120V 3PH 4W UPS DIST PANEL	TFS BLDG #48	**LP148-ES-00000501**	TR-LP1-48-01-A-2	SWITCH PANEL **(UPS)**	TFS BLDG #48	**LP148-ES-00000501**	FIELD ROUTED	**55**	**N/A**	
1	25	**48RP/000002-TRLP14801A1-P**	**N/A**	**3X8_1X10 GND_CU_XHHW-2_TC-ER**	48-RP/00-00-02	208/120V 3PH 4W GEN. SERVICE POWER PANEL	TFS BLDG #48	**LP148-ES-00000501**	TR-LP1-48-01-A-1	CABLING PANEL **(NORMAL)**	TFS BLDG #48	**LP148-ES-00000501**	FIELD ROUTED	**55**	**N/A**	
1	26	**48RP/000002-TRLP14801A2-P**	**N/A**	**3X8_1X10 GND_CU_XHHW-2_TC-ER**	48-RP/00-00-02	208/120V 3PH 4W GEN. SERVICE POWER PANEL	TFS BLDG #48	**LP148-ES-00000501**	TR-LP1-48-01-A-2	SWITCH PANEL **(NORMAL)**	TFS BLDG #48	**LP148-ES-00000501**	FIELD ROUTED	**55**	**N/A**	
1A	27	**48MCC/000001-YP0631ADS-P**	**N/A**	**3X8_1X10 GND_CU_XHHW-2_TC-ER**	48-MCC/00-00-01	RAIL CAR LOADING - 480V MCC	TFS BLDG #48	**LP1Y-EP-0009A005**	LP1Y-P-6031A-DS	UNLOADING PUMP DISCONNECT SWITCH	UNLOADING RACK - GROUND LEVEL	**LP1Y-EP-0009A005**	FIELD ROUTED IN CABLE TRAY, AS DETAILED ON PLANS	**300**	**N/A**	**K215 SCOPE**
1A	28	**YP0631ADS-YP0631A-P**	**N/A**	**3X8_1X10 GND_CU_XHHW-2_TC-ER**	LP1Y-P-6031A-DS	UNLOADING PUMP DISCONNECT SWITCH	UNLOADING RACK - GROUND LEVEL	**LP1Y-EP-0009A005**	LP1Y-P-6031A	UNLOADING PUMP	UNLOADING RACK - GROUND LEVEL	**LP1Y-EP-0009A005**	FIELD ROUTED	**15**	**N/A**	**K215 SCOPE**
1A	29	**48MCC/000001-YWH0610DS-P**	**N/A**	**3X2_1X6 GND_CU_XHHW-2_TC-ER**	48-MCC/00-00-01	RAIL CAR LOADING - 480V MCC	TFS BLDG #48	**LP1Y-EP-0009A005**	LP1Y-WH-0610-DS	WATER HEATER DISCONNECT SWITCH	UNLOADING RACK - GROUND LEVEL	**LP1Y-EP-0009A005**	FIELD ROUTED IN CABLE TRAY, AS DETAILED ON PLANS	**340**	**N/A**	**K215 SCOPE**
1A	30	**YWH0610DS-YWH0610-P**	**N/A**	**3X2_1X6 GND_CU_XHHW-2_TC-ER**	LP1Y-WH-0610-DS	WATER HEATER DISCONNECT SWITCH	UNLOADING RACK - GROUND LEVEL	**LP1Y-EP-0009A005**	LP1Y-WH-0610	WATER HEATER	UNLOADING RACK - GROUND LEVEL	**LP1Y-EP-0009A005**	FIELD ROUTED	**15**	**N/A**	**K215 SCOPE**
1A	31	**48UPP/000001-25-YIO000048001-P**	**N/A**	**2X10_1X10 GND_CU_XHHW-2_TC-ER**	48-UPP/00-00-01-25	208/120V 3PH 4W UPS DIST PANEL	TFS BLDG #48	**LP1Y-EP-0009A005**	LP1Y-IO-00-0048-001 CHARM I/O	CHARM I/O PANEL	UNLOADING RACK - GROUND LEVEL	**LP1Y-EP-0009A005**	FIELD ROUTED IN CABLE TRAY, AS DETAILED ON PLANS	**270**	**N/A**	**K215 SCOPE**
1A	32	**48RP/000002-28-YIO000048001-P**	**N/A**	**2X10_1X10 GND_CU_XHHW-2_TC-ER**	48-RP/00-00-02-28	208/120V 3PH 4W GEN. SERVICE POWER PANEL	TFS BLDG #48	**LP1Y-EP-0009A005**	LP1Y-IO-00-0048-001 CHARM I/O	CHARM I/O PANEL	UNLOADING RACK - GROUND LEVEL	**LP1Y-EP-0009A005**	FIELD ROUTED IN CABLE TRAY, AS DETAILED ON PLANS	**270**	**N/A**	**K215 SCOPE, NORMAL**
1A	33	**48RP/000002-30-YIO000048001-P**	**N/A**	**2X10_1X10 GND_CU_XHHW-2_TC-ER**	48-RP/00-00-02-30	208/120V 3PH 4W GEN. SERVICE POWER PANEL	TFS BLDG #48	**LP1Y-EP-0009A005**	LP1Y-IO-00-0048-001 CHARM I/O	CHARM I/O PANEL	UNLOADING RACK - GROUND LEVEL	**LP1Y-EP-0009A005**	FIELD ROUTED IN CABLE TRAY, AS DETAILED ON PLANS	**270**	**N/A**	**K215 SCOPE, UTILITY**
1A	34	**48UPP/000001-27-YIPP000048001-P**	**N/A**	**2X10_1X10 GND_CU_XHHW-2_TC-ER**	48-UPP/00-00-01-27	208/120V 3PH 4W UPS DIST PANEL	TFS BLDG #48	**LP1Y-EP-0009A005**	LP1Y-IPP-00-0048-001	IPP PANEL	UNLOADING RACK - GROUND LEVEL	**LP1Y-EP-0009A005**	FIELD ROUTED IN CABLE TRAY, AS DETAILED ON PLANS	**270**	**N/A**	**K215 SCOPE**
1A	35	**48RP/000002-27-YIPP000048001-P**	**N/A**	**2X10_1X10 GND_CU_XHHW-2_TC-ER**	48-RP/00-00-02-27	208/120V 3PH 4W GEN. SERVICE POWER PANEL	TFS BLDG #48	**LP1Y-EP-0009A005**	LP1Y-IPP-00-0048-001	IPP PANEL	UNLOADING RACK - GROUND LEVEL	**LP1Y-EP-0009A005**	FIELD ROUTED IN CABLE TRAY, AS DETAILED ON PLANS	**270**	**N/A**	**K215 SCOPE, NORMAL**
1A	36	**48RP/000002-29-YIPP000048001-P**	**N/A**	**2X10_1X10 GND_CU_XHHW-2_TC-ER**	48-RP/00-00-02-29	208/120V 3PH 4W GEN. SERVICE POWER PANEL	TFS BLDG #48	**LP1Y-EP-0009A005**	LP1Y-IPP-00-0048-001	IPP PANEL	UNLOADING RACK - GROUND LEVEL	**LP1Y-EP-0009A005**	FIELD ROUTED IN CABLE TRAY, AS DETAILED ON PLANS	**270**	**N/A**	**K215 SCOPE, UTILITY**
1A	37	**48MCC/000001-YP0631ADS-C1**	**N/A**	**1PRX16_600V_SHLD_CU_PVC_TC**	48-MCC/00-00-01	RAIL CAR LOADING - 480V MCC	TFS BLDG #48	**LP1Y-EP-0009A005**	LP1Y-P-6031A	UNLOADING PUMP CONTROLS	UNLOADING RACK - GROUND LEVEL	**LP1Y-EP-0009A005**	FIELD ROUTED IN CABLE TRAY, AS DETAILED ON PLANS	**300**	**N/A**	**K215 SCOPE**
1A	38	**48MCC/000001-YP0631ASHDS-P**	**N/A**	**2X10_1X10 GND_CU_XHHW-2_TC-ER**	48-MCC/00-00-01 (120V POWER VIA 48-RP/00-00-01-15)	RAIL CAR LOADING - 480V MCC	TFS BLDG #48	**LP1Y-EP-0009A005**	LP1Y-P-6031A-SH-DS	MOTOR SPACE HEATER DISCONNECT SWITCH	UNLOADING RACK - GROUND LEVEL	**LP1Y-EP-0009A005**	FIELD ROUTED IN CABLE TRAY, AS DETAILED ON PLANS	**300**	**N/A**	**K215 SCOPE**
1A	39	**YP0631ASHDS-YP0631ASH-P**	**N/A**	**2X10_1X10 GND_CU_XHHW-2_TC-ER**	LP1Y-P-6031A-SH-DS	MOTOR SPACE HEATER DISCONNECT SWITCH	UNLOADING RACK - GROUND LEVEL	**LP1Y-EP-0009A005**	LP1Y-P-6031A-SH	MOTOR SPACE HEATER	UNLOADING RACK - GROUND LEVEL	**LP1Y-EP-0009A005**	FIELD ROUTED	**15**	**N/A**	**K215 SCOPE**
1A	40	**48RP/000001-42-YSSEW15062101-P**	**N/A**	**2X10_1X10 GND_CU_XHHW-2_TC-ER**	48-RP/00-00-01-42	208/120V 3PH 4W GEN. SERVICE POWER PANEL	TFS BLDG #48	**LP1Y-EP-0009A005**	LP1Y-SSEW-15-0621-01	SAFETY SHOWER EYEWASH	UNLOADING RACK - GROUND LEVEL	**LP1Y-EP-0009A005**	FIELD ROUTED IN CABLE TRAY, AS DETAILED ON PLANS	**400**	**N/A**	**K215 SCOPE**
1A	41	**48RP/000001-40-YSSEW15062102-P**	**N/A**	**2X10_1X10 GND_CU_XHHW-2_TC-ER**	48-RP/00-00-01-40	208/120V 3PH 4W GEN. SERVICE POWER PANEL	TFS BLDG #48	**LP1Y-EP-0009B005**	LP1Y-SSEW-15-0621-02	SAFETY SHOWER EYEWASH	UNLOADING RACK - PLATFORM LEVEL	**LP1Y-EP-0009B005**	FIELD ROUTED IN CABLE TRAY, AS DETAILED ON PLANS	**360**	**N/A**	**K215 SCOPE**
1A	42	**48UPP/000001-29-YLCP0000640001-P**	**N/A**	**2X10_1X10 GND_CU_XHHW-2_TC-ER**	48-UPP/00-00-01-29	208/120V 3PH 4W UPS DIST PANEL	TFS BLDG #48	**LP1Y-EP-0009B005**	LP1Y-LCP-000-0640-001	LEVEL CONTROL PANEL	UNLOADING RACK - PLATFORM LEVEL	**LP1Y-EP-0009B005**	FIELD ROUTED IN CABLE TRAY, AS DETAILED ON PLANS	**450**	**N/A**	**K215 SCOPE**
1A	43	**48UPP/000001-30-YLCP0000650001-P**	**N/A**	**2X10_1X10 GND_CU_XHHW-2_TC-ER**	48-UPP/00-00-01-30	208/120V 3PH 4W UPS DIST PANEL	TFS BLDG #48	**LP1Y-EP-0009B005**	LP1Y-LCP-000-0650-001	LEVEL CONTROL PANEL	UNLOADING RACK - PLATFORM LEVEL	**LP1Y-EP-0009B005**	FIELD ROUTED IN CABLE TRAY, AS DETAILED ON PLANS	**450**	**N/A**	**K215 SCOPE**
1A	44	**48LP/000001-02-K215/LIGHTING-P**	**N/A**	**2X10_1X10 GND_CU_XHHW-2_TC-ER**	48-LP/00-00-02-01	480/277V 3PH 4W LIGHTING POWER PANEL	TFS BLDG #48	**LP1Y-EL-00090001**	MULTIPLE LIGHTS (NA)	NORMAL LIGHTING - UNLOADING RACK	UNLOADING RACK	**LP1Y-EL-00090001**	FIELD ROUTED IN CABLE TRAY, AS DETAILED ON PLANS	**600**	**N/A**	**K215 SCOPE, MULTIPLE LIGHT FIXTURES, LENGTH BASED OFF LONGEST CIRCUIT STRETCH**
1A	45	**48MCC/000001-Y48WR000002-P**	**N/A**	**3X4_1X6 GND_CU_XHHW-2_TC-ER**	48-MCC/00-00-01	RAIL CAR LOADING - 480V MCC	TFS BLDG #48	**LP1Y-EP-0009B005**	48-WR/00-00-02	WELDING RECEPTACLE	UNLOADING RACK - PLATFORM LEVEL	**LP1Y-EP-0009B005**	FIELD ROUTED IN CABLE TRAY, AS DETAILED ON PLANS	**400**	**N/A**	**K215 SCOPE**
1A	46	**48RP/000001-38-K215/RECEPTS1-P**	**N/A**	**2X10_1X10 GND_CU_XHHW-2_TC-ER**	48-RP/00-00-01-38	208/120V 3PH 4W GEN. SERVICE POWER PANEL	TFS BLDG #48	**LP1Y-EP-0009A005**	MULTIPLE RECEPTACLES (NA)	CONVENIENCE RECEPTACLES - UNLOADING RACK - GROUND LEVEL	UNLOADING RACK - GROUND LEVEL	**LP1Y-EP-0009A005**	FIELD ROUTED IN CABLE TRAY, AS DETAILED ON PLANS	**500**	**N/A**	**K215 SCOPE, MULTIPLE RECEPTACLES, WP, DUPLEX, LENGTH BASED OFF FULL CIRCUIT RUN**
1A	47	**48RP/000001-36-K215/RECEPTS2-P**	**N/A**	**2X10_1X10 GND_CU_XHHW-2_TC-ER**	48-RP/00-00-01-36	208/120V 3PH 4W GEN. SERVICE POWER PANEL	TFS BLDG #48	**LP1Y-EP-0009B005**	MULTIPLE RECEPTACLES (NA)	CONVENIENCE RECEPTACLES - UNLOADING RACK - PLATFORM LEVEL	UNLOADING RACK - PLATFORM LEVEL	**LP1Y-EP-0009B005**	FIELD ROUTED IN CABLE TRAY, AS DETAILED ON PLANS	**500**	**N/A**	**K215 SCOPE, MULTIPLE RECEPTACLES, WP, DUPLEX, LENGTH BASED OFF FULL CIRCUIT RUN**
1A	48	**48RP/000001-35-GMD/000621A-P**	**N/A**	**2X10_1X10 GND_CU_XHHW-2_TC-ER**	48-RP/00-00-01-35	208/120V 3PH 4W GEN. SERVICE POWER PANEL	TFS BLDG #48	**LP1Y-EP-0009A005**	LP1-GMD/00-0621A	GROUNDING MONITORING SYSTEM PANEL	UNLOADING RACK - GROUND LEVEL	**LP1Y-EP-0009A005**	FIELD ROUTED IN CABLE TRAY, AS DETAILED ON PLANS	**270**	**N/A**	**K036 SCOPE [DESIGN HOLD]**
1A	49	**48RP/000001-37-GMD/000631A-P**	**N/A**	**2X10_1X10 GND_CU_XHHW-2_TC-ER**	48-RP/00-00-01-37	208/120V 3PH 4W GEN. SERVICE POWER PANEL	TFS BLDG #48	**LP1Y-EP-0009A005**	LP1-GMD/00-0631A	GROUNDING MONITORING SYSTEM PANEL	UNLOADING RACK - GROUND LEVEL	**LP1Y-EP-0009A005**	FIELD ROUTED IN CABLE TRAY, AS DETAILED ON PLANS	**270**	**N/A**	**K036 SCOPE [DESIGN HOLD]**
1A	50	**48RP/000001-39-GMD/000640A-P**	**N/A**	**2X10_1X10 GND_CU_XHHW-2_TC-ER**	48-RP/00-00-01-39	208/120V 3PH 4W GEN. SERVICE POWER PANEL	TFS BLDG #48	**LP1Y-EP-0009A005**	LP1-GMD/00-0640A	GROUNDING MONITORING SYSTEM PANEL	UNLOADING RACK - GROUND LEVEL	**LP1Y-EP-0009A005**	FIELD ROUTED IN CABLE TRAY, AS DETAILED ON PLANS	**450**	**N/A**	**K036 SCOPE [DESIGN HOLD]**
1A	51	**48RP/000001-41-GMD/000651A-P**	**N/A**	**2X10_1X10 GND_CU_XHHW-2_TC-ER**	48-RP/00-00-01-41	208/120V 3PH 4W GEN. SERVICE POWER PANEL	TFS BLDG #48	**LP1Y-EP-0009A005**	LP1-GMD/00-0651A	GROUNDING MONITORING SYSTEM PANEL	UNLOADING RACK - GROUND LEVEL	**LP1Y-EP-0009A005**	FIELD ROUTED IN CABLE TRAY, AS DETAILED ON PLANS	**450**	**N/A**	**K036 SCOPE [DESIGN HOLD]**

1A